

Welcome to

MLL 104

Characterization of Materials Lab

2022-23 Even Semester

3 – 0 – 3

Team

Krishna Balasubramanian

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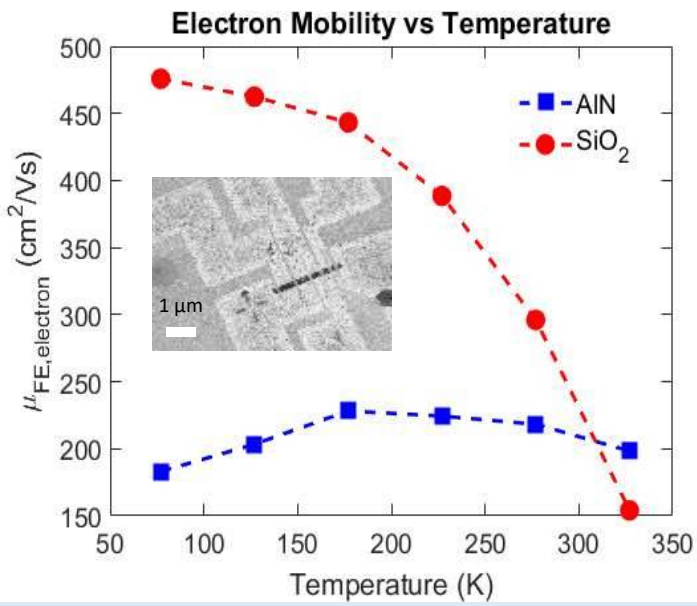
Leena Nebhani

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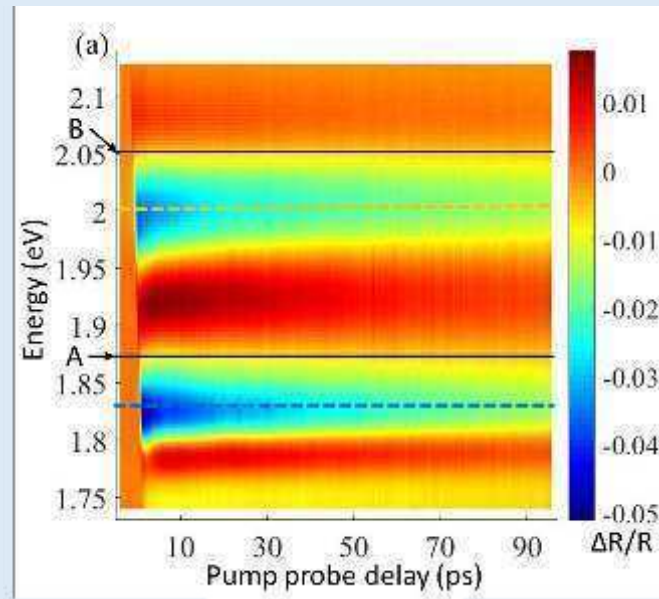
Teaching Assistants: (11)

Dr. Nagaraj, Durgesh Banswar, Ingita Indu, Priyanka Goel, Malar Vadani, Subhashree Subhasmita Pradhan, Chhotrai Soren, Mahesh Ranjan Dash, Monomoy Chatterjee, Sarthak Rautela

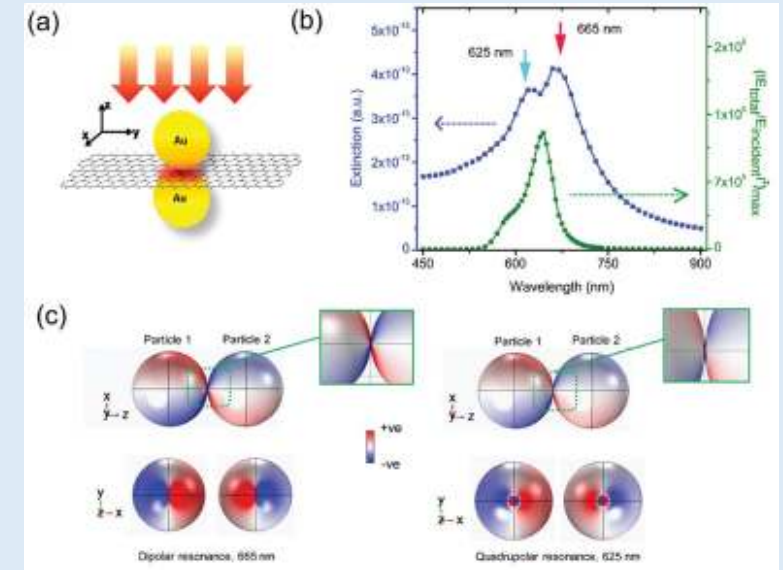
Graphene on III-V substrates



Ultra-fast dark exciton dynamics – TMD MoS₂

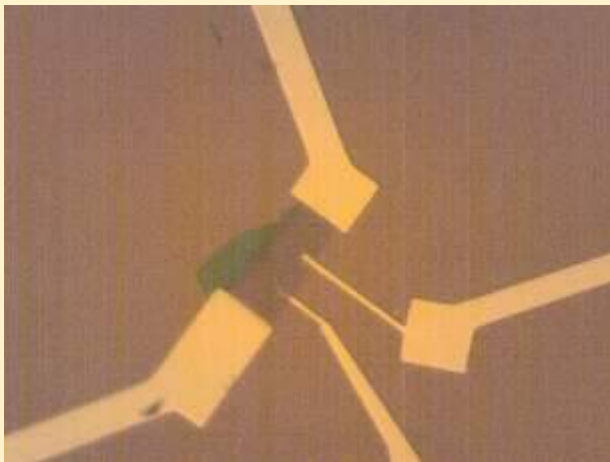


2D-nanoparticle plasmonics

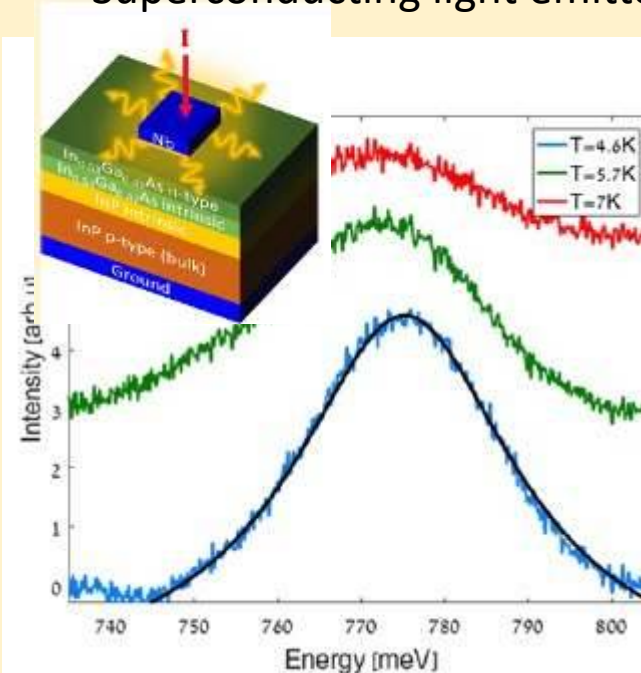


2D Electron transport and optics

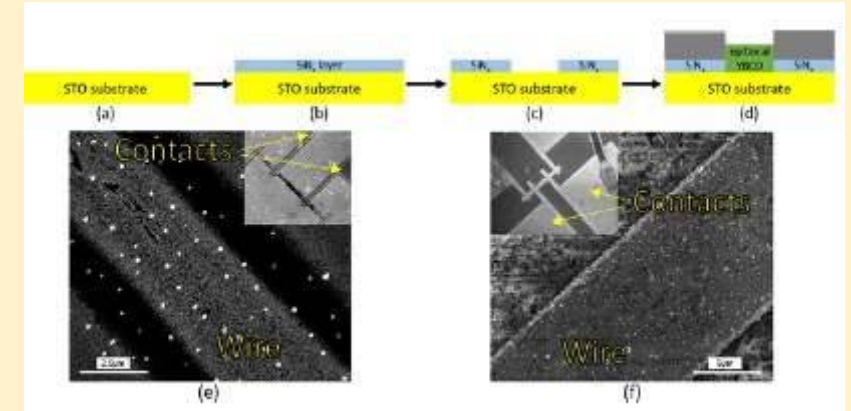
2D – Superconductor interfaces



Superconducting light emitters



Superconducting nanowire photon detector



Nature communications 10 (1), 1-9
 Optics Express 27 (23), 33427-33435
 physica status solidi (a) 217 (16), 1900949
 Journal of Physics D: Applied Physics 49 (26), 265301
 Nanotechnology 27 (20)

Superconductivity

About the course

- Introductory course on material characterization techniques.
- Introduce the concept of measurement, observations, qualification and characterization.
- Start with fundamental principles – measurement technique - Instrumentation
- The measurement techniques will involve several fundamental physical properties which every engineer, particular a materials engineer should master.
- The course has **three hours of lecture and three hours** of lab session per week and has a massive **4.5** credits.

Content

- Fundamental of measurement
- Optical Spectroscopy Techniques
- Optical microscopy
- X-ray diffraction and applications
- Scanning electron microscopy
- Thermal analysis
- Electrical characterization

Course Information

Course Timing:

- Lectures
Slot E, Tuesday, Wednesday, and Friday 10 AM – 11 AM
- Labs
Mondays 2 – 5 PM. All students will need to assemble in the lab hall (to be decided)

Prerequisites:

- Basic first year Math, Physics – Calculus, Electromagnetics.
- Inquisitiveness to learn!

Lecture Hosting

- Lecture delivery:
 - All sessions in-person at LH604
 - Attendance will be recorded. In fact participation in lectures is mandatory.
- Homework to be submitted before the class sessions. Could involve simple coding exercising.
 - Submissions are binary: either you submit before time or leave without marks. Sorry, no concessions for any amount of delay!
- All exams and evaluation will be offline. The answer sheets will be evaluated by TA assigned.
- Dishonesty will be very seriously dealt with. We are a huge team of very experienced professionals. So, any malpractice will be identified, and the student will be given a 'F'. Remember, this is much worse than just dropping the course!

Lab Hosting

- Experiments will happen at different places in the department/Institute
- A lab file is to be maintained by all. Each lab you will bring the file with sheets.
- In lab session,
 - We will start with an introductory briefing by a faculty member (Either myself/Prof. Leena)
 - A question list will be provided which needs answering
 - We will divide you ppl into sufficiently small groups and each group will be independently taken for experiments.
 - The group will return back and complete the lab sheet.
 - The lab sheets are individual exercises. They are to be treated as exam papers and filled individually.
 - You are to bring several A4 sheets, graph papers, stapler, calculator and necessary stationaries.
 - No one is allowed to leave before 5 PM.
 - The sheets are expected to be completed on the same day and submitted to the TA. It will not be collected later on.
 - After evaluation, the sheets will be returned back and it needs to be filed.
 - A final viva-voce will be conducted on the last day of the course and your lab file be collected for evaluation.

References

- Multiple references for each topic. Will be announced during the lecture time. General references are as follows:
- Pavia, Lapman and Kris, Introduction to spectroscopy, Brooks Cole 2000
- B. D. Cullity and S. R. Stock, Elements of X-ray Diffraction, Prentice Hall; 3 ed. 2001R
- J. Goldstein, Scanning electron microscopy and X-Ray Microanalysis
- R. Haynes, Optical Microscopy of Materials, Springer, 1984
- Michael E. Brown and Patrick K. Gallagher, Handbook of Thermal Analysis and Calorimetry, Elsevier, 2003.
- Peter J. Haines, Thermal methods of Analysis, Principles, Applications and Problems, Springer

Evaluation policy

Evaluation:

- Relatively graded course with the following split up:

Lecture part:

- **35% minor, 45 % major, 10% homework/assignments and 10 % in-class interactions**

- Practical part:

- **60% - lab sheet evaluations, 30% Viva-Voce and 10% participation**

- **Final score: 60% Lectures and 40 % Practical**

- **Audit Pass is 35%**

Academic Honesty

All students registering to this course voluntarily commit to the academic integrity, research and Ethics policy, IIT Delhi (https://academics.iitd.ac.in/sites/default/files/registration/forms/10_FORM%20H.pdf). This is an under-graduate level course in the country's most premier institute, and I expect the students to know their responsibility. If any document submitted to us is plagiarized (copied verbatim either from textbook/ online resources/ fellow students...), the student will be deregistered and promptly handed to institute authorities.

Discussion is okay for homework, but the submission needs to reflect the student's work. No discussion/references are allowed in lab sheets, quizzes/minors/final exams.

Characterization, Measurement and Metrology

Character:

person: thinks, feels and behaves

Material: ?

Physical: density, thermal/electrical conductivity, optical transparency/reflectivity, hygroscopicity

Mechanical: Hardness, tenacity, elasticity, ductility, malleability,

Chemical: Reactivity, catalytic property, corrosion resistance, flammability

Measurement:

An act of quantification of the character

The International Bureau of Weight and Measures states that: "Metrology is the science of measurement, embracing both experimental and theoretical determinations at any level of uncertainty in any field of science and technology."

Measurement

I believe in evidence. I believe in observation, measurement, and reasoning, confirmed by independent observers. I'll believe anything, no matter how wild and ridiculous, if there is evidence for it. The wilder and more ridiculous something is, however, the firmer and more solid the evidence will have to be.³¹ (Isaac Asimov)

Count what is countable, measure what is measurable and what is not measurable, make measurable (Galileo Galilei)

An experiment is a question which science poses to Nature, and a measurement is the recording of Nature's answer.⁴⁶ (Max Planck)

If someone separated the art of counting and measuring and weighing from all the other arts, what was left of each (of the others) would be, so to speak, insignificant.⁴⁷ (Plato)

I often say that when you can measure what you are speaking about, and express it in numbers you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind: it may be the beginning of knowledge, but you have scarcely advanced to the stage of science whatever the matter may be [2]. (Lord Kelvin)

What is measurement (Historical Perspective)



Among the earliest known measurement is of length and weight, As early as 2900 BC.

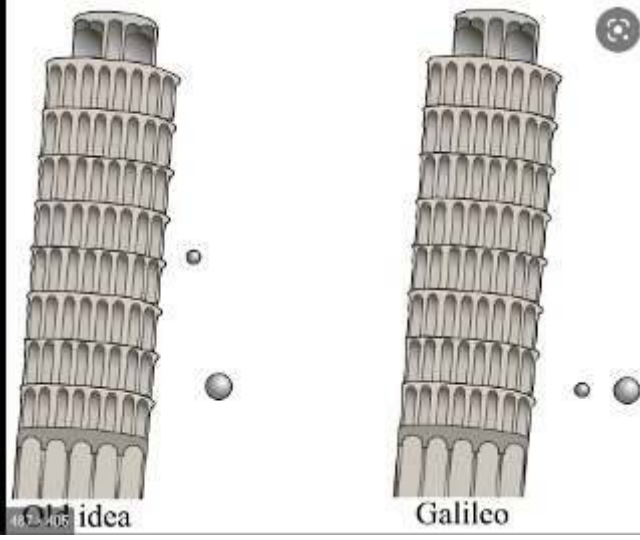


No side of the Pyramid's square base deviated from the 9000 inch value by 1/20 Inch

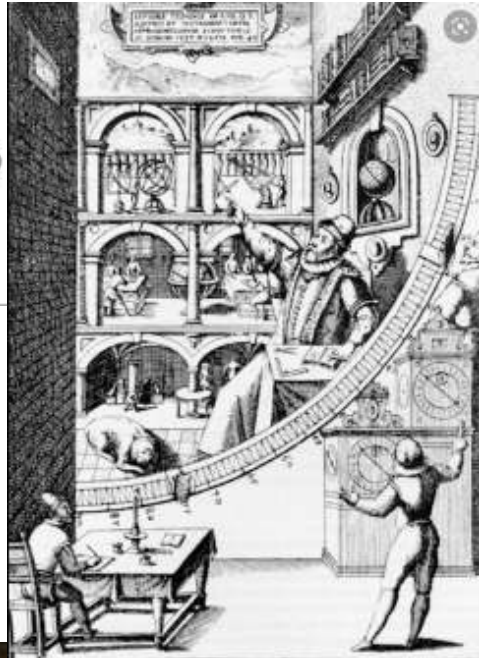
The right angle never deviated greater than 3/1000 a degree.

750-year-old sun dial in Konark, Odisha

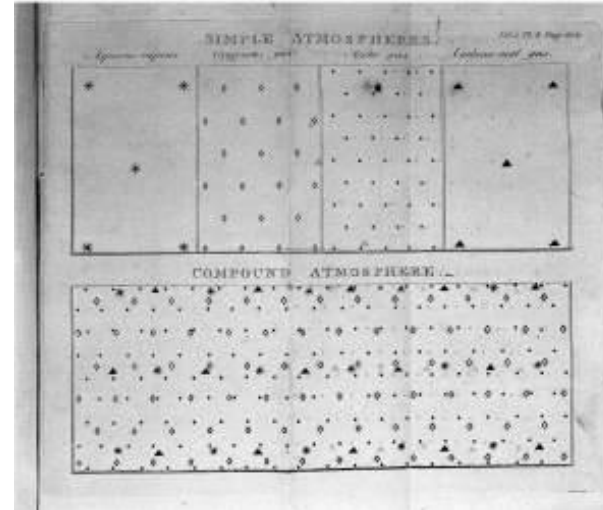




Galileo's gravitational experiment



Astronomical observations
– Tycho Brahe



Observation of chemical reactions
J. Dalton



X-rays through Au foil, Atomic structure

What is measurement ?

- Experimentally assign to a quantity a number for comparison and calculations.
- Many a times, we define a 'standard' for a particular measure and compare the object to the standard (unit)
- Necessary conditions of measurement:
 - The act of measurement does not affect the object
 - Preferably, the measuring apparatus does not degrade and can be reused infinitely.
- Qualities of a measurement:
 - Accuracy
 - Precision
 - Resolution
 - Sensitivity
 - True Value
 - Error

Types of measurement

SCALE	DEFINITION	EXAMPLE
Nominal	Only the presence/absence of an attribute; can only count items	go/no go; success/fail; accept/reject
Ordinal	Can say that one item has more or less of an attribute than another item; can order a set of items	taste; attractiveness
Interval	Difference between any two successive points is equal; often treated as a ratio scale even if assumption of equal intervals is incorrect; can add, subtract, order objects	calendar time; temperature
Ratio	True zero point indicates absence of an attribute; can add, subtract, multiply and divide	elapsed time; distance; weight

DIY technique



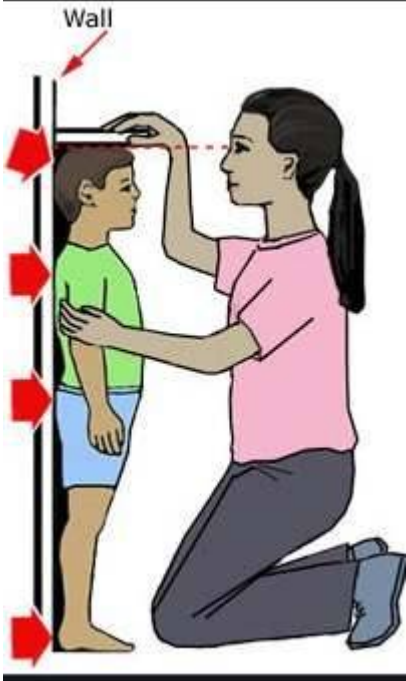
Call a friend



Go to a professional



Call a Scientist



What ever you do, you will finally come out with a singular value.

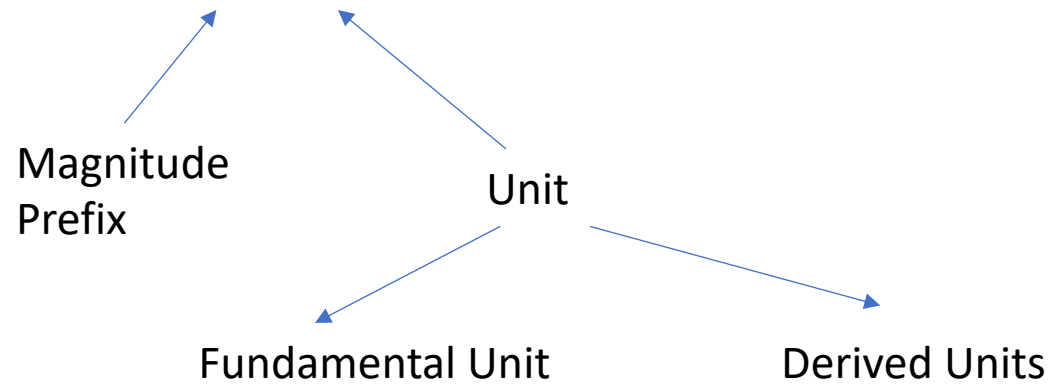
Say the exact height of the person is 5 feet, 10 inches.

The act of measurement on an object's attribute

- The act observes the right attribute (validity)
- The act observes the attribute to the closest value possible (accuracy)
- The act does not alter the object
- Repeated acts results in similar results (repeatability, stability)
- An object (n-times the original) should preferably result in n-times the originally observed (linearity)

Lets break the measurement down

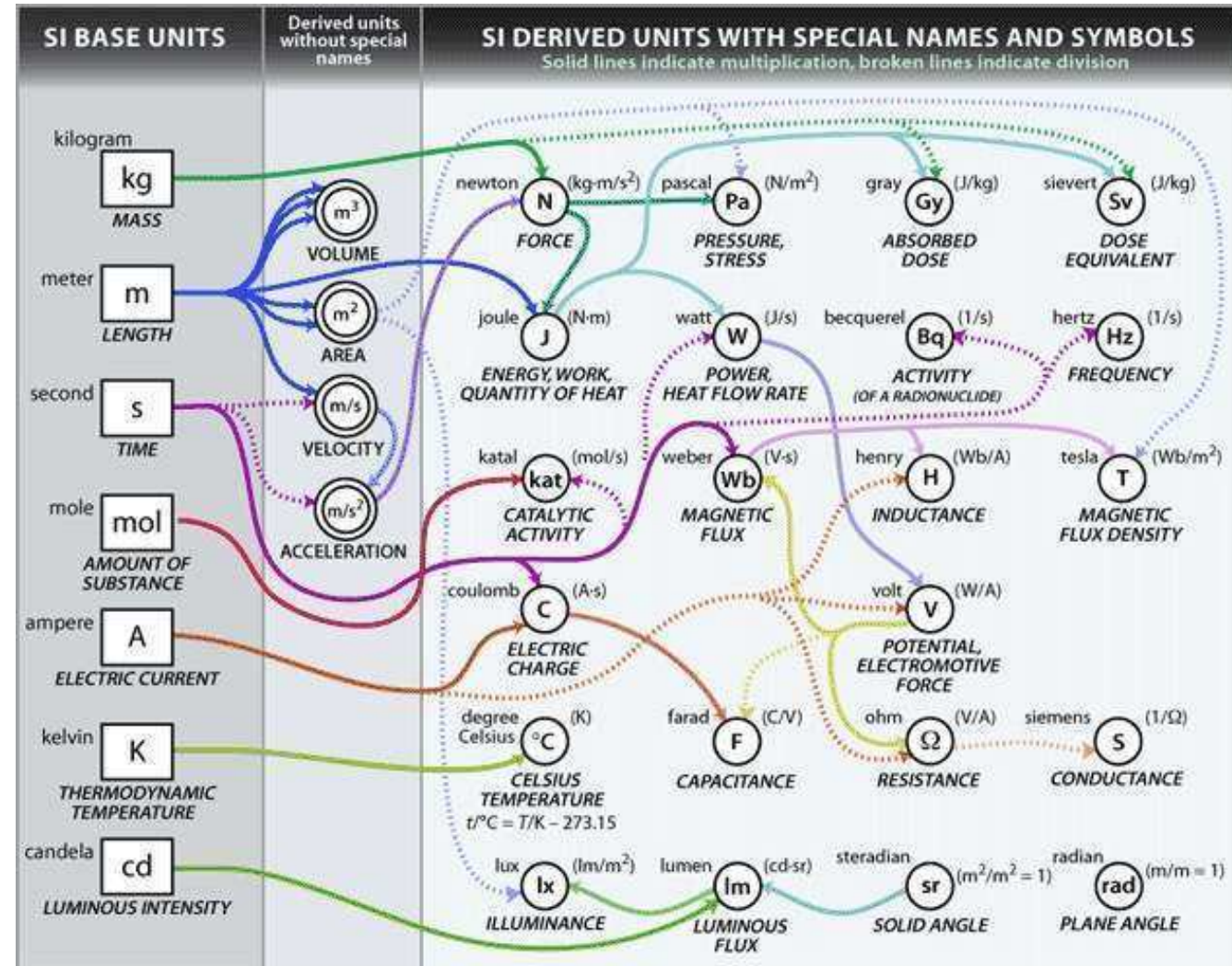
The height is 5 ft 10 inches



Any measure is expressed as a product of numeric with a measurable quantity

Units and Standards

- Earlier units like foot, inch, yard, furlong are very hard to compare.
 - First introduced by Egyptians, 1ft = 12 inches, yard = 3 ft and mile about 5000 ft. [things were later changed]
 - Defeats the purpose of measurement
- In 1960, Systeme International Denominations (SI) gave the standards for the units



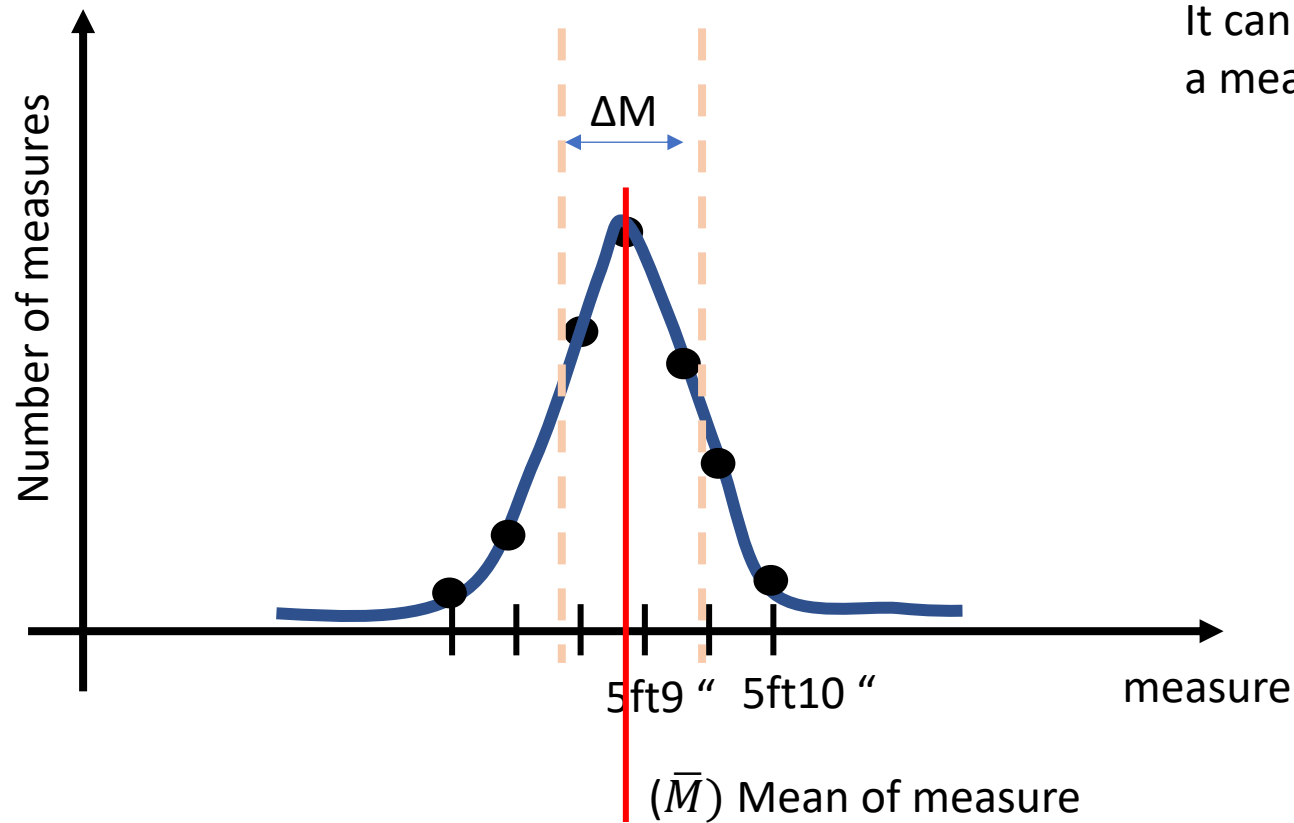
Fundamental Units

- These are fundamental jumps in our understanding
- **Meter (m)**, is the length of the journey path from the light in the vacuum in a range of $1/299\,792\,458$ seconds;
- **Second (s)**, is the duration of 9,192,631,770 periods of radiation corresponding to the transition between two hyperfine levels of the fundamental state of the Cesium Atom 133;
- **Kilogram (kg)**, is the mass unit; It is equal to the mass of the international prototype of the kilogram >> prototype made of Iridium Platinum in 1889 and conserved in Sevres;
- **Amperes (a)**, is the intensity of a constant electric current which, kept in two parallel straight conductors of infinite length, of negligible circular section, placed at the distance of one meter from each other in the void, would produce between these conductors A force equal to 2×10^{-7} Newton on each metre of length;
- **Kelvin (K)**, thermodynamic temperature unit, is the fraction $1/273,16$ of the thermodynamic temperature of the triple point of the water;
- **Mole (MOL)**, is the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kg of carbon 12. When using the mole, the elementary entities must be specified; They can be atoms, molecules, ions, electrons, other particles, or specified groupings of such particles;
- **Candle (CD)**, is the luminous intensity, in a given direction, of a source emitting a monochromatic frequency radiation 540×10^{12} hertz and whose energy intensity in that direction is $1/683$ watts to the Steradian.

Quality of a measure

- How reliable is your prefix/magnitude ?
- Q1: Is your measurement valid ?
- Q2: How accurate is your measurement
- Q3: What now happens if there were multiple attempts at the measurement using the same equipment and the object ?





If each of the measurement event is uncorrelated (independent)
It can be proved that the outcome will be normally distributed with a mean and variance

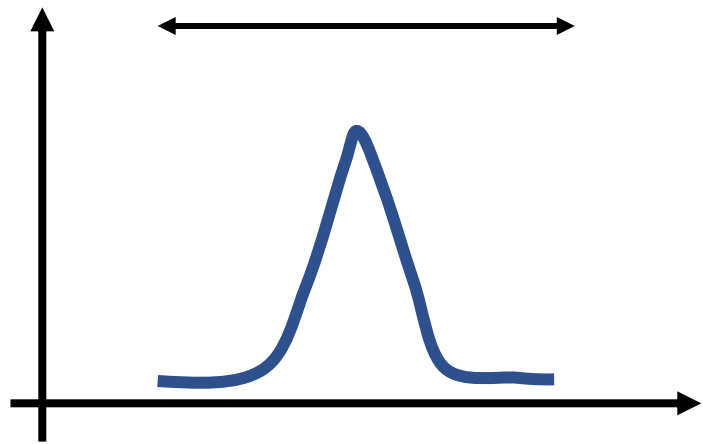
A single act of measurement gives a value (say M)

Least significant digit of your measure,
A band of values which you cannot distinguish (δM)
is called as resolution.

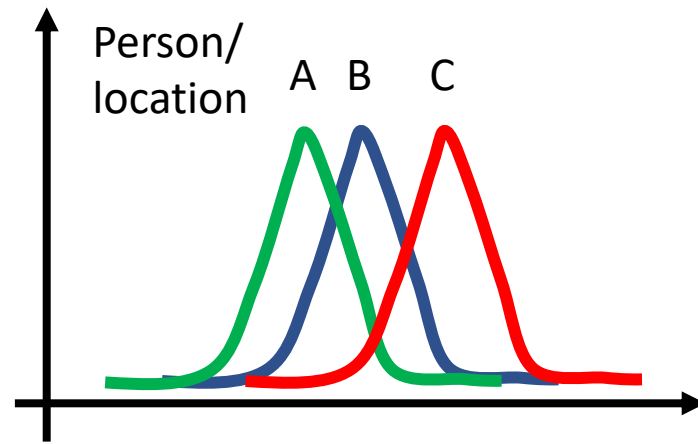
If numerous such measurements are made, (say $M_1, M_2, \dots, M_n$)

One can plot a histogram and obtain a mean of all the measurements $(\bar{M})$ $\bar{M} = \frac{\sum M_i}{n}$

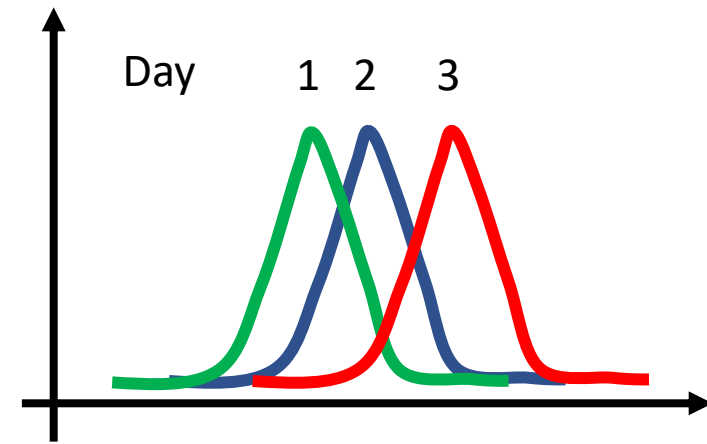
Precision of the measure is given by the width of the normal distribution $(\Delta M) = \pm \sqrt{\frac{\sum (M - \bar{M})^2}{n}}$



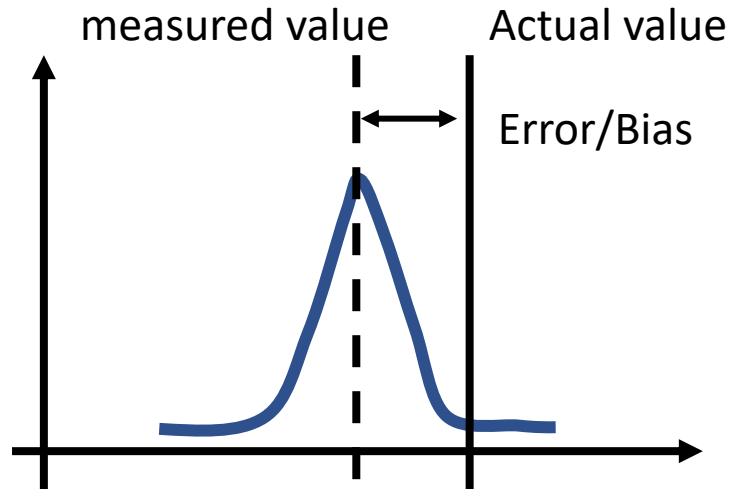
Repeatability of a measurement:



Reliability of a measurement:

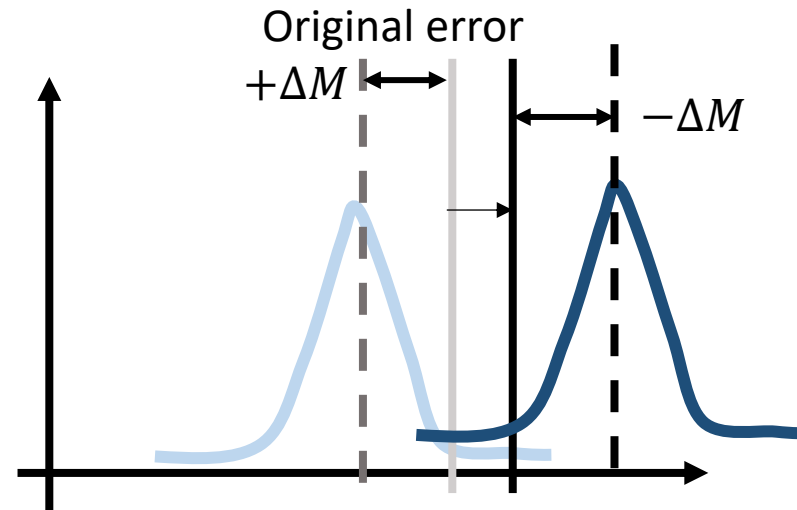


Stability of a measurement:



Error in the measurement

Difference between the measured and actual value is the error in the measurement



Linearity

If the actual value is changed by a small value, linear measurement will have a proportional change in the measured value and hence the error

Remember all these aspects as we conduct several measurements using very different techniques.

'Light' As Characterization Tool

Need to Characterize Materials:

Say I hand you a material

Phase of the material:	Solid, liquid, gaseous or the plasma
Physical properties	Density, electrical conductivity, magnetism, optical transparency, thermal conductivity,...
Mechanical properties	Strength, elasticity, stiffness, fracture characteristics, weldability, malleability, ductility...
Structure	Constituents, phases, stoichiometry, polymorph,...

Are there correlations between the properties ?

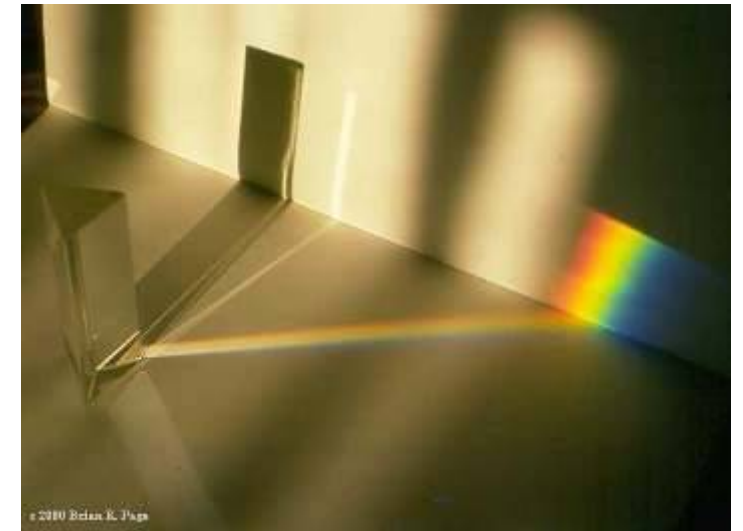
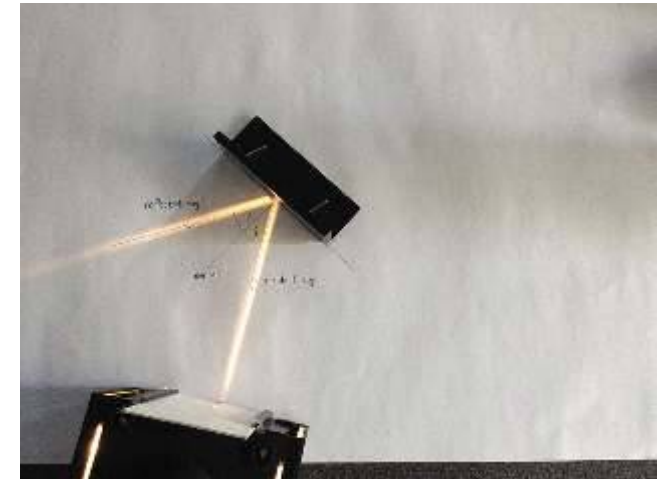
How to characterize ?

- To study/observe any property of the material, we perturb the material with an external energy source
 - Electromagnetic
 - Mechanical
 - Thermal
 - Chemical
- Measure the response of the material
- A good measure has to satisfy certain qualities mentioned before

Measurements using Electromagnetic systems

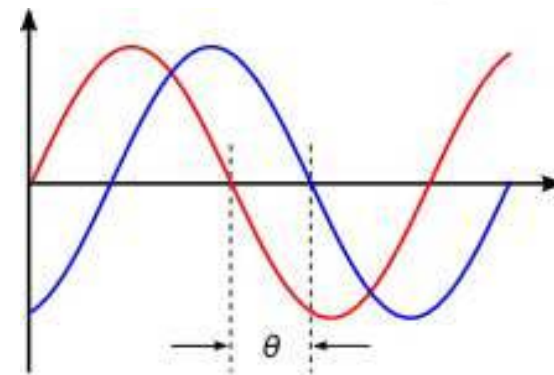
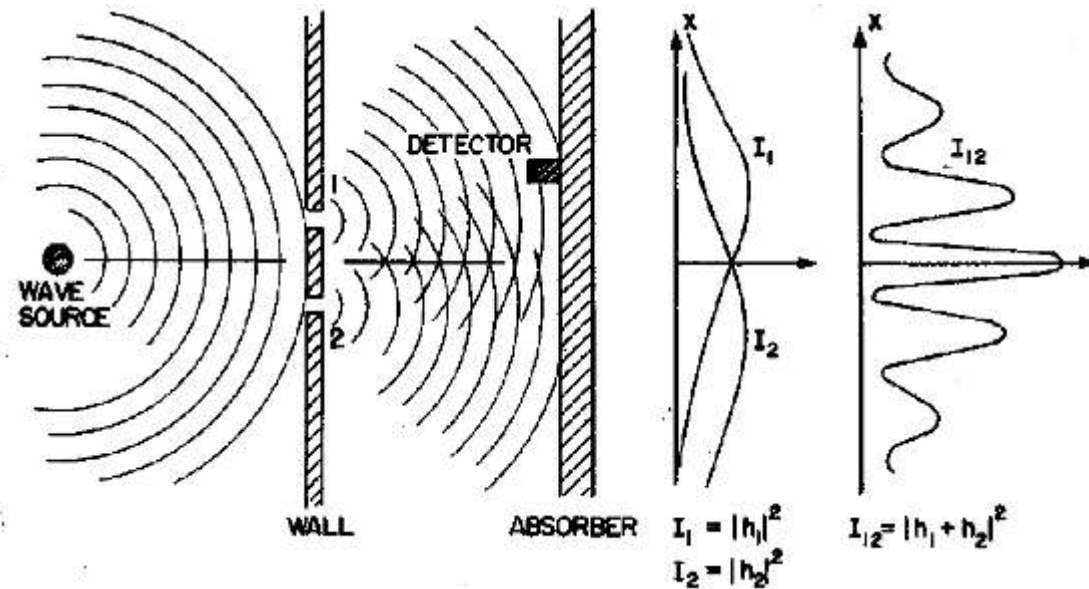
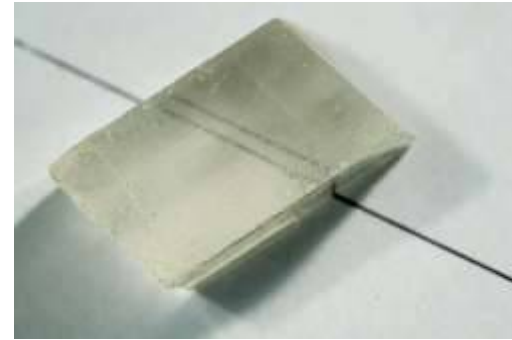
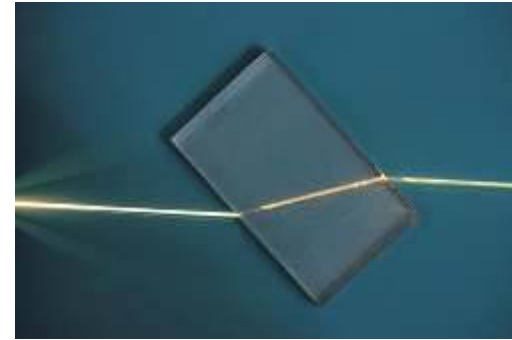
What do we know about light ?

- Light (for now forget the form) has a source.
- It emanates from the source, and travels at incredible speeds.
- Certain objects can stop the transport, form shadows.
- Light beams are reflected (sent back) such that they follow a rule $\theta_{inc} = \theta_{ref}$
- Light beams at the interface of two materials bends (refract) such that $\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_1}{n_2}$
- Velocity of light in medium $v_1 = \frac{c}{n_1}$
- The 'color' of light determines the index 'n' – Reason why prism splits different colors



What do we know about light ?

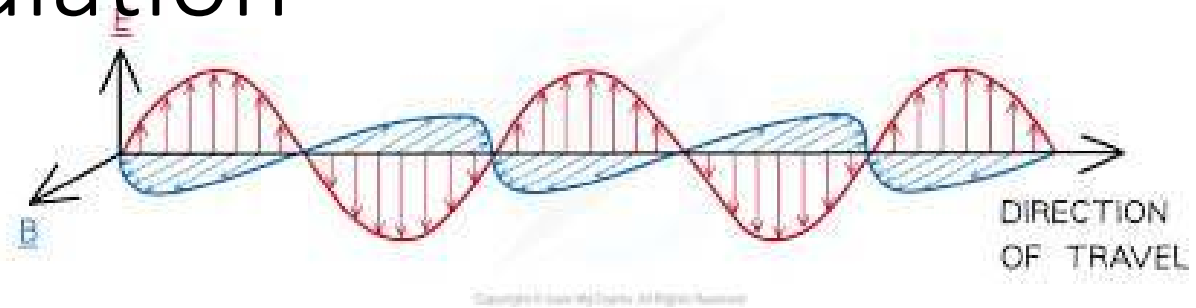
- In certain crystals, there is evidence of double refraction: evidence of at least two polarizations
- Light radiations diffract: they seem to bend around obstacles.
- When there are two near by sources, they seem to interact – wave like.
- In addition to intensity, and polarization, light also has property: Phase



$$Ae^{i(kx - \omega t + \phi)}$$

Light as electromagnetic radiation

- All electric and magnetic fields obeys:



$\nabla \cdot E = \frac{\rho}{\epsilon_0}$	(1)	Gauss' law
$\nabla \cdot B = 0$	(2)	Magnetic monopoles
$\nabla \times E = -\frac{\partial B}{\partial t}$	(3)	Faraday's law
$\nabla \times H = J + \frac{\partial D}{\partial t}$	(4)	Ampere-Maxwell law

The solutions of these differential equations are sinusoidal E and M waves

- which are perpendicular to each other
- Which travel with a velocity $V = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 299792458 \frac{m}{s}$
- Energy travels perpendicular to both E and B, along the Poynting's vector $S = E \times H$

All these seems very similar to Light: -> Light must be an electromagnetic wave! With E always perpendicular to B

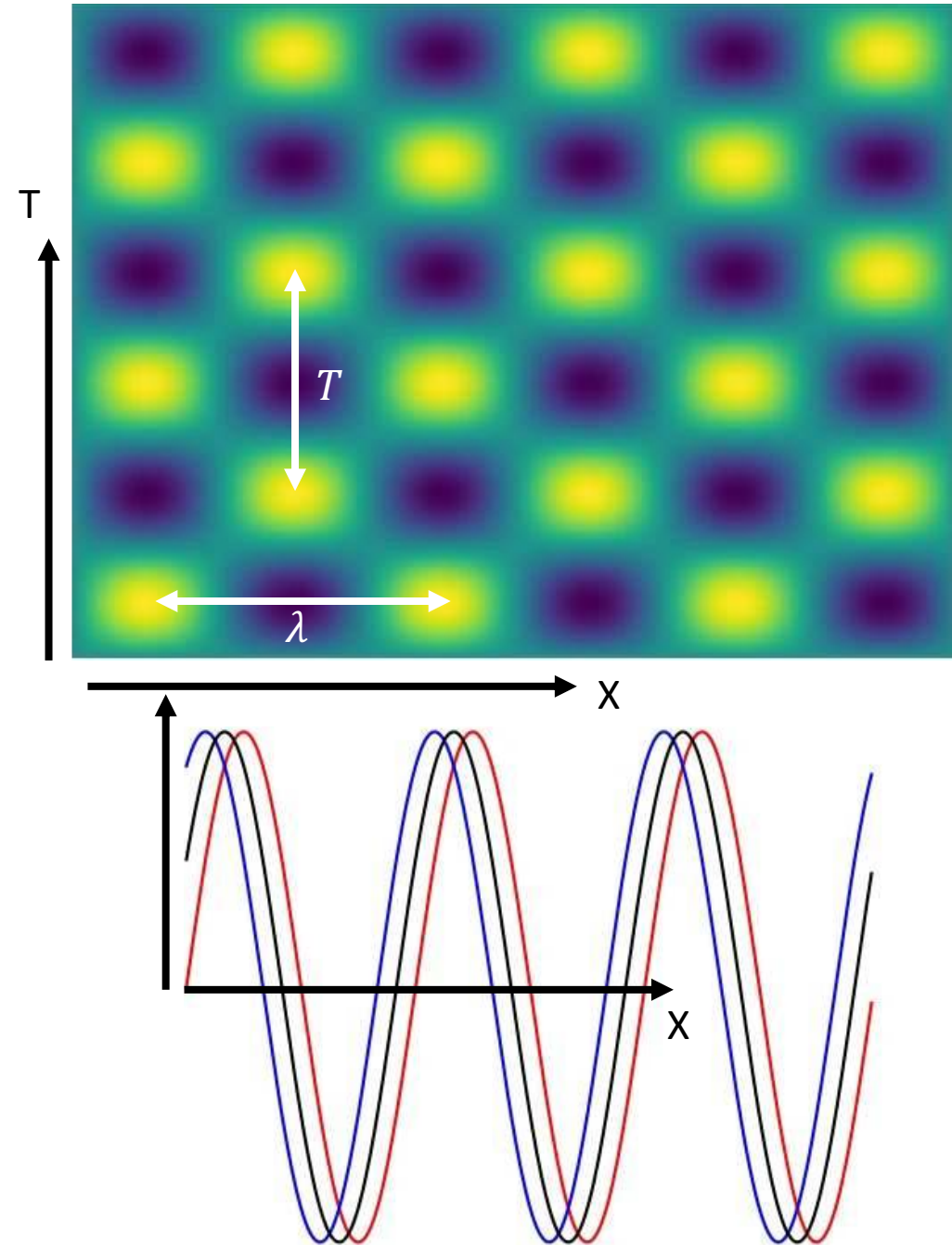
Properties of a wave

- The wave is periodic in space and in time!

- The spatial periodicity : wave vector $k = \frac{2\pi}{\lambda}$
- The temporal periodicity: frequency $\omega = \frac{2\pi}{T}$

Final wave equation $\psi = A \sin(kx - \omega t + \phi)$

- Velocity of the wave: $\frac{\partial x}{\partial t} = \frac{\frac{\partial \psi}{\partial t}}{\frac{\partial \psi}{\partial x}} = \frac{\omega}{k}$
- This velocity is also called as the phase velocity
The velocity at which the phase of the wave travel



Energy in a wave (Light)

- Many a times, we want to measure the energy in a wave
 - Typical experiments use light and one want's to measure what the object did to light

Power in electromagnetic radiation defined by ψ

$$p(x) \propto \frac{\partial \psi}{\partial x} \frac{\partial \psi}{\partial t}$$

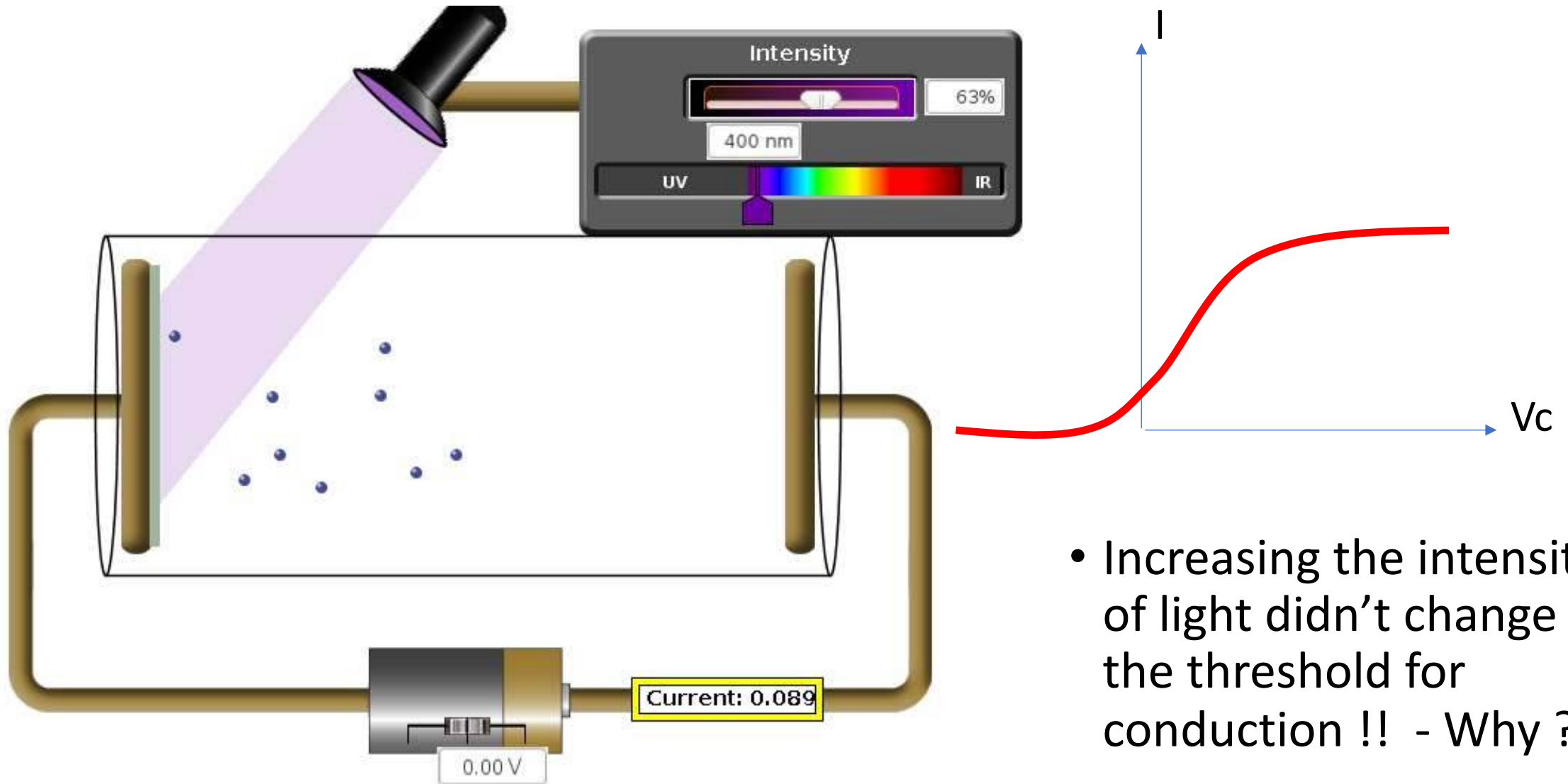
This gives the power per unit area $P(x) \propto k\omega A^2 \cos^2(kx - \omega t)$

Finally a time averaged power in an EM wave is given by $I \propto k\omega \frac{1}{T} \int_0^T \cos^2(kx - \omega t)$

$$\text{Intensity of an EM wave } I = \alpha A^2$$

$$\text{Where } \alpha = \frac{1}{2} k\omega$$

Surprise Googly – Photo electric effect



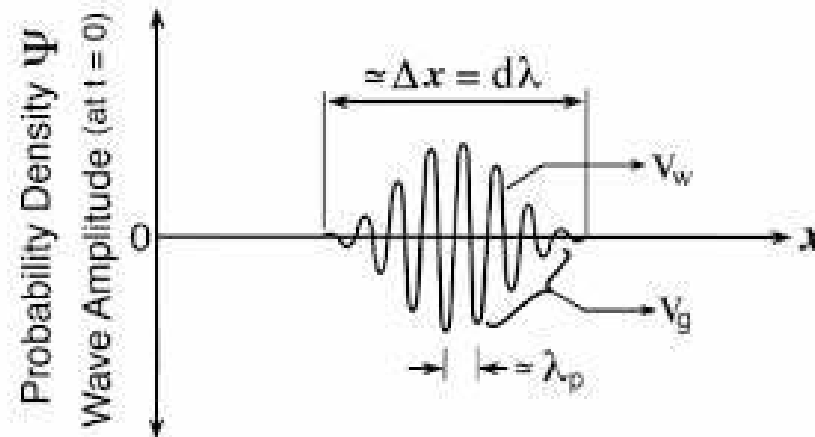
Duality conundrum

- Photons, which were once wave like with intensity decided only by the amplitude, also behaves like a particle!
- Each particle interacts with other particles in a collision kind-of-behavior, which can kick electrons out of the electrode!
- No particle can diffract, cause interference! So how do we reconcile ?

Wave-Particle Duality

- A Plane wave: $A = A_0 e^{i(kx - \omega t)}$ oscillates in space and time.
 - Is present everywhere in space !
- A particle on the other hand, has precise position in space and has a definite velocity
 - Much like a ball whose forces are known.
- So now, what is light ? What's the confusion all about?

The idea of a wave packet



Properties of such waves

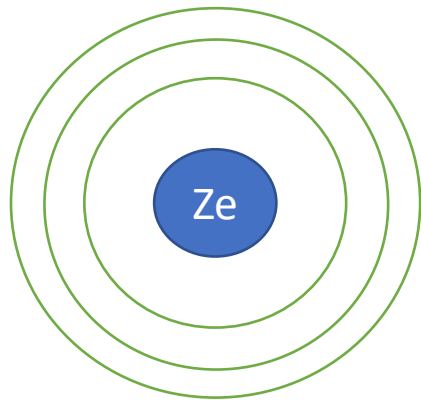
- We can define a wavelength λ , frequency ν and a velocity v such that
- Velocity = distance /time $V = \lambda \nu$
- The energy of the particle moving at velocity is give by $E = h\nu$
 - Where h is called as the Planck's constant
- The packet of light which propagates is called as the 'quanta' – minimum measurable physical quantity
 - Though thought to be used in atomic/low energy physics, quantum and quanta has been in usage by Physicians long before!

How does light interact with
Matter ?

Interaction of EM waves with Matter

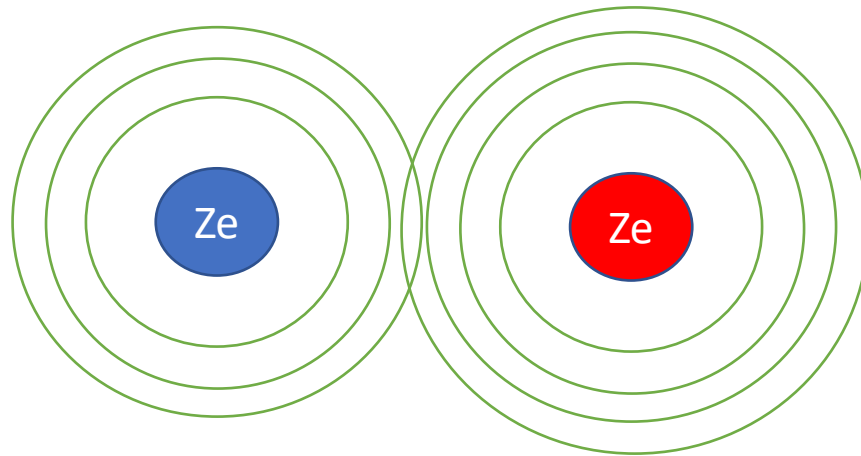
- For simplicity, we consider matter as either atom/molecule or a solid.
- What is the difference between them ? What similarities they share ?
- For that we need to understand energy levels:

Sodium Atom

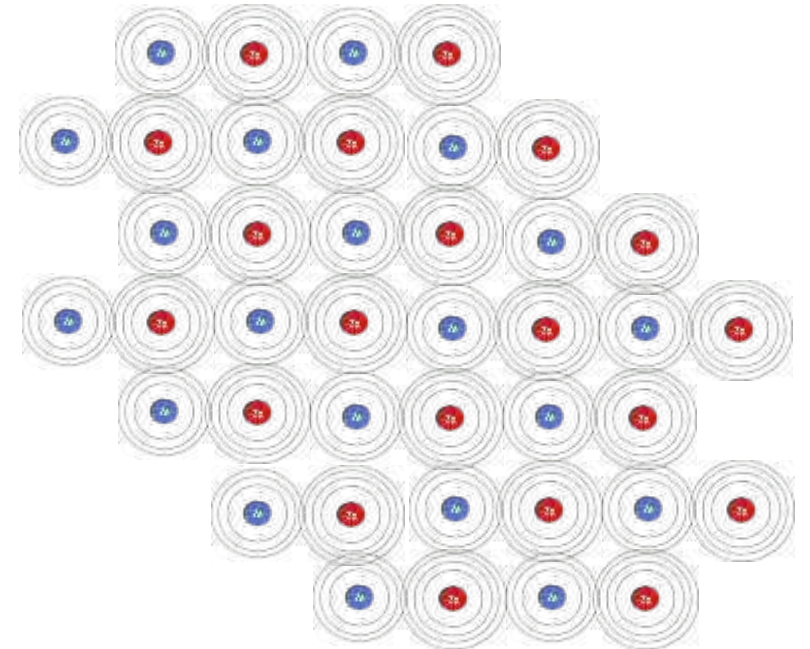


Atomic number Z- 11

Chlorine Atom



Atomic number – Z=17

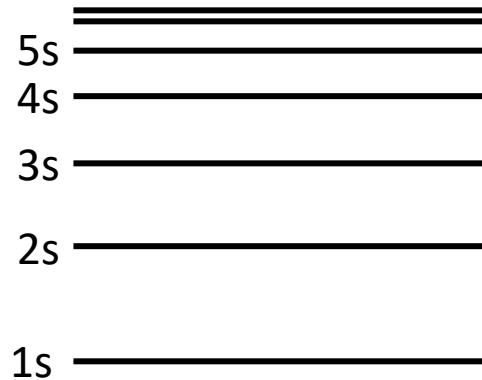


Energies and Energy levels in matter

Atom

Energy levels in an atom

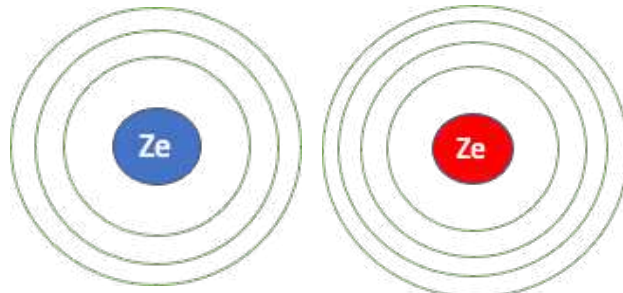
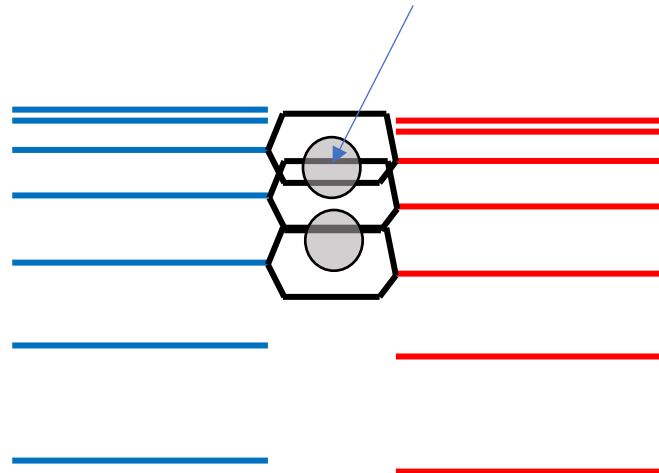
$$E_n = -\frac{hcR_y Z^2}{n^2}$$



Molecule

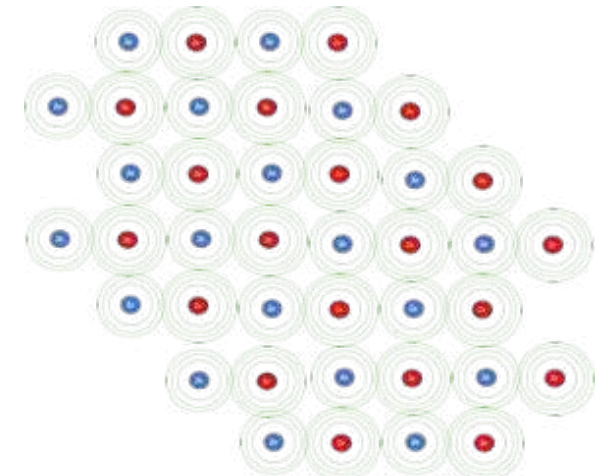
Discrete energy levels separated from each other

Interacting energy levels/orbitals



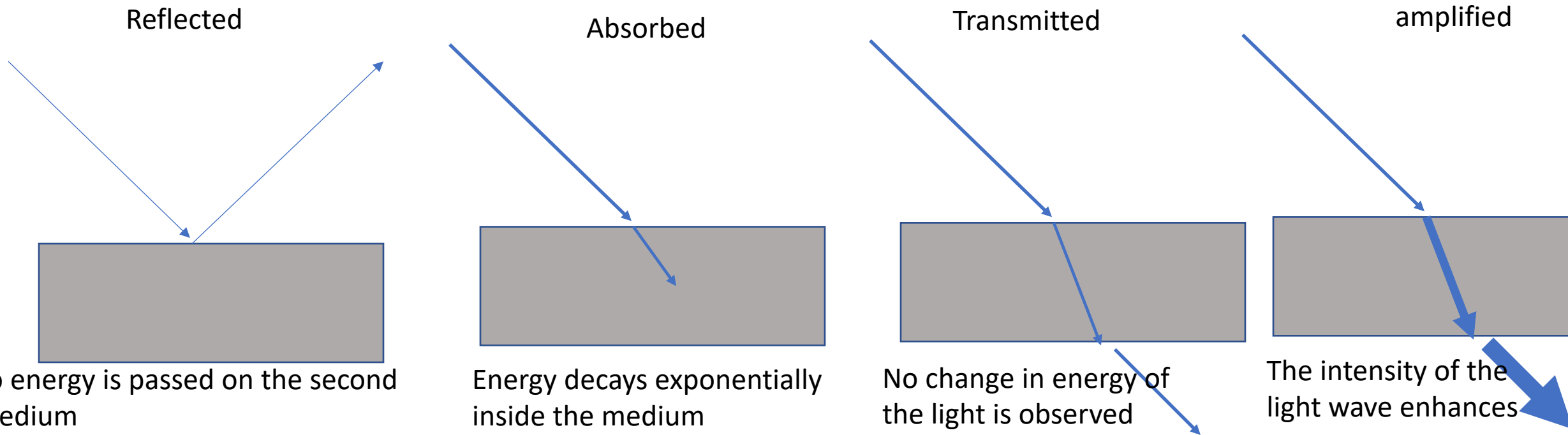
Solid

Continuum of energy levels called bands, separated from each other



Light – Matter Interaction

- Consider a light of energy of frequency ν , energy $E = h\nu$
 - Wavelengths are not used here as it is material dependent.
- Different modes of interaction with matter:

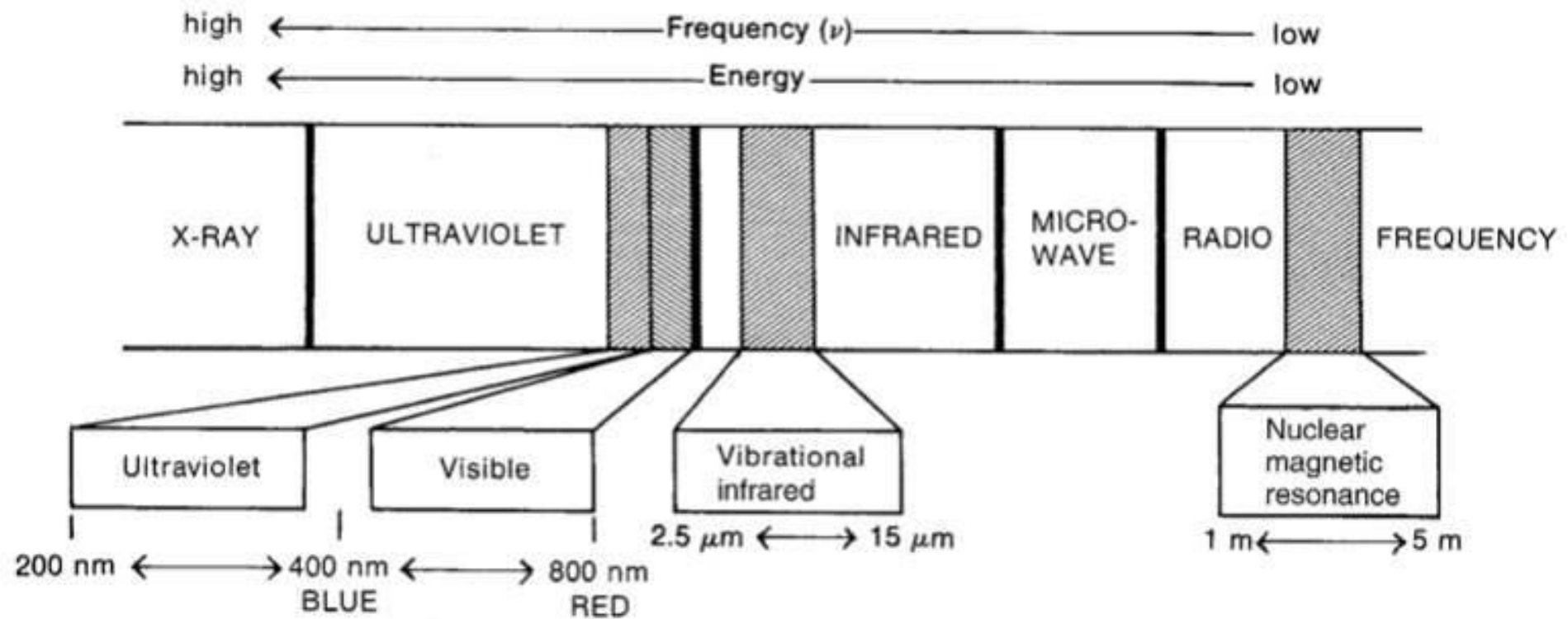


Selection Criteria for Interaction

- A material has a finite probability to absorb an electromagnetic radiation with energy E , iff
 1. $\Delta E = E_f - E_i = h\nu$
 2. E_f has empty electronic levels
 3. E_i has filled electronic levels

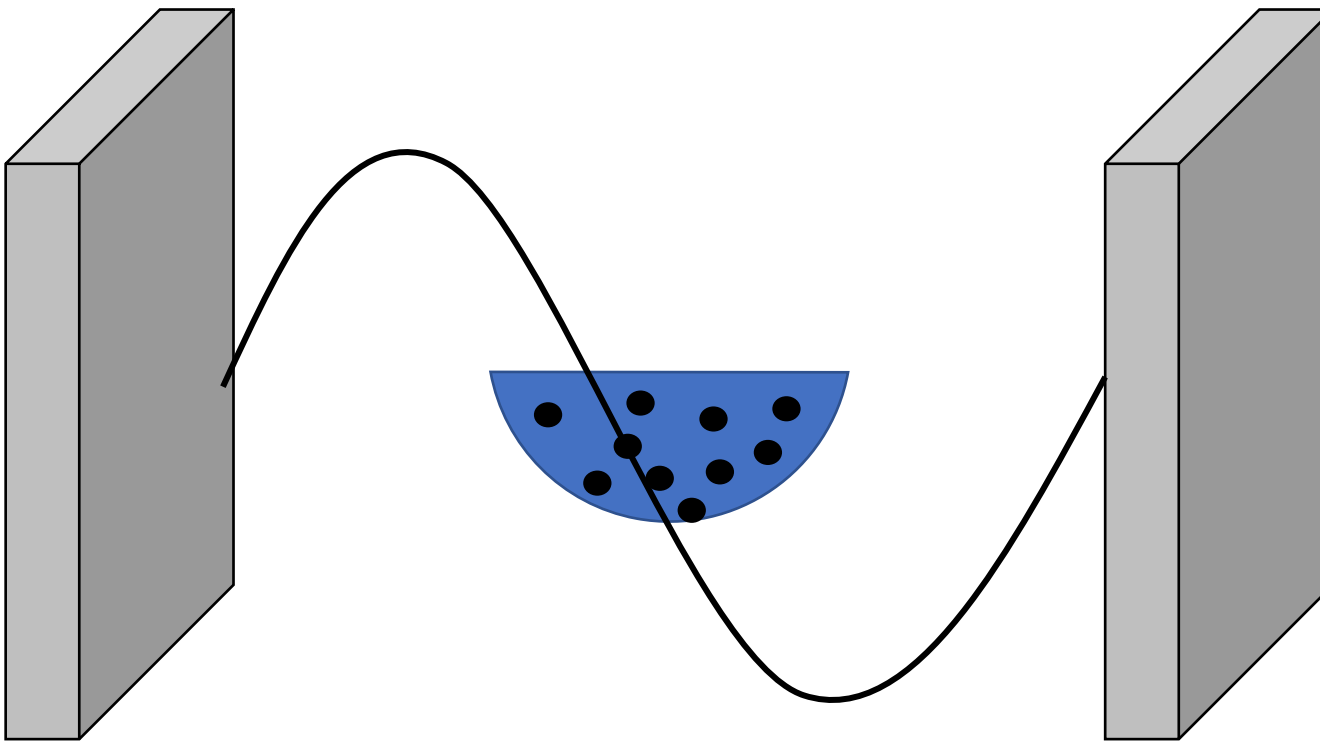
Now can you explain, why CuSO_4 solution appears blue ? What color of light does it absorb ?

Choice of Light



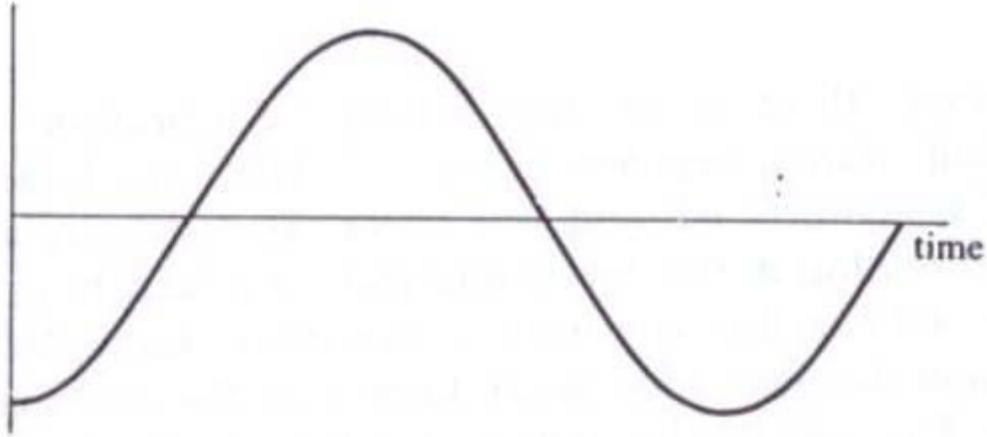
Type of radiation	Energy	Frequency	Type of transitions
X-Rays	>1 keV	> 10^{18}	Bond breaking
Ultraviolet	10 – 100 eV	> 10^{16}	Electronic transitions
Infra-red light	10^{-2} - 1	> 10^{12}	Vibrations
Microwave	10^{-4} – 10^{-2}	> 10^8	Rotational transitions
Radio frequencies	10^{-8} – 10^{-4}	> 10^4	Nuclear spin resonances and Electron spin resonances

Interaction of Light with Low-Energy Light



- From the schematic, how do we see the microwave interact with the mole of water ?
 - Water has an Avogadro number of molecules
 - Each molecule interacts with a single photon
 - The molecule gets excited to one of the accessible excited states

Rotation in molecules



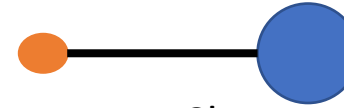
An oscillating electromagnetic field

A simple molecule with no dipole

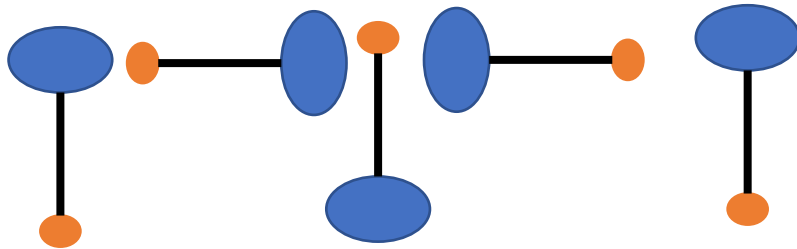


O₂, H₂ ...

Molecule with permanent dipole



H-Cl



Rotation of a polar molecule with the frequency of the incident radiation.

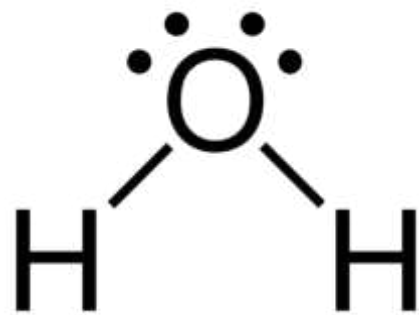
If you have two masses m_1 and m_2

Moment of inertia about a point C is given by $I = m_1 r_1^2 + m_2 r_2^2$

Rotational energy levels of a rigid diatomic molecule can be calculated to be $E_J = \frac{h^2}{8\pi I} J(J + 1)$

Discrete J values leads to quantized rotational states.

$J = 0, 1, 2, \dots$



Rotational transitions with $\Delta J = \pm 1$
are only possible

J

3

2

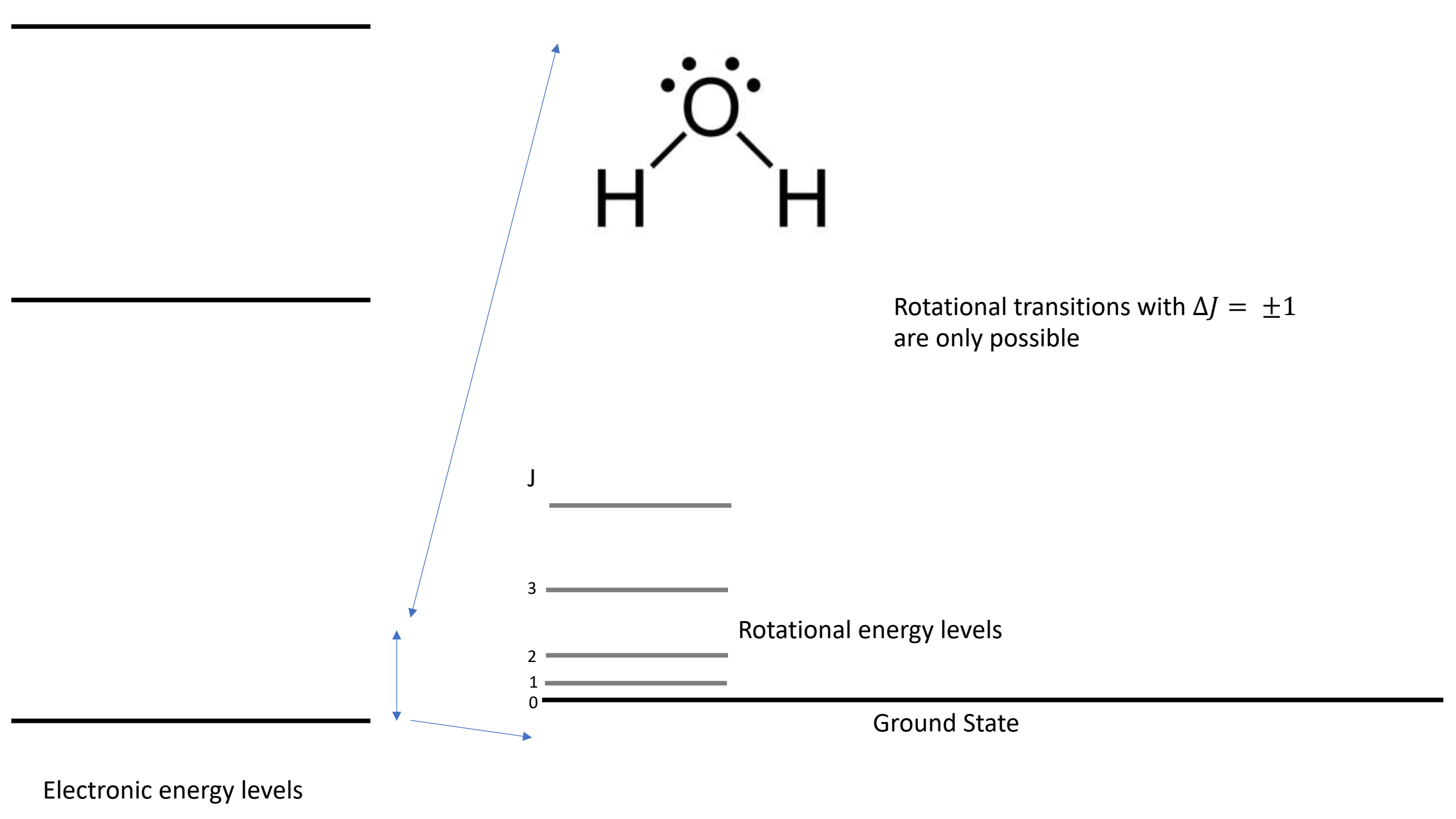
1

0

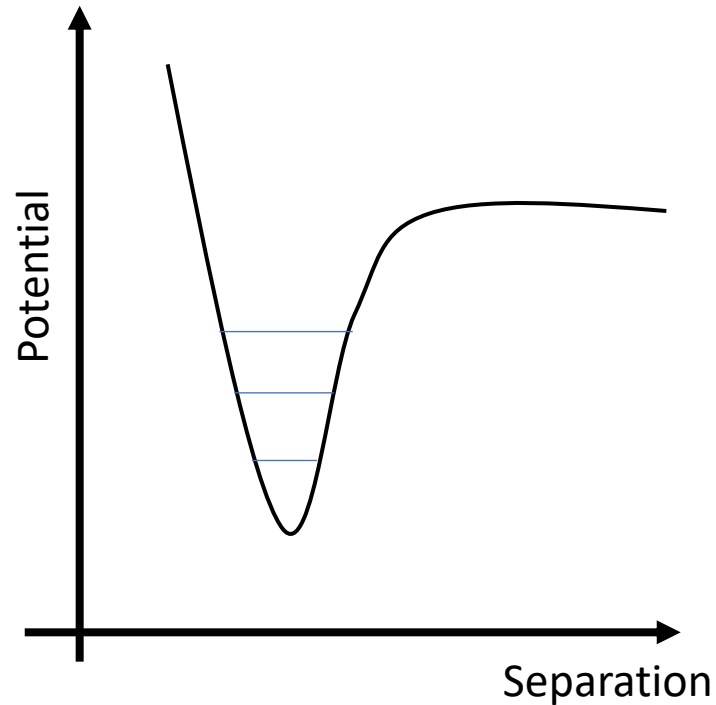
Rotational energy levels

Ground State

Electronic energy levels



Vibrational Energy levels



A restoring force acts on molecules separated away from the eq. distance r_{eq}

$$f = -k(r - r_{eq})$$

If this is the force, the potential tends to be parabolic

$$U = \frac{1}{2}k(r - r_{eq})^2$$

This is a simple harmonic oscillator

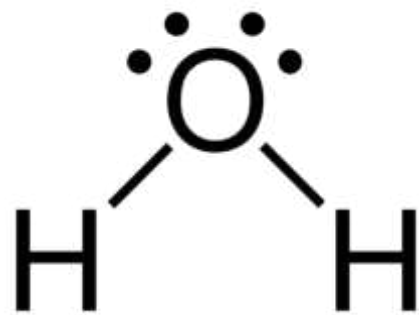
The two atoms will vibrate at a frequency

$$\omega = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

Quite like the rotational system, the vibrational states are also discrete.

No atom can have 0 vibrational energy!

$$E_v = \left(v + \frac{1}{2}\right) h\omega \quad v = 0, 1, 2, 3, \dots$$



Vibrational energy levels (ν 's)

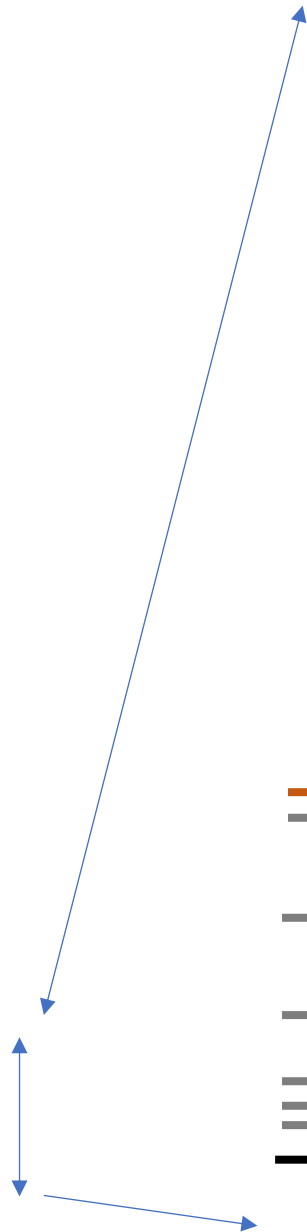


Rotational energy levels (J 's)



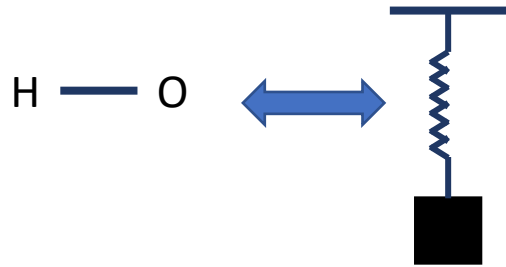
Ground State

Electronic energy levels



Characteristic Vibrational Energy levels

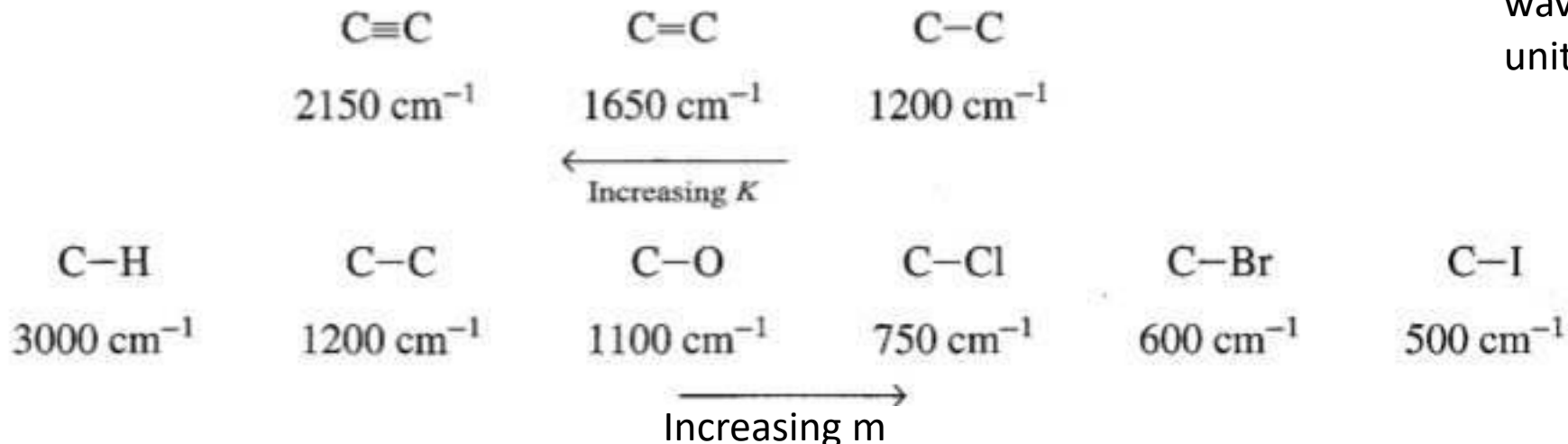
Any chemical bond can be represented as a spring



A spring with a strength K , develops a force $F = -Kd$

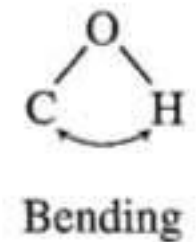
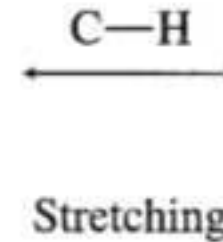
The spring has natural frequency $\omega = \sqrt{\frac{K}{m}}$

Remember, energy can also be written in terms of wavenumber k , with the units of cm^{-1}



Absorption in a molecule

- An isolated molecule/large well-defined phase/long chain
 - All have characteristic bonding between constituent atoms
- Each atomic bond has well defined states
 - Rotational
 - Vibrational
 - Stretching – symmetric, asymmetric
 - Bending – scissoring, rocking, wagging, twisting
 - Electronic transition



If a photon of energy matching the energy difference is provided, an excited state reached in which the bonds either vibrate/rotate

Typical energy scales:

Rotation - 1- 20 cm^{-1} , Vibration – 2000 – 4000 cm^{-1} , Electronic - 10000 – 50000 cm^{-1}

Types of Vibrations

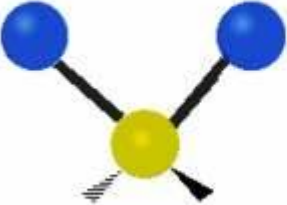
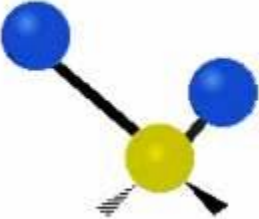
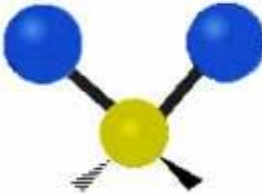
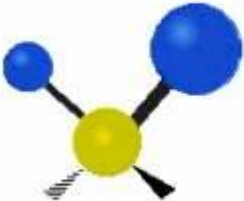
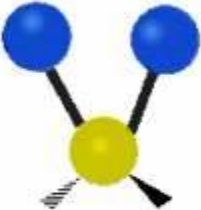
Symmetric Stretching	Asymmetric Stretching	Wagging
		
Twisting	Scissoring	Rocking
		

Figure 1: Six types of Vibrational Modes. Images used with permission (Public Domain; Tiago Becerra Paolini).

Vibrational Modes and Selection Rules

If a molecule has N atoms:

-> If we are worried only about translational in 3D world, $\text{DOF} = 3 * N$

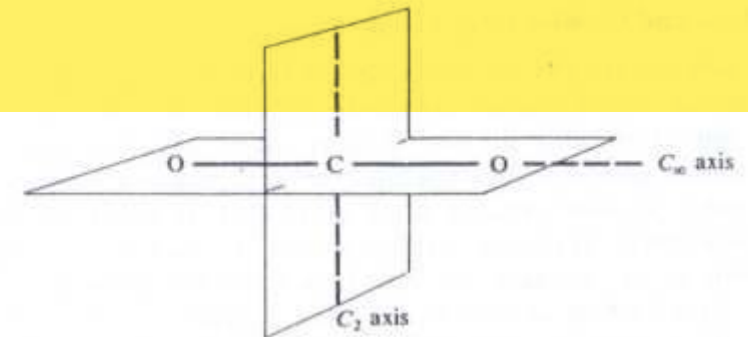
-> If molecules are bonded, then some of the translations become rotational (because of constraints)

- Any rotation can be described by 3 rotational axes.

- So **total $\text{DOF} = 3*N - 6$ (3 rotational DOF x 2 chirality)**

-> However, if the molecule is linear, then rotational along the linear axis leaves the molecule unchanged.

- so $\text{DOF} = 3*N - 5$

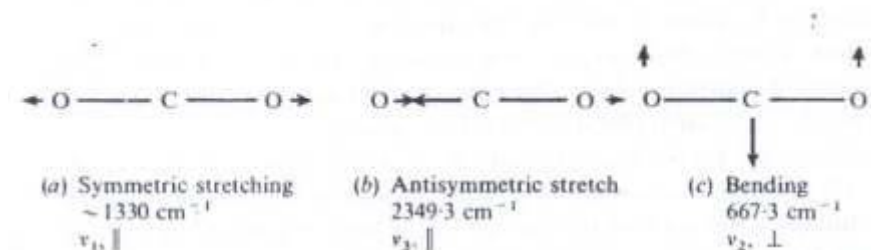


So, how many vibrational modes for CO_2 ?

$$3 * 3 - 5 = 4$$

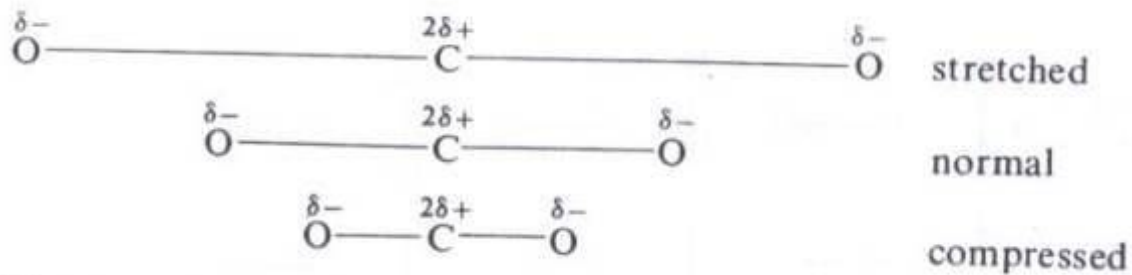
For methane :

$$3 * 5 - 6 = 9$$

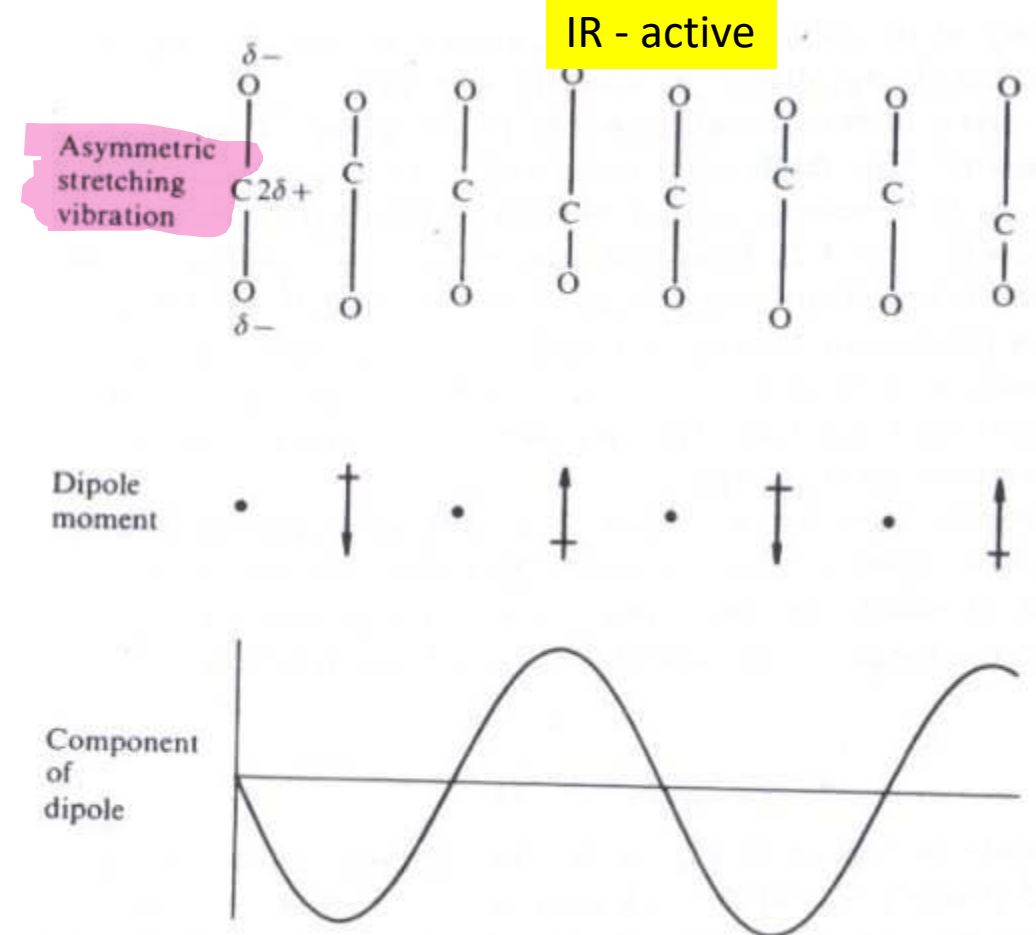


Selection Rule for IR activeness

- A vibration is said to be IR-active if dipole moment changes during the vibration.



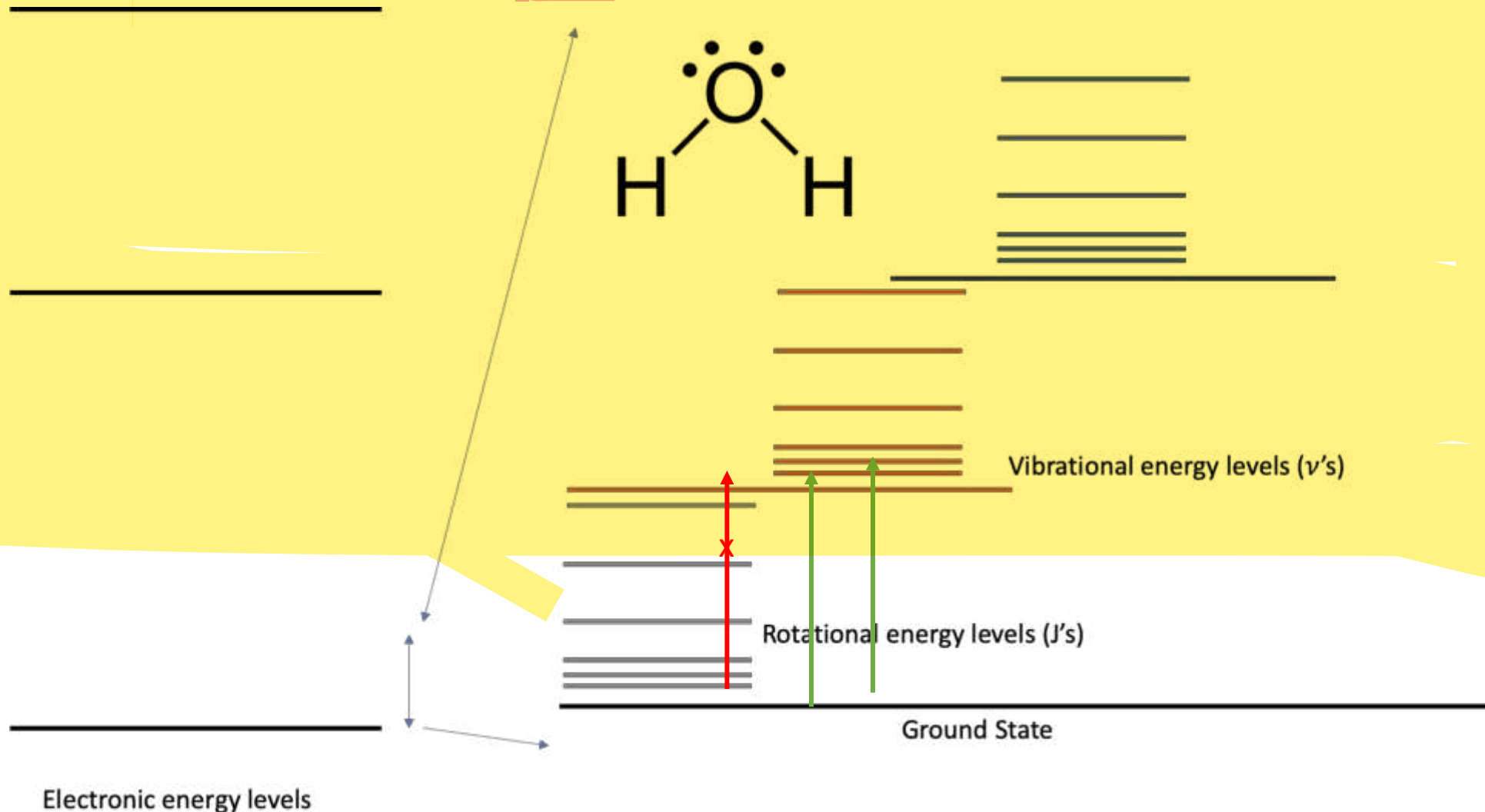
IR - inactive



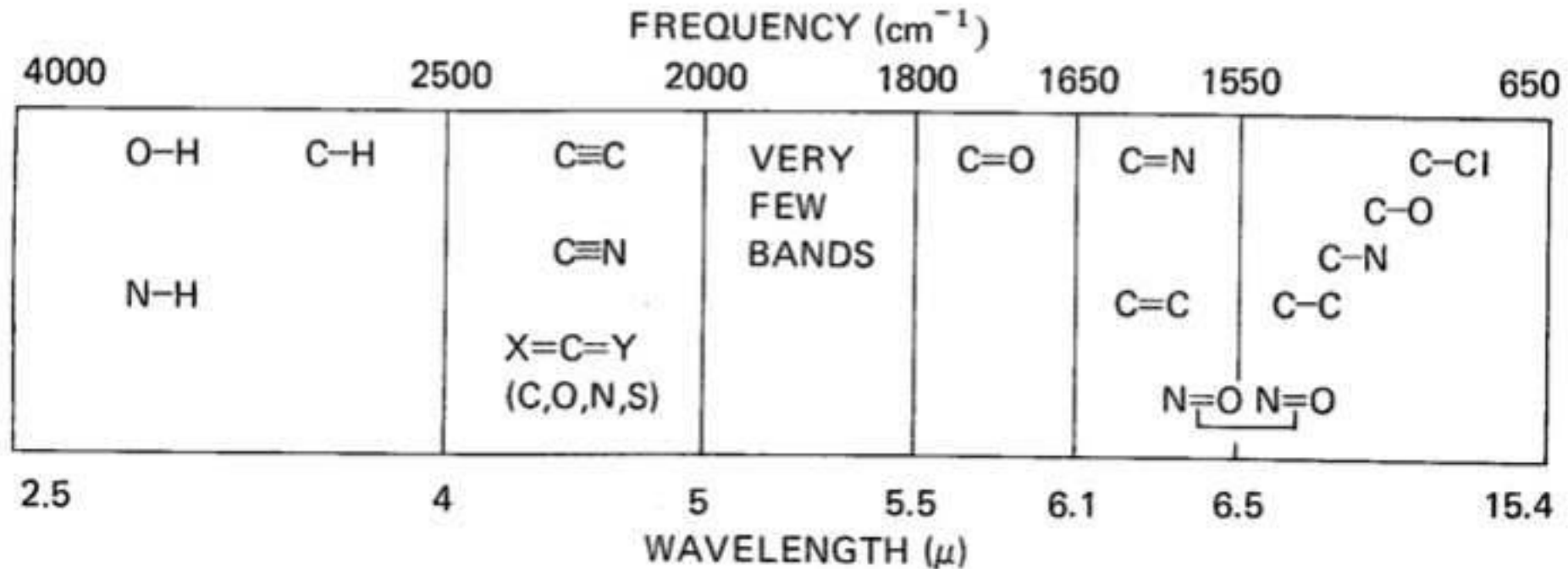
Selection Rule:

This means, you cannot have a vibrational energy change without rotation.

$$\Delta v = \pm 1 \text{ and } \Delta J = \pm 1$$

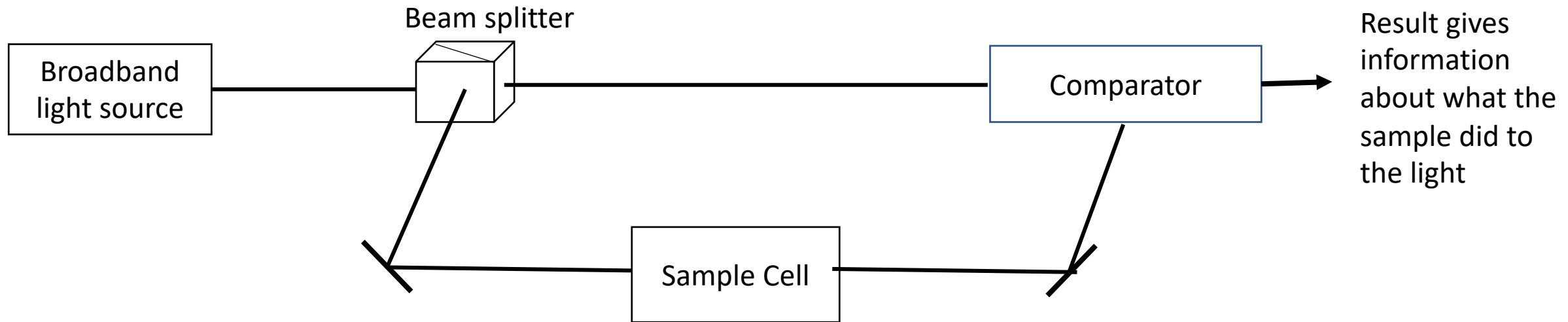


Infrared spectroscopy



- Most common bonds have characteristic frequencies between 2.5 – 25 μm
- This lies in electromagnetic spectrum called Infra-red (infra – meaning lower/below)

General Idea of Spectrometer



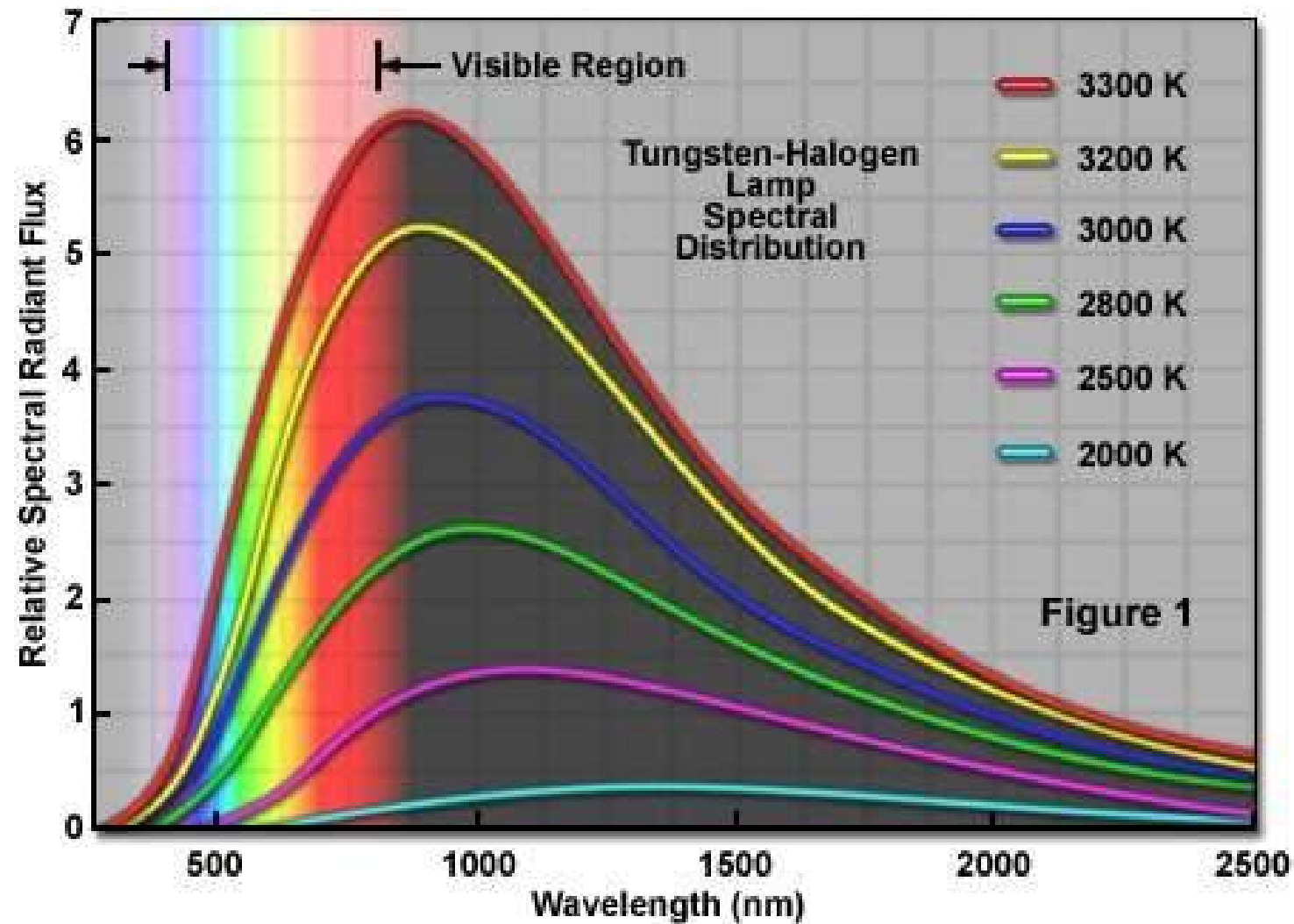
Broadband light: A electromagnetic source which has many wavelength contributions.

Eg: Tungsten bulb.

A typical specification of this source is its intensity distribution vs wavelength

Black-Body Radiation

Eg: Tungsten halogen lamp



Homework: Get the solar spectrum plot! How does it compare with the T-H lamp and why ?

Design of Absorption cell - Beer-Lambert Law

- If I_R is the intensity of the reference, and I_S is the intensity of the light through the sample cell, the % transmittance is given by

$$T = \frac{I_S}{I_R} * 100$$

From theory, transmittance through a sample is given by

$$T = e^{-\tau} = 10^{-A}$$

Where τ is called as optical path, A is called as absorbance.

Absorbance is dependent on specific details of sample cell. So, we use Beer-Lambert law which is given by

$$A = \epsilon lc$$

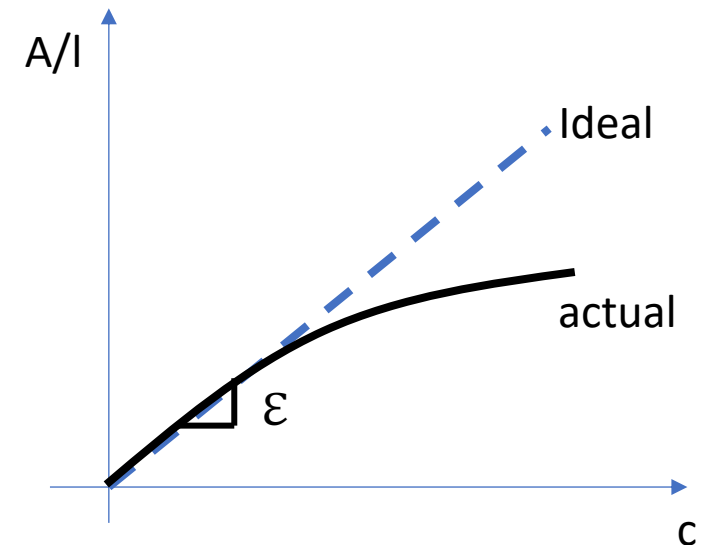
l – optical path length

c – concentration of the absorbing species

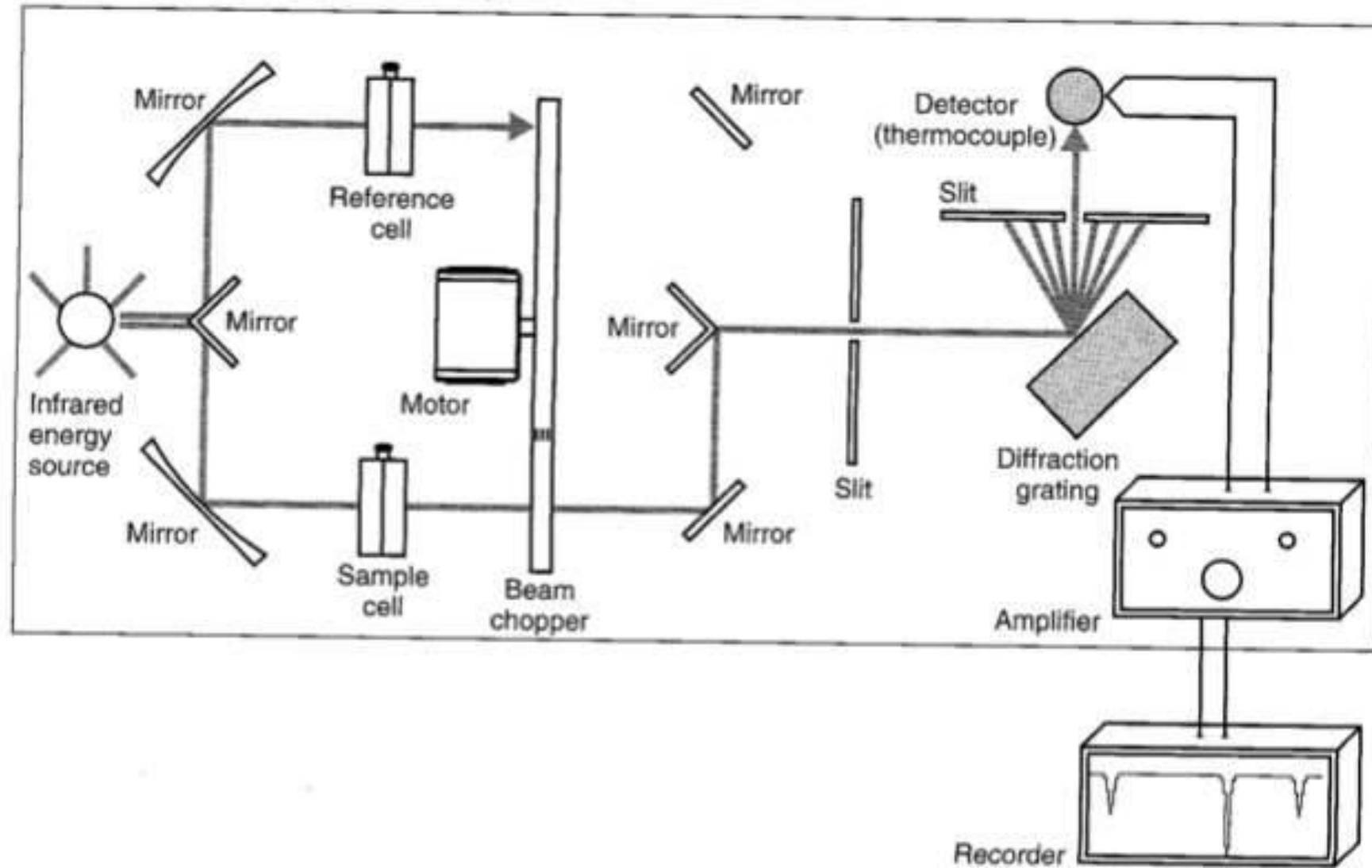
ϵ - molar attenuation coefficient or absorptivity

Properties of Absorbance

- Absorbance is additive
 - A mixture of n components has total absorbance $A_T = \sum_i A_i$
 - Extent of linearity
 - Absorptivity depends on lot of factors
 - Polarizability and rate of polarizability
 - Interaction cross-section
 - Energy levels and correlation with energy of photon

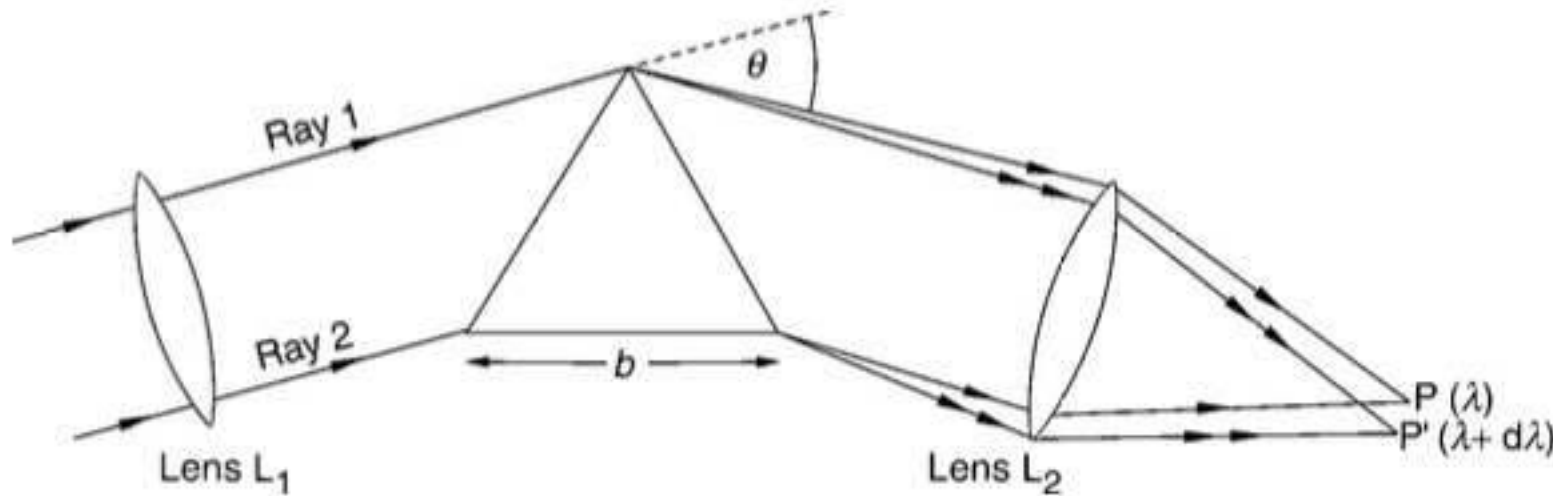


A simple setup



Dispersing Element

- Disperse: Separate in space light with differing wavelength



Simplest method: A prism with base 'b'

If we assume the light is incident on an entire face as shown in the figure,

n – refractive index of the material.

We define dispersion power/wavelength resolution as R

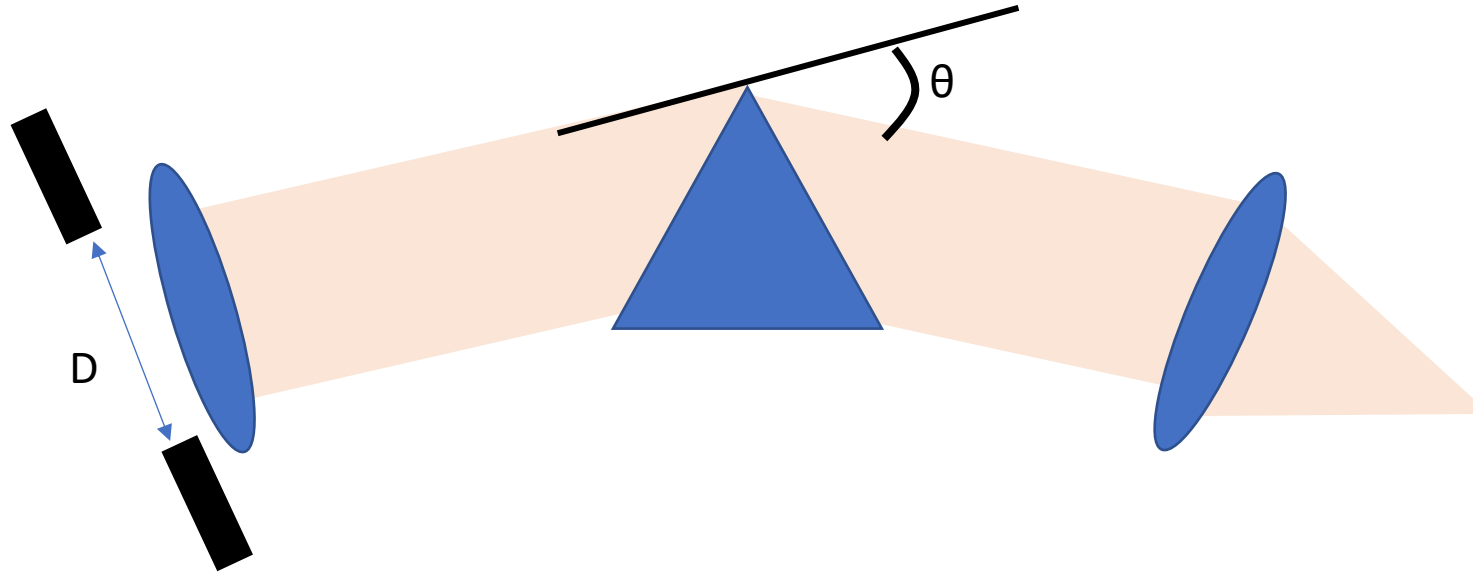
$$R = \frac{\lambda}{d\lambda}$$

Where $d\lambda$ is the smallest change in wavelength that is visibly separated.

How small can the light be focused ?

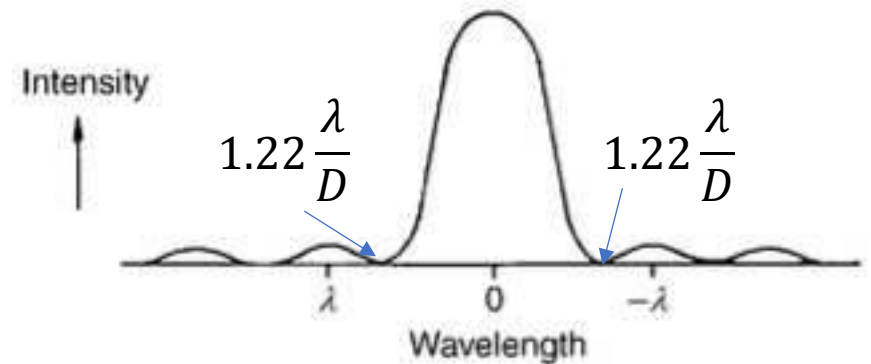
HW: information about Airy's function

They are solutions to a very commonly occurring equation

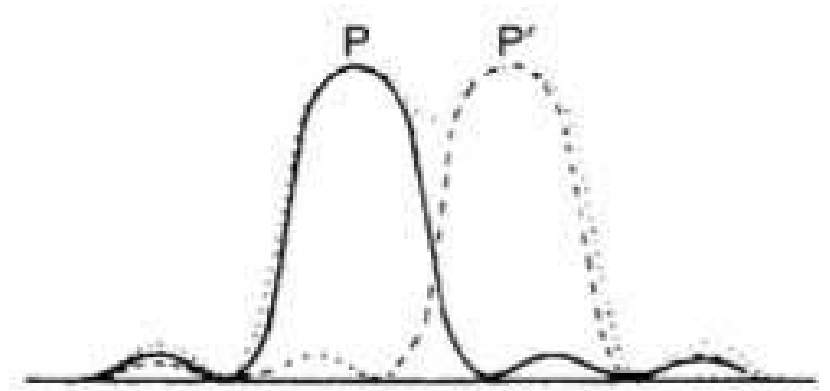
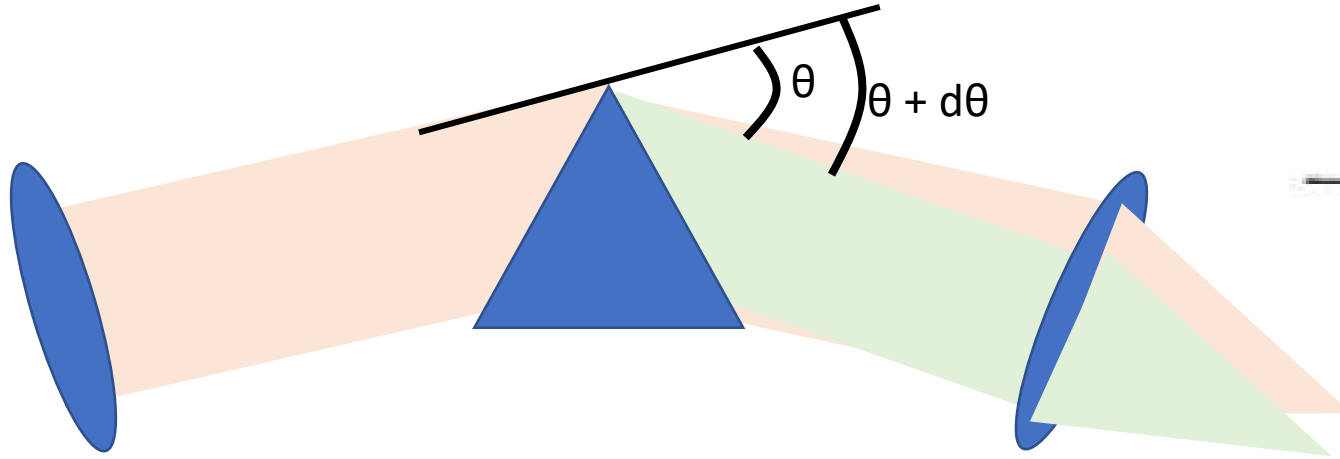


- Limit of visible separation of two light spots is given by Rayleigh's criteria:

If the light is obtained from a slit of width 'D', then the first minima in the Airy disk formed by the focused light is obtained at angle $\theta = \frac{1.22\lambda}{D}$



What is visible separation ?

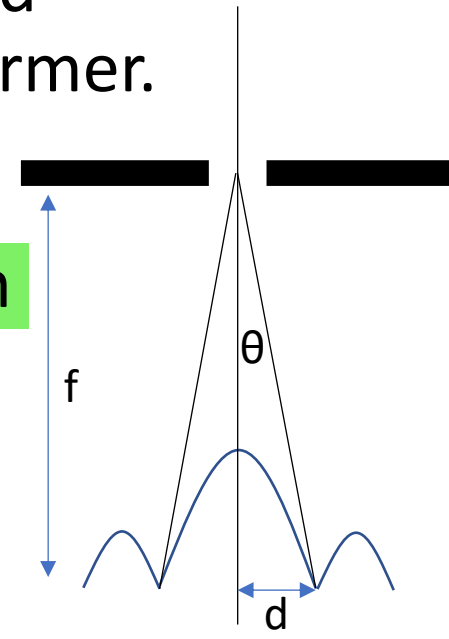


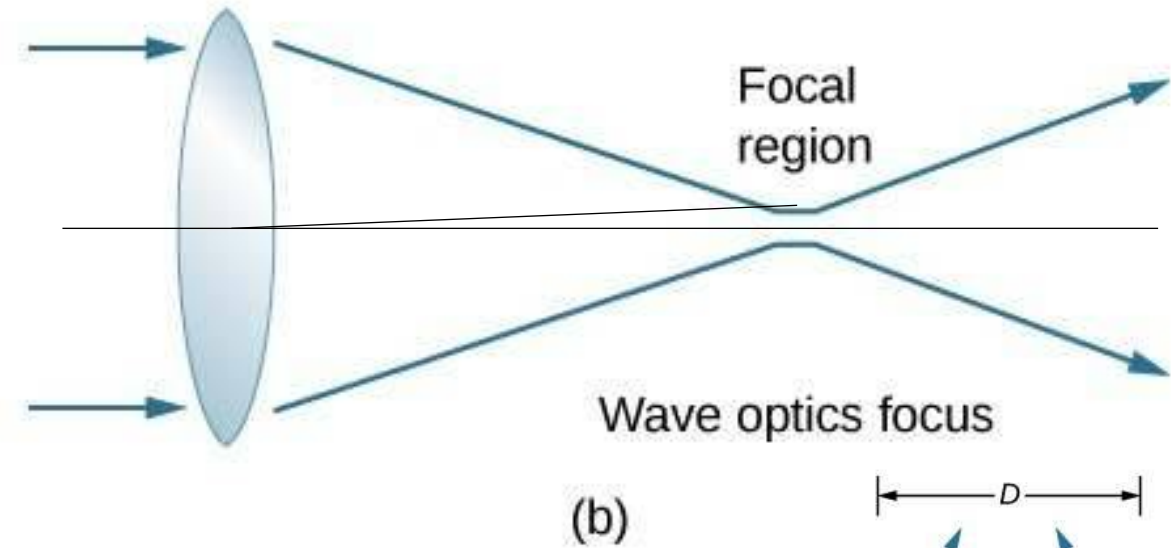
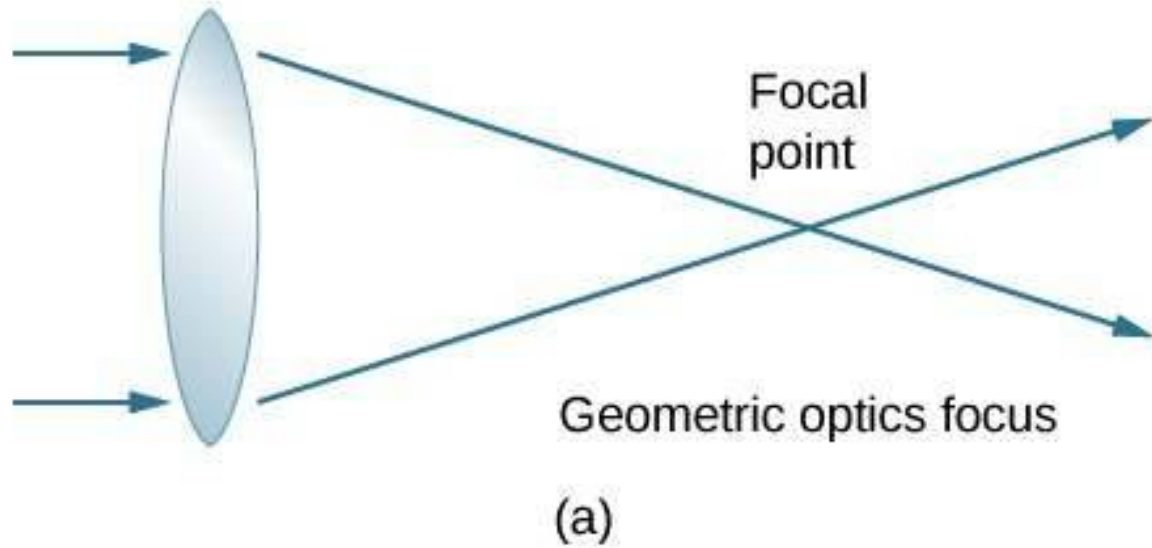
Lord Rayleigh: Two Airy disks can be resolved if they are separated such that the maxima of one co-incides with the minima of the former.

If f is focal length of the lens, interparticle separation at maximum resolution is given by

$$d = f * \tan \theta$$

In small angle approximation $d = f\theta = \frac{1.22\lambda f}{D}$





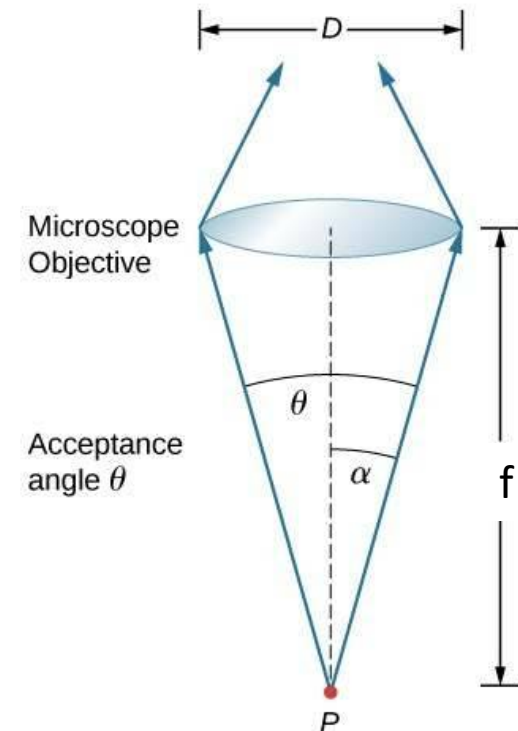
- The point is broadened by the size of the disk as calculated before

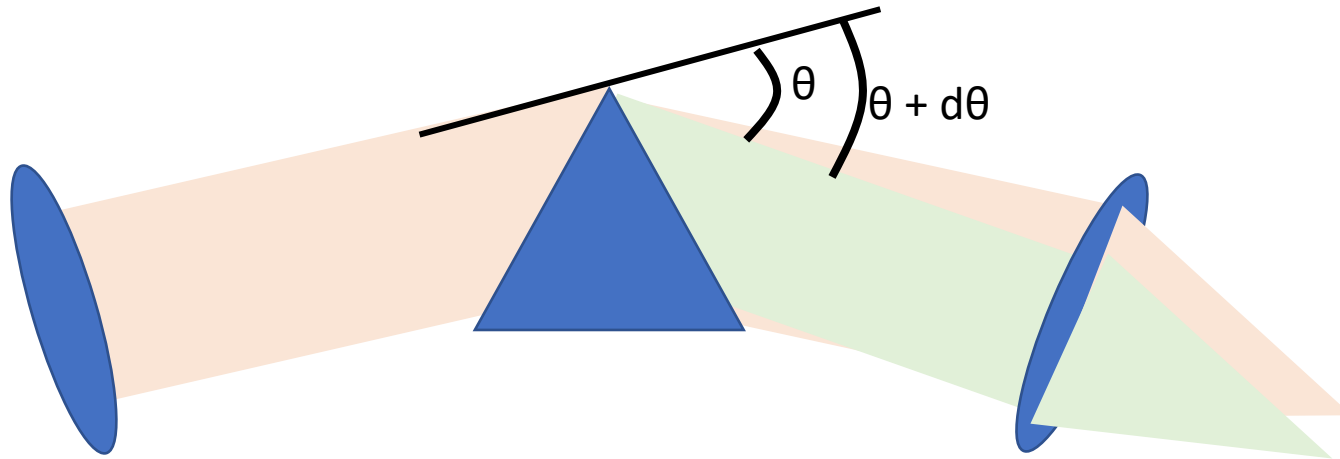
$$R = 1.22 \frac{\lambda f}{D}$$

We define a term Numerical aperture NA as the acceptance angle of a lens

$$NA = n \sin \alpha$$

By simple trigonometry, we get $R = 1.22 \frac{\lambda}{2 \tan \alpha} = 1.22 \frac{\lambda}{2 \sin \alpha} = 0.61 \frac{\lambda n}{NA}$

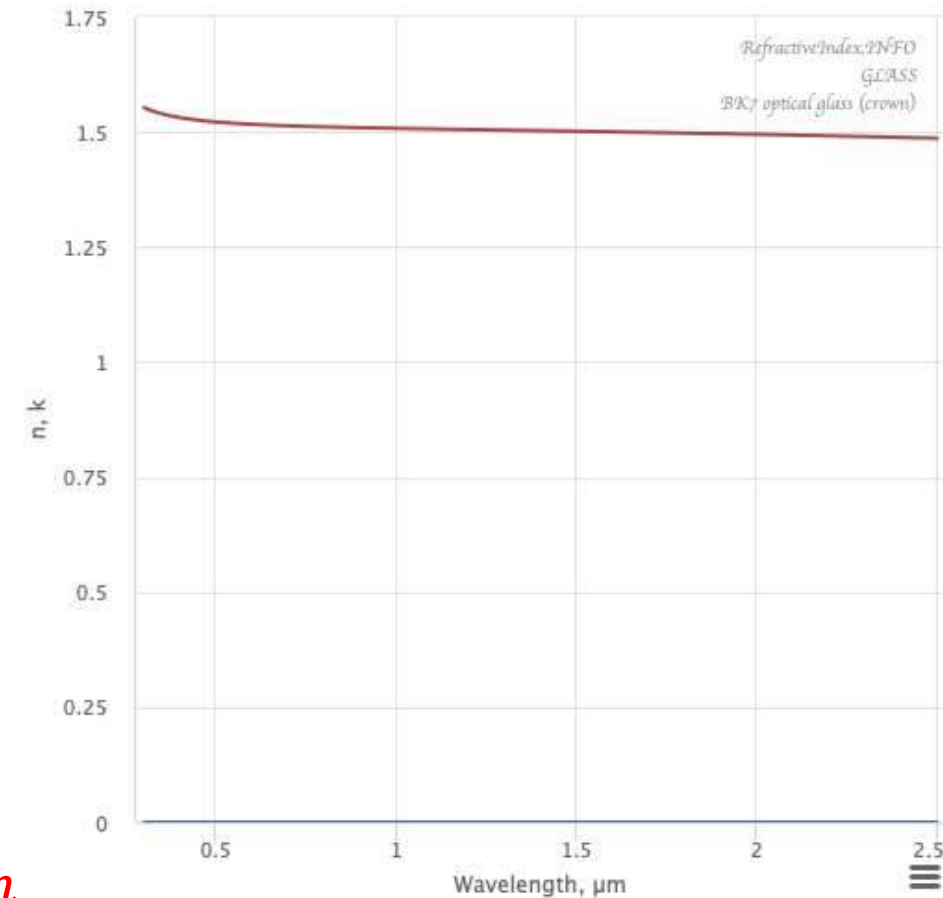




- Prism dispersion = $\frac{d\theta}{d\lambda}$

Finally, the prism dispersion also depends on $\frac{d\theta}{dn} \frac{dn}{d\lambda}$

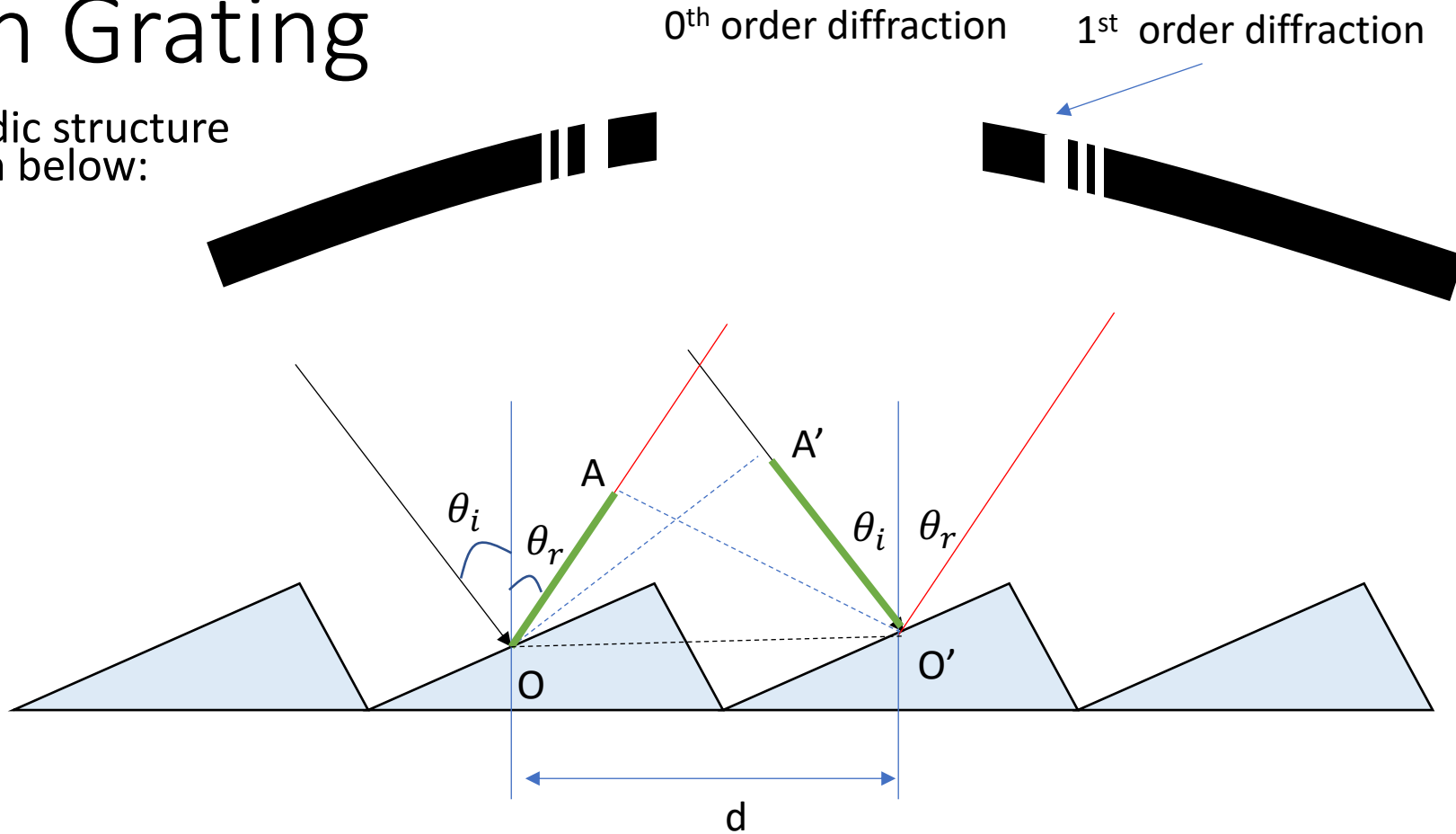
Glass prisms are not great for IR radiations as the index rarely changes in the IR region!



Diffraction Grating

If we have a periodic structure such as one shown below:

With a period 'd'



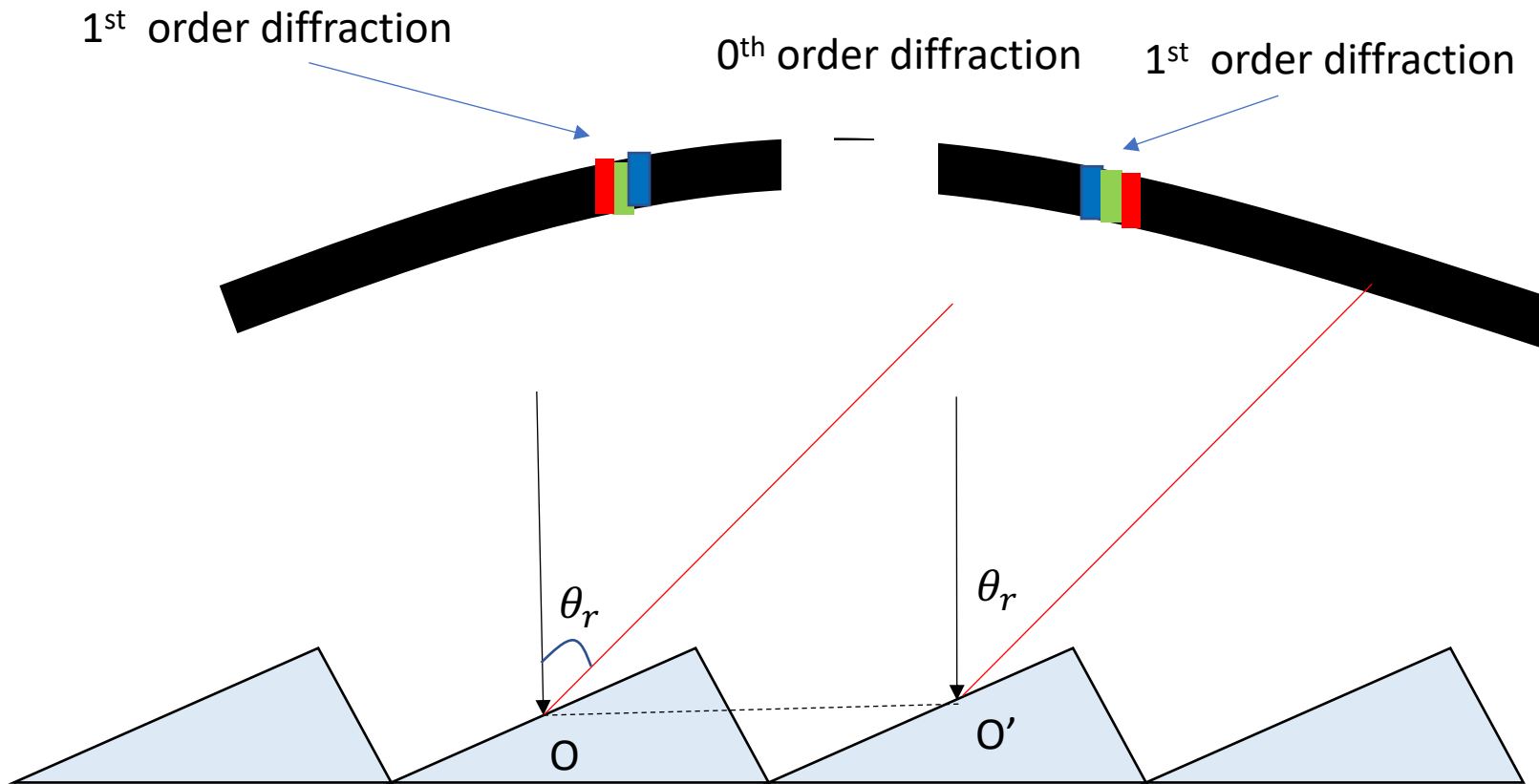
Total path difference between the two radiations: $OA - O'A' = d \sin \theta_r + d \sin \theta_i$

As before, we get constructive interference when the path length is equal to integer multiples of λ

$$d (\sin \theta_r - \sin \theta_i) = n\lambda$$

If the incident angle is 90° , the equation simplifies to $d \sin \theta_r = n\lambda$

White light incidence on diffraction grating



Consider normal incidence

$$d \sin \theta_r = n\lambda$$

0th order, the beams with different wavelengths does not have angular separation.

So, a bright white spot is obtained.

For the first order, when $n = 1$,

The dispersion

$$\frac{d\theta_r}{d\lambda} = \frac{n}{d \cos \theta_r}$$

Higher dispersion is observed, when smaller 'd' is taken;

However, also remember, that no solution for constructive interference can be obtained when $d < \lambda$

Better dispersion can also be obtained when we take higher orders;

However, this has two fold problems:

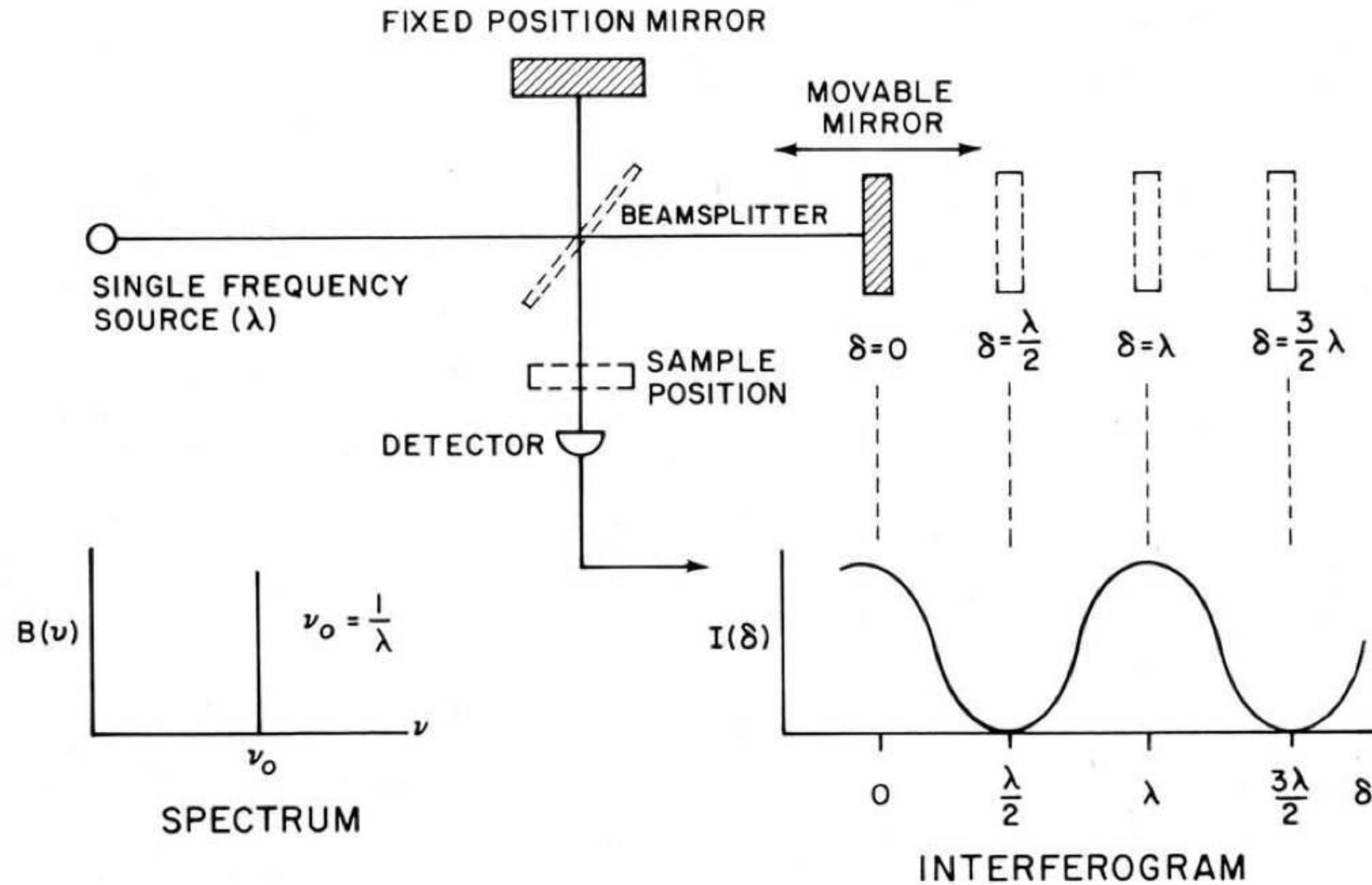
1. The light intensity is significantly lower in higher orders
2. There will be significant overlap between the orders

General sources are weak in this part of spectrum

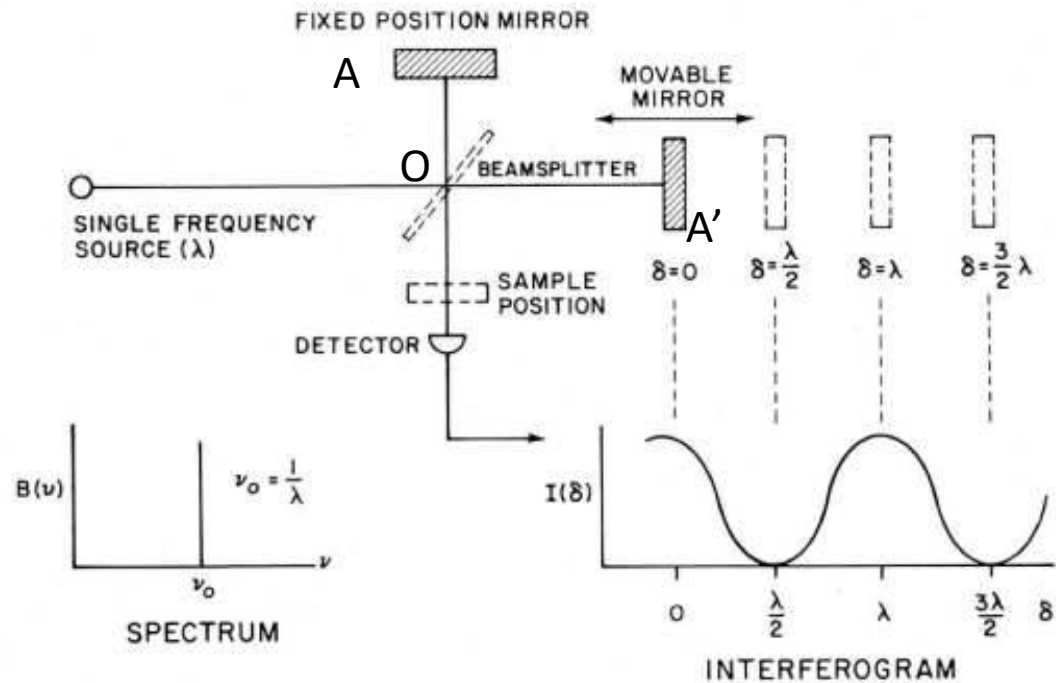
Solution is therefore FTIR

Detectors are weak.

Michelson Interferometer



Michelson Interferometer



Light is split at point O into two paths.

First one with fixed mirror

Second one with a movable mirror.

Constructive interference, when the total path difference is an integer multiple

Lets define $\delta = OA - OA'$

If the source is a monochromatic (single wave)

It can be proved that the intensity of the wave at the detector: $I \propto B[\cos k\delta]$

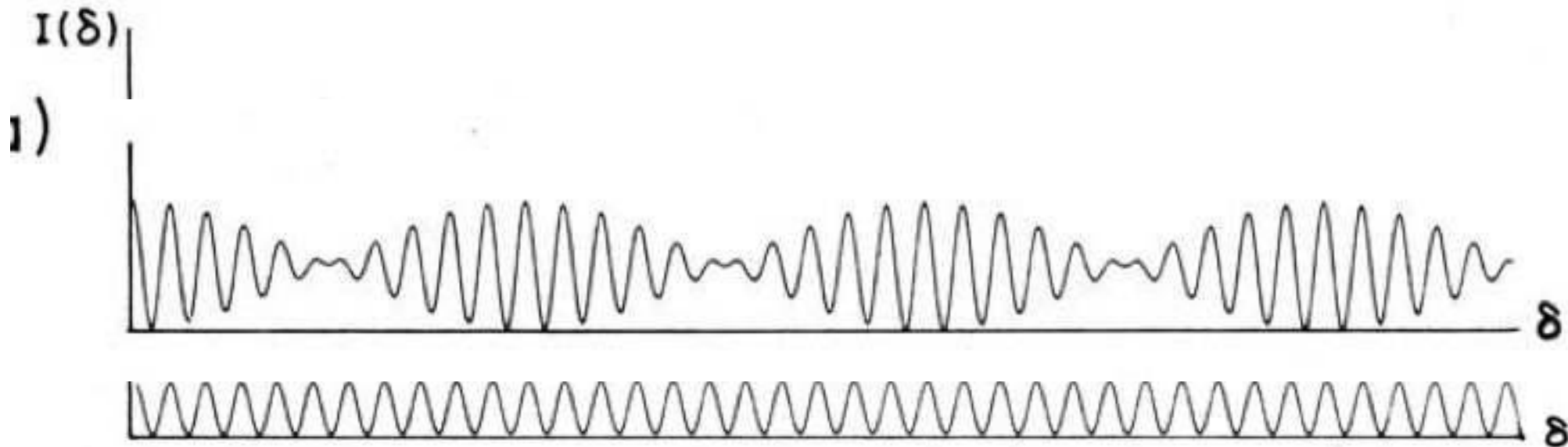
HW

Where $k = \frac{2\pi}{\lambda}$

Interferometer with many colors:

- Lets say, if the interferometer is made incident with light with many colors.
- The different colors do not interfere. So, the net intensity is given by

$$I(\delta) = \sum_{\lambda_i} B(\lambda_i) \cos \frac{2\pi\delta}{\lambda_i}$$



If we have a continuum of wavelengths, we can use the 'chemist' notations for wavenumber $\bar{\nu} = \frac{1}{\lambda}$

$$I(\delta) = \int_0^{\bar{\nu}} B(\bar{\nu}) \cos 2\pi\delta\bar{\nu} d\bar{\nu}$$

Does this expression look familiar to you ?

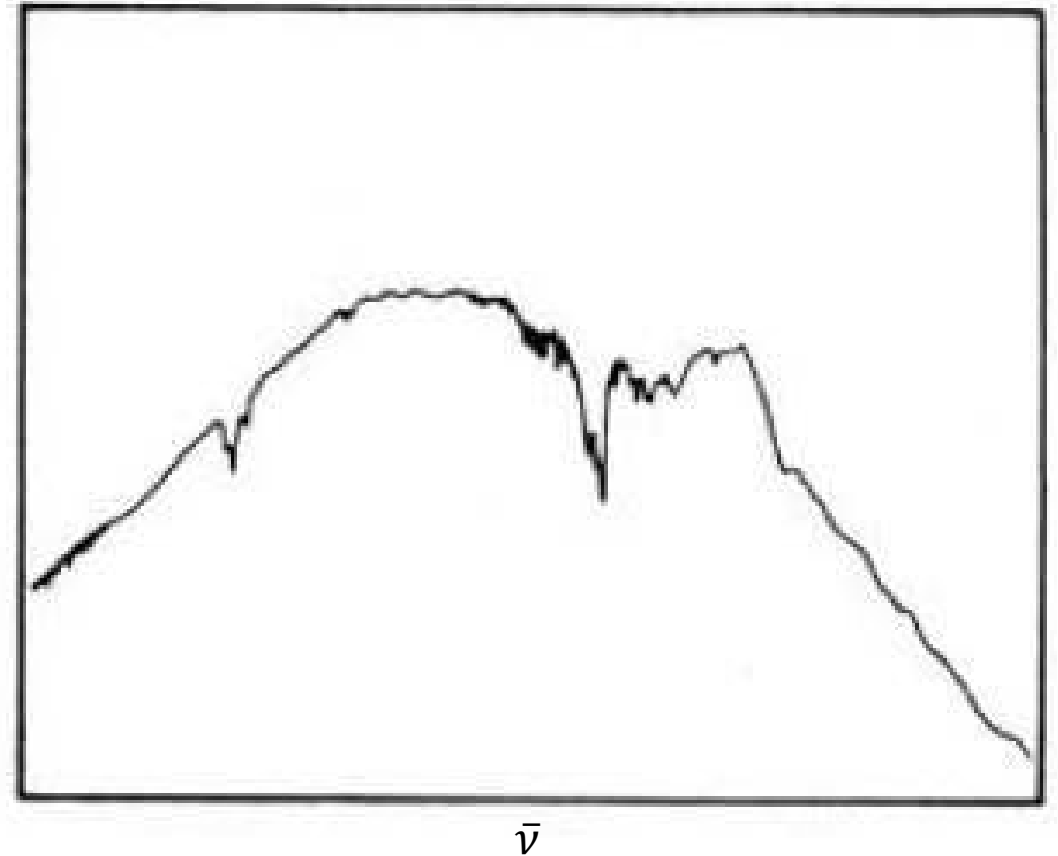
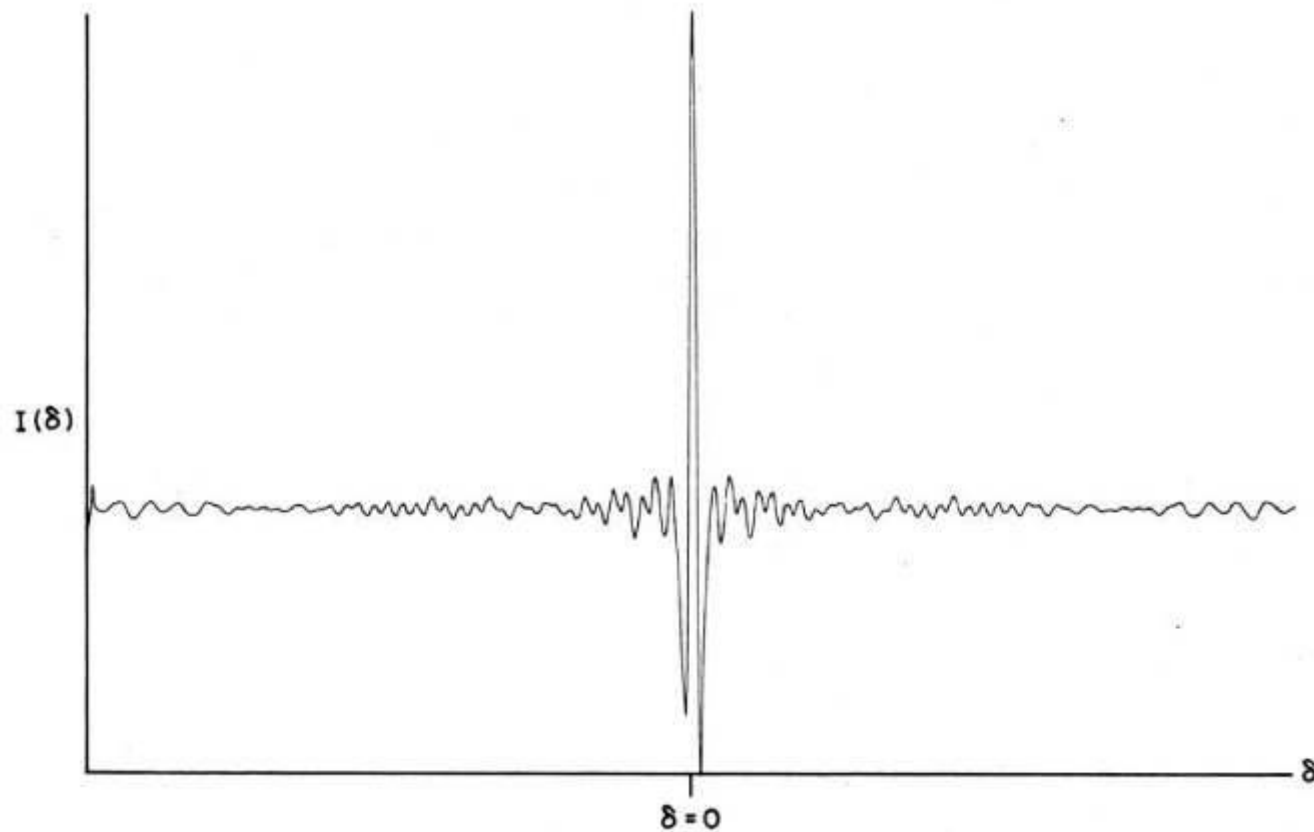
We can do inverse Fourier transform of the measured intensity signal in each frequency

$$B(\bar{\nu}) = \int_{-\infty}^{\infty} I(\delta) \cos 2\pi\nu\delta d\delta$$

Thus, if we record, the intensity as a function of different mirror positions $I(\delta)$, we can extract the intensity of the light as a function of wave-vector ($\bar{\nu}$), which is the frequency spectrum!

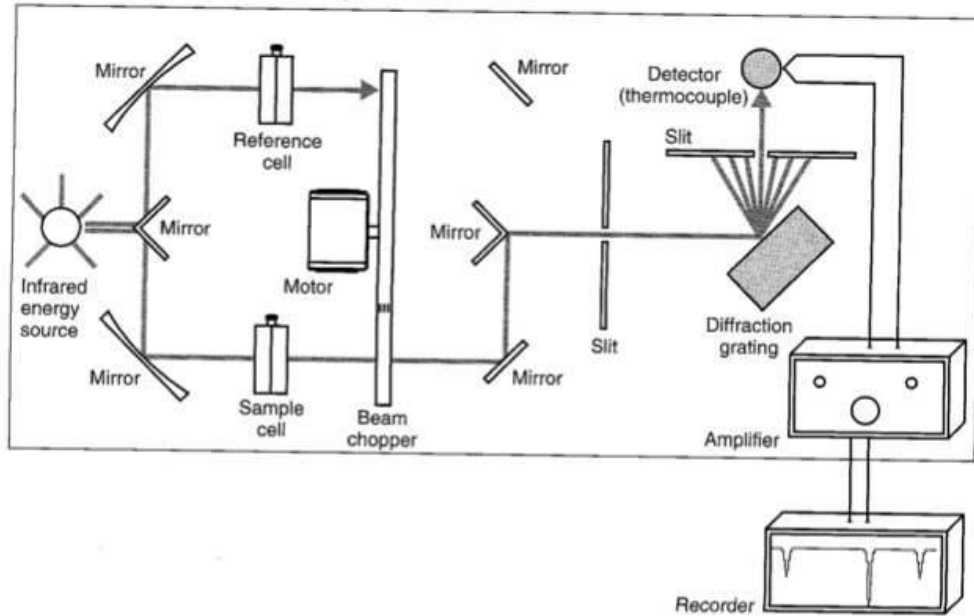


A typical measured intensity as a function of the path delay obtained from the interferometer



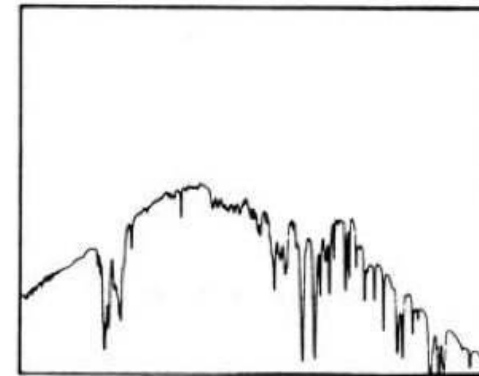
The spectrum (intensity vs wavelength/frequency/energy) gives a typical distribution of energy in a radiation from a black body. In case the path of the light has some absorbents like moisture in the atmosphere and other gaseous absorbents like CO_2

Measurement from a sample:



INDENE

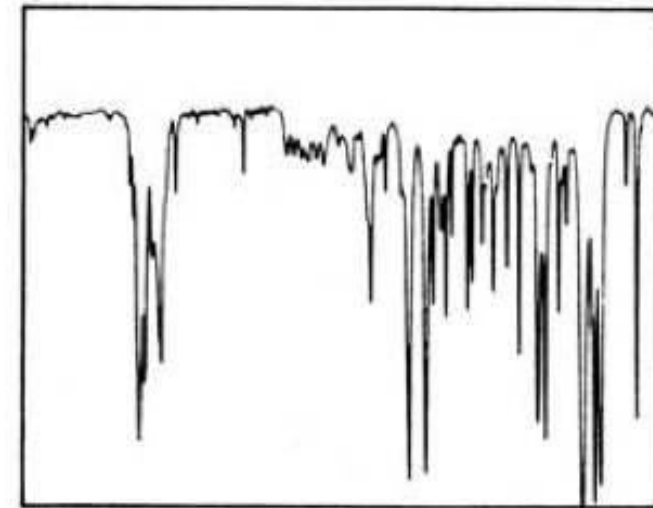
— FFT —→



INDENE
(SINGLE BEAM)

Final a percentage transmission $\%T(\bar{\nu}) = \frac{T(\bar{\nu})}{T^B(\bar{\nu})}$

%T



ν

A single scan of the mirror path is enough to record the complete spectral response.

The FT-IR does a complete spectral scan in a matter of seconds that other dispersive techniques will take several minutes.

Instrumentation

Source: The source is mostly a heated element. Most spectrosopes operate at temperatures close to 1000 – 1500K.

The radiant power falls of accord to Plank's law of black body radiation:

Intensity of radiation at frequency ν is given by $B(\nu) = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}$

B - power per unit solid angle per unit area normal to the propagation of light

Biggest problem in IR instrumentation is every component absorbs IR. Glass has strong absorption in IR (HW: get glass IR spectrum)

So, specialized components have to be used which have negligible absorption in the IR range used to measure samples. s

Ideal materials for usage:

1. KBr – has almost 100% transmittance in the range of $4000 - 400 \text{ cm}^{-1}$.
2. Even smaller energies can be observed by switching to CsI or Mylar films.
3. Thermopiles/pyro-electric detectors are typically used

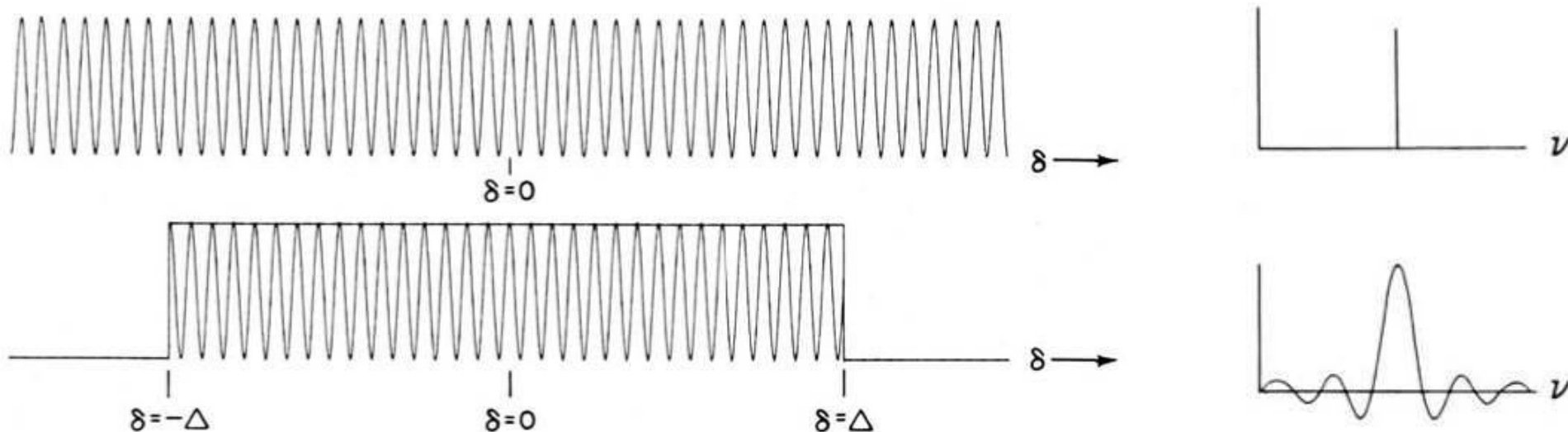
Most common detector is Deuterated triglycine sulphate (DLGS)

Alternate detectors are mercury cadmium telluride (MCD) which requires lower operational temperatures

4. The beam splitter is typically a thin Ge film deposited on KBr bulk crystals.
5. An important aspect of extracting the intensity in each frequency is the inverse Fourier transform

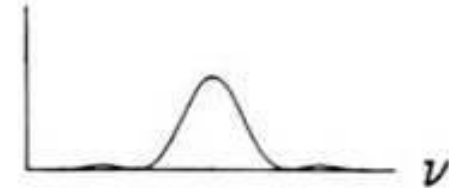
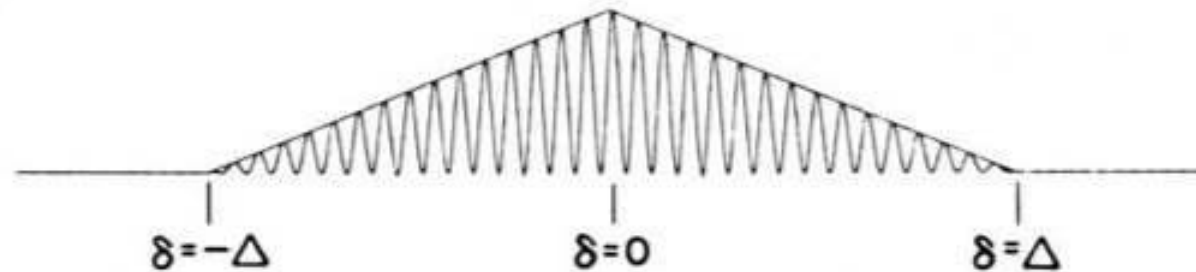
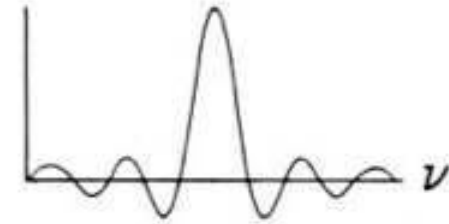
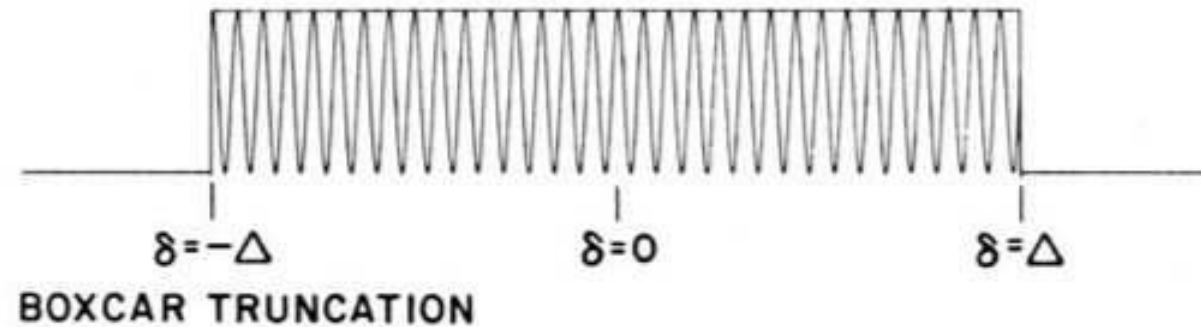
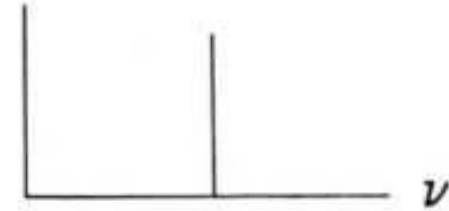
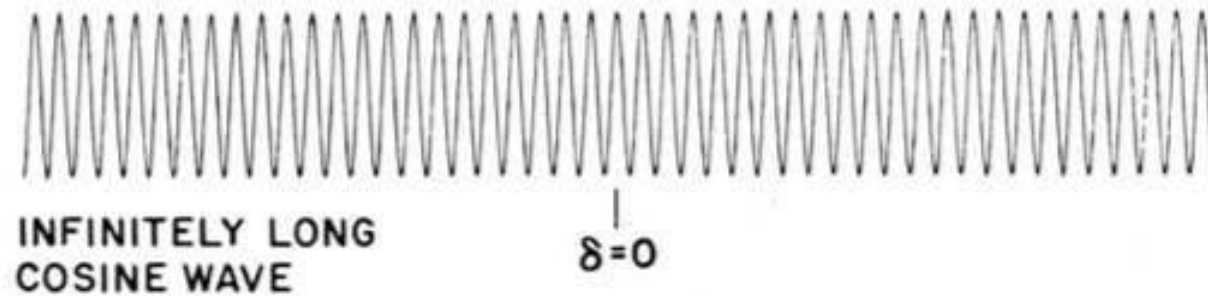
(i) However, true inverse transform requires intensity record for infinite δ

(ii) Practical considerations the path difference obtainable in experiments are limited.



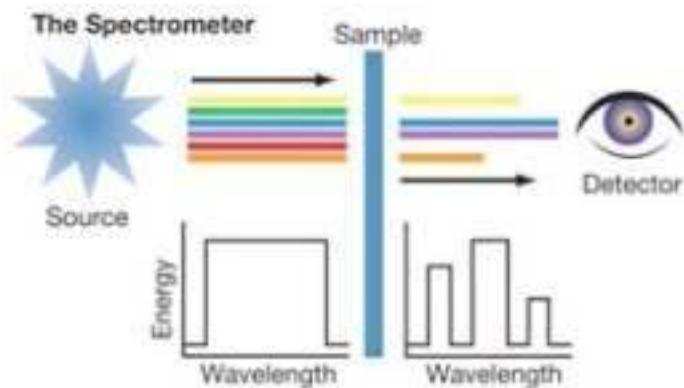
Apodization

Meaning in Greek to remove the feet, ie the ripples in the Fourier transform,

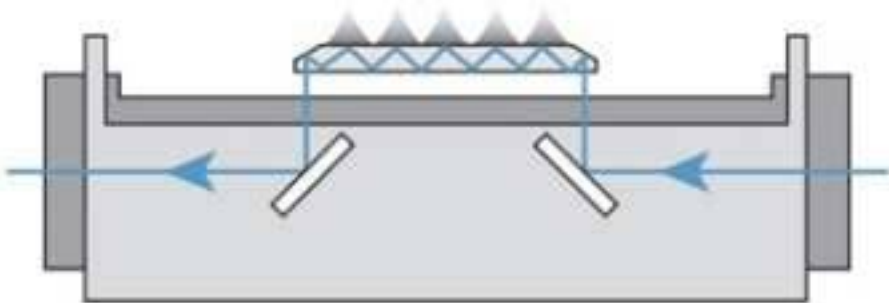


Sample Collection techniques

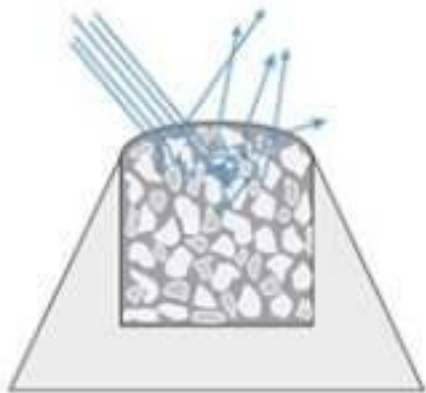
Transmission



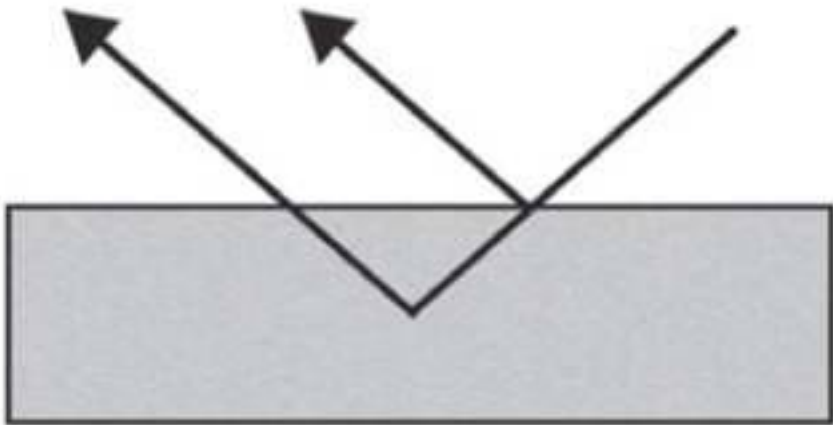
Attenuated Total Reflectance (ATR)



Diffuse Reflectance (DRIFTS)



Specular Reflectance



Examples

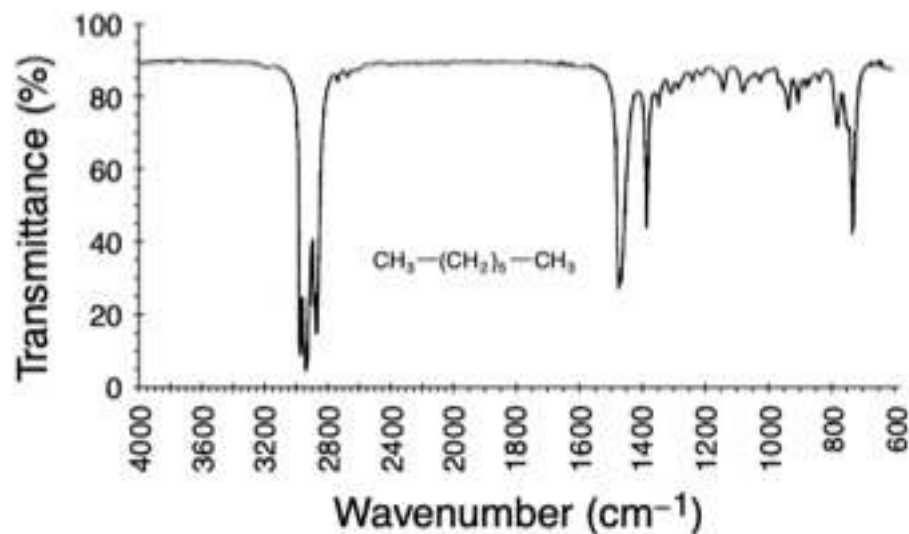


Figure 3 Attenuated total reflectance (ATR) spectrum of *n*-heptane. Copyright Coates Consulting.

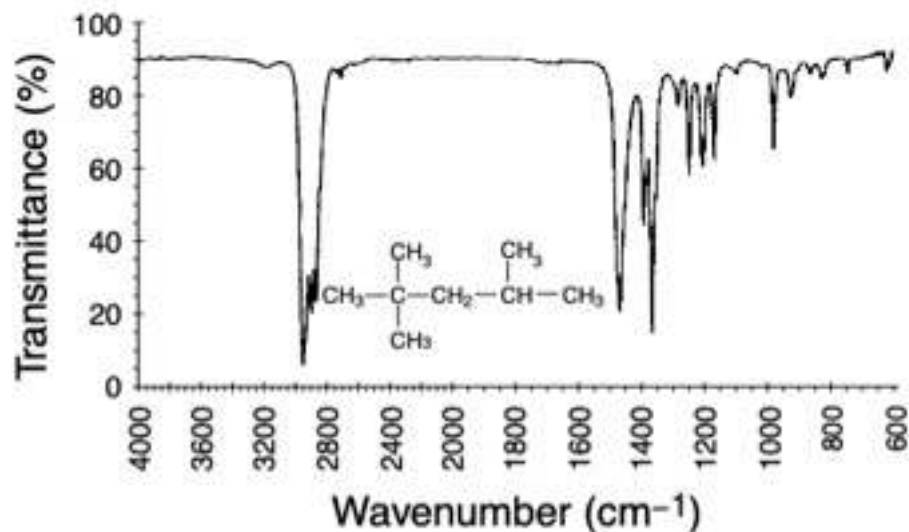


Table 1 Saturated aliphatic (alkane/alkyl) group frequencies

Group frequency (cm ⁻¹)	Functional group/assignment
Methyl (–CH₃)	
2970–2950/2880–2860	Methyl C–H asym./sym. stretch
1470–1430/1380–1370	Methyl C–H asym./sym. bend
1385–1380/1370–1365	<i>gem</i> -Dimethyl or “iso”- (doublet)
1395–1385/1365	Trimethyl or “ <i>tert</i> -butyl” (multiplet)
Methylene (>CH₂)	
2935–2915/2865–2845	Methylene C–H asym./sym. stretch
1485–1445	Methylene C–H bend
750–720	Methylene –(CH ₂) _n – rocking (<i>n</i> ≥ 3)
1055–1000/1005–925	Cyclohexane ring vibrations
Methyne (>CH–)	
2900–2880	Methyne C–H stretch
1350–1330	Methyne C–H bend
1300–700	Skeletal C–C vibrations
Special methyl (–CH₃) frequencies	
2850–2815	Methoxy, methyl ether O–CH ₃ , C–H stretch
2820–2780	Methylamino, N–CH ₃ , C–H stretch

Interpretation of Infrared Spectra, A Practical Approach

John Coates
Coates Consulting, Newtown, USA

Examples - II

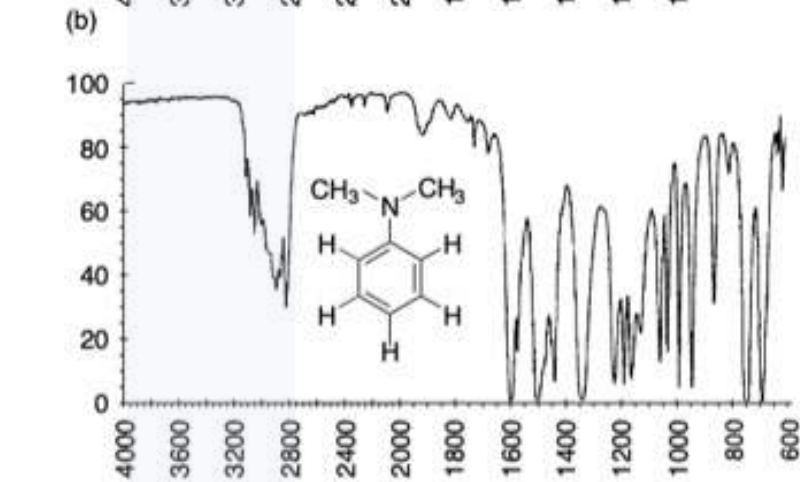
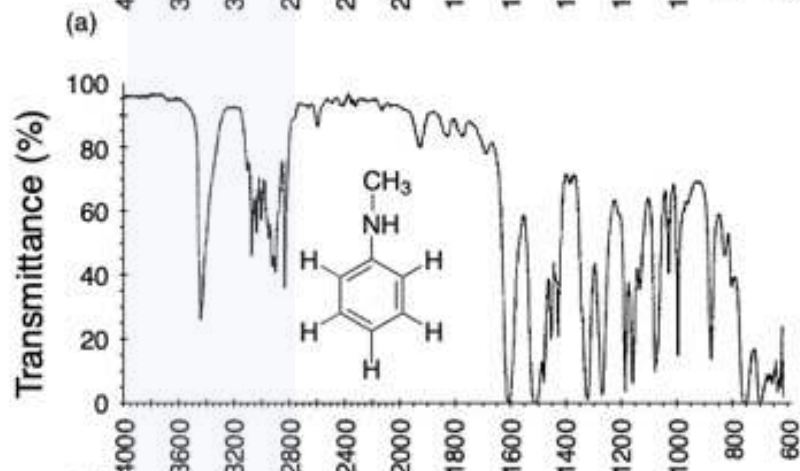
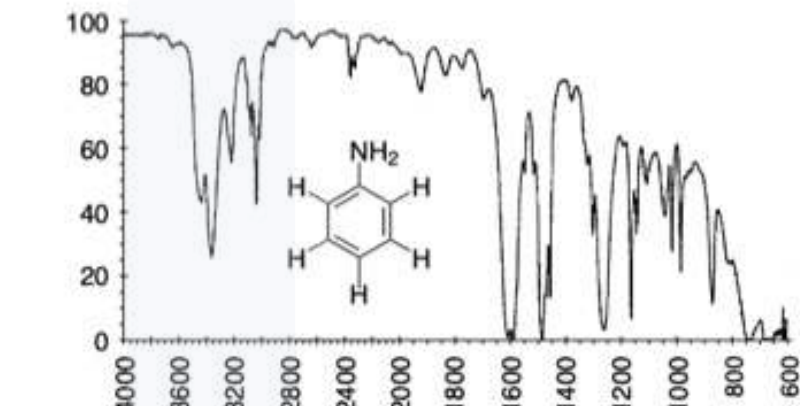


Table 8 Amine and amino compound group frequencies

Origin	Group frequency, wavenumber (cm ⁻¹)	Assignment
N-H	3400–3380 +3345–3325	Aliphatic primary amine, NH stretch
N-H	3510–3460 +3415–3380	Aromatic primary amine, NH stretch
N-H	1650–1590	Primary amine, NH bend
C-N	1090–1020	Primary amine, CN stretch
>N-H	3360–3310	Aliphatic secondary amine, NH stretch
>N-H	~3450	Aromatic secondary amine, NH stretch
>N-H	3490–3430	Heterocyclic amine, NH stretch
=N-H	3350–3320	Imino compounds, NH stretch
>N-H	1650–1550	Secondary amine, NH bend
C-N	1190–1130	Secondary amine, CN stretch
C-N	1210–1150	Tertiary amine, CN stretch
C-N	1340–1250	Aromatic primary amine, CN stretch
C-N	1350–1280	Aromatic secondary amine, CN stretch
C-N	1360–1310	Aromatic tertiary amine, CN stretch

Table 3 Aromatic ring (aryl) group frequencies

Origin	Group frequency, wavenumber (cm ⁻¹)	Assignment
C=C-C ^a	1615–1580	Aromatic ring stretch
C=C-C ^a	1510–1450	Aromatic ring stretch
C-H	3130–3070	Aromatic C-H stretch
C-H	1225–950 (several)	Aromatic C-H in-plane bend
C-H	900–670 (several)	Aromatic C-H out-of-plane bend
	770–730 + 710–690	Monosubstitution (phenyl)
	770–735	1,2-Disubstitution (ortho)
	810–750 + 900–860	1,3-Disubstitution (meta)
	860–800	1,4-Disubstitution (para)
“Combi” ^b	2000–1660 (several)	Aromatic combination bands

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Coates Consulting, Newtown, USA

Lastly, width of every band

- Rotational states around vibrational levels – Selection rules
- Collision broadening
- Doppler broadening
- Heisenberg uncertainty
- Detector uncertainty

Visible, Ultra-Violet Spectroscopy

Electronic transitions in molecules

Energy levels of molecules

Total Energy

$$E_T = E_{elec} + E_v + E_{rot}$$

Any transition leading to an energy change

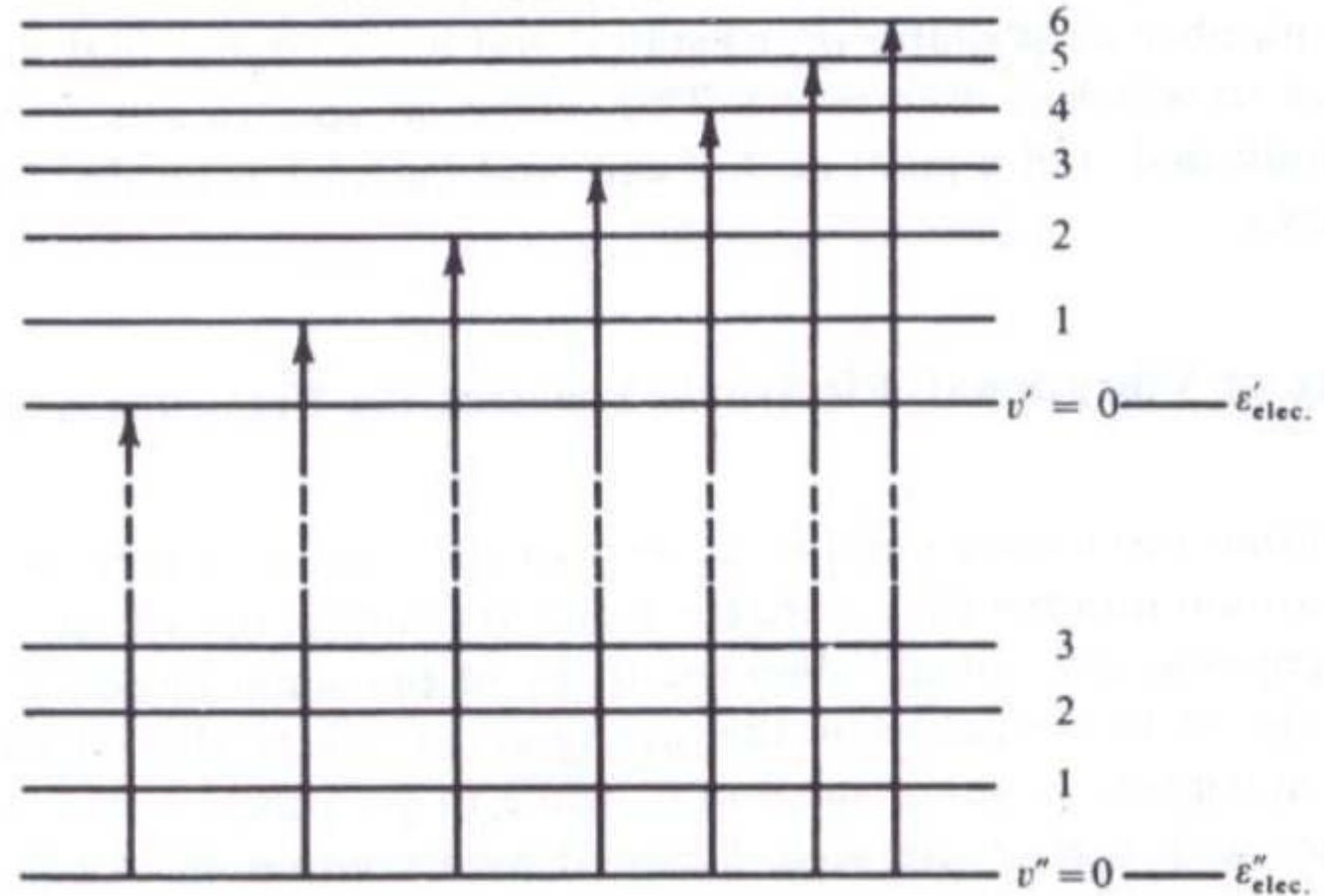
$$\Delta E_T = \Delta E_{elec} + \Delta E_v + \Delta_{rot}$$

This is Born-Oppenheimer approximation.

Each of the degree of freedom can be independently considered.

Frank-Condon Principle

- No selection criteria for electronic transition with regards to vibrational states.
- Electronic transitions take place so quickly
 - Vibrational mode change is seldom associated.
 - Most probable transition is from zero to zero vibrational state.

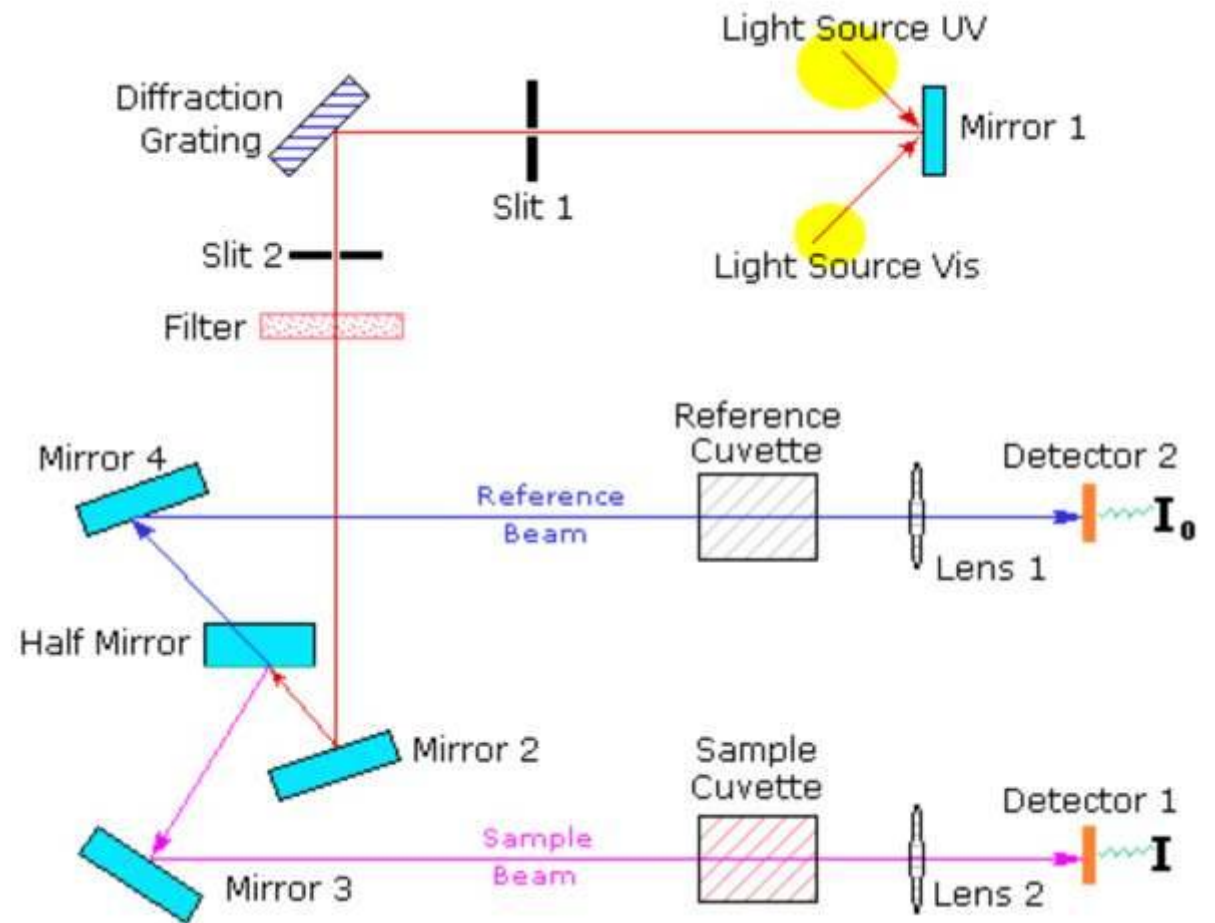


UV-Visible region

- 200 – 400 nm – UV
- 400 – 800 nm – Visible

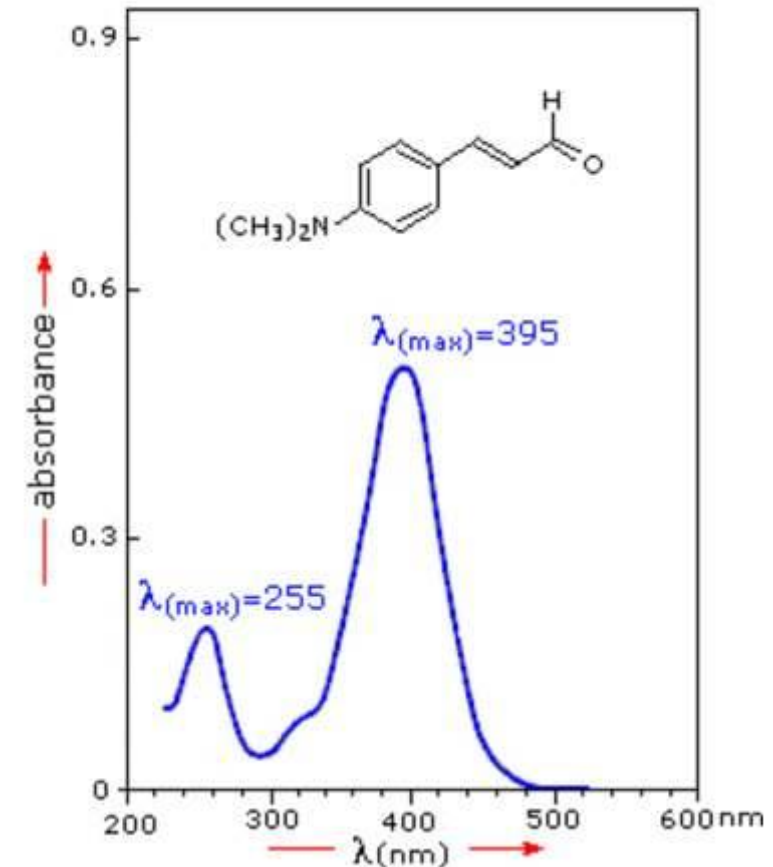
Remember, $A = \log I/I_0$

Molar Absorptivity = A/cl



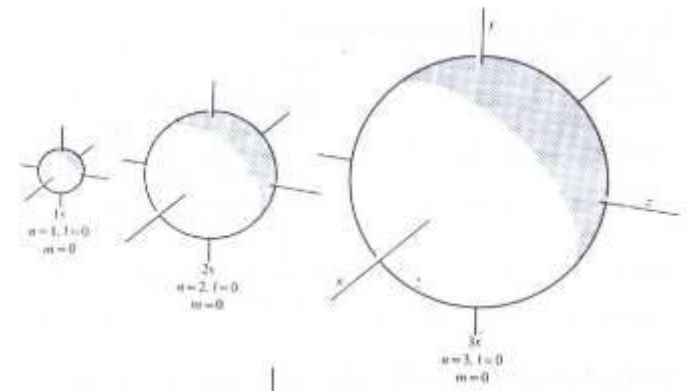
A typical measurement

- 0.249 g of aldehyde
- In ethanol
- Can we estimate the absorptivity at 395 and 255
- $\epsilon > 10000$ are strong absorbers and $\epsilon < 100$ are weak absorbers



Molecular Orbital Theory

Electrons in a Atom



- ψ – a complex valued function such that $|\psi^* \psi| (r)$ gives the probability to find an electron at r .
- ψ for any atom requires only three integers to uniquely define its shape.
- n – principle quantum number $\psi_n(r) = f(r)e^{-r/na_0}$

The probability to find an electron dies exponentially with r

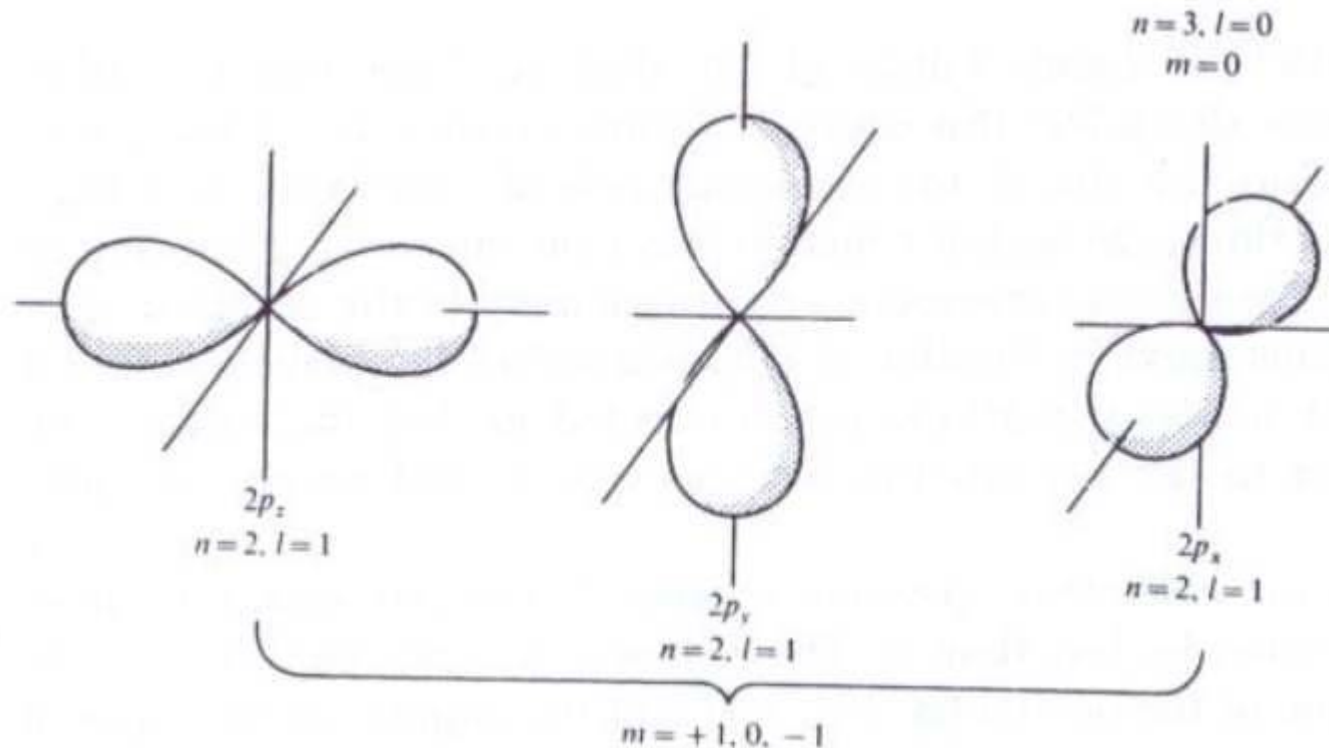
This fall is modulated by n . For larger n , the electrons fall slowly.

The region in space where electron spends >95% of time is called orbital.

Orbital Angular Momentum

Solutions to the spherically symmetric potential well also have solutions with angular nodes!

I.e., the electrons chose to stay away from certain angles



The situation where

L – number of angular nodes

$L = 1$.

Has three degenerate ways

Designated by $m = 0, \pm 1$

Higher order wavefunctions

- When $l = 1$, you have two lobes separated by a node
- When $l = 2$, the wavefunction has two nodes.
 - This gives 5 possible ways to distribute them in 3D space.
 - We call these D-diffused orbitals.
- When $l = 3$, we have three nodes
 - There 7 different m-values.

Energy levels in a molecule

Energy levels in an atom

$$E_n = -\left(\frac{\mu e^4}{32\pi^2 \epsilon^2 \hbar^2}\right) \frac{1}{n^2}$$

For $n = 1, 2, \dots$

Another way to write the same:

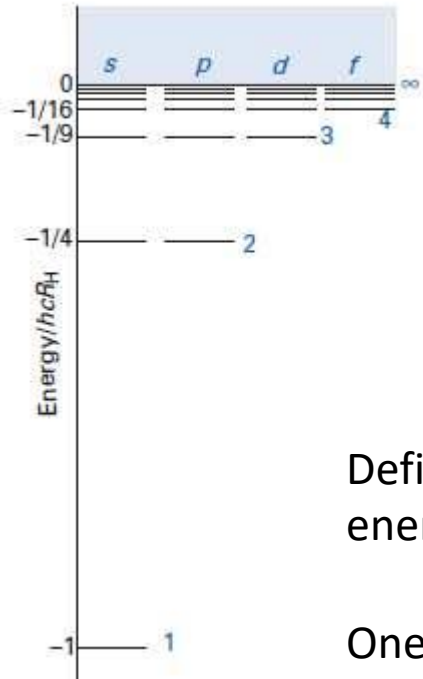
$$E_n = -\frac{hcR_H}{n^2} \text{ where } R_H = \frac{\mu e^4}{8\epsilon^2 h^3 c}$$

R_H is called as Rydberg constant

Defined in this fashion, since the expression of light energy $E = hc\bar{\nu}$ can be compared with atomic energy lines.

One can then expect light emission/absorption such that

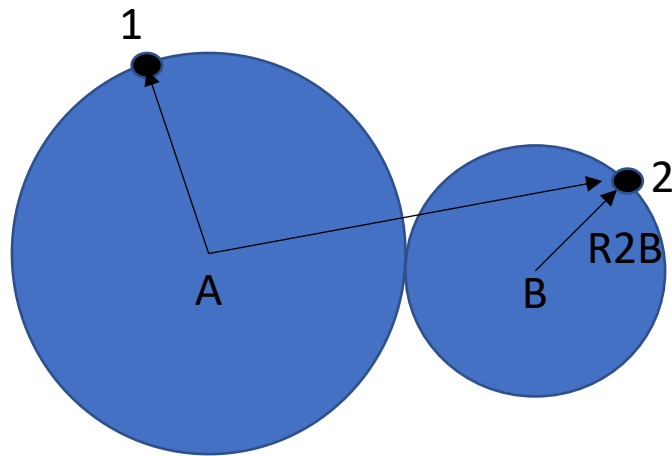
$$\bar{\nu} = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) R_H$$



Initial State	Final State	Series
2,3,...	1	Lyman
3,4,...	2	Balmer
4,5,...	3	Paschen
5,6,...	4	Brackett

Electrons in a molecule

- Biggest difference: Electrons in a shared potential of the atomic cores as well as other electrons



Hamiltonian of the Hydrogen molecule:

$$H = -\frac{h^2}{8\pi^2m} [\nabla_1^2 + \nabla_2^2] + \frac{e^2}{R} - \frac{e^2}{r_{1A}} - \frac{e^2}{r_{2A}} - \frac{e^2}{r_{2B}} - \frac{e^2}{r_{1B}} + \frac{e^2}{r_{12}}$$

If ϕ_A is the 1S atomic orbital of Atom A
And ϕ_B is the 1S atomic orbital of Atom B,

A trial solution to Schrodinger's equation can be obtained by a linear combination of atomic orbitals

One solution is a direct linear combination of atomic orbitals

$$\psi^1 = c_1\phi_A + c_2\phi_B$$

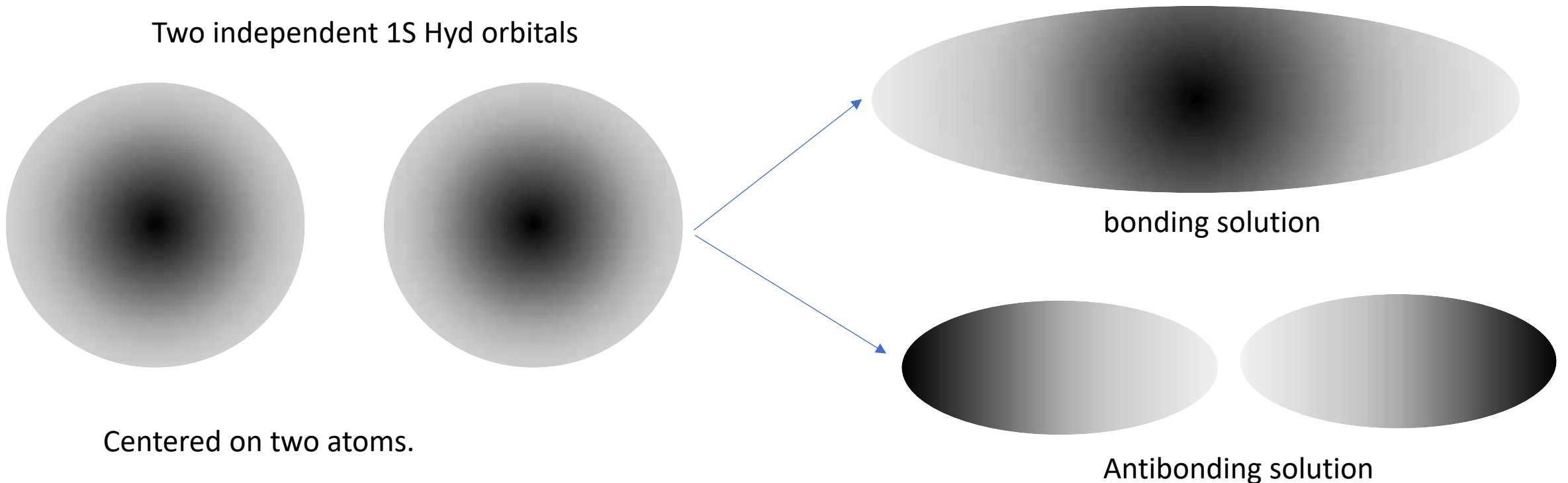
A second solution can be obtained by

$$\psi^2 = c_3\phi_A - c_4\phi_B$$

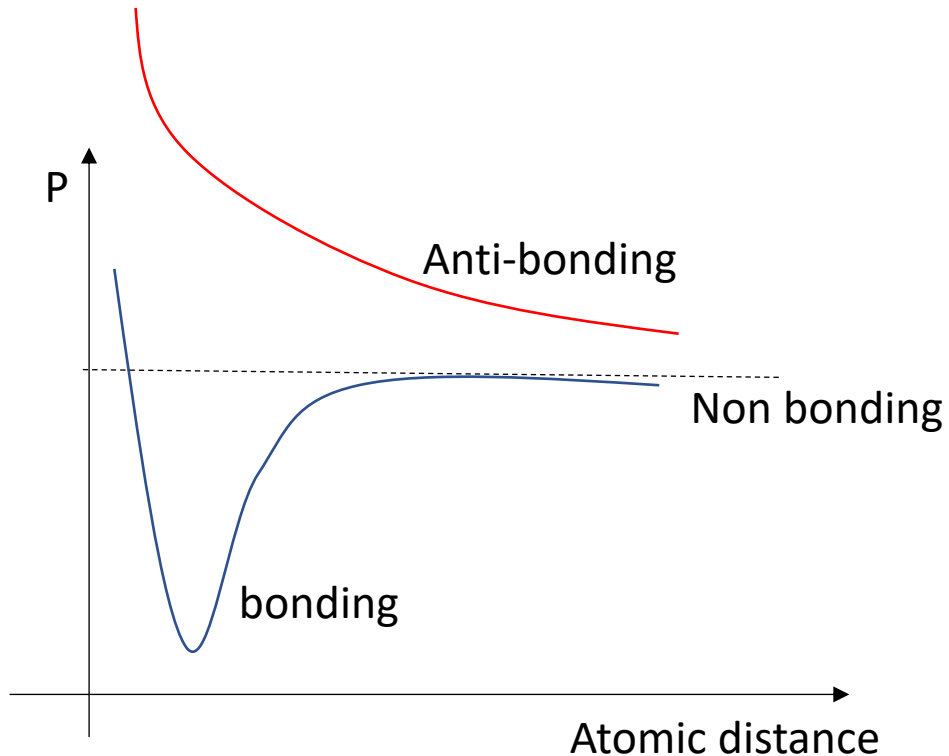
.

In the bonding case, the wavefunctions add up with the probability at the regions of overlap higher than other regions.

In the anti-bonding case, the wavefunctions cancel each other with the probability at overlap the lowest.



Bond Energetics



As two atoms are brought together, the coulombic attraction due to bonding allows for the total energy to reduce.

-> This leads to a dip in potential energy with a minima at a distance called as the bond length

-> This curve denotes the bonding curve which typically goes as $\psi = c_1\phi_A + c_2\phi_B$

-> The anti-bonding orbital, in which the contributions of the two atoms are added out-of-phase, has energies much higher than the bonding ones.

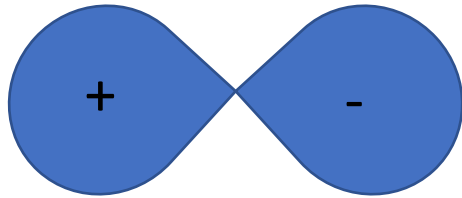
-> There is no energetic minimum for the non-bonding case.

During bonding, there is mixing of atomic orbitals!

Hydrogen is among the simplest of molecules.

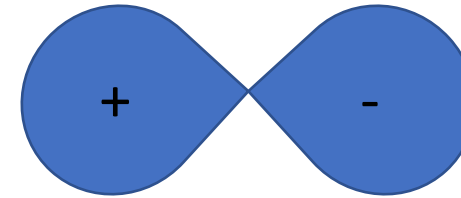
What happens, when there is mixing between orbitals, which are not spherically symmetric ?

Consider an electron in p orbital



Px orbital

Mixing with another px

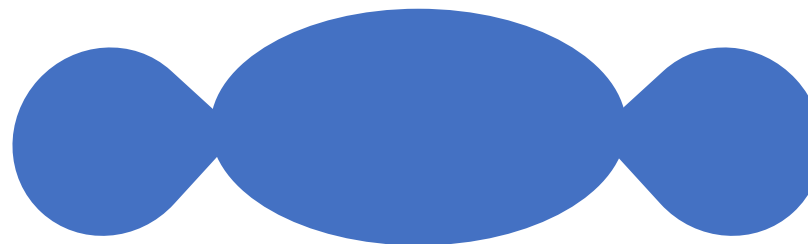


Px orbital

one option:

$$\psi_{2p} = \psi_{2p}^A - \psi_{2p}^B$$

One sort of bonding is obtained when the orbitals overlap in an 'end on' mode

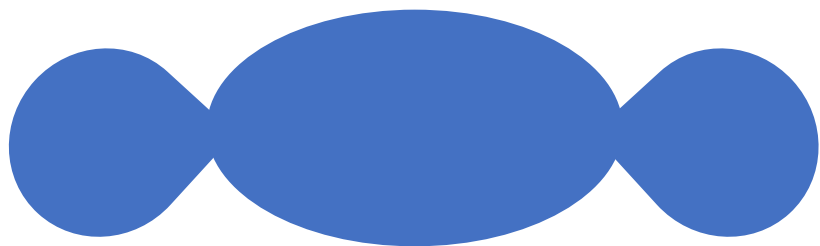


Such bonds are called as sigma bonds

Interestingly, the difference will lead to bonding scenario and the summation will lead to anti-bonding scenario

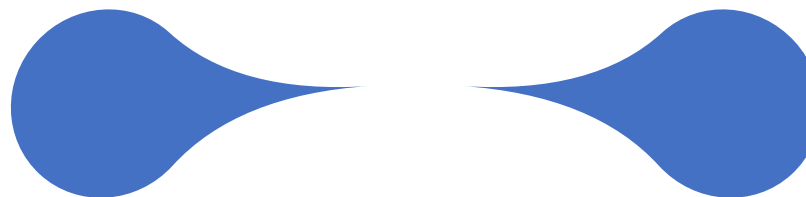
Sigma Bonding orbitals

$$\psi_{2p} = \psi_{2p}^A - \psi_{2p}^B$$

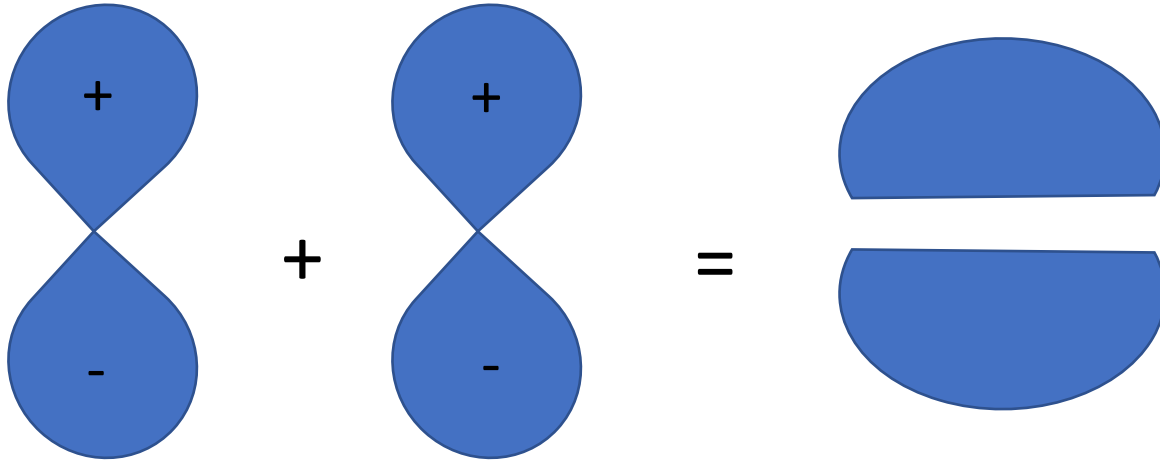


Sigma anti-bonding orbitals

$$\psi_{2p} = \psi_{2p}^A + \psi_{2p}^B$$



Mixing of two p-orbitals in sideways



$$\psi_{2p} = \psi_{2p}^A + \psi_{2p}^B$$

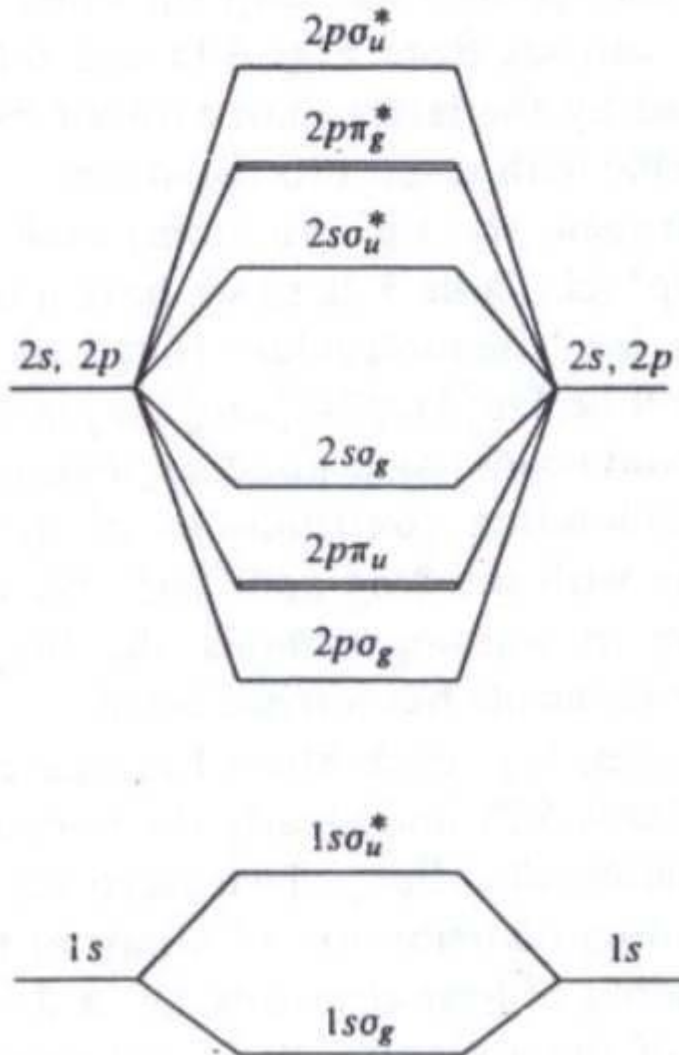
Such an overlap/mixing is called as Pi bonding

Remember, for each of the bonding orbital, there exists an equivalent anti-bonding orbital.

For σ bonds, you have an anti-bonding orbital called as σ^* bonds

For π bonds, you have an anti-bonding orbital called as π^* bonds.

Case of a diatomic molecule

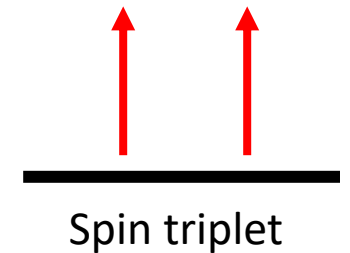
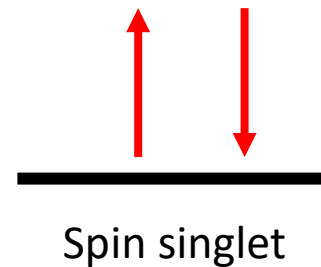


For H₂, has 2 electrons.

Both these electrons will sit in the $1s\sigma$ state.
This is the ground state of the hydrogen molecule.

Pauli's exclusion says, no two electrons with same spin can be in the same energy.

So, the ground state of the hydrogen atom is a spin singlet.



In a diatomic molecule, a huge collection of excitation is possible with discrete energies

Homework: Why He₂ molecule doesn't form ?

HYDROGEN SPECTRUM



400nm

500nm

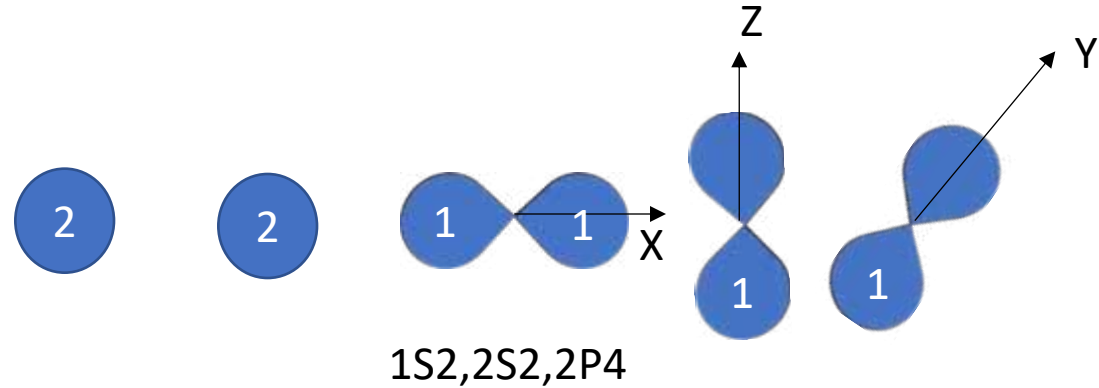
600nm

700nm

Poly-atomic molecules

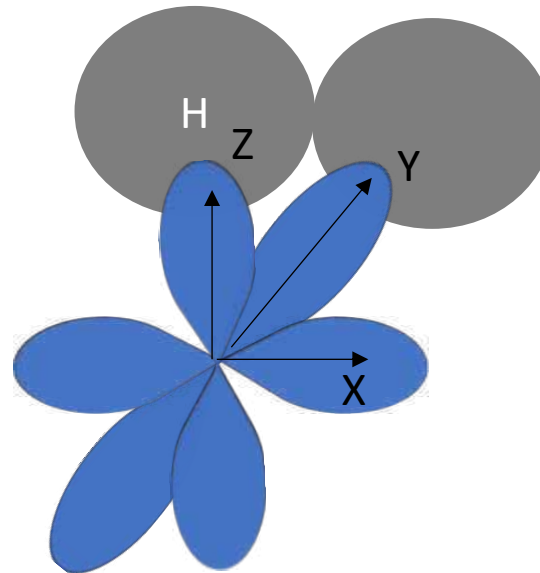
Example: Say water.

Oxygen has electronic configuration:



Hydrogen with $1s$ orbitals can overlap with any orbitals in an end-on way due to spherical symmetry.

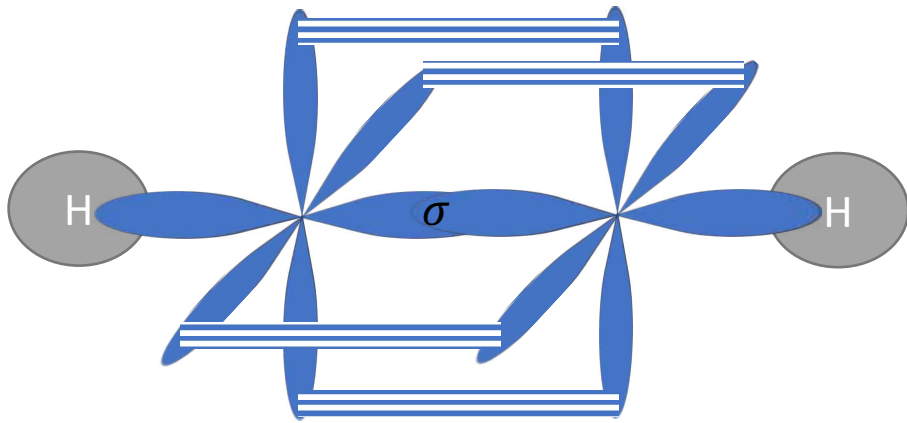
So water forms two hydrogen molecular sigma bonds with p_x and p_z orbitals.



If p_y and p_z orbitals are orthonormal,

What is the angle between the water bonds ?

Bonding picture of Acetylene



Atomic orbitals of certain atoms are special.

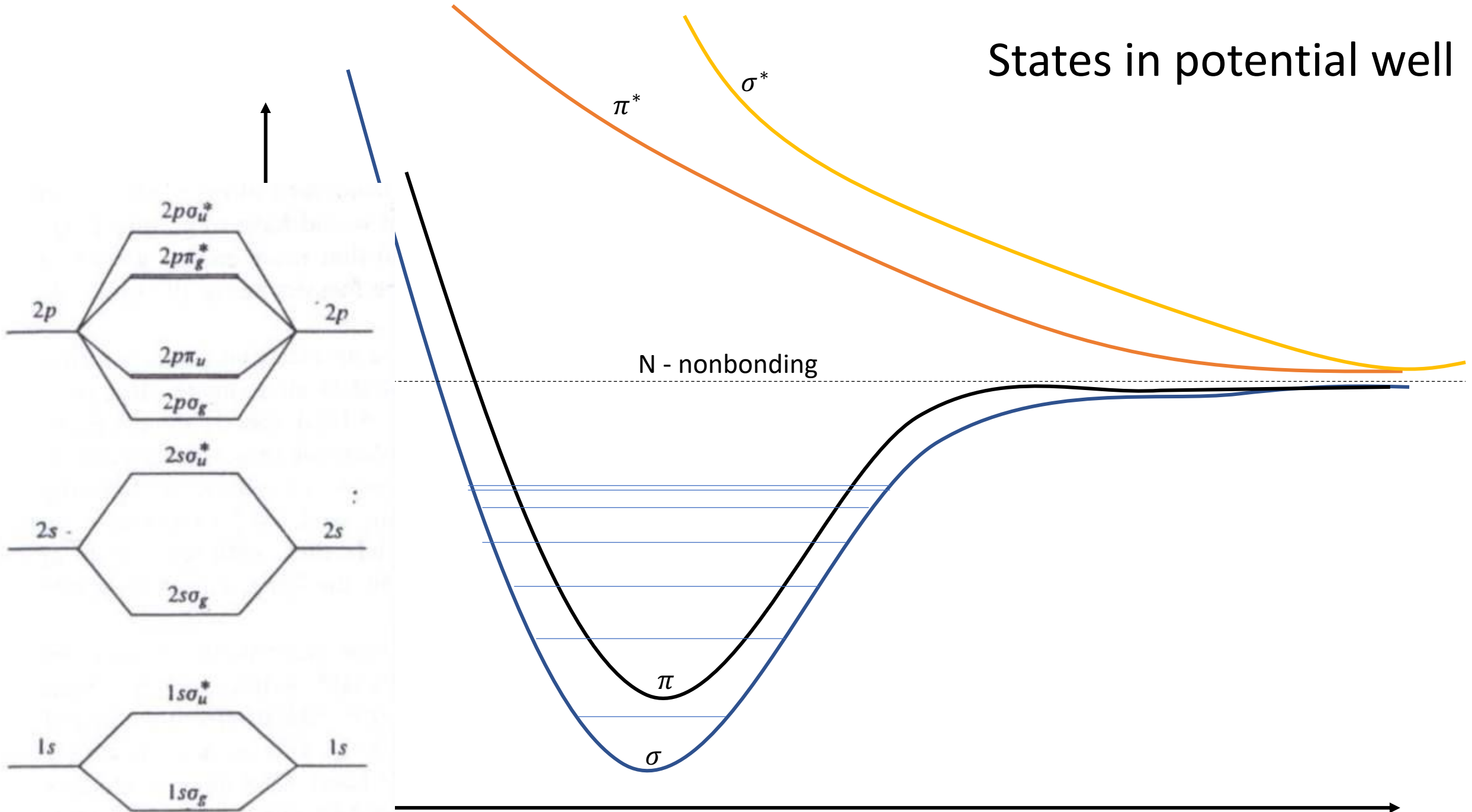
For instance, those in carbon hybridize within the atom. The valence of carbon has 2S², and 2P² electrons.

In this example of Acetylene, the S orbitals and 1P orbital of carbon hybridize (mix together) forming two SP orbitals.

They form bond with hydrogen and another carbon atom as shown in the figure.

Free p_y and p_z orbitals of carbon overlap forming the pi bonds.

States in potential well

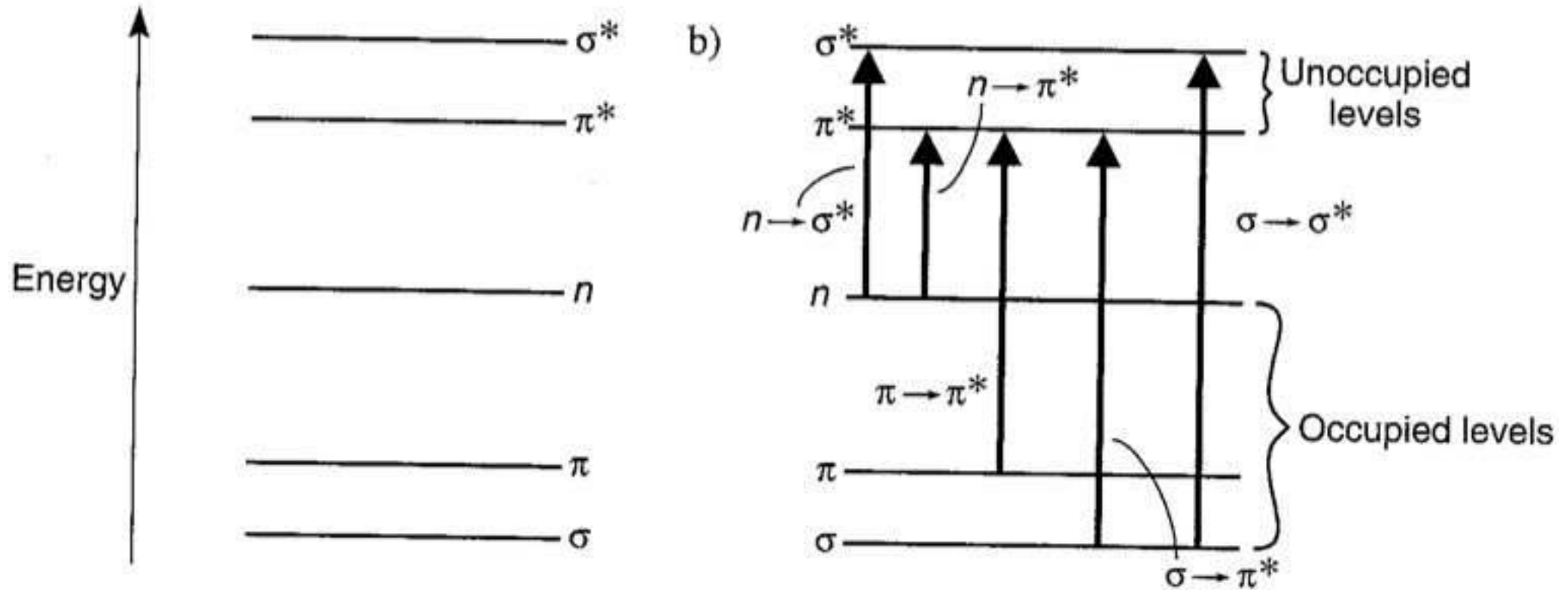


Types of electrons in molecule

- Sigma bonded electrons as in C-C, C-H, O-H etc.
- Pi electrons in C=C, $C \equiv C$, C=N etc.
- Atoms to the right of C in the periodic table O, N and halogens have unbonded valence electrons.

Energies and excitation with high energy radiations

Typical energy difference: 150 – 600 kJ/mol



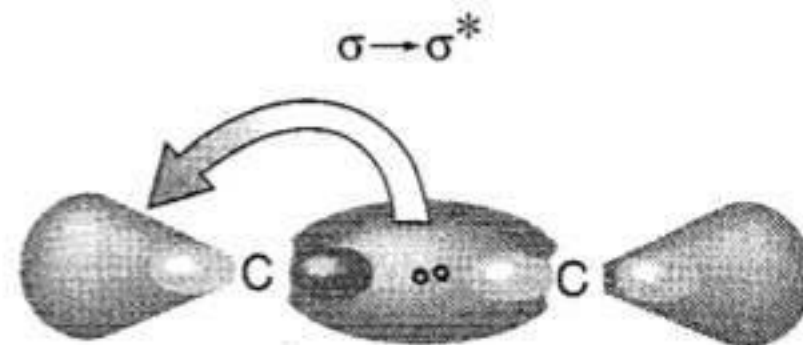
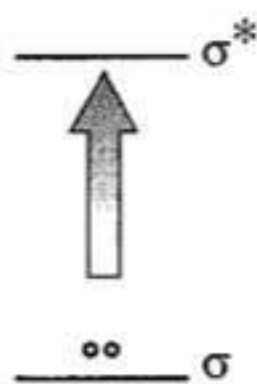
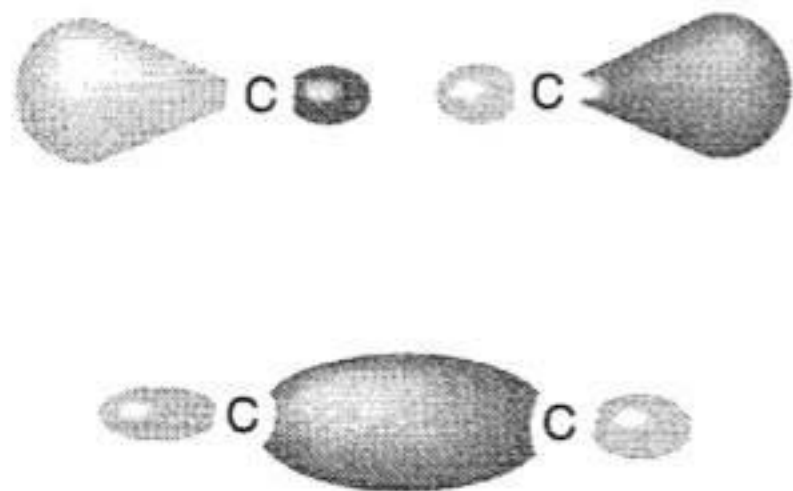
Before the interaction with light, electrons in a molecule are either in the bonding orbitals (σ , π) or non-bonded (atom localized). Anti-bonding states are un-occupied energy levels.

The high energy excitations have very small interaction time: $\sim 10^{-15}$ s. Thus, they do not interact with much slower nuclear motion ($\sim 10^{-13}$ s)

Selection Rules for Transition:

1. No change in electron spin state is allowed. A singlet state to remain singlet and a triplet a triplet
2. $\sigma \rightarrow \pi^*$ transitions and $\pi \rightarrow \sigma^*$ transitions are forbidden
3. $n \rightarrow \pi^*$ are also symmetry forbidden, but can sometimes be seen
4. $\pi \rightarrow \pi^*$ and $\sigma \rightarrow \sigma^*$ are commonly observed.

Possible transitions in Alkanes: (Say Ethane $\text{CH}_3\text{-CH}_3$)



Only possible transitions are high energy $\sigma \rightarrow \sigma^*$ requiring about 135 nm

If we introduce an atom containing non-bonded electrons like N

Consider a molecule such as $CH_3 - NH_2$

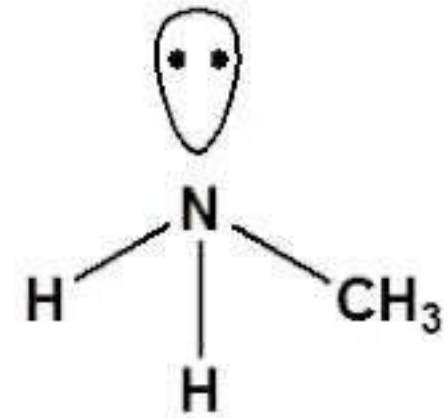
Two different absorption peaks are observed:

1. $\sigma \rightarrow \sigma^*$ at 170 nm
2. $n \rightarrow \sigma^*$ at 213 nm

Two observations here:

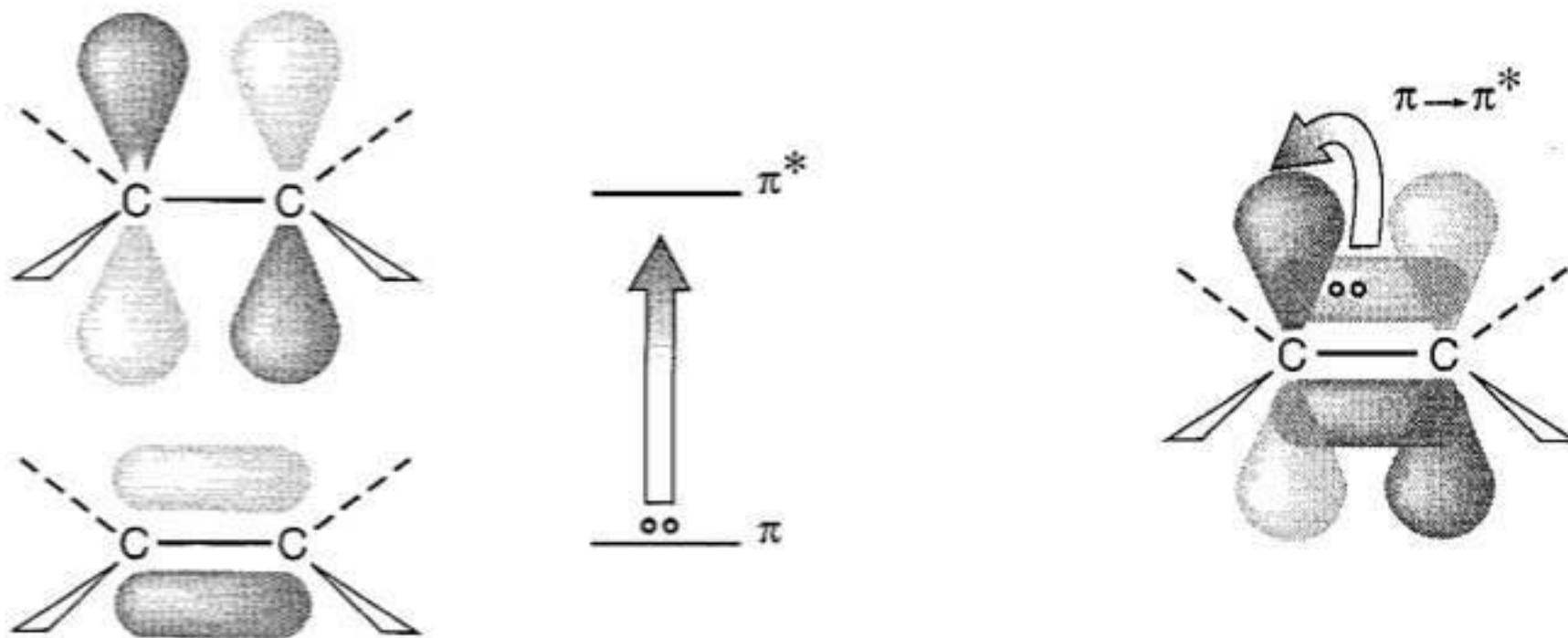
Inclusion of another atomic cluster

- changed the sigma transition energies
- Increased the lowest energy transition



Methylamine

Transitions in molecules with π - bonding



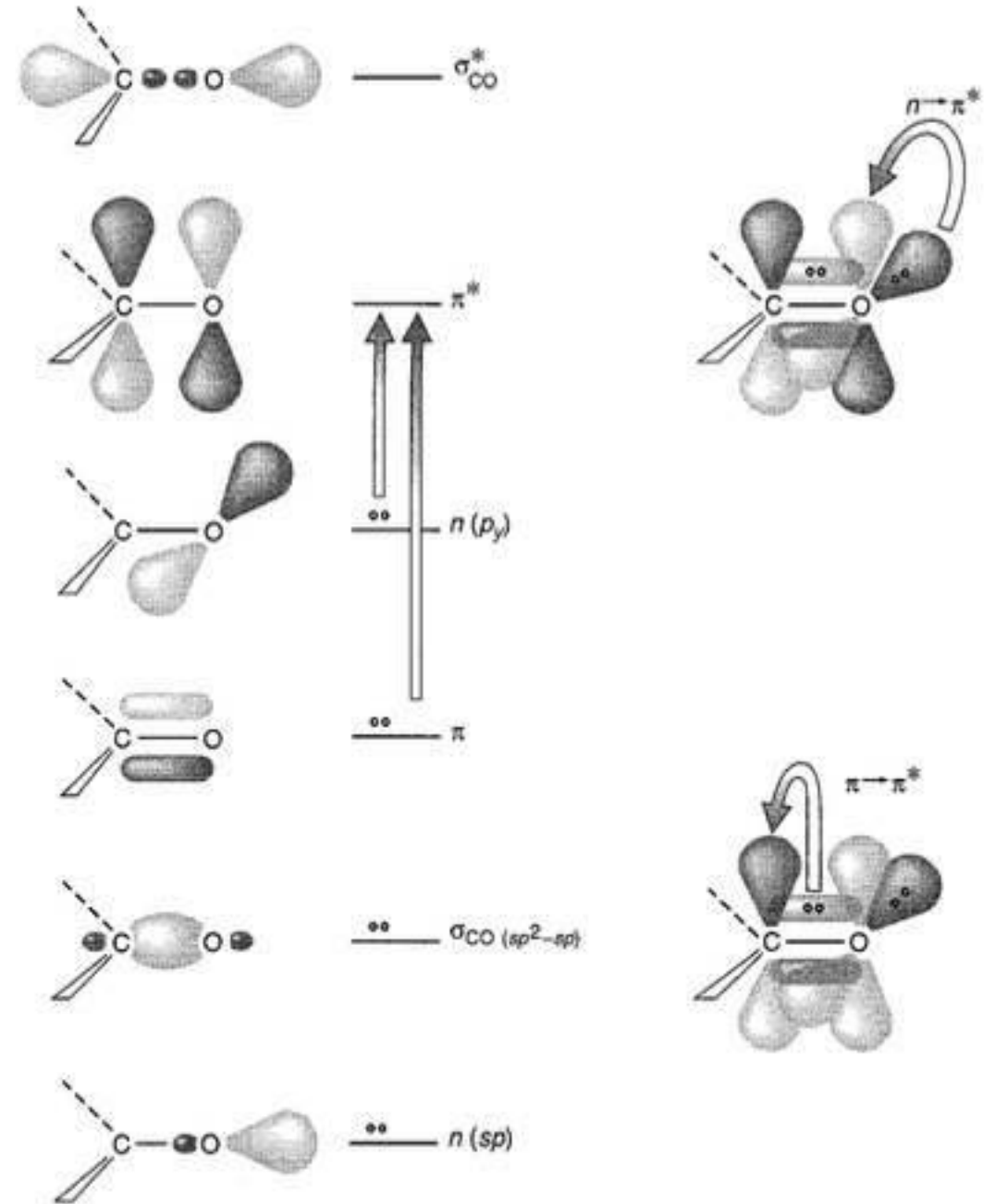
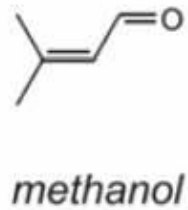
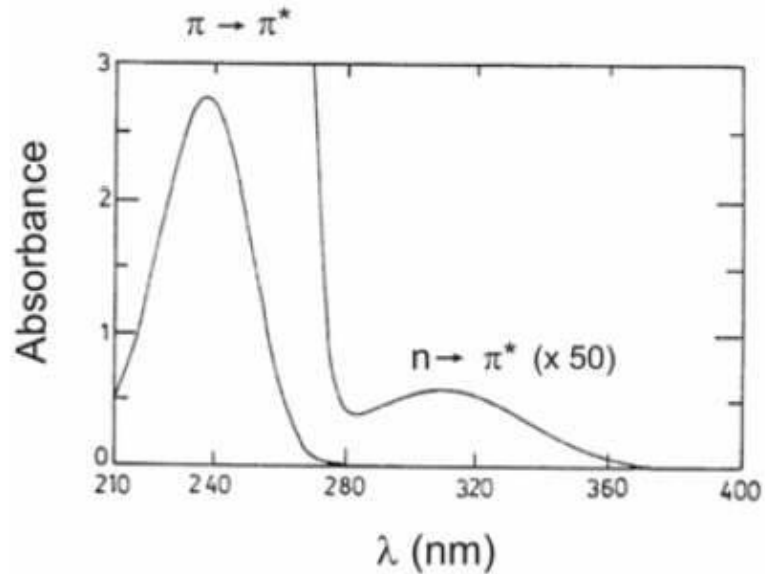
These transitions though also a bit high in energy, is also accessible under deep UV illuminations in certain molecules

Carbonyl compounds

Molecules having oxygen/nitrogen

Have unbonded electrons, which can undergo $n \rightarrow \pi^*$ such transitions have absorption peaks around 380 – 300 nm

Existence of π bonded electrons also allow for $\pi \rightarrow \pi^*$



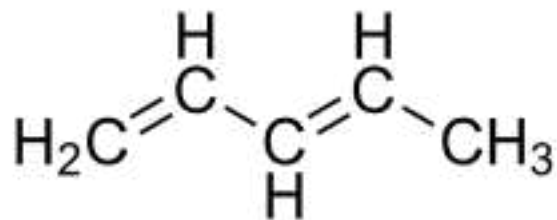
<i>Chromophore</i>	<i>Compound</i>	<i>Transition</i>	$\lambda_{\max}$ (nm)	ϵ
C-H	CH ₄	$\sigma \rightarrow \sigma^*$	122	
C-C	C ₂ H ₆	$\sigma \rightarrow \sigma^*$	135	
C=C	C ₂ H ₄	$\pi \rightarrow \pi^*$	103	15000
			174	5500
C=C=C	C ₃ H ₄	$\pi \rightarrow \pi^*$	170	4000
			227	630
C≡C	R-C≡C-R'	$\pi \rightarrow \pi^*$	178	10000
			196	2000
			223	160
C-O	R-O-R	$n \rightarrow \sigma^*$	180	500
C-O	R-O-R'	$n \rightarrow \sigma^*$	180	3000
C-N	Amino	$n \rightarrow \sigma^*$	190-200	2500-4000
C-S	R-S-H	$n \rightarrow \sigma^*$	195	1800
C-S	R-S-R	$n \rightarrow \sigma^*$	235	180
C=O	Aldehyde/Ketone	$n \rightarrow \sigma^*$	166	16000
		$\pi \rightarrow \pi^*$	189	900
		$n \rightarrow \pi^*$	270	10-20
C=O	Carboxylic acid	$n \rightarrow \pi^*$	200	50
C=O	Carboxylate	$n \rightarrow \pi^*$	210	150
C=O	Ester	$n \rightarrow \pi^*$	210	50
C=O	Amide	$n \rightarrow \pi^*$	205	200
C=N	(NH ₂) ₂ C=NH	$n \rightarrow \pi^*$	265	15
C≡N	CH ₃ C≡N	$\pi \rightarrow \pi^*$	<170	
N=N	Me-N=N-Me	$n \rightarrow \pi^*$	350-370	15
N=O	Me ₃ NO	$n \rightarrow \pi^*$	300	100
			665	120
N=O	Me ₃ NO ₂	$n \rightarrow \pi^*$	276	27
C=C=O	Et ₂ C=C=O	$\pi \rightarrow \pi^*$	227	360
		$n \rightarrow \pi^*$	375	20
C-Cl		$n \rightarrow \sigma^*$	173	200
C-Br		$n \rightarrow \sigma^*$	208	300
C-I		$n \rightarrow \sigma^*$	259	400

Common molecules and their absorption peak position with Stength

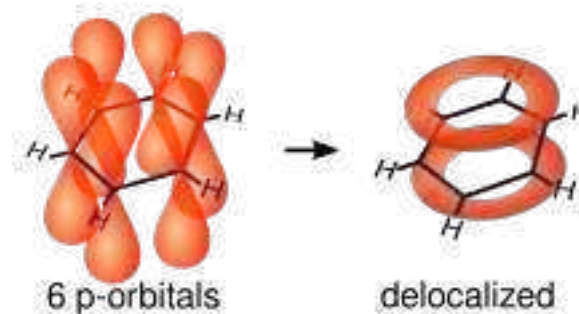
Chromophore	System	λ_{max} (nm)	ϵ_{max}
Aldehyde	—CHO	210	strong
		280-300	11-18
Amine	—NH ₂	195	2800
Bromide	—Br	208	300
Carbonyl	>C=O	195	1000
		270-285	18-30
Carboxyl	—COOH	200-210	50-70
Disulfide	—S—S—	194	5500
		255	400
Ester	—COOR	205	50
Ether	—O—	185	1000
Ethylene	—C=C—	190	8000
Iodide	—I	260	400
Nitrate	—ONO ₂	270 (shoulder)	12
Nitrile	—C≡N	160	-
Nitrite	—ONO	220-230	1000-2000
		300-400	10
Nitro	—NO ₂	210	strong
Nitroso	—NO	302	100
Sulfoxide	>S=O	210	1500
Benzene		184	46700
		204	6900
		255	170
Diphenyl		246	20,000



Effect of conjugation



Penta 1-3- diene



Delocalized orbitals of Benzene

Collection of delocalized p-orbitals leads to strong $\pi \rightarrow \pi^*$ transitions with low energy excitations.

When interfaced with non-bonded electrons (O,N), strong absorbers in visible region are observed.

Chromophore:

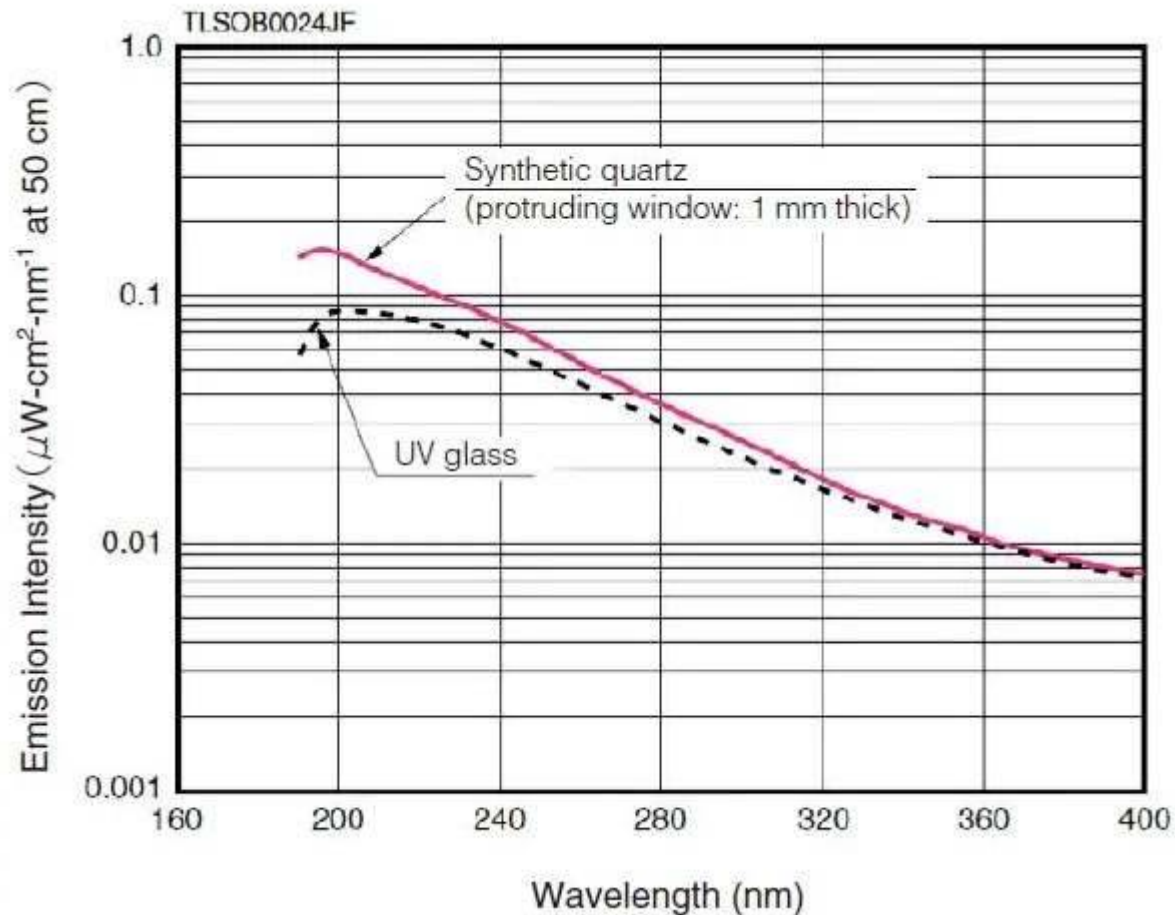
The group of atoms which decide the absorption behavior of bonded electrons is called as Chromophore.

Such chromophores typically used in dyes are made of such cyclic conjugated molecules.

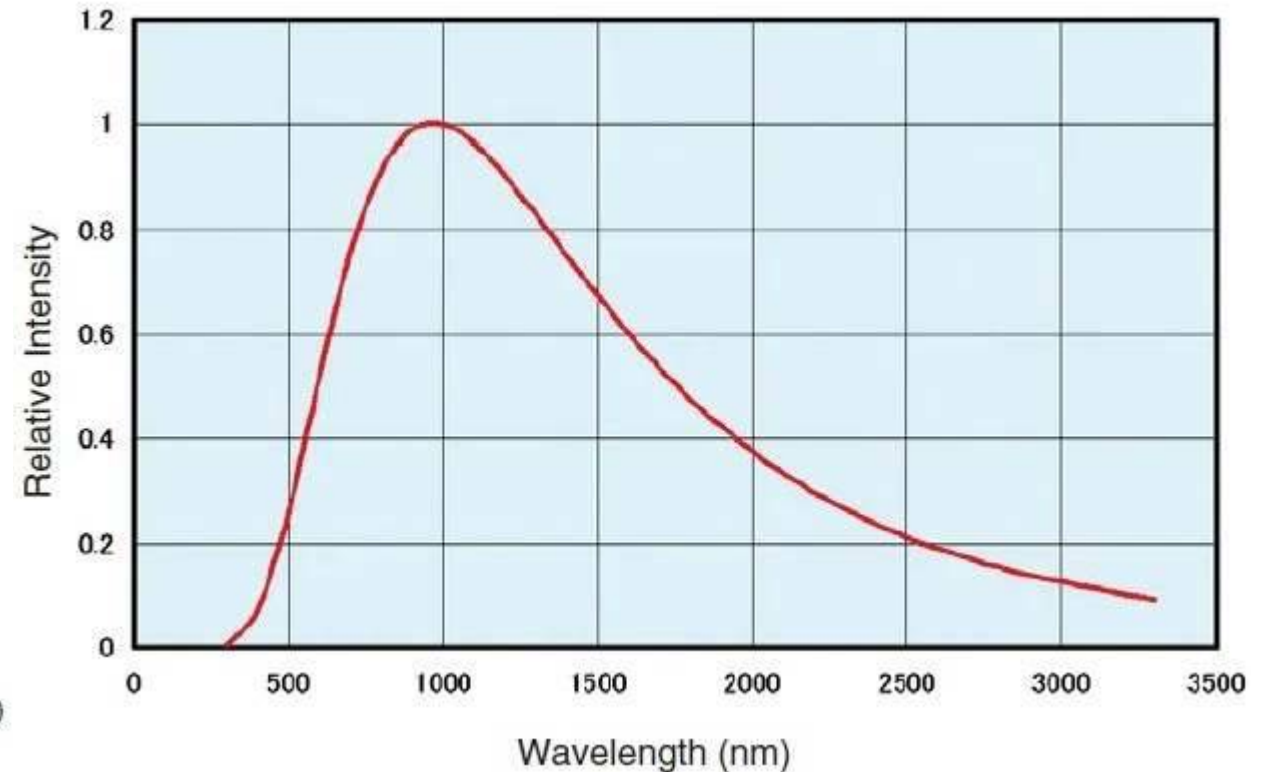
Instrumentation

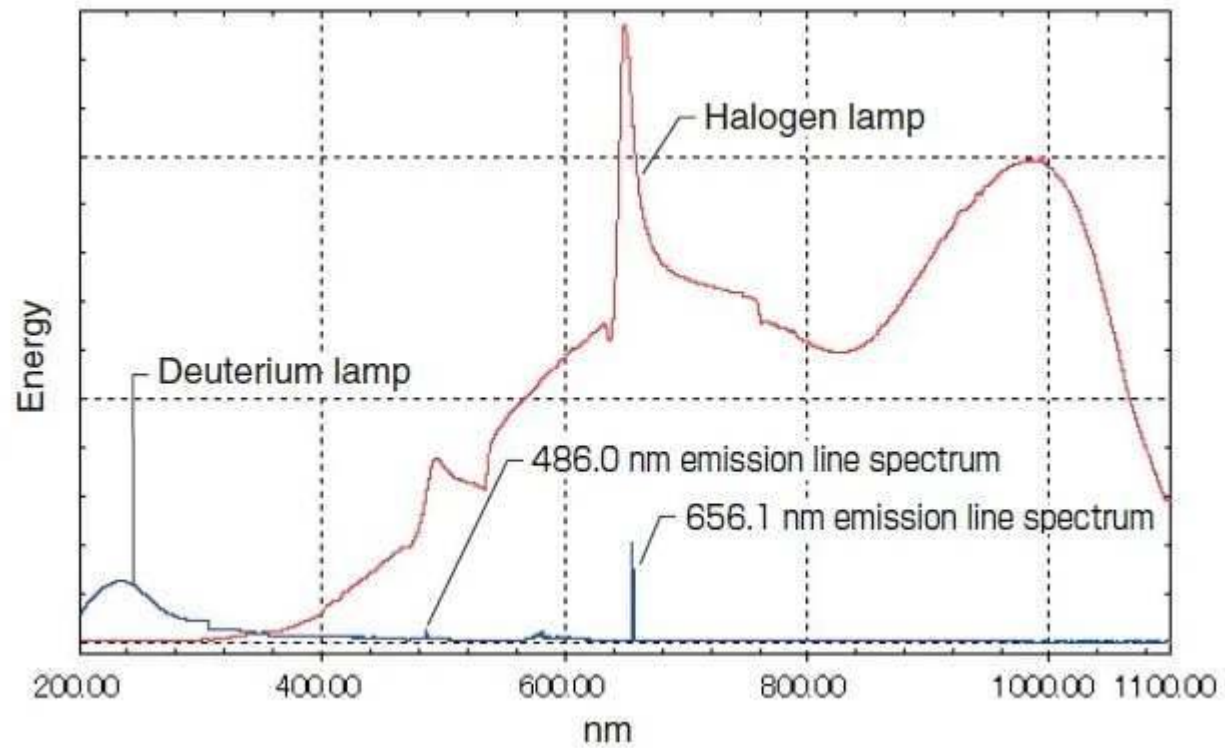
UV light source:

Typically used element is Deuterium arc source



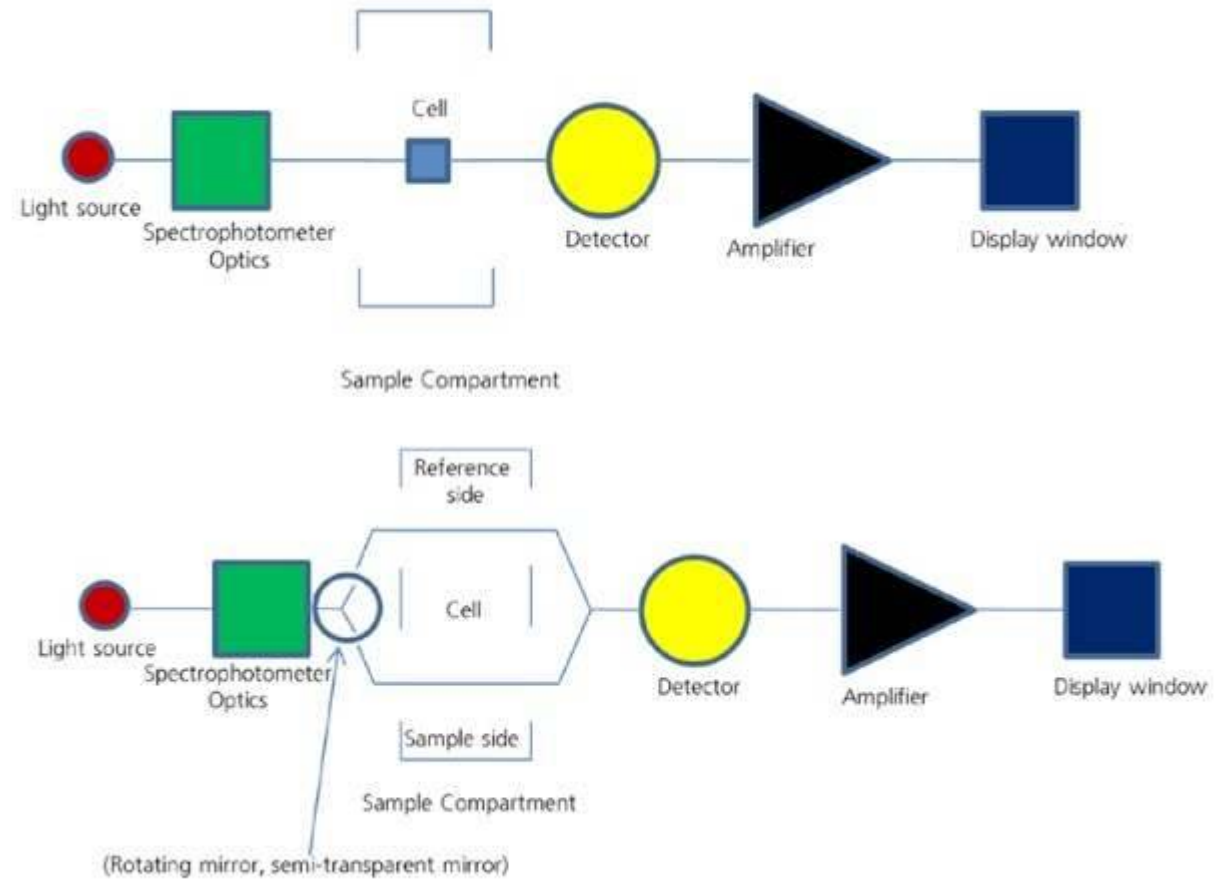
Visible-IR light source: Tungsten halogen lamp





Light source is switch at around 300 – 350 nm where the light intensity of the two sources are approximately equal.

General spectrometer setup:

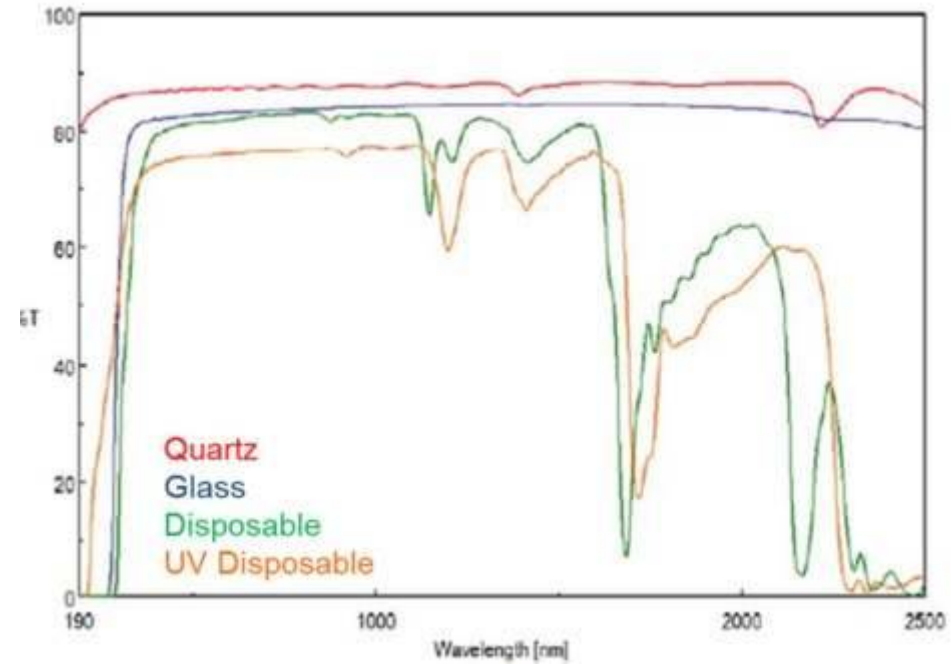


Sample holders:

Cuvettes:



Cuvettes transparency



Cuvette volumes: macro cells – 2.5 – 4 ml
micro cells – 250 – 1000 μ L

Other custom sizes and volumes can also be designed as necessary.

Influence of Solvents:

An ideal solvent does not interfere not absorb in the region of interest.

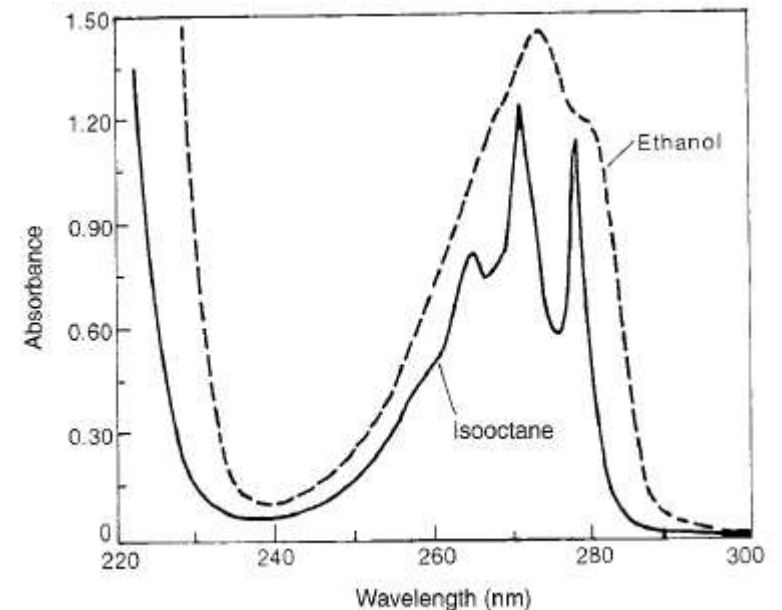
Given below is a list of commonly used solvents and their UV cut-offs

SOLVENT CUTOFFS			
Acetonitrile	190 nm	<i>n</i> -Hexane	201 nm
Chloroform	240	Methanol	205
Cyclohexane	195	Isooctane	195
1,4-Dioxane	215	Water	190
95% Ethanol	205	Trimethyl phosphate	210

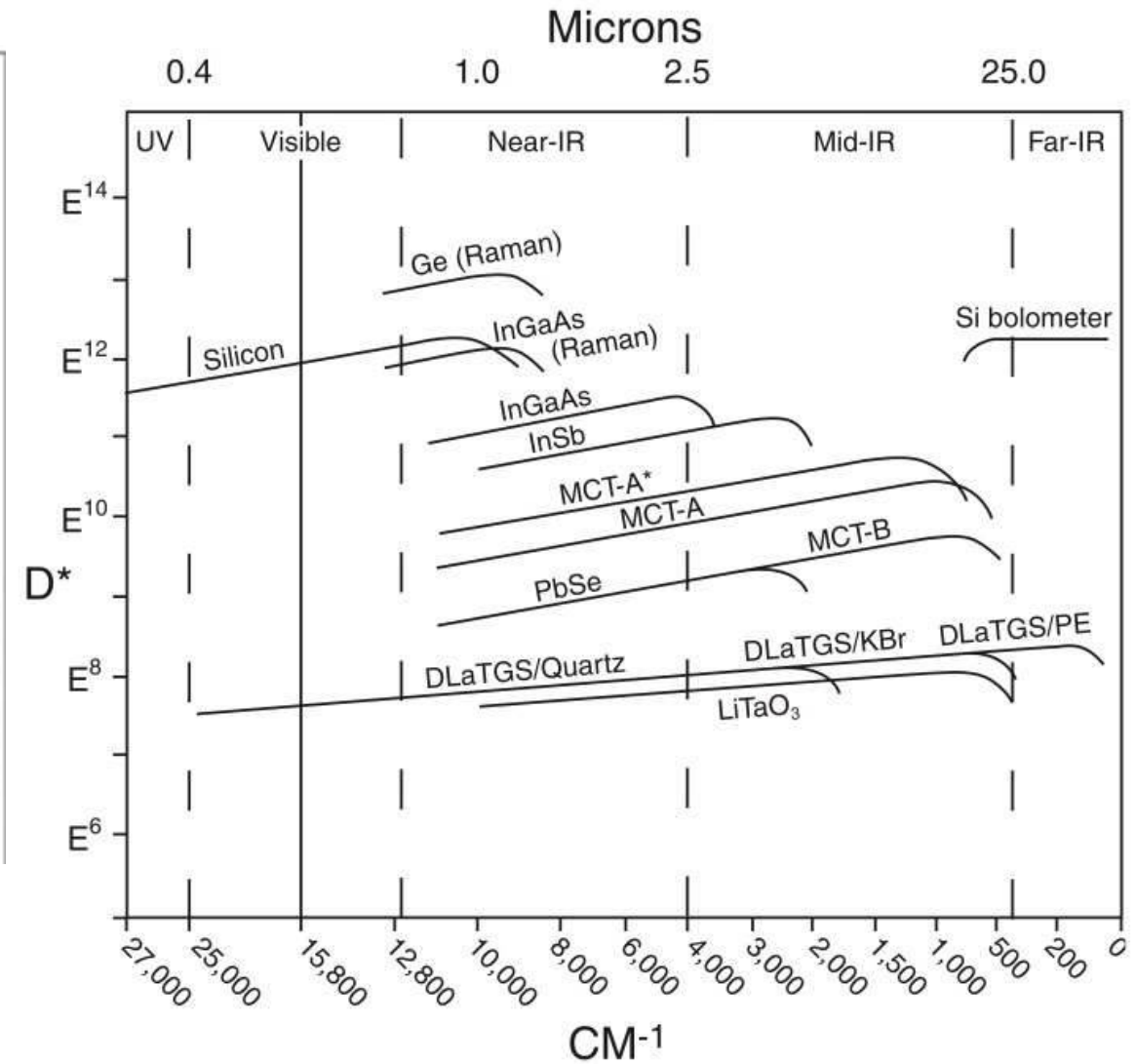
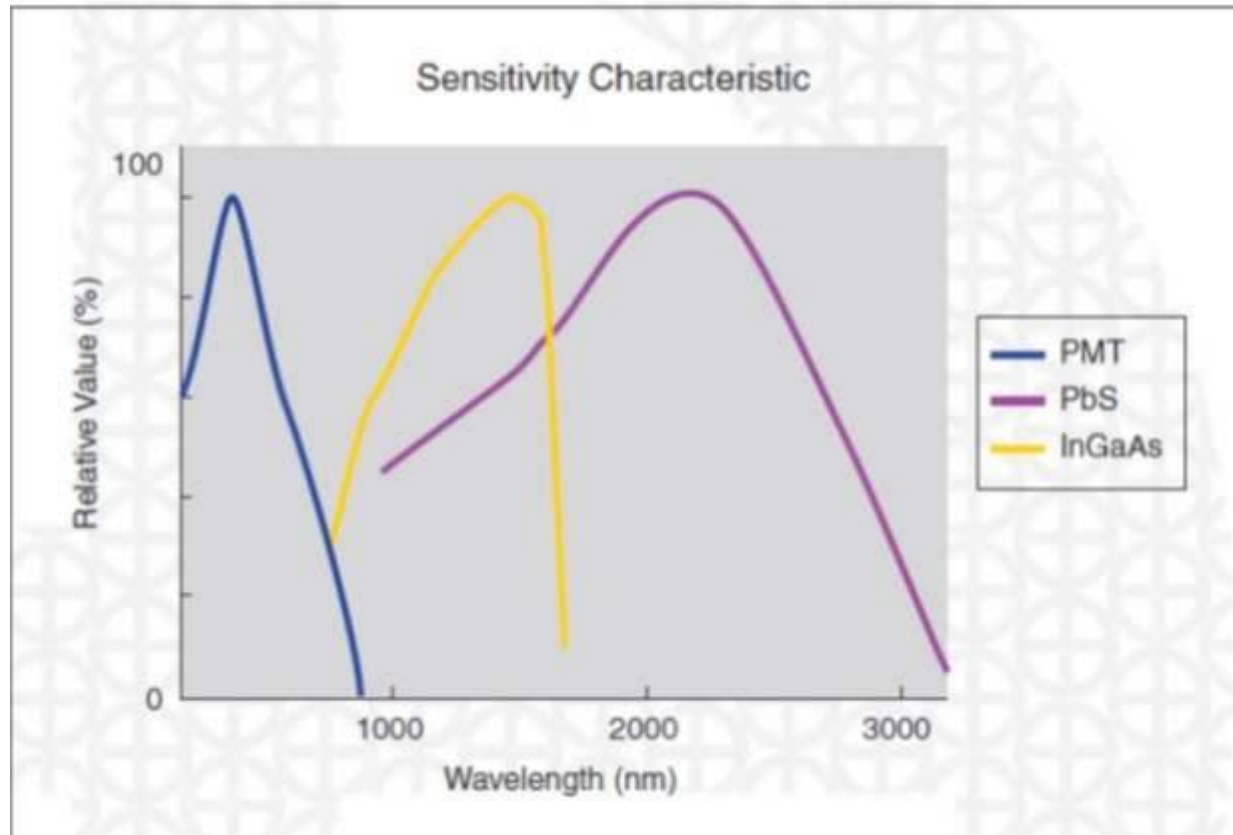
While water starts absorbing below 190 nm, ethanol absorbs below 205 and such as shown.

This can hide several characteristic peaks occurring in the region.

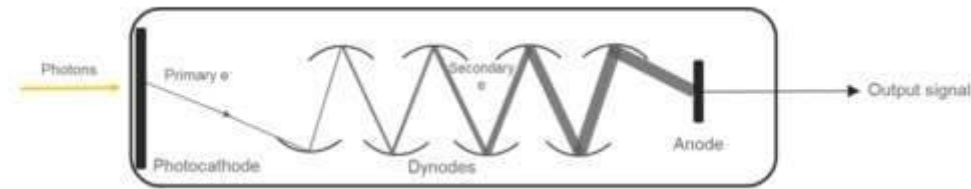
Polar molecules form hydrogen bond with the solute. This obstructs independent absorption smearing sharp features



Detector set for spectroscopic applications



Operation of a Photo-multiplier Tube



A primary detector finds the photons and generates photo dependent electrons.

The electrons are subsequently accelerated by several collisions in dynodes

Finally, a large avalanche of electrons are generated from a single photo generated electron.

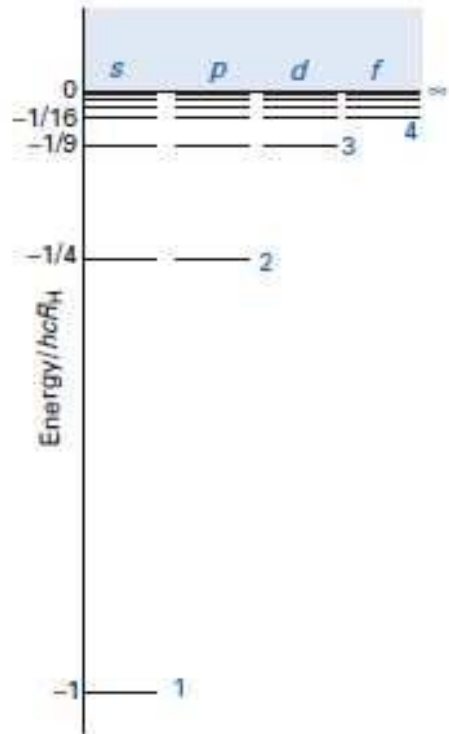
Pro: Highly sensitive. Can be extended to single photon detection.

Con: Susceptible to noises. Stray electrons are also multiplied.

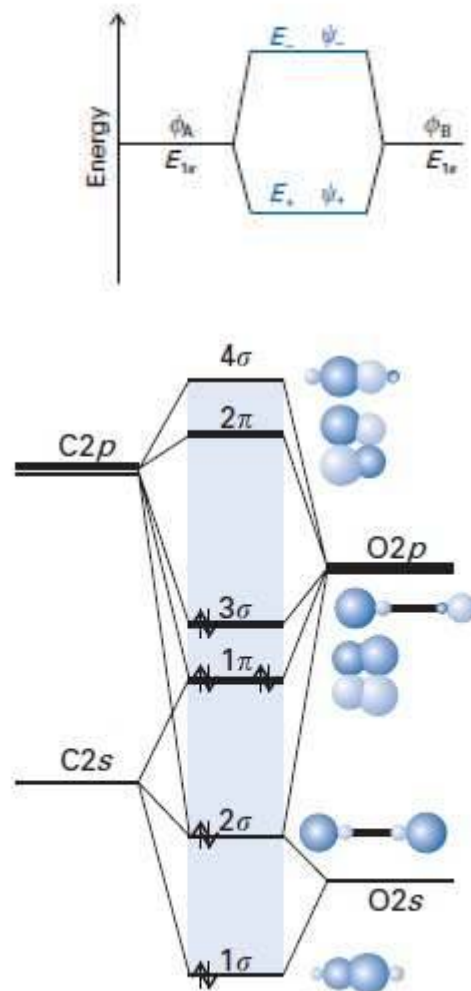
Large jitter, turn around time for generation of electric pulses to a single photon input.
The detector is down for a significant portion of time. Limiting photo detection bandwidth

Introduction to Light microscopes

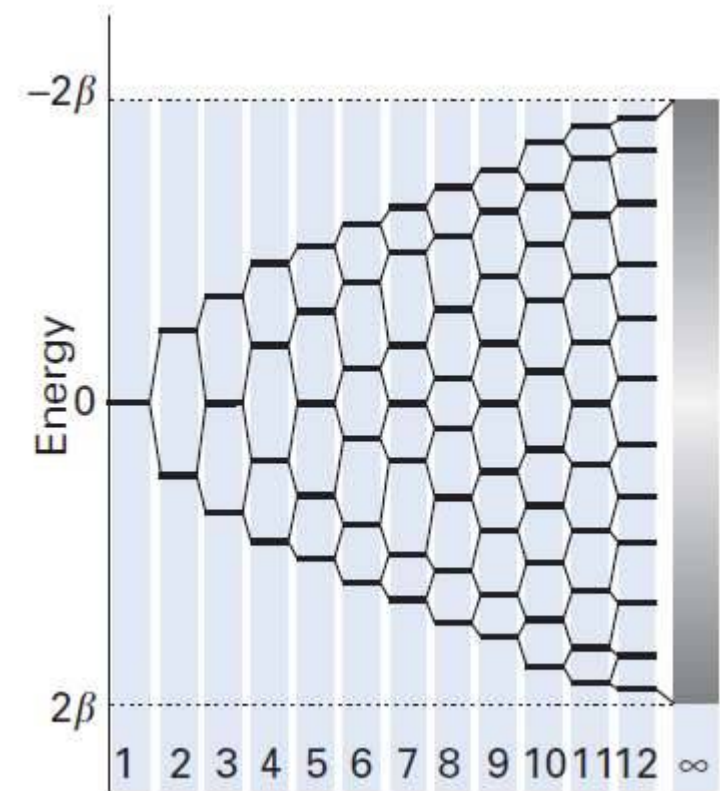
Atomic States



Molecular orbitals



String of N atoms – leading to solids



Lowest energy corresponds to completely bonded orbital with no nodes.
 k th energy level is formed with $k-1$ nodes.

All of Maxwell's equation can be narrowed down to

$$\nabla^2 E = \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla^2 B = \epsilon_0 \mu_0 \frac{\partial \vec{B}}{\partial t}$$

Remember, E, B are vectors $E = \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$

Hence the Maxwell's equation will come to

$$\begin{aligned} \frac{\partial^2 E_x}{\partial^2 x^2} + \frac{\partial^2 E_x}{\partial^2 y^2} + \frac{\partial^2 E_x}{\partial^2 z^2} &= \epsilon_0 \mu_0 \frac{\partial E_x}{\partial t} \\ \frac{\partial^2 E_y}{\partial^2 x^2} + \frac{\partial^2 E_y}{\partial^2 y^2} + \frac{\partial^2 E_y}{\partial^2 z^2} &= \epsilon_0 \mu_0 \frac{\partial E_y}{\partial t} \\ \frac{\partial^2 E_z}{\partial^2 x^2} + \frac{\partial^2 E_z}{\partial^2 y^2} + \frac{\partial^2 E_z}{\partial^2 z^2} &= \epsilon_0 \mu_0 \frac{\partial E_z}{\partial t} \end{aligned}$$

The velocity of wave can be deciphered by comparing with standard wave equation

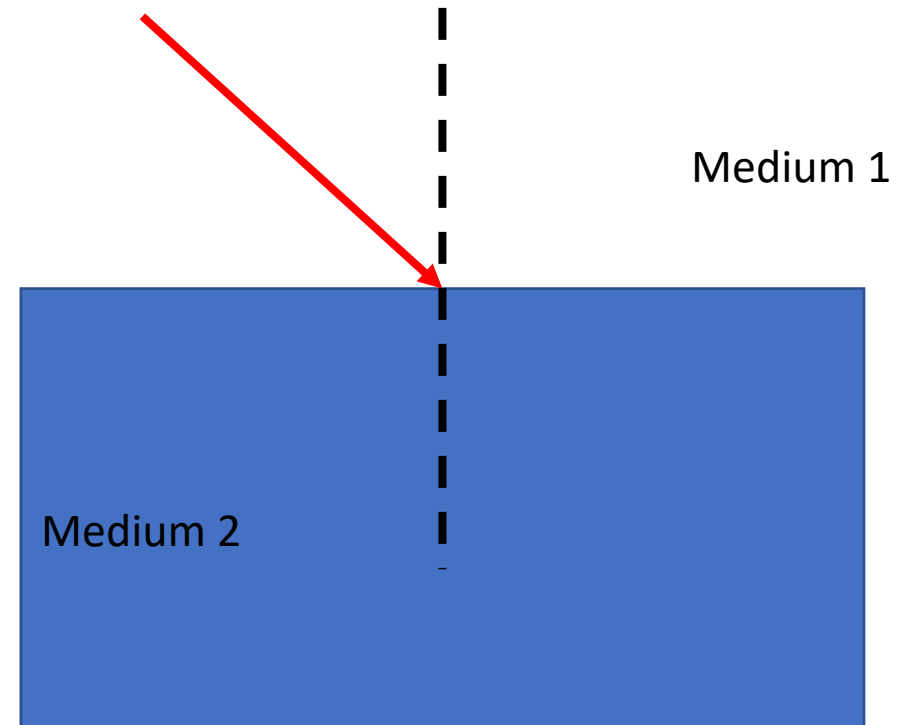
$$v = \frac{1}{\sqrt{\epsilon_m \mu_m}}$$

If the wave is moving in free space, $\mu_0 = 4\pi \cdot 10^{-7} \text{ m} \frac{\text{Kg}}{\text{C}^2}$, $\epsilon_0 \mu_0 = 11.12 \times 10^{-18} \text{ s}^2/\text{m}^2$

This gives the velocity of light in free space to be $C = 2.99792458 \times 10^8 \frac{\text{m}}{\text{s}}$

Light at the interface of materials

- We saw that the material determines the velocity, wavelength and such.
- So, what happens to the direction of propagation when light meets an interface between two materials ?
- What decides ?



The velocity of light is now different.

$$v = \frac{1}{\sqrt{\epsilon_m \mu_m}}$$

For each material, we can define a number called as refractive index –n

$$n = \frac{C}{v} = \sqrt{\frac{\epsilon_m \mu_m}{\epsilon_0 \mu_0}} = \sqrt{K_E K_M}$$

It is known that $K_M \cong 1$, for most materials and do not deviate from few parts in 10^4 .

TABLE 3.2 Maxwell's Relation		
Gases at 0°C and 1 atm		
Substance	$\sqrt{K_E}$	n
Air	1.000294	1.000293
Helium	1.000034	1.000036
Hydrogen	1.000131	1.000132
Carbon dioxide	1.00049	1.00045
Liquids at 20°C		
Substance	$\sqrt{K_E}$	n
Benzene	1.51	1.501
Water	8.96	1.333
Ethyl alcohol (ethanol)	5.08	1.361
Carbon tetrachloride	4.63	1.461
Carbon disulfide	5.04	1.628
Solids at room temperature		
Substance	$\sqrt{K_E}$	n
Diamond	4.06	2.419
Amber	1.6	1.55
Fused silica	1.94	1.458
Sodium chloride	2.37	1.50

But what happens when the light gets absorbed ?

We defined Absorbance

$$A(\lambda) = \epsilon l c = \log_{10} \frac{I_0}{I}$$

This means, the intensity of light passing through a material will decay exponentially $I = I_0 e^{-A}$

If we can take the length (l) outside, we rewrite the same equation as $I = I_0 e^{-\alpha t}$

α attenuation coefficient

t – thickness through which the light is propagating

The electric field in the light is taken to be $E = E_0 e^{i(kx - \omega t)}$

and $k = \frac{2\pi\tilde{n}}{\lambda}$

$$\tilde{n} = n - i\kappa$$

$$\kappa = \frac{\alpha\lambda}{4\pi}$$

The electric field in the light is taken to be $E = E_0 e^{i(kx - \omega t)}$

$$\text{and } k = \frac{2\pi\tilde{n}}{\lambda}$$

$$\tilde{n} = n - i\kappa$$

$$\kappa = \frac{\alpha\lambda}{4\pi}$$

$\tilde{n}$ is the complex refractive index

Such that the wavelength in the medium $\lambda = \frac{\lambda_0}{n}$ where, n is the refractive index

κ , the complex part, is proportional to the attenuation coefficient.

n and κ are related by Kramer-Kronig relation

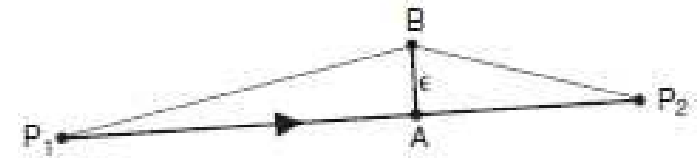
$$n(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\kappa(\omega')}{\omega' - \omega} d\omega'$$

Light propagation:

- Rectilinear propagation

Between two points light is assumed to travel in a straight line by Fermat's principle of least time

4.3.1

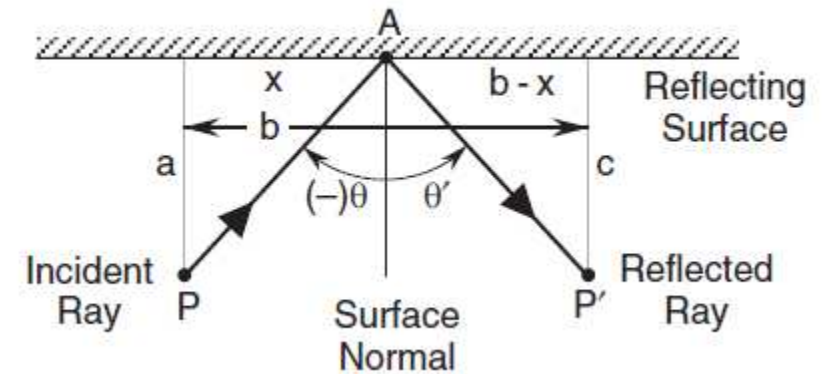


- Reflection:

At the interface between two media, the light gets reflected such that the incident ray and reflected ray are coplanar,

The angle of incidence = angle of reflection

$$\theta = -\theta'$$



Snell's law

Velocity of light $v_2 n_2 = v_1 n_1 = C$

Where C – velocity of light in vacuum
 $n = 1$ is the index of vacuum.

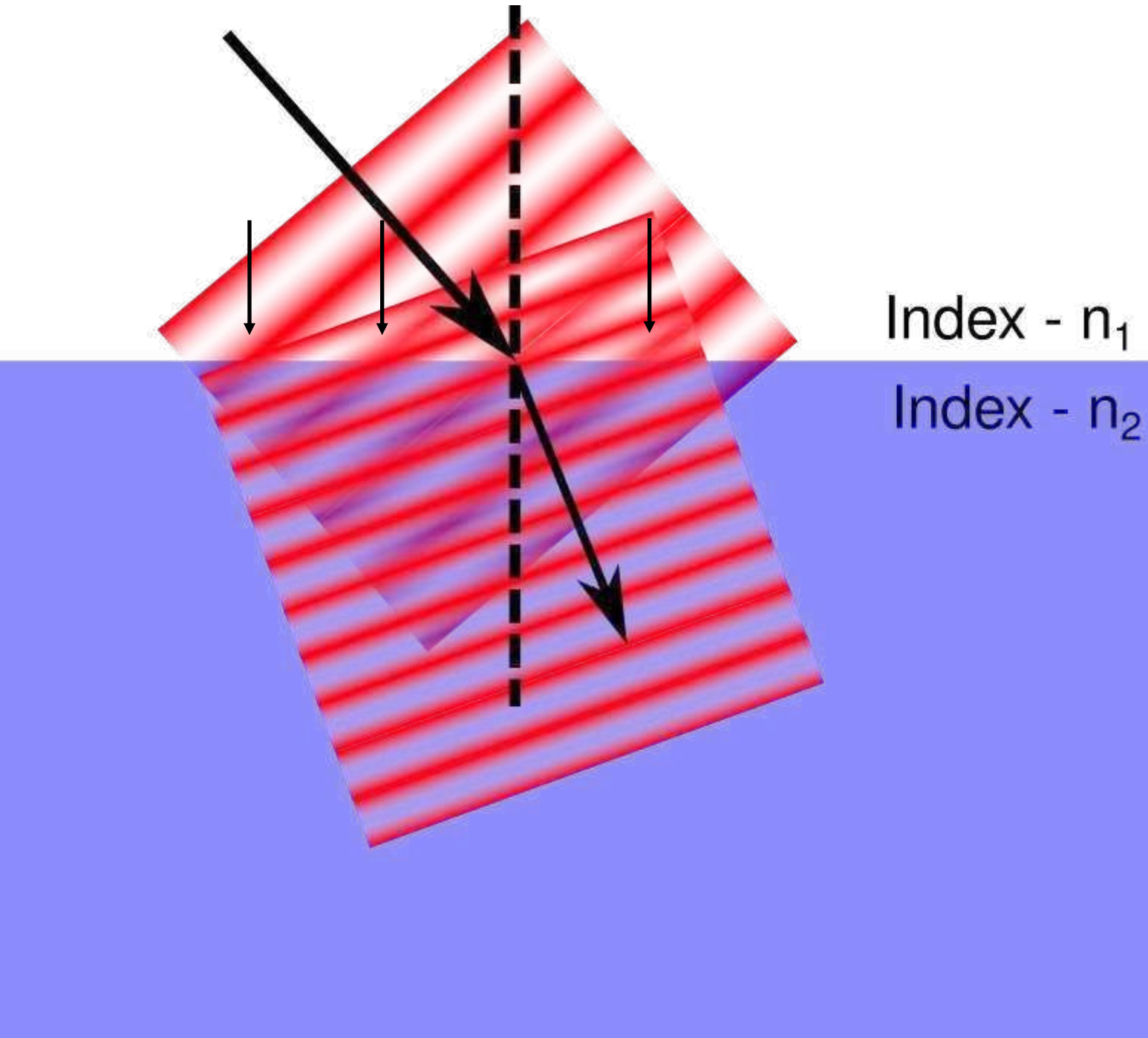
The process is elastic (No loss of energy)

So $E_1 = h\nu_1 = E_2 = h\nu_2$

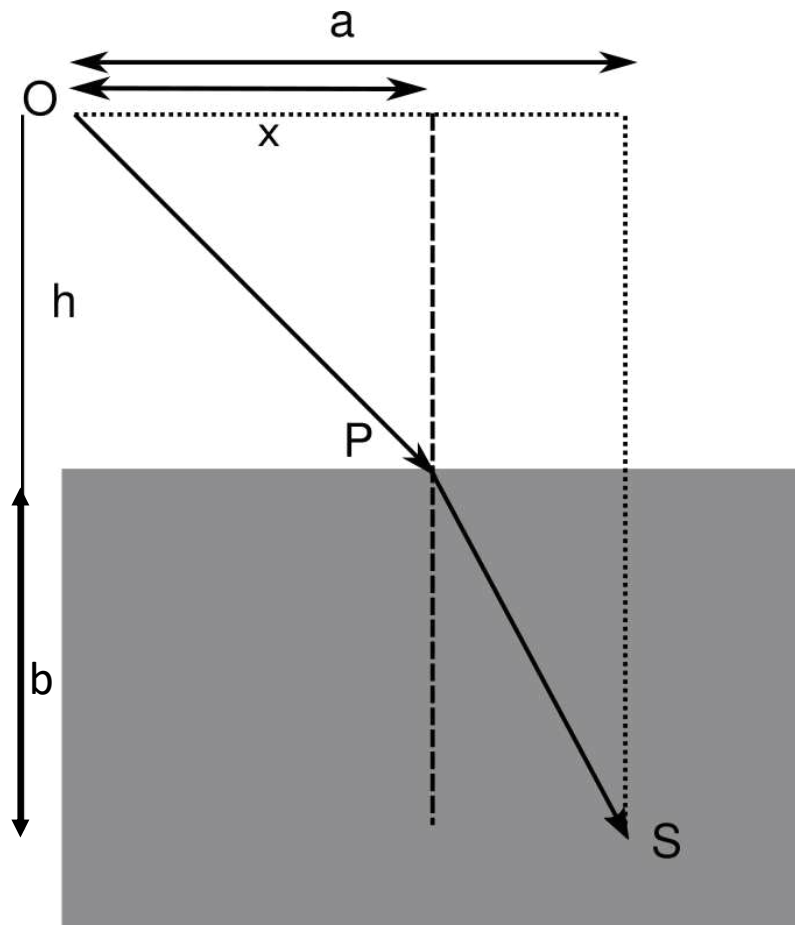
Therefore wavelength $\lambda_i = \frac{v_i}{\nu} = \frac{C}{n_i \nu}$

The light waves have to bend to accommodate for the change in the wavelength in phase velocity in different medium.

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$



Derive Snell's law from Fermat's principle of least time



Fermat's principle:

Between $O \rightarrow S$, light travels the path that requires the least time

$$T = \overline{OP} + \overline{PS}$$

$$T = \frac{\sqrt{h^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (a - x)^2}}{v_2}$$

Minima can be obtained when $\frac{dT}{dx} = 0$ and we can arrive at the Snell's law.

$$\frac{dT}{dx} = \frac{x}{v_1 \sqrt{h^2 + x^2}} - \frac{a - x}{v_2 \sqrt{b^2 + (a - x)^2}} = 0$$

$$\frac{\sin \theta_i}{v_1} = \frac{\sin \theta_2}{v_2}$$

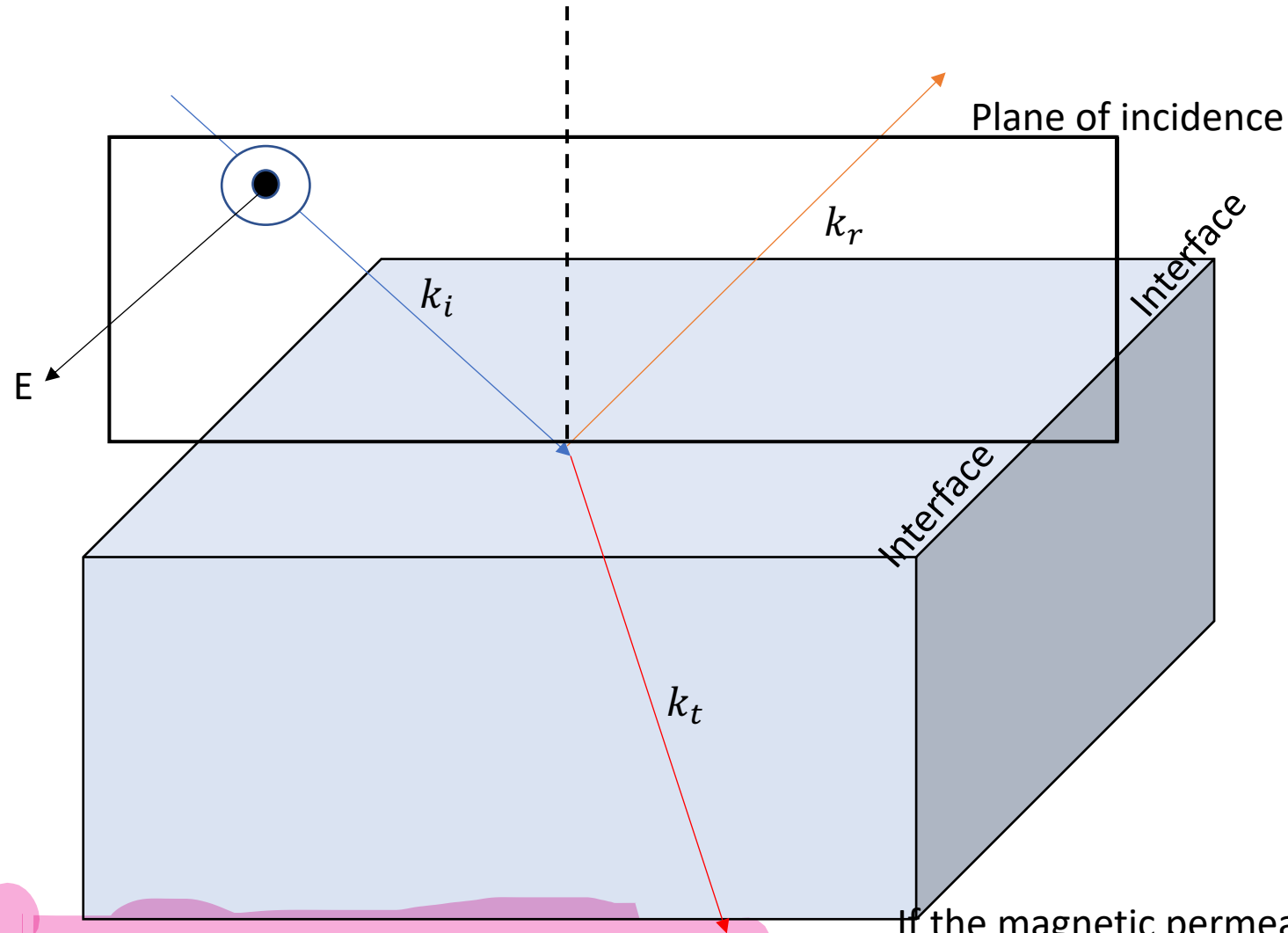
Intensities

Plane of incidence carries both the medium

Assume incident radiation is

$$E_i = E_{0i} \cos \vec{k}_i \cdot \vec{r} - \omega_i t$$

Electric field normal to the plane of incidence



If the magnetic permeabilities does not change

$$r_{\perp} = \frac{E_{0r}}{E_{0i}} = \frac{\left(\frac{n_i}{\mu_i} \cos \theta_i - \frac{n_t}{\mu_t} \cos \theta_t \right)}{\frac{n_i}{\mu_i} \cos \theta_i + \frac{n_t}{\mu_t} \cos \theta_t}$$

$$r_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}$$

When you have incidence with Electric field perpendicular to plane of incidence

$$r_{\perp} = \frac{E_{0r}}{E_{0i}} = \frac{\left(\frac{n_i}{\mu_i} \cos \theta_i - \frac{n_t}{\mu_t} \cos \theta_t\right)}{\frac{n_i}{\mu_i} \cos \theta_i + \frac{n_t}{\mu_t} \cos \theta_t}$$

$$T_{\perp} = \frac{E_{0r}}{E_{0i}} = \frac{\frac{2n_i}{\mu_i} \cos \theta_i}{\frac{n_i}{\mu_i} \cos \theta_i + \frac{n_t}{\mu_t} \cos \theta_t}$$

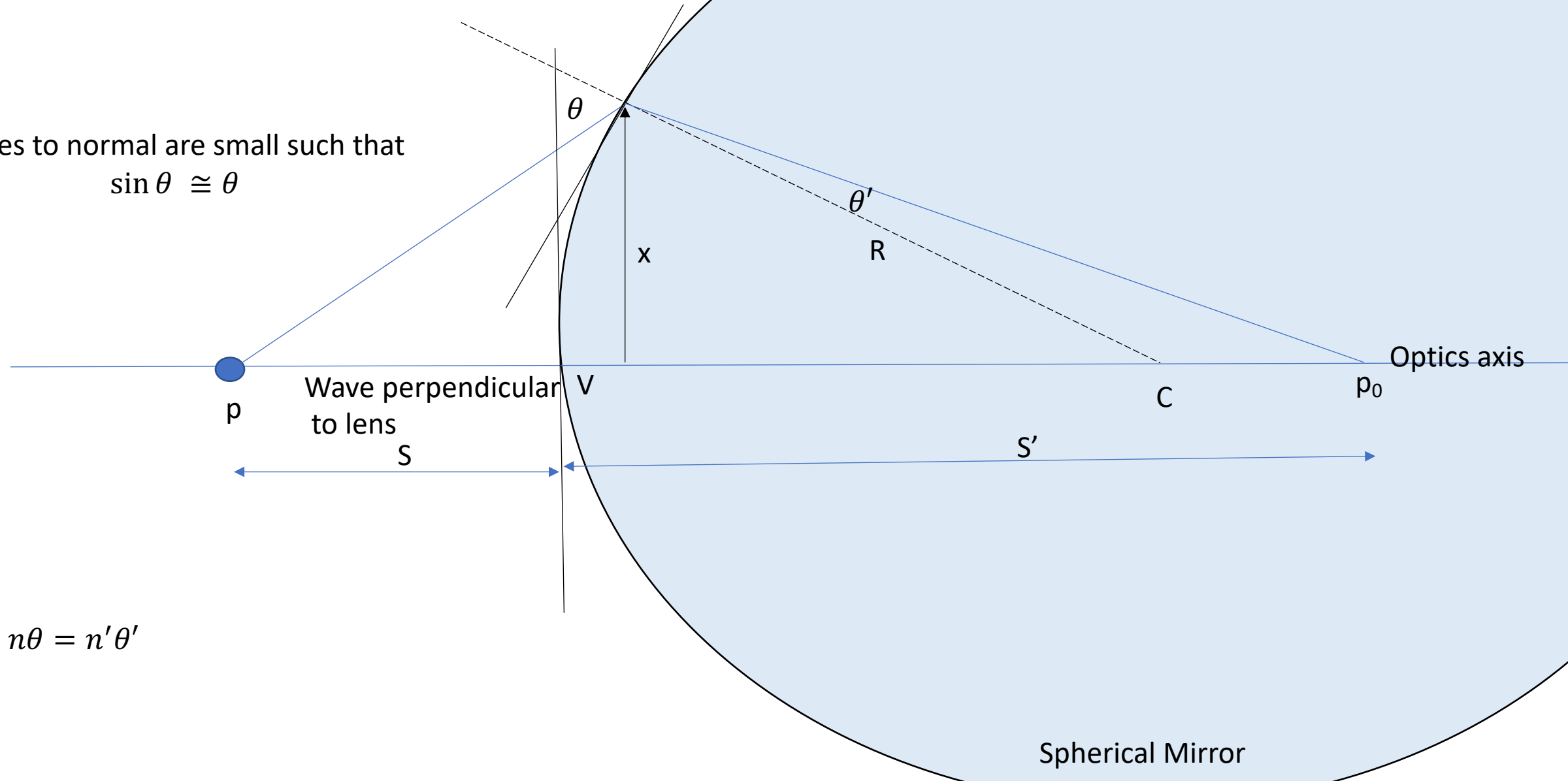
When you have incidence with Electric field parallel to plane of incidence

$$r_{\parallel} = \frac{E_{0r}}{E_{0i}} = \frac{\left(\frac{n_t}{\mu_t} \cos \theta_i - \frac{n_i}{\mu_i} \cos \theta_t\right)}{\frac{n_i}{\mu_i} \cos \theta_t + \frac{n_t}{\mu_t} \cos \theta_i}$$

$$r_{\parallel} = \frac{E_{0r}}{E_{0i}} = \frac{\left(\frac{2n_i}{\mu_i} \cos \theta_i\right)}{\frac{n_i}{\mu_i} \cos \theta_t + \frac{n_t}{\mu_t} \cos \theta_i}$$

Geometric Optics [Gaussian Optics]

Angles to normal are small such that
 $\sin \theta \cong \theta$



$$n\theta = n'\theta'$$

Rules / Sign convention

- S positive to the left of V
- S' positive to the right of V
- R positive if it is right of V

$$\frac{n'}{S'} + \frac{n}{S} = \frac{n' - n}{R}$$

Gaussian imaging equation

The equation is independent of x.

I.e. in Gaussian optics,

All rays, independent of X, will meet at the Optics axis at point P₀

$$\frac{n'}{S'} + \frac{n}{S} = \frac{n' - n}{R}$$

First focal length/Object focal length:
 object if imaged at ∞ , $s' \rightarrow \infty$

The object then is at a point $S = \frac{n}{n' - n} R$ also called as f_o , object focal length/first focal length

If the object is at infinity, $S \rightarrow \infty$

Then, the image is formed at a distance $S' = \frac{n'}{n' - n} R$ otherwise called as image focal length f_i , second focal length

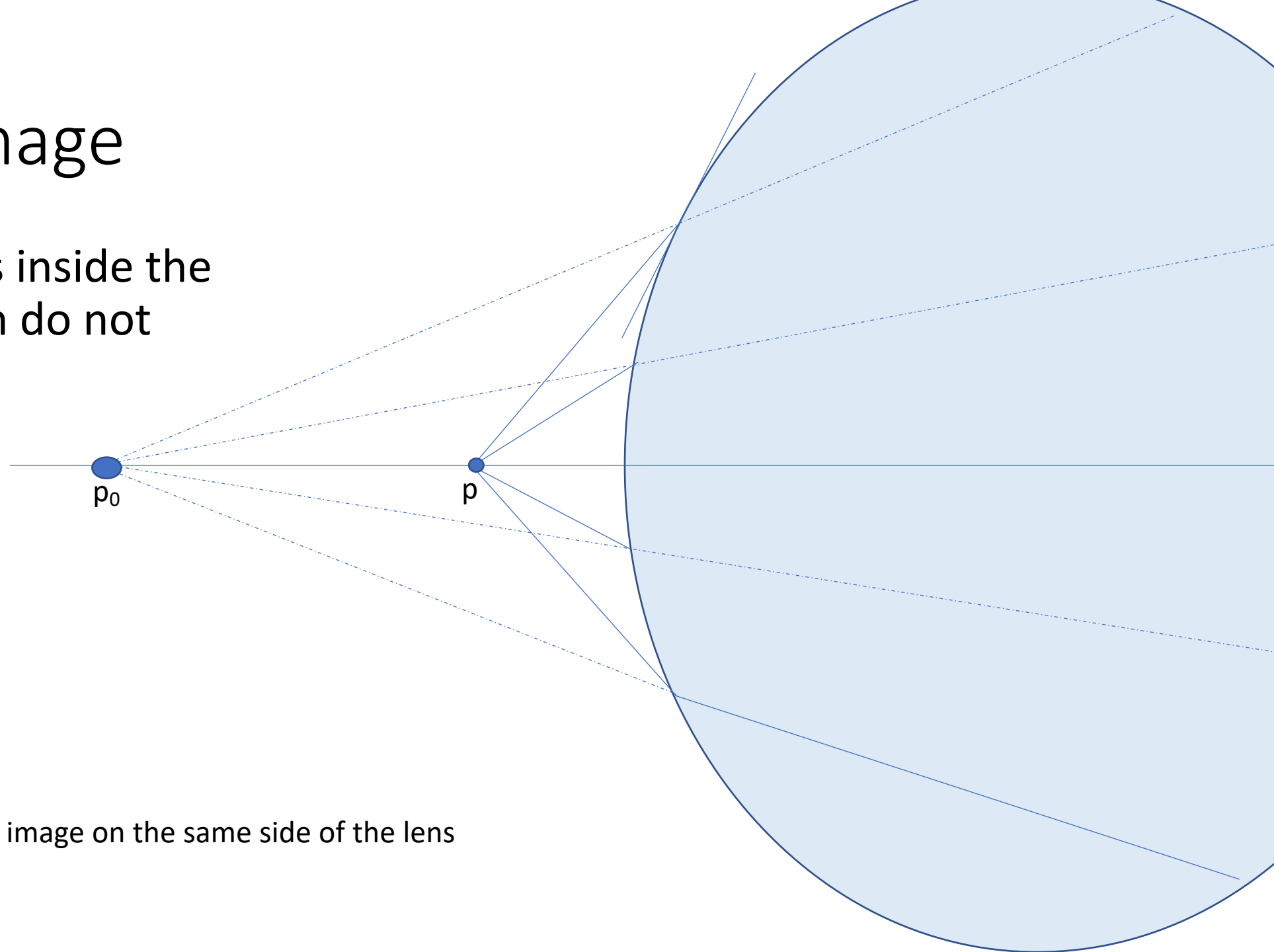
If the object $S < \frac{n}{n' - n} R$, then no positive solutions are available for the image

This means, the light diverges inside the second medium.

A virtual image can only be obtained. $S' < 0$

Virtual Image

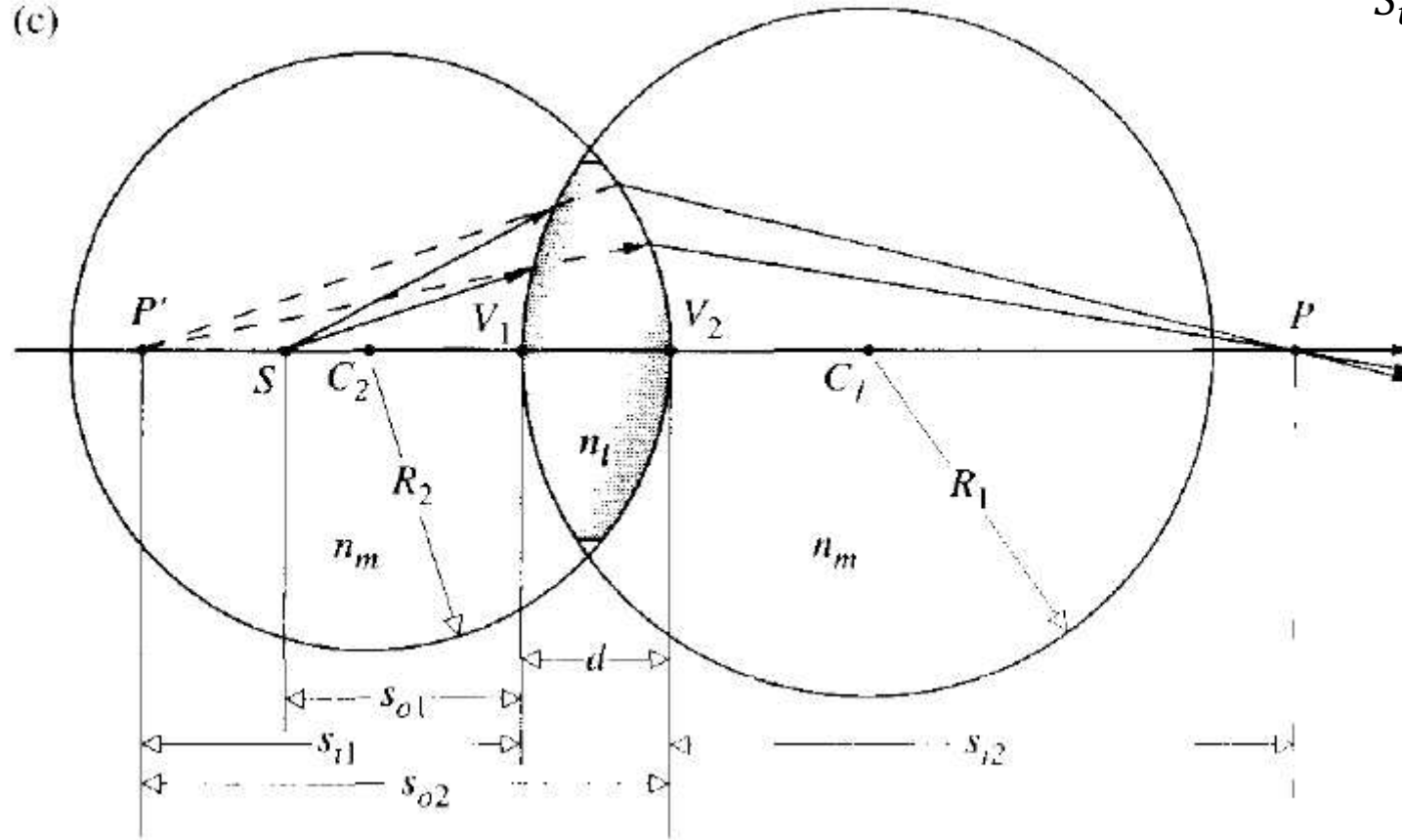
- What if the rays inside the second medium do not converge ?



You have an virtual image on the same side of the lens

Case for a thin lens

$$\begin{aligned} d &\rightarrow 0 \\ S_{01} &\rightarrow S \\ S_{t2} &\rightarrow S' \end{aligned}$$



Governing equation:

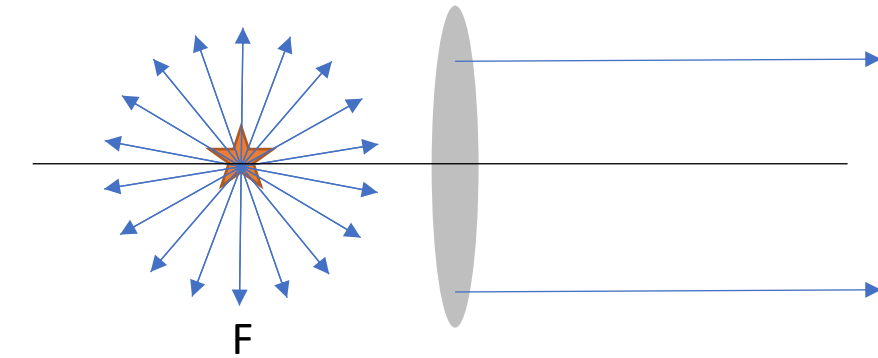
$$\frac{1}{S'} + \frac{1}{S} = (n_l - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

We call $\frac{1}{f} = \left[\frac{1}{R_1} - \frac{1}{R_2} \right] (n_l - 1)$

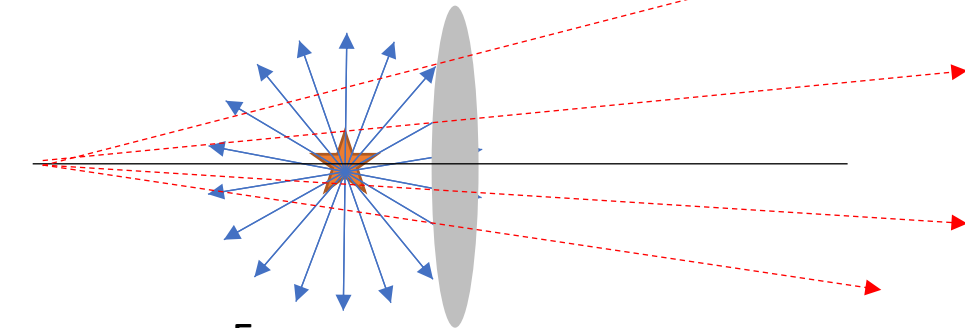
As focal length

Gives Thin-lens equation/ Lensmaker's equation $\frac{1}{S'} + \frac{1}{S} = \frac{1}{f}$

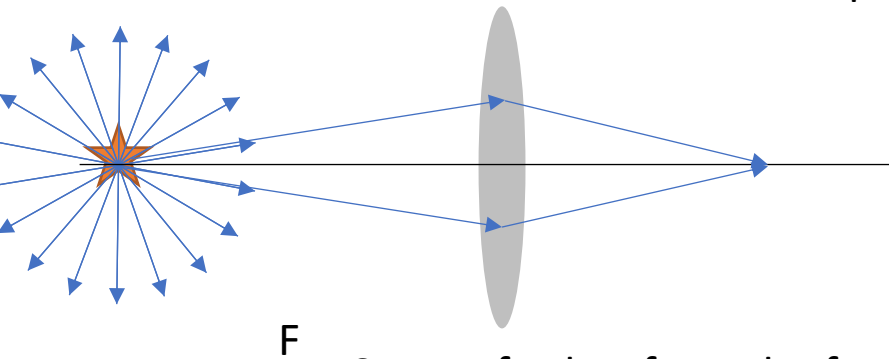
Lets us look objects away from optic axis



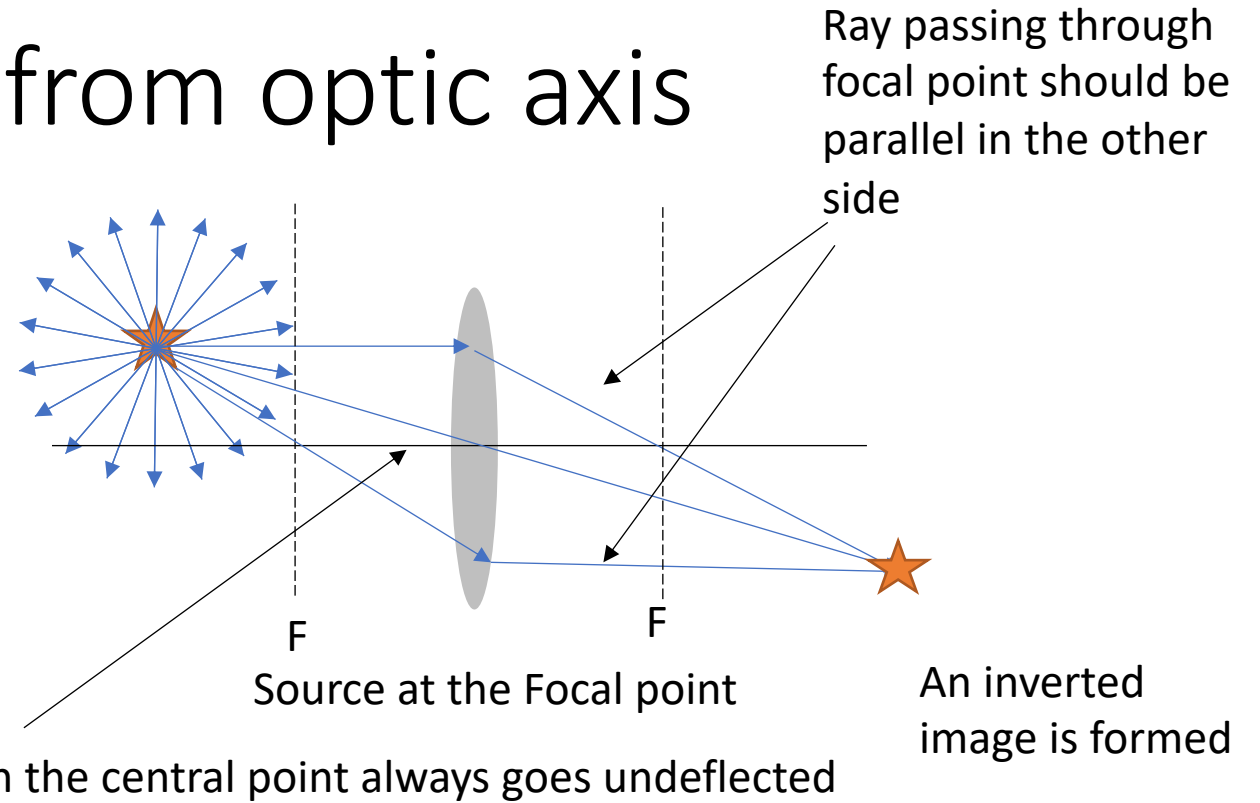
Source at the Focal point



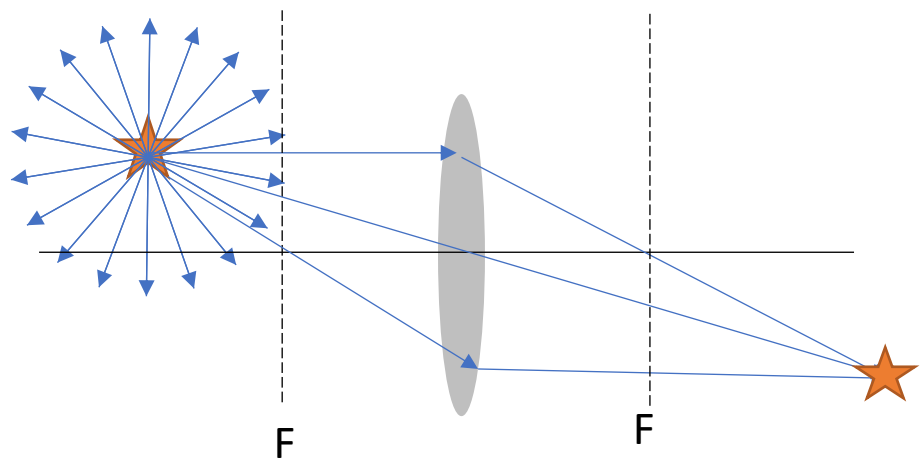
Source closer than the focal point



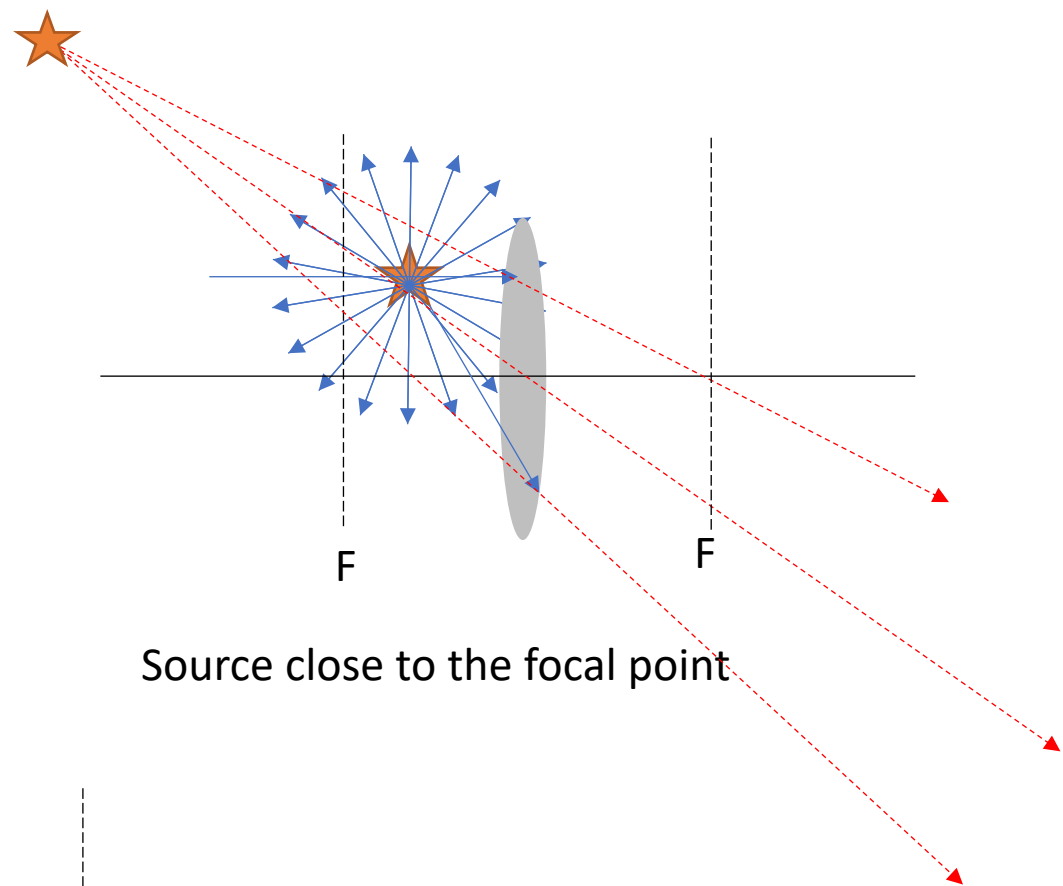
Source farther from the focal point



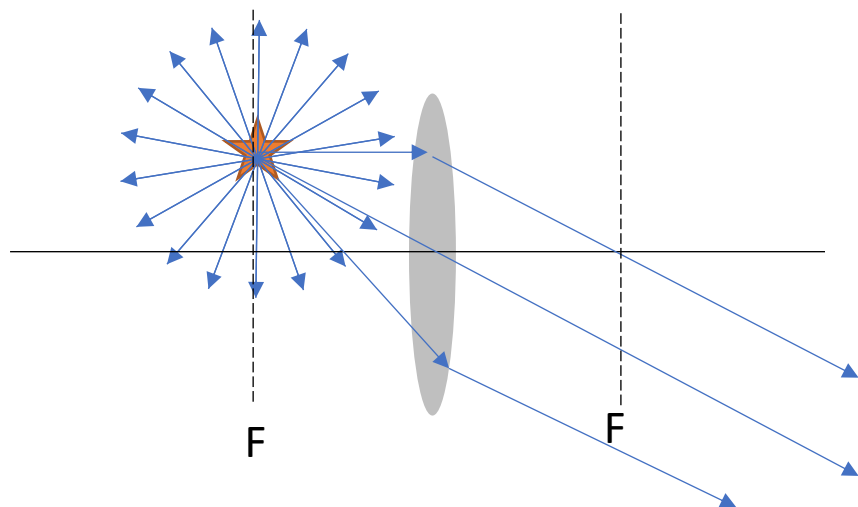
Remember, the point is called as image as at this point, the light rays again diverge seeming like a source



Source farther from the focal point



Source close to the focal point

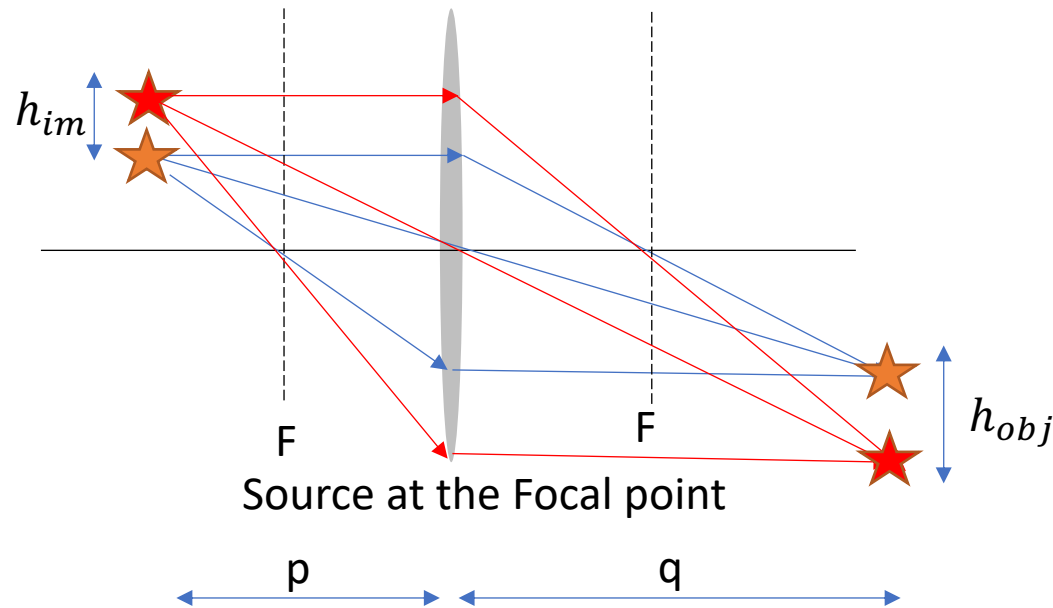


Source farther from the focal point

No image is formed.
Image at ∞

So what happens at the object at the focal point ?

Object, Image and Magnification



Magnification of this single lens is given

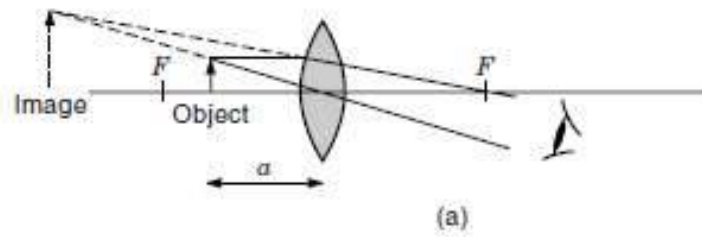
$$|m| = \frac{h_{img}}{h_{obj}}$$

Homework: Just using simple geometric arguments:
prove that magnification:

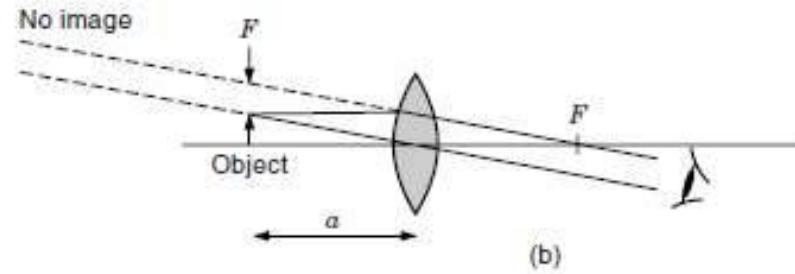
$$m = \frac{q}{p}$$

- An object – Collection of point sources.
- A plane containing point sources normal to optic axis gives an image plane normal to optic axis!.
- Each point in the object will have a corresponding image point
- The image-point is obtained by coincidence of at least two rays.
- Ray 1 – which passes through the lens center – undeflected.
- Ray 2 – a line parallel to optic axis has to pass through the focal point on the otherside.

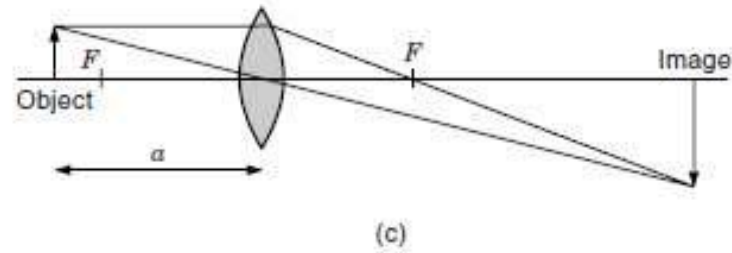
Summary



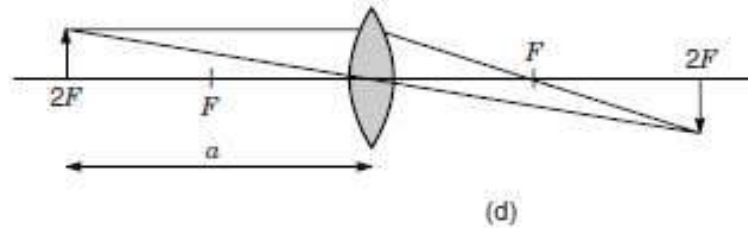
$a < f$
Virtual
magnified
image



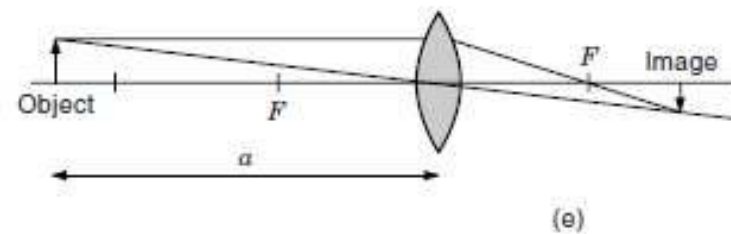
$a = f$
Image
distance ∞



$2f > a > f$
Real magnified
image



$a = 2f$
Real,
magnified = 1x



$a > 2f$
Real,
demagnified

Table of reference:

Object	Type	Location	Orientation	Relative size
$\infty > p > 2f$	Real	$f < q < 2f$	Inverted	Minified
$p = 2f$	Real	$q = 2f$	Inverted	Same size
$f < p < 2f$	Real	$\infty > q > 2f$	Inverted	Magnified
$p = f$		$\pm\infty$		
$p < f$	Virtual	$q > p$	Erect	Magnified
Anywhere	Virtual	$q < f, p > q$	Erect	Minified

$$\text{Magnification } |M| = \frac{q}{p} = \left(\frac{q}{f} - 1 \right)$$

For virtual image

$$\text{Magnification } |M| = \frac{q}{p} = \left(\frac{q}{f} + 1 \right)$$

y_i is negative by convention due to the downward direction of the image

Triangles AOF_i and $P_1P_0F_i$ are similar

$$\text{Therefore: } \frac{y_0}{y_i} = \frac{f_i}{q - f_i}$$

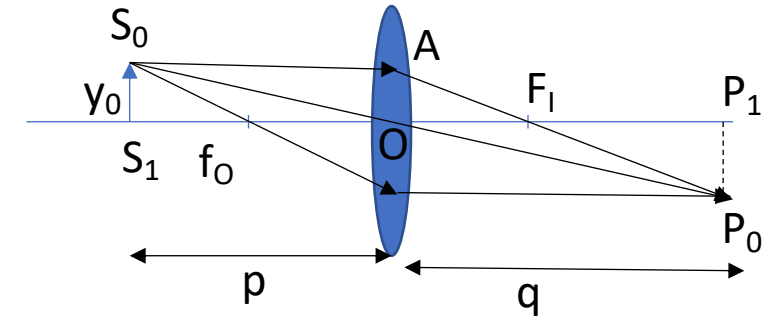
From the similarity of S_0S_1O and OP_1P_0

$$\frac{y_0}{y_1} = \frac{p}{q}$$

Also from the thin lens equation $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$

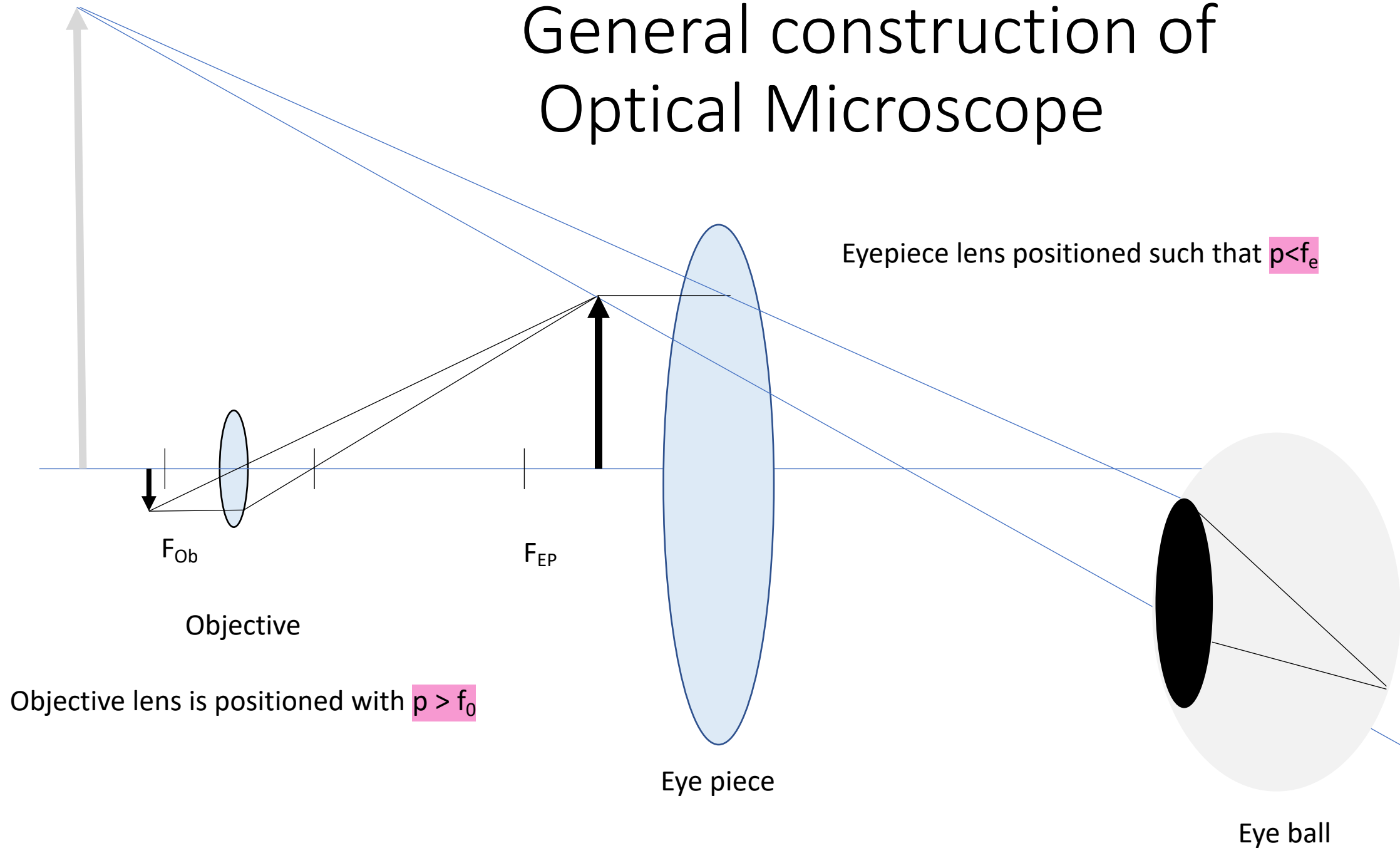
If we denote $x_0 = p - f$ and $x_i = q - f$

We can obtain the Newton's equations for magnification: $M = \frac{y_i}{y_o} = \left(-\frac{q}{p}\right) = -\frac{x_i}{f} = -\frac{f}{x_0}$



$$f^2 = x_0 x_i$$

General construction of Optical Microscope



Magnification in a microscope

Magnification produced by the objective lens $M_o = \frac{q}{p} = \frac{q}{f_o} - 1 = \frac{q_o - f_o}{f_o}$

Magnification of the eye-piece $M_e = \frac{q_e + f_e}{f_e}$

We call the term $q_o - f_o = t$ the tube length of the microscope.

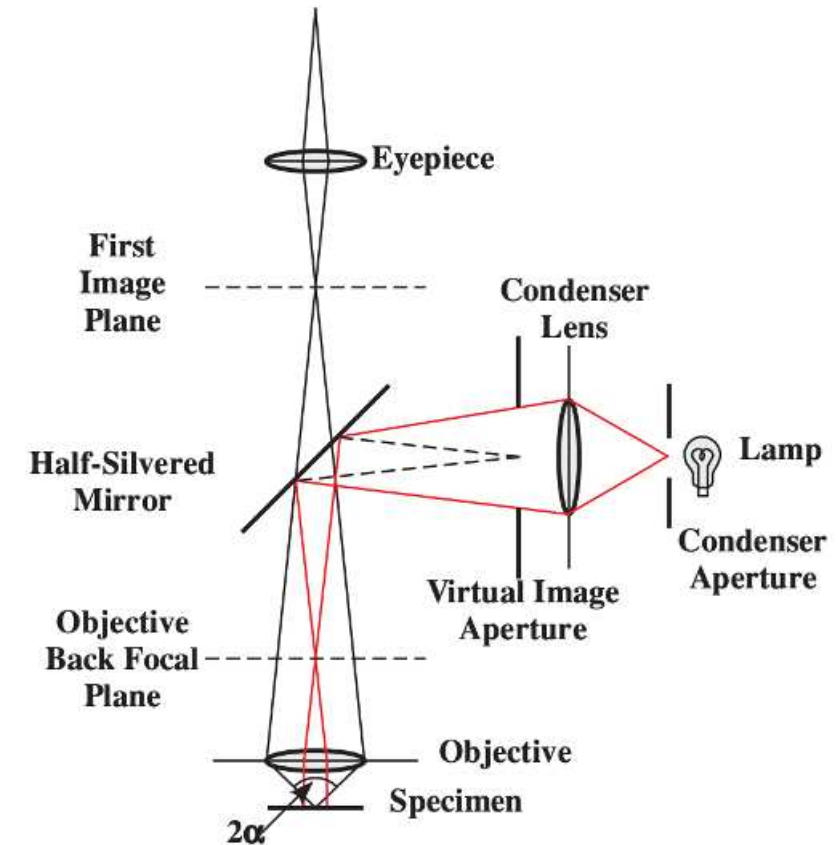
Human eye has a number called the distance of distinct vision: D

If we place the eye at the exit of eye piece, $D = f_e + q_e$

And the eye piece magnification can be written as $M_e = \frac{D}{f_e}$

• The total magnification: $M = M_o M_e = \frac{tD}{f_o f_e}$

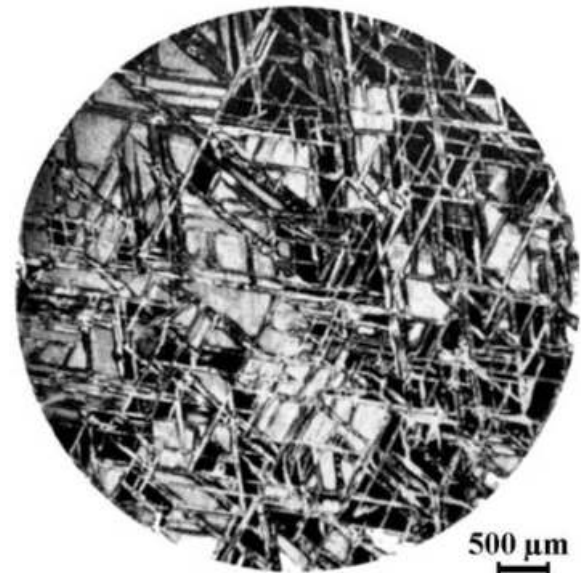
- Thus in-principle, one can achieve extra-ordinarily high magnification just with optical microscope!
- Why then are we seeing only 100X microscopes every where ? Why not higher ? Why do we need even sophisticated instruments ?



Optical microscope

- Primary tool for morphological characterization
 - In biological sciences thin slices of tissues are viewed in transmission mode
 - Contrast enhancements using fluorescent enhancement, dark field , differential interference contrast and phase contrast are employed
 - Metallurgical samples are typically investigated in reflection
 - Contrast are topological, differential absorption, reflection with optical interference

A “Widmanstatten” microstructure in steel. Specimen prepared by Henry Sorby (ca. 1864).



Examples of microscopic images

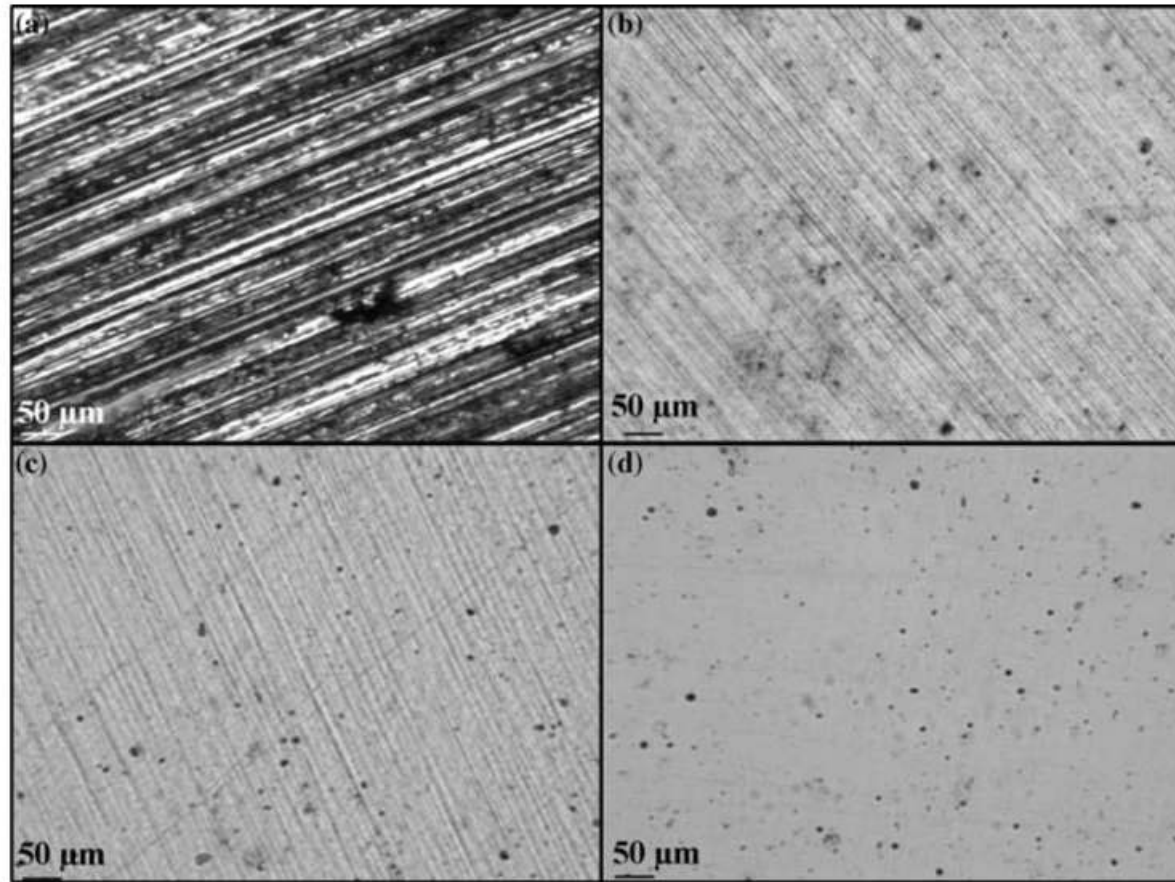


Figure 3.39 Optical micrographs of 1040 steel after polishing with a sequence of diamond grits: (a) Rough-grinding to achieve a planar surface; (b) after polishing with $6\text{ }\mu\text{m}$ diamond grit; (c) after polishing with $1\text{ }\mu\text{m}$ diamond grit; (d) after polishing with $1/4\text{ }\mu\text{m}$ diamond grit.

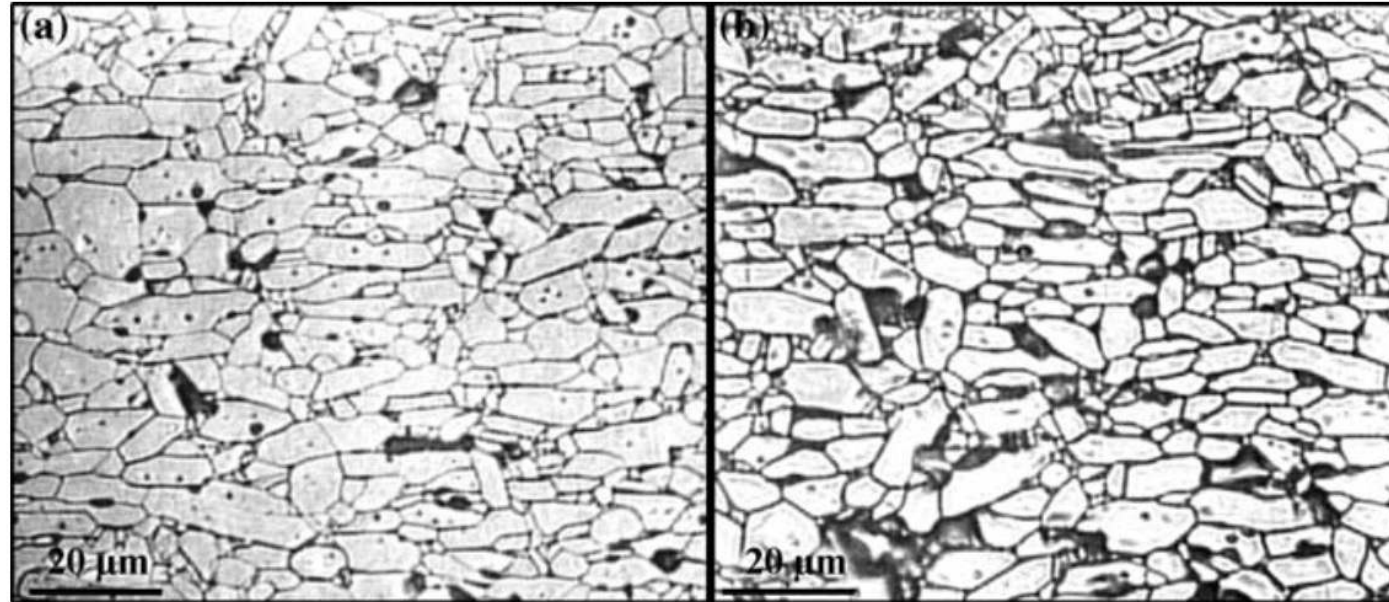
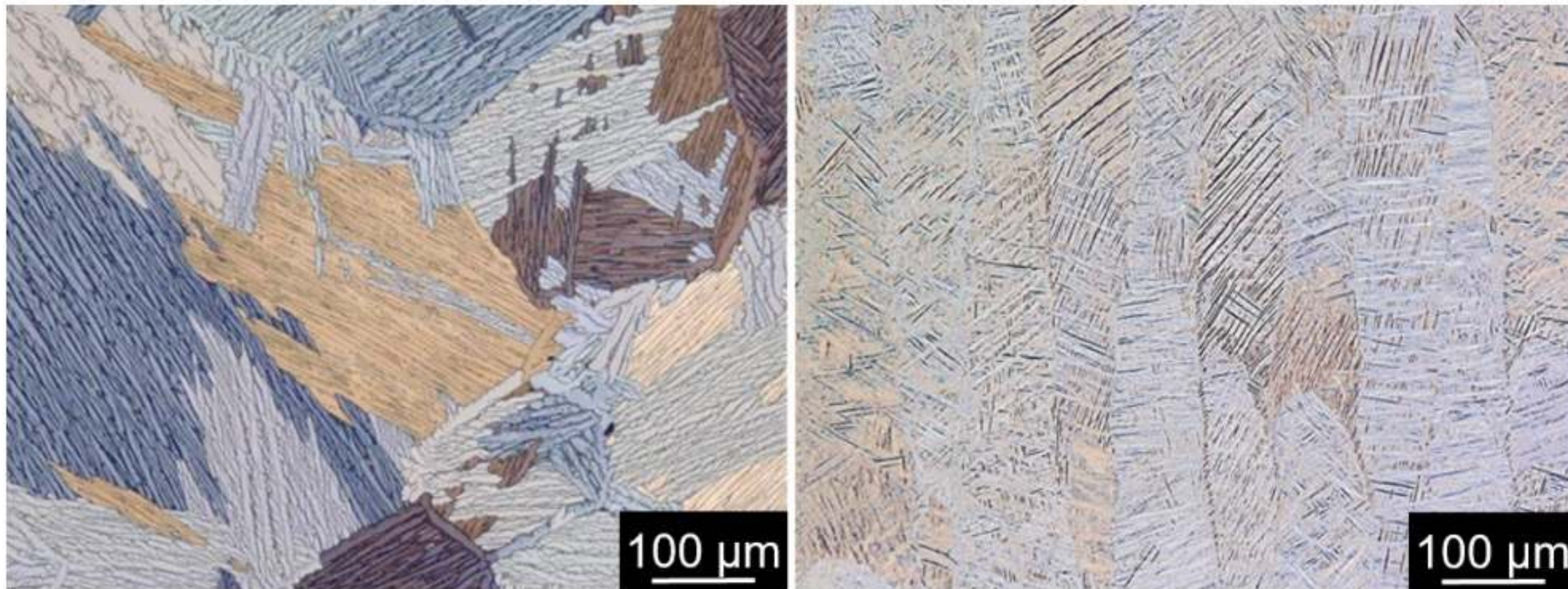


Figure 3.42 Polished alumina thermally etched at 1200°C for 30 min shows poor contrast and fails to reveal all the boundaries (a). Thermal etching for 2 h at the same temperature clearly reveals all the grain boundaries but with some loss of resolution for the finest grains (b).

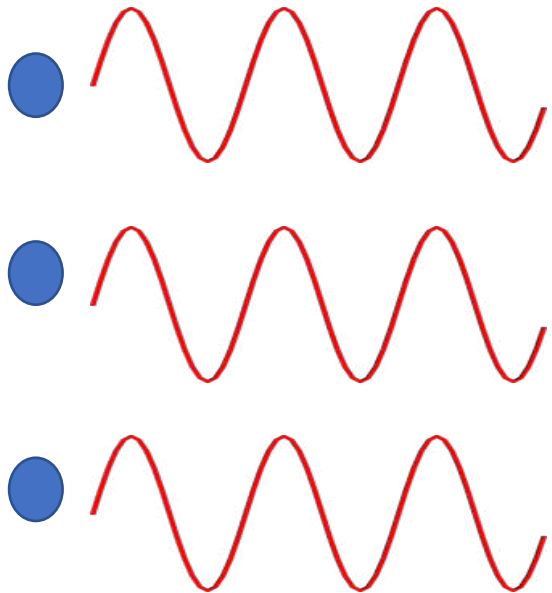


Light microscopic photograph of Ti-6Al4 prepared by KOH etch to show basket weave structure on the left and martensitic microstructure on the right.

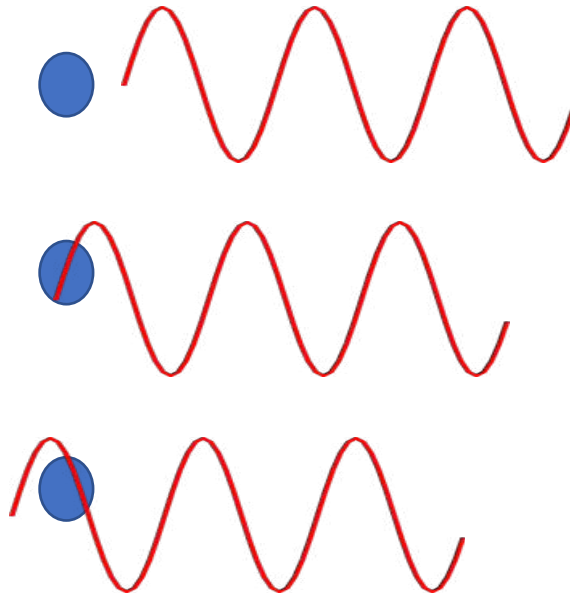
What can the microscope
can/cannot do ?

Wave properties of light:

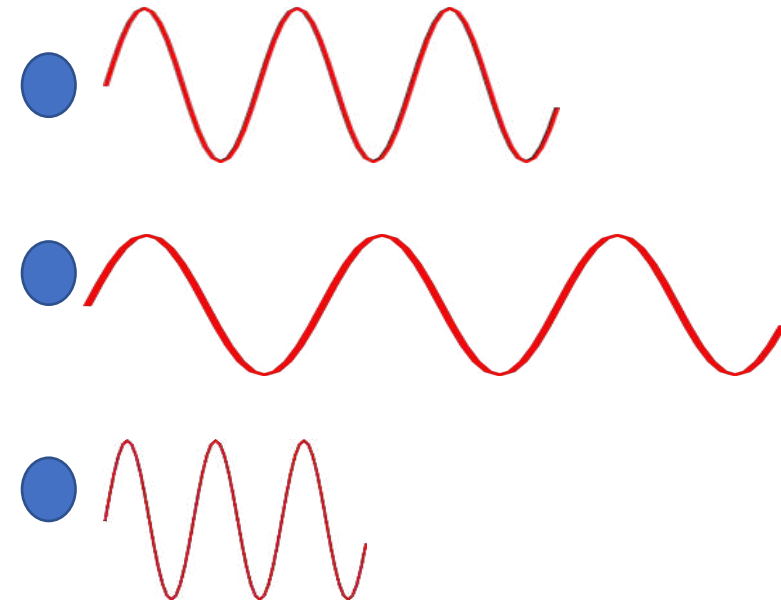
- Coherence:
 - A property of the source of light
 - Consider an object: composed of multiple point light sources
 - These point sources can have the following property:



Monochromatic and coherent

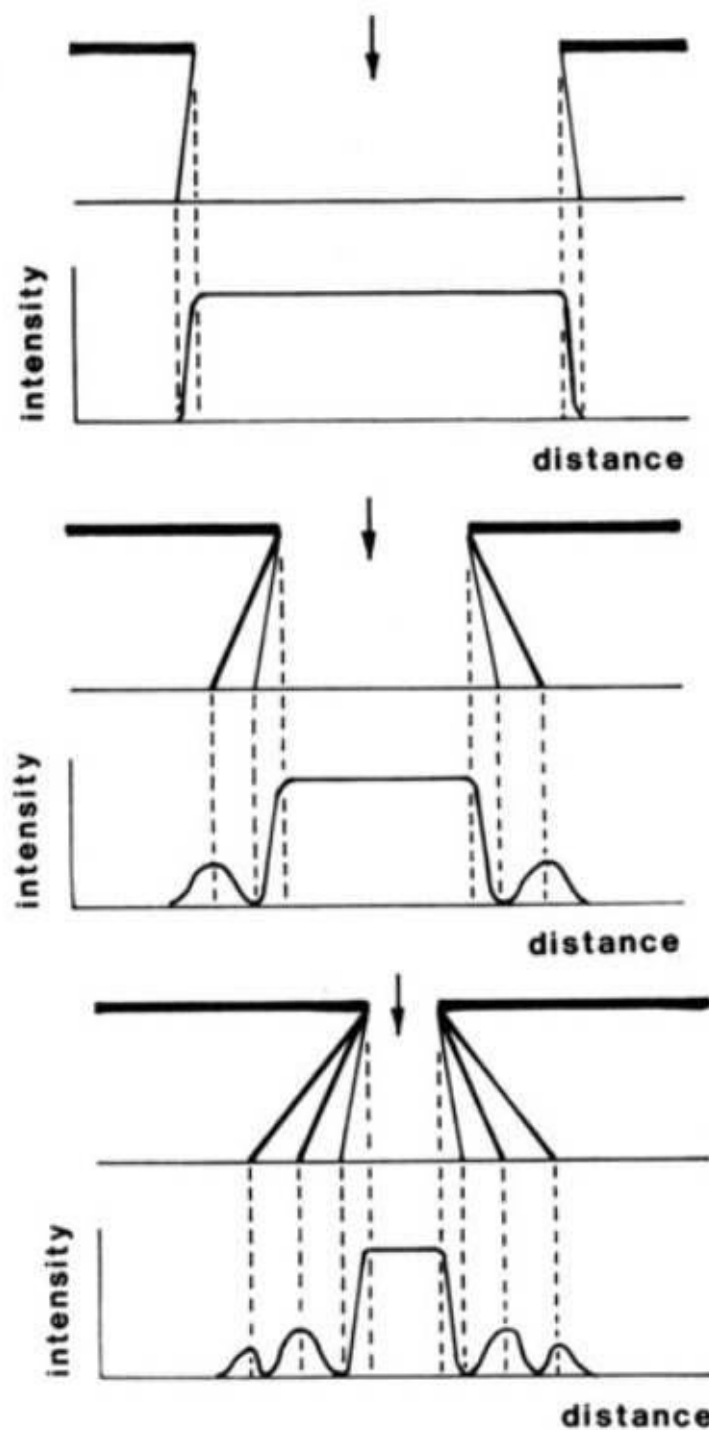
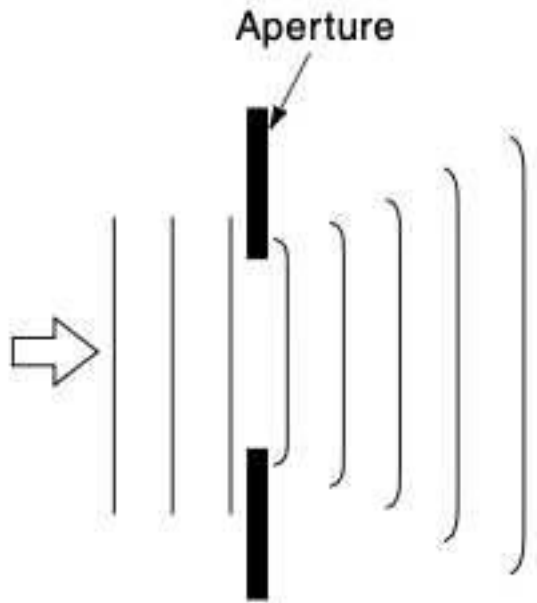


Monochromatic and non-coherent



Non-monochromatic

Diffraction



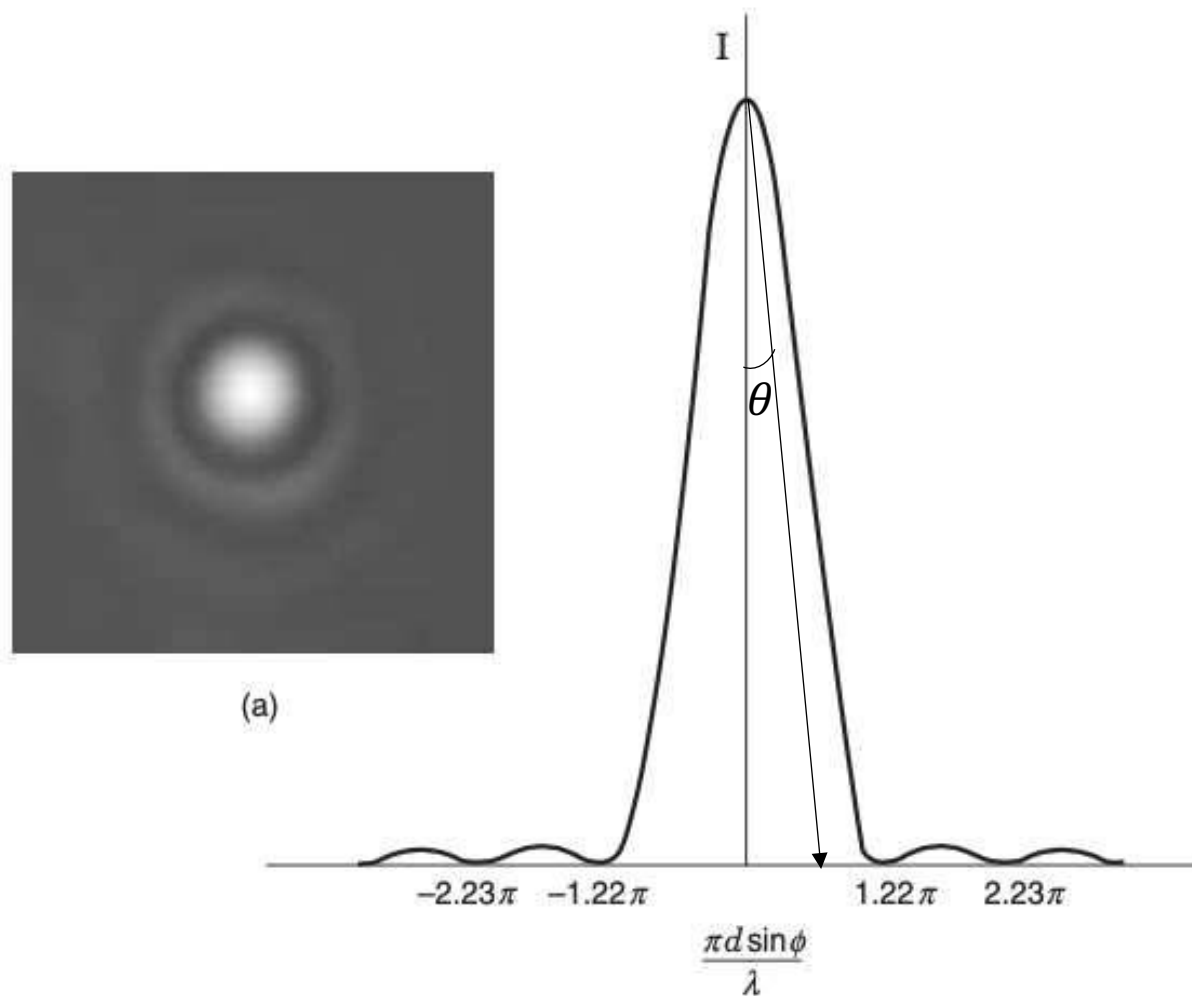
Light, being a wave, hugs an obstruction, much like a water wave.

As the width of the slit decreases, the extent of diffraction, wave bending also increases

This causes, interference between waves reaching to a point from different sources

Diffraction and interference are inevitable properties of waves

Diffraction from a circular slit



The intensity profile is an Airy disk

$$I(\theta) = I_o \left(\frac{2J_1\left(\frac{2\pi a}{\lambda} \sin \theta\right)}{\frac{2\pi a}{\lambda} \sin \theta} \right)^2$$

Where a is the radius of the aperture.

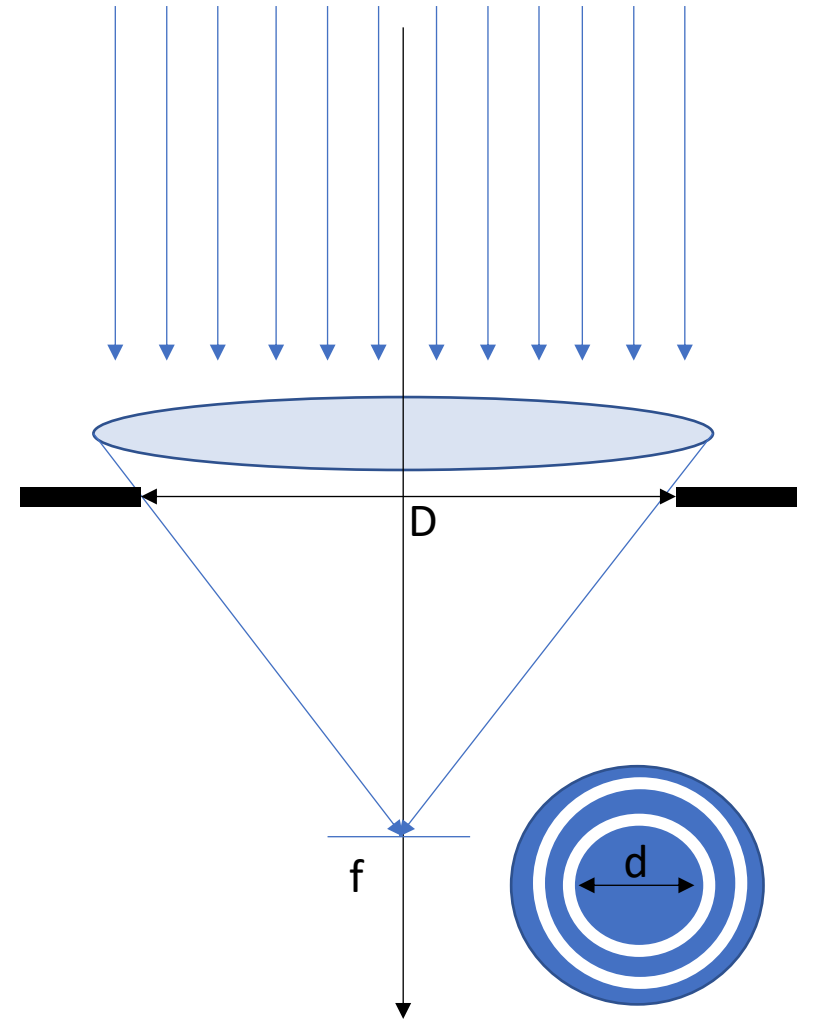
HW: plot the function for different disk openings ' a '

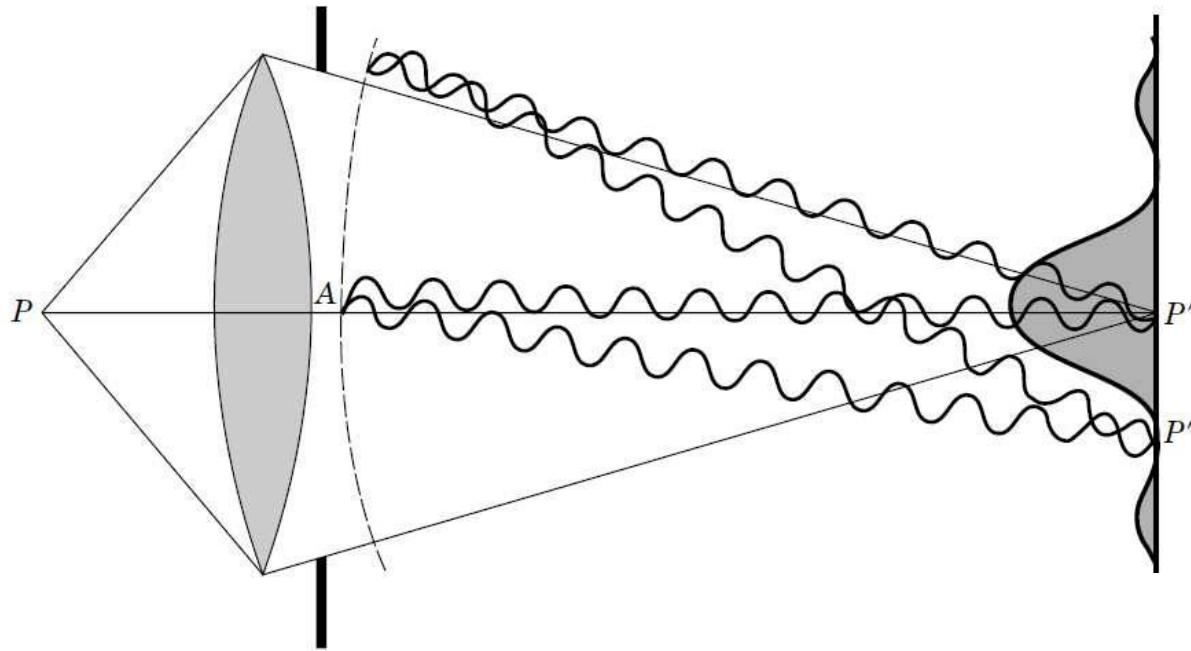
Diffraction in a lens

- Similar to the previous situation, when a collinear collection of rays are incident on a lens, it should focus the rays to the focal point as seen before!
- However, the point is not a point due to the diffraction effects just discussed. The point gets to be a disk! - Particularly an Airy disk of minimum diameter
- The size of the disk $d = 1.22 \frac{\lambda f}{D}$

$$\text{If } \sin \alpha \cong \frac{D}{2f}$$

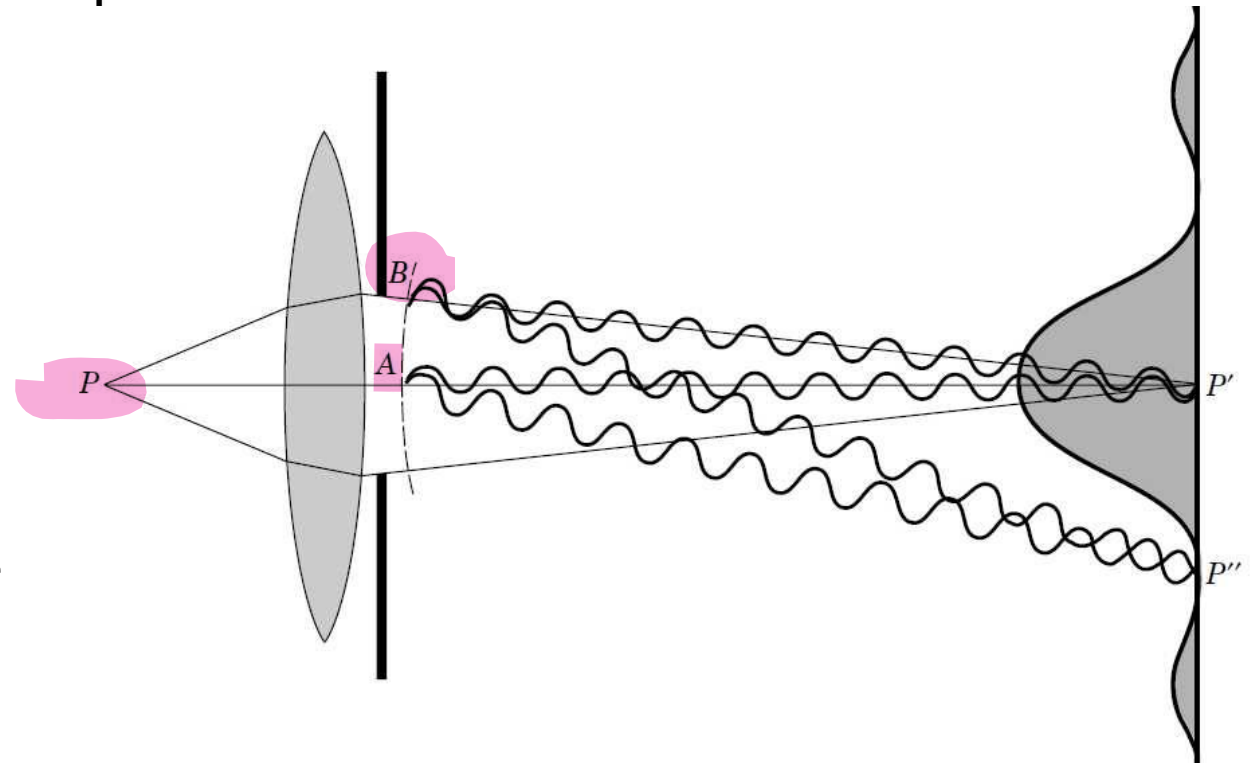
$$d = 0.61 \frac{\lambda}{\sin \alpha}$$



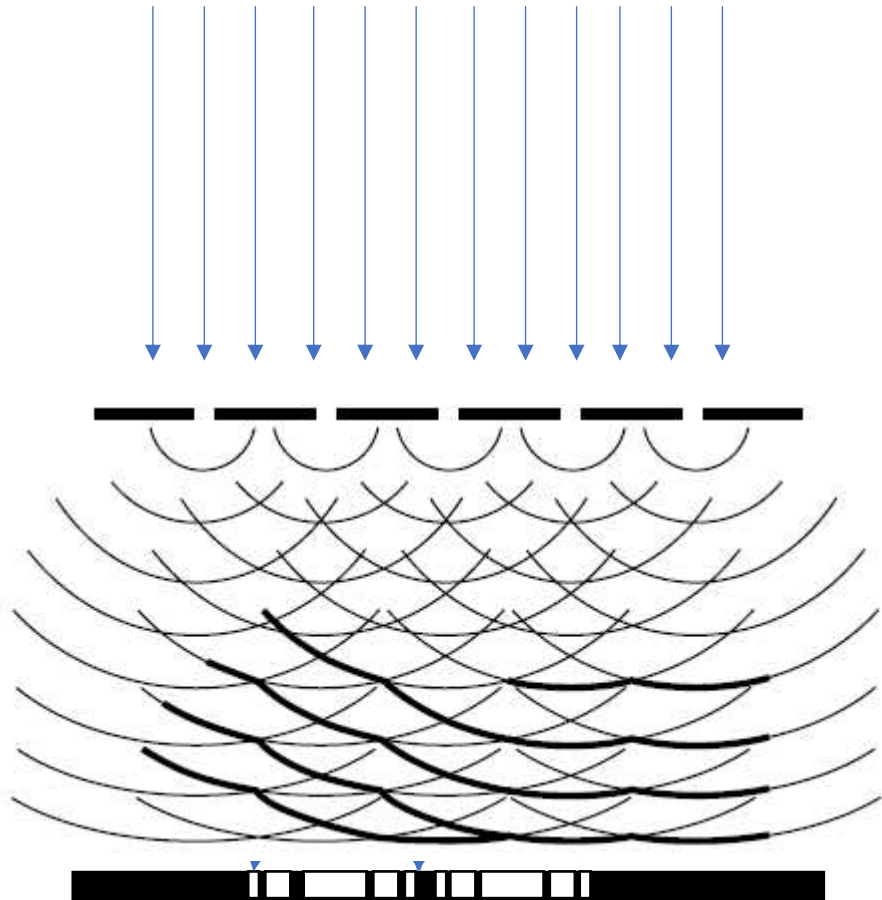


Effect of aperture opening

- As the width of aperture is reduced, number of light waves falling on a single point is reduced.
- The dots are less well defined
- It can also be looked at from the collection angle:
 - Larger the opening, larger the collection of incident radiation by the lens.

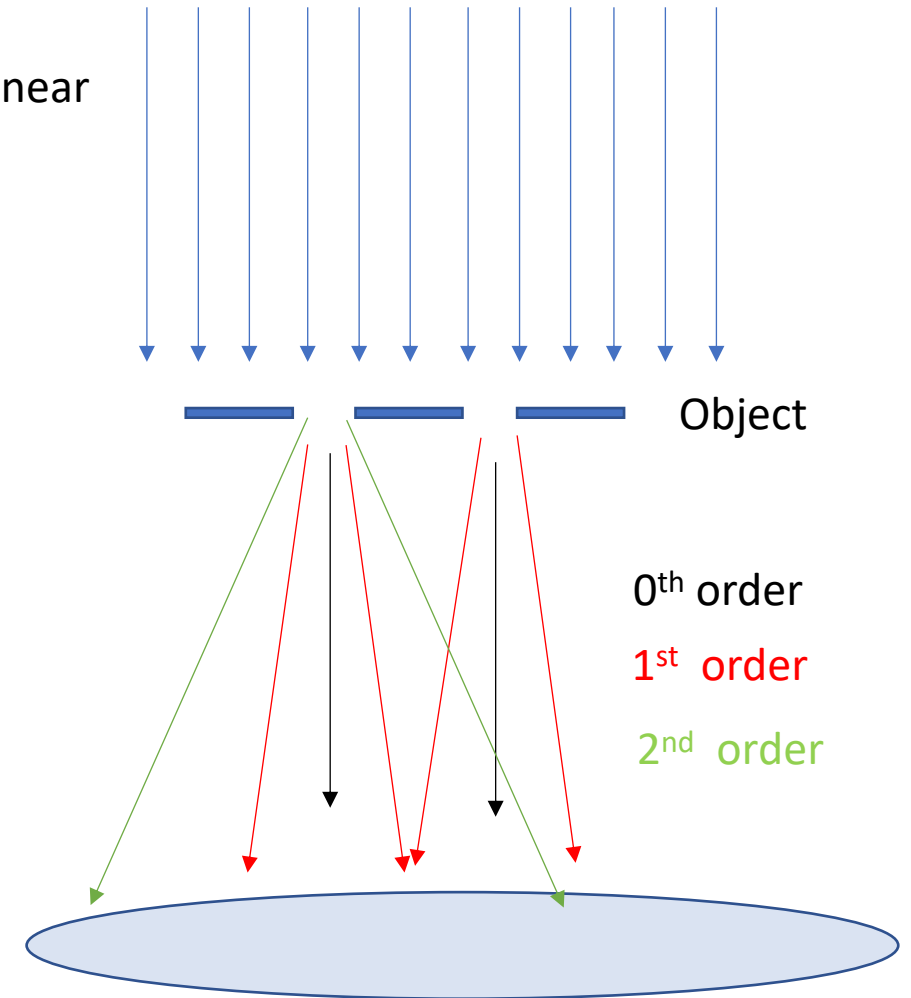


Diffraction from closely spaced particles



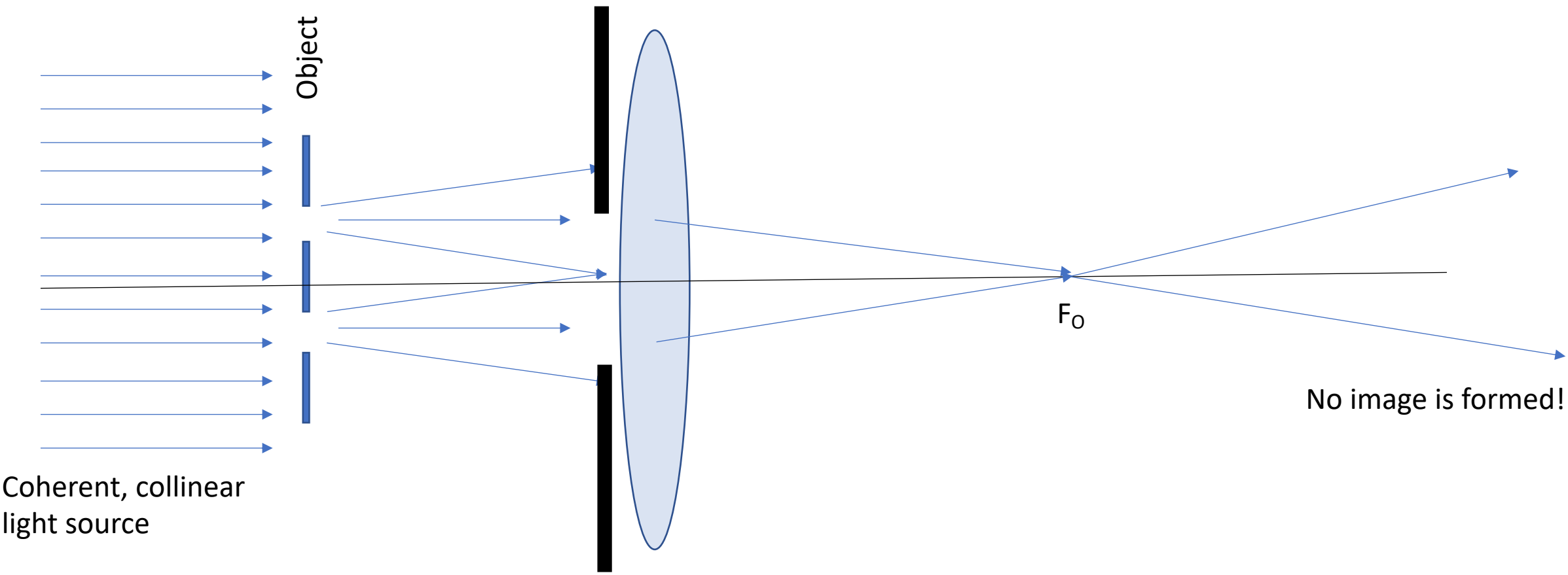
The famous double slit diffraction experiments
Photo screen

Coherent, collinear
light source



What happens when we converge such diffracted beams ?

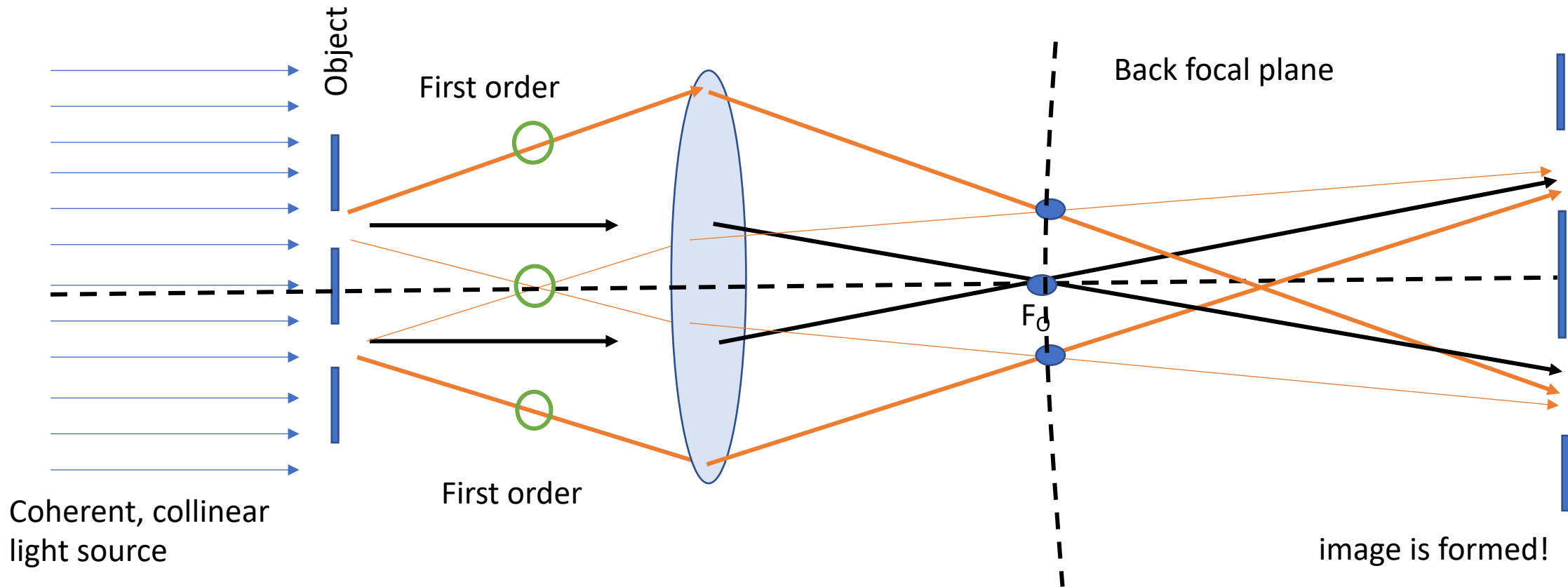
Suppose your objective lens is small/ you have a small aperture



Coherent, collinear
light source

So, if we only collect the undiffracted (not deflected) light rays, No image is formed!

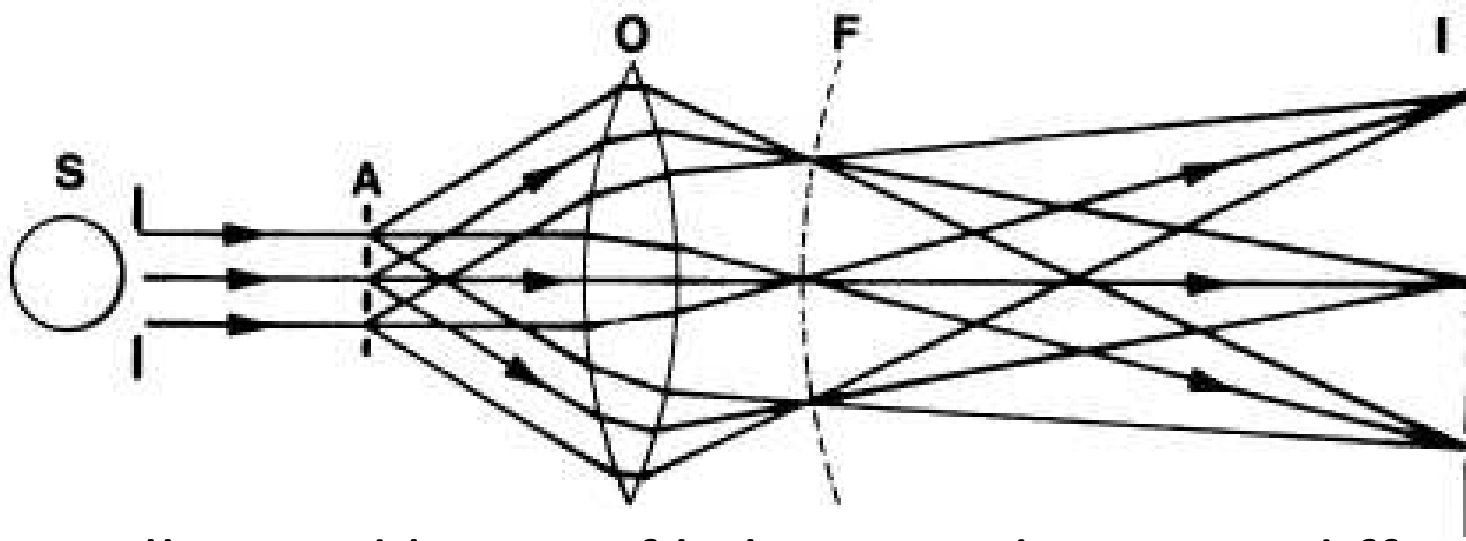
Suppose your objective lens is small/ you have a small aperture



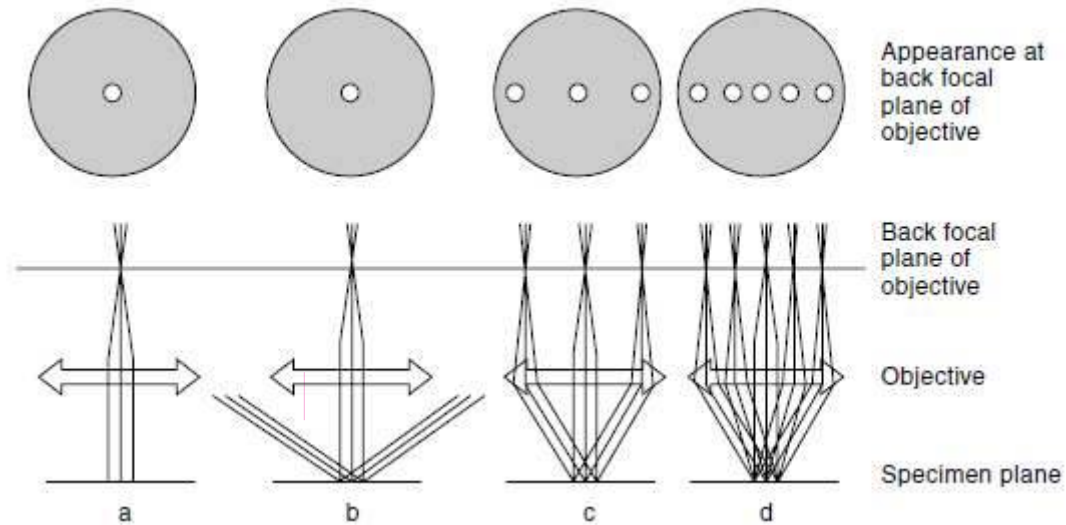
So, if we only collect the undiffracted (not deflected) light rays, No image is formed!

An image of the object is formed when atleast two diffracted beams are collected!

Properties of Image Formation



- When a collimated beam of light is incident on a diffracting object:
 - N-orders of diffracted collimated beams are formed.
 - 0th order – non-diffracted light called as bright field. Collection of the 0th order will form no image.
 - An 1st order diffracted beams are all parallel
 - Parallel beams get focused to a point – focal plane
 - 1st order beams differ by exactly 1 wave-length, hence are in phase
 - Angular separation between the diffracted beams decrease with the order
 - Angular separation between the beams increase as the object separation decreases



- The 1st order diffracted beams are all parallel
 - Parallel beams get focused to a point – focal plane
 - 1st order beams differ by exactly 1 wave-length, hence are in phase
 - They interact constructively giving corresponding bright spots in the focal plane
 - The dark areas behind the lens (focal plane) are regions of destructive interference.

If d is the gap spacing in the object, the spacing in the focal plane: $D \cong \frac{1}{d}$

Abbe's criteria

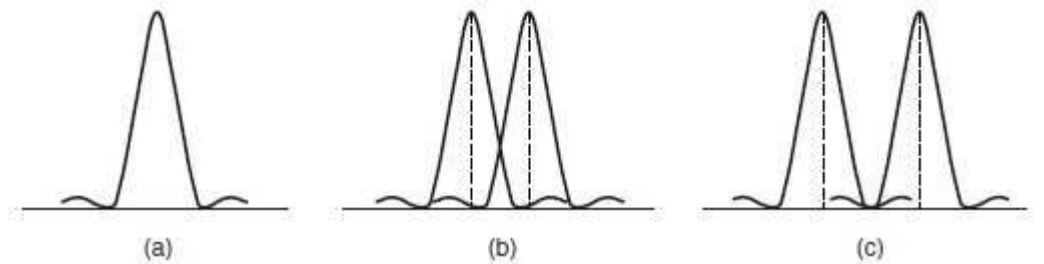
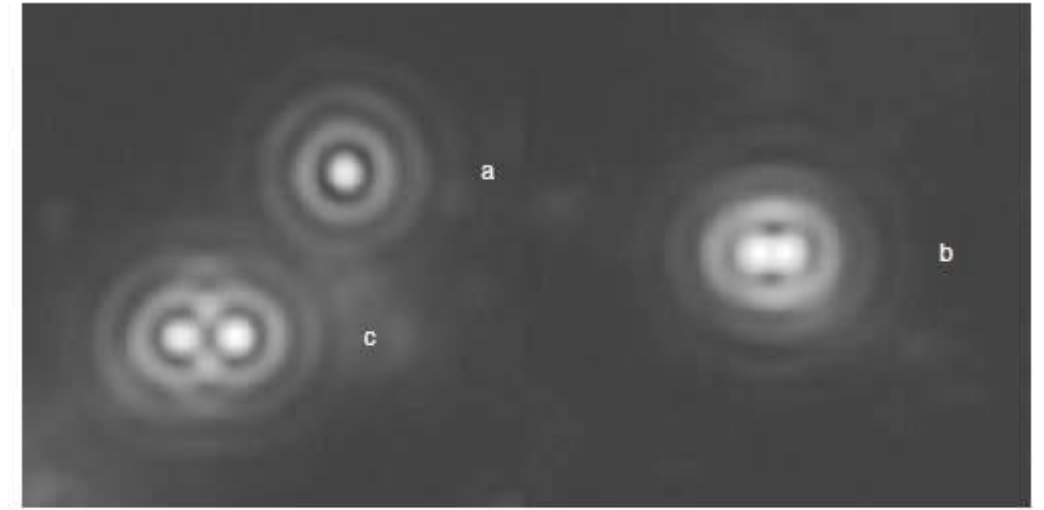


- Bright field (0^{th} order diffracted beam cannot form a distinct image)
- Collection of at least the first order forms a defined image
- Higher the number of diffracted beams captured, better is the resolution.
- Fortunately, the intensity of diffracted beams fall quickly with order, so first few orders are the most prominent
- However, smaller the particle size, larger the angle of diffraction
 - This means, even the first order diffracted line will be farther for smaller particles

If an object doesn't diffract light/ the objective lens does not collect diffracted light, no image is formed.

Spatial Resolution of a microscope

- A point in the image plane (due to converging light beams of λ) will have a minimum diameter $0.61 \frac{\lambda}{NA}$
- Two bright points in the image plane is considered to be resolved, when the peak of the second coincides with the first minima of the first source.



Remember, the limit is not about resolving the smallest particle!

It is about identifying two close lying particles are two separate particles.

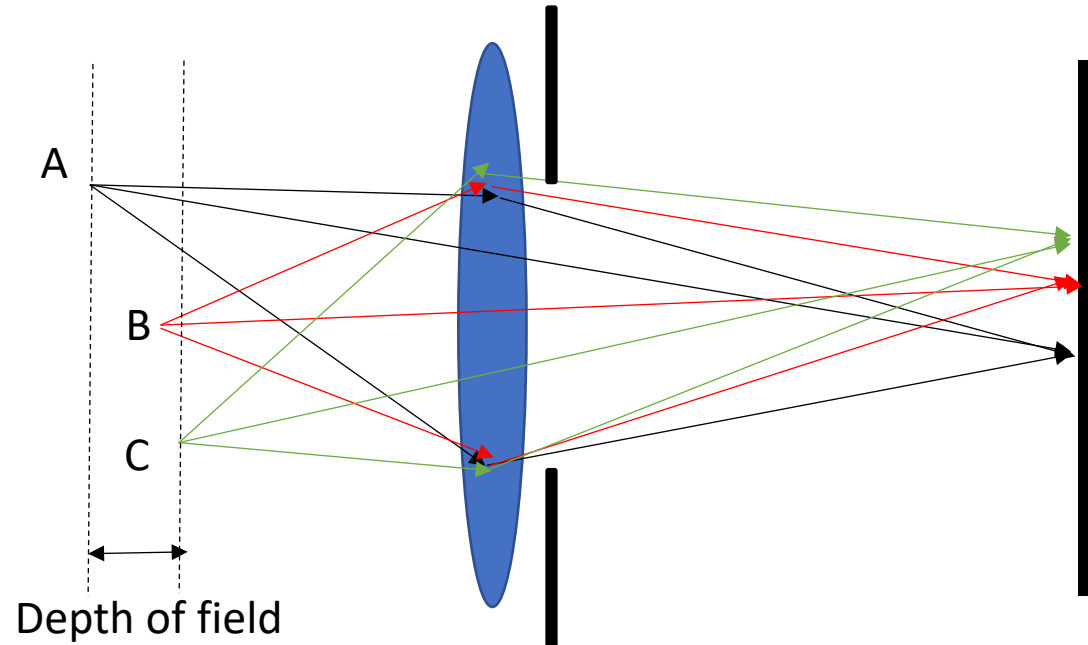
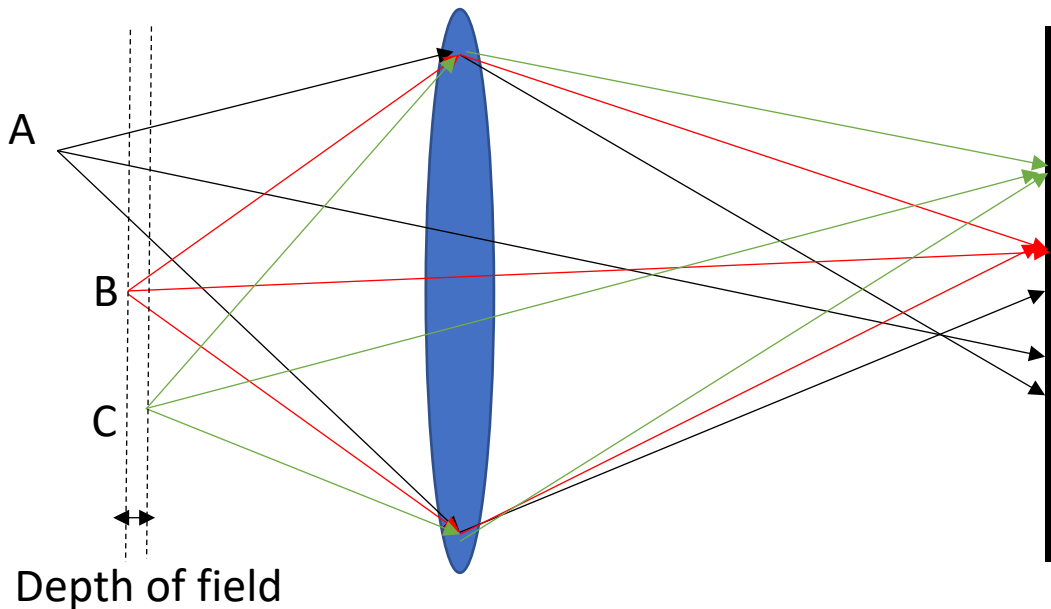
Controlling the spatial resolution.

- Spatial resolution is limited by $d \cong 0.61 \frac{\lambda}{NA}$
 - Smaller resolution is obtained by increasing NA or reducing λ
 - Typically wavelength of illumination is not largely modified because
 - A blue and lower, the optical components start absorbing
 - UV light sources are typically more expensive
 - Human eye has very low sensitivity to blue (green is the most sensitive to eye)
 - NA can be increased by
 - Choosing a higher refractive index medium (Oil immersed lenses)
 - Increasing the acceptance angle (larger lenses)

Depth of Field

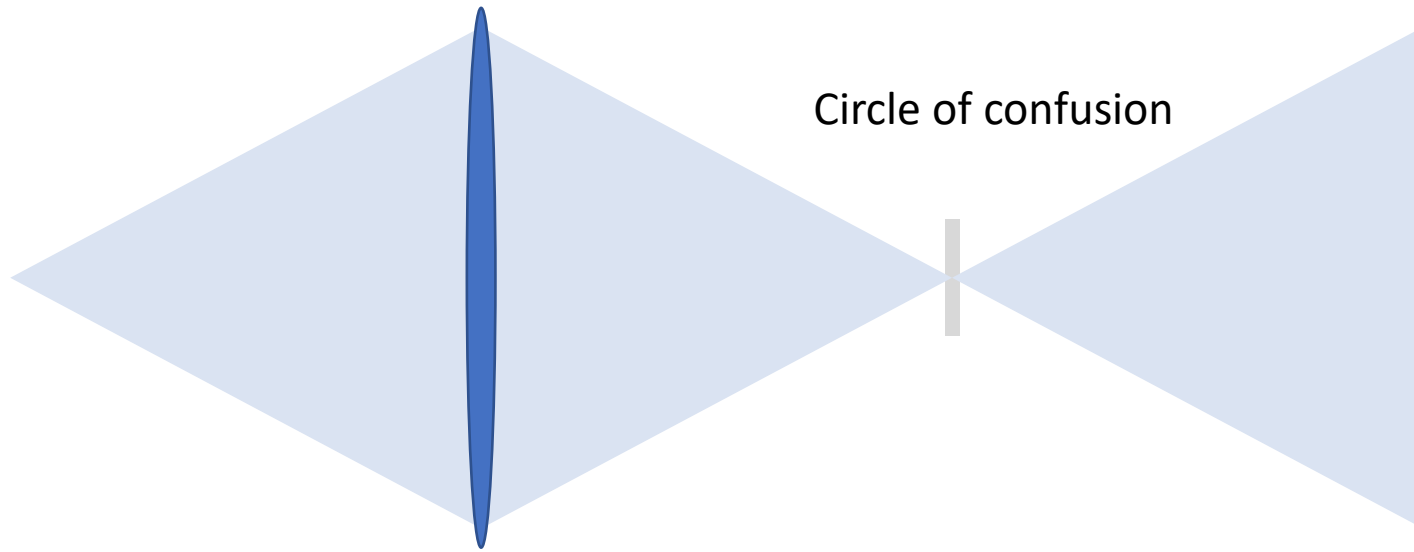
- This is the thickness of the optical section along the Z-axis within which the specimen are in focus

$$Z = \frac{n\lambda}{NA^2}$$

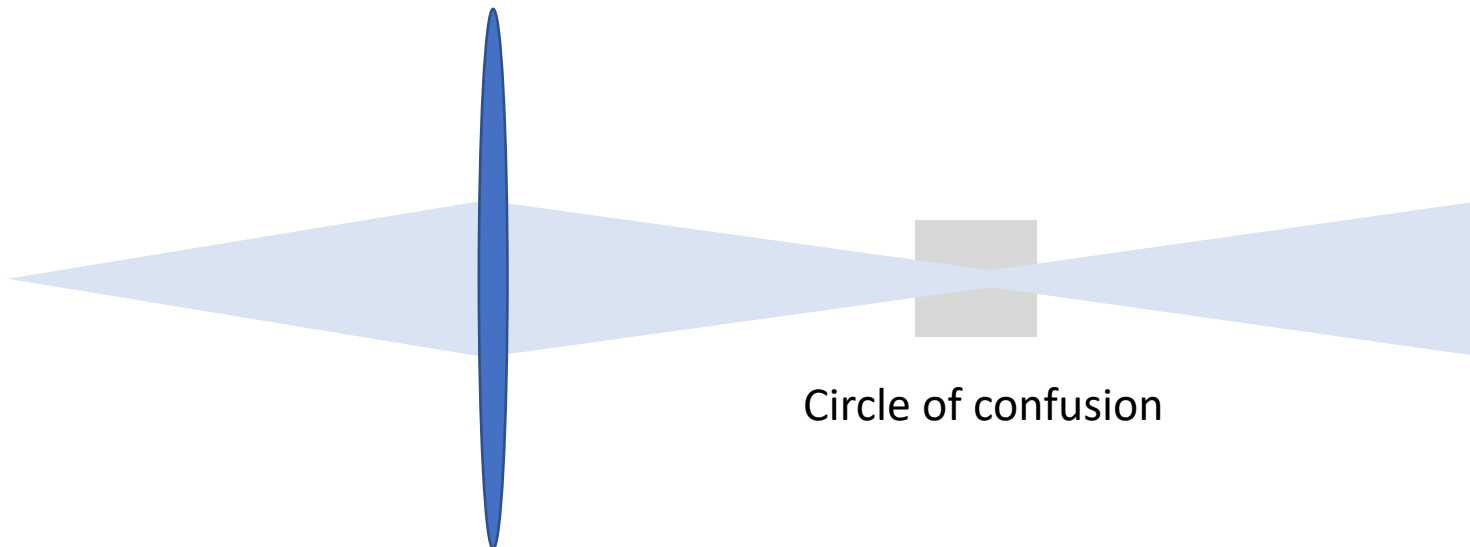


Depth of focus

Smaller apertures and stronger lens (smaller f) causes larger depth of focus



Larger aperture has steeper angles, the region in which the image will appear to be focused will be small



Smaller aperture openings leads to a larger circle of confusion, region in which the image will appear to be in focus

Resolution vs Depth of Field Trade-off

Maximum spatial resolution is obtained by having the largest NA (collection inside the lens).

This has the largest aperture opening.

However,

Largest NA, also degrades (1) depth of field. Only a very small region (thickness) is imaged.
(2) Has poorer contrast.

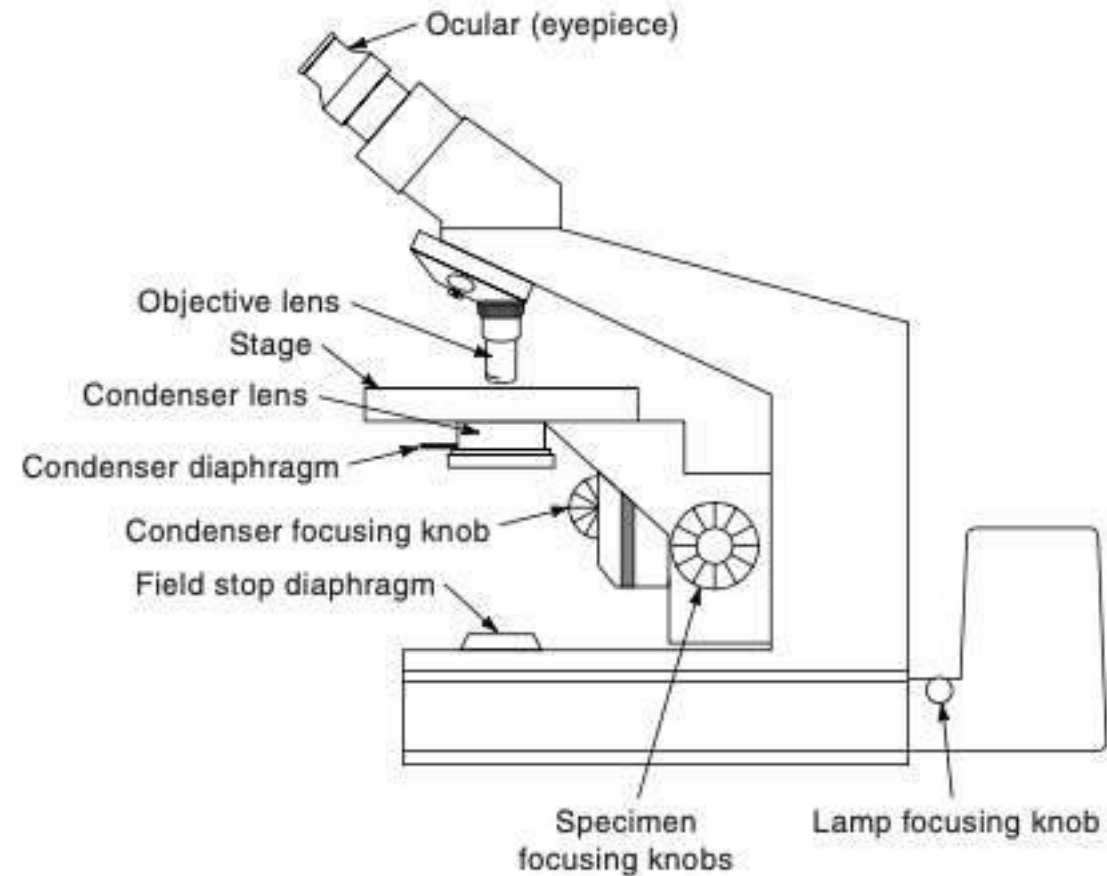
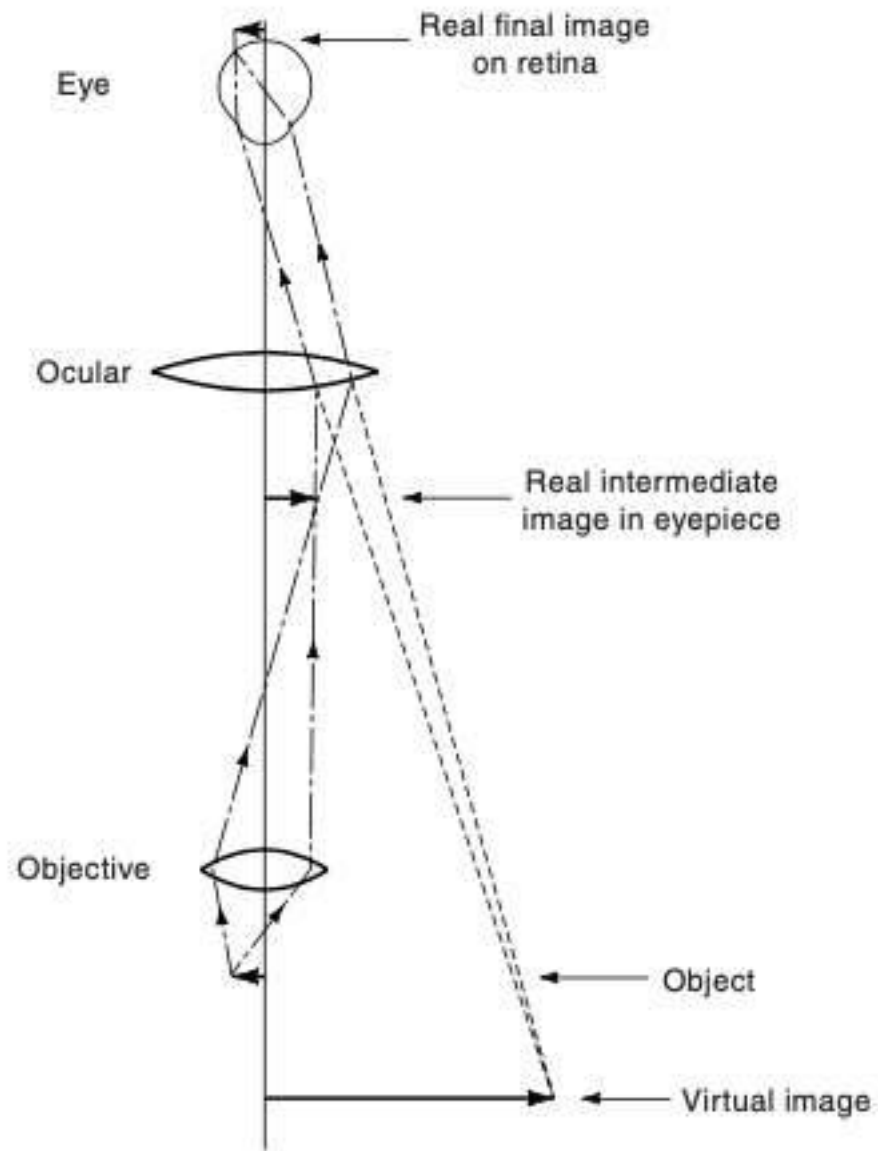
Lack of contrast in large NA objectives is due to two fold reasons:

1. Large NA also collects stray light
2. Large NA, has interference with light with large varying coherence. (Smaller NA, smaller ray bunches, better coherence)

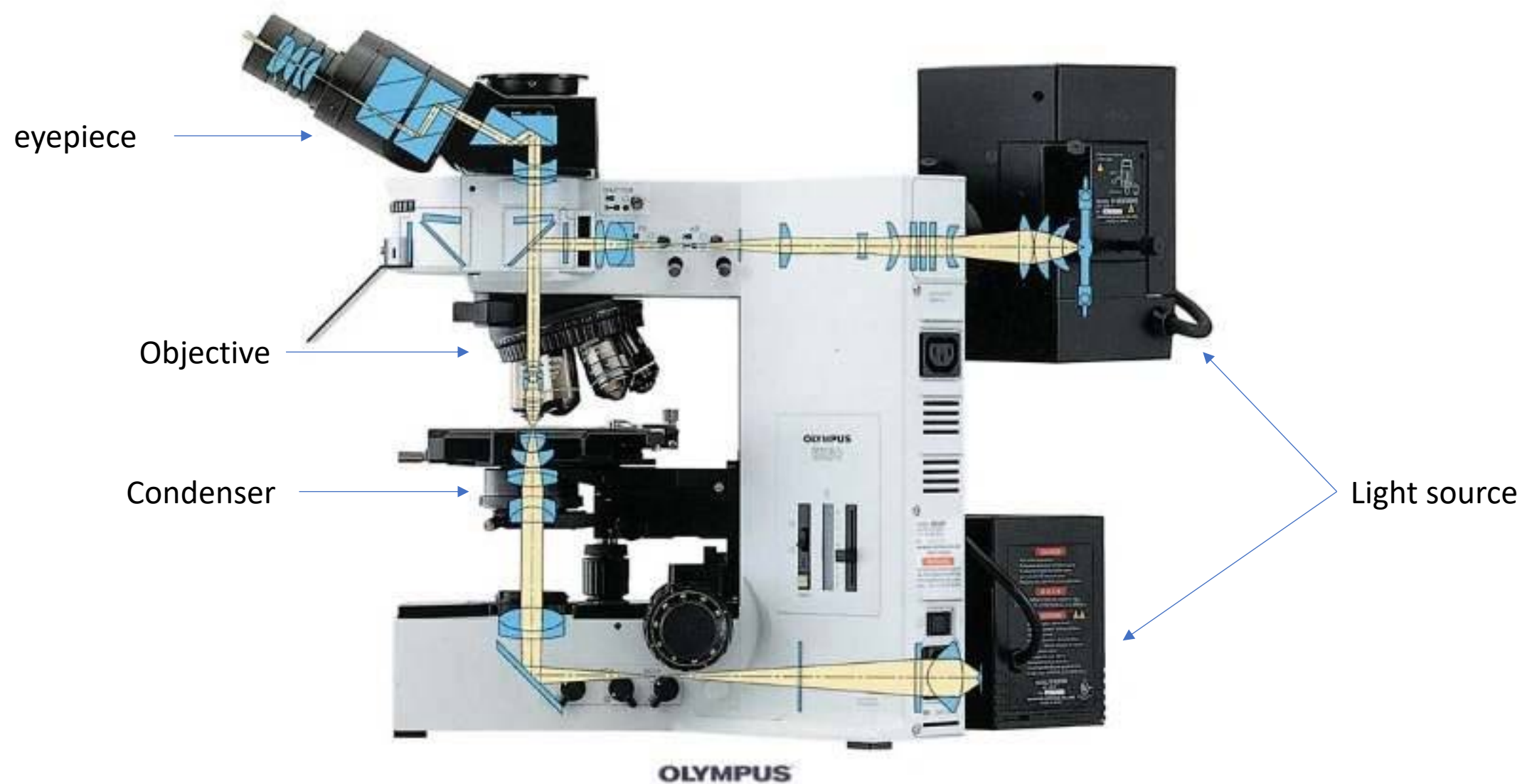
Microscope

Parts and components

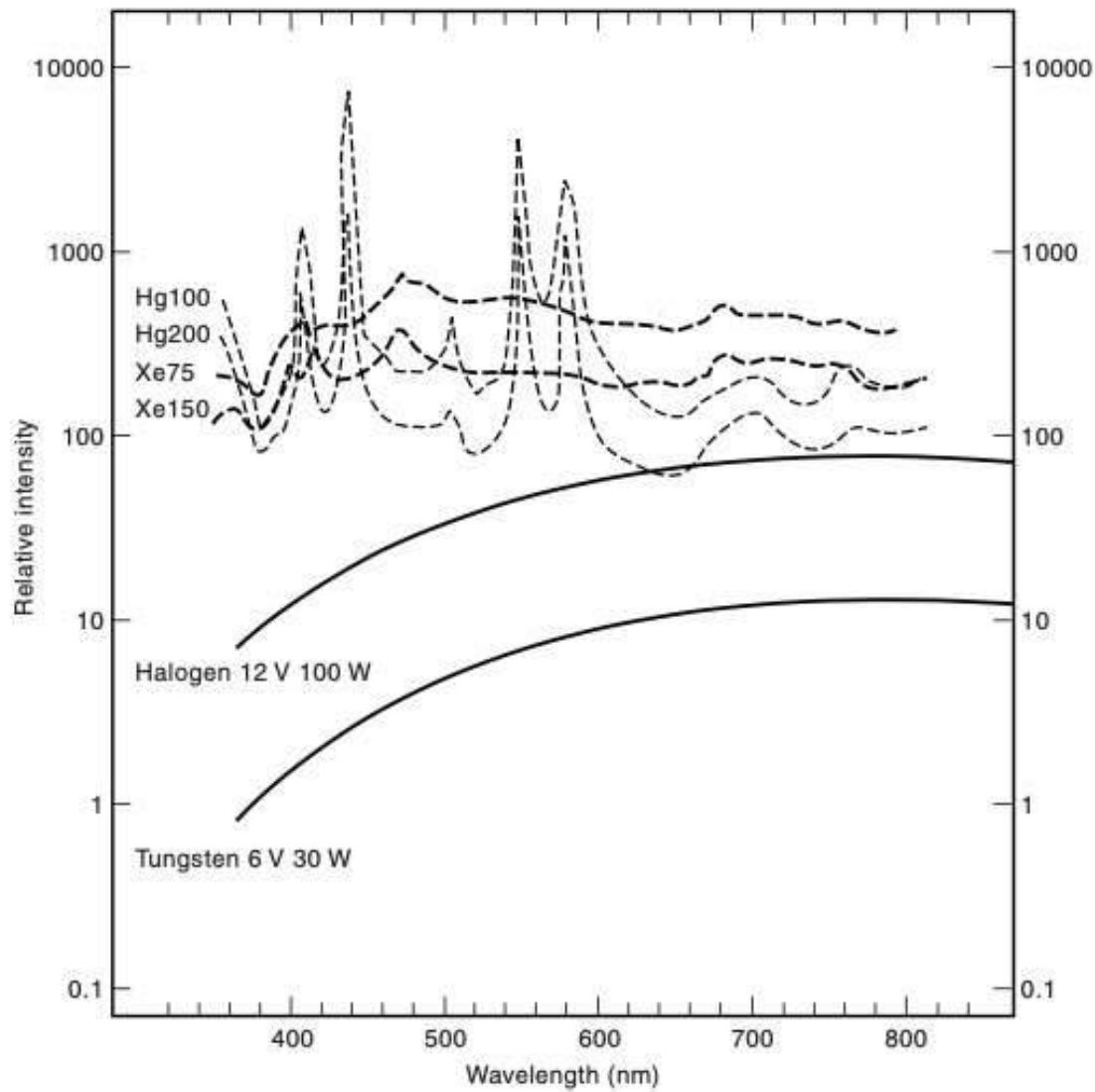
Ray diagram of a light microscope



Modern Metallurgical Microscope

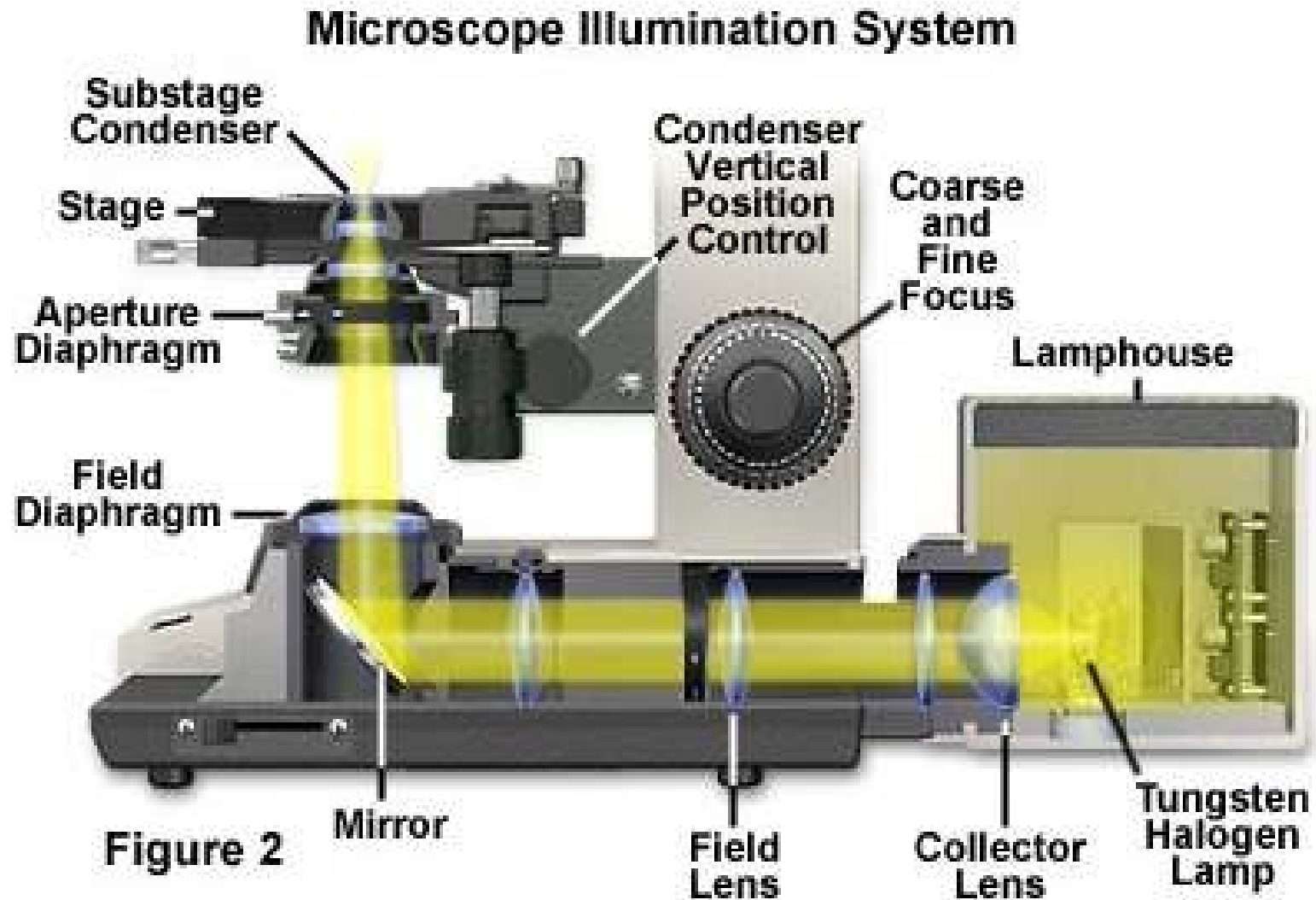


Illumination sub-system

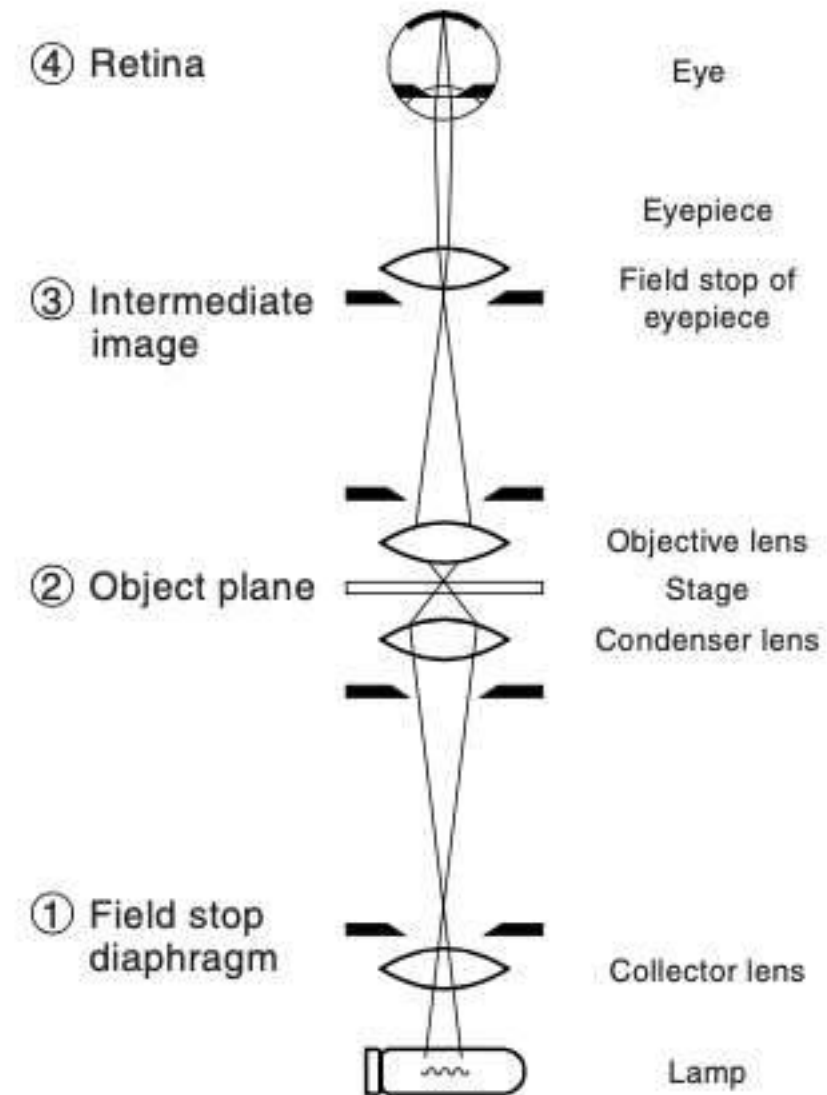


- Heated elements are among the most basic light emitters
- Ion arc lamps (Xe/Hg) are 10 – 100 times brighter
- However, such ion emissions have some characteristic emission lines which requires specific notch filters.
- Typical lamps also have IR filters to avoid those light which just harm sensitive samples such as biological specimens.

Object Illumination



- Collector lens:
 - Has the lamp filament at its first focus
 - Forms a collimated beam of light with uniform intensity
- Field lens
 - Focusses the lamp light into the condenser
- Field Diaphragm:
 - Limits the intensity of light falling into the sample
- Condenser Aperture: (hole/opening in an opaque mask)
 - Controls the NA of the illumination



Transmitted Light Microscopy Optical Pathways



Outside

Field Diaphragm



Aperture Diaphragm



Illumination Intensity : 100 %

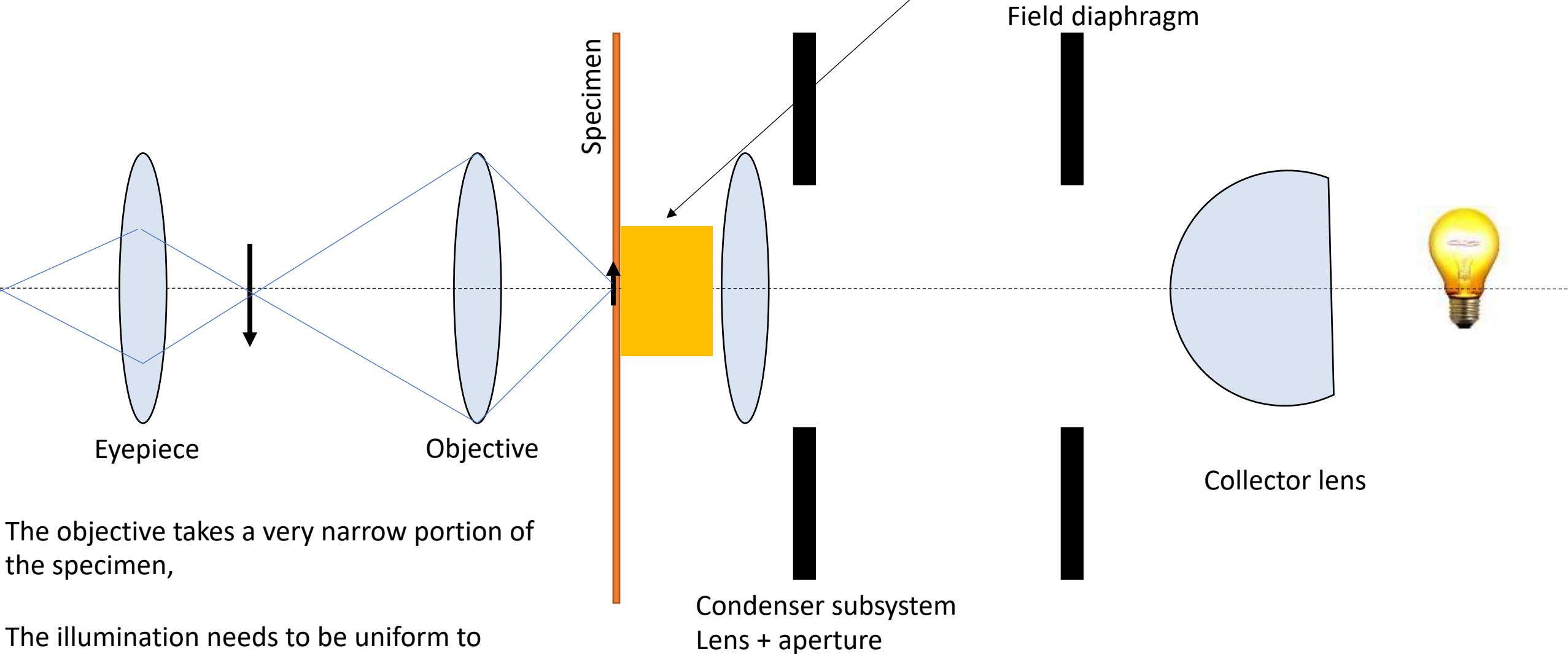


Koehler Illumination

The objective is to illuminate the entire specimen uniformly such that any intensity change is associated with the object and not to the illumination system

Problem: The light sources (bulbs) are not spatially uniform.

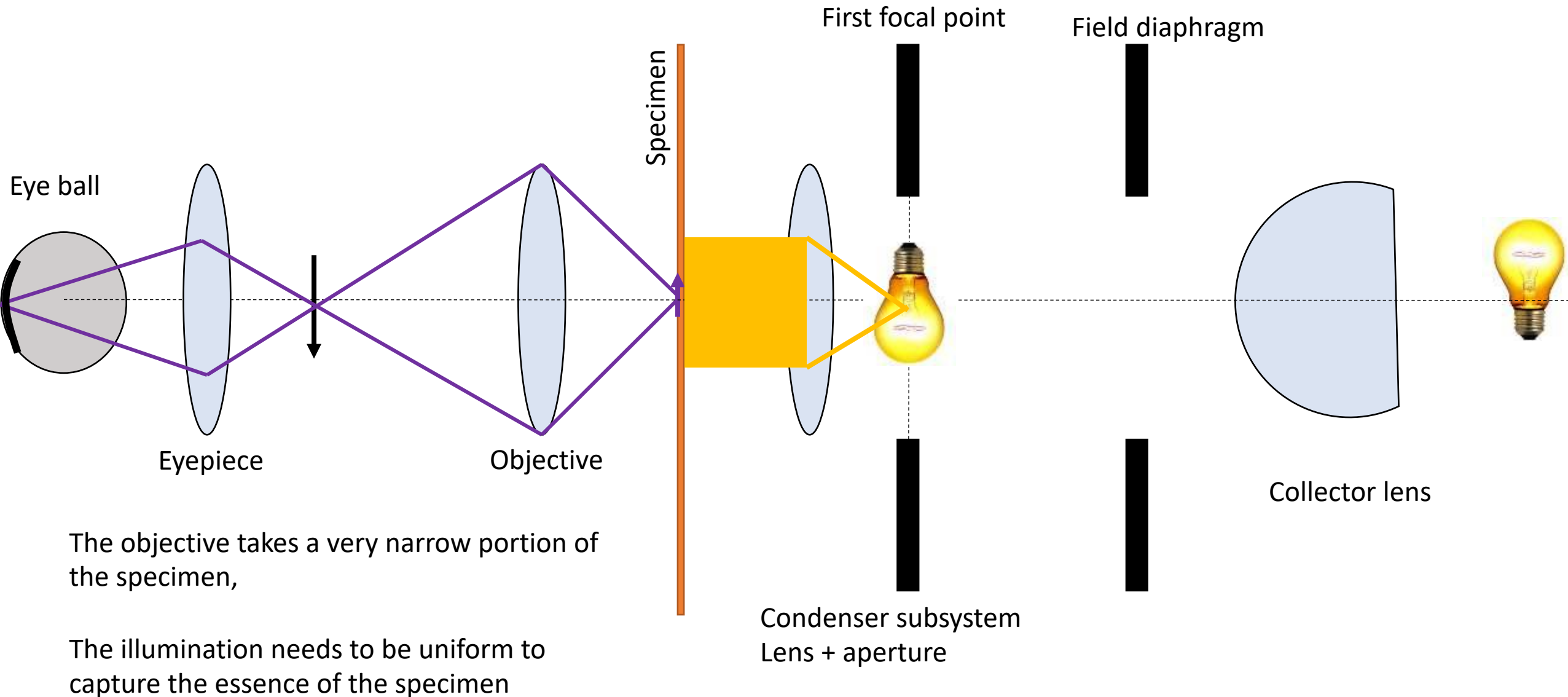
Illumination subsystem



The objective takes a very narrow portion of the specimen,

The illumination needs to be uniform to capture the essence of the specimen

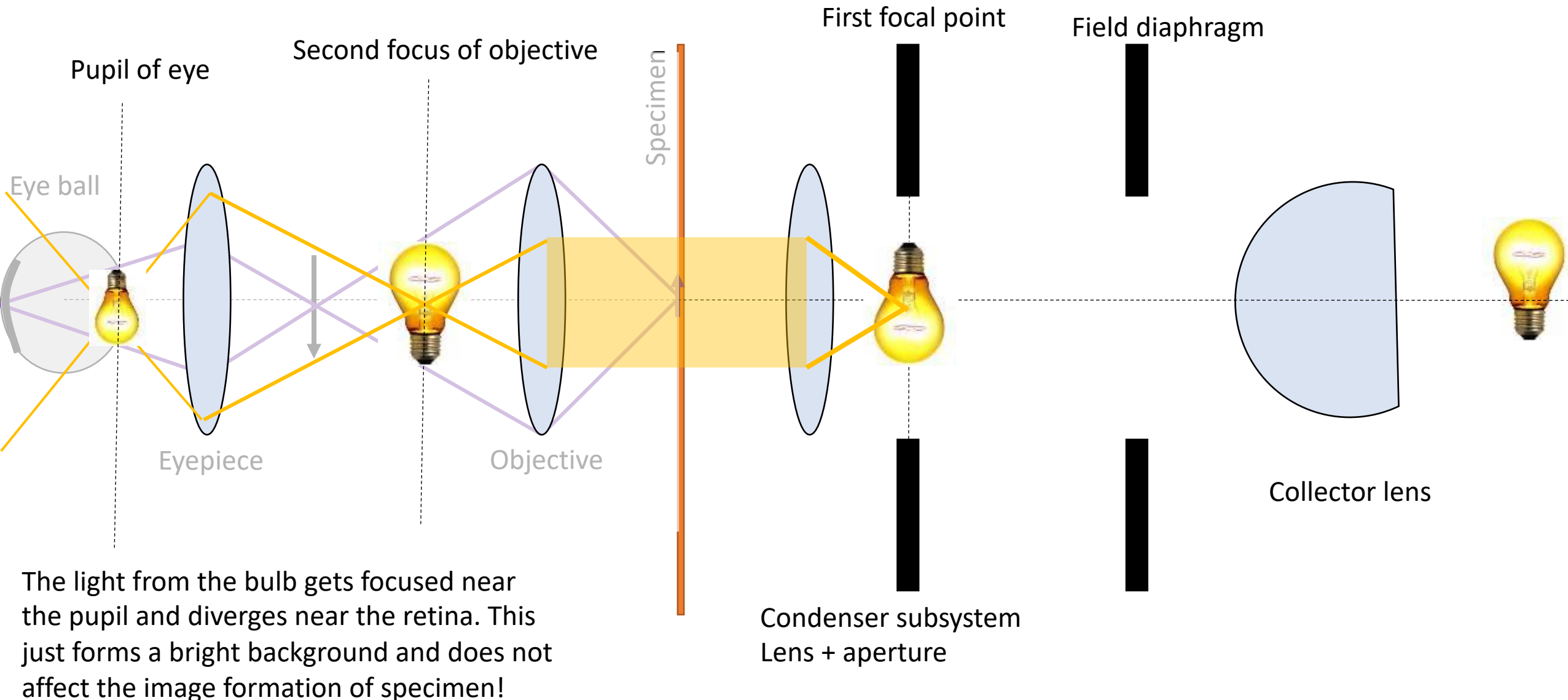
The idea is to place the image of bulb on the first focal point of condenser



Conjugate illumination planes

Set of planes having the same focus are called as conjugate planes

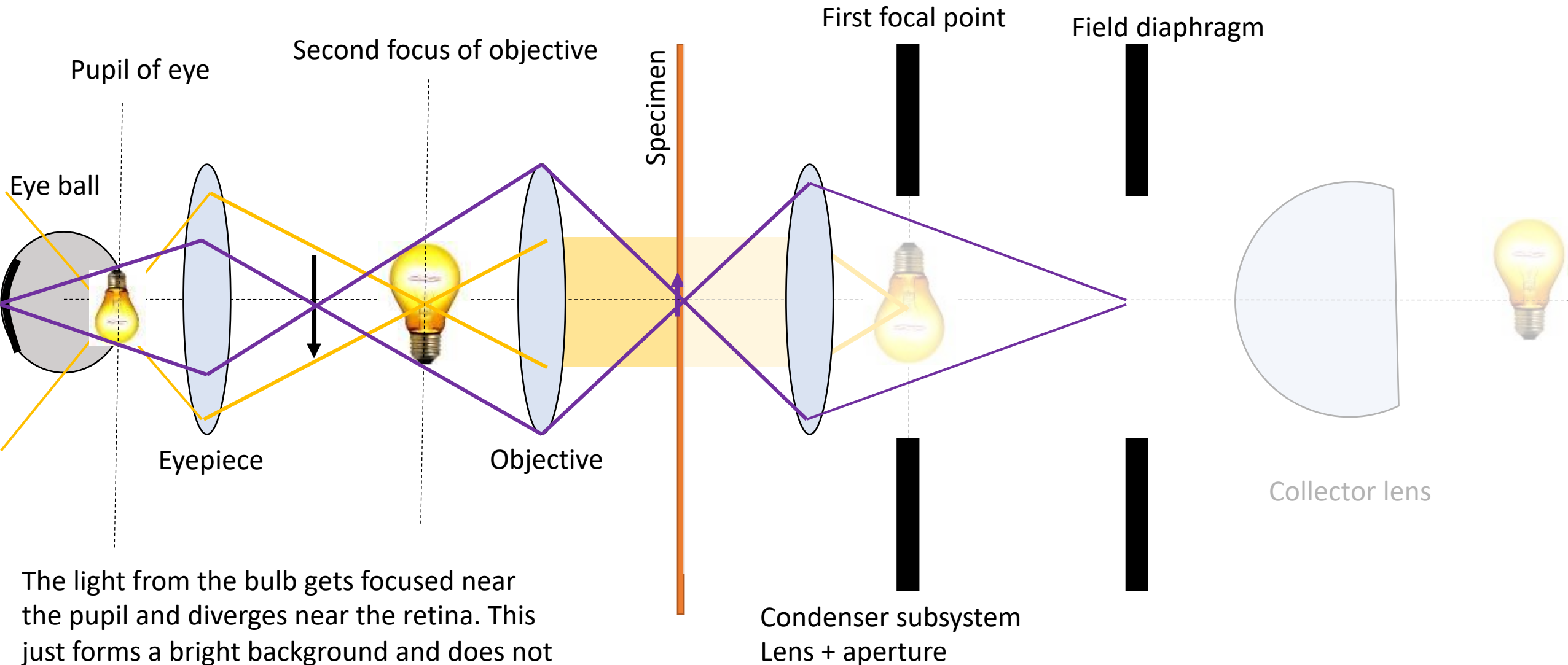
Here, since they form the image of the bulb, they are called the conjugate illumination planes



Conjugate focal planes

Set of planes having the same focus are called as conjugate planes

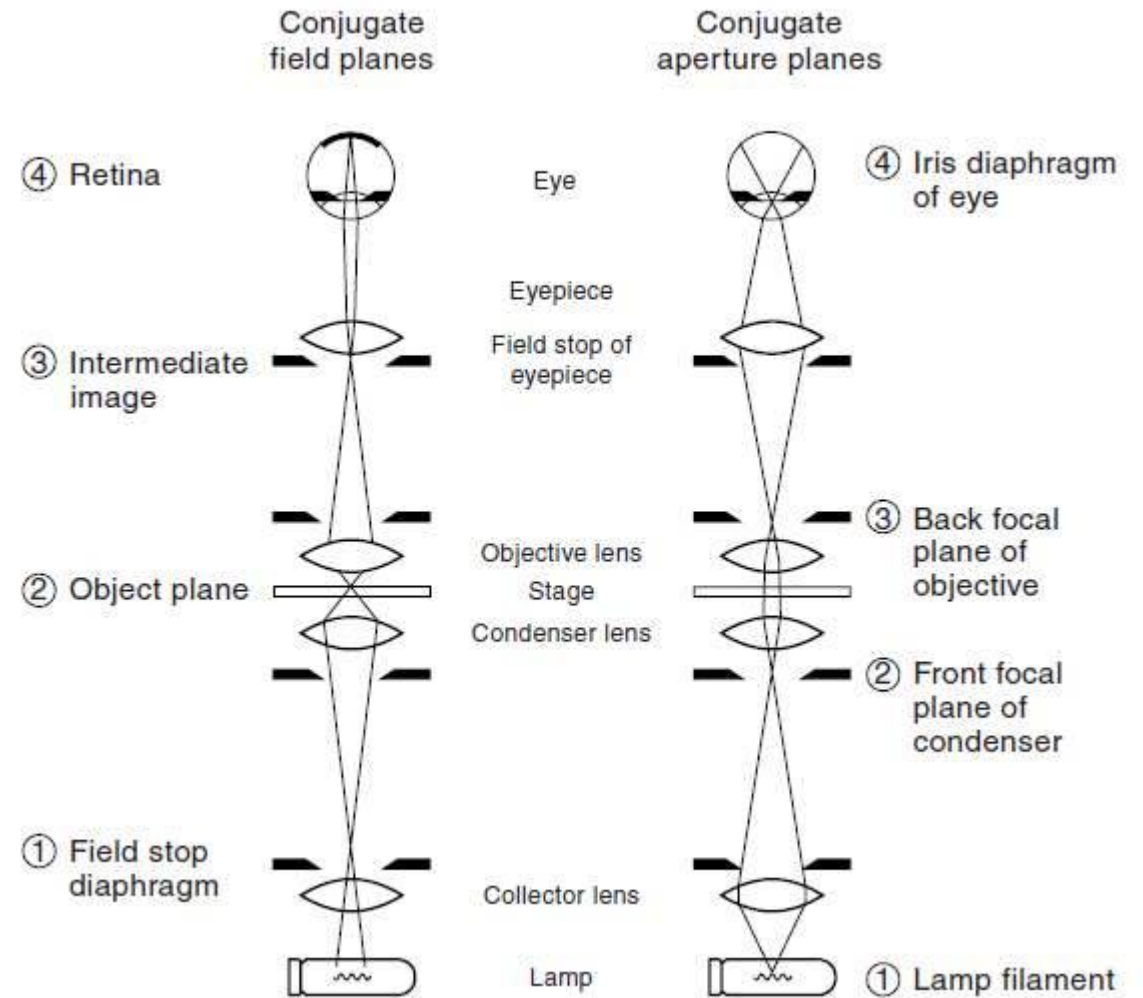
Here, specimen, field diaphragm, retina all will have the same image.



The light from the bulb gets focused near the pupil and diverges near the retina. This just forms a bright background and does not affect the image formation of specimen!

Steps for Koehler illumination

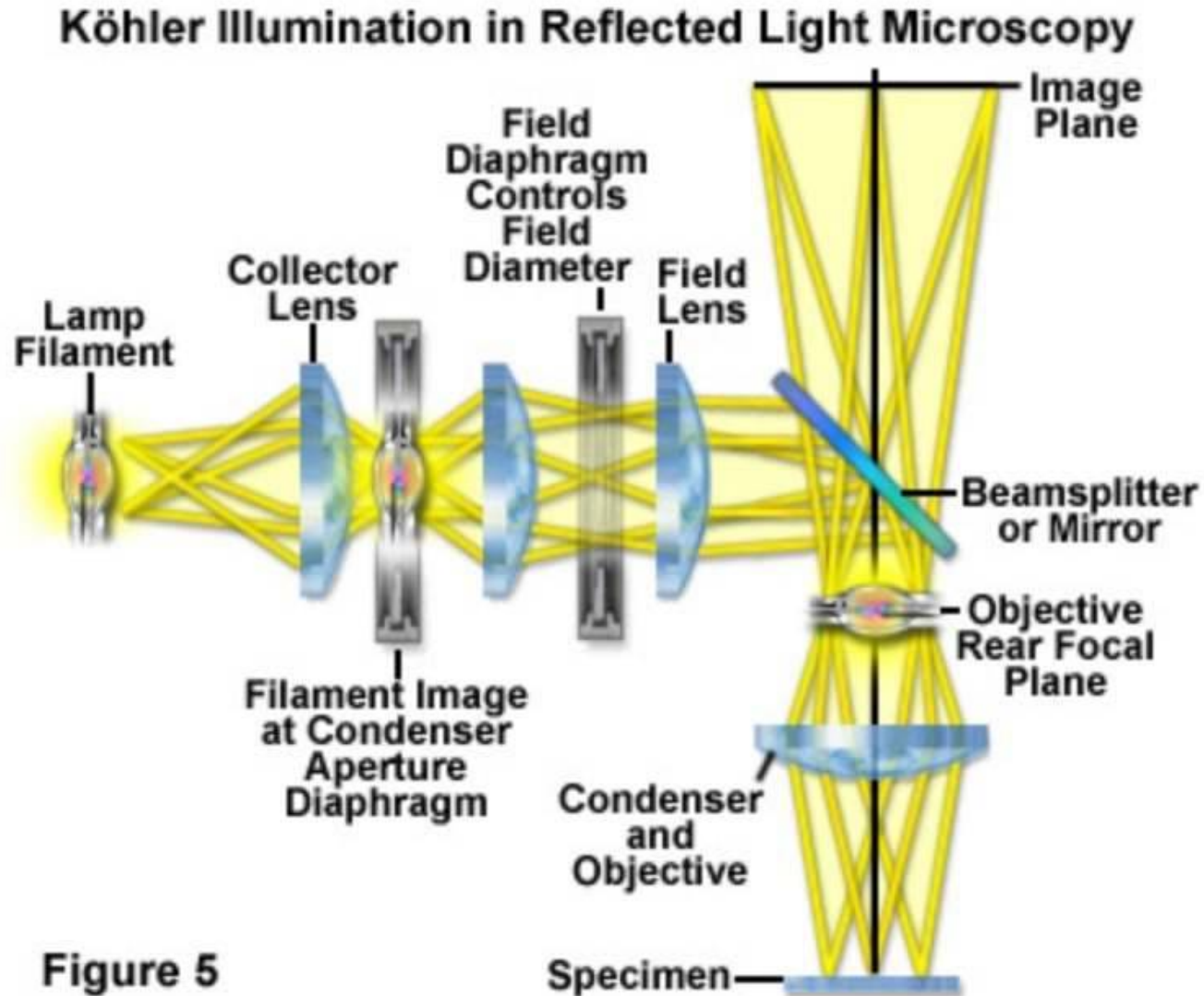
1. Focus the objective on the sample stage
2. Now move the condenser (attached with its aperture) such that the field diaphragm is visible. This sets the condenser position.
3. Remove the eye piece. You should now see the first focal plane of the condenser. Now close the aperture such that you see the complete image of the lamp. Now you are set in Koehler illumination!



Condenser Sub-stage:

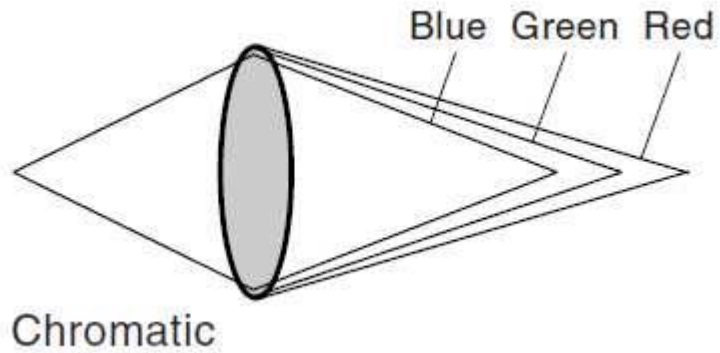


How is Koehler illumination obtained in reflection microscope ?



Aberration in lenses

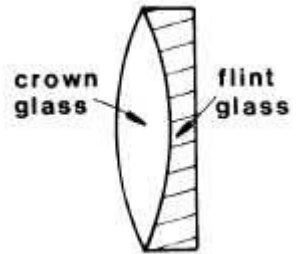
Chromatic Aberration:



Different components of white light converging on different points on optic axis

Refractive index of glass in blue is higher than red. This causes blue to be bent further!

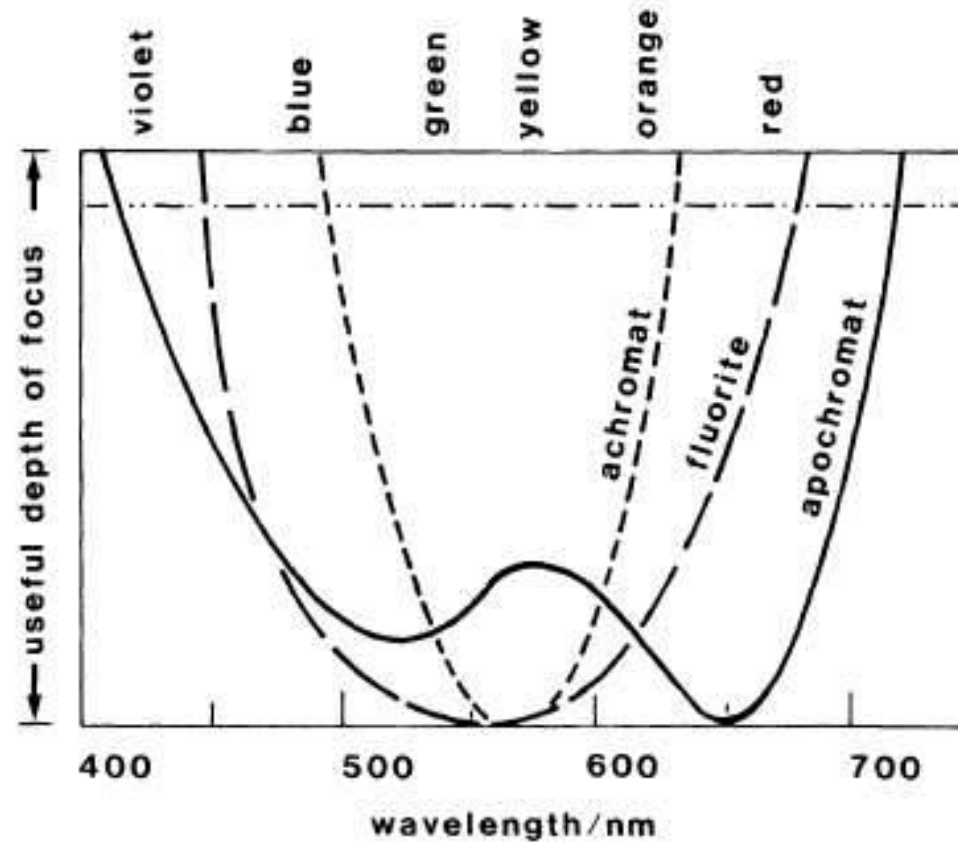
Clue: Colored halo, color changing upon different focus, image never becomes sharp – atleast not in one wavelength



Solution:

Couple glasses with different refractive indices. Match for differing paths for different wavelengths.
Eg: Couple crown glass with flint glass to form a lens combination called as achromatic doubled lens

Variation in depth of focus for different kinds of objectives



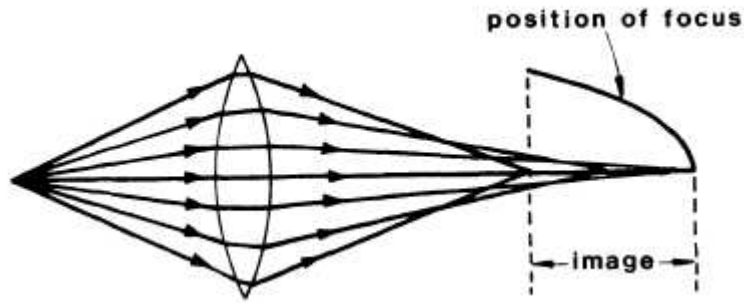
Achromat – simple doublet

Fluorite- Fluospar like materials which have low dispersion $\frac{dn}{d\lambda}$ than optical glasses

Apochromat –Has a combination of positive Fluospar (convex) and negative special optical materials

Spherical Aberration

Monochromatic rays!



Rays that pass-through lens extremes get focused at different point on optic axis

Gets focused closer to the lens.

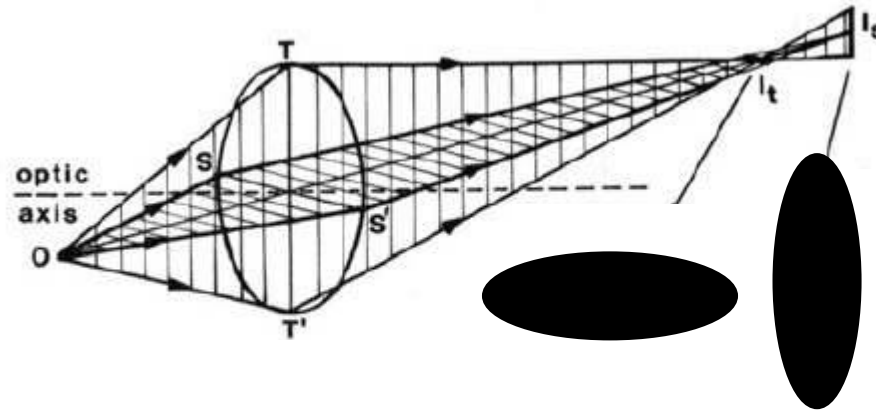
The aberration gets severe for higher NA lenses

One possible solution:

Use compound lenses – positive and negative lenses with differing thickness

Correction is often effective for a well calibrated object position, presence/absence of cover-slip, refractive index of medium (oil/air)

Astigmatism



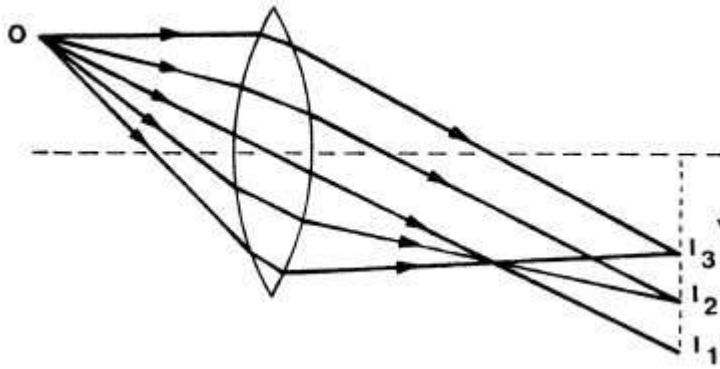
An object which is off-axis

The rays which go along the Y-axis (TT') is focused at a different point, in comparison with the rays which went along X-axis (SS').

Thus, when the focus is set at I_t , the X direction is not focused, seems like an elongated disk along X

When the focus is set at I_s , the Y-direction is not focused, causes an elongated disk along Y.

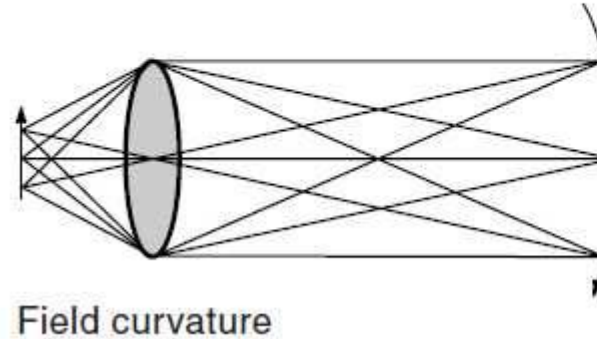
Coma and Distortions



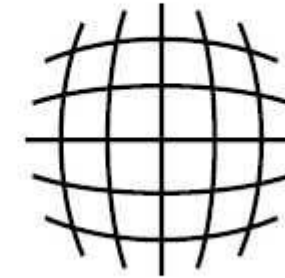
Another off-axis issue (non-axial spherical aberration).

Images have a comet's tail normal to optics axis

Rays which pass through the edge of lens are focused to a point closer to the optic axis.



Distortion



Barrel distortion



Pincushion distortion

Caused due to different in magnification, which is the ratio of image to object distance.

This is a symptom, and is often corrected by addressing the other aberrations.

Construction of Objective:

Achromat: red – blue corrected (chroma aberration), spherical aberrations are corrected for yellow-green ~ 540 nm

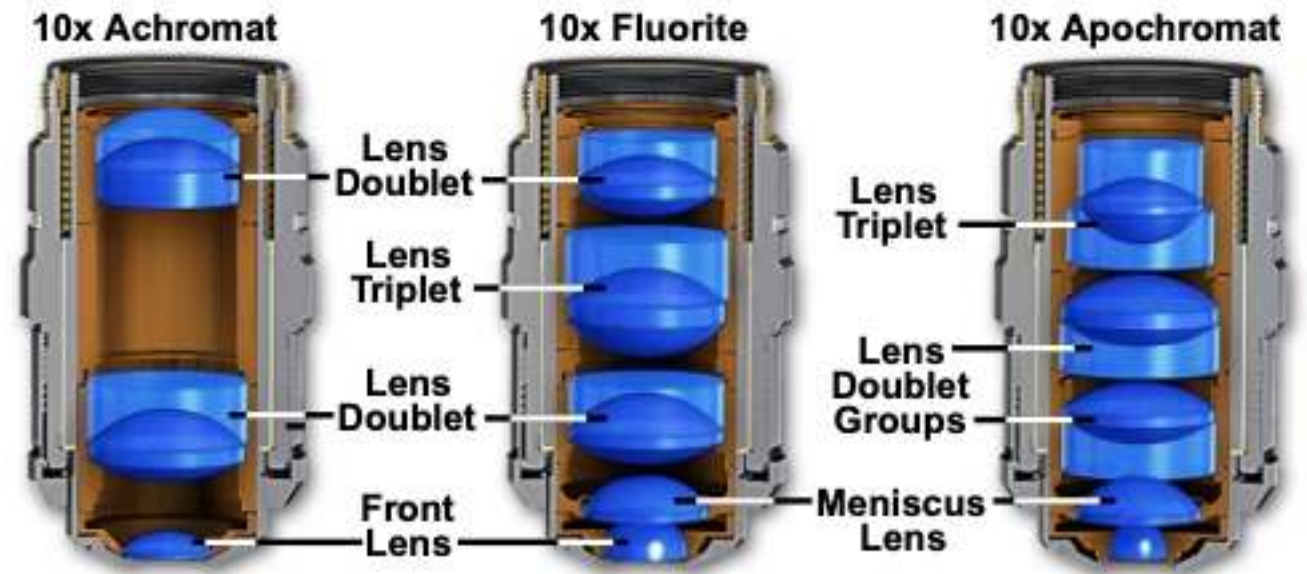
Fluorite/semiapochromat – have CaF_2 or LaF_3

Have good correction for chroma, field curvature, high transparency in UV

Apochromat: Most expensive. A combination of many such correctors. Useful in wide range of wavelengths.

However, suffers from field curvature (only a small region can be brought to focus).

Plano-apochromat: also has the field curvature corrected.



Objective marking



Construction of Condenser lenses

Abbe Condenser (Numerical Aperture = 1.25)

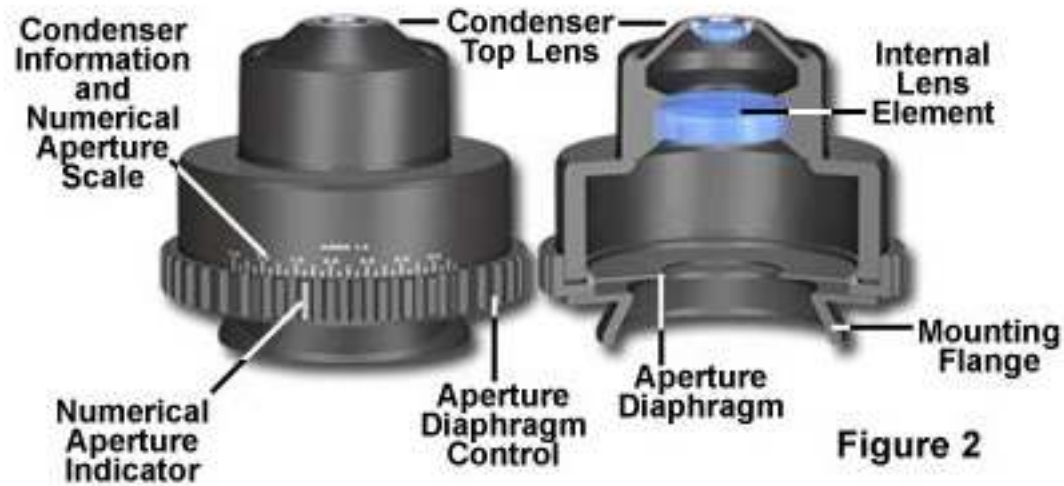


Figure 2

Achromatic Condenser (Numerical Aperture = 0.95)

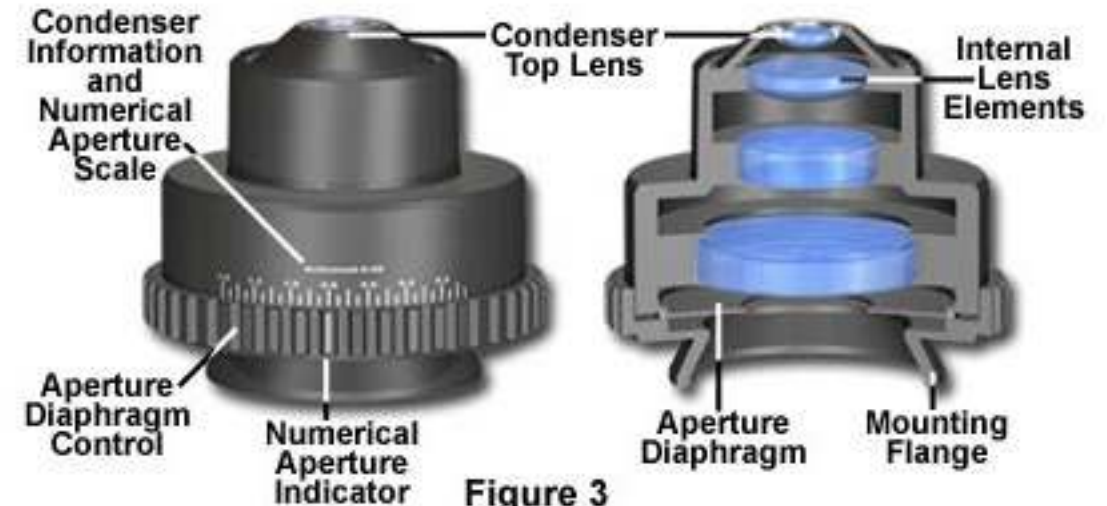


Figure 3

Simplest condenser: With no correction for chroma/spherical

Achromat/Aplanat Condenser (Numerical Aperture = 1.38)

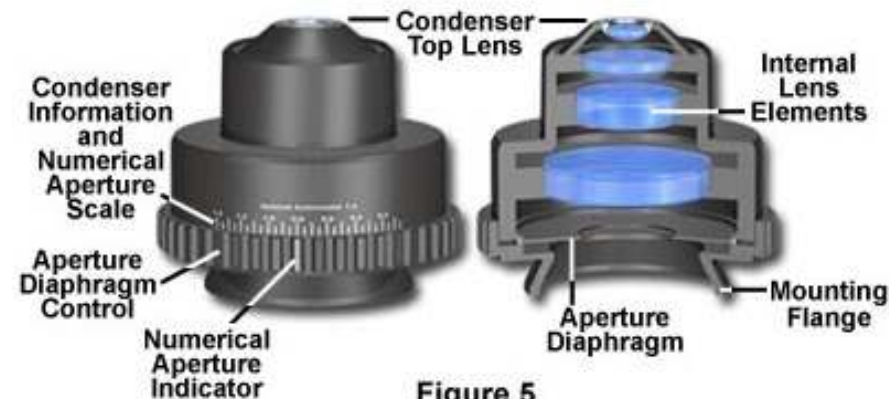


Figure 5

A microscope design problem:

The human eye can distinguish two points that subtend at least 1 minute of arc at the retina.

If distance of distinct vision is about 25 cm, eye can distinguish two points separated by

$$\frac{d}{25} = \sin\left(\frac{1}{60}\right)$$
$$d = 145 \mu m$$

If we use a light of wavelength 520 nm, close to ideal NA = 1

Then the minimum resolution that we can achieve $0.62 * 520 = 317.2 \text{ nm}$

The minimum magnification required for comfortable viewing of the fundamental limit of separation

$$M = \frac{145000}{317.2} = 457$$

This is the minimum magnification required to resolve the fundamental optical limit.

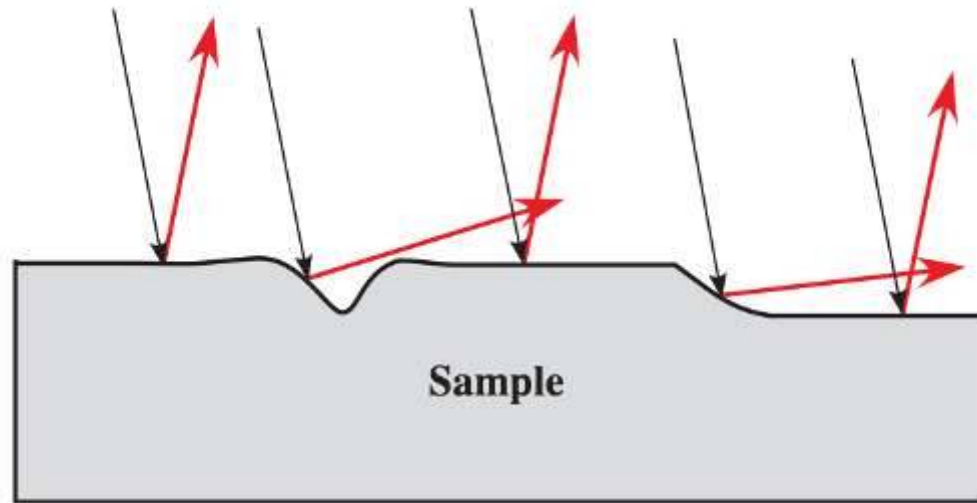
However, comfortable viewing for humans require about twice as much magnification.

Contrast Mechanisms

Different contrast mechanisms

- Bright field
- Dark field
- Phase contrast
- Differential interference
- Modulation
- Fluorescence
- Polarization

Contrast in bright field



Contrast allows for identification of features in a microscopic image.

Contrast:

Intensity difference between neighboring resolved points

$$C = \ln \frac{I_1}{I_2}$$

$$\Delta C = \frac{\Delta I}{I}$$

A minimum of $\Delta C = 0.14$ is required for human resolution of neighboring particles

Dark Field Illumination

- An opaque plate is used to block major portion of incident light.
 - Causes bright field to escape objective
 - Image is formed from diffracted rays
- Images with high contrast is observed. Very high NA objectives are rarely used.
- There is loss of resolution
 - Small NA of the objective

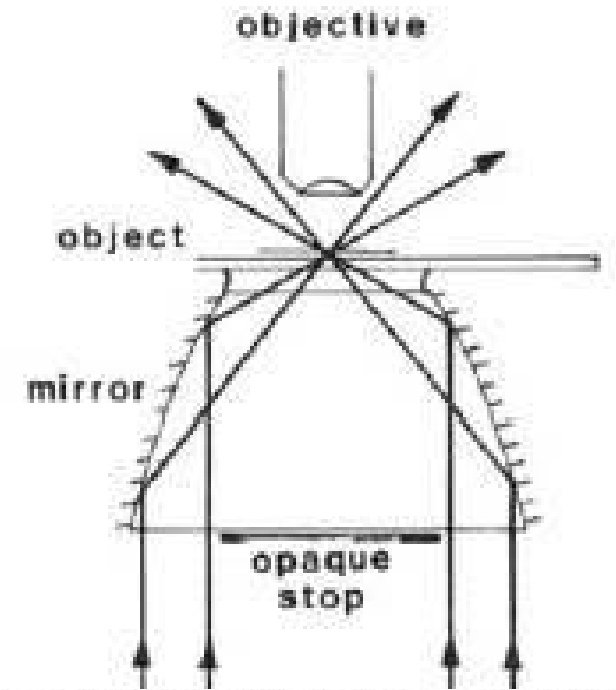
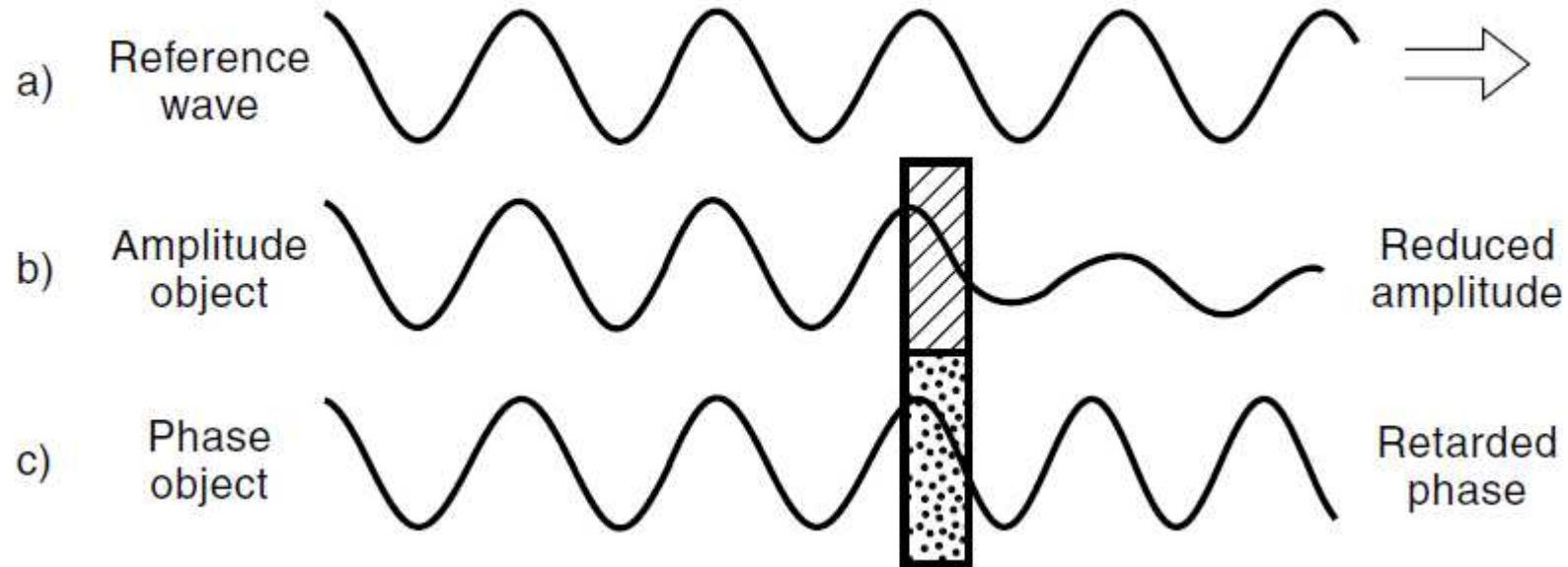


Image of pearlite
with dark field
illumination

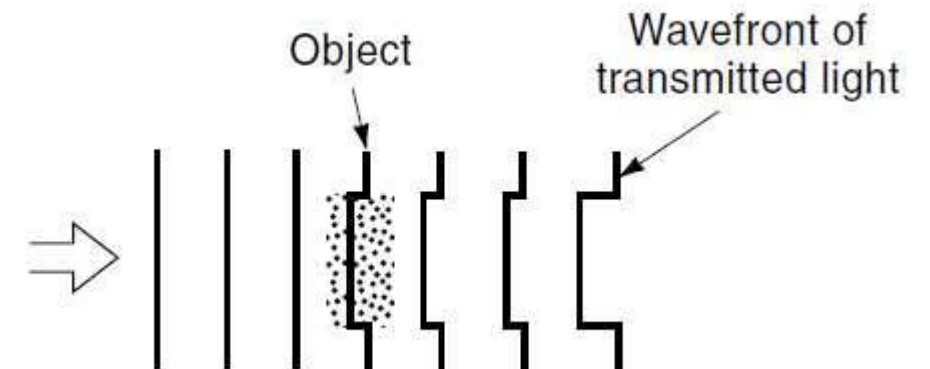
Phase contrast

Human eyes are sensitive to only intensity.

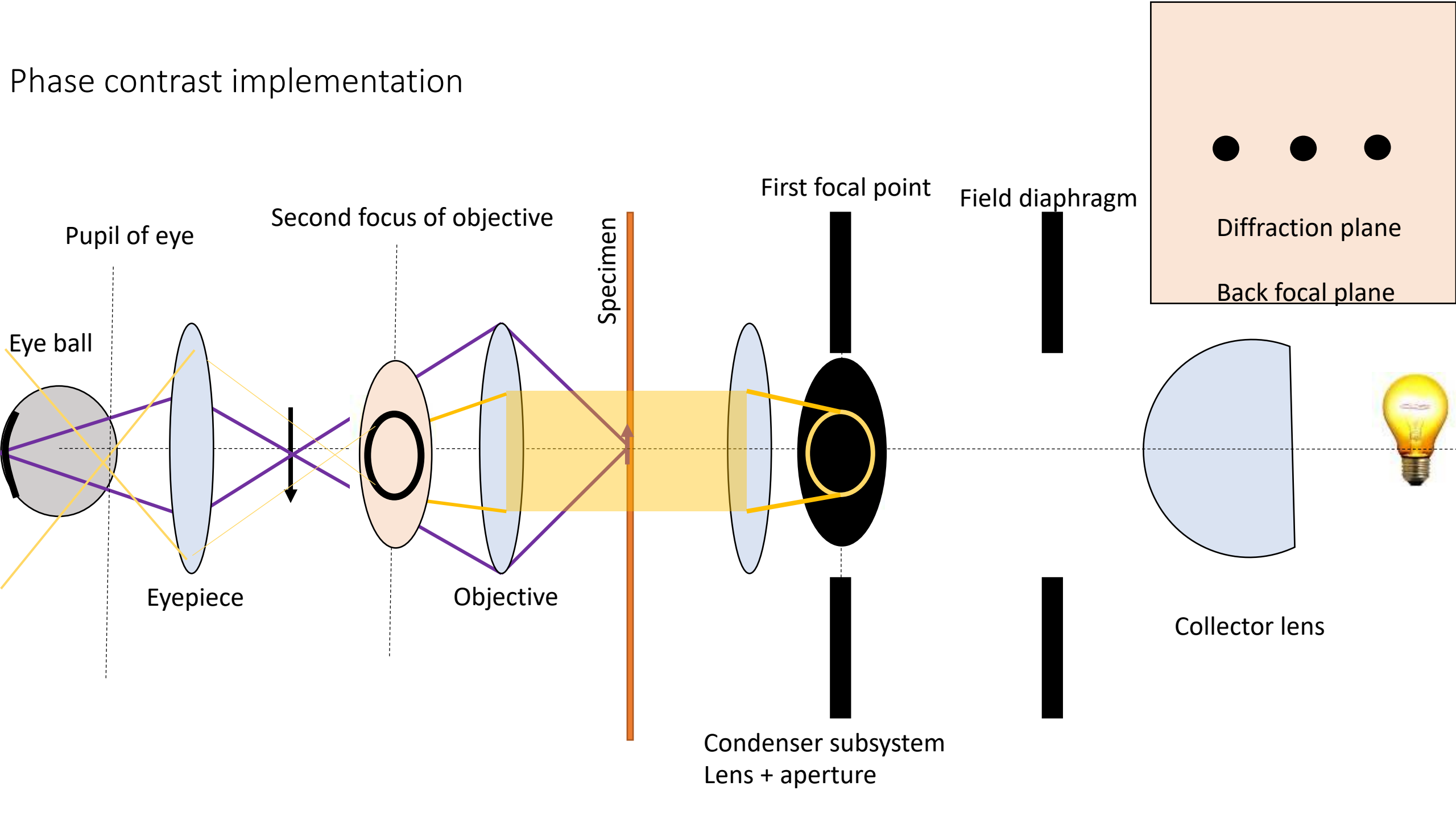
Phase microscope converts phase difference into intensity difference



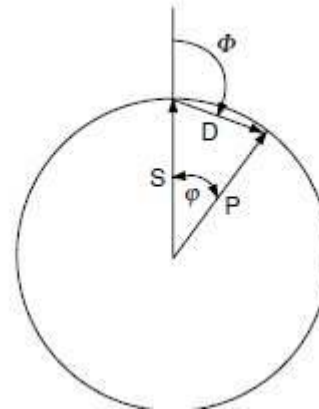
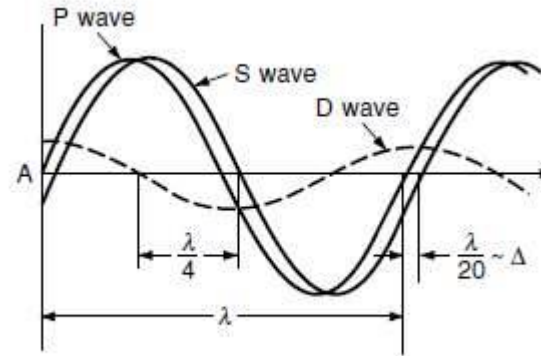
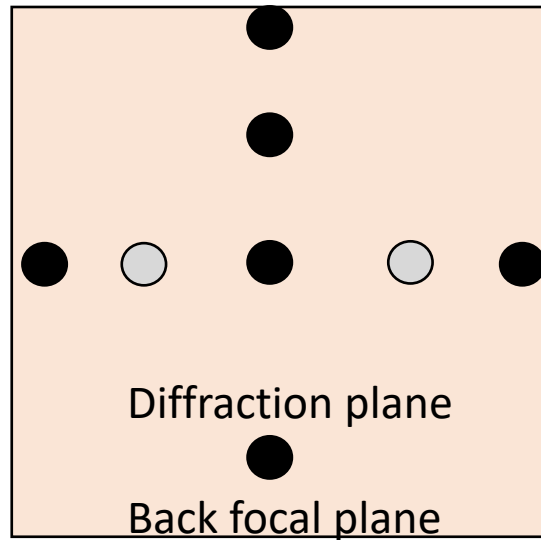
Amplitude vs phase object.



Phase contrast implementation

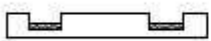


How to implement ?



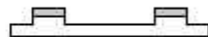
- A condenser annulus is used as in dark field
- A phase plate with complementing transparency is used.
- The un-diffracted light is reduced in amplitude.
- The resulting image is formed with intensity phase addition with background and the diffracted beams.

Positive phase contrast

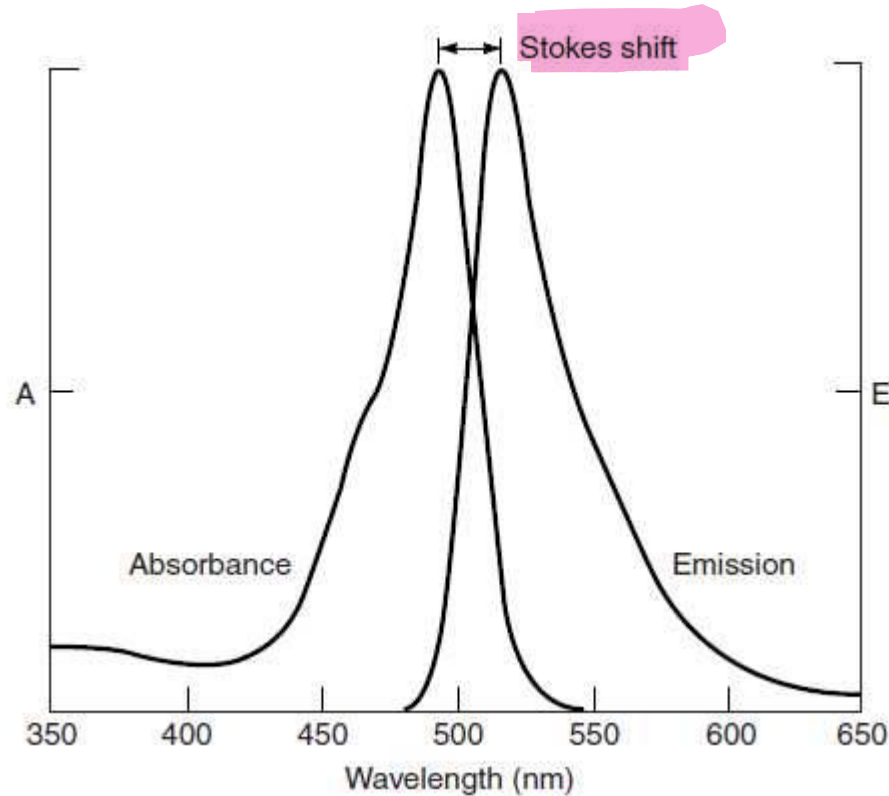


Phase plates

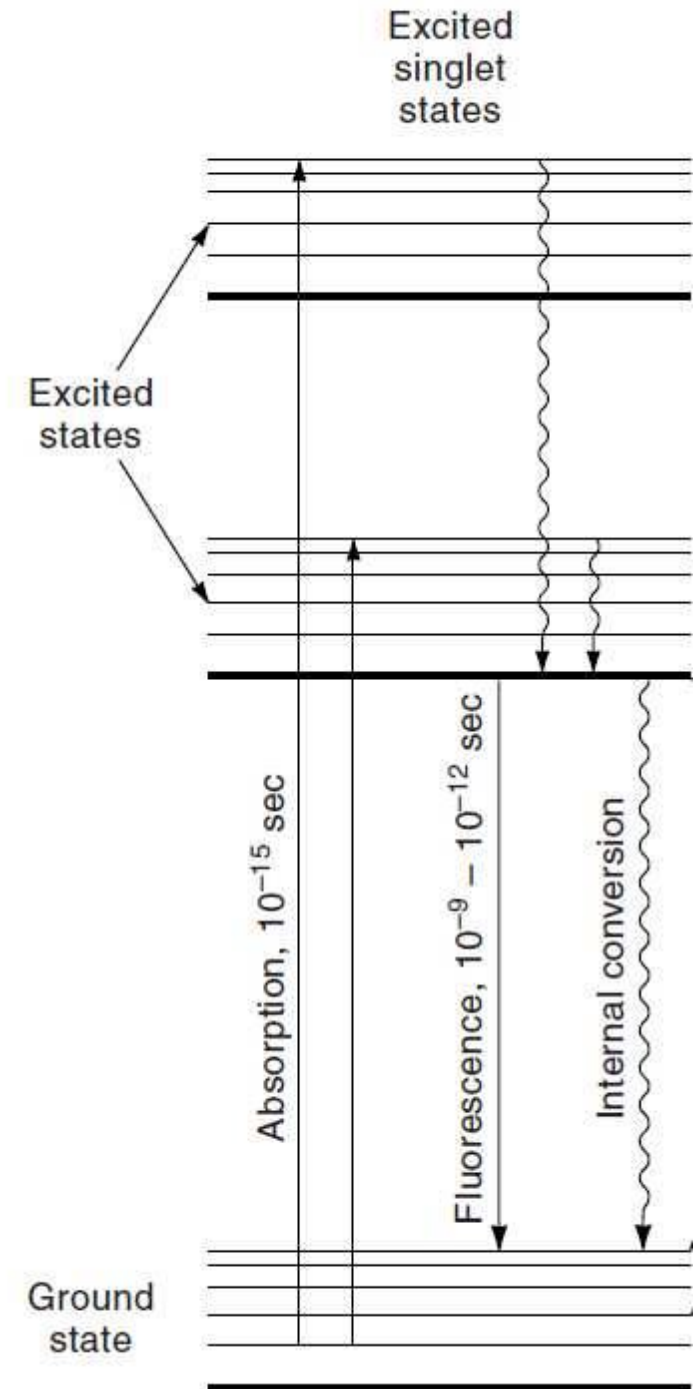
Negative phase contrast



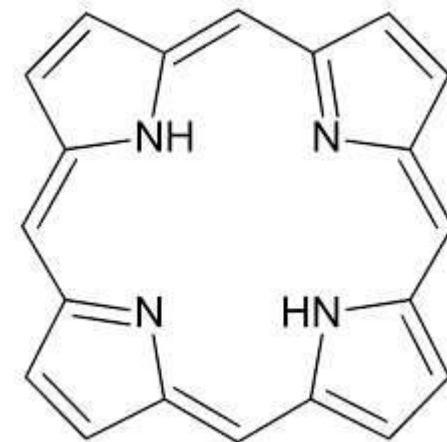
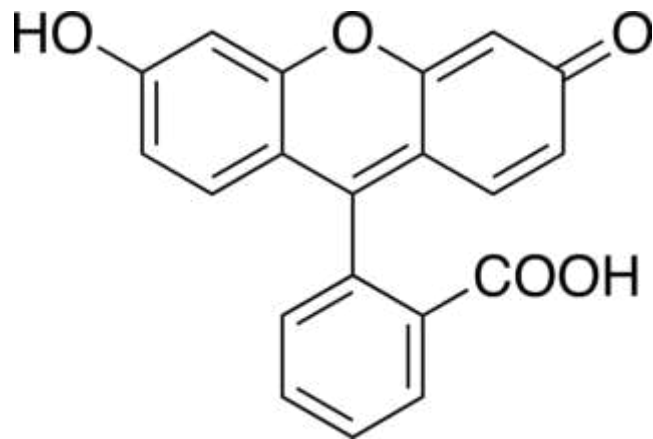
Fluorescent microscope



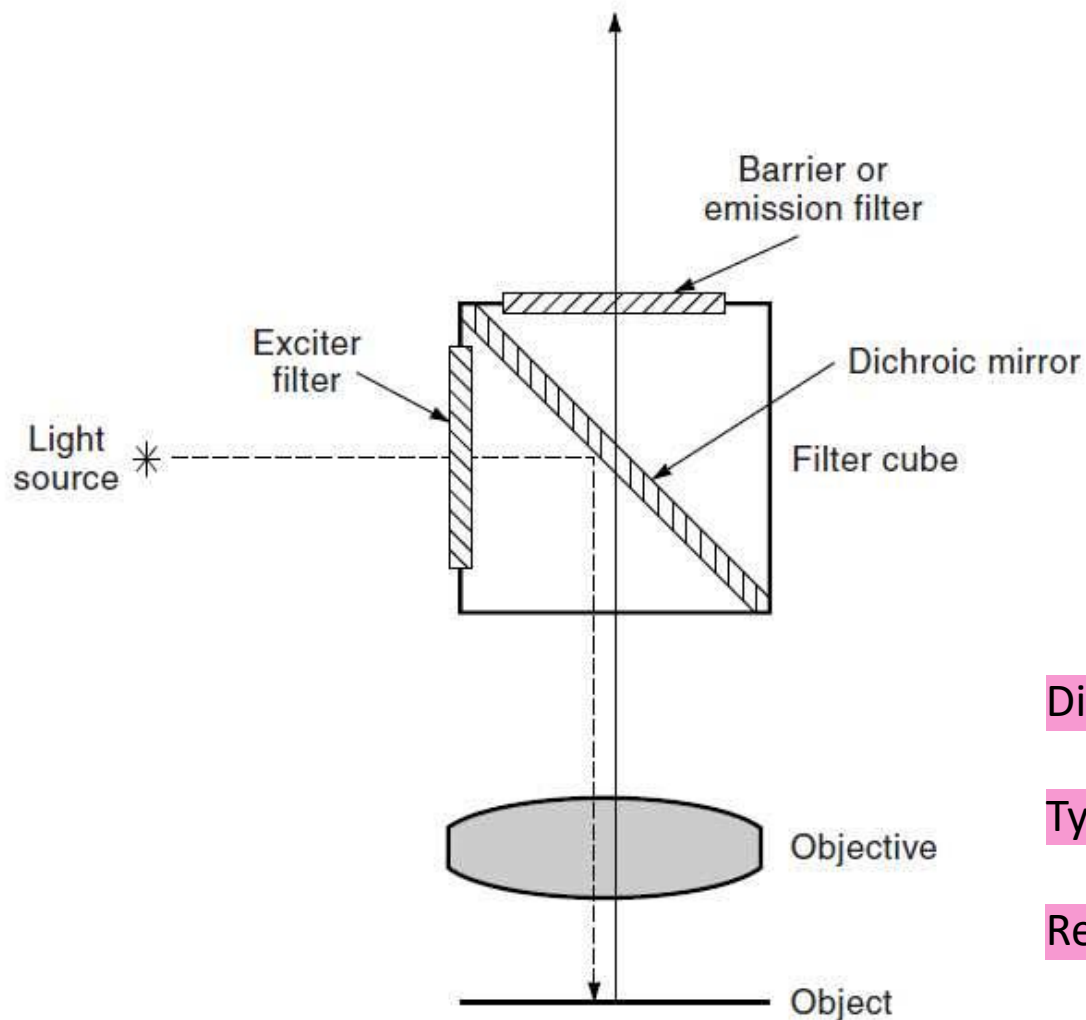
- A fluorescent molecule absorbs energy to excite electrons to higher energy states
- The electrons returns back to ground state by emission of photons of lower energy



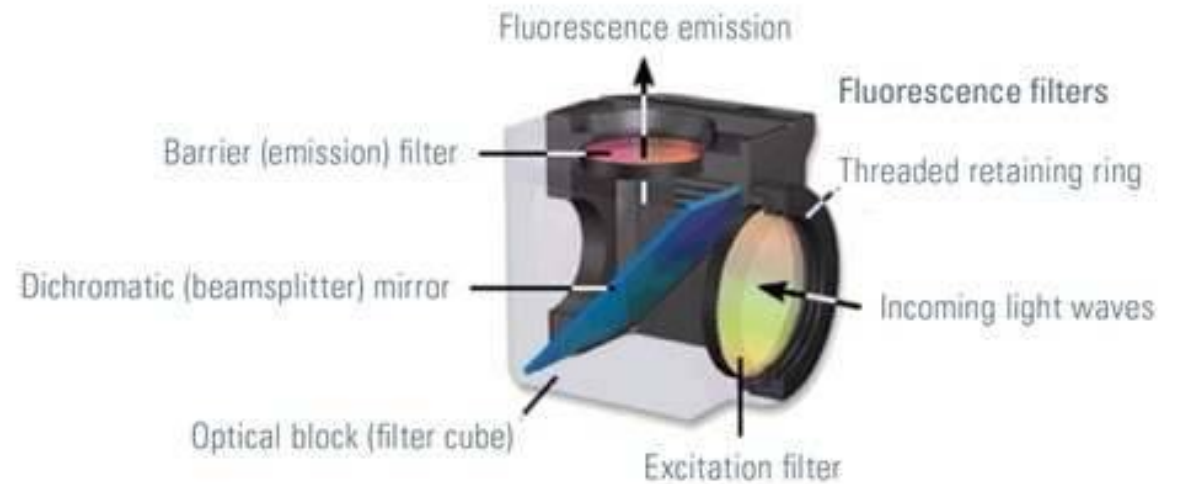
- Difference in energy between absorption peak and emission peak is called as Stokes shift.
- Molecules with large shift is suitable for molecule tagging
 - Eg: fluorescein has a shift of 20 nm, and that of porphyrin molecule has a shift of about 200 nm.



Implementation



Practical filter cube



Dichroic mirror:

Typically made of multiple layers of dielectric: like interference filters.

Reflects shorter wavelengths and transmits longer wavelengths!

Filters and chroma selectors

- Types of filters

- Neutral density filters – Thin sheet of metal with controlled optical attenuation

$$OD = \log_{10} \frac{1}{T}$$

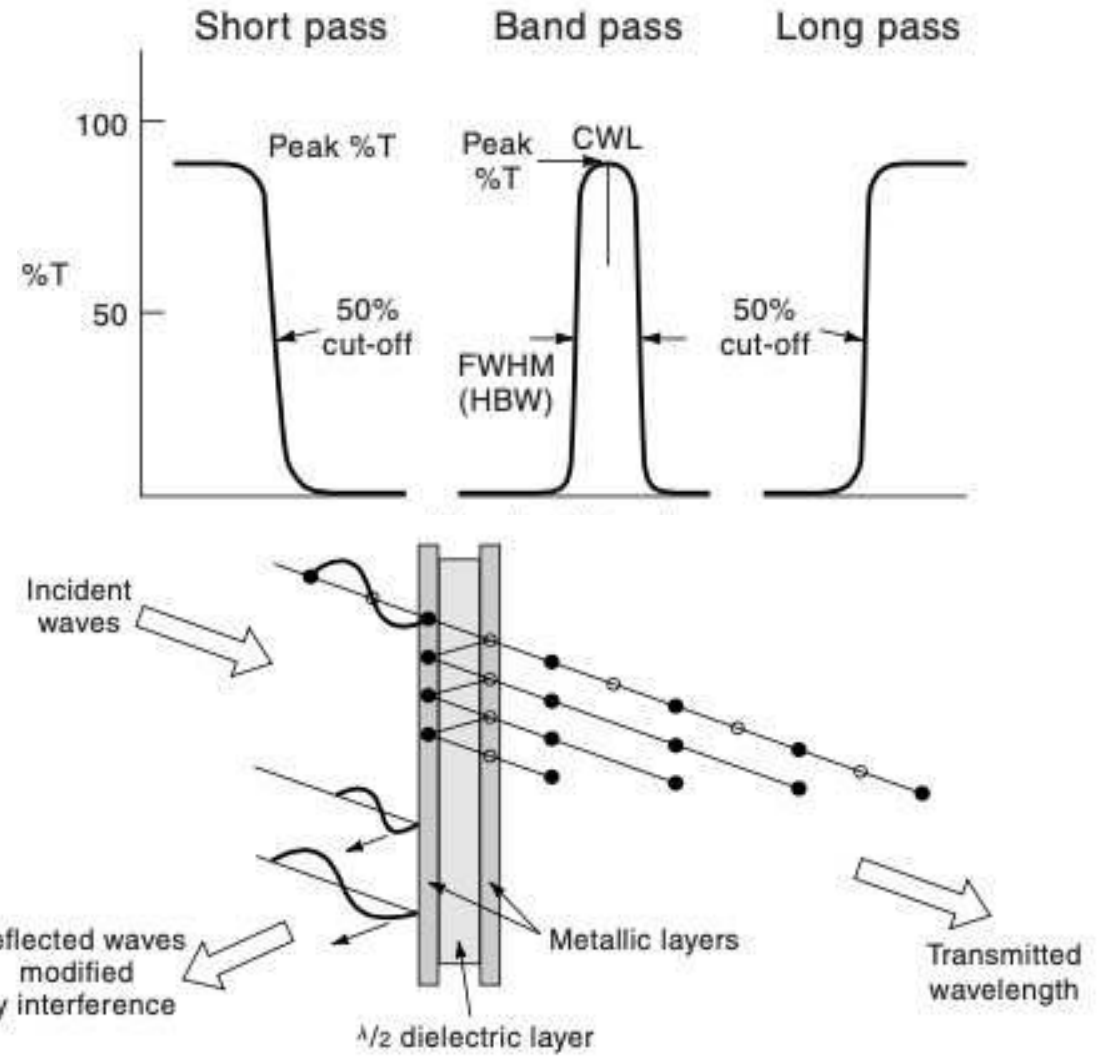
Where T is the transmittance.

- Color glass filters:

- They are molecular absorbers impregnated into glass
- Certain molecules have bands of absorption

- Interference Filters

- Have much steeper roll-off (sharper wavelength selection).
- Dielectric material sandwiched between metallic partial reflectors



Quantitative Studies

Image Capturing

Intensity of light on the image plane is a continuous function

$$I = f(x, y)$$

As in the older, photographic plate, the image intensity is in analogue format.

The image plane is then sampled to obtain the digital version of it.

Each sampled point on the image plane is a pixel.

If δ is the minimum resolution of your microscope and M is the magnification setting.

$$\Delta x, \Delta y < M\delta$$

A typical rule of thumb is that $\Delta x < \frac{M\delta}{3}$. That is 3x3 pixel array for every resolvable point

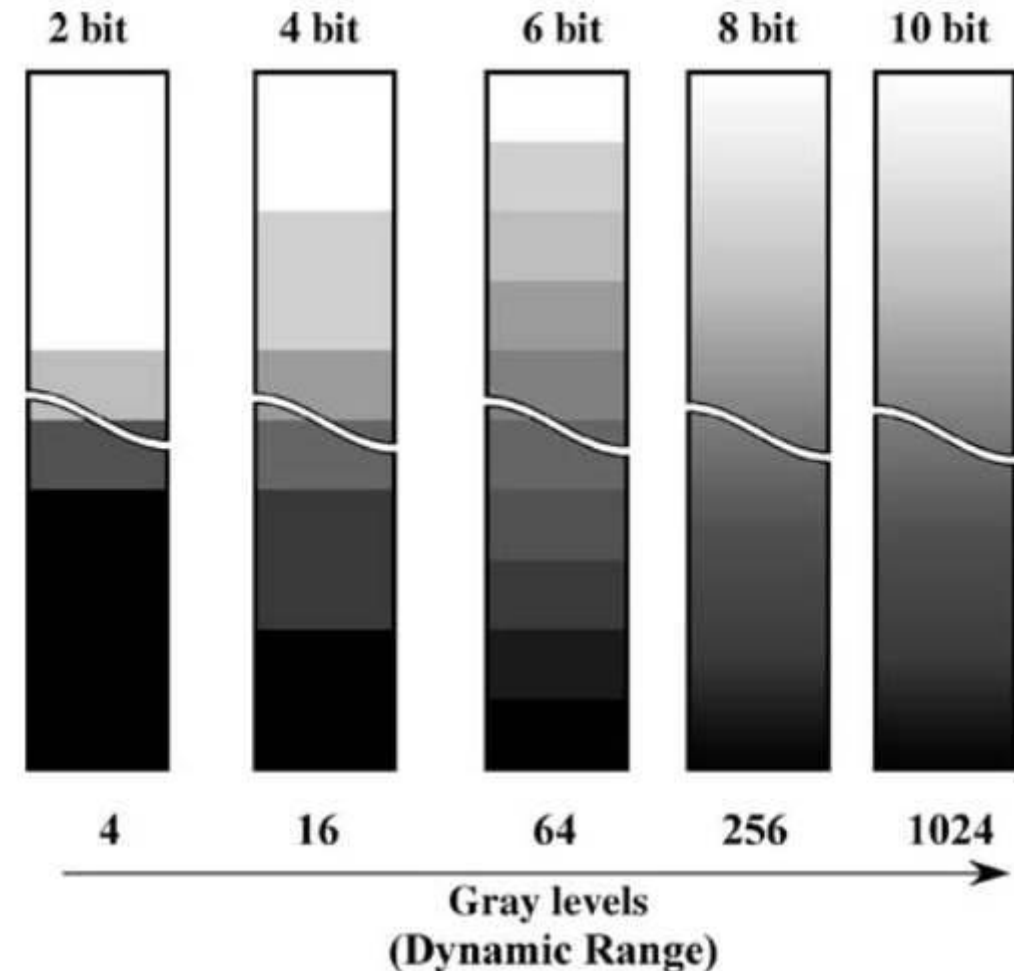
Intensity in each pixel is stored digitally in several bits.

Say we have a 512 x 512 pixel array

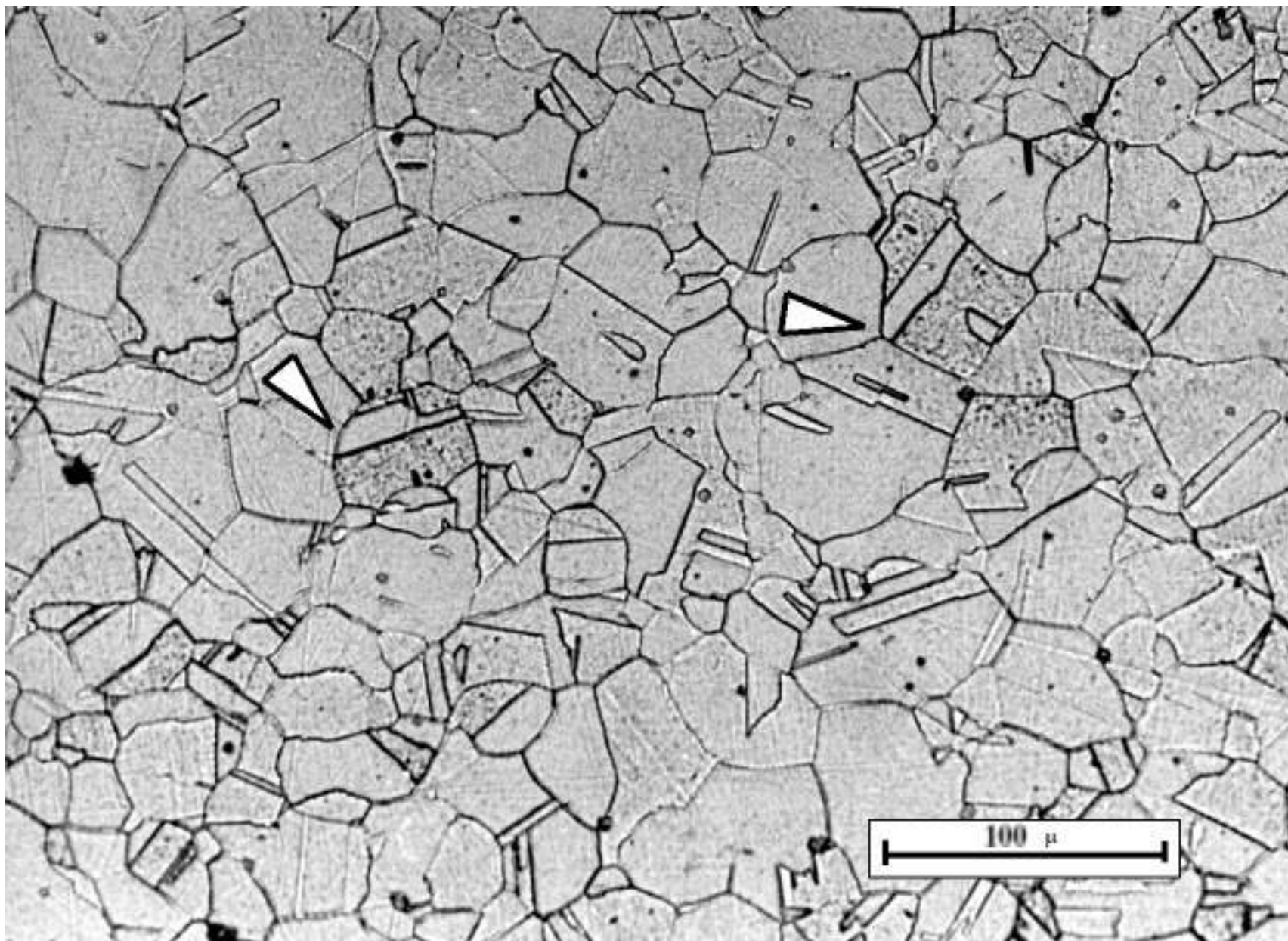
Each pixel intensity is stored in 8 bits.

The image size is 256 kB

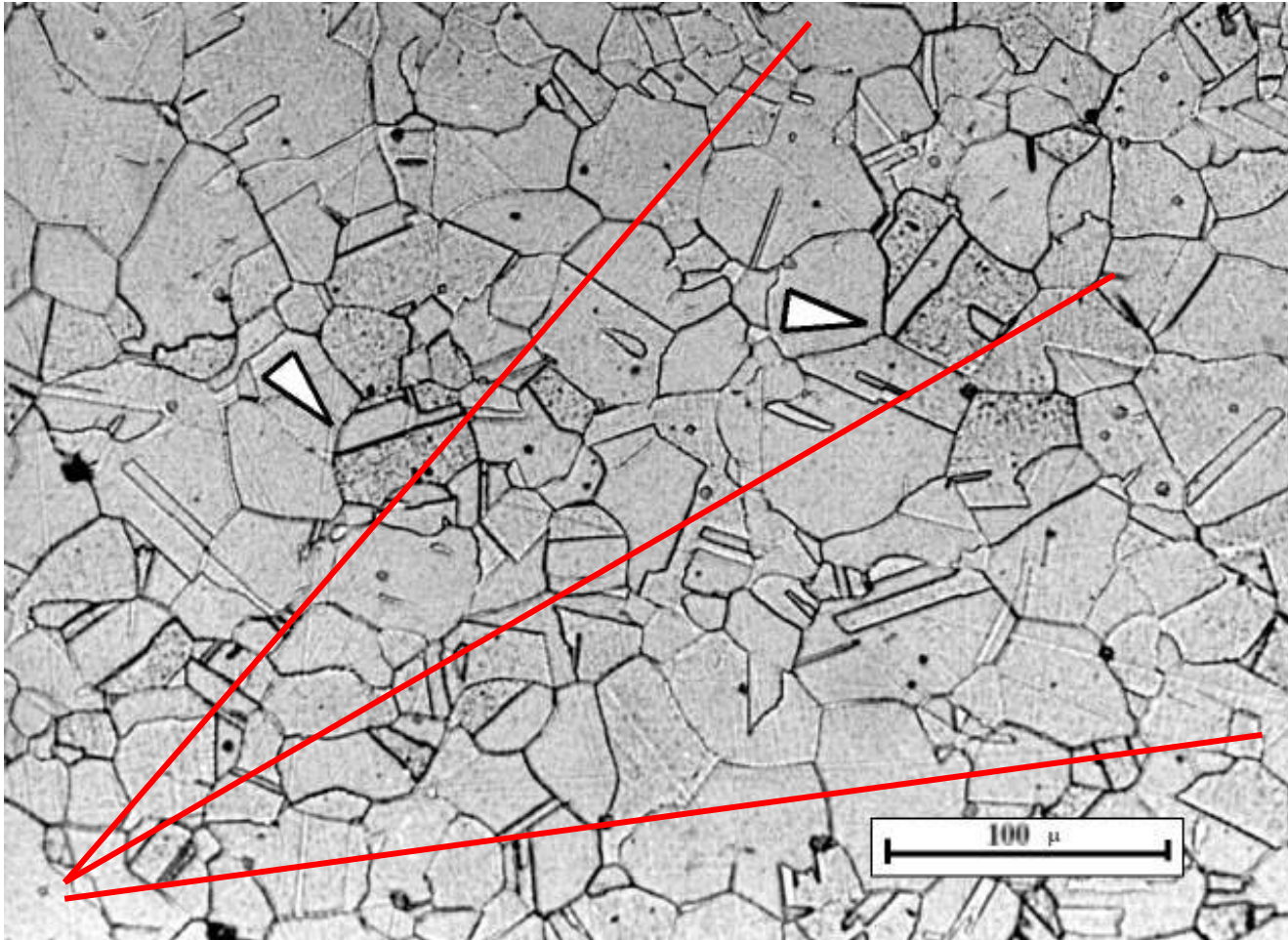
However, many softwares use image compression algorithms such as jpg, TIFF, etc.



Example image



How to measure grain size ?



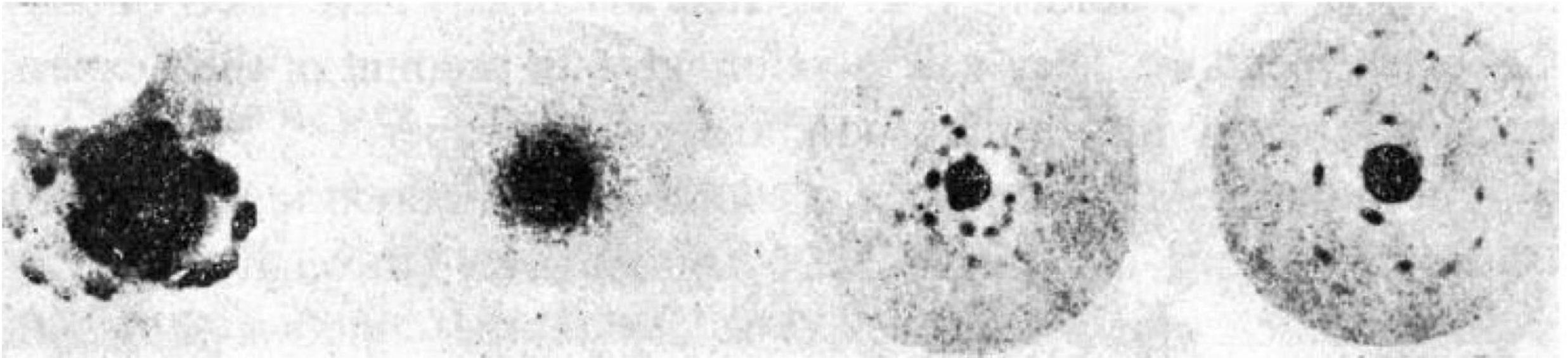
INTERCEPT PROCEDURE:

1. Draw random oriented lines
2. Find $\bar{l}$ - average intercept length
3. Grain size is a function of $\bar{l}$

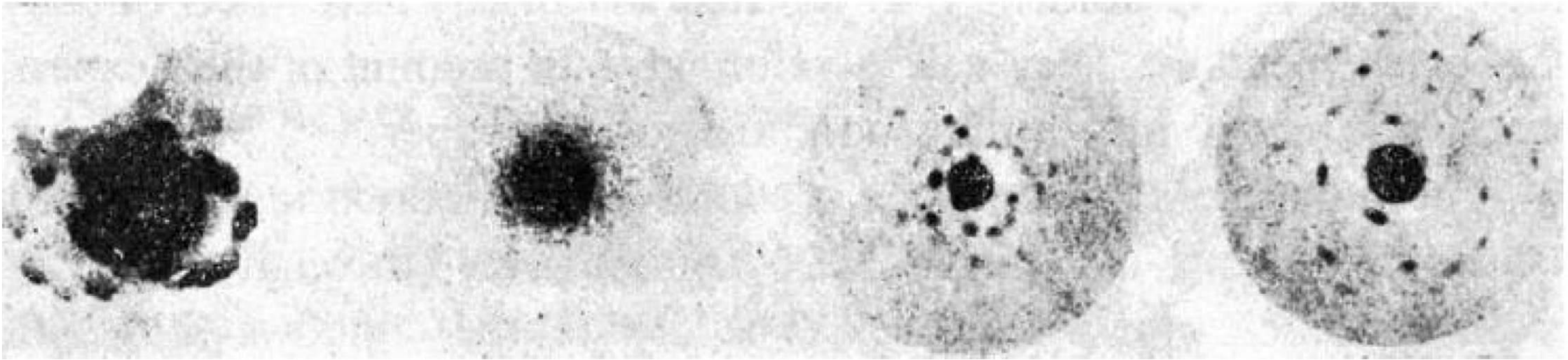
How do we find the mean segment length ?

Von Laue - Theoretical physicist claiming X-Rays are waves

Elements of XRD Analysis



First X-Ray pattern by Walter Friedrich and Paul Knipping on ZnS, 1912



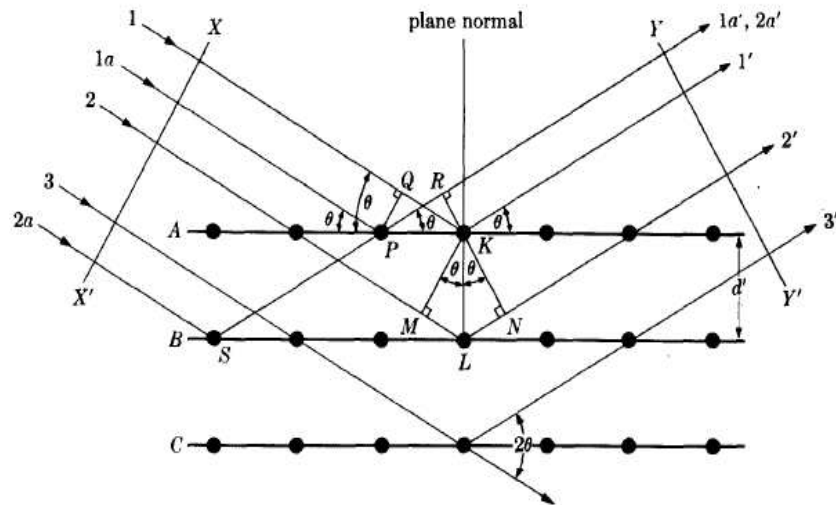
Laue: ZnS crystal is an infinite collection of cubes with Zn and S occupying alternate columns!

L. and W. Bragg: Look there are spots which are not so bright! – There should also be atoms on the walls of the cubes!

H. Armstrong: 'This is repugnant to sense! - Absurd to nth degree. It seems you will say that the rock salt does not contain molecules of 'NaCl' but a simple extension of N-Cl atoms?

Xray study components:

$$\lambda = 2d_{nh,nk,nl} \sin \theta$$



$$\text{Intensity } I = |F|^2 p \left(\frac{1 + \cos^2 2\theta}{\sin^2 \theta \cos \theta} \right) e^{-2M}$$

$$F = \sum_1^n f_n e^{2\pi i(hu_n + kv_n + lw_n)}$$

Atomic scattering factor (f_n) and phase interference

Intensity along Bragg's angle $I \propto FF^*$

Other factors:

Polarization

Multiplicity

Lorentz

Absorption

Temperature

Somewhere around two-dozen nobel prizes for work on X-Rays. Venki Ramakrishnan the latest among them for structure of R

Crystals and Structure

Solids in our universe is almost a singularity. It is a region with extreme densities of atoms with sharp boundaries with almost no densities.

More surprising is the occurrence of 'order' in the packing of atoms in the solids.

Therefore, a common analysis (characterization) of a solid material:

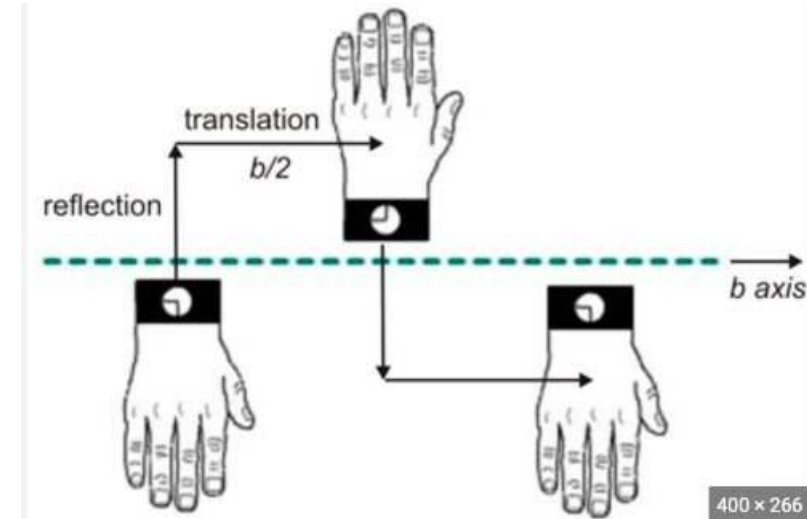
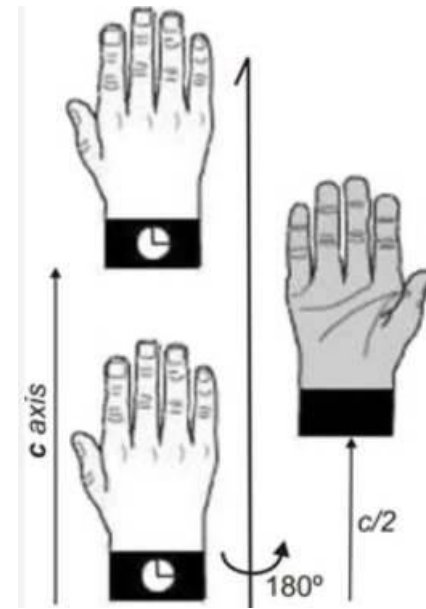
1. Investigation of the geometry of the repeating scheme
2. Investigation of the shape of the motif

Symmetry operations

Repetition upon

- Translation
- Rotation
- Screw motion – reflection + translation
- Reflection
- Glide-reflection – reflection + translation
- Roto-reflection

A complete set of combinations of these symmetry operations leads to 230 space groups.

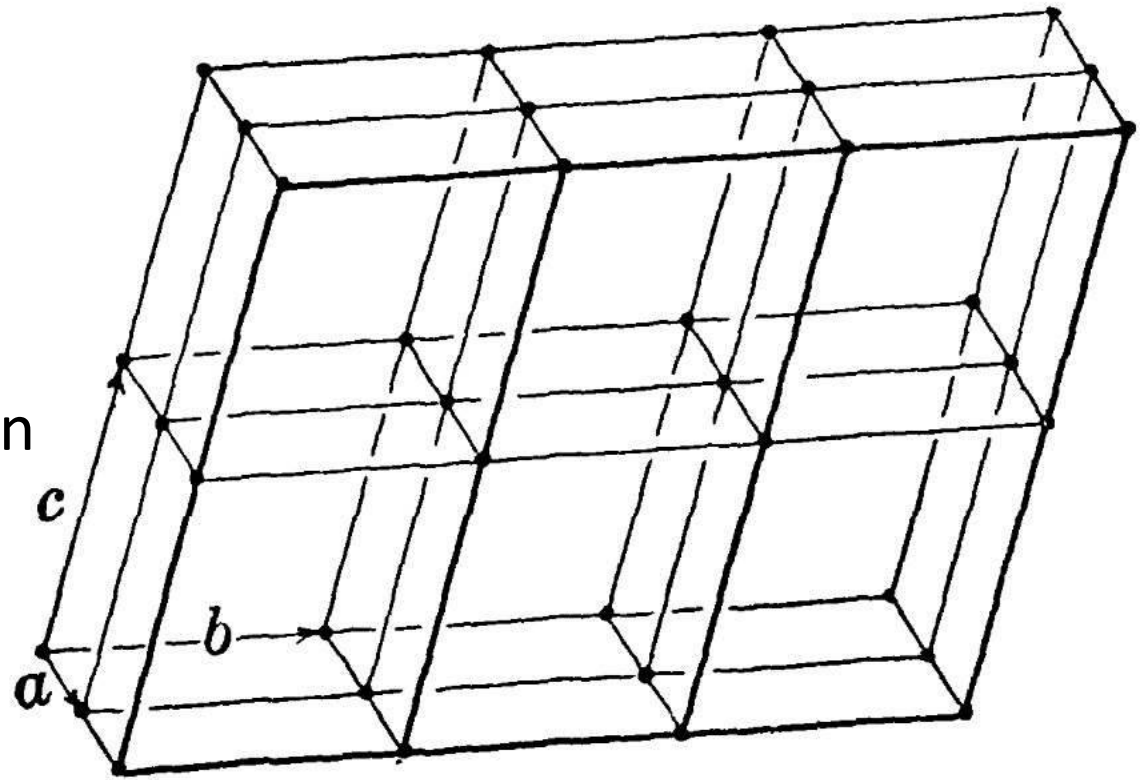


A crystal system may or may not have many of these symmetry operations; but by definition should at least have translation!

Translation is a vector: A magnitude with direction

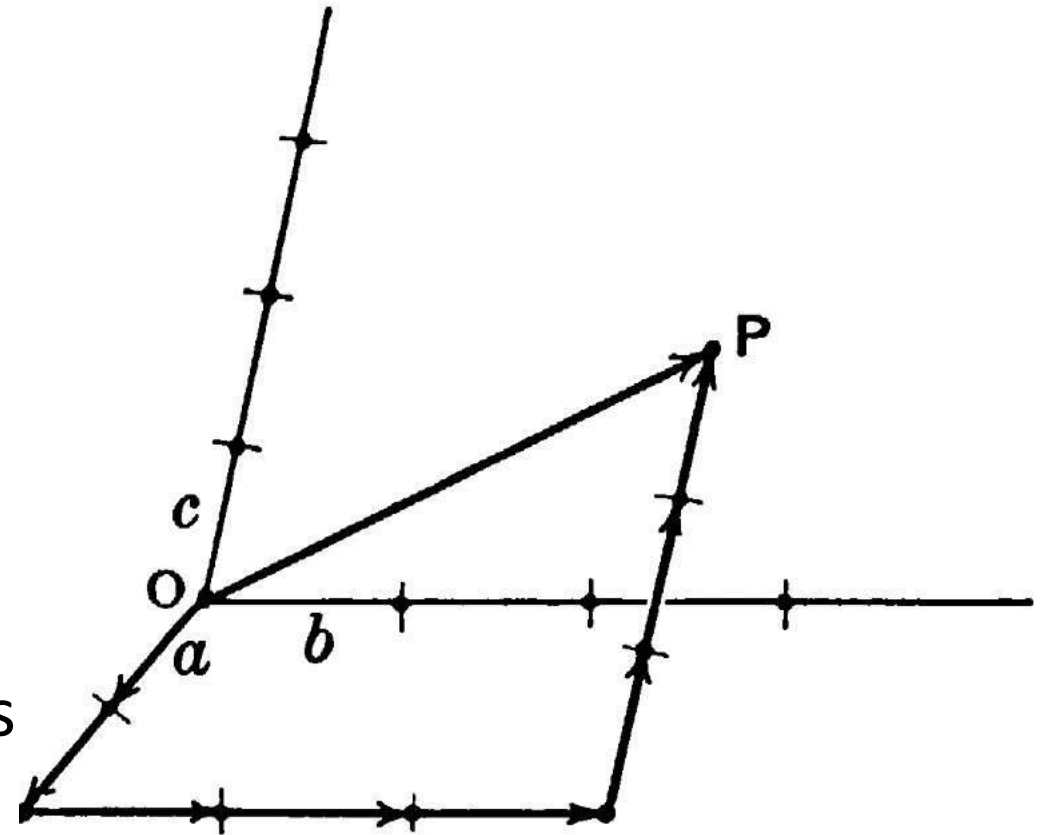
Any point in a crystal when translated with the vector will give an infinite set of points called as 'point space lattice'

An enclosed volume that spans the space in the crystal is called a cell; and a cell which encloses only one lattice point is called as primitive cell.



- Given a point lattice,
with a translational vector $O\vec{P}$

We define three vectors and three integers



$$O\vec{P} = u\vec{a} + v\vec{b} + w\vec{c}$$

Represented by $[u,v,w]$

Crystal Systems

All translational vectors in 3D space is described by 3 unique non-collinear vectors:

$$\vec{a}, \vec{b} \text{ and } \vec{c}$$

System	Axial lengths and angles	Bravais lattice	Lattice symbol
Cubic	Three equal axes at right angles $a = b = c, \alpha = \beta = \gamma = 90^\circ$	Simple Body-centered Face-centered	P I F
Tetragonal	Three axes at right angles, two equal $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$	Simple Body-centered	P I
Orthorhombic	Three unequal axes at right angles $a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$	Simple Body-centered Base-centered Face-centered	P I C F
Rhombohedral*	Three equal axes, equally inclined $a = b = c, \alpha = \beta = \gamma \neq 90^\circ$	Simple	R
Hexagonal	Two equal coplanar axes at 120° , third axis at right angles $a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	Simple	P
Monoclinic	Three unequal axes, one pair not at right angles $a \neq b \neq c, \alpha = \gamma = 90^\circ \neq \beta$	Simple Base-centered	P C
Triclinic	Three unequal axes, unequally inclined and none at right angles $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ$	Simple	P

Rational plane

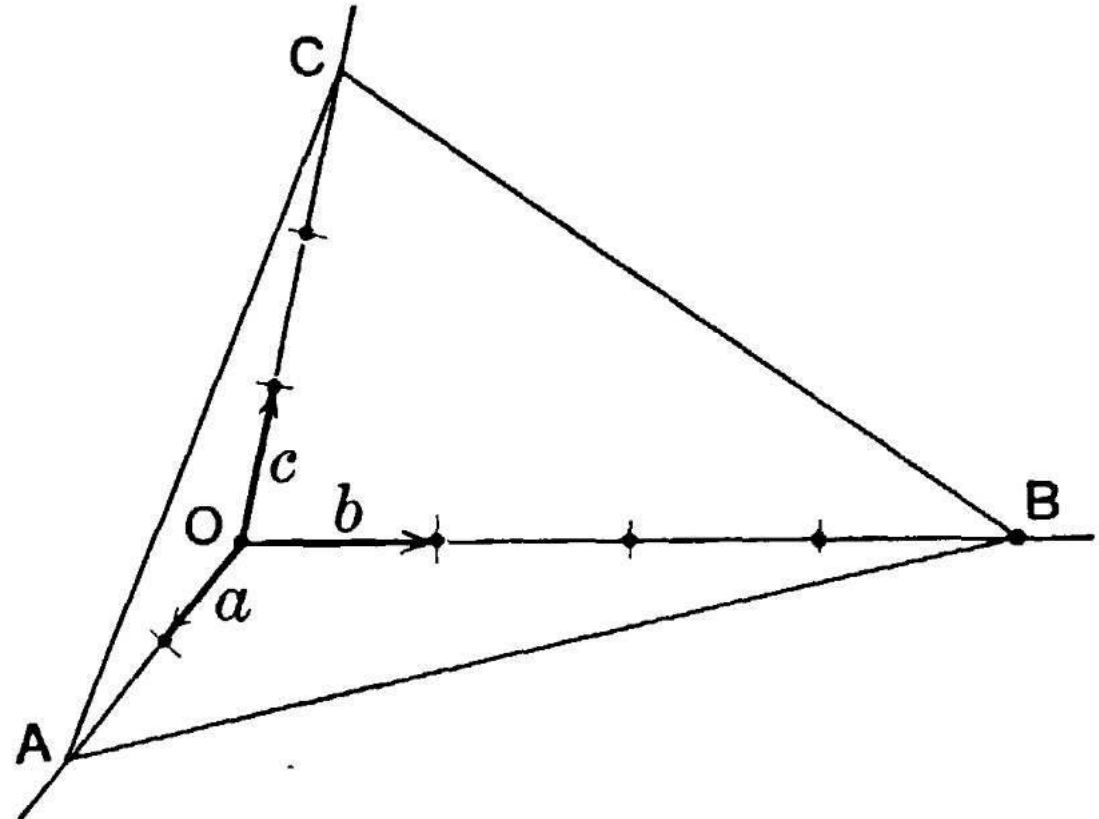
A plane containing lattice points not all collinear.

If the plane intercepts $\vec{a}$ axis at A units, $\vec{b}$ direction at B units

It can be written in the intercept form as

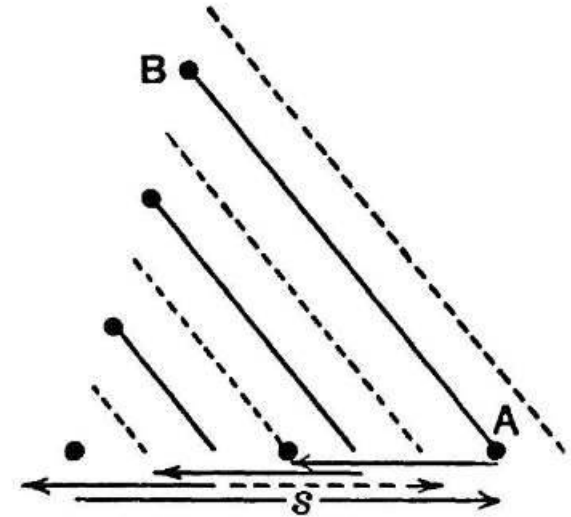
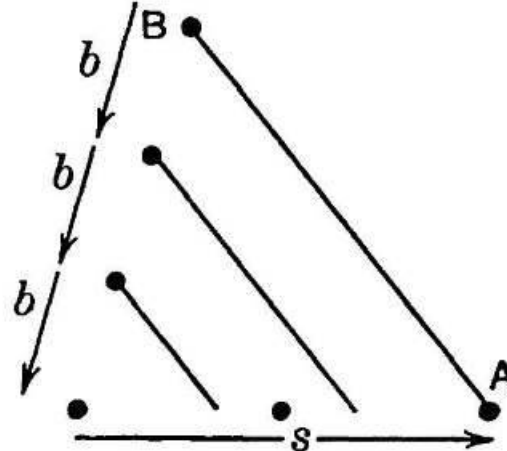
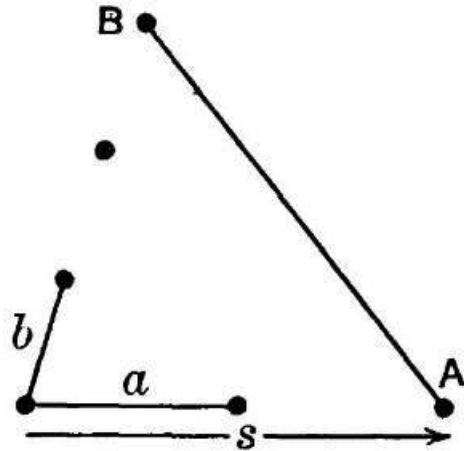
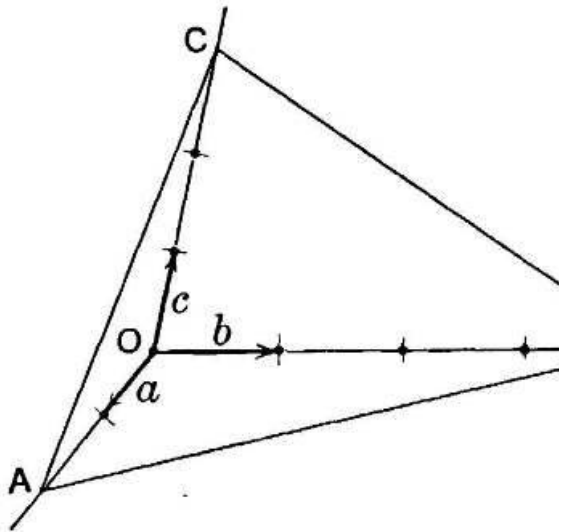
$$\frac{x}{A} + \frac{y}{B} + \frac{z}{C} = 1$$

* x,y and z are along the $\vec{a}$, $\vec{b}$ and $\vec{c}$ directions.



Parallel planes

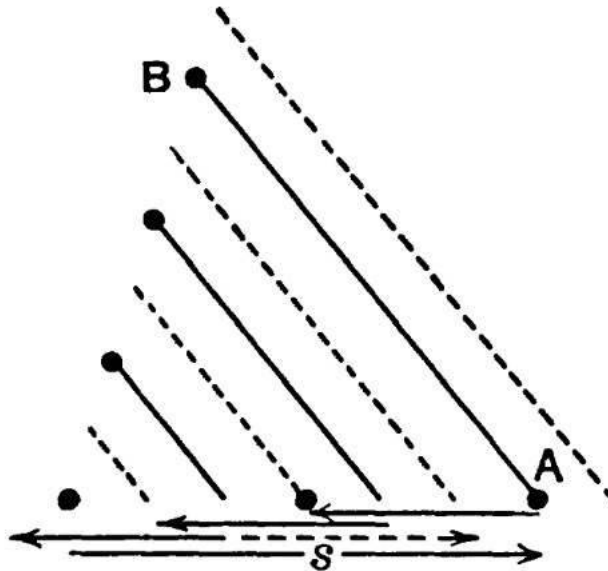
- How many (rational) planes parallel can we construct with the previous plane ?



Lets look at this in 2 dimensions

So, if $A = 2\vec{a}$ and $B = 3\vec{b}$ and $C = 4\vec{c}$

- In 2D, there are $2 \times 3 = 6$ lines between origin and the said plane.
- In 3D, you will have $A \times B \times C$ lines, if there are no common factors.



- Lastly,

If A, and B has a common factor t

Then the number of lines between AB and origin is $A \times B/t$

Following this, If B and C has common factor r, and C and A has common factor s.

The total number of planes between ABC and origin is $ABC/(rst)$

Length of the intercept of the first plane from origin = $L/(ABC/rst)$

Equation of this plane is then:

$$\frac{\frac{x}{A}}{\frac{ABC}{rst}} + \frac{\frac{y}{B}}{\frac{ABC}{rst}} + \frac{\frac{z}{C}}{\frac{ABC}{rst}} = 1$$

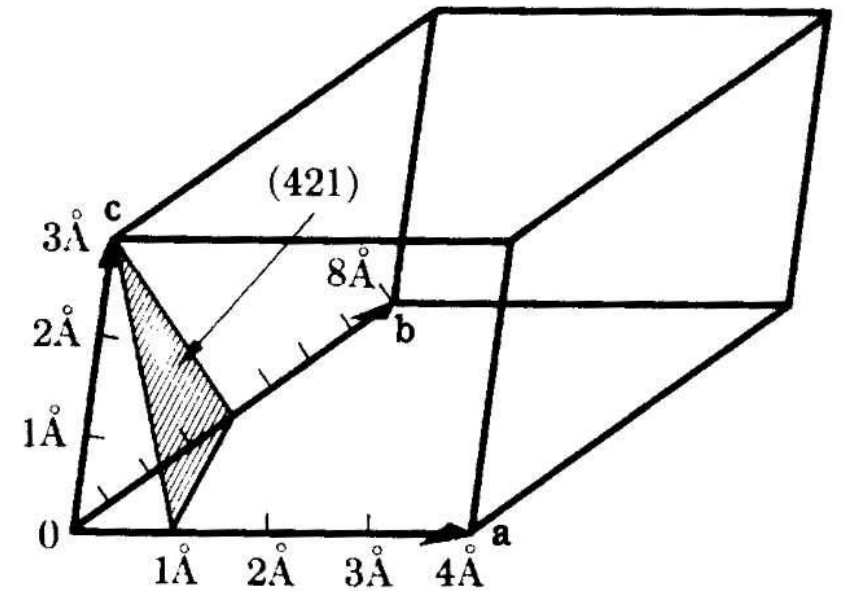
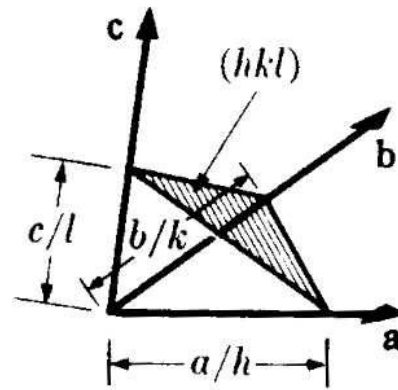
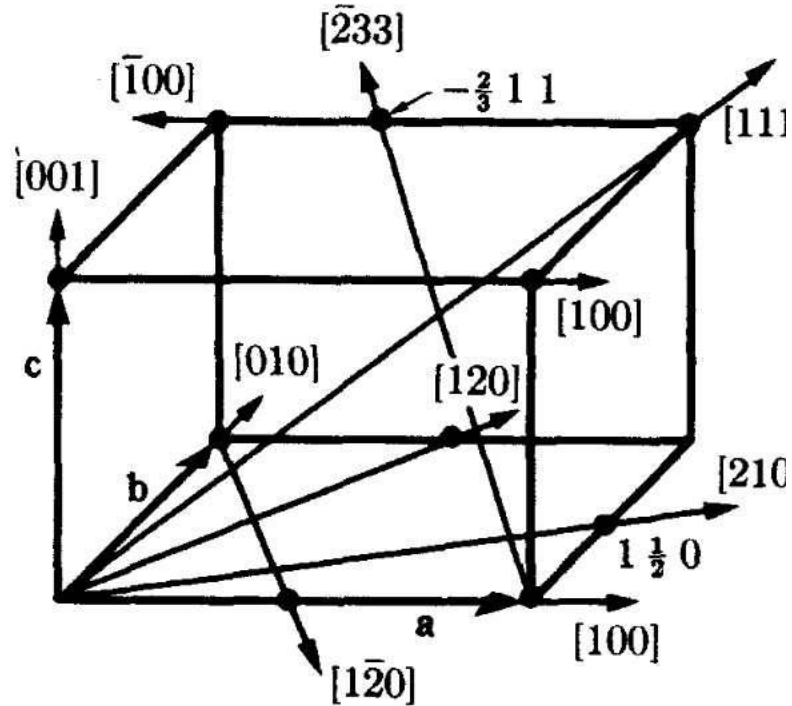
Recasting this:

$hx + ky + lz = 1$ is the equation of the first plane from the origin

Where $h = \frac{BC}{rst}$ and $k = \frac{AC}{rst}$ and $l = \frac{BA}{rst}$

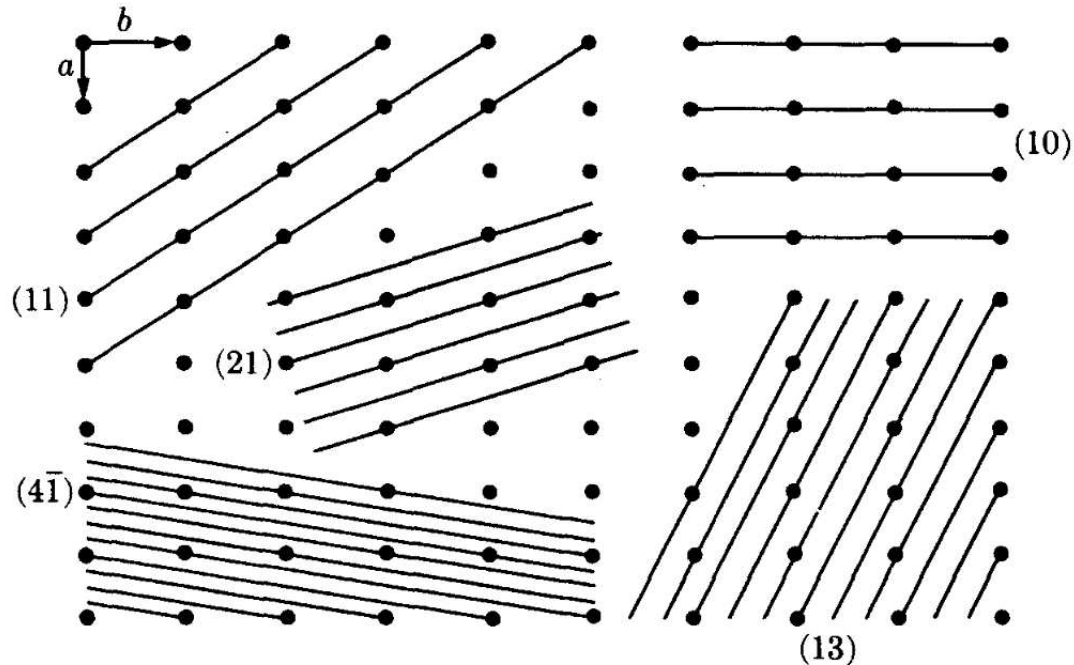
So, we represent this set of planes with (h,k,l) whose intercepts are 1/h, 1/k and 1/l

Examples of indices



For a cubic system, the set of planes designated by (h,k,l) are separated by

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$



Larger h,k,l values, smaller interplanar spacing

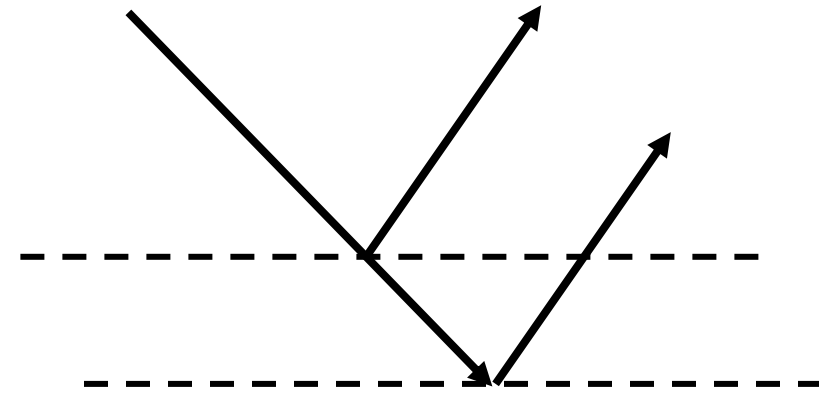
Remember, [h,k,l] will always be normal to (h,k,l) in cubic systems

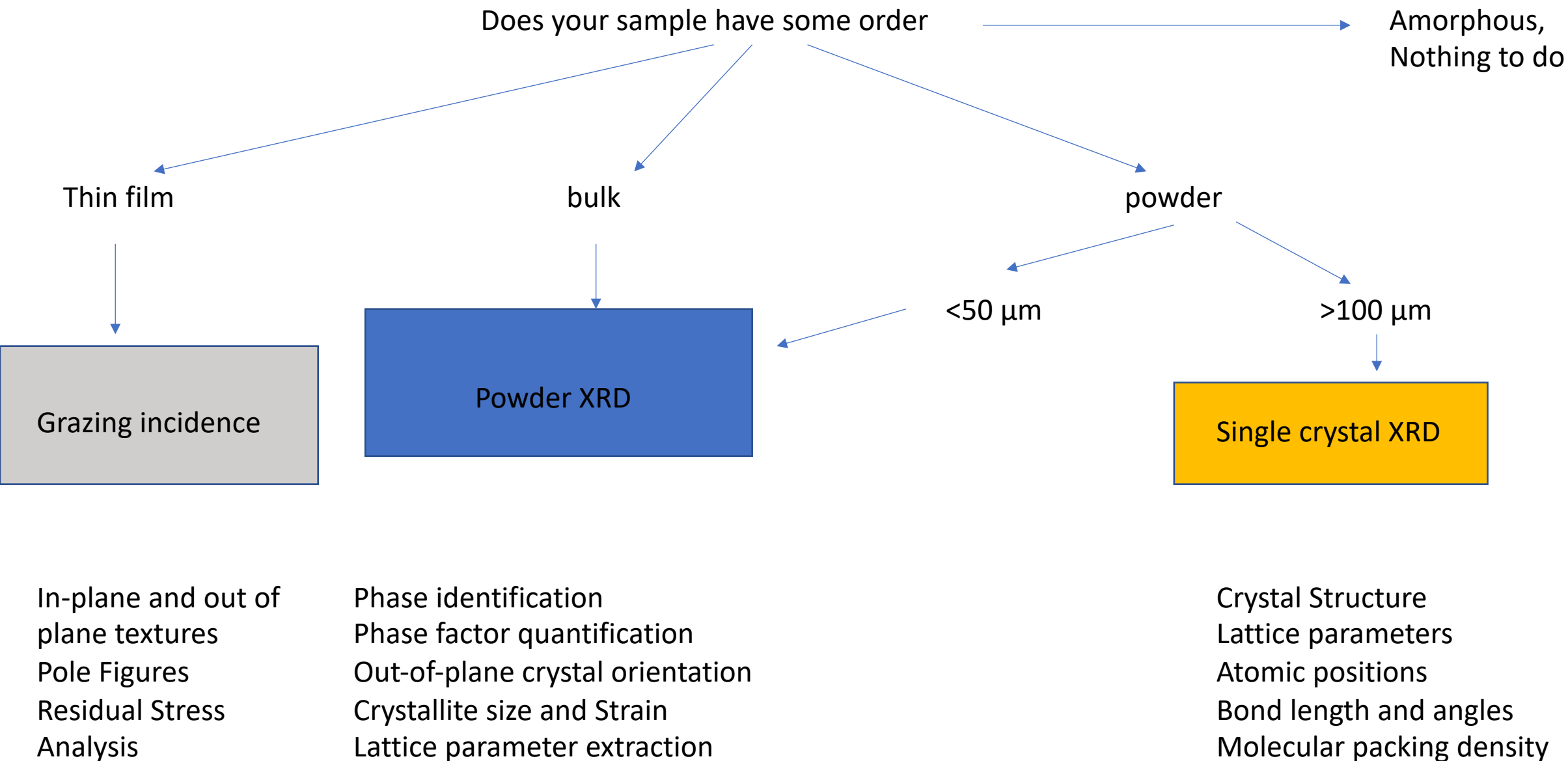
The Idea

- In crystals with atomic ordering,

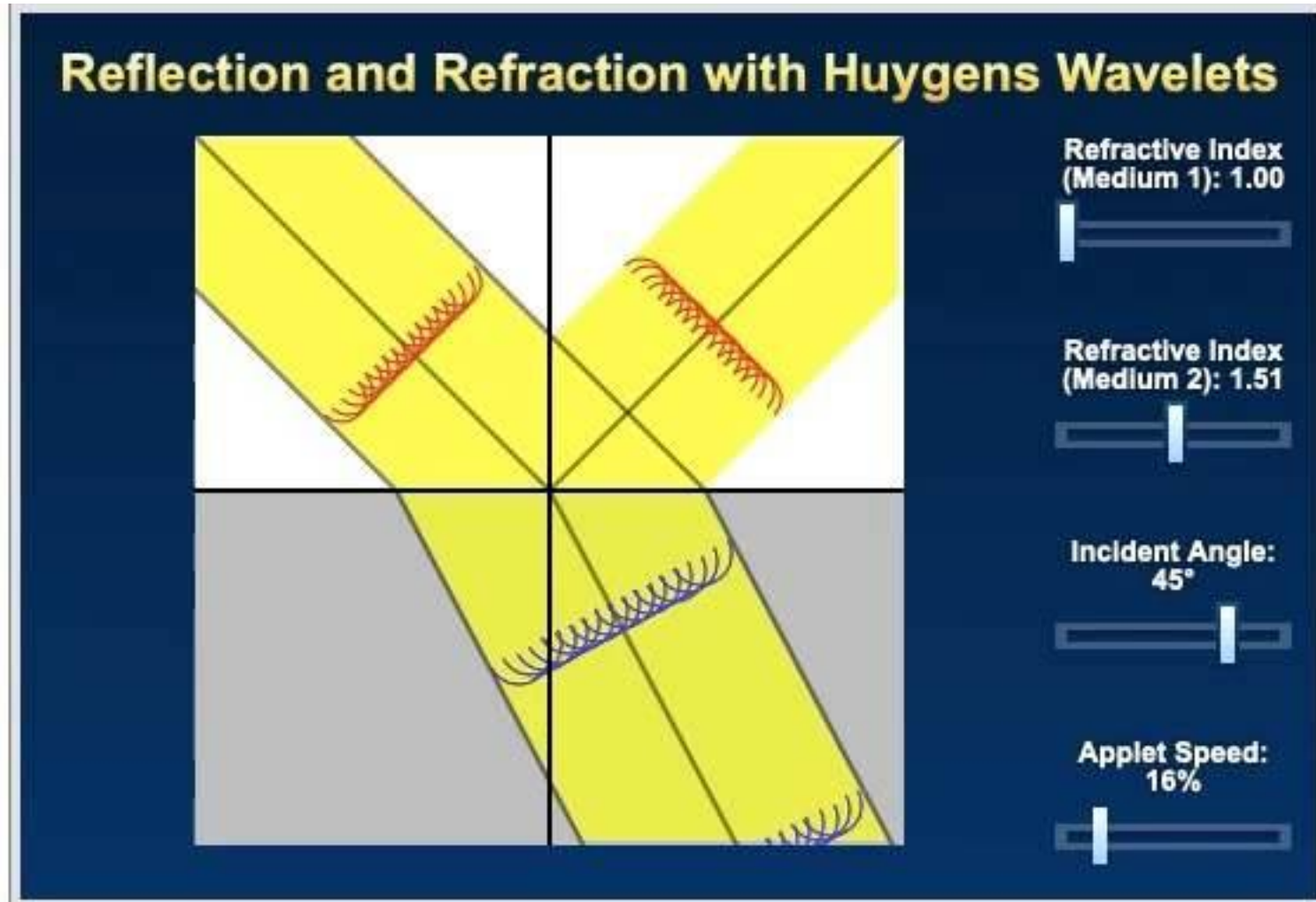
Radiations reflected by independent planes have path differences proportional to the interplanar spacing!

If the path difference is integer multiple of wavelength, then 'diffraction' information can be used to inspect the interplanar spacing!





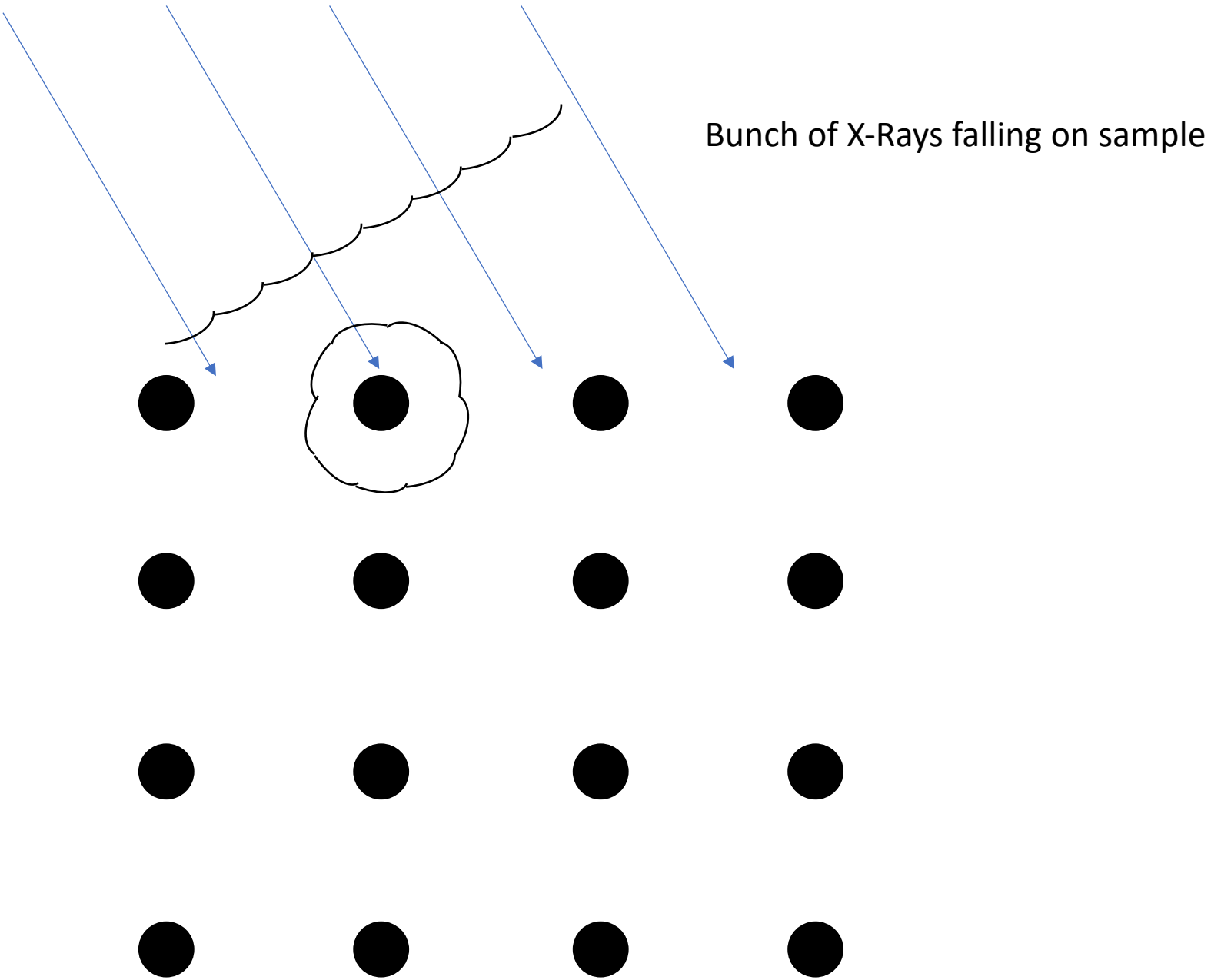
Normal Reflection and refraction



A normal reflection is observed when wavelets which are generated at the region of illumination destructively intersect in all direction except the one which has the name angle with the surface normal.

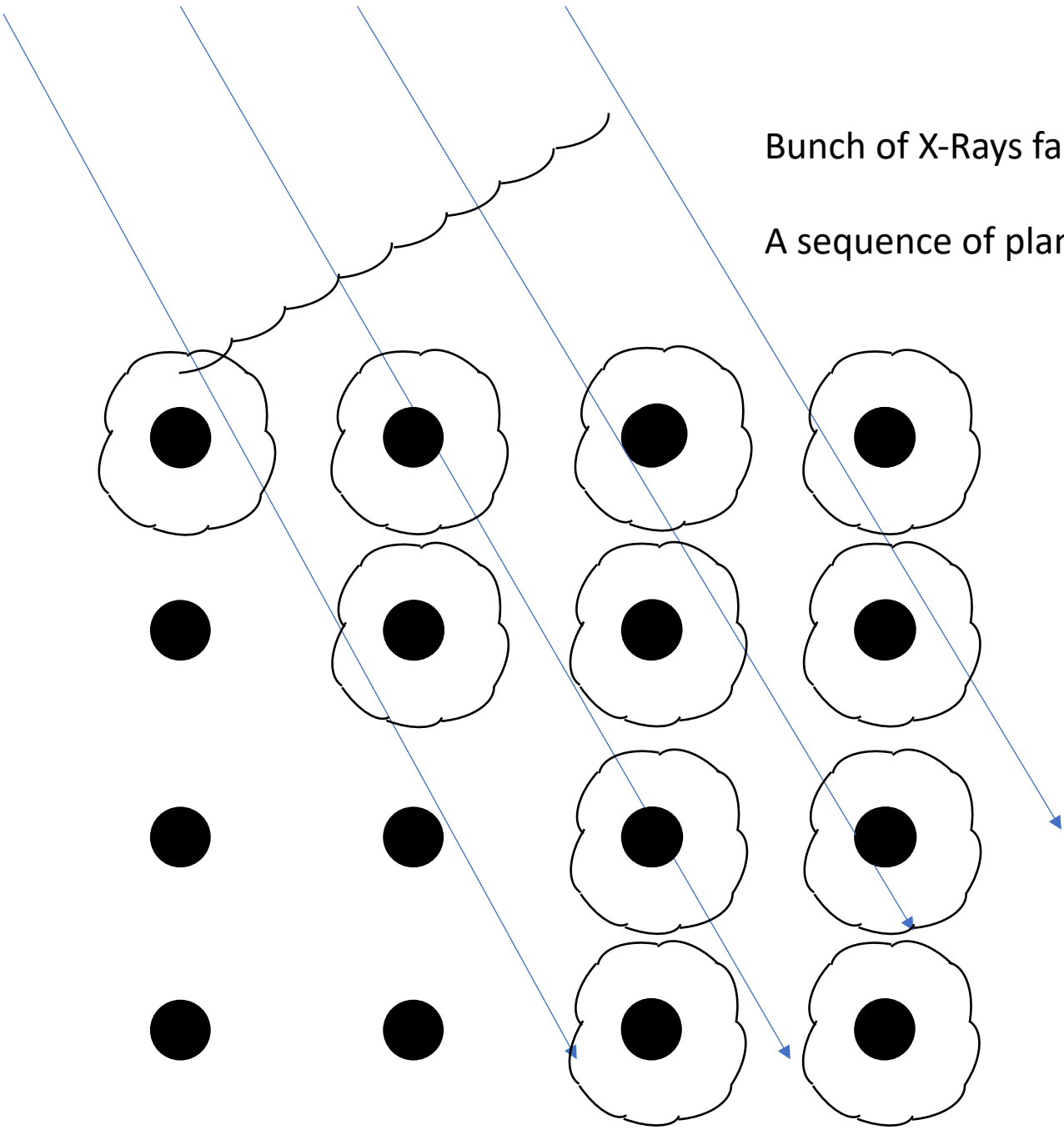
The surface atoms of the interface acts as secondary light sources.

Consider a periodic lattice



Bunch of X-Rays falling on sample

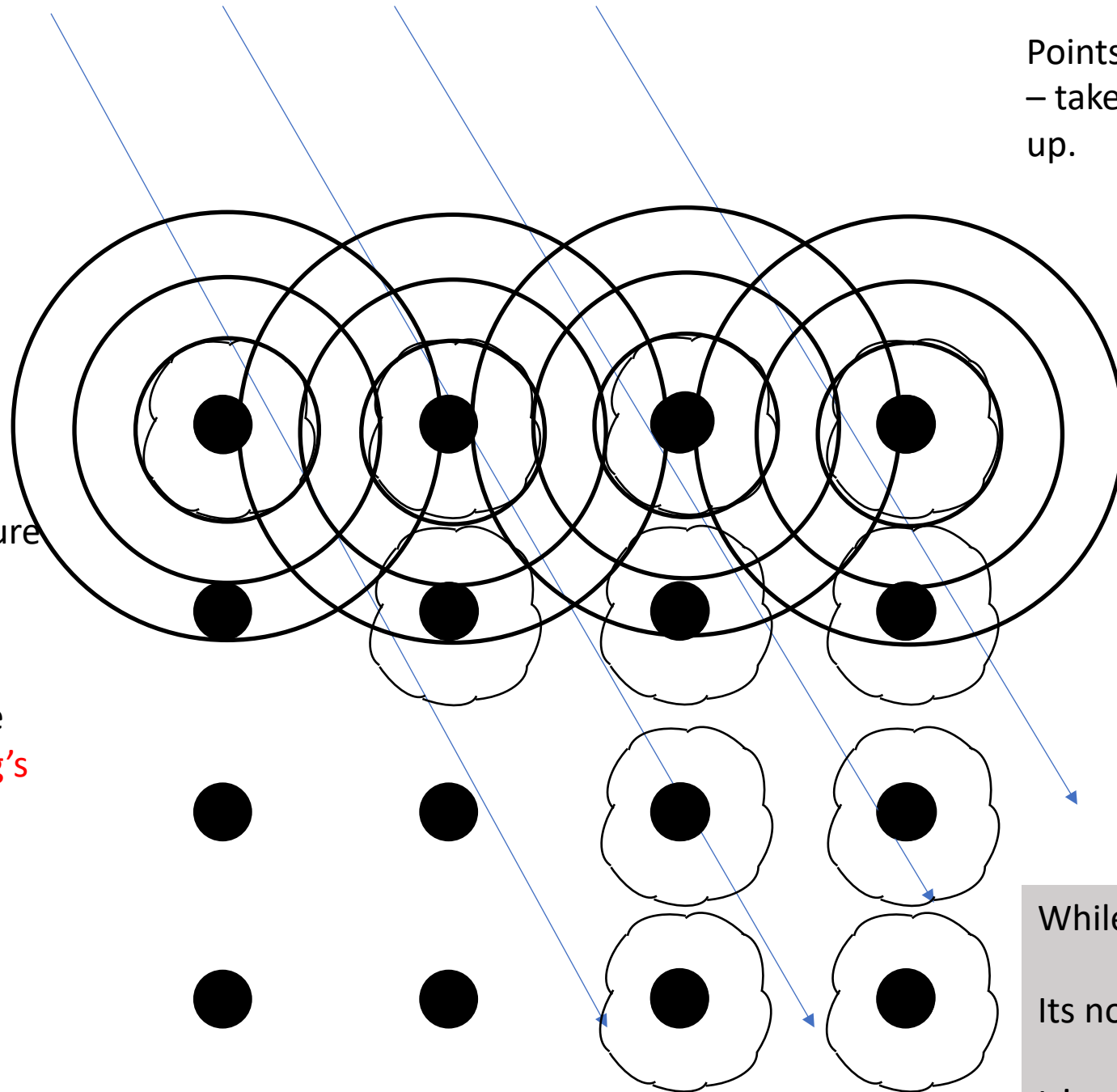
Consider a periodic lattice



Bunch of X-Rays falling on sample.
A sequence of planes act as light sources.

Consider a periodic lattice

Points of intersection of the circles
– taken here to be say crests, add
up.



While the surface atoms ensure
that the reflection law is
satisfied,

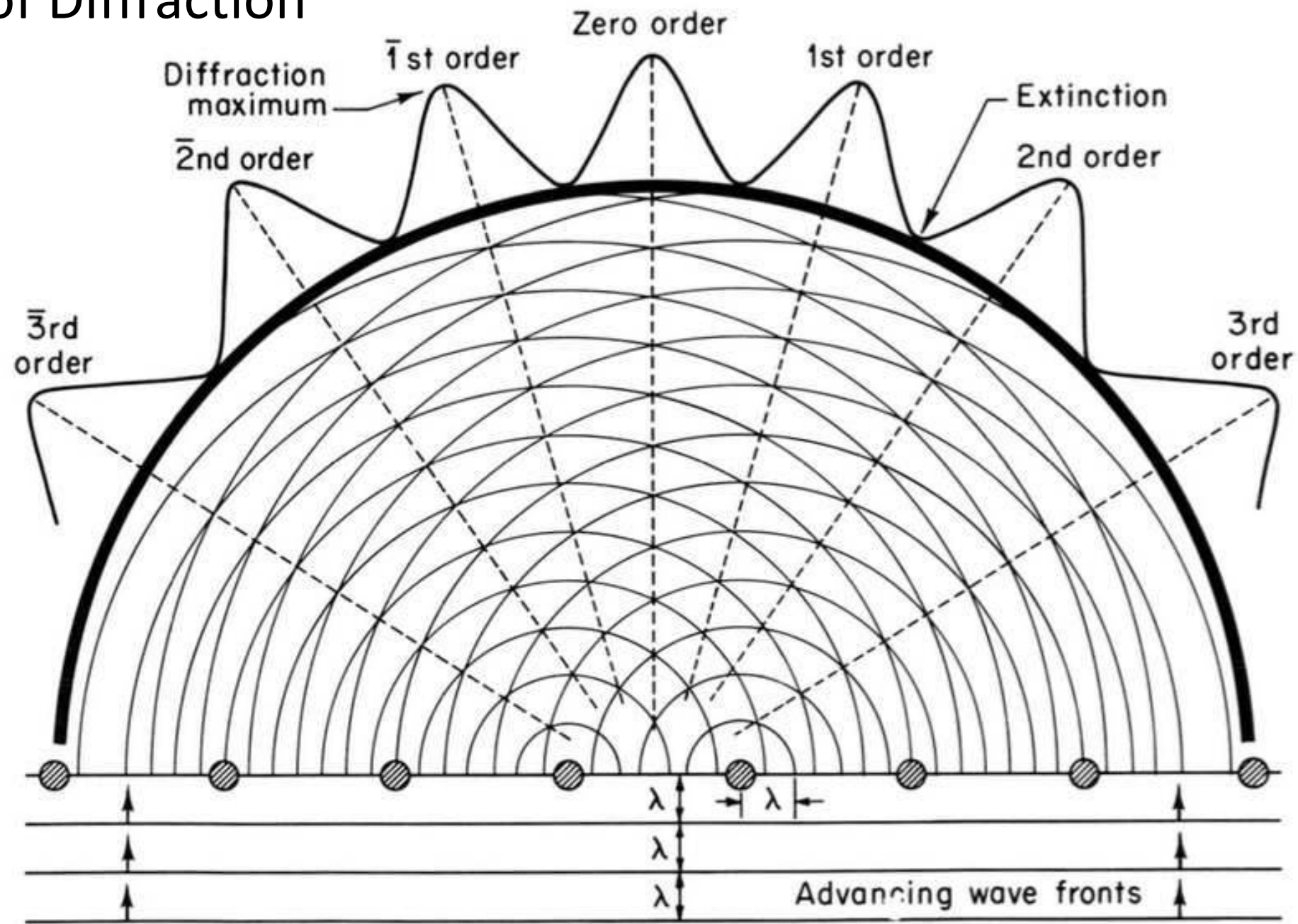
The subsequent planes in the
sample ensure that **the Bragg's
law is satisfied**

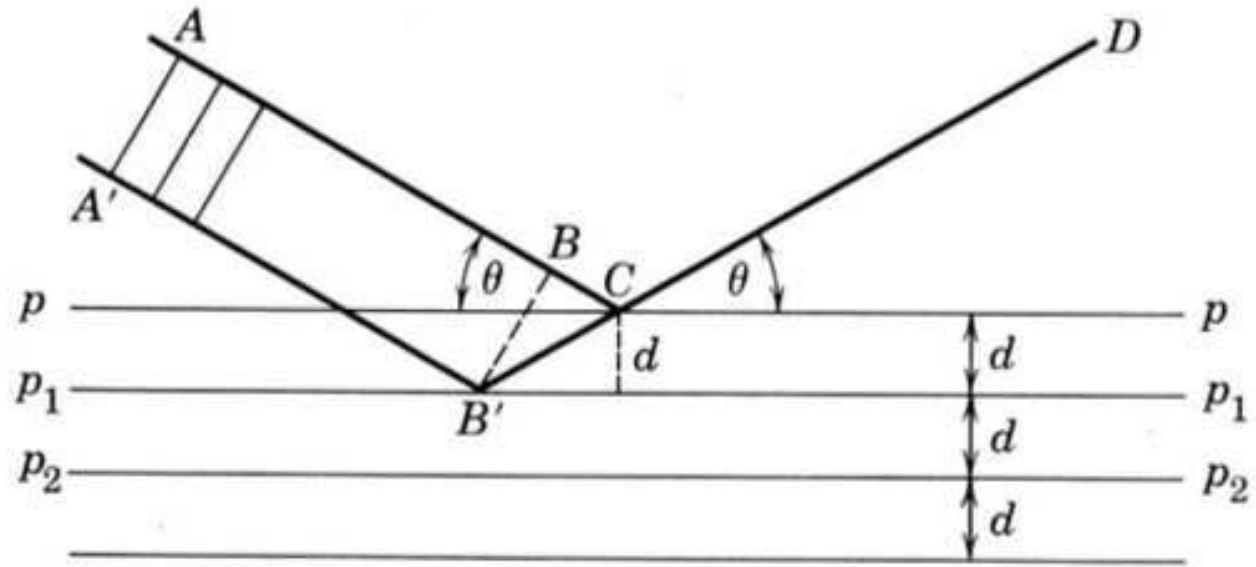
While, its is indeed reflection.

Its not purely reflection.

It's a diffraction!

Orders of Diffraction





Path difference:

$$\begin{aligned}
 B'C - BC &= \frac{d}{\sin \theta} [1 - \cos 2\theta] \\
 &= \frac{d}{\sin \theta} 2 \sin^2 \theta
 \end{aligned}$$

The condition for constructive interference: $B'C - BC = n\lambda$

$$2d \sin \theta = n\lambda$$

The essential condition for diffraction

So, we write the Bragg's law for reflection as

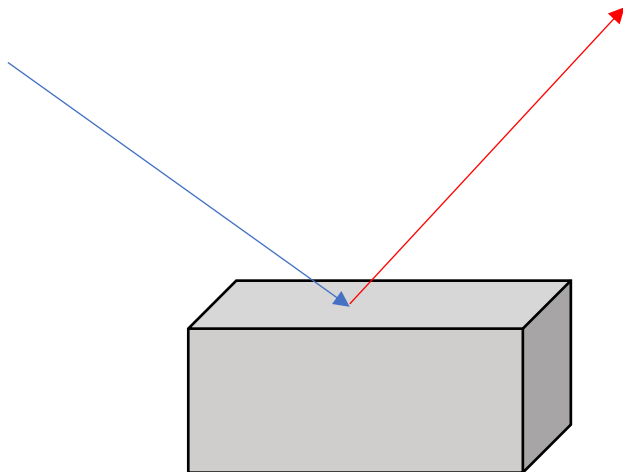
$$\lambda = 2d_{nh,nk,nl} \sin \theta$$

For cubic systems

$$\text{Where } d_{nh,nk,nl} = \frac{a}{\sqrt{n^2h^2+n^2k^2+n^2l^2}} = \frac{d_{h,k,l}}{n}$$

Remember h,k,l are inverse of planar intercepts.

Higher orders (nk,nl,nh) have a smaller d-spacing



For every diffracted beam with angle θ , we associate a set of planes with spacing d .

From here to identification of the plane is the idea of solving the structure.

Bragg's law dictates the angle at which diffracted beam can be found.

Can we estimate the intensity ?

Diffracted beam intensity depends on

1. Polarization factor
2. Lorentz/velocity factor
3. Atom scattering factor
4. Scattering inside a unit cell (structure factor)
5. Multiplication factor
6. Lorentz factor
7. Absorption factor
8. Temperature factor

Electron Scattering factor

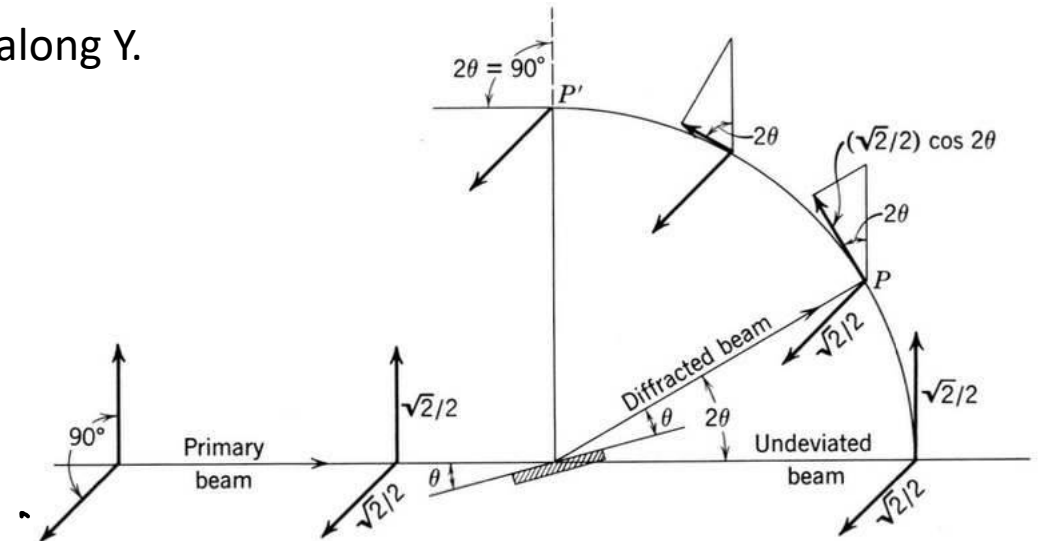
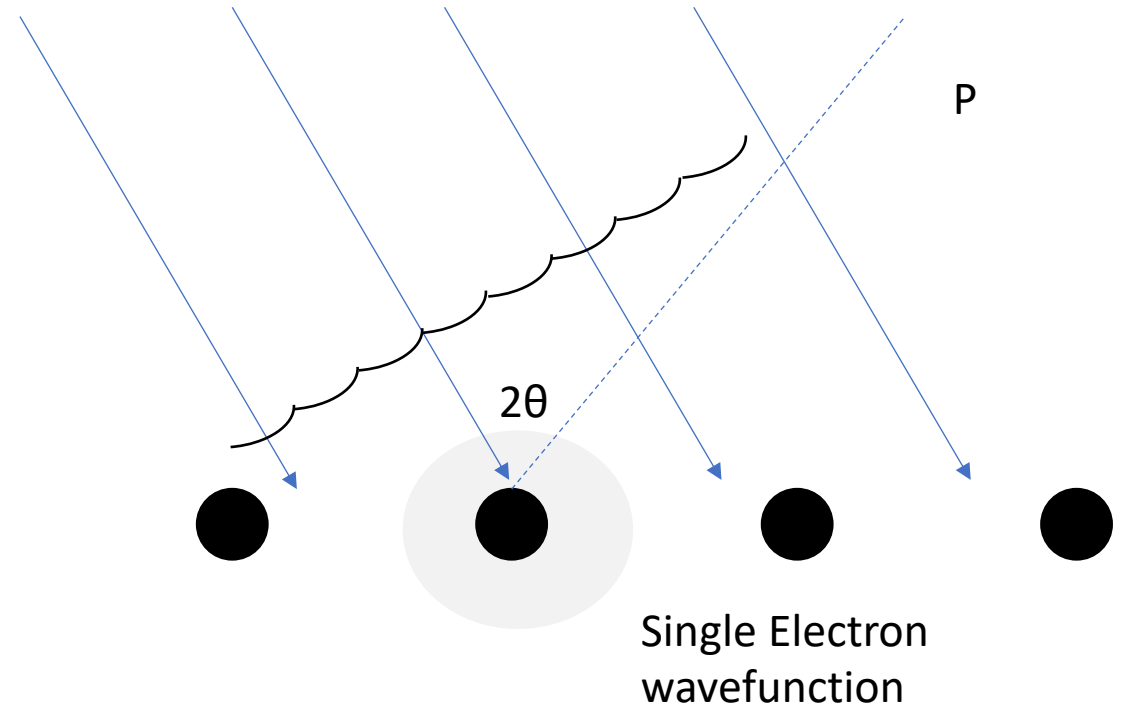
Polarization factor:

Incident beam of light is unpolarized.

However, the scattered light is polarized at higher angles.

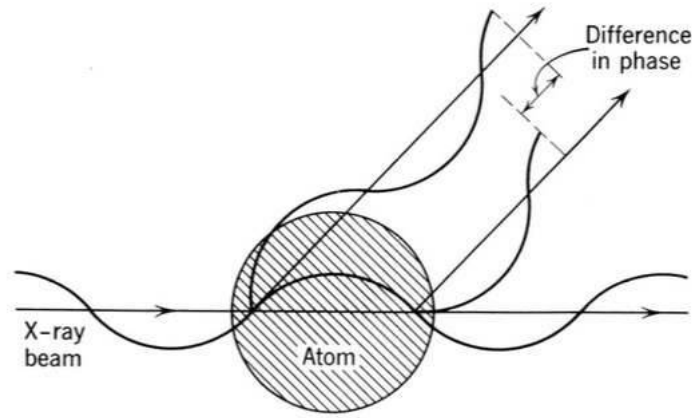
Reason:

If electric field in incident light is along y axis,
X-Rays scattered close to y-axis direction will have reduced E field along Y.



$$I(2\theta) = \frac{I_0 K}{2r^2} (1 + \cos^2 2\theta)$$

Atomic Scattering Factor

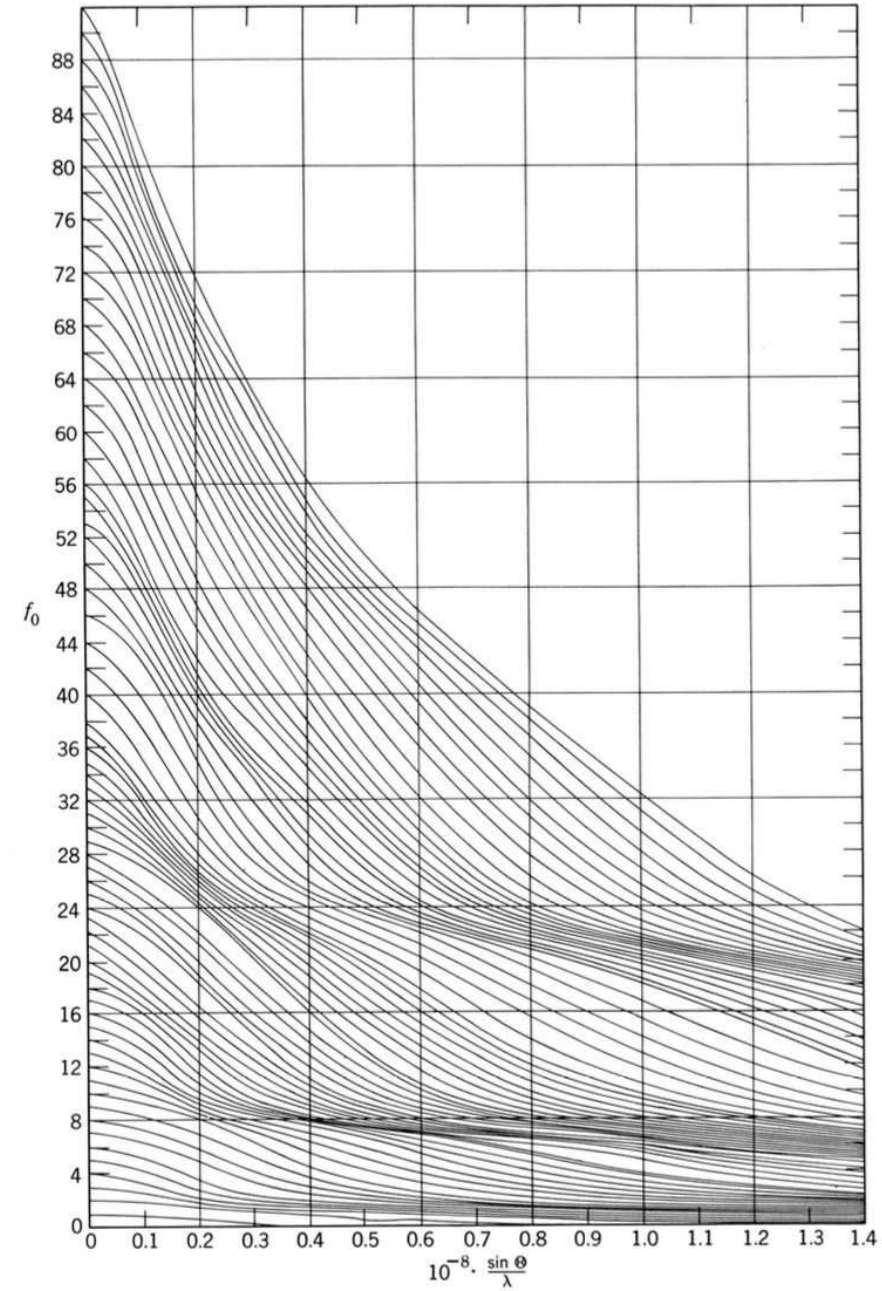


Atoms are not point sources!

If there are N electrons in an atom, the efficiency of all those electrons contributing to X-Ray scattering is represented by f_0

f_0 - ratio of scattering from the designated atom to that of a single electron at the same point.

$f_0 = Z$ at very small angles. It reduces sharply at larger angles.



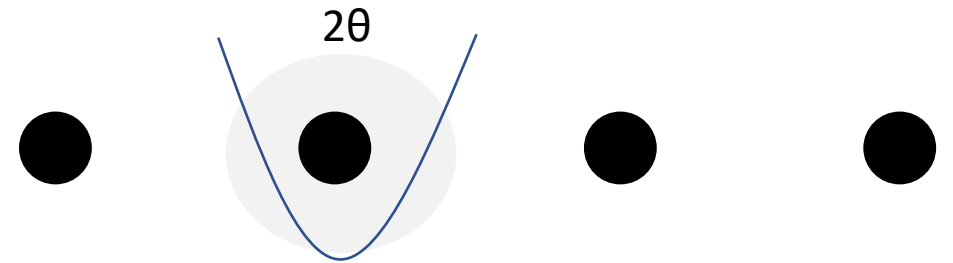
Lorentz Factor

The fraction of the sample interfacing with the X-Ray depends on θ

This is due to the slits, the fraction of crystals that can have their normals according to the Bragg's condition

$$\text{Lorentz Factor } L = \frac{1}{4 \sin^2 \theta \cos \theta}$$

Temperature Factor

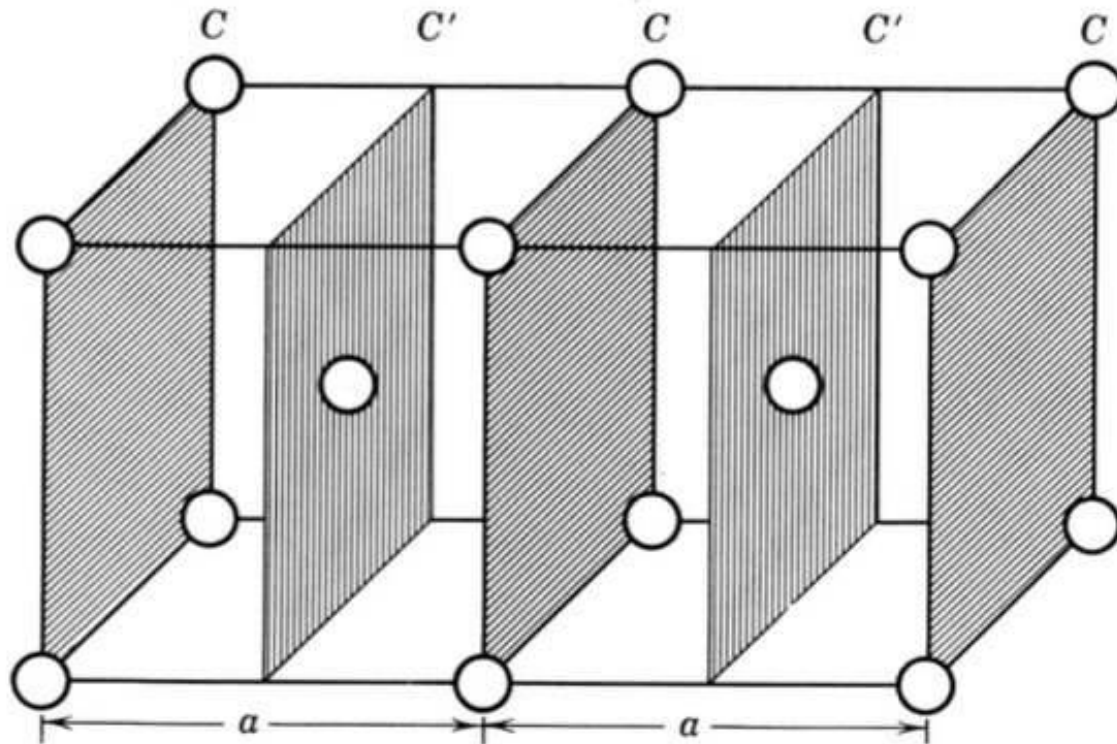


At any non-zero temperature, the atoms are vibrating about their mean position. Harmonic potential

This disturbs the interplanar spacing and hence the total intensity of radiations in constructive interference.

Therefore, the atomic scattering factor can be scaled as $f_N \rightarrow F_N e^{-M}$

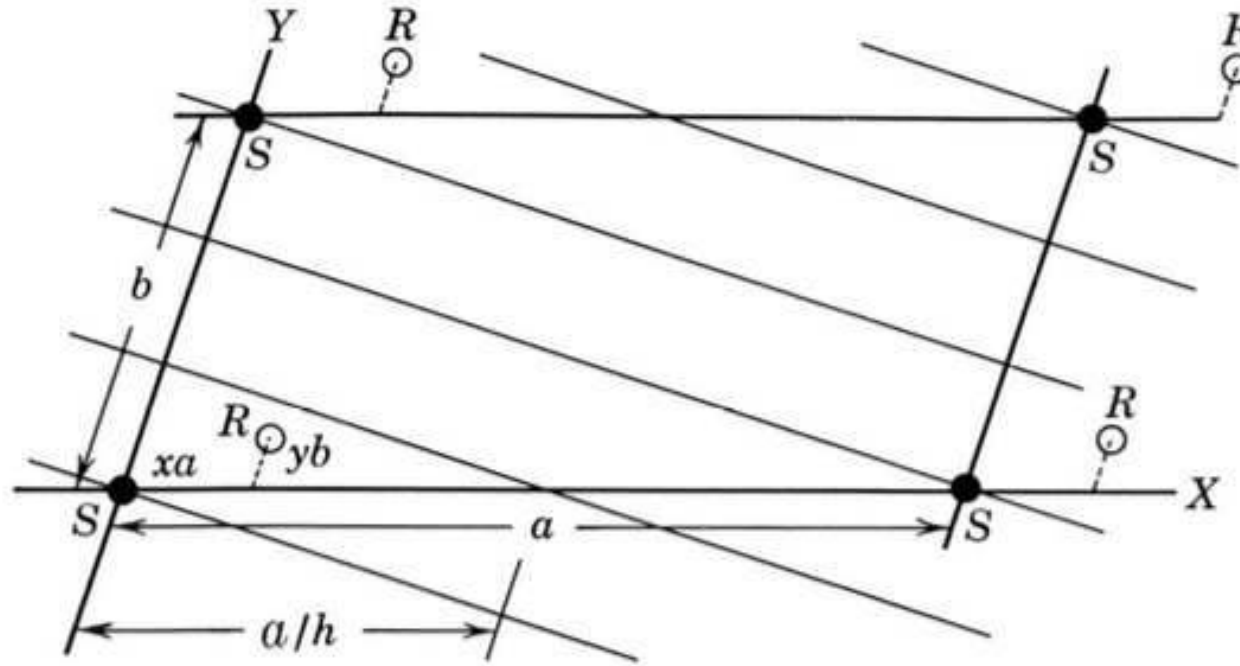
The curious case



Missing (100) planes ?

Table 1. Calculated 2θ angles for internal standards.
CuK α , $\lambda = 1.5405981 \text{ \AA}$

hkl	Tungsten $a=3.16524 \text{ \AA}$ $\pm .00004$	Silver $a=4.08651 \text{ \AA}$ $\pm .00002$	Silicon $a=5.430825 \text{ \AA}$ $\pm .000011$ (SRM 640a)
110	40.262		
111		38.112	28.443
200	58.251	44.295	
211	73.184		
220	86.996	64.437	47.304
310	100.632		
311		77.390	56.124
222	114.923	81.533	
321	131.171		
400	153.535	97.875	69.132
331		110.499	76.378
420		114.914	
422		134.871	88.033
511/333		156.737	94.955
440			106.712
531			114.096
620			127.550
533			136.900
444			158.644



Consider a simple example in 2D:

Atom S, separated by lattice constants **a** and **b**

For any plane with index (h, k) seen in the example as $(2, 3)$ – divides the lattice constants by the same number.

For constructive interference – a phase difference of 2π is needed along $\vec{a}$ direction in distance $\frac{a}{h}$
 a phase difference of 2π is needed along $\vec{b}$ direction in distance $\frac{b}{h}$

If we insert an additional atom at (x_a, y_a)

Phase difference added at Bragg's condition is : $x_a * \frac{2\pi}{a/h}$ along $\vec{a}$

We define the additional phase to be $\phi_a = 2\pi h * x_a/a$

In three dimensions, then the extra phase included due to an atom at x_a, y_a, z_a is

$$\phi_R = 2\pi(hx + ky + lz)$$

Where $x = x_a/a$ - is the lattice constant normalized x position of the new atom.

Structure Factor

Consider the situation where

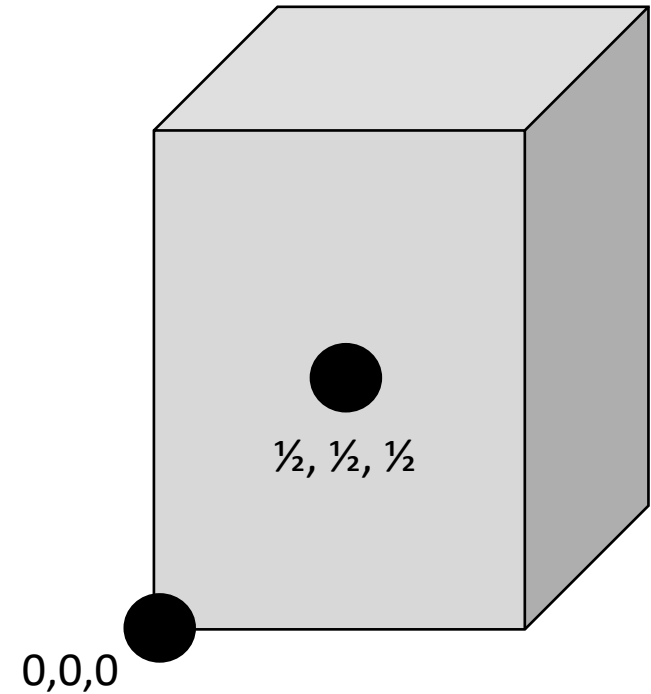
If there are n atoms in a unit cell;

With normalized offsets (u_i, v_i, w_i) from the cell origin.

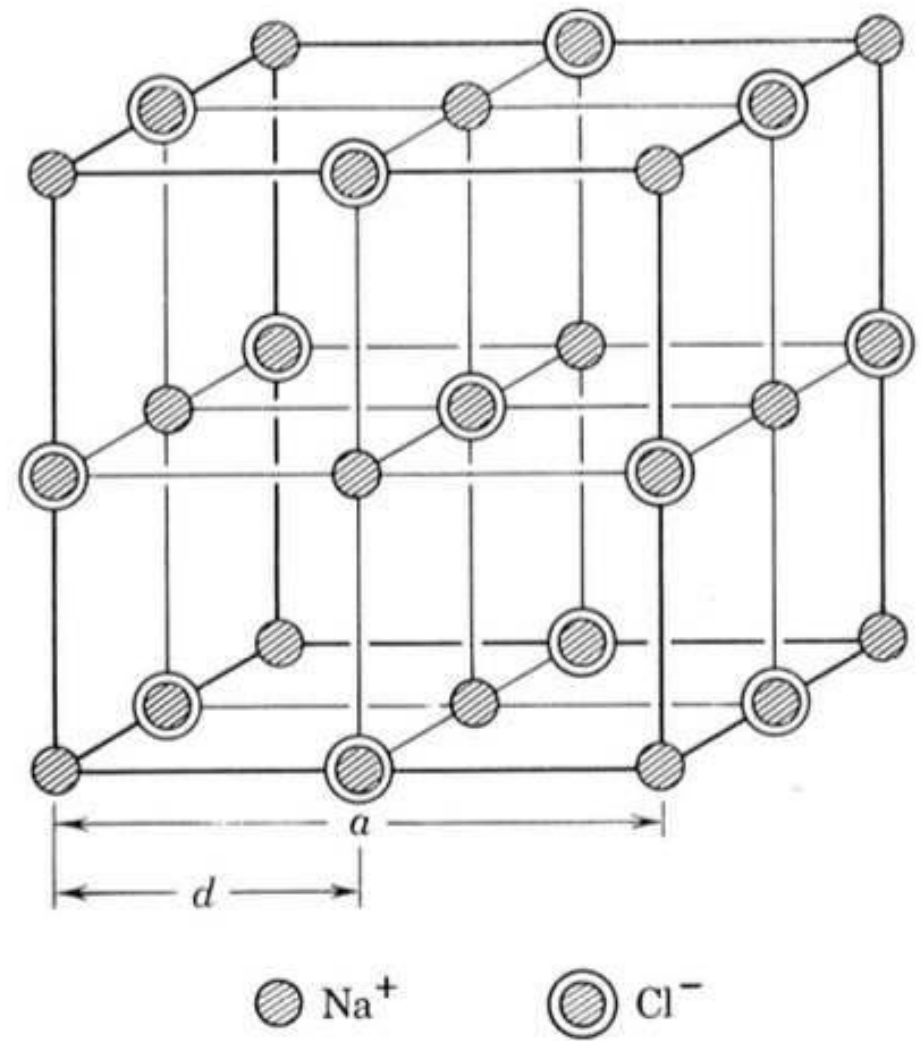
The phase difference $\phi_i = 2\pi(hu_i + kv_i + lw_i)$

Total phase difference $F = \sum_1^n f_n e^{i\phi_i}$

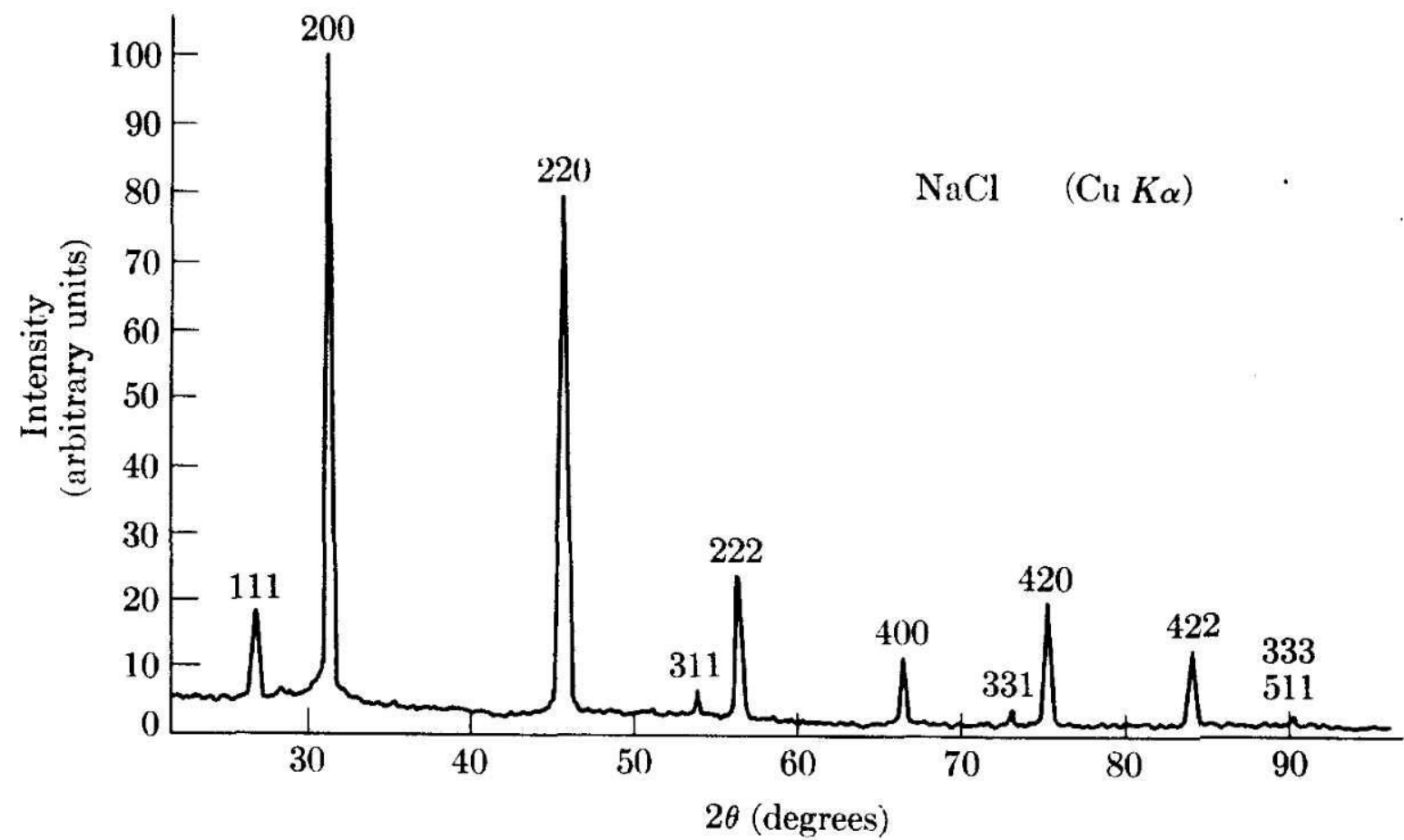
Therefore, for $h,k,l = 100$, the structure factor for BCC $\rightarrow 0$



Structure factor for NaCl:



X-Ray Measurements on NaCl



Multiplicity in polycrystal

In a polycrystalline film, different grains with varying orientation are possible.

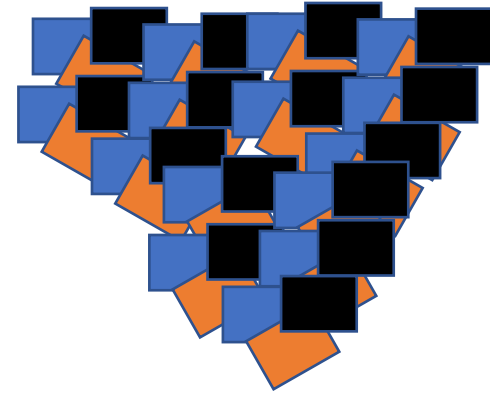
In a truly random mixture, $(100), (010), (001), (-1,0,0), (0,-1,0), (0,0,-1)$ will occur with equal probability.

However, independent of what is the orientation, all of them will diffract at same 2θ .

Thus, the multiplicity $p\{100\} = 6$

On similar lines the multiplicity of $P\{111\} = 8$

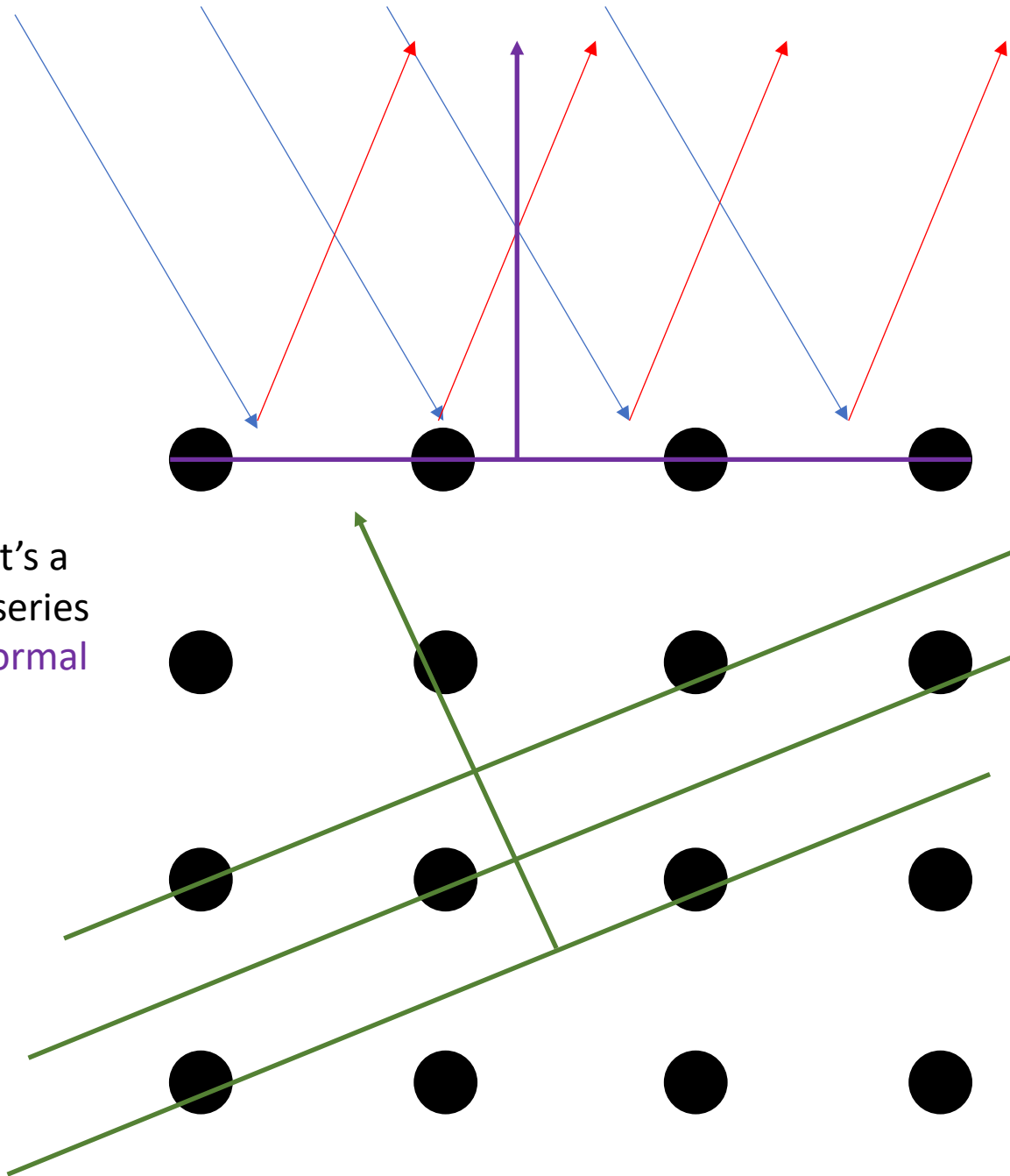
The ratio of peak intensity of $111 / 100 = 4/3$



The total intensity is then given by $I = |F|^2 p \frac{(1 + \cos^2 2\theta)}{\sin^2 \theta \cos \theta} e^{-2M}$

Bragg's Idea:

A set of plane diffracts such that it's a specific case of **reflection** from a series of planes, with a defined **plane normal**



Is that the only plane that can diffract as if its reflection ?

NO!

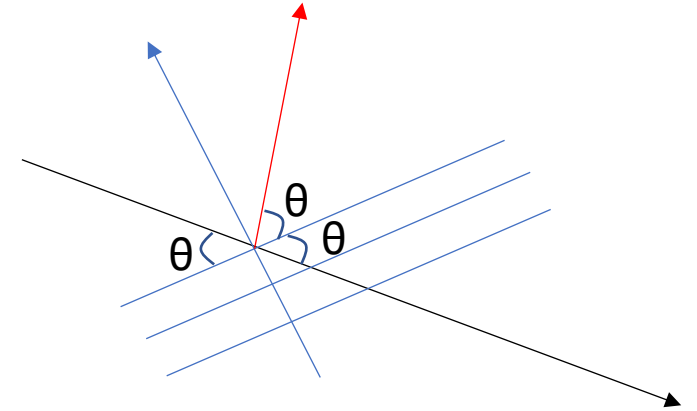
There can be infinitely many possibilities. Each of which has a different plane normal and the diffracted beam appears to be specularly reflected.

Experimental XRD

Bragg's condition is necessary!

The incident beam is reflected at some plane normal!

If θ is the angle between incident ray with the planes, the diffracted beams are always at 2θ to the transmitted beam.



How do we now construct a device to identify all the planes in the system ?

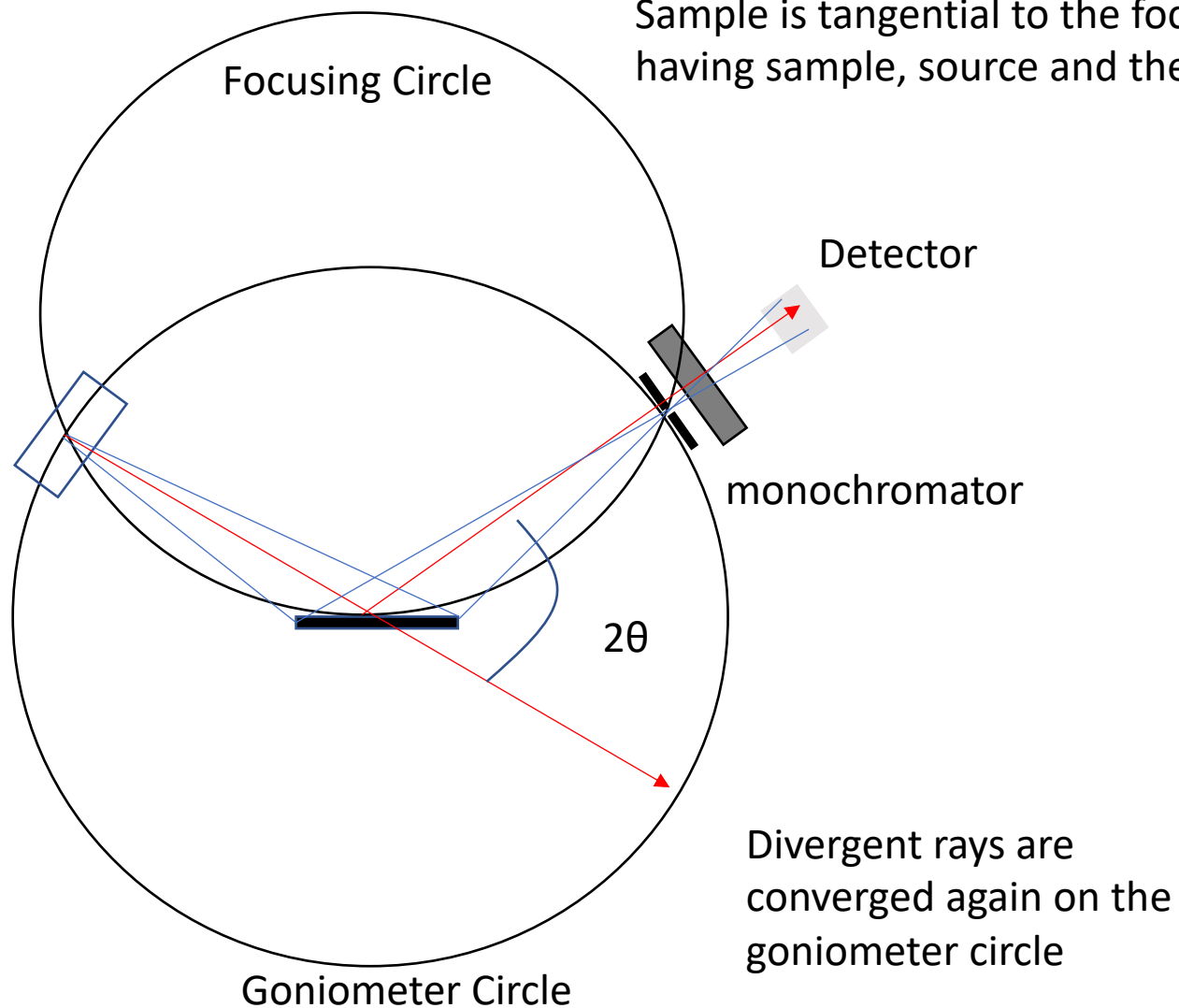
1. Set the source and detector always at Bragg's condition and look for the beam
2. Keep the source constant and rotate a crystal to expose all the planes
3. Send in a broad band X-Ray source and record on a photographic plate!

Geometries of measurement

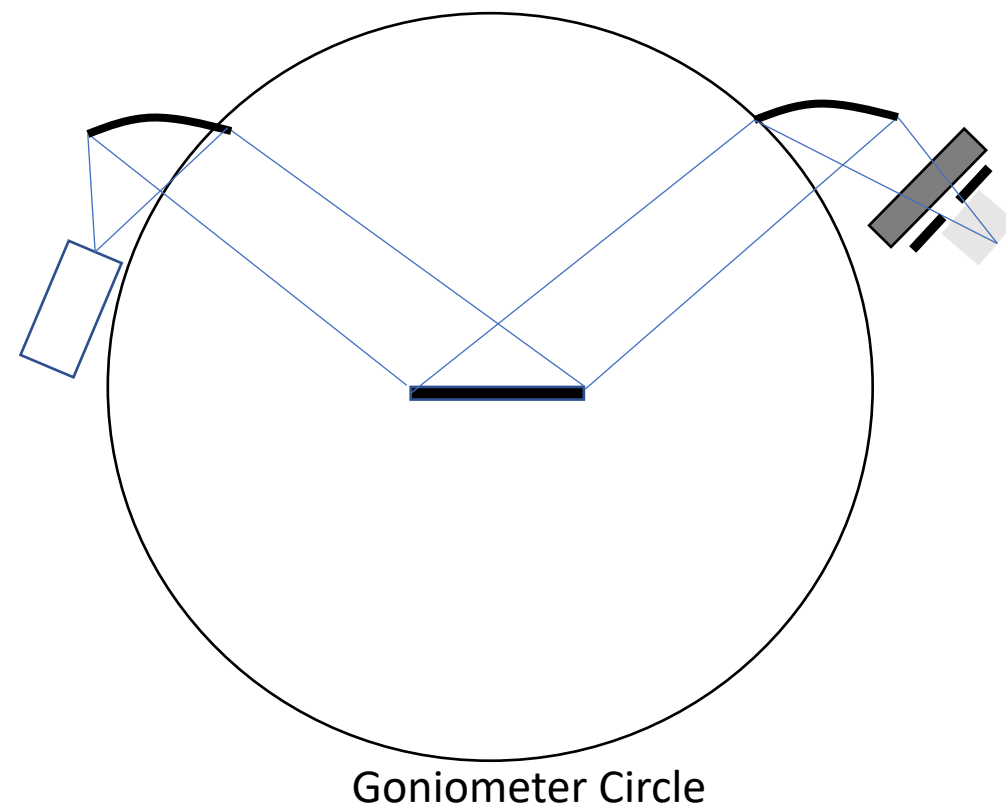
The source and the detector moves on the goniometer circle

Bragg-Brentano

Sample is tangential to the focusing circle having sample, source and the detector slits



Parallel beam

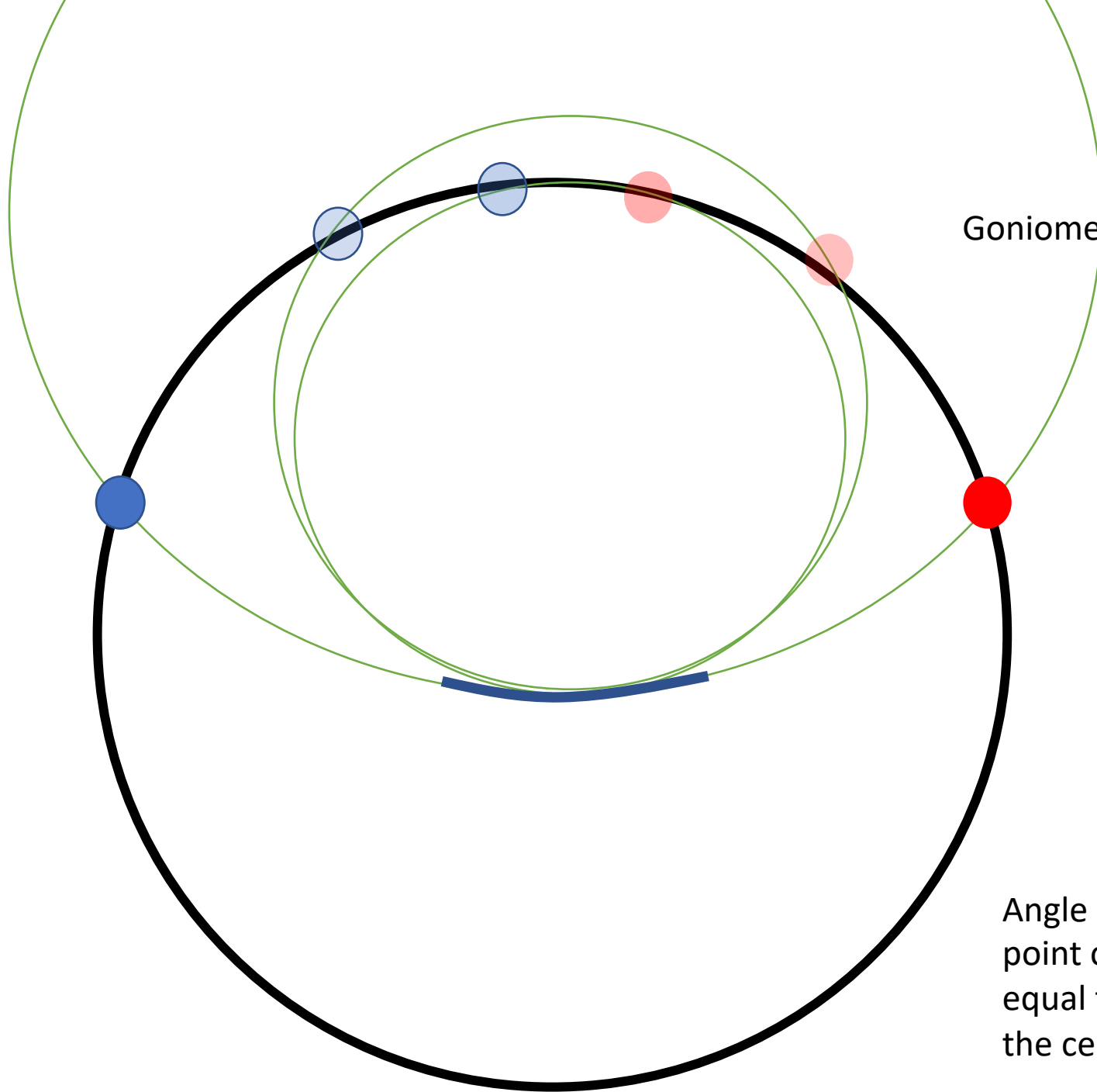


Parafoocusing

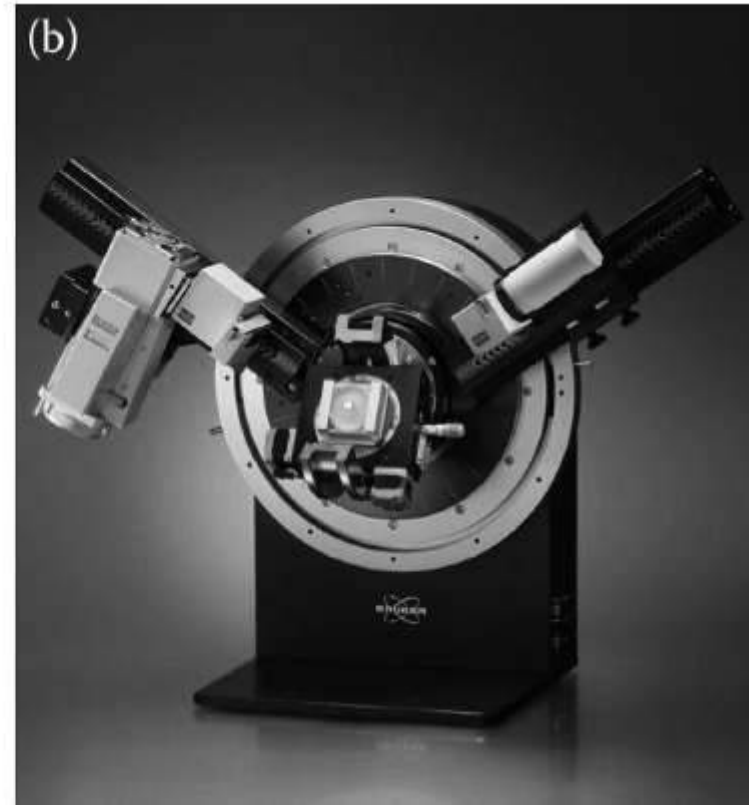
Goniometer Circle

The incident beam though
diverging subtends closely
similar angles to the entire
sample

Angle subtended by an arc on any
point on the remaining circle is
equal to half that it subtends at
the center

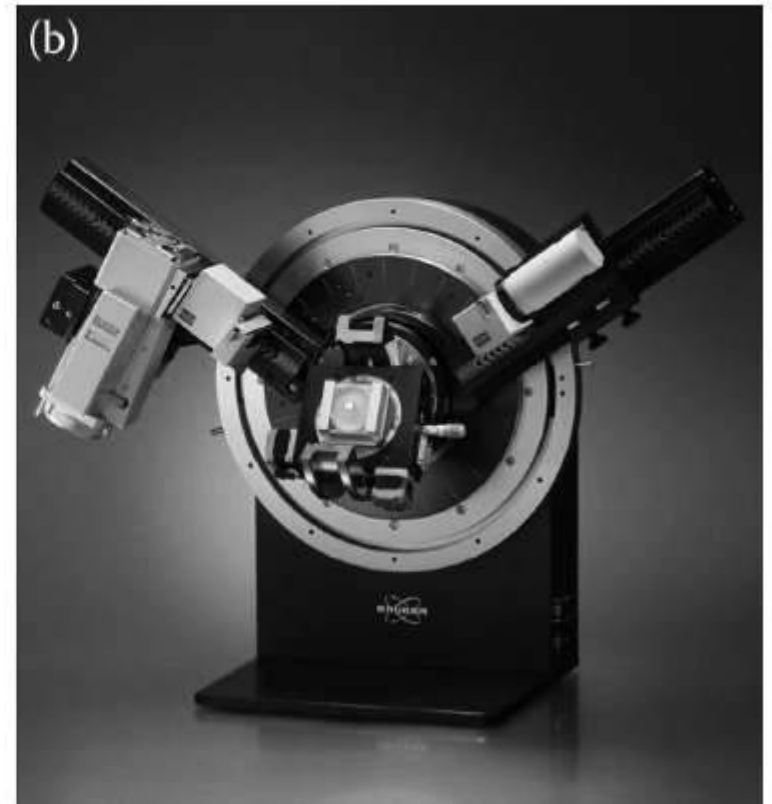


Typical commercial X-Ray diffractometer

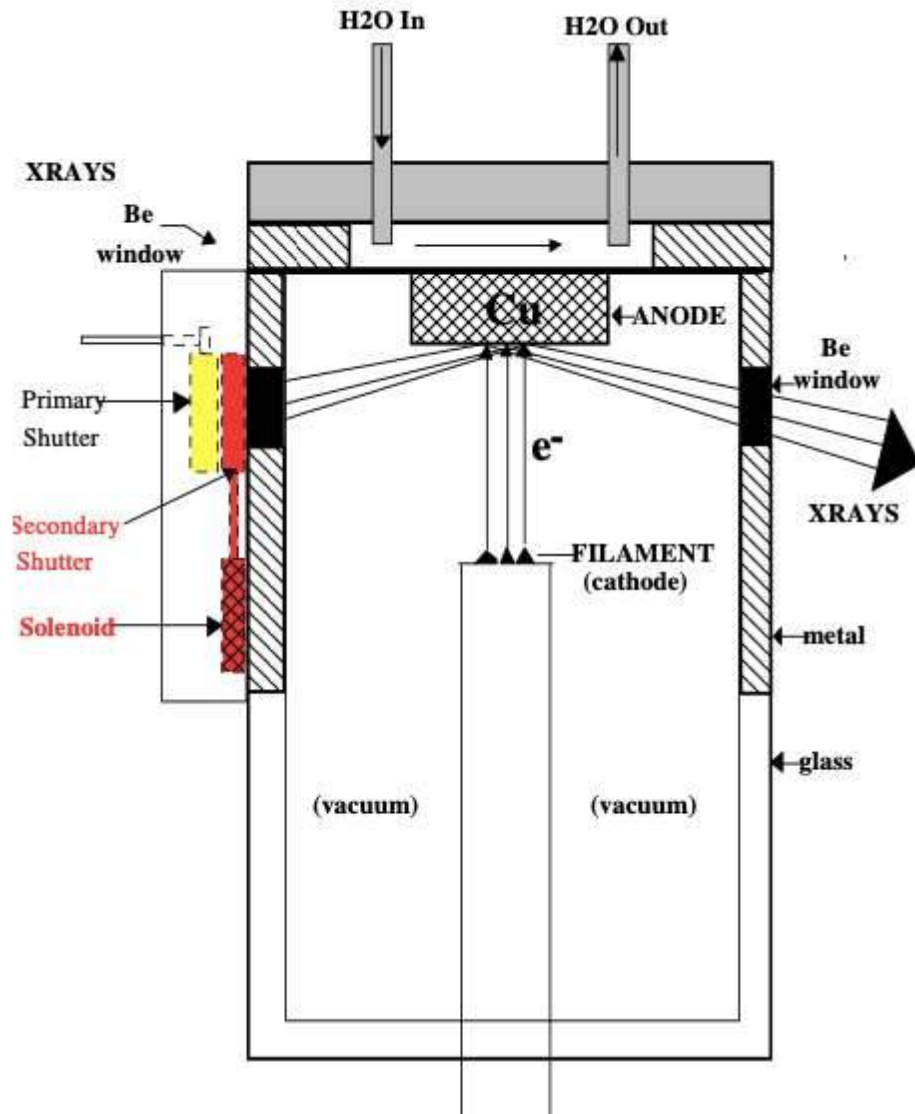


Parts of a typical X-Ray diffractometer

- X-Ray source
- Monochromators
- Ray optics
- X-Ray detectors
- Stage holders



Xray tube



- Xrays exit the tube via the Be windows.
 - Because Be has low mass absorption coefficient
 - Shutters are present at the exit to protect the users.
 - The walls, windows are cooled to prevent radiation damage.

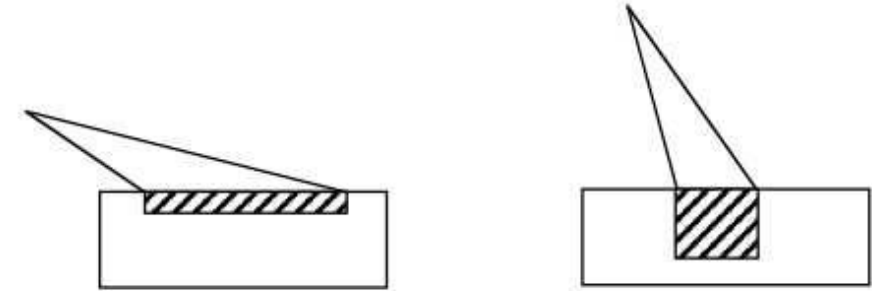
Beam Conditioners

X-Rays are divergent from the source and has different wavelengths.

Divergence is controlled by divergence slits

1. Divergence slits are used to limit the extent of X-Rays
 1. Smaller slit widths leads to smaller divergence, better resolution but lower intensity

Divergence also leads to difference x-ray interaction volume at different angles.

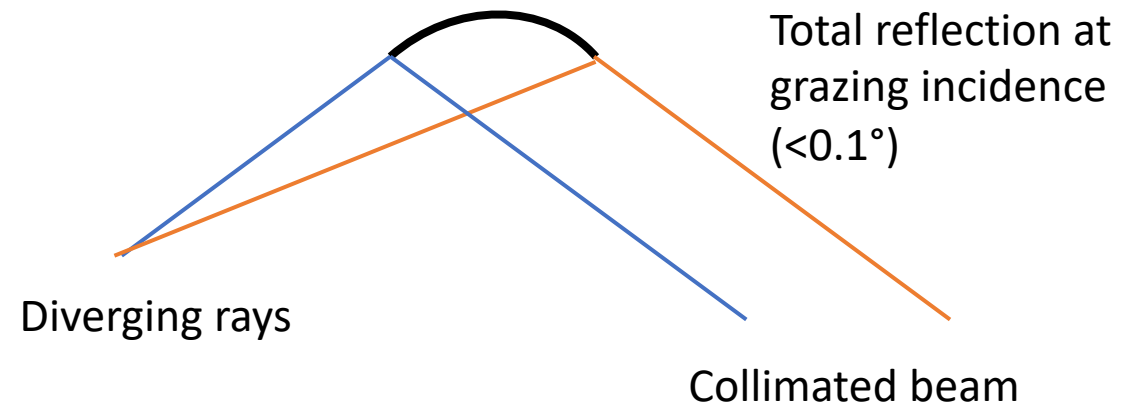


A diverging ray can be made parallel by several means.

parallel plate collimators

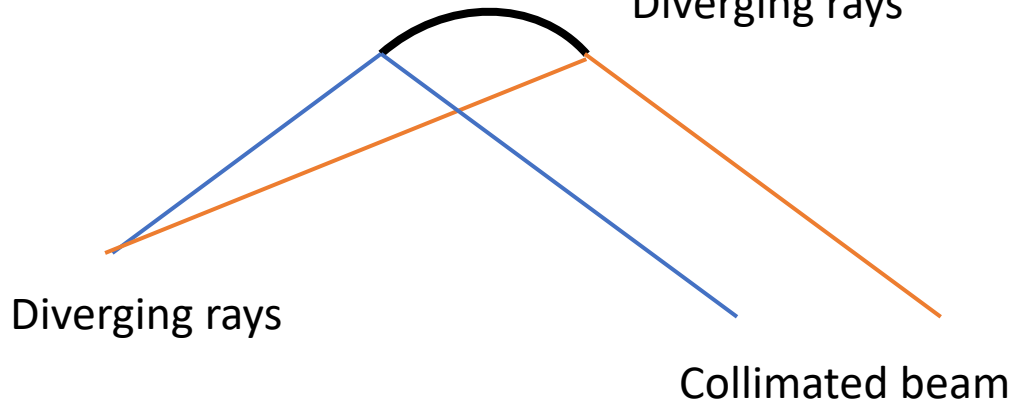
Soller slits

Parabolic mirrors and capillary lenses

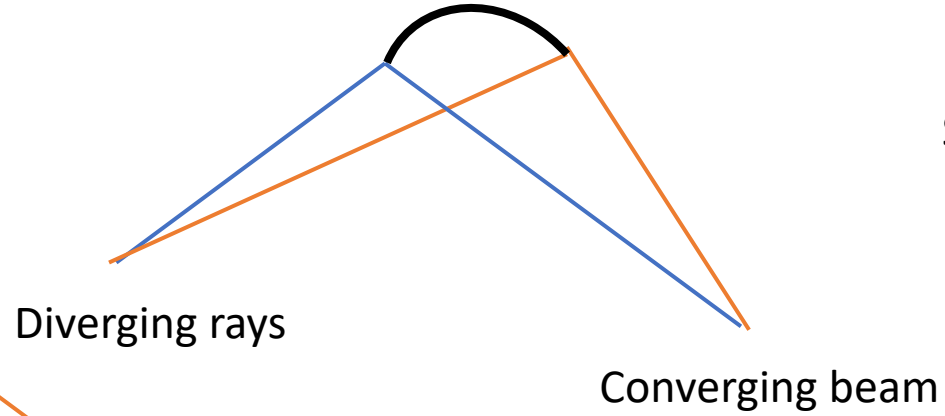


Collimators

Parabolic Mirrors



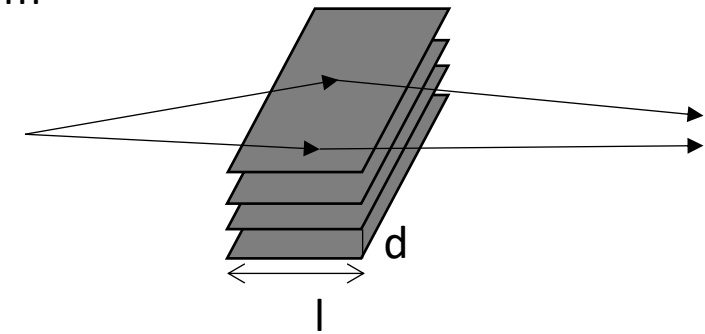
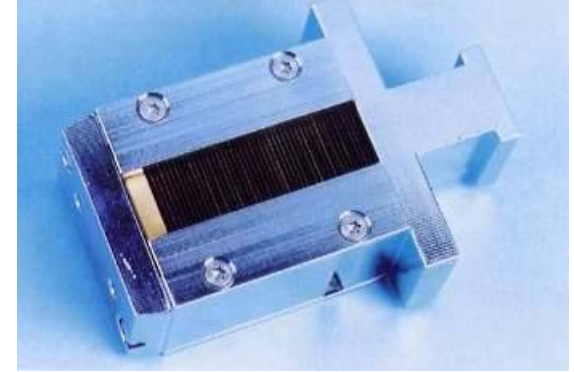
Elliptical Mirrors



Total reflection at grazing incidence ($<0.1^\circ$)

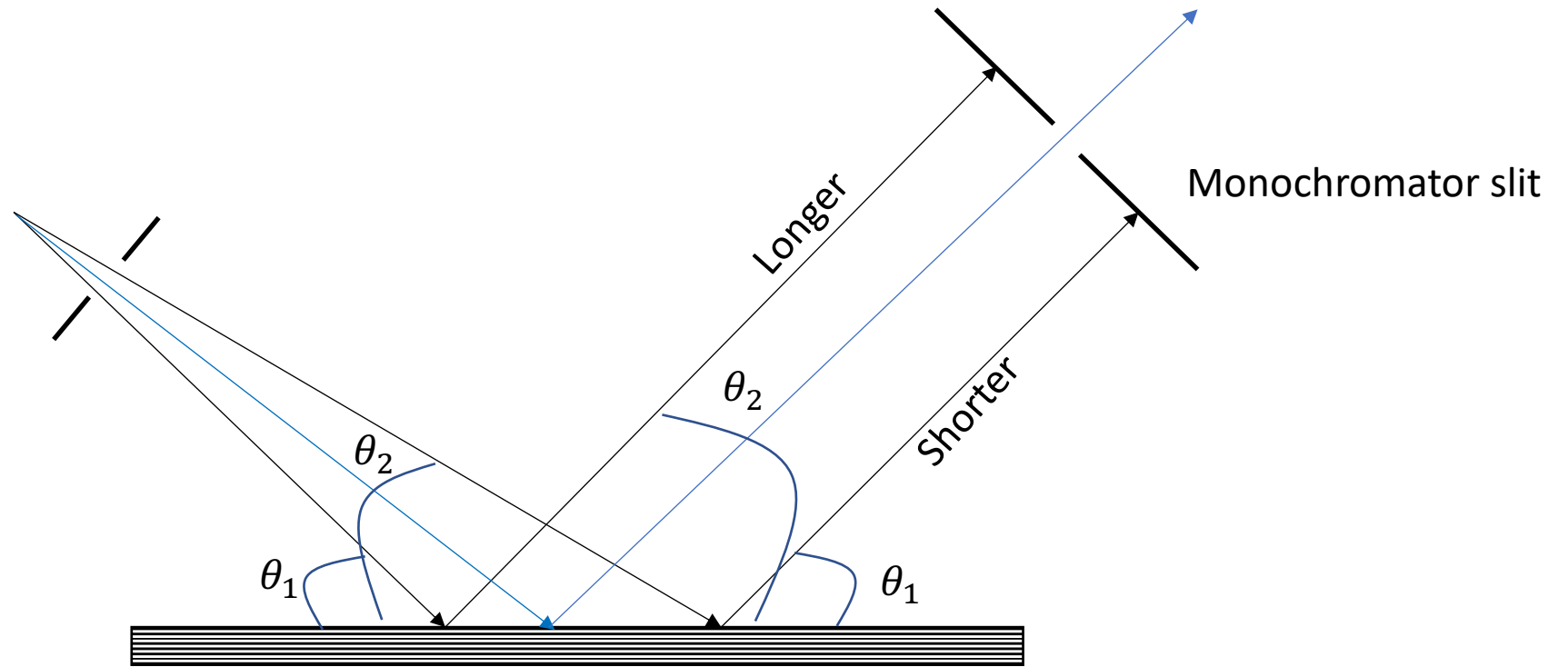
Soller Slits:

Reduce vertical divergence



Typical beam divergence $\alpha = \frac{2d}{l}$

Monochromators



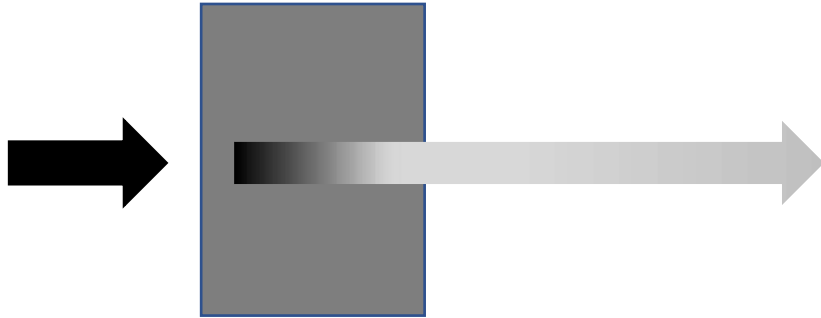
A properly aligned substrate such that its in Bragg's angle $2d\sin\theta = \lambda$

Beam Absorbers and Attenuators

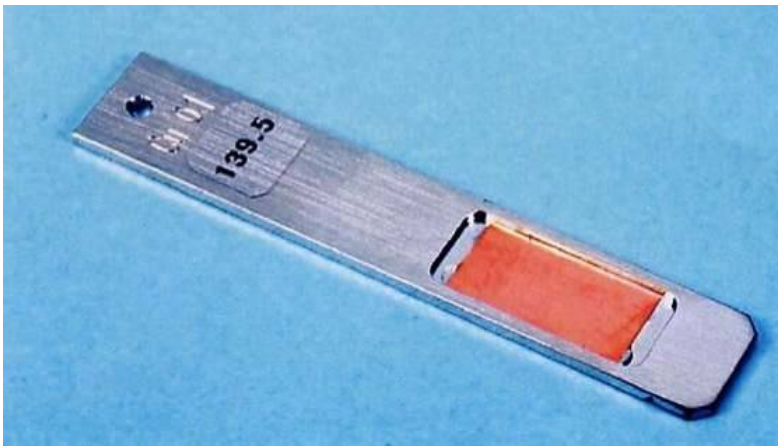
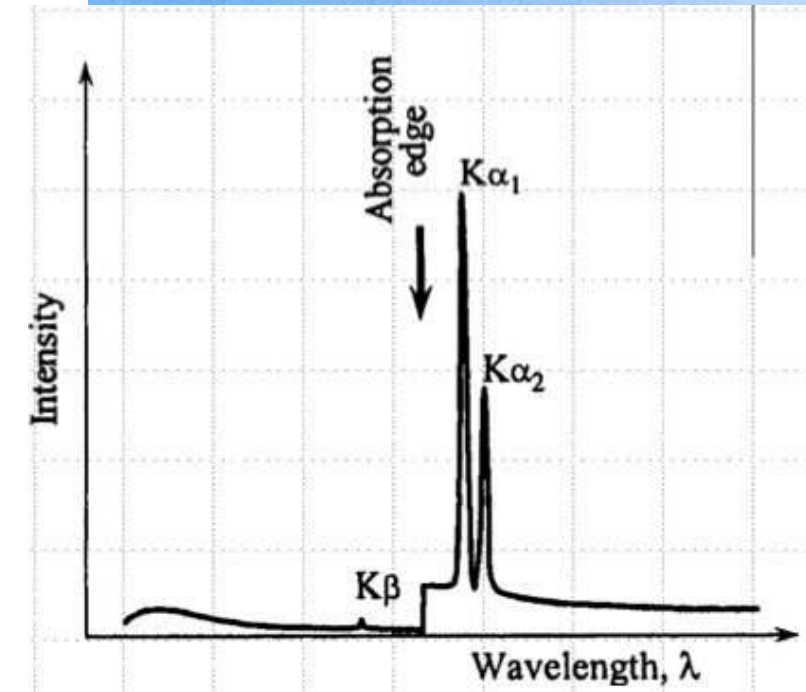
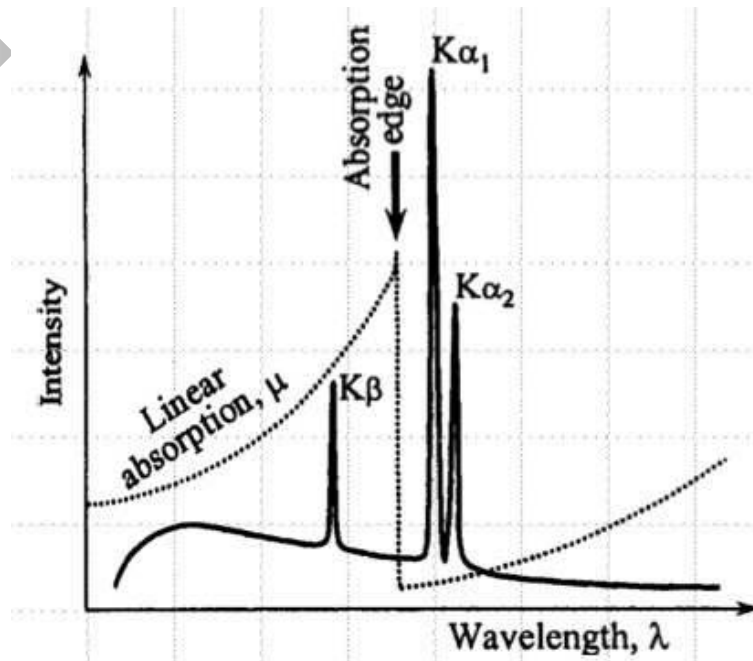
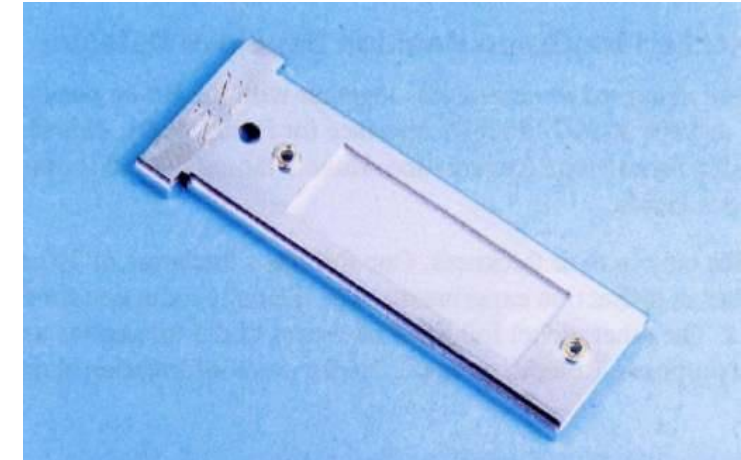
Selective filters

If μ is the absorption coefficient

$$I = I_0 e^{-\mu x}$$



β filters



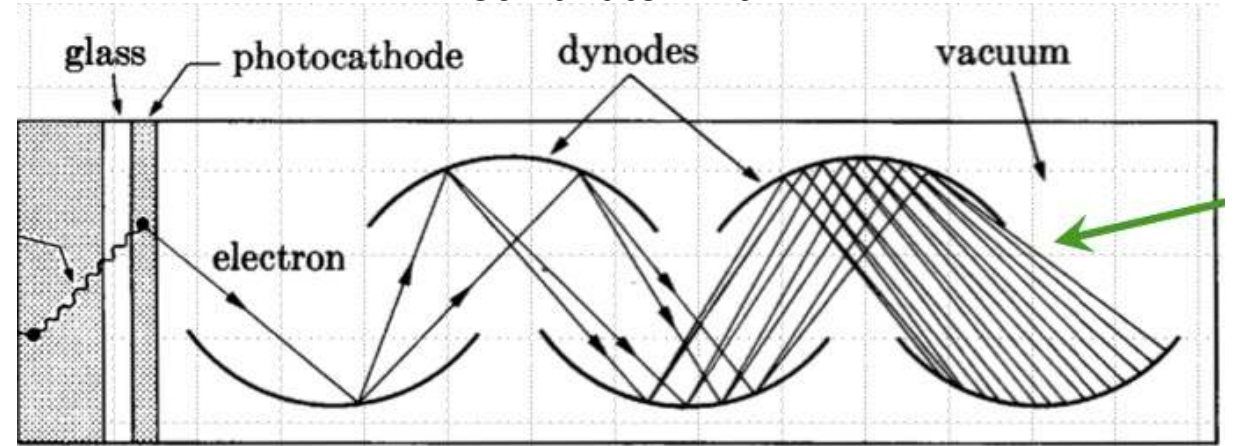
Xray filters

Gun Material	Filter	Thickness (μm)	K β Reduction	K α Reduction
Mo	Zr	75	97	54
Cu	Ni	20	99	58
Co	Fe	16	99	51
Cr	V	13	98	45

X-Ray Detectors

- Single photon detectors
 - Scintillators
 - Proportional counters
 - Semiconductor detectors
- Detectors with spatial resolution
 - Charge coupled device (CCD)
 - Wire detectors

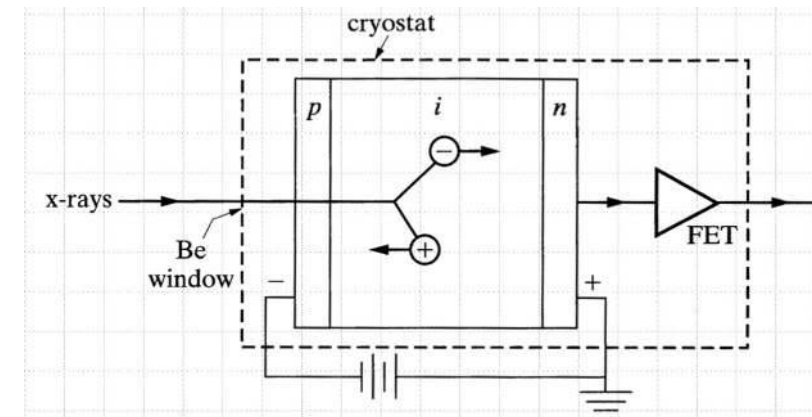
Scintillator with PMT



Proportional Wire detector



Semiconductor detector



Shared

Locations

Krishna's...

OneDrive

Tags

P-10-DU

Orange

Purple

Blue

Red

All Tags...

Media

normalReflection.mov

04-Mar-2022 at 7:31 PM

2 MB QuickTime movie

Cancel

Open

Pio, 2020-2021

prepared using blender

music: creative minds from Bensound.com



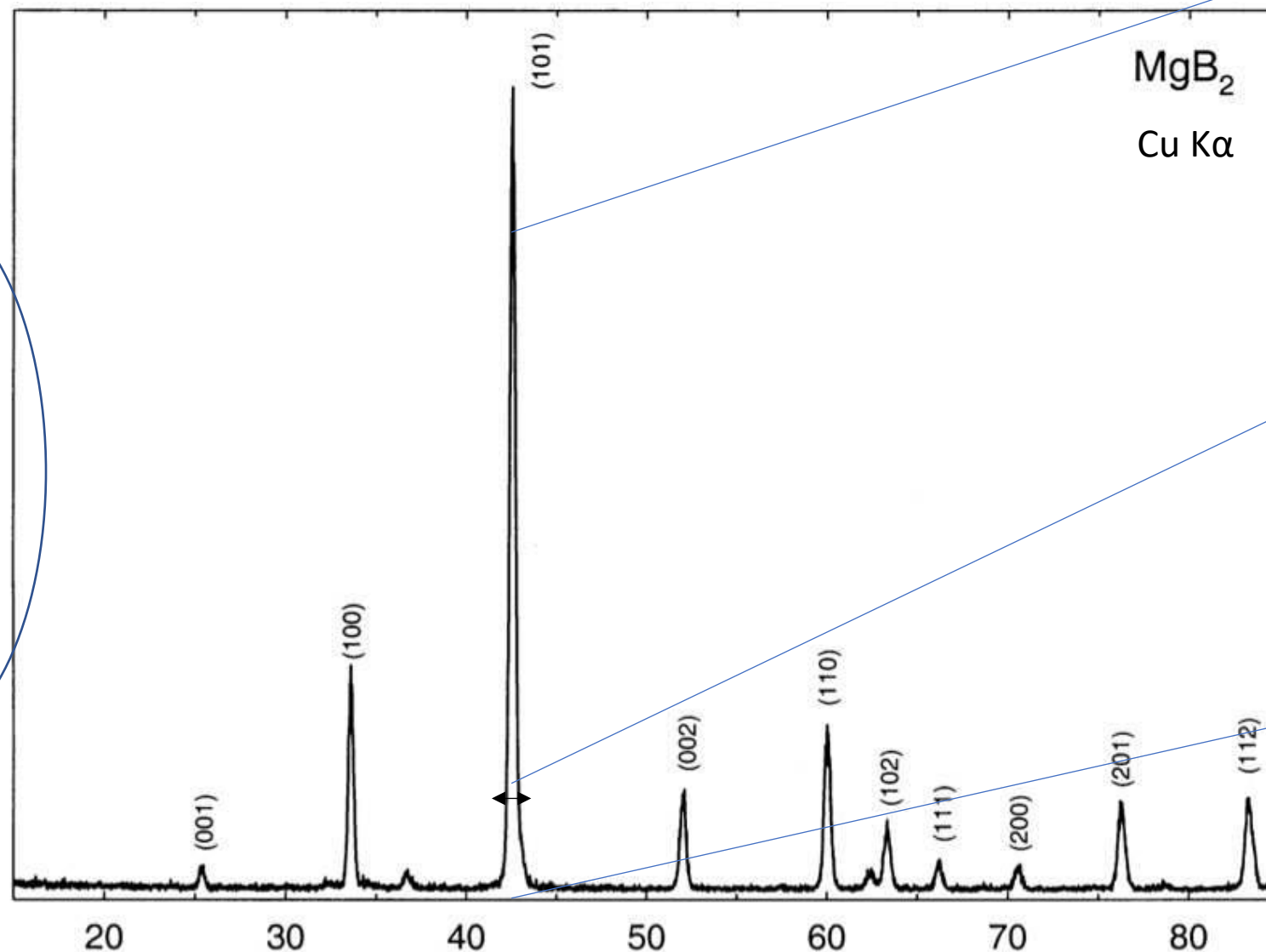
0:03 / 3:02



A sample measurement

Site occupancy and
phase fraction

Intensity (arb. units)



Crystallite size and
micro strain

2θ position of the
peak

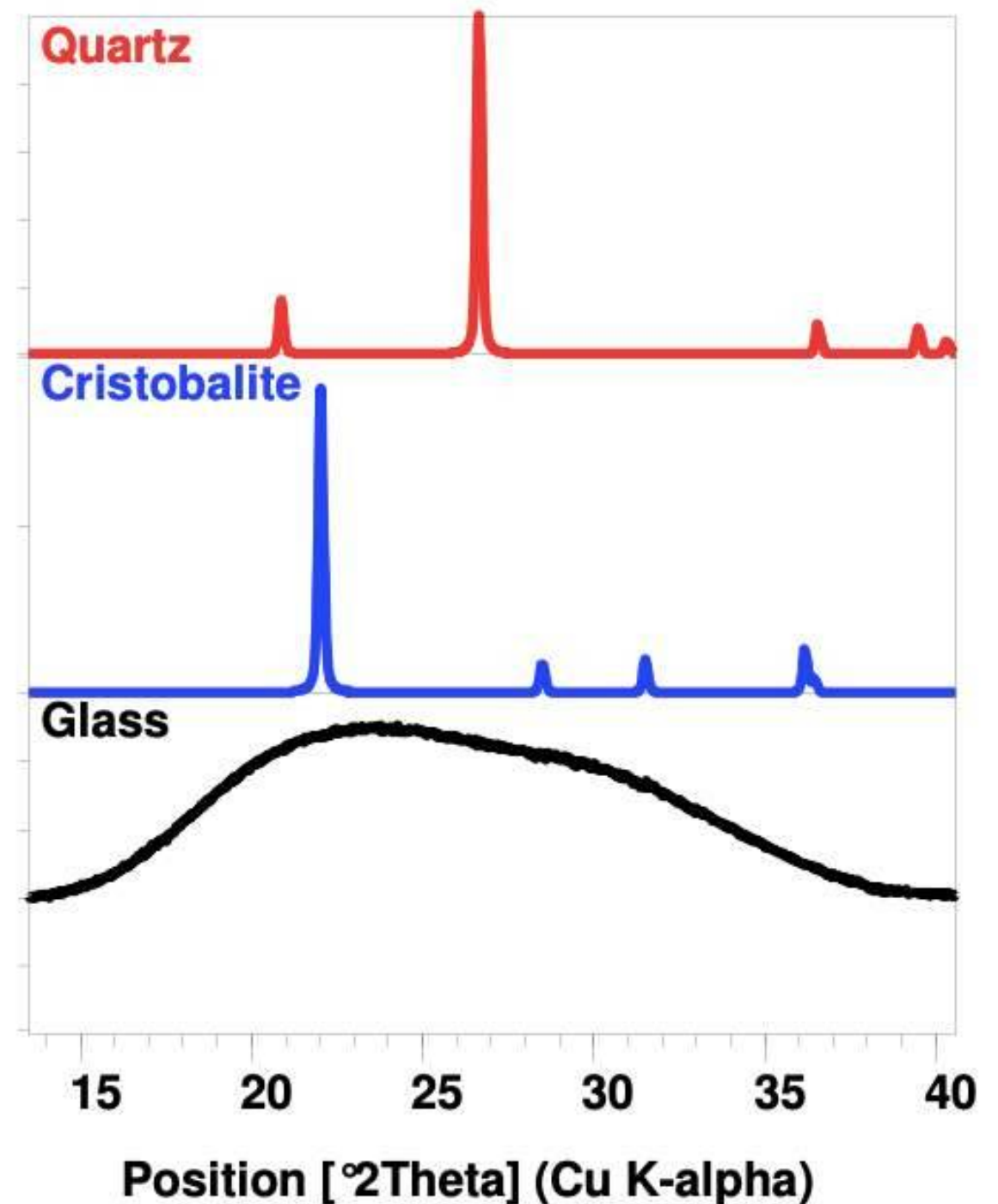
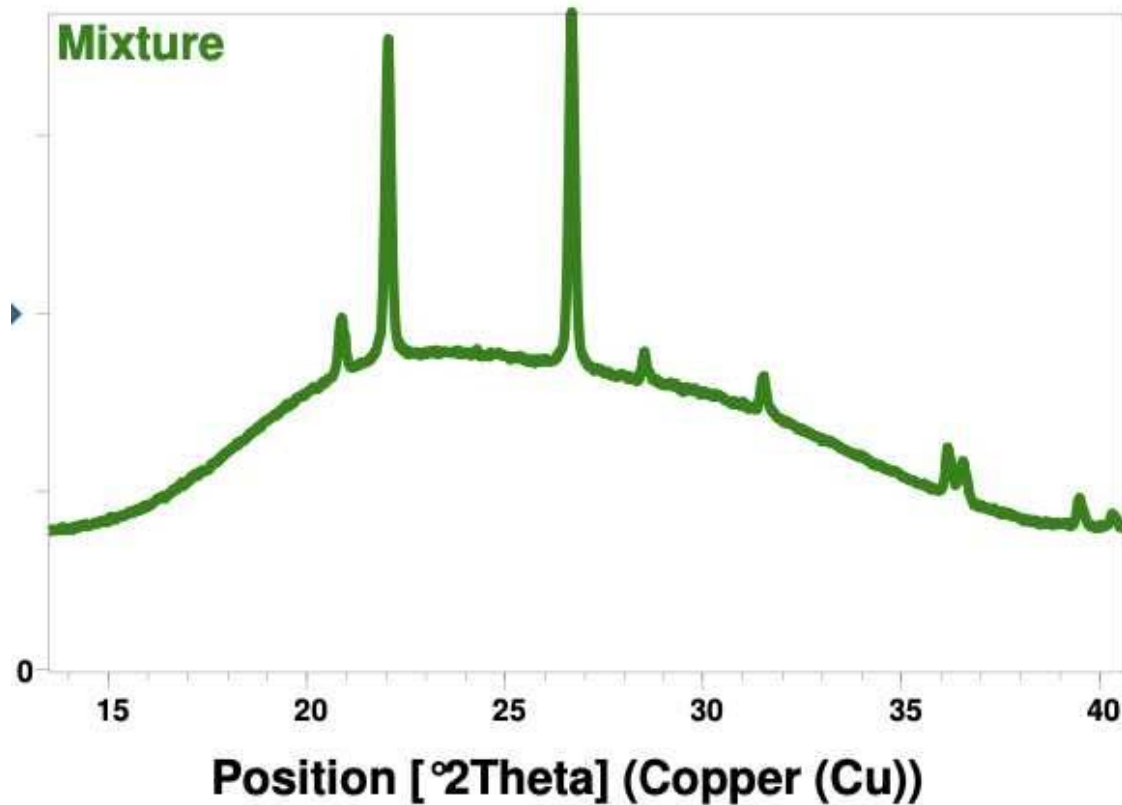
Gives spacing
between planes

$$2d \sin \theta = \lambda$$

Phases and composites

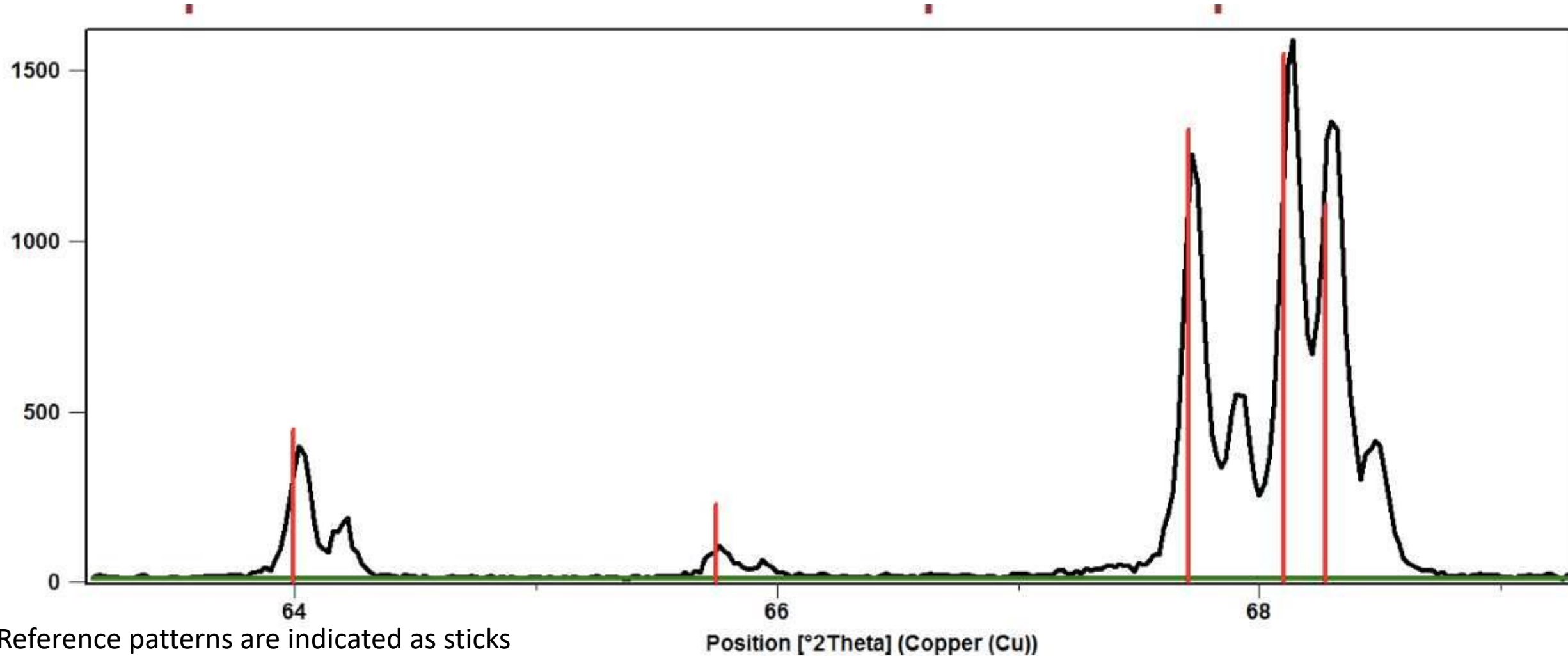
All these are different phases of SiO_2

Chemically identical, but with different atomic arrangement

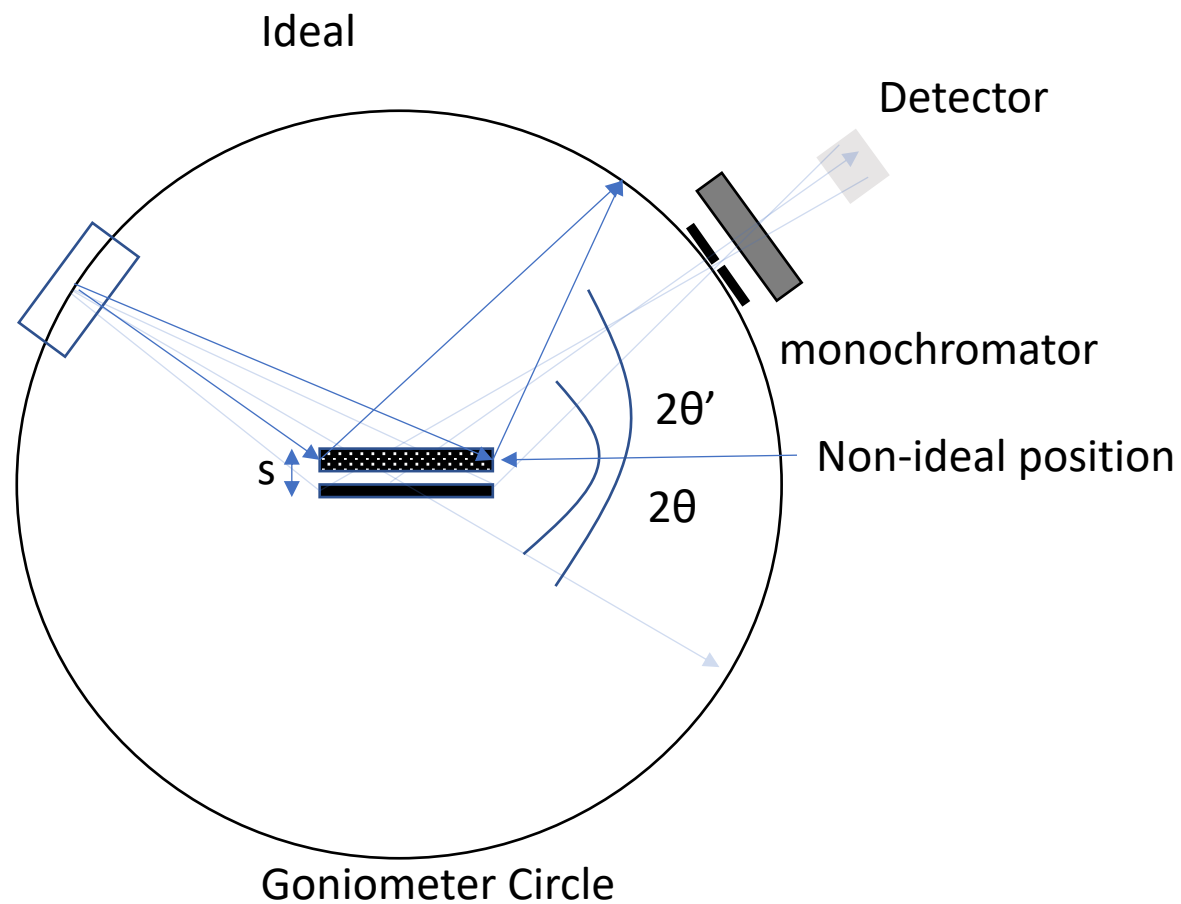


Quantitative Analysis

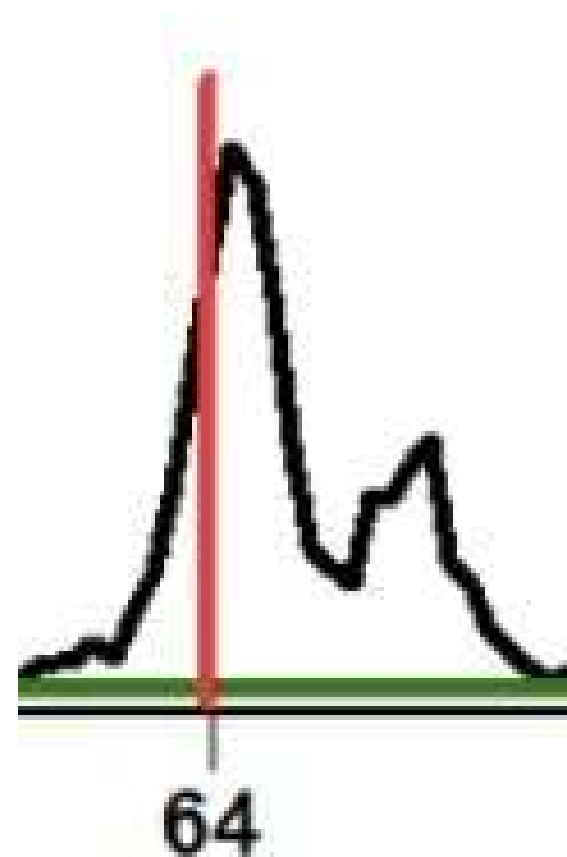
For the material, the peak intensities and the position should match



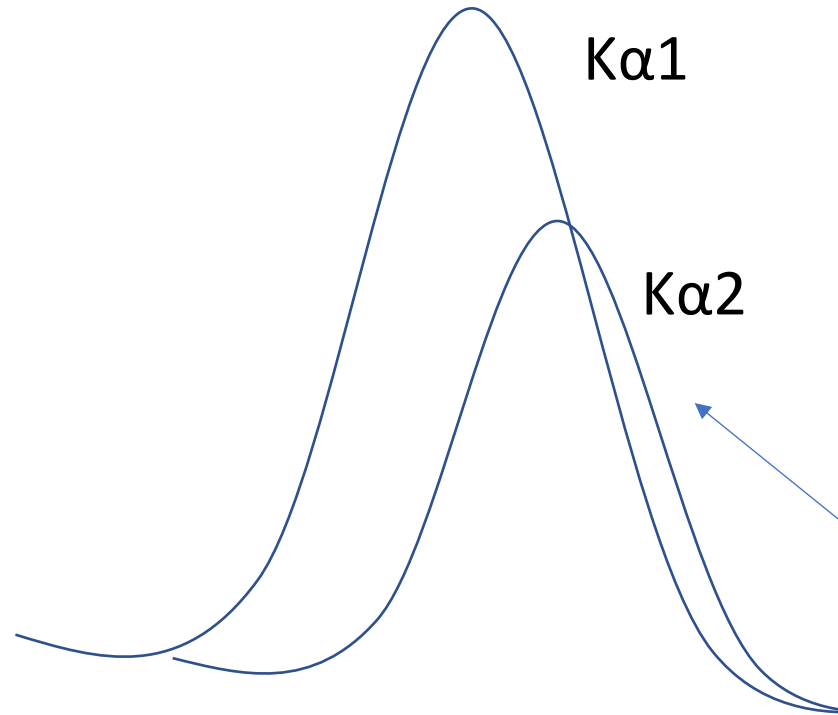
Specimen position errors



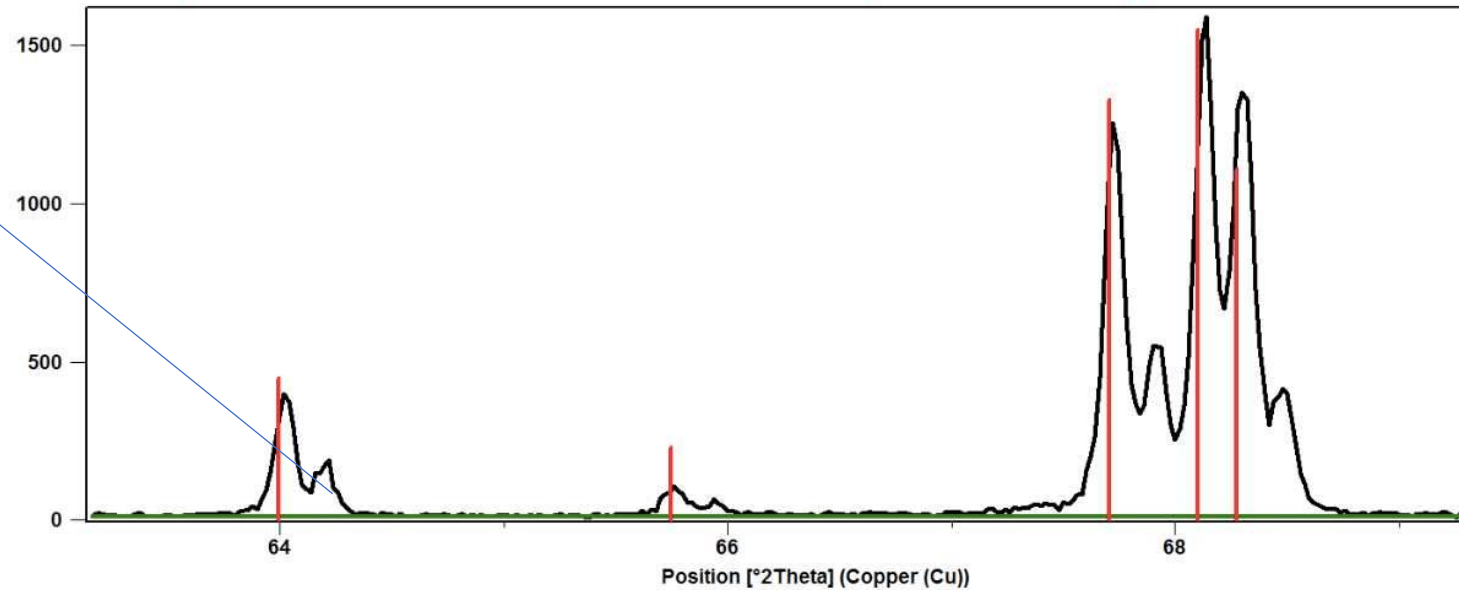
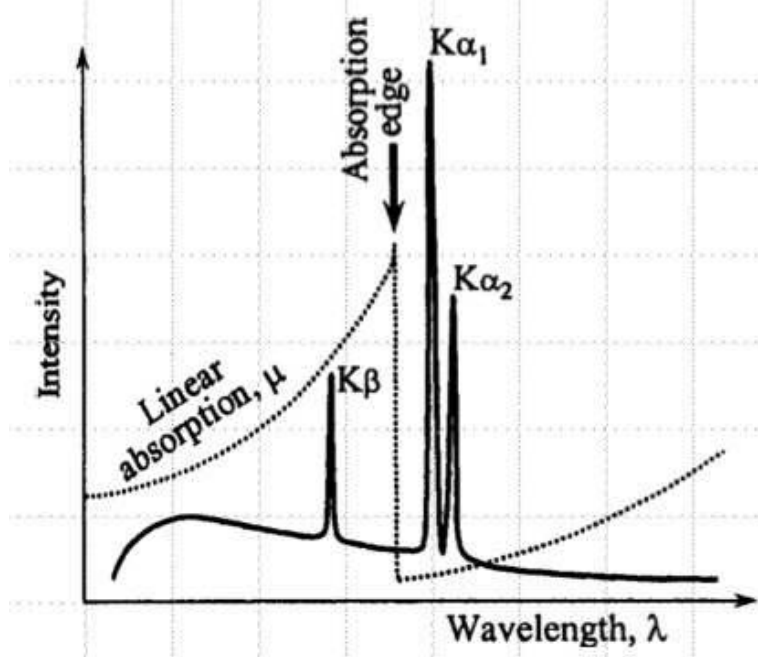
$$\theta' - \theta = \frac{S \cos \theta}{R}$$



Truly non-monochromatic light

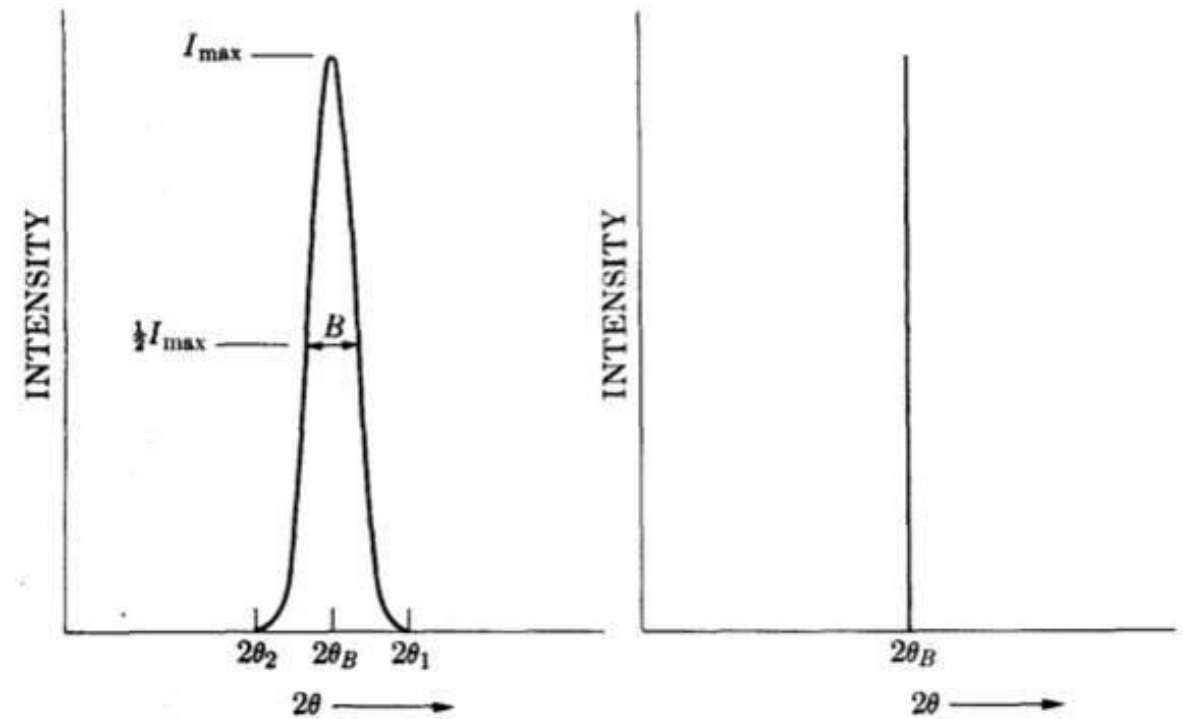
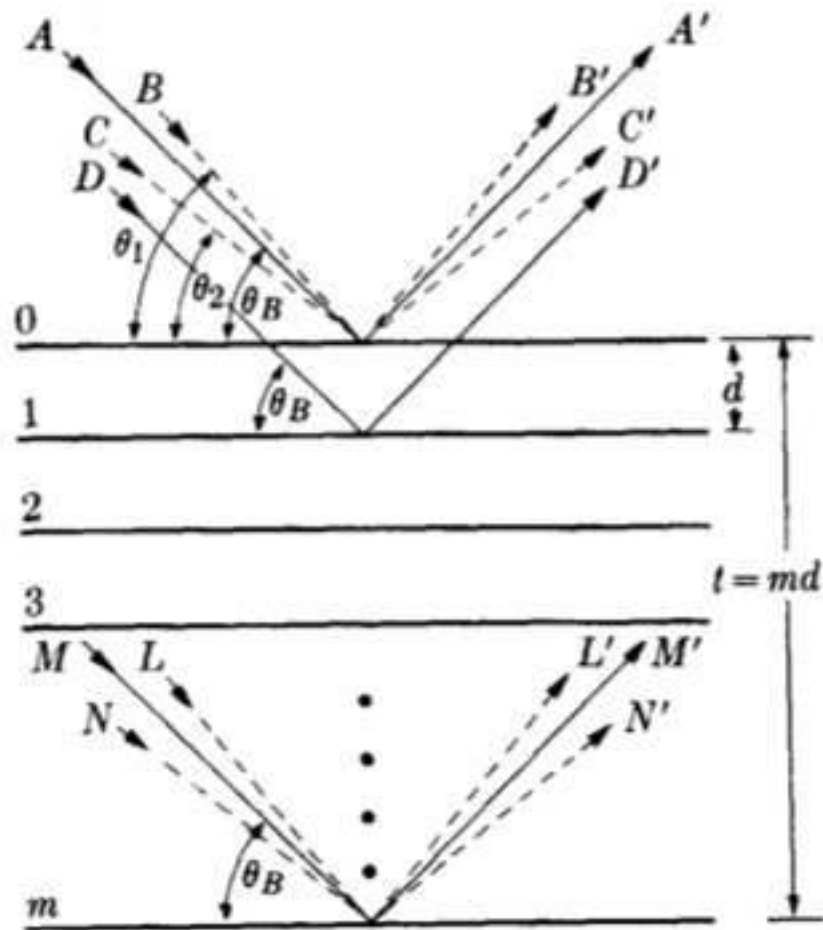


$$d = \frac{\lambda_{K\alpha_1}}{2 \sin \theta_1} = \frac{\lambda_{K\alpha_2}}{2 \sin \theta_2}$$



Peak Width

The diffraction peaks are broadened due to lack of infinite atomic planes to produce perfect destructive interference.



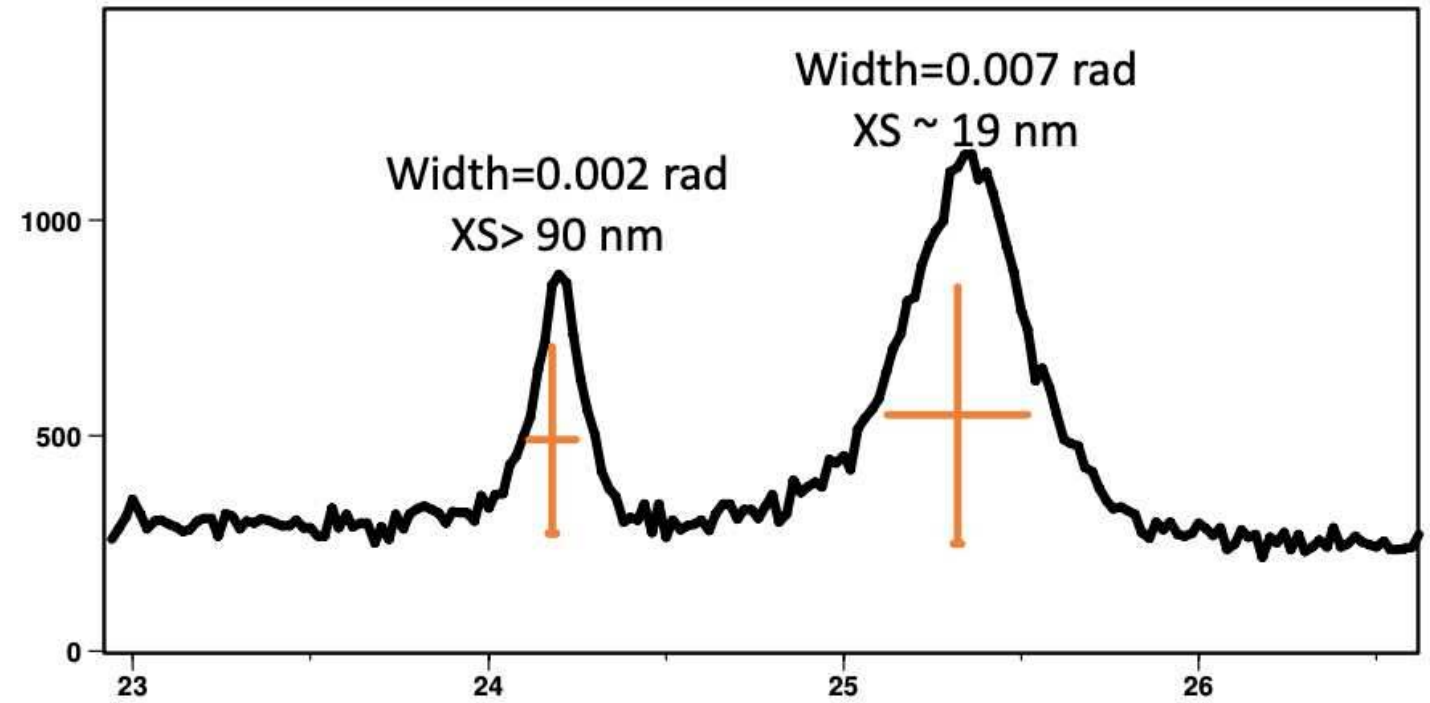
Lack of destructive interference leads to peak broadening

Size of crystallite size (Scherrer's Formula)

$$\tau = \frac{k\lambda}{B \sin \theta_B}$$

w – full width half maximum of the peak

k – constant ~ 0.9



However, non-uniform lattice strain can also lead to peak broadening

Hence one needs to fit a Gaussian/Lorentzian function to obtain the true FWHM.

Therefore from Powder XRD,

Collect the peak list (position and intensity).

1. Unit cell lattice parameters from the peak positions
2. Observed peaks can be indexed and absences to be noted. This gives the crystal system
3. Get the reference pattern and compare to get phase composition
4. Crystallite size and microstrain: Peak width and/or shape

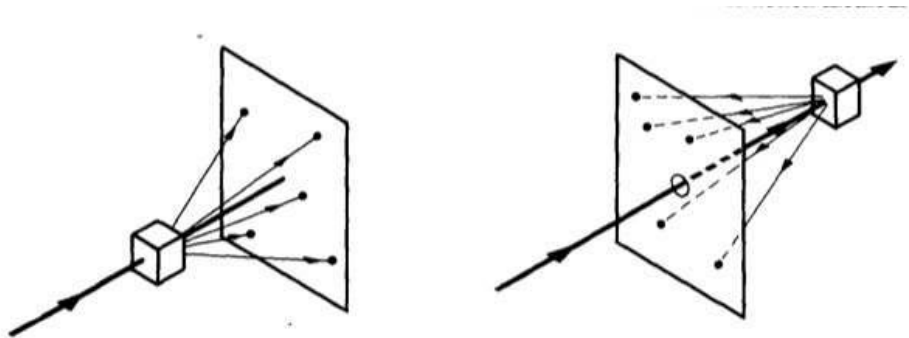
Elements of Single crystal XRD

The necessary condition for XRAY peak – Bragg reflection.

The source, plane normal and the detector should be in a plane!

In a pure single crystal, there atomic planes are perfectly aligned, a single strong normal exists.

The difficulty now is in getting this normal in the plane of the source and the detector.



Electron Microscope

Reference: J. Goldstein et. al., Scanning electron microscope and X-Ray microanalysis

Origin of Alternate Microscopic Instrument

Limitations with Light microscopes:

1. Spatial resolution limited by the wavelength sensitive to eye
2. Depth of field extremely low at higher magnifications
3. Optical components cannot be tuned in-field (aberrations are determined by various factors not easily tuned)
4. Higher magnifications required to visualize sub-micron features are not practical due to different trade-offs.

SOLution is to use radiations with smaller wavelengths!

Solution: Electron microscope

Why electrons ?

Electrons in material can be ejected, accelerated to required velocities.

Fundamentally all systems have wave-particle duality. Electron velocities can be adjusted to have waves with sufficiently low wavelength

Electron beams can be converged, diverged and suitably chiseled using electrically tuned components 'electron optics'

Immense experience exists thanks to CRT televisions

Electron Gun

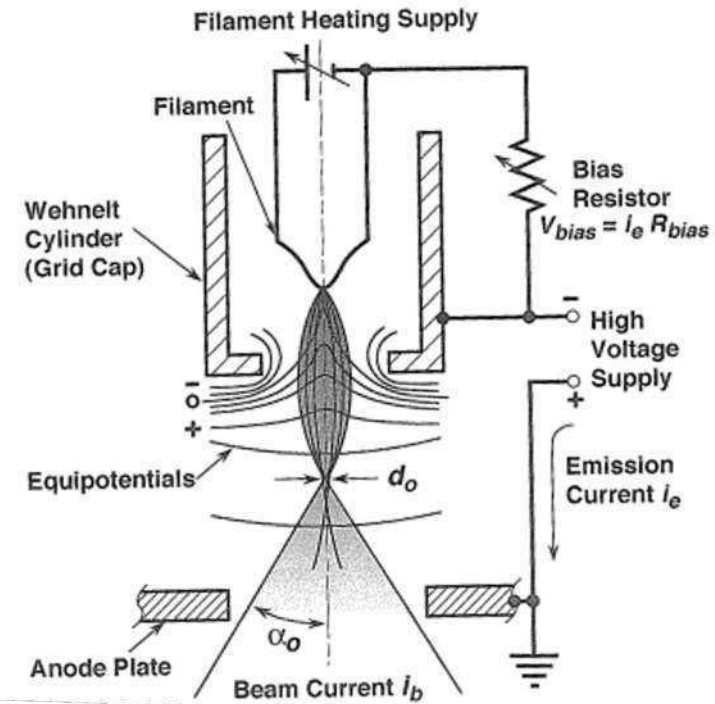
Otherwise called a Cathode Ray Tube:

Cathode is typically a metallic wire (W) typically bent like a 'V' shape

Wire heated to about 2000 - 2700 K by resistive heating.

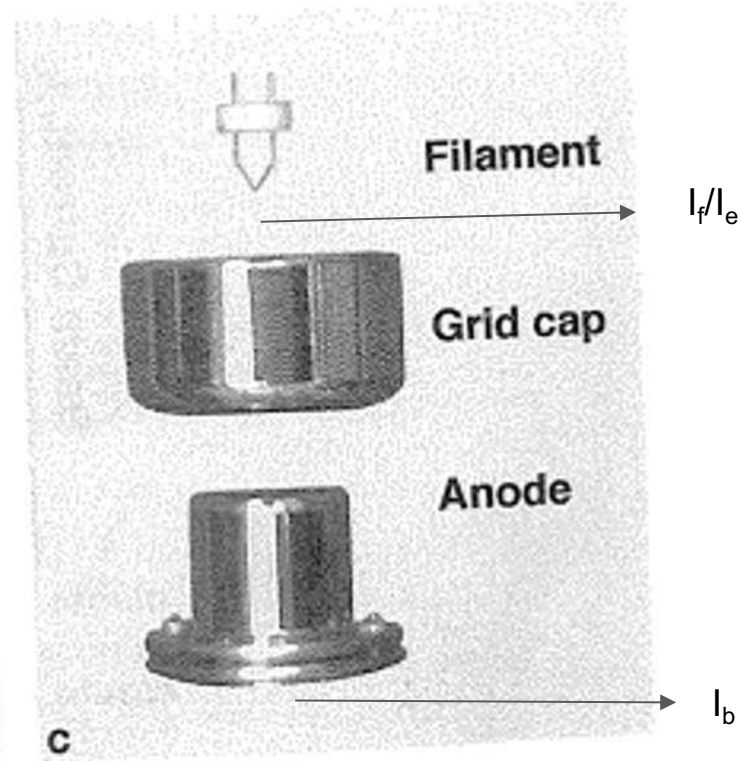
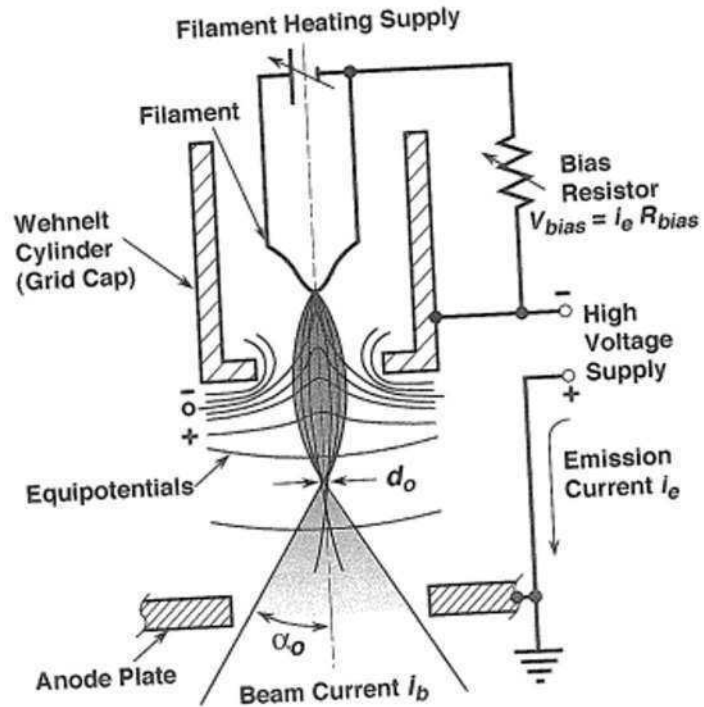
Sharpest region - largest current density - region of significant electron emission

Grid cap - enclosure tied to negative potential. Equipotential lines shown in the image focuses the electron beams to lesser-negative regions



Finally, the Anode plate is grounded. The cathode is set to about -20kV, the large potential difference leads to electron acceleration and a beam current below the anode plate

Currents in electron sources



Electron Lenses

Electron motion in magnetic field is determined by force

$$F = -e(E + v \times B)$$

In a purely magnetic field

$$F = evB \sin \theta$$

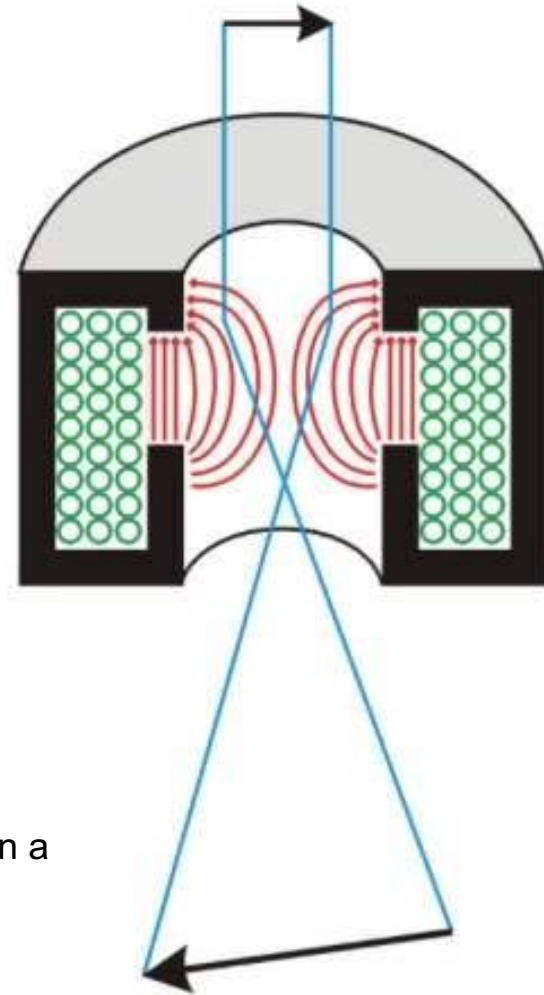
θ angle between v and B

Rotationally symmetric magnetic field is inhomogeneous

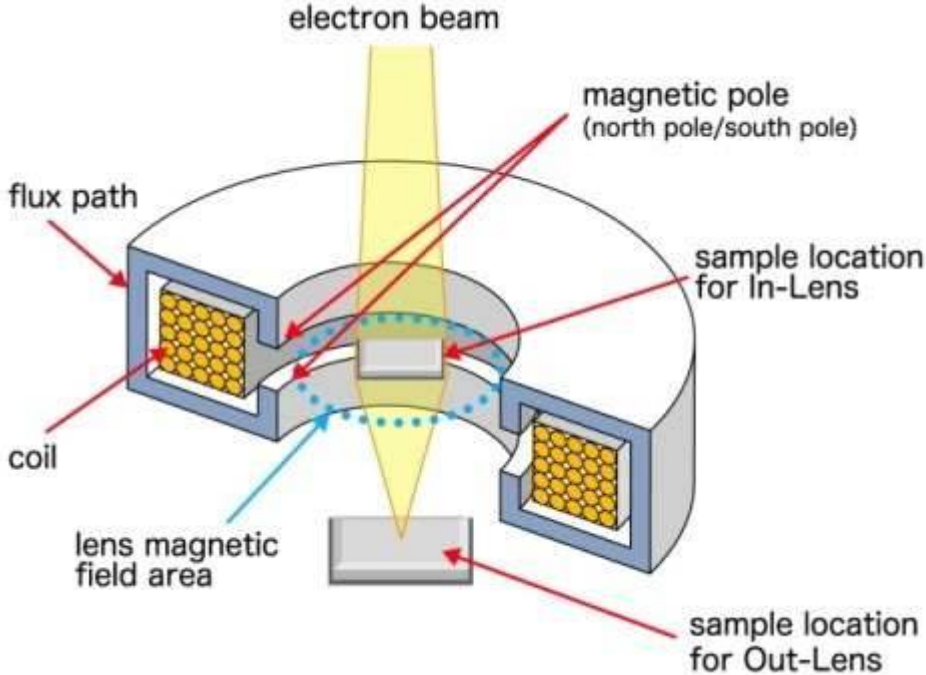
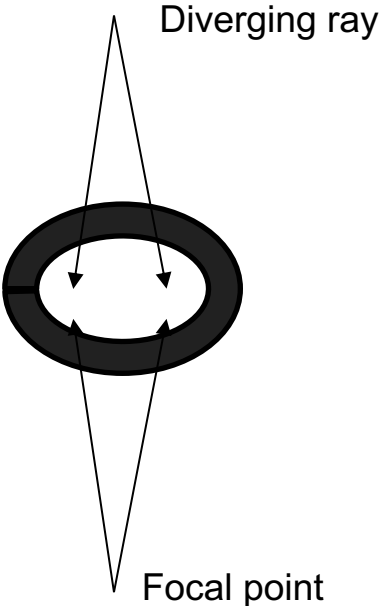
- field is stronger near the core and weaker near the center of the bore.

- This makes the electrons near the coils bend towards the center, in a circular fashion

- The extent of bend/deviation depends on the magnetic field (B) which can be controlled by the current through the lens coils.



Electron path inside a lens

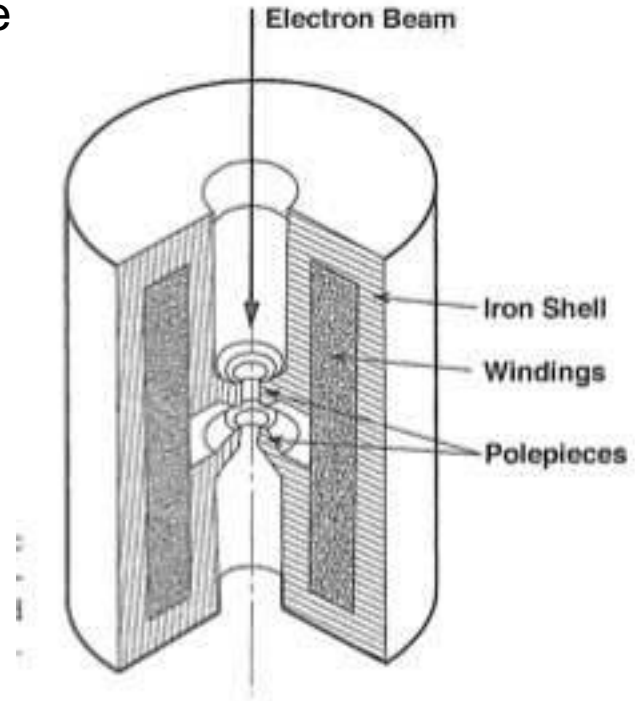


Focusing in electron optics

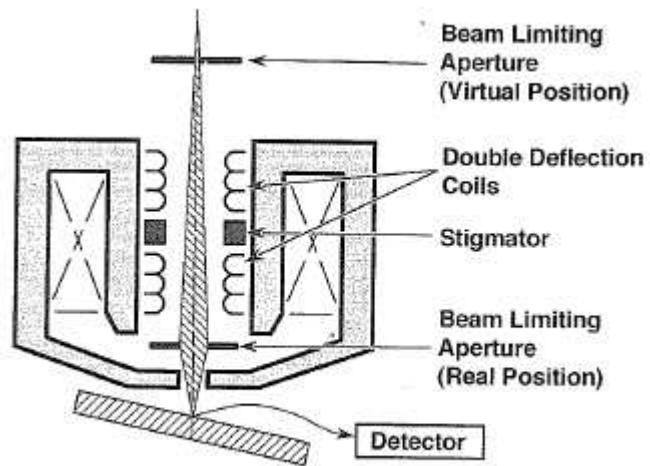
- The focal point is defined as the distance from the point at which the rays begin to bend to the point where it intercepts the axis.
- Typically $f = \frac{V_0}{(NI)^2}$, where V_0 is the accelerating voltage, I current inside the N turn coil.
- Following geometric optics

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q}$$

Where p is the object distance, q is the image distance and f is the focal length of lens

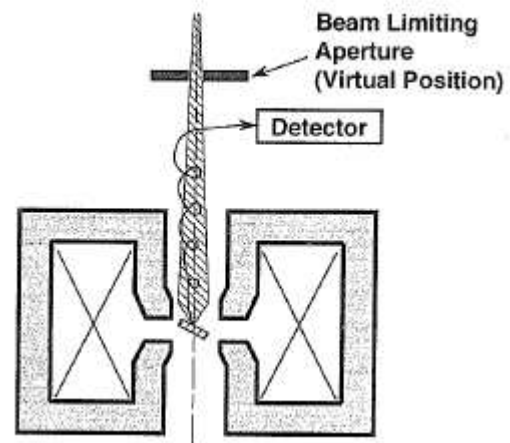
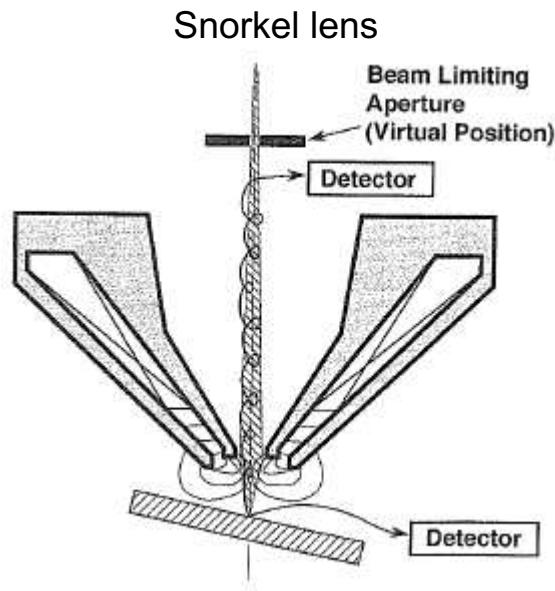


Important point to remember: the focal length of the lens is tunable



Asymmetric pin hole lens

Specimen outside,
Stronger aberrations



Symmetric immersion lens

Specimen inside lens bore

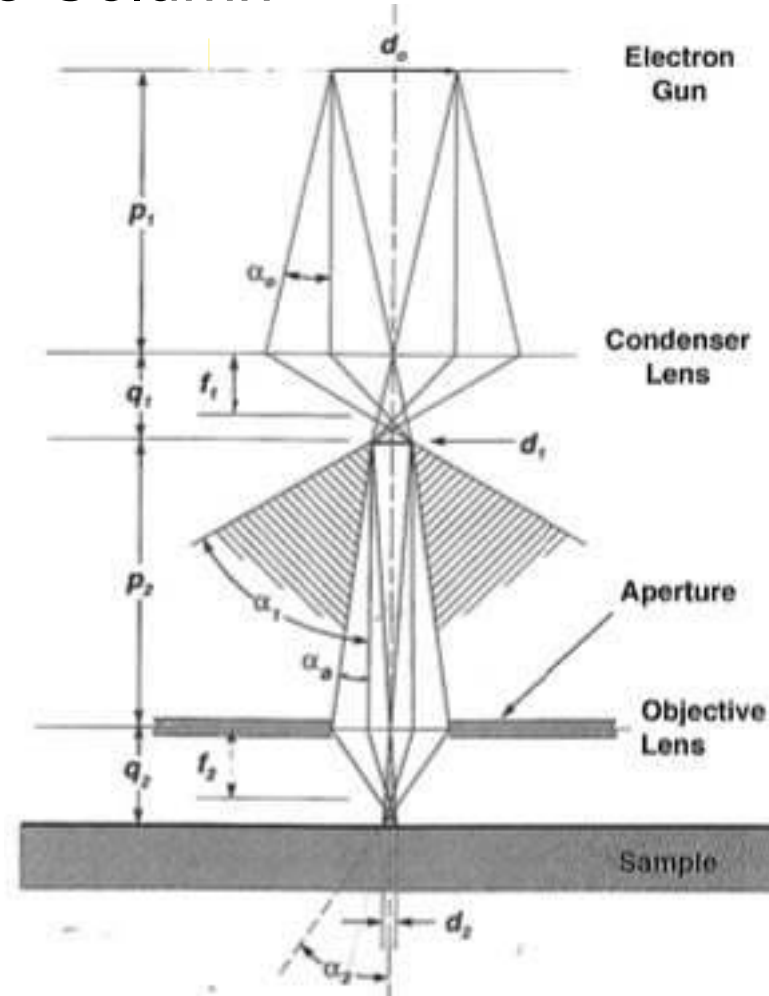
Typical Design of Electron Microscope Column

Object – electron gun typically 10 – 50 μm

Condenser lens accepts rays with an angle α , focusses it down and forms a demagnified image

A objective aperture is used to restrict beam current and limit aberrations

Objective lens converges the beam to form a spot on the sample of disk size d



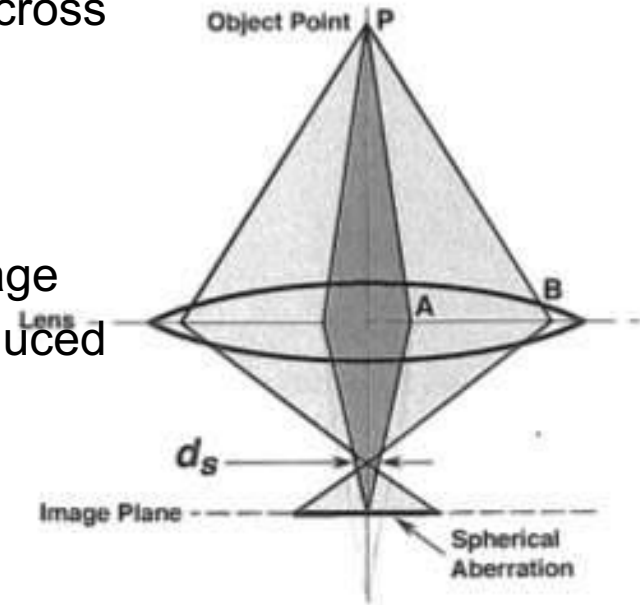
Aberrations in electron lenses

Spherical Aberration

Electrons closer to the coil are bent further than those near the axis.

This causes spread in the image to be spread across along the 'optic axis'

This leads to a disk formation instead of a point. Smallest disk is typically formed just before the right image plane. Disk called disk of least confusion (d_s) can be reduced by reducing the acceptance angle α of the lens.

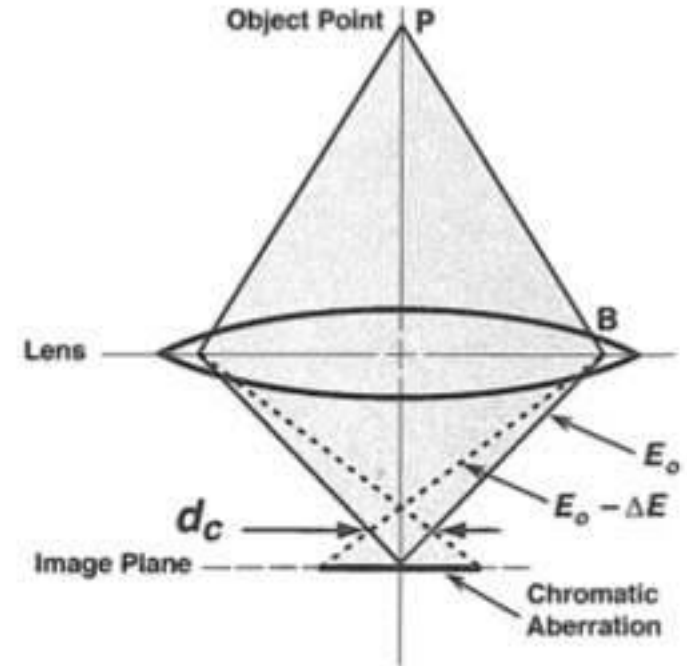


Chromatic Aberration

The electron deflection in lenses also depends on the energy (velocity).

So beams with different energy (E) will be focused differently.

If the incident beam has a central energy E_0 and a spread ΔE , then a disk of least confusion is given by $d_c = C\alpha \left(\frac{\Delta E}{E_0} \right)$



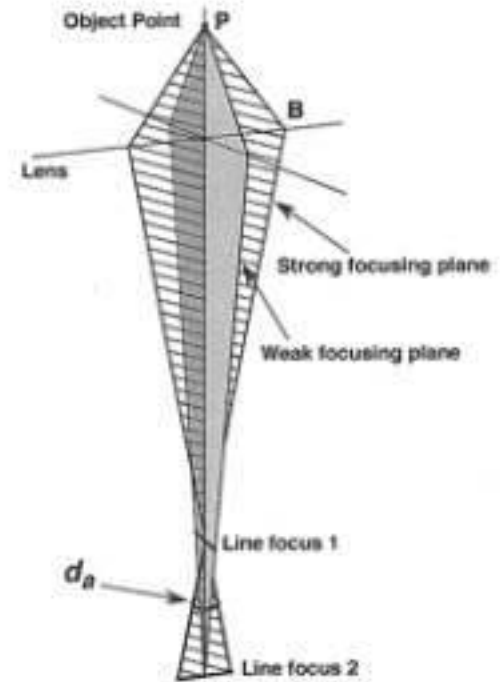
Astigmatism

Machining errors in the lenses, inhomogeneity of pole piece, asymmetry in the lens winding – deviation from perfect cylindrical hole, causes astigmatism.

Image points are stretched either along the x-plane or along the y-plane.

Way to observe:

As you move from out-of-focus to in-focus to out-of-focus, the image will stretch from one direction to its perpendicular



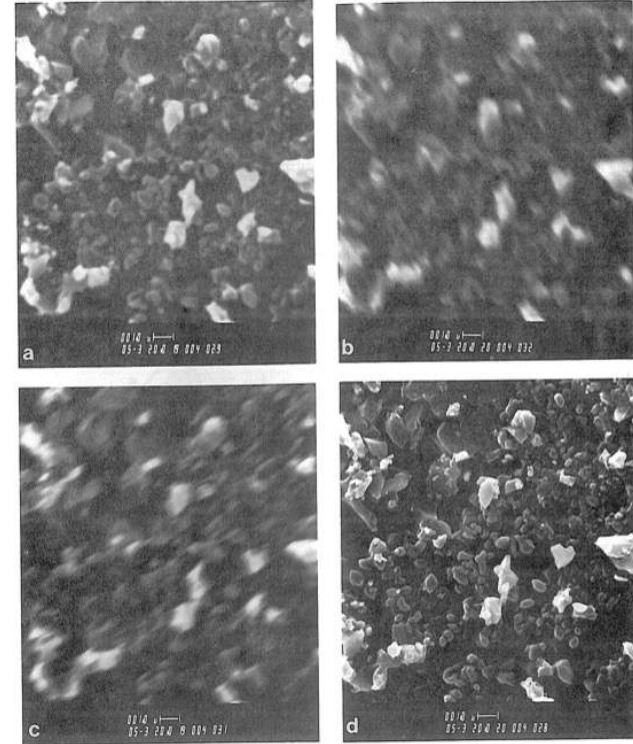
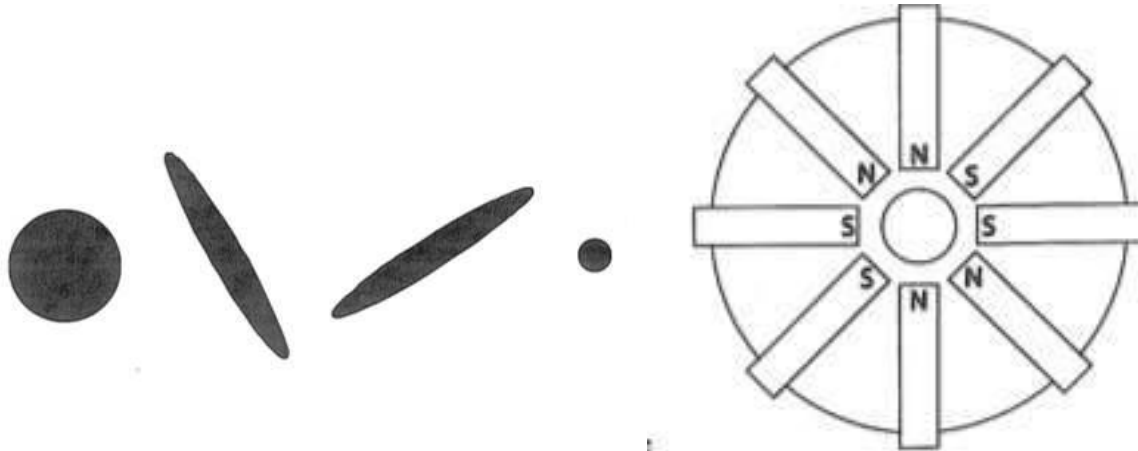
Among the most common of issues with electron lenses

Astigmatism correction:

An octupole magnet is used to correct for astigmatism.

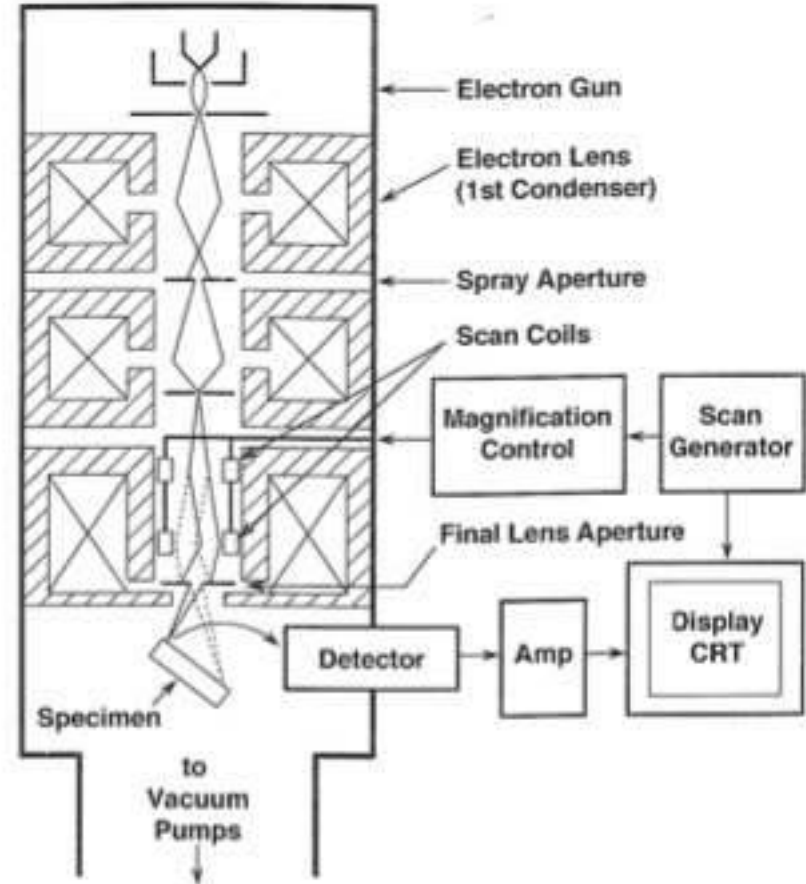
A typical correction cycle (x-stigmator, focus, y-stigmator, focus) should be employed to correct.

Eventually, the beam cross section will be the smallest

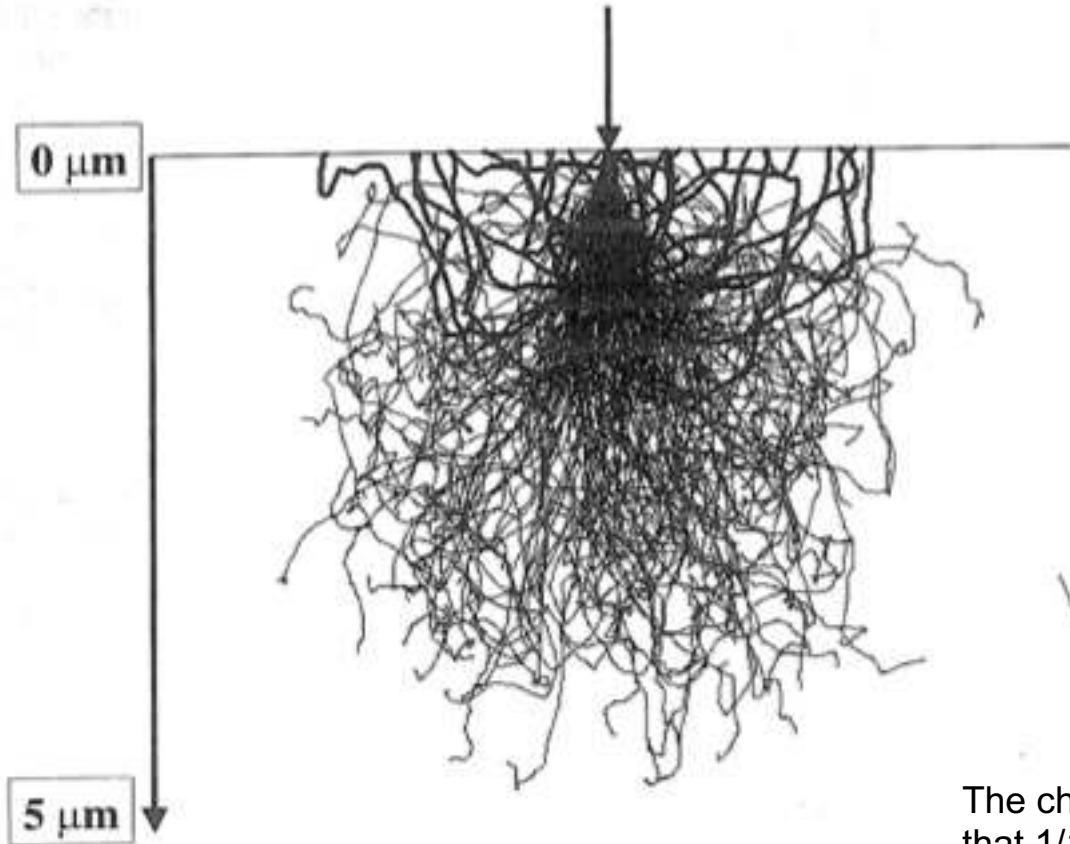


Electron – Sample Interaction

- Electrons are emitted from filament,
- Accelerated by the anode-cathode potential difference
- Diverging rays are converged onto the sample stage using condensing lenses and objective lenses.
- The beams converge on the sample and now we need to understand the e-beam sample interaction.



E-Beam accelerated at 20 kV, striking Si



- Assume the filament has a brightness of $1 \times 10^8 \text{ A/cm}^2 \text{ Sr}$
- Say the anode is at 20 kV higher than filament.

The electron beam that incidents the sample has a dia of 1nm, divergence $5 \times 10^{-3} \text{ sr}$, and a current of 60 pA.

The chamber is at 10^{-7} Torr , which makes sure that 1/10000 electrons collide with a gas atom

Electron – Material Scattering physics

As accelerated electrons interact with specimen, two kinds of scattering can happen

1. Elastic – (No transfer of energy/mass)
2. In-elastic – (energy transfer from the electron -> atoms)

Probability to scatter elastically $Q(> \phi) = C \frac{Z^2}{E^2} \cot^2 \frac{\phi}{2}$ [cm^2] –scattering cross section

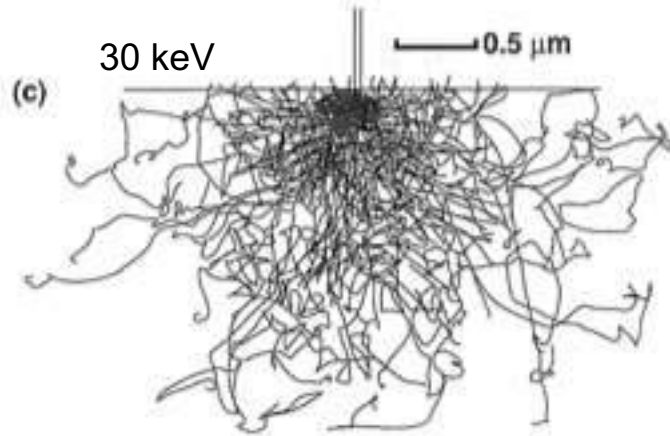
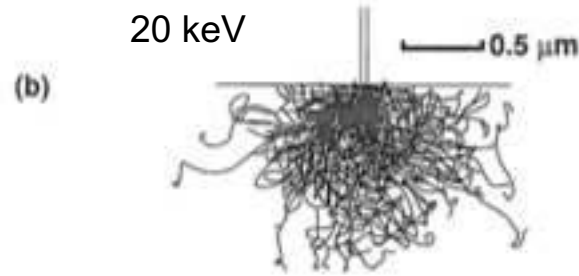
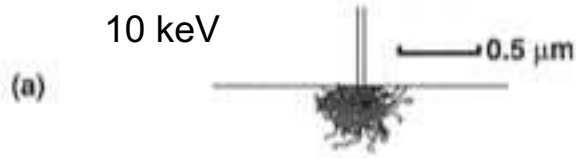
Z – atomic number, E – electron energy, ϕ – scattering angle.

In similar spirit, the rate of energy loss for inelastic scattering is given by Bethe equation

$$\frac{dE}{ds} = C \frac{Z\rho}{AE_i} \ln \frac{1.66E_i}{J}$$

ρ – density, J average loss of energy/event

Variation of Interaction volume with electron acceleration



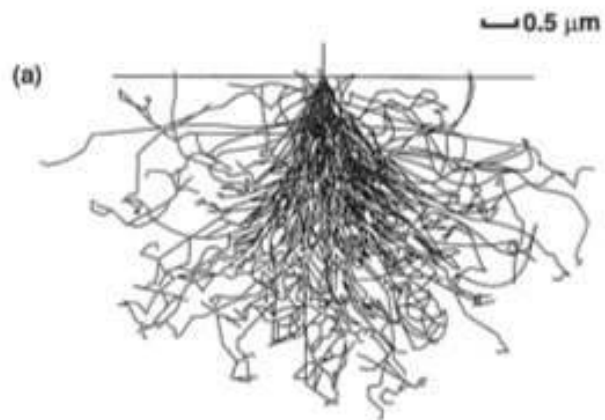
Elastic scattering rate inversely proportional to electron energy

$$\text{Electron energy loss } \frac{dE}{ds} \propto \frac{1}{E}$$

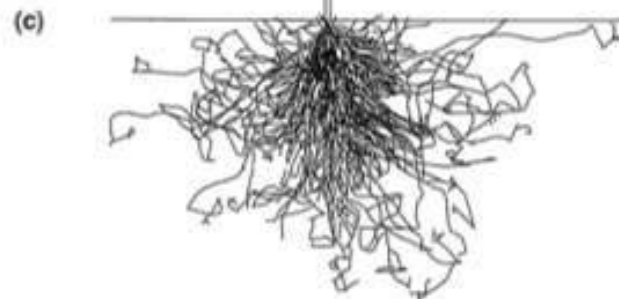
So beam penetrates further inside the material.

Some trajectories exist the material surface –
They are called **back scattered electrons** of the beam!

Z = 6 Carbon

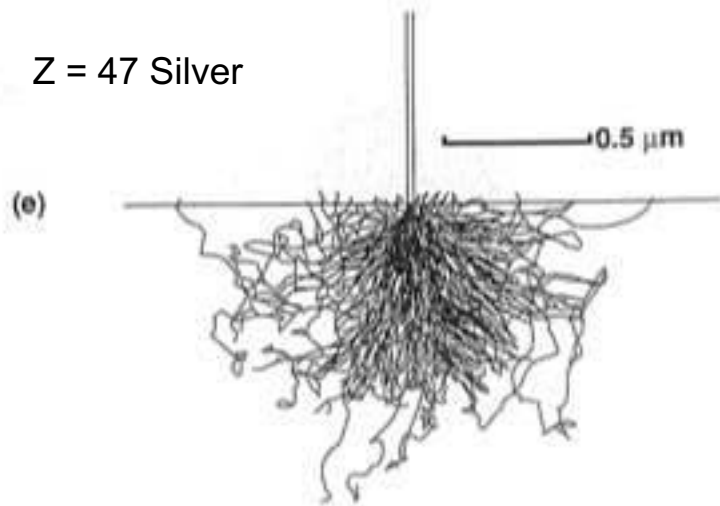


Z = 26 Iron

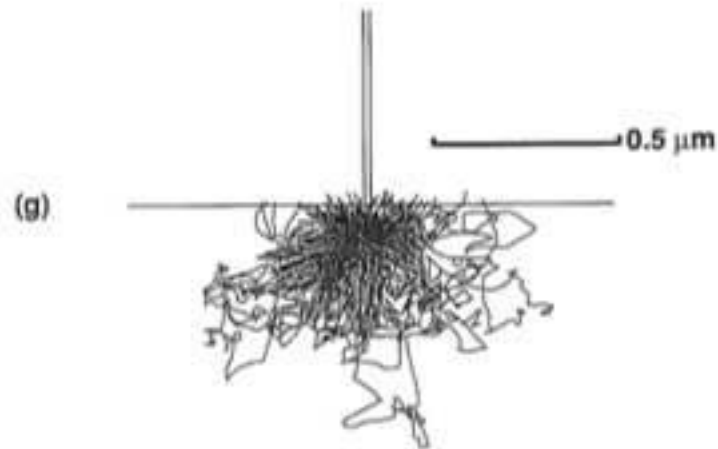


Beam Energy 20 kV

Z = 47 Silver



Z = 92 Uranium

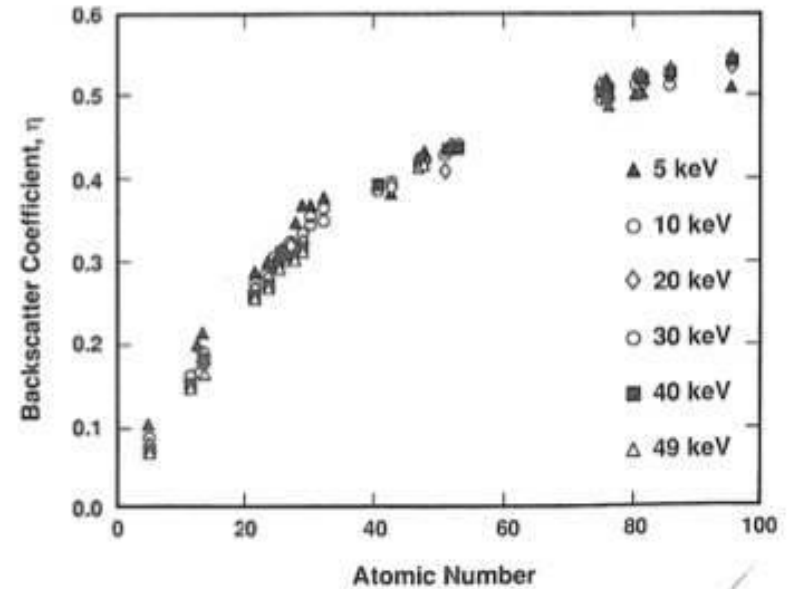
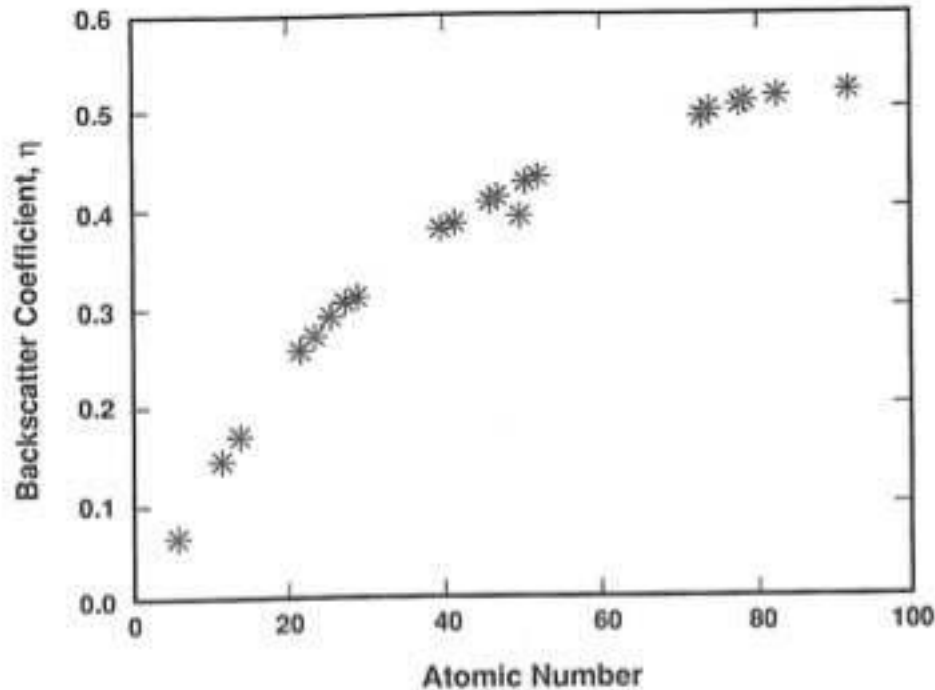


- As Z increases, the cross section (probability) for large angle elastic scattering increases. $Q \sim Z^2$
- For lower atomic number, the scattering angles are not large, the beam penetrates deeper into the sample, decaying slowly. Smaller percentage of electrons exit the material. Interaction volume – pear shaped
- For higher atomic number, the interaction volume is smaller, yield of BSE higher. Interaction volume plane truncated hemisphere.

We define back scattering coefficient $\eta = \frac{n_{BSE}}{n_B}$

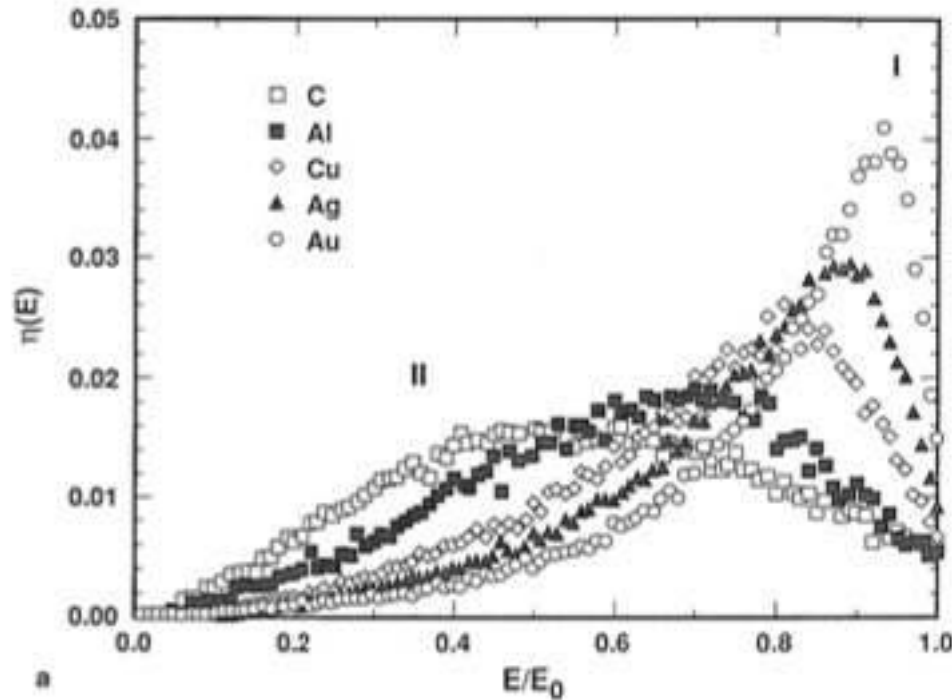
Where n_{BSE} is the number of back scattered electrons and n_B the number of electrons in the beam incident on the sample

η dependence on energy and sample material



Atomic number is the most dominant contributor to BSE yield. Thus, a great method to obtain material contrast.

Energy of BSE



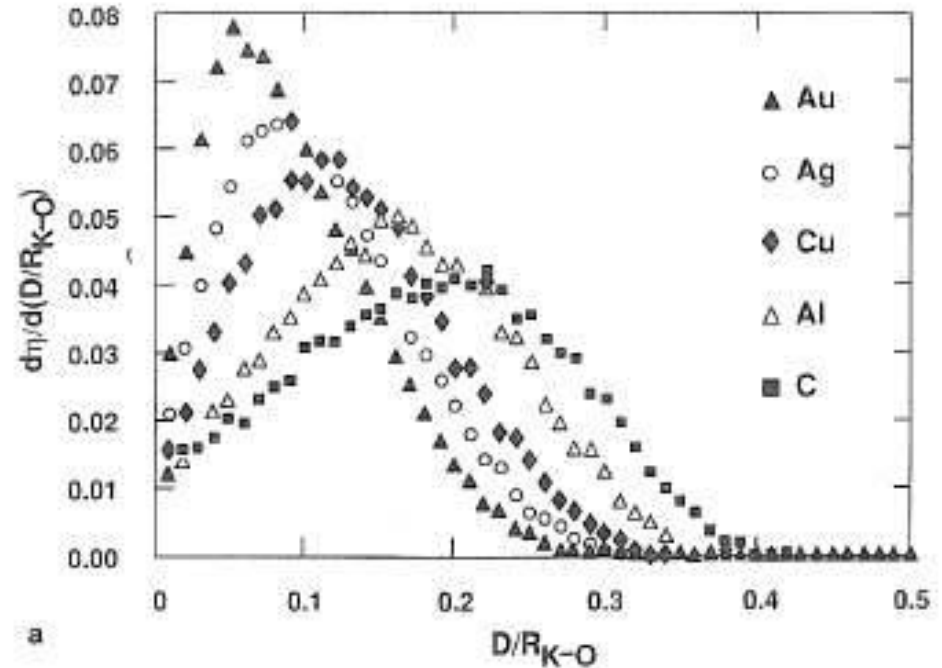
The energy of BSE electrons are clearly separated.

High Z substrates has BSE yield with energy very close to incident beam (Region 1)

Low Z substrates has poor BSE yield and even then their energies are low (region II).

Thickness dependence of BSE yield

1. Heavier atoms give maximum yield
2. The yield is obtained mostly closer to the surface.
3. Lighter elements give generally a smaller yield
4. Most of the exiting electrons are from certain distance from the surface

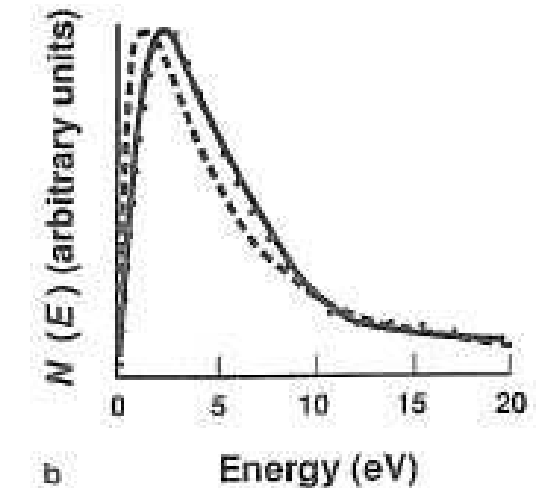
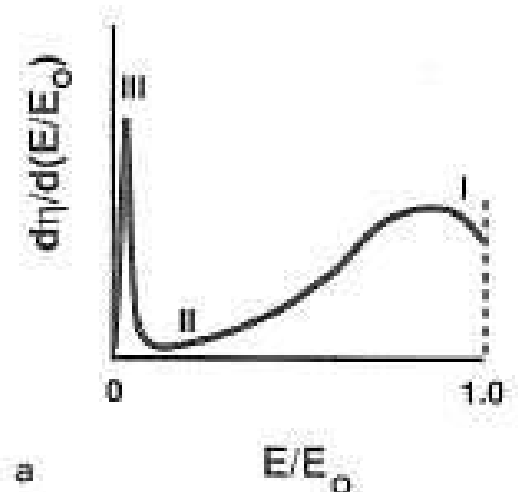


Effect of sample tilt (Given to self-reading)

Very intuitive.

Secondary electron

- The largest (integrated) yield of electrons in SEM are from elastic scattering (similar to the beam energy)
- However, the yield of electrons at very low energy (<50 eV) is higher in numbers than the high energy electrons.
- These are secondary electrons (electrons with energy less than 50 eV (arb. Choice)).
- 90% of the electrons are ejected with energy 0 – 10 eV.



Thickness dependence

The probability of electron to escape depends on the distance from surface

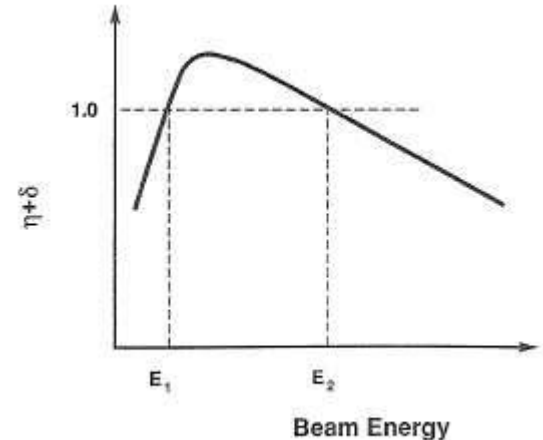
$$p = e^{-Z/\lambda}$$

Z – distance from surface, λ is mean free length of electron

Typical maximum distance of SE yield 5λ

If we define $\delta = \frac{n_{SE}}{n_{BSE}}$ and $\eta = \frac{n_{BSE}}{n_{beam}}$

There exists a window of maximum electron yield

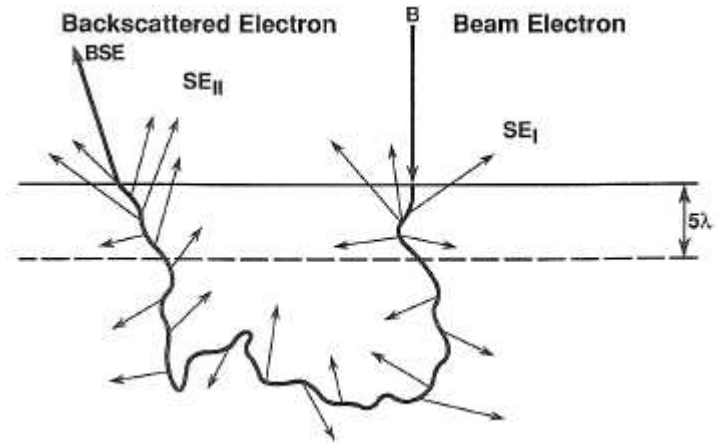


Types of secondary electrons

- SE – I emitted electrons at the beam incidence
- SE – II emitted electrons due to the back scattered electrons

SE – 1 which emerges near the beam position depends purely on the surface properties. It has high spatial resolution (electron emerging largely from the beam position).

However, SE- II which has its origin in BSE, has material dependence as well. Can have poor spatial resolution because the electrons exit from very different positions on the surface.

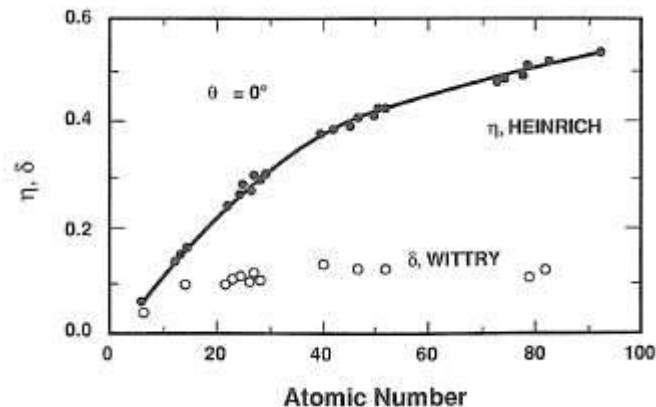


Compositional difference

Total secondary electron count $\delta_T = \delta_1 + \delta_2\eta$

δ_2 has its origin in the BSE and hence is multiplied by η

In typical scenarios, $\delta_2 = (3 - 4)\delta_1$
ie, BSE is far more efficient in generating SE.



While BSE has almost linear relation with atomic number, the total SE is largely unaffected.

Detectors

- Electron detectors
 - Back-scattered electron detector (Electrons with energy $0.7 - 0.9 E_b$)
 - In-lens detector (secondary electron detector ($0 - 10$ eV)
 - Everhart-Thornley detector
- Photon Detectors
 - X-Ray detectors with EDS/WDS
 - Photomultipliers

Everhart-Thornley detectors

Main components:

Scintillator – capable of luminescence

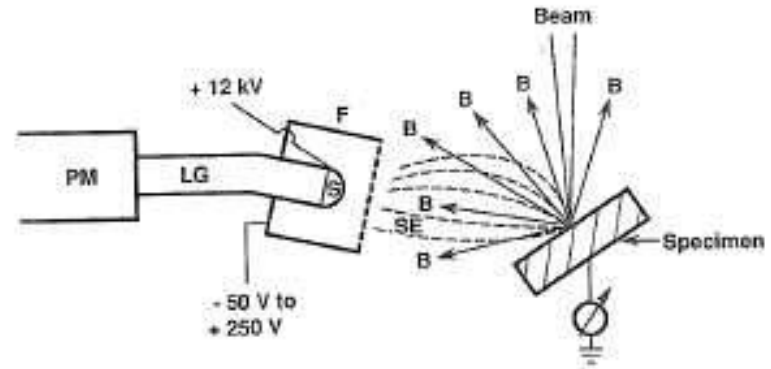
Eg: doped plastic/glass, doped CaF_2

High energy electrons/Xrays incident on scintillator leads to photo emission.

A positive bias is applied to accelerate the electrons when its incident on the material

A faraday cage (metal sheet) can be set to a bias potential to select electron energies.

Photons guided by light guide to photomultiplier

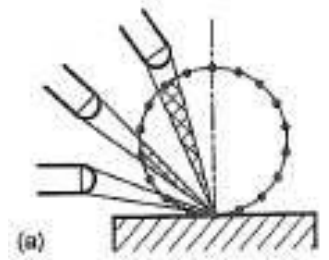


- A large potential is set close to the scintillator material
 - The metal sheet (faraday cage) shields the electric field from affecting the electron beam)
 - The metal sheet can also be slightly biased to select the electron energy.

When the metal sheet is biased negatively

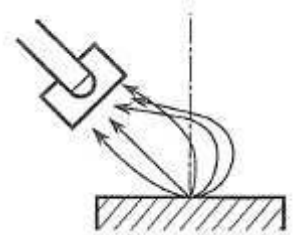
Electrons with small energies are repelled away (SE electrons).

High energy electrons are collected with line-of-sight



When positive potentials are applied, all electrons are collected.
However, BSE count is not changed largely (those with-in the line of sight),

But there is a huge increase in in-directly generated electrons



In short:

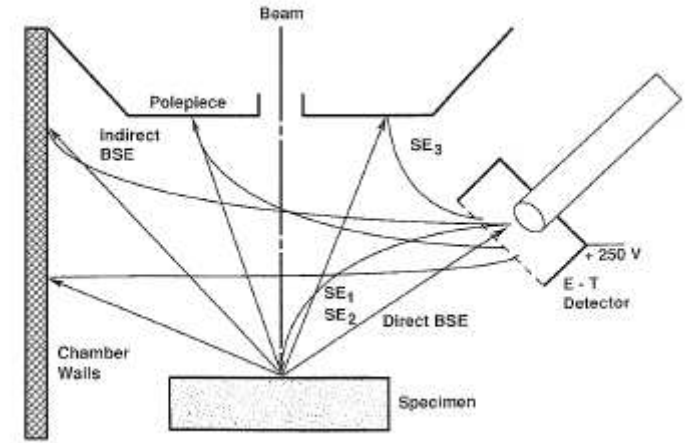
In ET detectors with –ve cage bias

- BSEs are largely collected.

In ET detectors with +ve cage bias

- both BSE + SE are collected.

+ve biased detectors are efficient detectors such that they can function even at very low acceleration potentials (~ 1 keV)



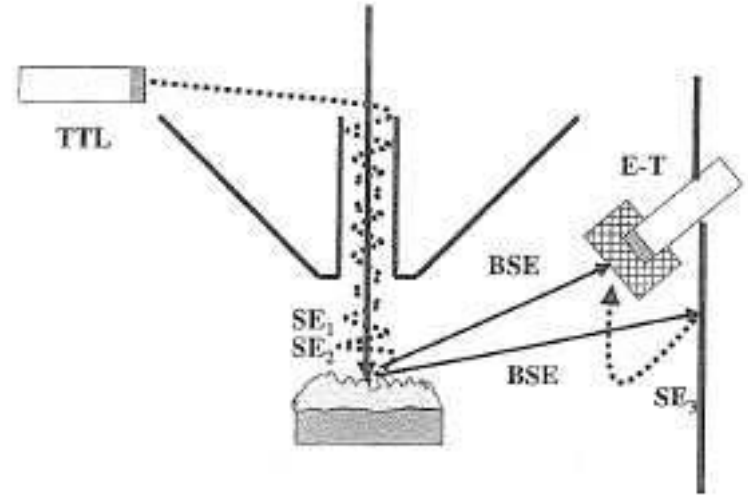
In-lens detectors

- In microscopes with Snorkel lenses
An additional detector can be placed inside the lens.

Magnetic field lines of the lens collect the electrons exiting the surface very near the beam entrance.

Such electrons are collected with high efficiency with positively biased scintillator.

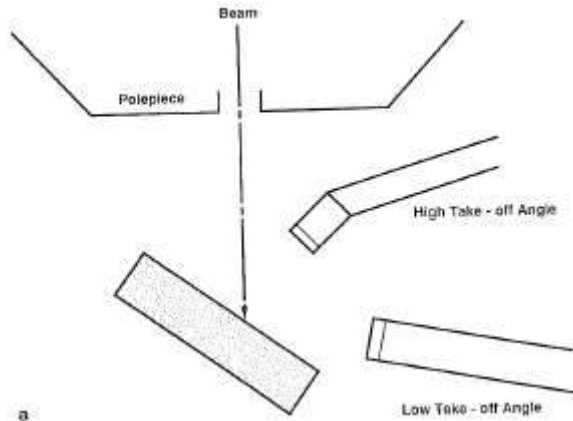
In-lens detector avoids SE3 and BSE efficiently.



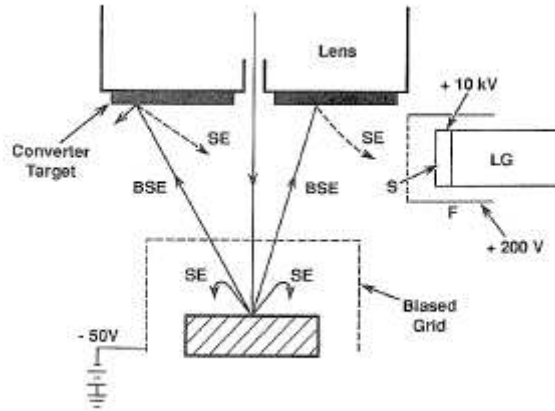
Kinds of BSD

- BSE are easier to detect due to their energy.

Direct Scintillator detector



BSE – SE detector



Annular BSE detector

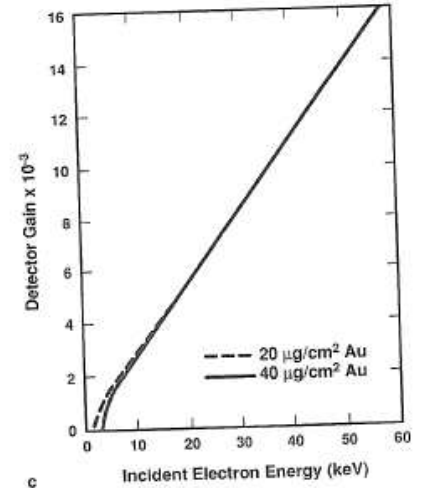
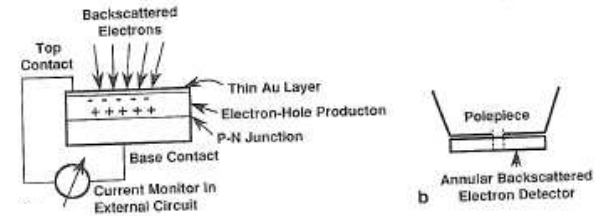


Image Formation

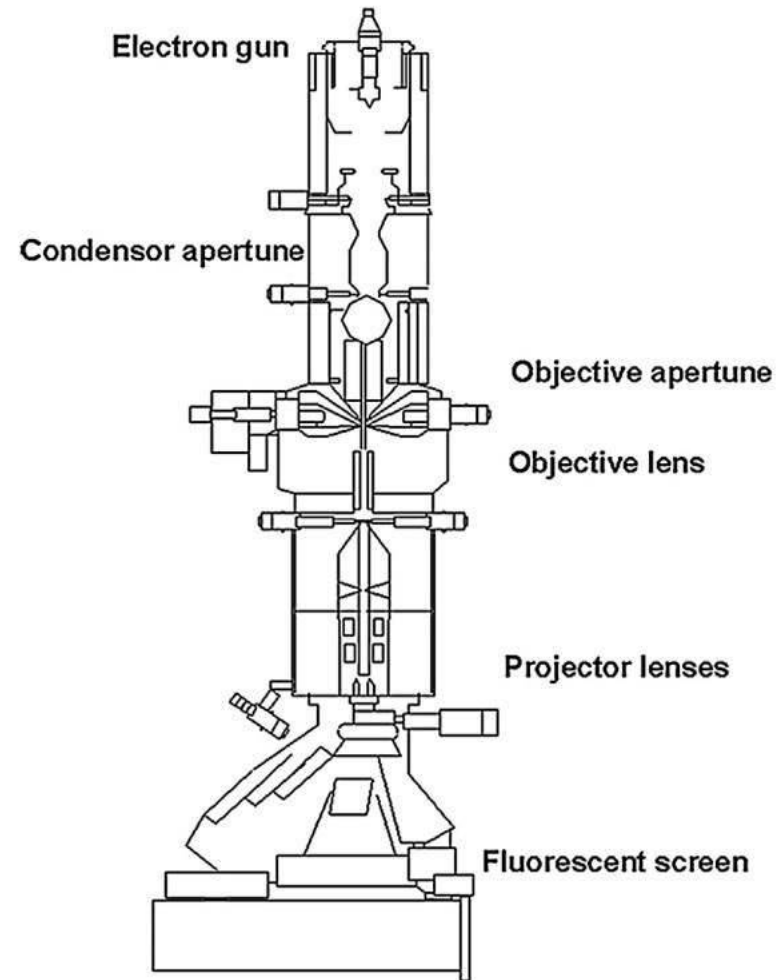
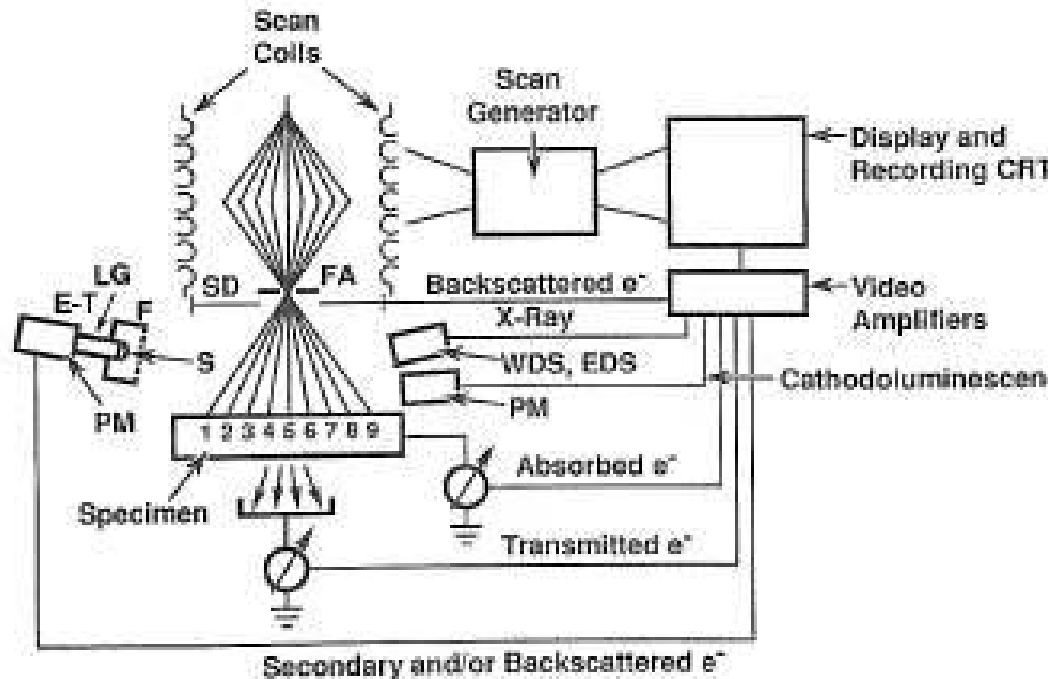
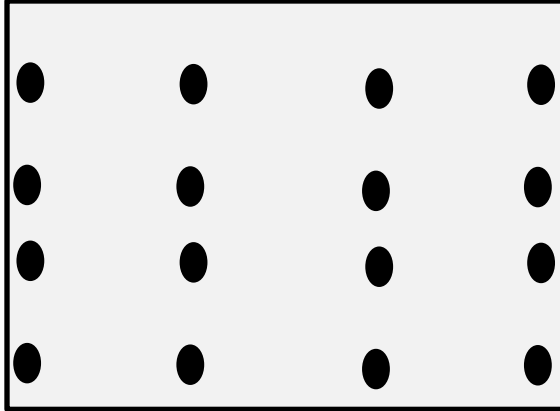
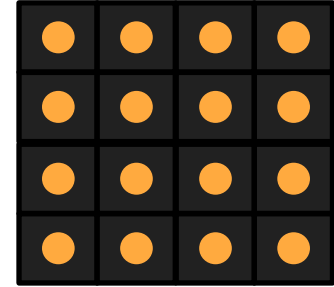


Image formation in a SEM



Beam scan points



Pixels on monitor

1 – to – 1 correspondence between beam points and pixels on screen

Pixels on monitor is set to gray scale. For instance, a range is set (0 – black and 256 for white).

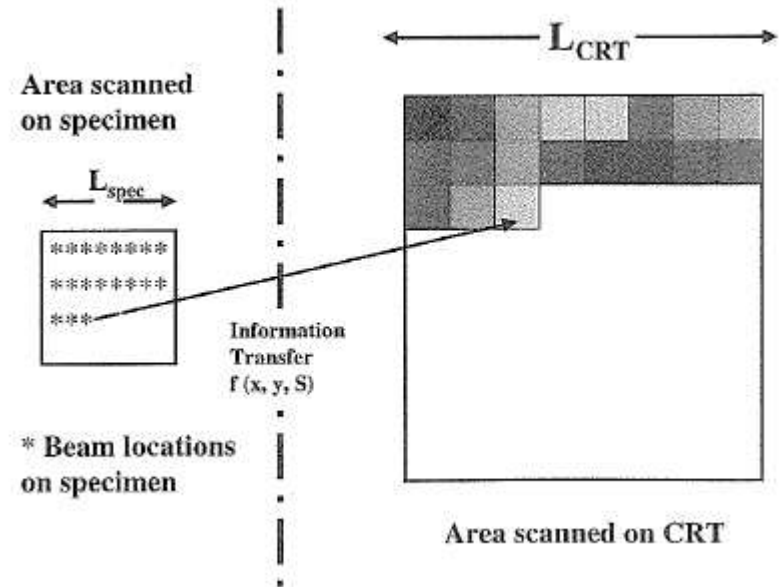
Depending on the intensity obtained in the detector, the gray scale is set.

A scanning electron microscope is mapping operation

- Each point on the sample is converted to a/sequence of points on display device
- Geometric rations need to be retained for faithful reproduction
- The mapping needs to be synchronous to be relevant

The image – grayscale bitmap can be stored digitally and recovered at need.

Each point on the sample is ‘scanned’ by rastering the electron beam on the sample sites.



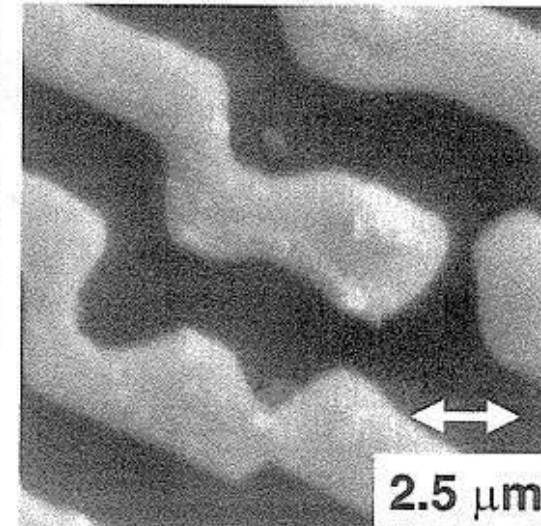
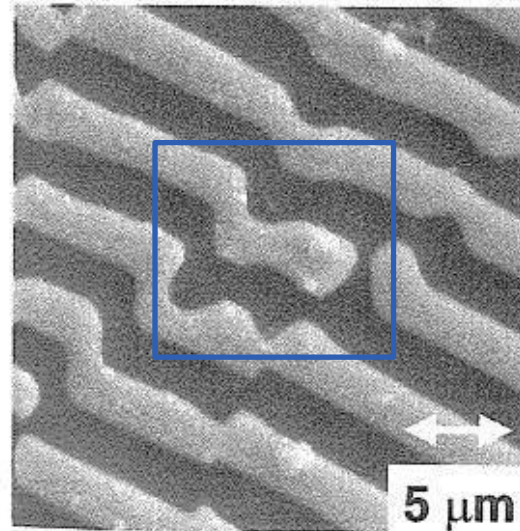
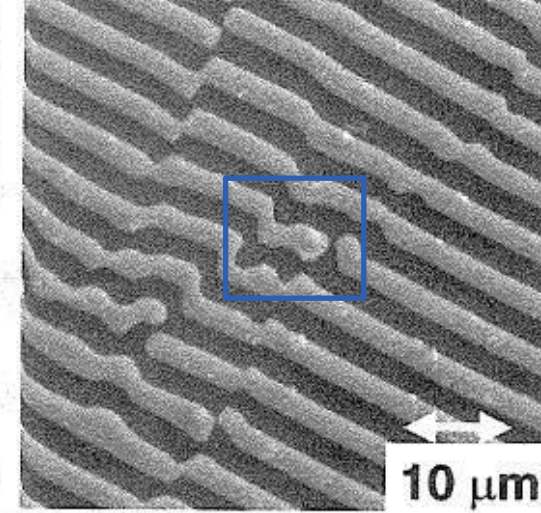
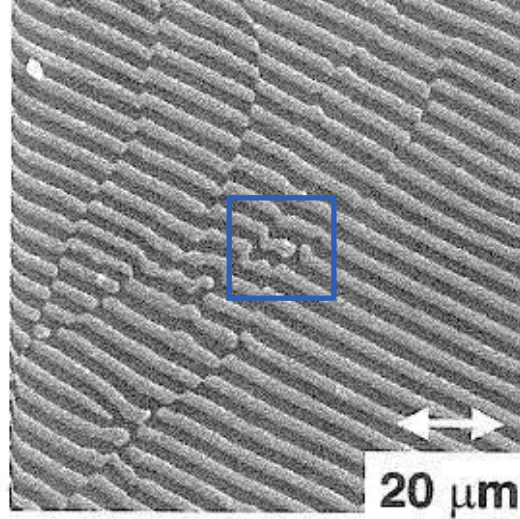
Magnification

Density of pixels on display device is fixed.

Magnification is fixed by the length of scan region.

Thus, as the magnification is increased, the spot size needs to reduce.

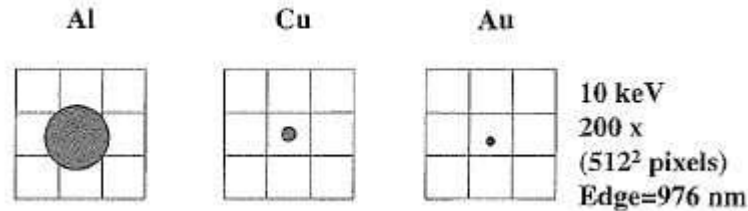
While changing magnification from 100x – 10000x, the area of the scan reduces 10^4 times !



We define a pixel element $D_E = \frac{L_S}{N_P}$ where L_S is the length of scan and N_P is the number of points along the length

Say, we are looking at display resolution of 1024x1024, as the magnification is increased, the pixel element gets narrower.

To have minimal overlap with neighbouring pixels, D_E should be equal to the area from which electron signals are detected



The pixel size (square length) is determined by the magnification. The region from which the signal is obtained in SEM depends on excitation potential and the sample material

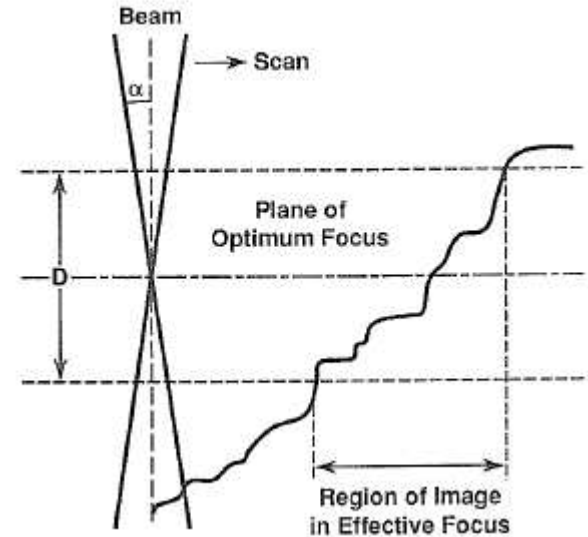
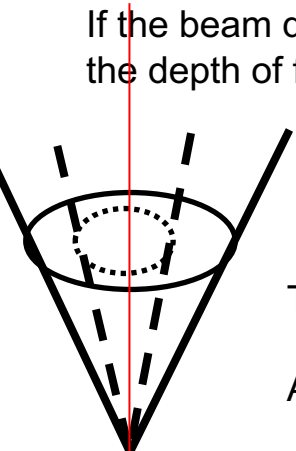
Depth of Field

The electron beam is converted to a point with angle α

We bring the sample to focus, when the surface is placed at the smallest beam diameter.

However, if the sample is rough, there can be regions where the beam is not in its sharpest form.

If the beam diameter becomes closer to the pixel size, we define the depth of focus there.



Thus, the deciding factor for depth of field is the convergence angle α

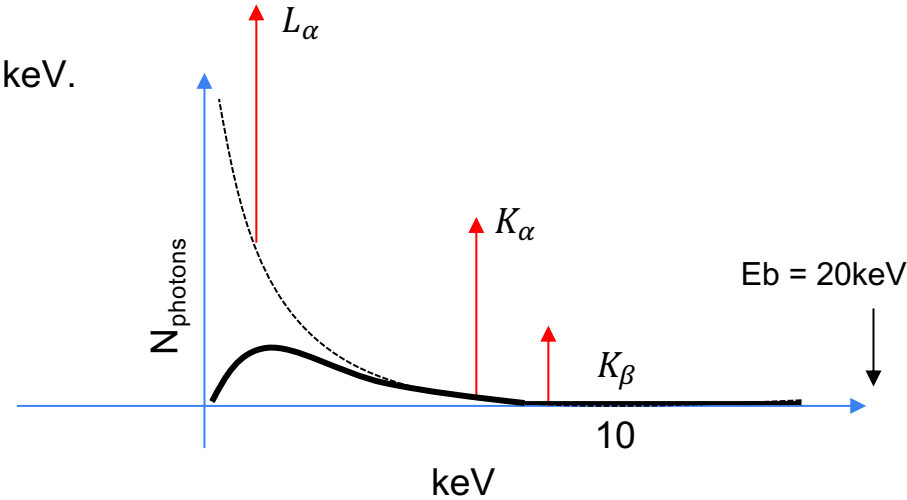
A smaller aperture opening, leads to smaller convergence angle and a higher depth of field!

Remember, the electrons in SEM are accelerated to few keV.

Electrons from beam lose energy by elastic collision.

The lost energy in the sample can be emitted back as photons

$$\Delta E = h\nu$$

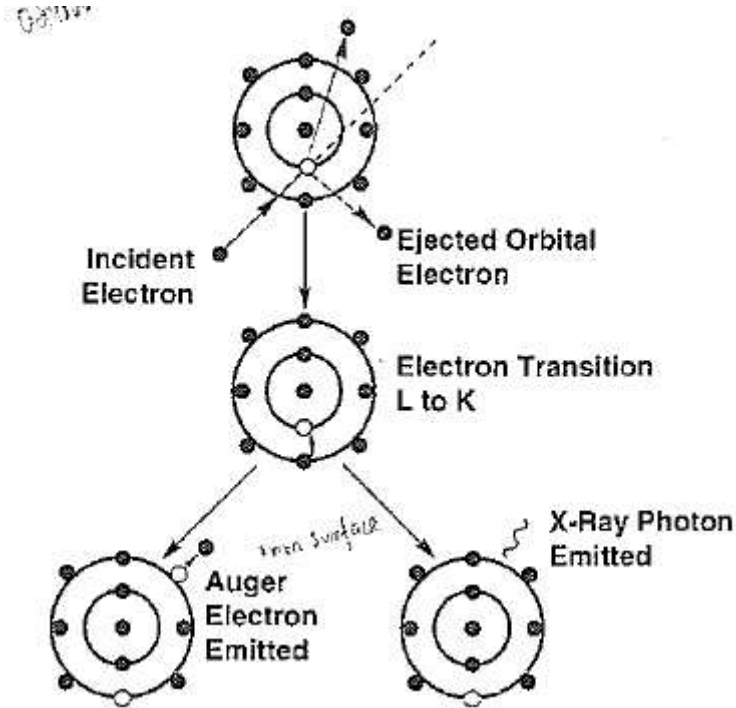


Take the example of Cu, an incident electron beam with energy 20 keV, will give a photon emission spectrum as shown in the dotted black line.

However, a typical measured photon spectrum looks a bit difference. Guess why ?

Allowed transitions in an atom

If interested, read
about Moseley,
Barkla, et. al.,



Auger – when energy is lost to
another electron in the atom

Characteristic X-rays can
be emitted when the
electrons are swapped from
outer to the core electrons

The diagram illustrates the Bohr-Mottelson model of the nucleus, showing concentric shells labeled K, L, M, N, O and various subshells like L_I , L_{II} , L_{III} , M_I , M_{II} , M_{III} , M_{IV} , M_V . It includes labels for nucleons (α , β , γ , δ) and specific energy levels (e.g., α_1 , α_2 , β_1 , β_2 , γ_1 , γ_2). The center is labeled 'NUCLEUS'.

Depending on atomic number, the number of possible lines in the Xray spectrum also increases.

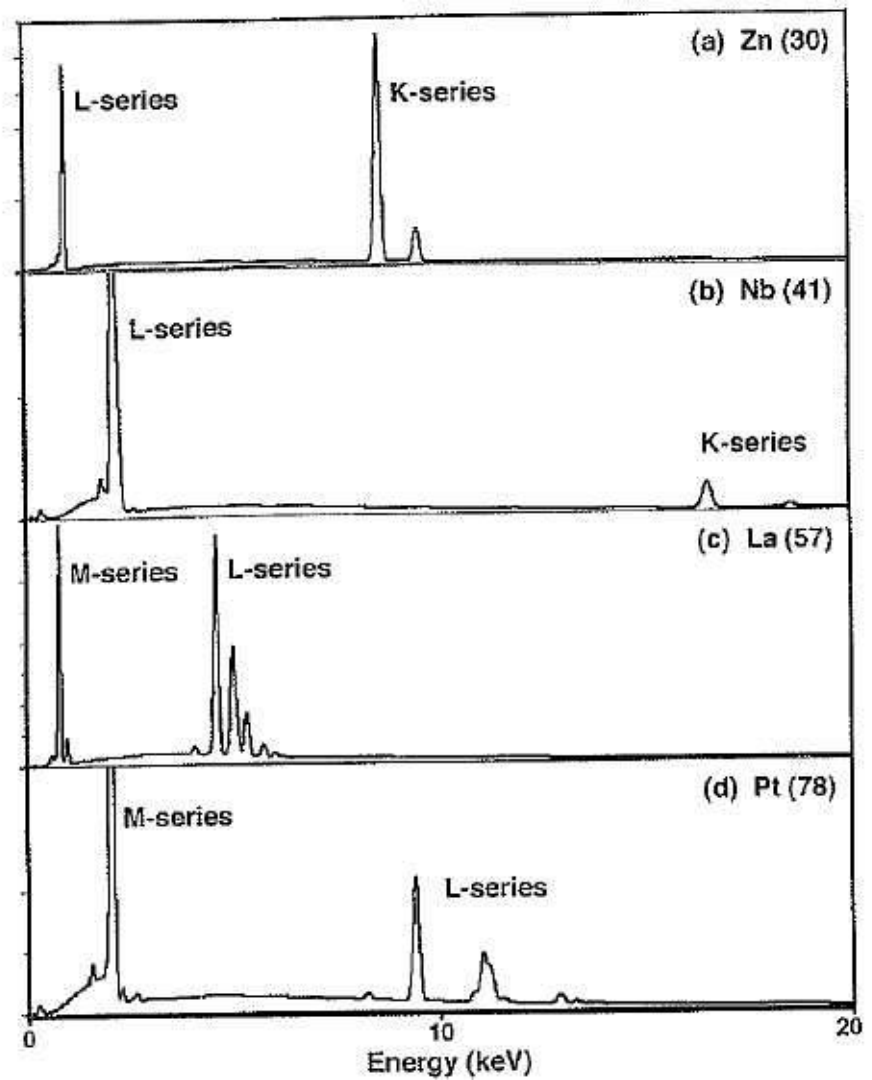
Peaks from different elements

Typical characteristic peaks have width much smaller than the peak energy

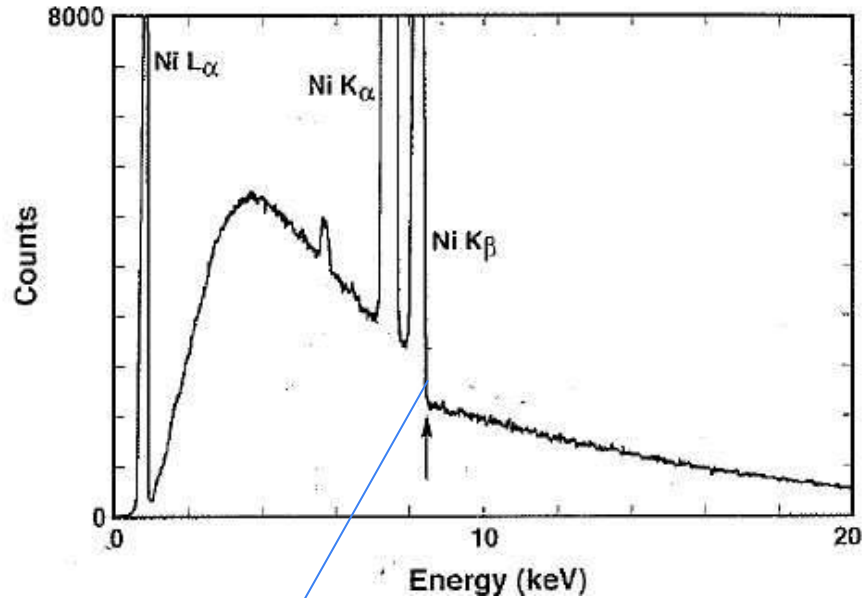
$$\Delta E \ll E_{peak}$$

While the L line is singular for smaller element,
The lines start to spread out as the Z increases.

If the energy of the X-ray can be measured along with the number of photons (spectrometer), the atoms under the electron beam can be uniquely identified.

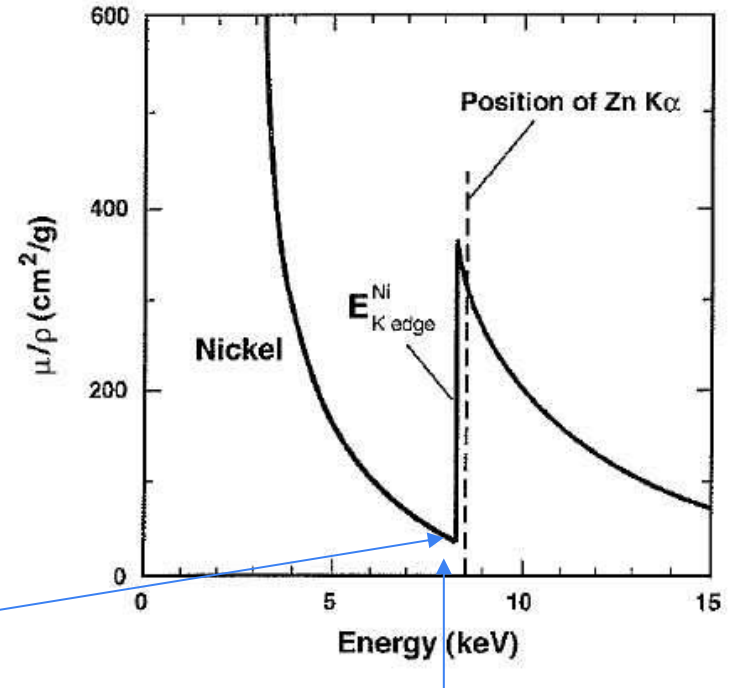


Emission spectra from Ni for excitation at 40 keV



Sudden dip in Bremsstrahlung

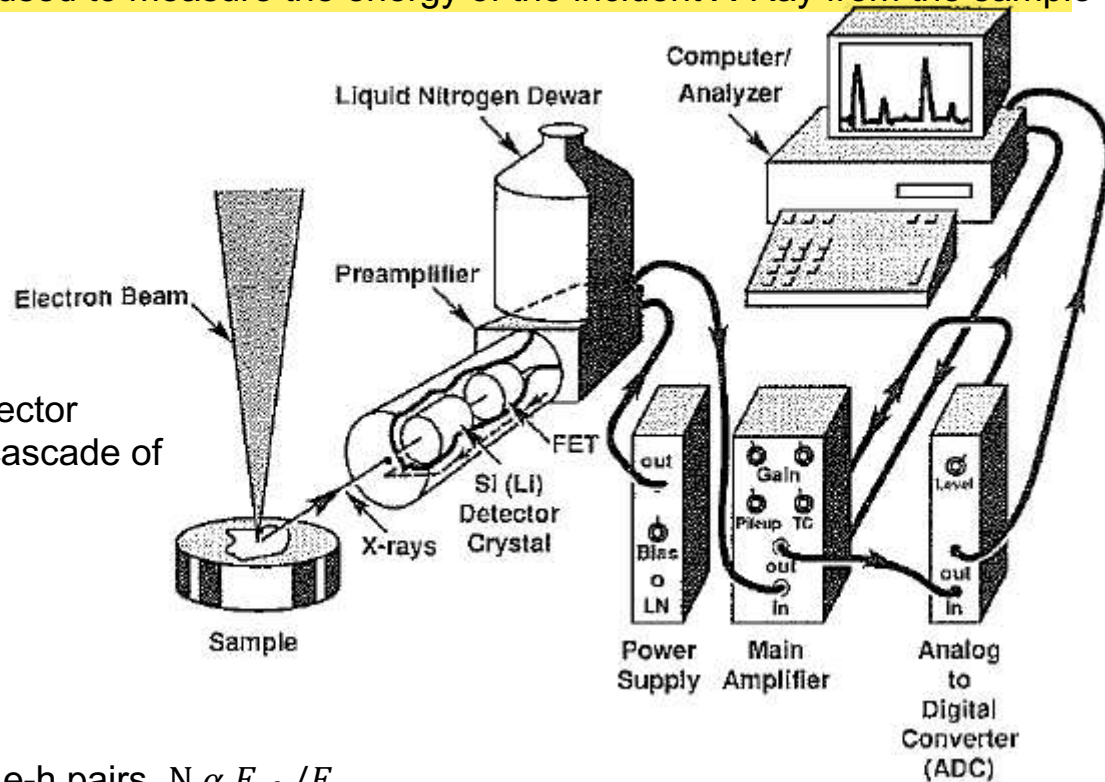
Mass absorption coefficient



This is due to strong increase in absorption at this point.

Typical Energy Dispersive Spectroscope

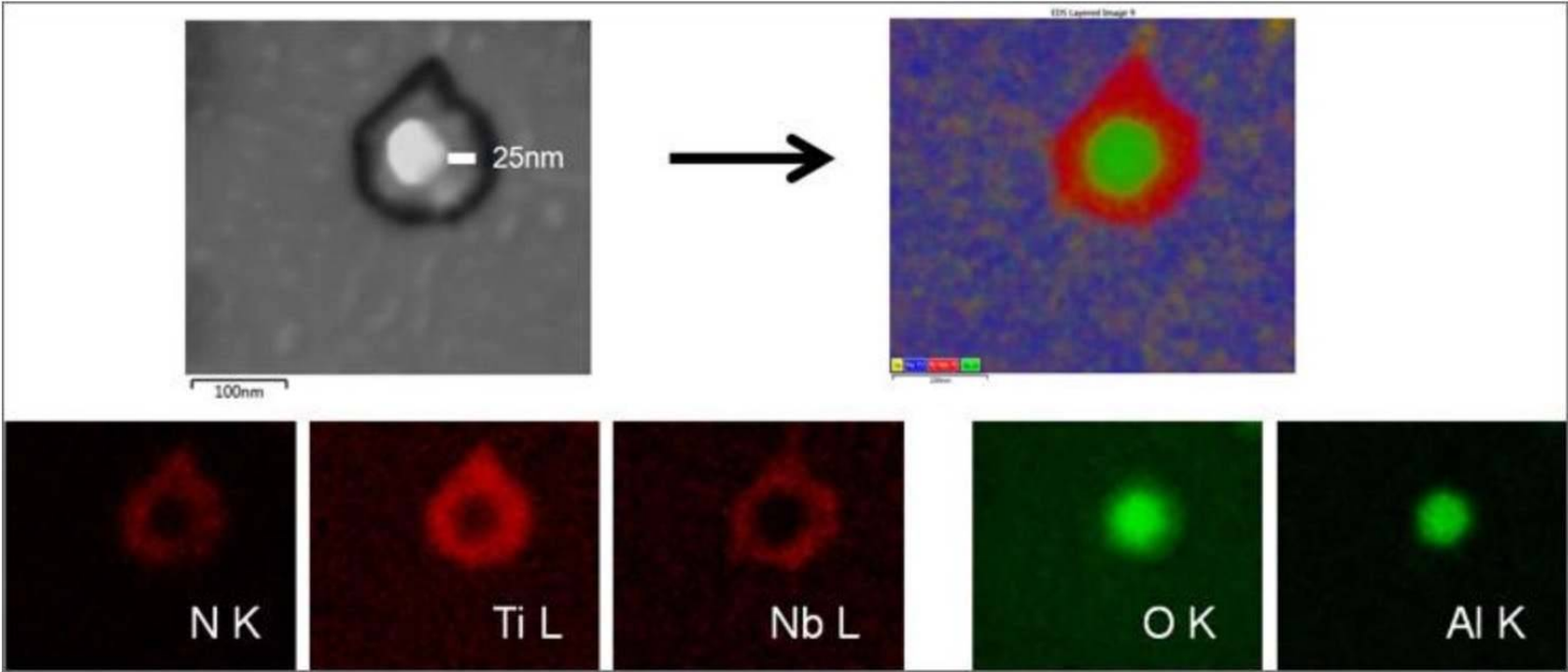
Energy dispersion is used to measure the energy of the incident X-Ray from the sample



A doped Si *p-i-n* detector
Is used to create a cascade of
e-h pairs

The total number of e-h pairs $N \propto E_{ph}/E_g$

Elemental Mapping in EDS



Thermal Analysis

We define thermodynamics... as the investigation of the dynamical and thermal properties of bodies, deduced entirely from the first and second laws of thermodynamics, without speculation as to the molecular constitution.

—J. Clerk Maxwell

Classical thermodynamics... is the only physical theory of universal content which I am convinced... will never be overthrown.

—Albert Einstein

Ice when heated, gives water which on heated boils!

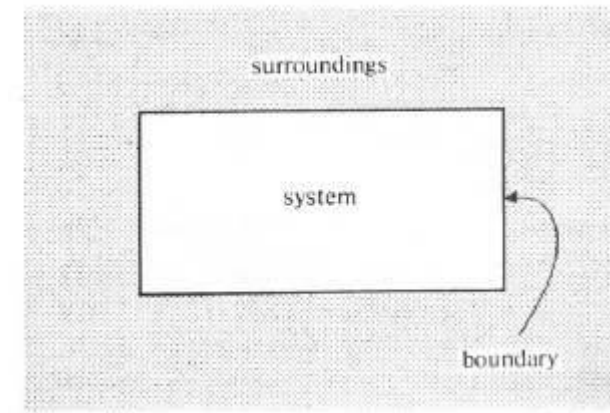
Which can then be cooled to return back to previous states!

Sugar when heated gives a brown caramel.
Heating coal gives gas, tar and coke.

Certain properties are obvious at appearance.

However detailed investigation can reveal interesting properties about material

Thermal properties such as heat capacity, enthalpy, entropy determine physical characteristics



What is Temperature ?

It follows from Zeroth law:

If two systems are in thermal equilibrium with third, then the two must be in thermal equilibrium.

So what is thermal equilibrium ?

For a portion of the universe that concerns us, called system, there exists a set of observables (measurables like pressure, volume, polarizability, displacement,...).

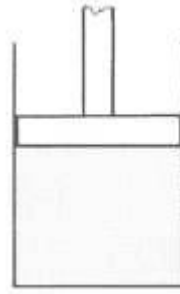
When our system is made to interact/ brought into contact with some surroundings/another system, a change of measurables is set in motion.

A state of thermal equilibrium is reached when the changes die down (No measurable change)

But still, what is temperature ?

Consider the example of frictionless piston P_1

At any time, its state is defined by variables P_1, V_1



The piston contains a fluid with uncountable number of particles. There are two free variables P, V

Imagine a reference piston P_R

We set the piston to a state with P_R, V_R

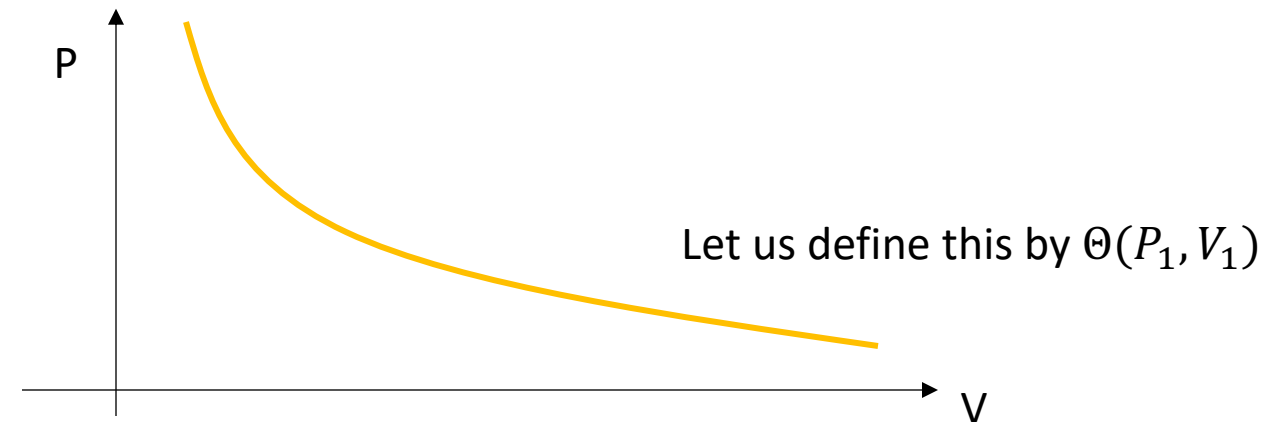
If we enforce thermal equilibrium between the two fluid pistons, then there will be only one free variable. If we take it to be V , then the pressure will be determined by the equilibrium condition

We can draw what are called as Isotherms

At any point on the curve, two systems are in equilibrium.

Each point is uniquely determined by P_1, V_1, P_R, V_R

We can define a function $\mathfrak{S}_1$ such that $\mathfrak{S}_1(P_1, V_1, V_R) = P_R$ for P_1 and V_1 along the line



Remember, P_2 is also in equilibrium with P_R

So, we can also then find another function $\mathfrak{T}_2(P_2, V_2, V_3) = P_R$

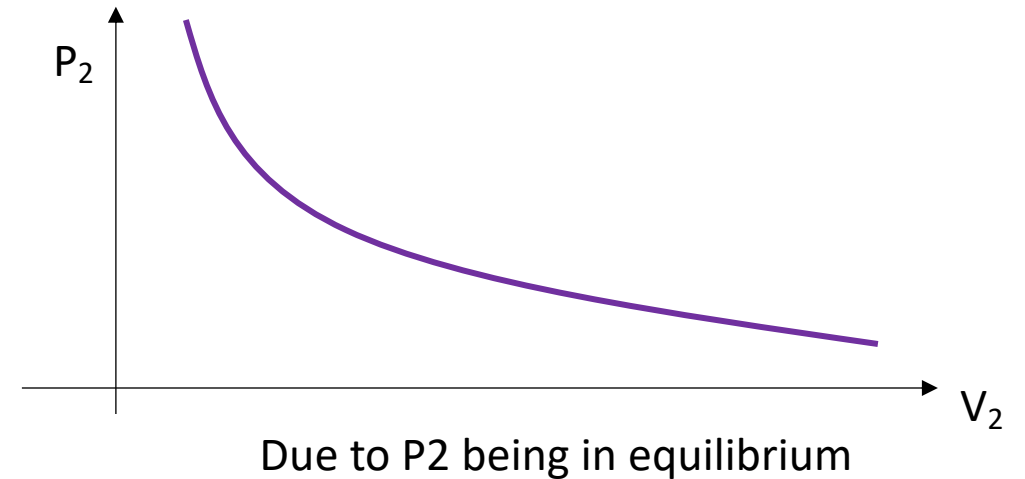
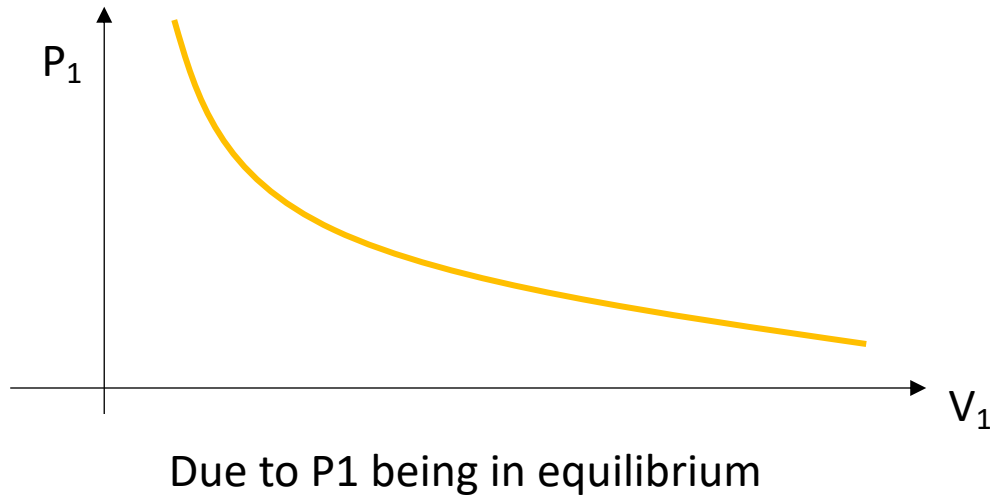
Obviously, $\mathfrak{T}_1(P_1, V_1, V_3) = \mathfrak{T}_2(P_2, V_2, V_3)$

Since, V_3 appears in both equations, coming from similar formulations, it will cancel out, giving you a form

$$\Theta_1(P_1, V_1) = \Theta_2(P_2, V_2)$$

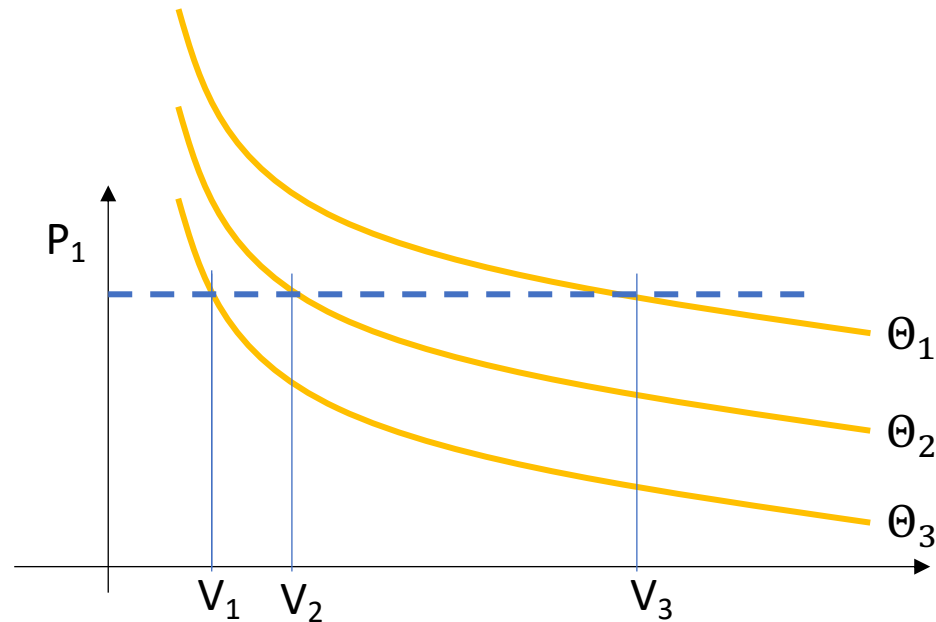
Thus, there exists function of pressure and volume of each systems, which remains the same in both the systems.

We call the function Temperature



We have still been very vague!

We need a numerical value of 'hotness'



You have three isotherms – defined by the 'invariant function Θ '

A numerical estimate of the function can be obtained as follows:

Take a gas – at low enough pressures, works like ideal gas.

$$\text{We defined } T = \lim_{p \rightarrow 0} \frac{pV}{R}$$

Lord Kelvin: let us choose R, such that at the triple point of water, the thermodynamic temperature $T = 273.16$ K.

The difference between melting point and boiling point of water was needed to be 100 units! (For calibration)

If you are wondering, why such weird number:

The ratio of 'hotness' of triple point to hotness difference from triple point – boiling point – 2.73!

That's why Kelvin's scale is an absolute scale. 0K is the lowest temperature.

Consider a situation when two systems are not in equilibrium and brought to contact.

So, two systems have two different temperatures. Under such situations, there seems to be an exchange – called heat.

Heat is a form of energy that is exchanged.

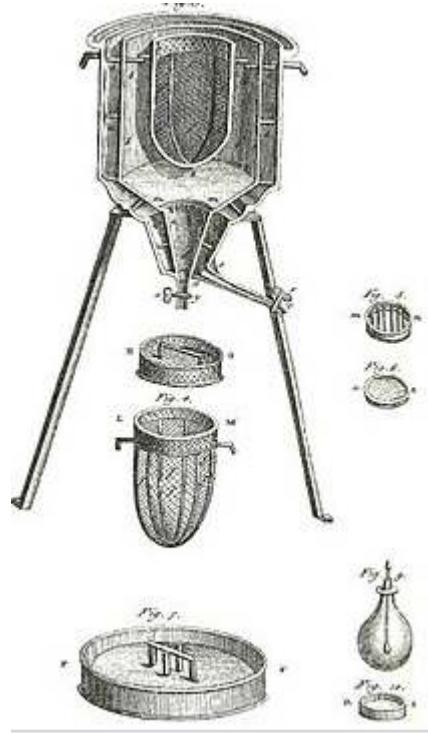
What is heat ?

Black's 1761 experiment:

Two theories existed:

Caloric theory

Heat is an indestructible fluid which permeated matter.
If flowed from hotter bodies to cooler bodies



Molecular motion theory

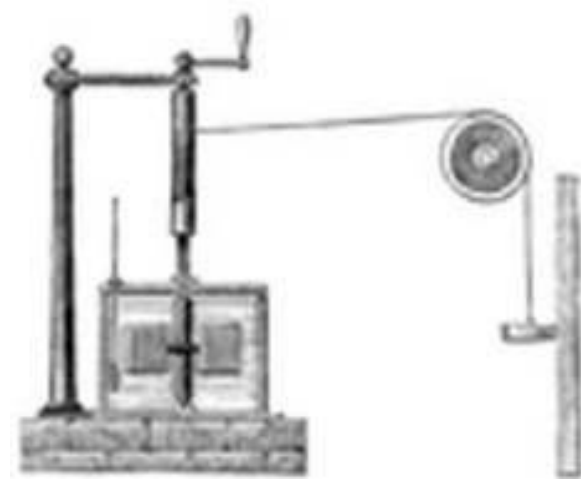
Heat was associated to rapid vibration of molecules.

When ice cold water was kept in hot room, the water temperature raised fast,
If an ice cube is inside the water, the ice cube melted, but the water temperature didn't rise!

If heat flowed from outside to inside, the heat in water should be more than that in the ice?

Joule's Heat (1840)

Heat doesn't have to flow from outside. He took thermally isolated systems and generated heat!

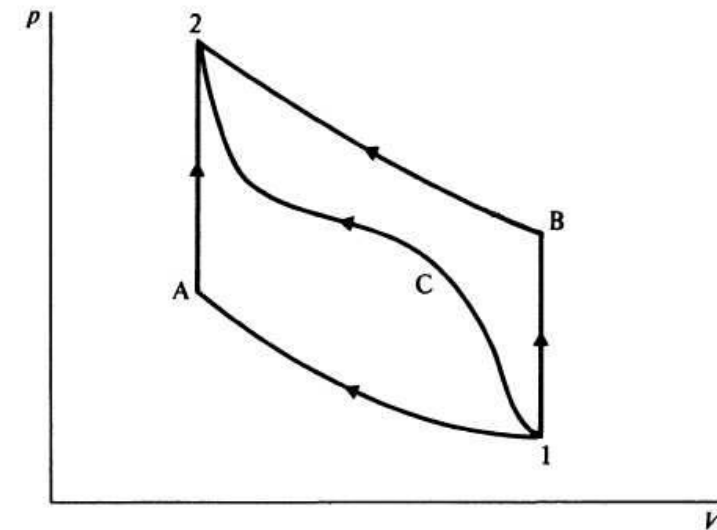


He showed temperature increase in isolated systems by

1. viscous dissipation in liquids
2. Friction in solids
3. Electric heating in wires.

'Work' needs to be done on the system to generate heat $\rightarrow$ rise temperature.

There has to be a proportion between heat and work.



1st law: If the state of an otherwise isolated system is changed by performance of work, the amount of work is solely decided by the change effected – Not by the means of work nor the intermediate stages through which the system passes between its initial and final states.

So, as before, a state of a system is defined by P,V in the simplest case of a piston.

We define U, (Internal Energy) a function of the state,

Such that when work is done **on** the state, there is change in the internal energy

$$\Delta U = W$$

However, this was the case when we used thermally isolated walls. (adiabatic)

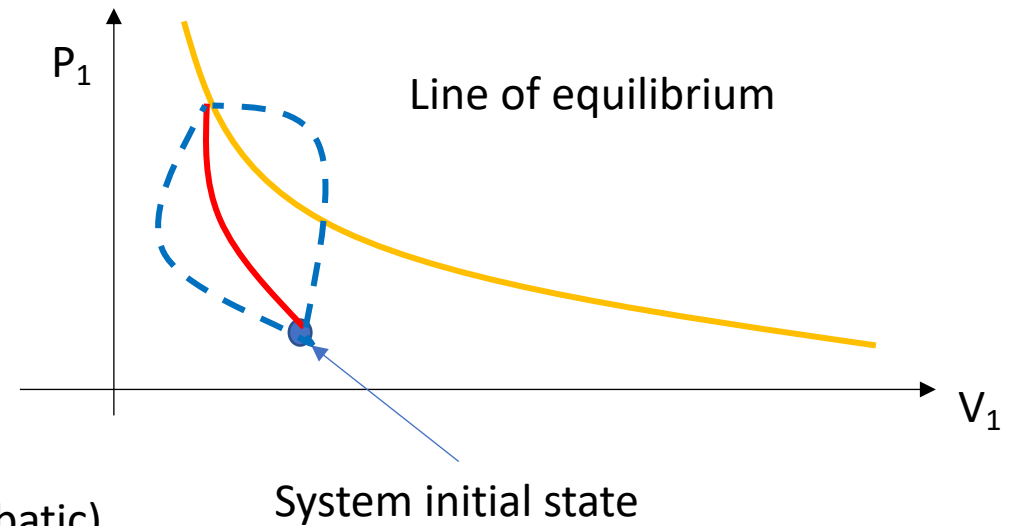
What happens if there can be heat flow (Q) from outside ?

$$\Delta U = Q + W$$

Remember, if work of magnitude W is performed **by** the system $\Delta U = Q - W$

For a small infinitesimal energy change dU can be obtained either from small heat flow dQ or small work dW

$$dU = dQ + dW$$



Examples 1: Work done under hydrostatic pressure

Piston with cross section A .

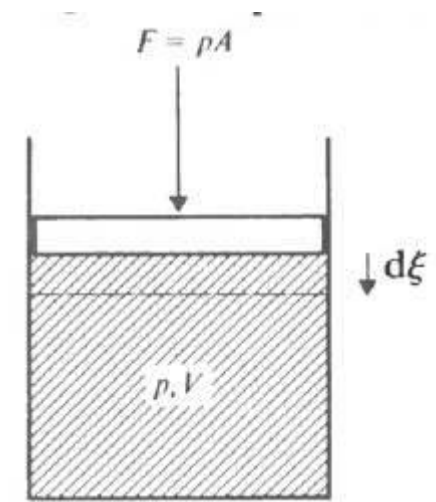
Force on the piston $F = pA$, p is applied pressure

If it causes a displacement $d\xi$, work done

$$dW = Fd\xi$$

Substituting $dW = pAd\xi = pdV$

In decreasing volume, work is done on the fluid, and in other case, work is done by the fluid.

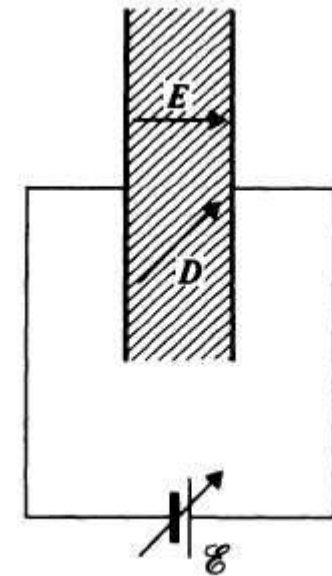


Example 2: Work done under electric field:

If E is the electric field perpendicular to the plates, causing a polarization

P is the dipole moment $p = q * d$ and $dp = q * dd$

Work done is then $dW = E \cdot dp$

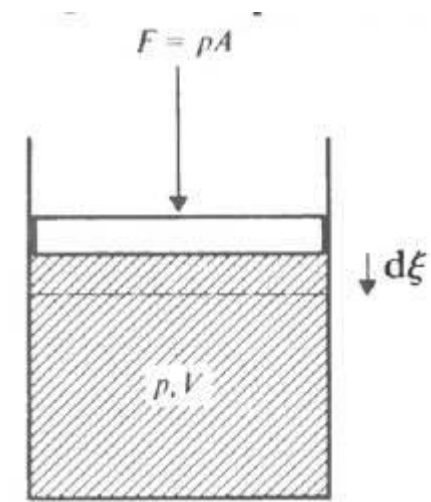


Consider the same hydrostatic pressure system

If a work W is done by the system

$$\Delta U + W = Q$$

In other words, for a work W done by the system, a heat Q is provided by the surroundings



Considering the case in which work is done under constant pressure P

$$W = \int_{V_1}^{V_2} p dV = p(V_2 - V_1)$$

$$Q_{1 \rightarrow 2} = U_2 - U_1 + pV_2 - pV_1$$

$$Q_{1 \rightarrow 2} = U_2 + pV_2 - (U_1 + pV_1) = H_2 - H_1$$

We call this term $H = U + pV$ as Enthalpy

Thus under constant pressure, heat provided to a system (closed) is the Enthalpy change

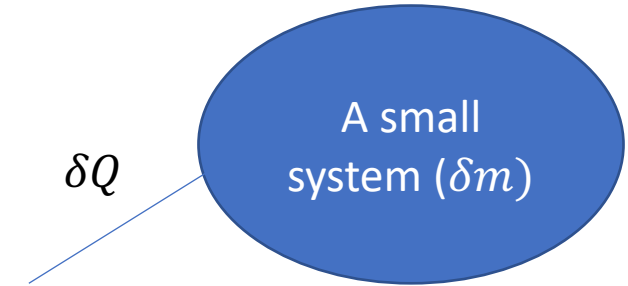
It is to say that if you need to expand (positive V change) under constant pressure, heat has to be supplied to the system.

Microscopically: the molecules now need to hit the walls at higher velocities

Heat Capacity

General heat capacity is the heat energy required to make a small change in a state observable α

$$C^{(\alpha)} = \left. \frac{\partial Q}{\partial \alpha} \right|_{\text{other variables}}$$



A common heat capacity is defined using temperature

$$C^{(T)} = \left. \frac{\partial Q}{\partial T} \right|_{V, P, \dots}$$

In the same hydrostatic pressure case

$$Q = U + p dV$$

This is not surprising: constant volume means no work is done!

$$C_V^{(T)} = \frac{\partial U}{\partial T}$$

Specific heat capacity at const volume $c_V = \frac{\partial u}{\partial T}$

Where $u = \frac{U}{\delta m}$

In similar ways, we can calculate $C_p^T = \left. \frac{\partial Q}{\partial T} \right|_P = \frac{\partial(U + p dV)}{\partial T} = \frac{\partial(U + pV)}{\partial T} = \frac{\partial H}{\partial T}$

Formal definitions of thermal analysis

Calorimetry:

Is the measurement of heat changes which occur during a process.

If calorimetric measurements are done under constant volume its called bomb calorimeter

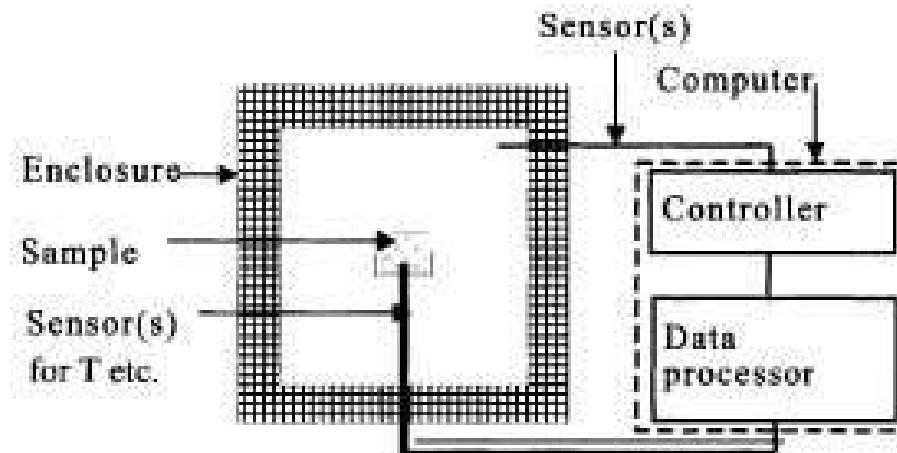
And those under constant temperature are called as isothermal calorimeter.

Thermal Analysis:

Is a group of techniques in which one property is studied as temperature is slowly varied.

1. E.g. sample may be heated or cooled at a fixed rate $\frac{dT}{dt} = \beta$ – eg: 10 K/min

Technique	Abbreviation	Property	Uses
Thermogravimetry or (Thermogravimetric analysis)	TG TGA	Mass	Decompositions Oxidations
Differential thermal analysis	DTA	Temperature difference	Phase changes, reactions
Differential scanning calorimetry	DSC	Power difference or heat flow	Heat capacity, phase changes, reactions
Thermomechanical analysis	TMA	Deformations	Mechanical changes
Dynamic mechanical analysis	DMA	Dimensional change Moduli	Expansion Phase changes, glass transitions, polymer cure as DMA
Dielectric thermal analysis	DETA	Electrical	Decompositions
Evolved gas analysis	EGA	Gases evolved or reacted	
Thermooptometry		Optical	Phase changes, surface reactions, colour changes
<i>Less frequently used methods</i>			
Thermosonimetry	TS	Sound	Mechanical and chemical changes
Thermoluminescence	TL	Light emitted	Oxidation
Thermomagnetometry	TM	Magnetic	Magnetic changes, Curie points



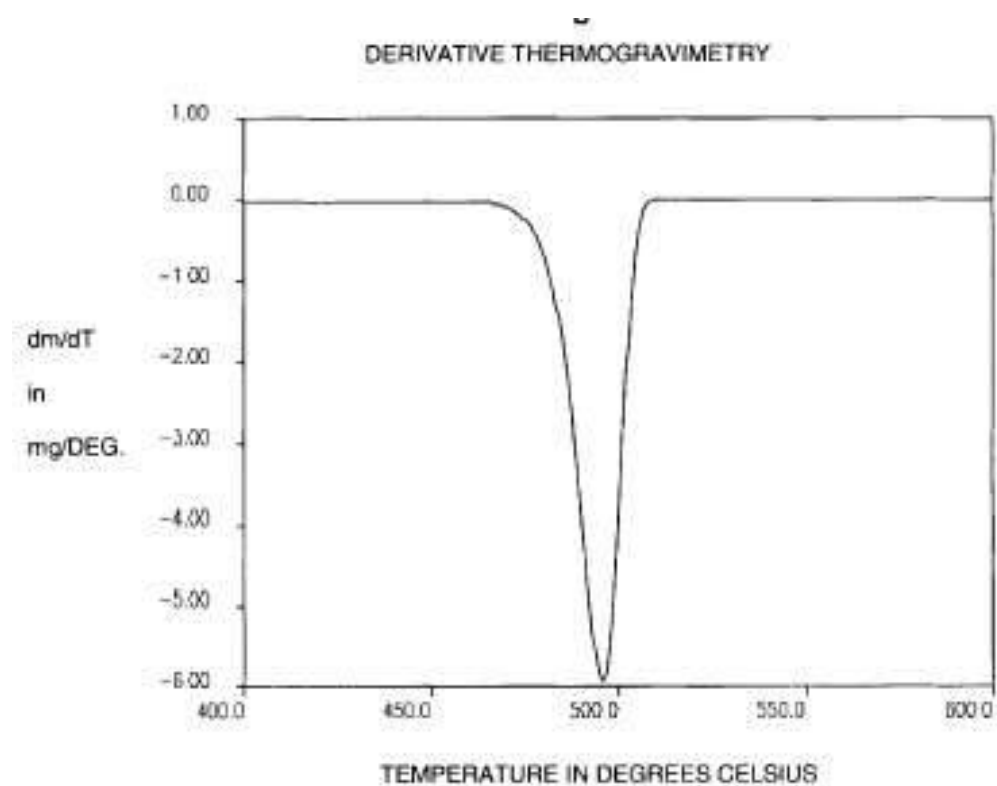
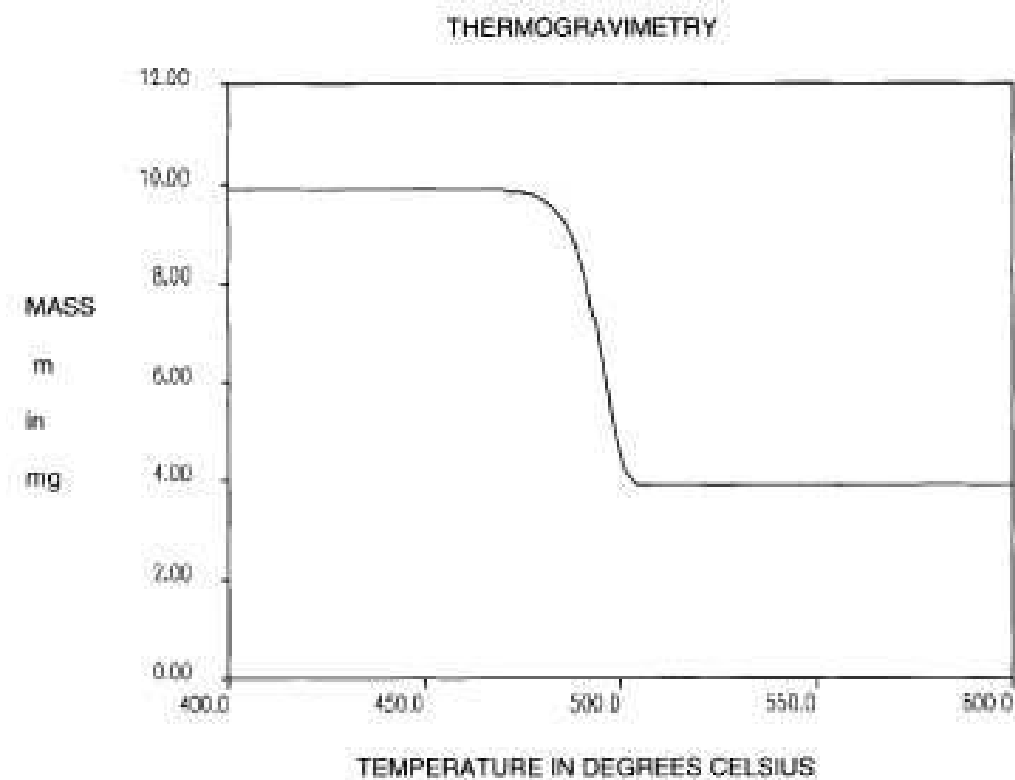
Thermal Transitions

Physical Phenomena	Chemical phenomena
Crystalline transition	Chemisorption
Second order transitions	Desolvation
Fusion	Decomposition
Vaporization	Oxidative transition
Sublimation	
Absorption	Solid-State reaction
Adsorption	Solid-gas reaction
Desorption	

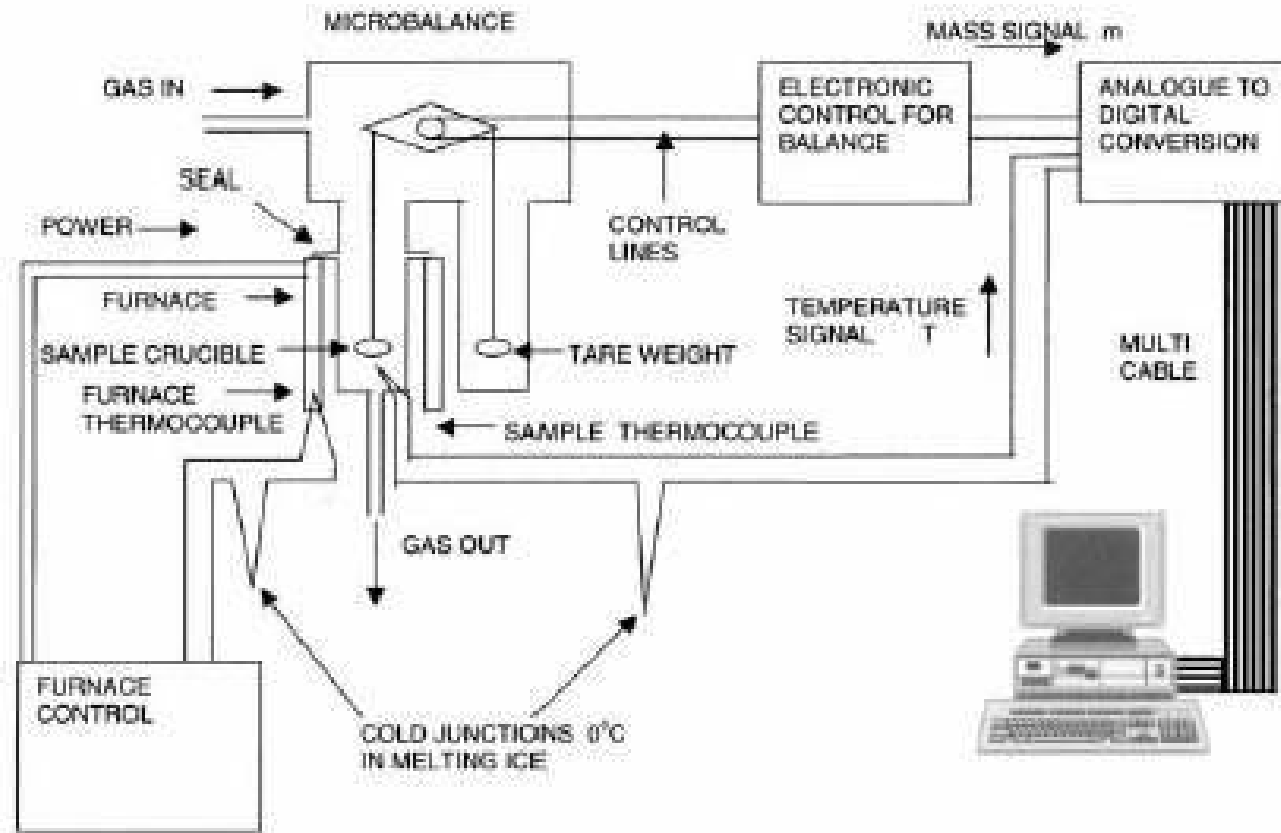
Thermal gravimetry analysis gives information relating to

Thermal Gravimetry:

Mass change of a substance is measured as a function of temperature whilst the substance is subjected to a controlled temperature program.



A typical thermobalance



Microbalance: The accurately measure weight

Furnace: Provide heat controllably
Typically can do -160 - 1600 °C

Atmosphere control: inert gas
ambient/oxidising ambient

Crucibles: Sample holders which doesn't get affected by heat/chemicals

Thermocouples: combination of two metals which provide a EMF based on the temperature.

Temperature control: This is typically using a feed back loop between the furnace and the thermocouples

Data Collection: This is typically computer automated. Measurements are typically made in terms of time or temperature. They are recorded using a computer and plotted as necessary.

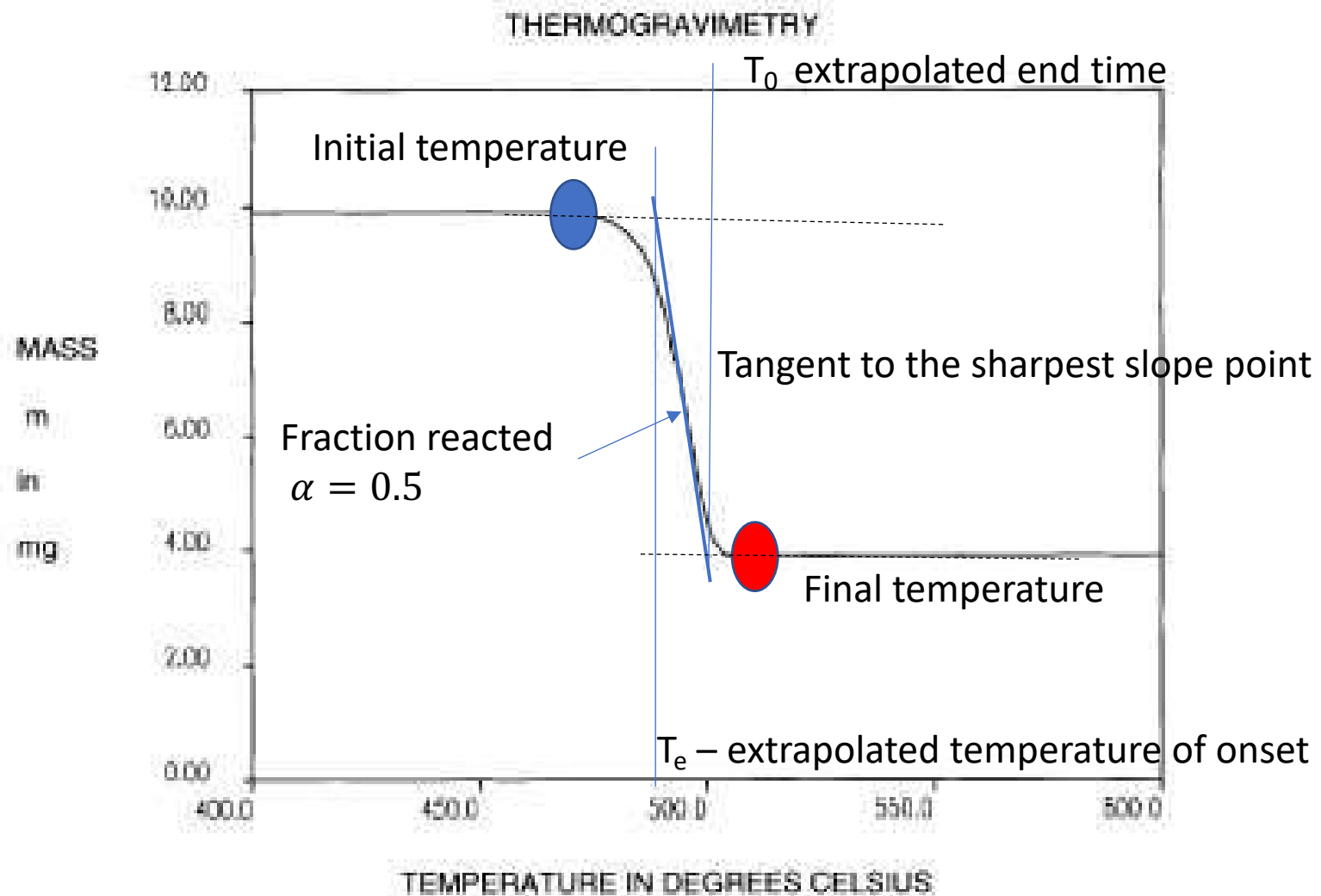
Types of thermogravimeter:

1. isothermal measurements
2. scanning/ramping/rising measurements

Calibration of mass and temperature:

1. Zeroed values: In the absence of the sample, the total mass should fall down to zero. This needs to be corrected before the experiment begins.
2. Calibration: This is typically done using a standard of known mass. The manufacturer of the equipment typically provides the mass standard.
3. For temperature, the thermocouple can be calibrated using different techniques
 1. Look for some known features that occur at predefined temperatures (such as melting of metals at STP)
 2. Calibrate unknown thermocouple with a known thermocouple.
 3. Locate the Curie point of metals – Say Nickel.

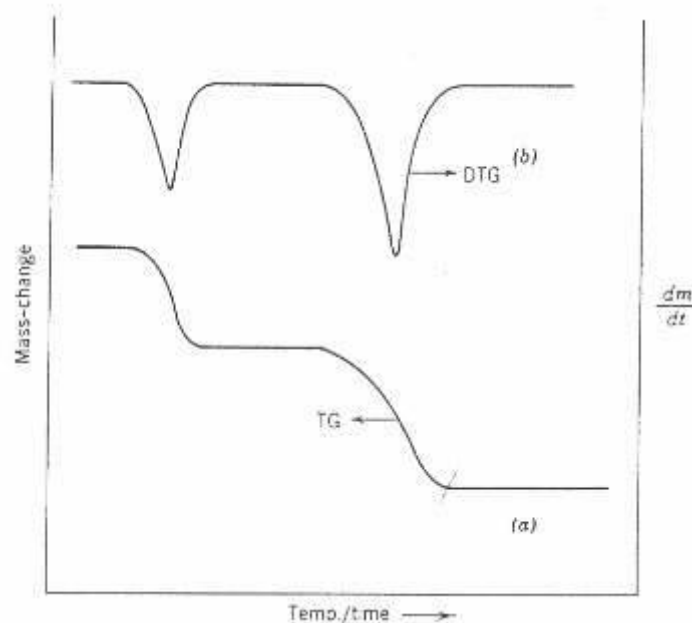
Analysing a measurement:



Differential TG

In traditional TG, mass is recorded as a function of Temp/Time,

$$m = f(T, t)$$



In derivative TG,

$$\frac{dm}{dt} = f(T, t)$$

When there is no mass loss, TG and DTG both have flat lines.

When there is change, peaks are observed in DTG.

The peak position tells you the inflection point – point at which the reaction rate is maximum

The area under the DTG is proportional to the mass change !

TG and DTG are invaluable experiments to observe desorption, decomposition , oxidation processes.

If you want to know the accurate temperatures for drying/dehydration, ignition of precipitates/solvent, TG/DTG is an ideal and simple technique

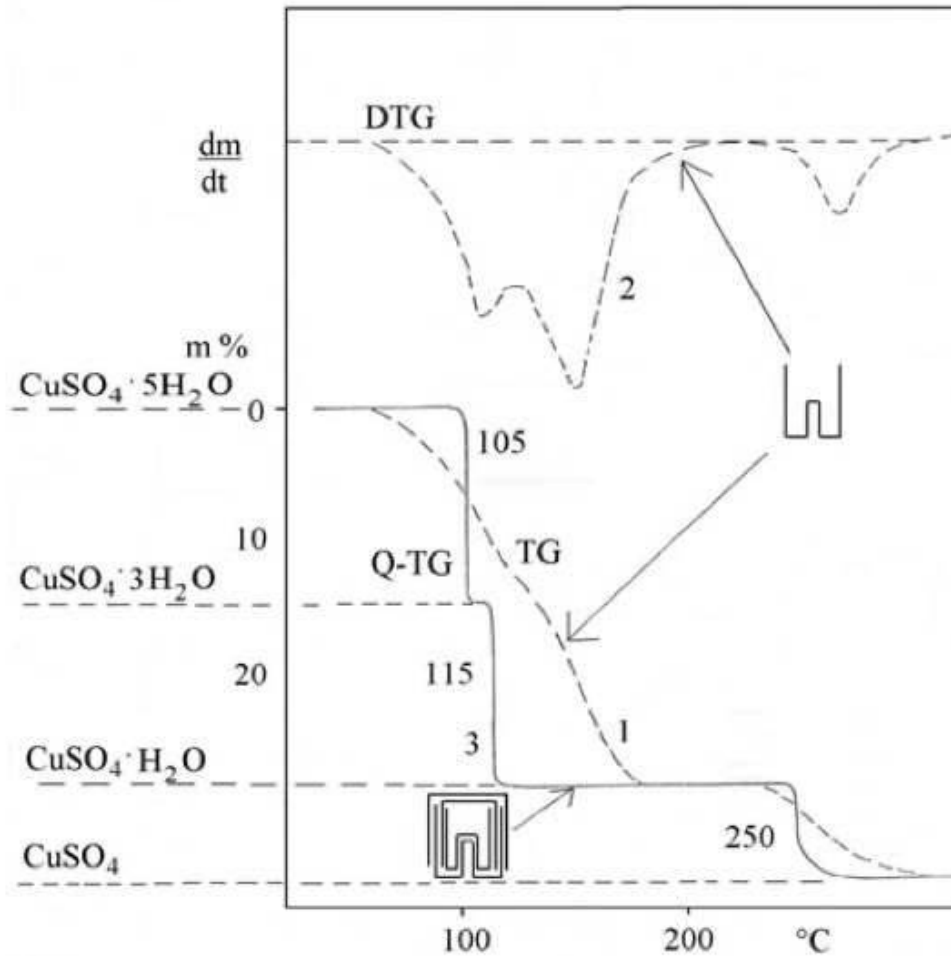
Main applications in characterization of

- Thermal stability
- Material purity
- Determination of humidity

Examination of

- Corrosion
- Gasification
- Kinetic processes

Examples of TGA: Dehydrogenation



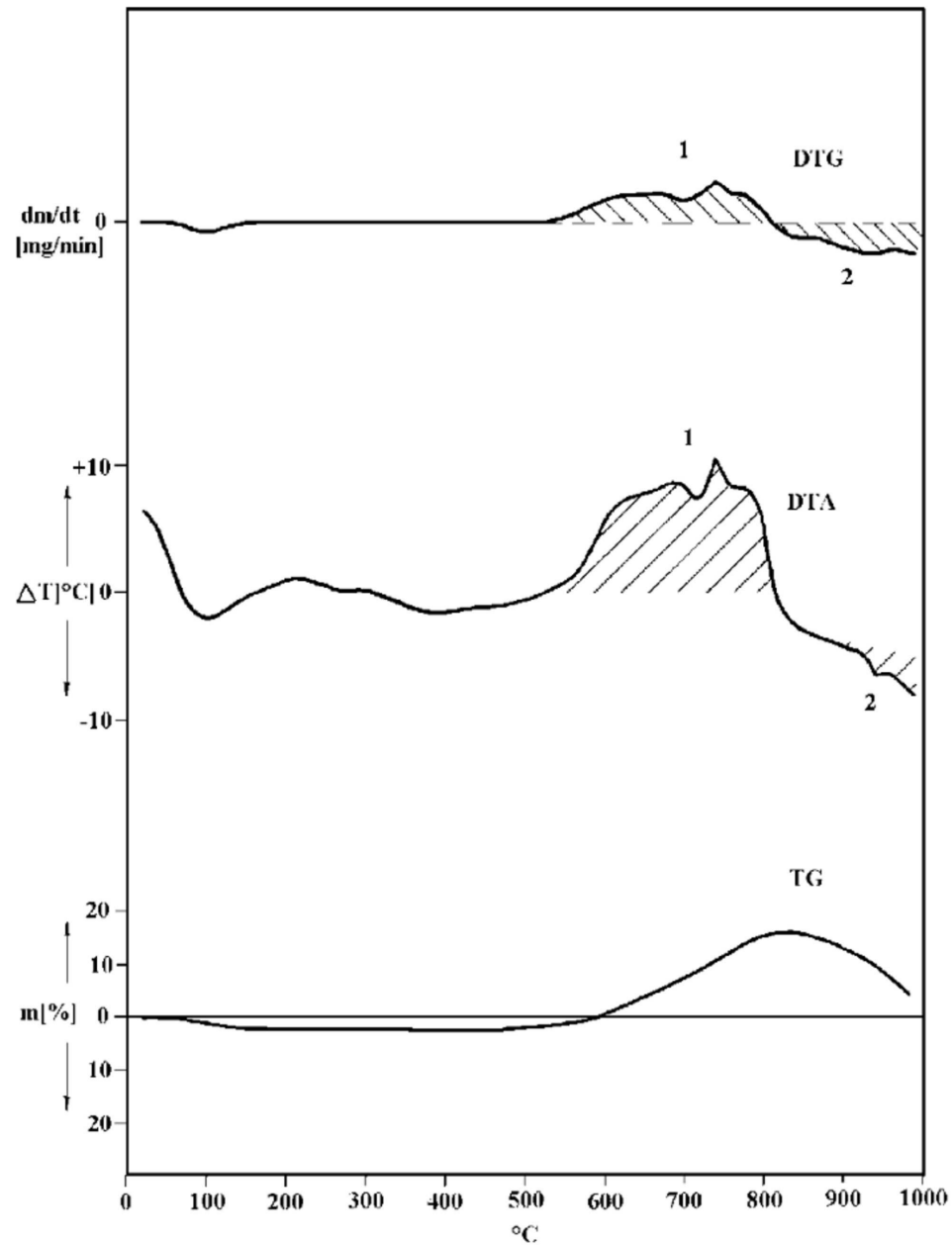
Water molecule coordinated with Cu^{2+} gets desorbed at different temperatures.

While the first two molecules are isothermal – starts almost at the same temperature,

The last molecule of water leaves much higher in temperature!

Reactions:

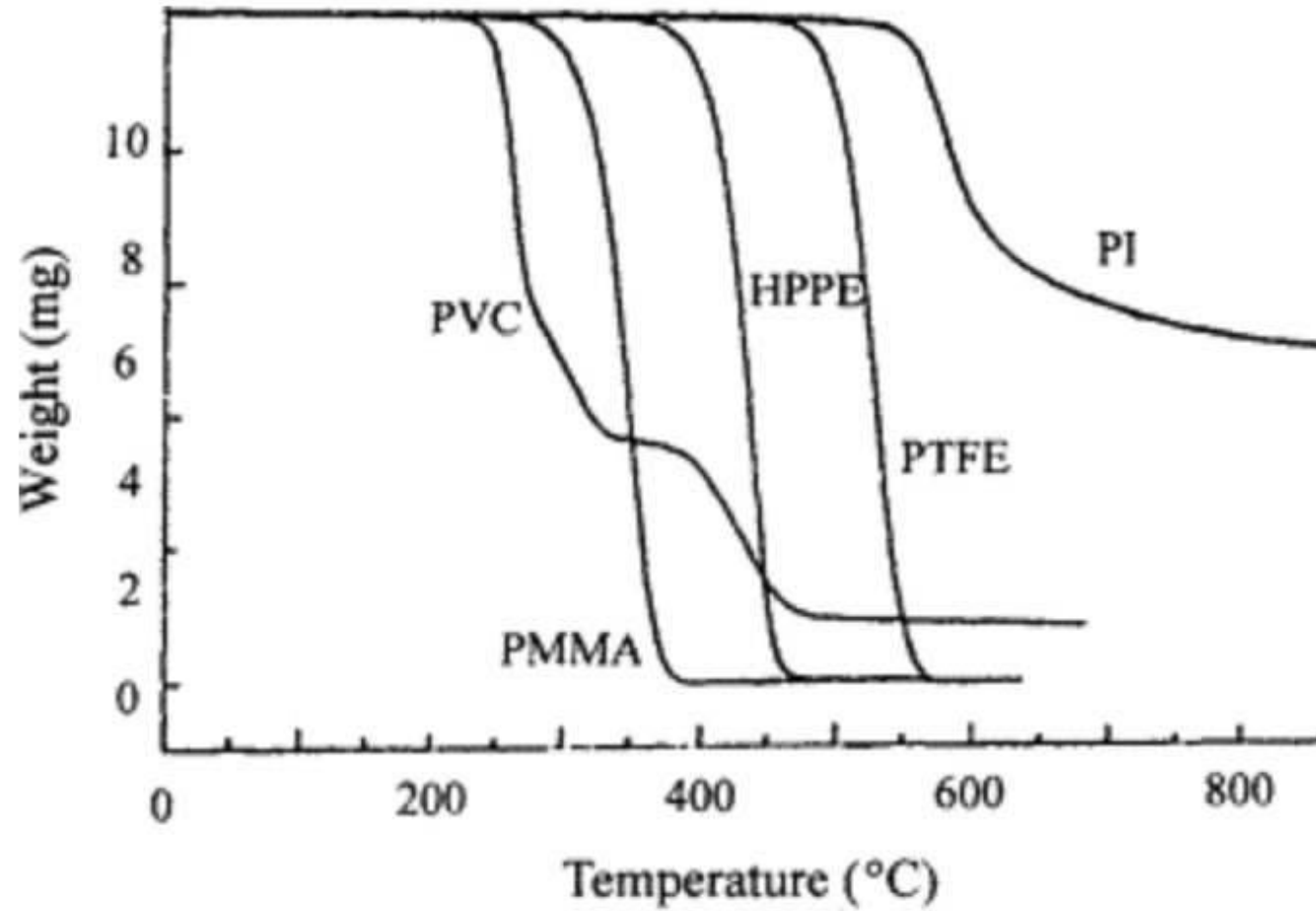
1. Between 600–820 °C: exothermic: oxidation with different internal products of PbO, PbSO₄, and PbO·PbSO₄.
2. Between 850–1100 °C: thermal decomposition, endothermic:
 $\text{PbSO}_4 \rightarrow \text{PbO} + \text{SO}_3 \uparrow$.



Reactions:

1. Between 600–820 °C: exothermic: oxidation with different internal products of PbO , PbSO_4 , and $\text{PbO} \cdot \text{PbSO}_4$.
2. Between 850–1100 °C: thermal decomposition, endothermic: $\text{PbSO}_4 \rightarrow \text{PbO} + \text{SO}_3 \uparrow$.

Example degradation of polymers



Factors affecting TGA measurements:

1. Heating Rate: If $T1$ is the reaction starting temperature, and $T2$ – reaction ending temperature

$$(T1)_f > (T1)_s \text{ and } (T2)_f > (T2)_s$$
$$(T2 - T1)_f > (T2 - T1)_s$$

2. Sample related:

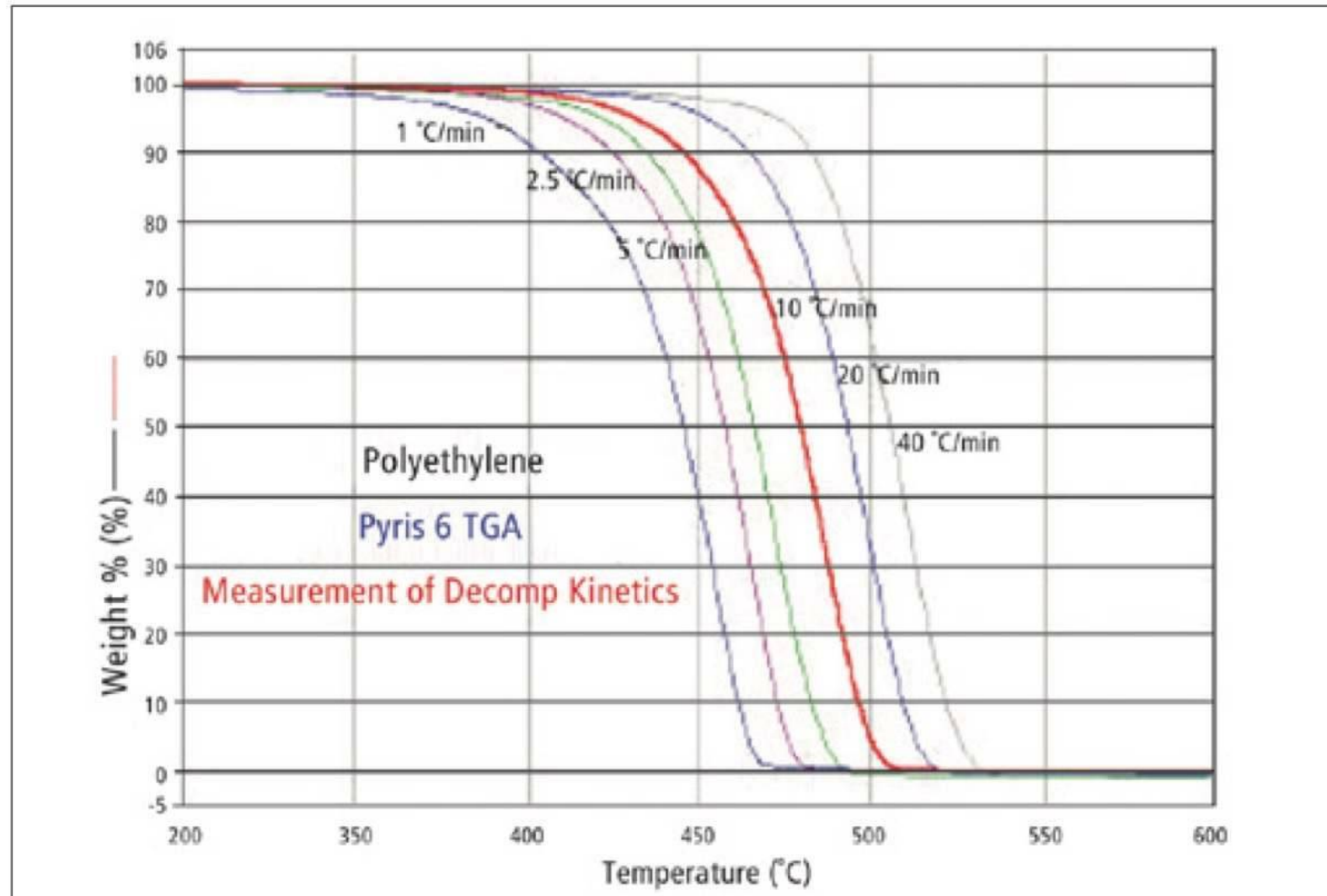
Reaction is endothermic/exothermic and its influence on the measurement is impacted by the mass of the sample used.

Thermal conductivity of the sample and the gas diffusivity effects the reaction temperature measurements.

Surface to volume rate has a dominant role to play

3. Geometry of the crucible
4. Atmosphere of the reaction

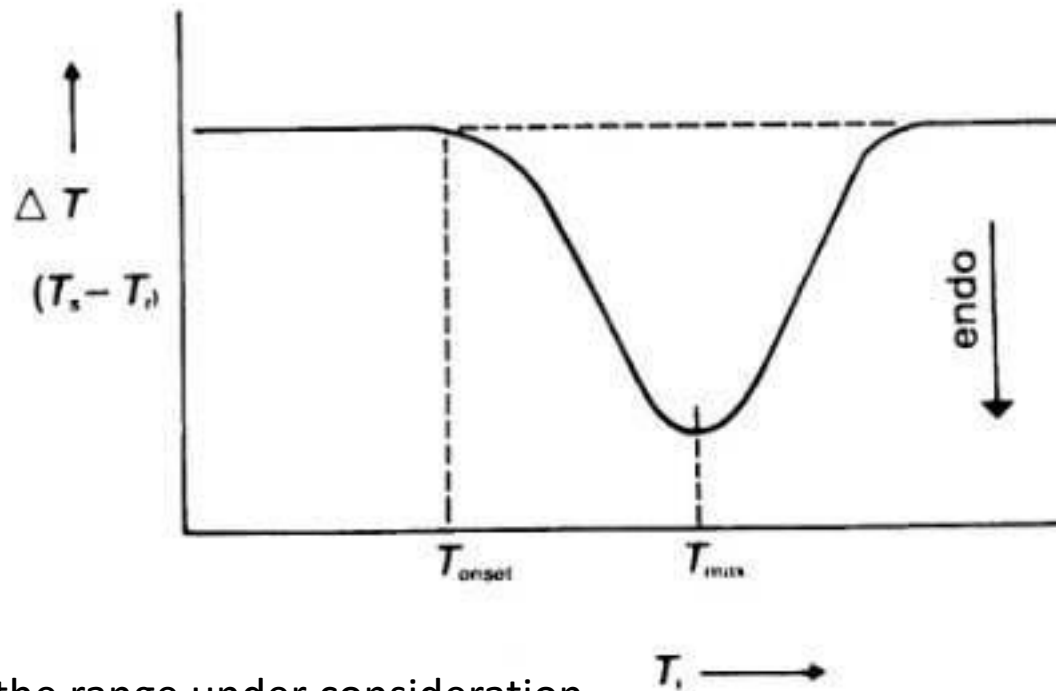
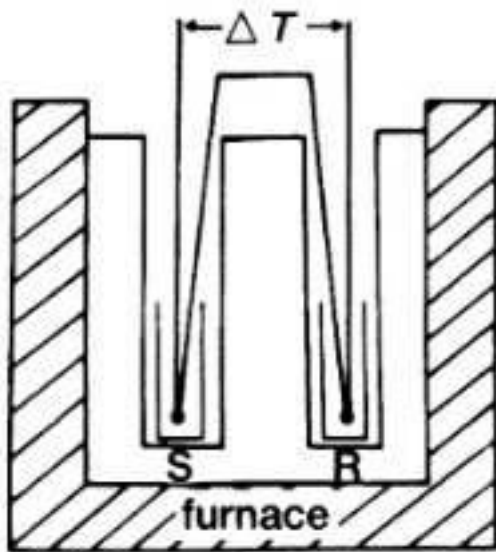
Example of effect of heating rate:



Differential Thermal Analysis and Differential Scanning Calorimetry

Being the most common technique:

It measures the temperature change difference between a sample and reference upon same delivery of heat.



A reference is chosen such that:

1. There should be no thermal event in the range under consideration
2. It shouldn't react with the sample holder/thermocouple
3. The thermal conductivity and heat capacity should be similar to the sample.

Alumina/carborundum
are commonly used
references

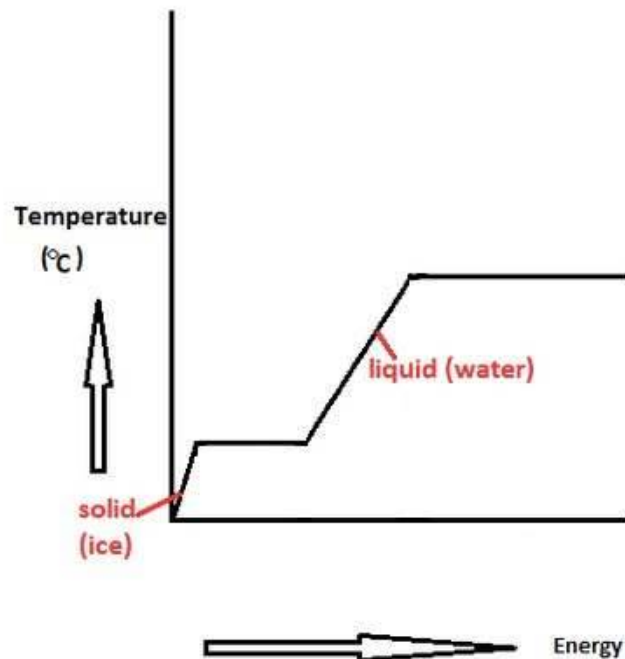
Remember, we defined Enthalpy H , and related $\Delta H = Q$ to the heat supplied to the system.

Take the example of ice



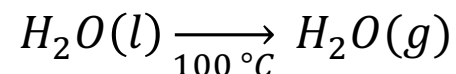
As the heat is supplied to ice, its temperature will rise till very near to zero. Beyond which a mole of water will not change temperature till ~6kJ of heat is provided.

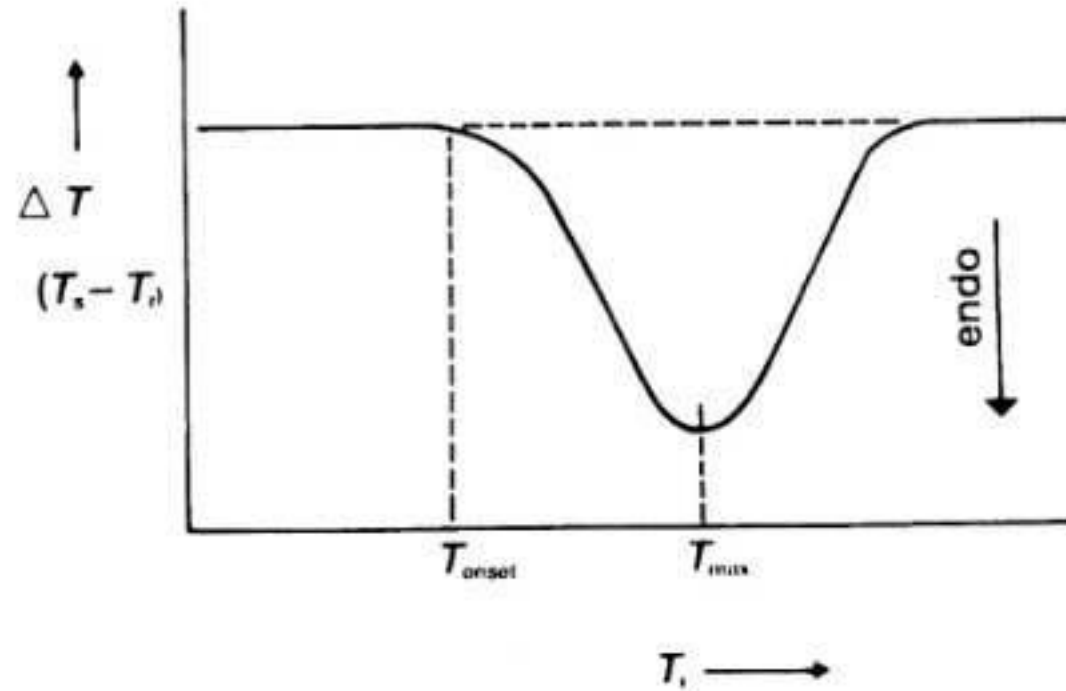
This heat energy, is called as enthalpy of fusion.



A similar transformation is observed when $100^\circ C$ is in vicinity.

A mole of water will not change in temperature for 40.67 kJ, called as enthalpy of vaporization.



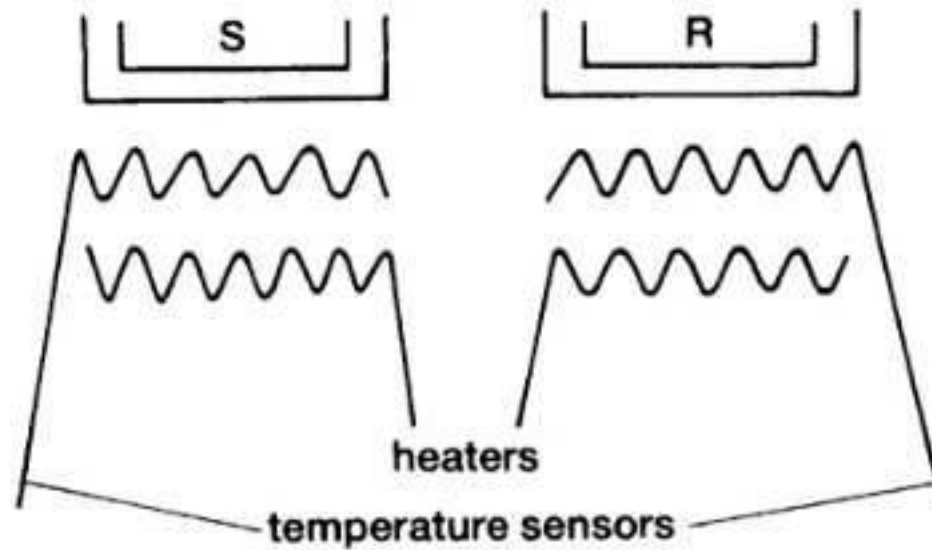


Since, in DTA, the plotted function in y-axis is $\Delta T = T_s - T_R$

When a thermal event such as melting is going to happen, the sample temperature doesn't change. A dip in the plot is obtained.

Since melting is an endothermic reaction in most cases, a dip is associated to endothermic reaction.

Differential Scanning Calorimetry

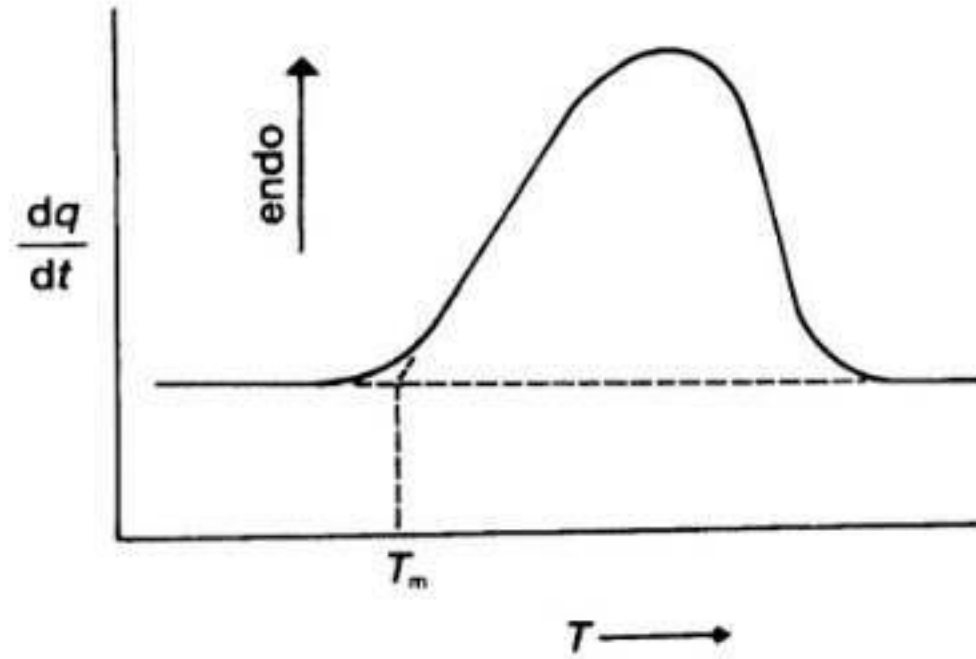


In a calorimetric setup, the heat supplied is measured instead of the temperature. The temperature of the sample and the reference is kept constant.

Such meters are called as power compensated DSCs.

$$\frac{dQ}{dt} = \left(\frac{dT}{dt} \right) (C_S - C_R)$$

A typical DSC plot for endothermic reaction

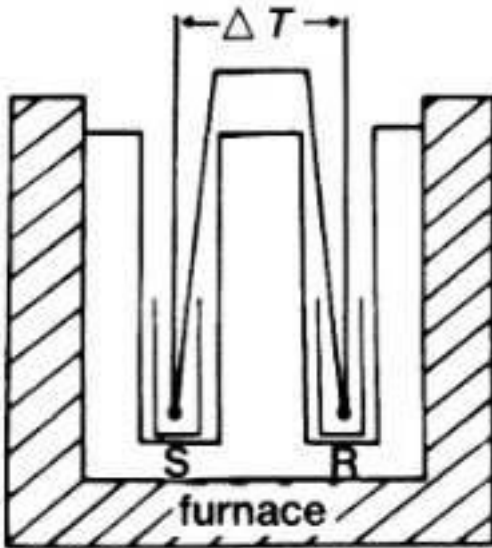


A typical measurement in which a melting transition is observed. Remember, here, again the differential is on the heat! Additional heat is required in the case of transition in the sample marking an endothermic reaction!

The Ohm's law for heat:

$$\Delta T = R * \frac{dq}{dt}$$

R – thermal resistance, q – heat supplied



In classical DTA –

The heat supplied is maintained a constant.

If the thermocouples remain the same in the sample and reference, thermal resistance between the sample and its holder and reference and its holder negligible

$$T_S - T_S^0 = \Delta T_S = R * \left(\frac{dq}{dt} \right) \text{ from the Ohm's law for the sample}$$

$$\frac{dq}{dt} = C_S \frac{dT_S}{dt}$$

$$\Delta T = T_S - T_R = R(C_S - C_R) \frac{dT}{dt}$$

The measured temperature difference depends on

heat capacity difference, heating rate as well as the thermal resistance (R)! R is very system specific!

Power compensated DSC:

The heat is supplied separately to sample and reference!

Such that $T_S = T_R \Rightarrow \Delta T = 0$

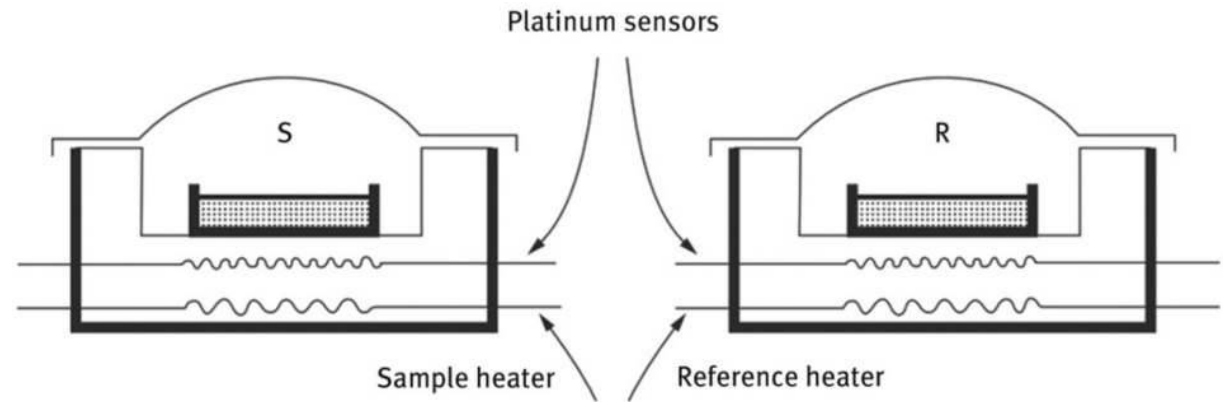
$$T_S - T_S^0 = \Delta T_S = R * \left(\frac{dq_S}{dt} \right)$$

$$T_R - T_R^0 = \Delta T_R = R * \left(\frac{dq_R}{dt} \right)$$

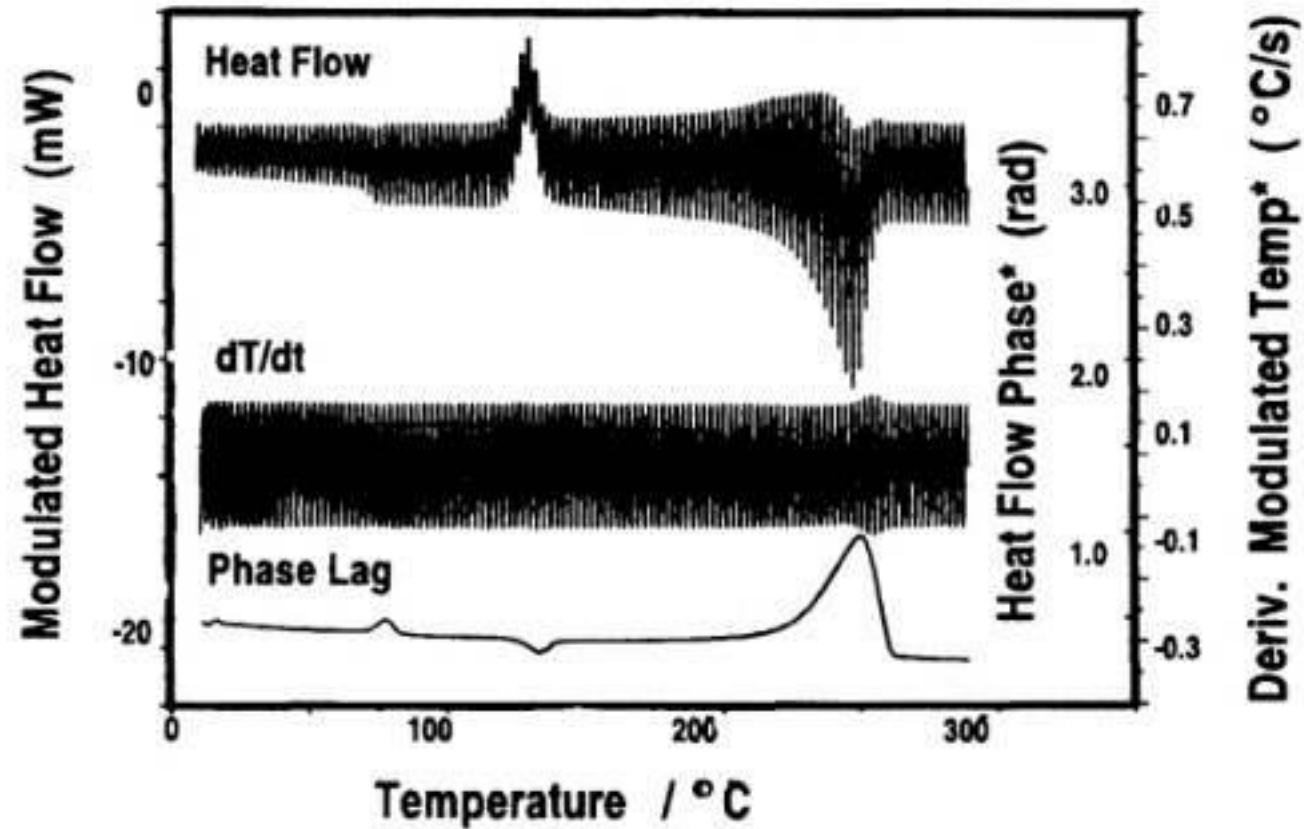
If the power is compensated such $T_S = T_R$,
 $\Delta T = T_S - T_R = 0 \Rightarrow R = 0$

Since the sample and reference are heating with some rate -
 $\frac{dq_S}{dt} \neq 0$

$$\text{From } \frac{dq}{dt} = C \frac{dT}{dt}$$
$$\Delta \frac{dq}{dt} = \frac{dT}{dt} (C_S - C_R)$$



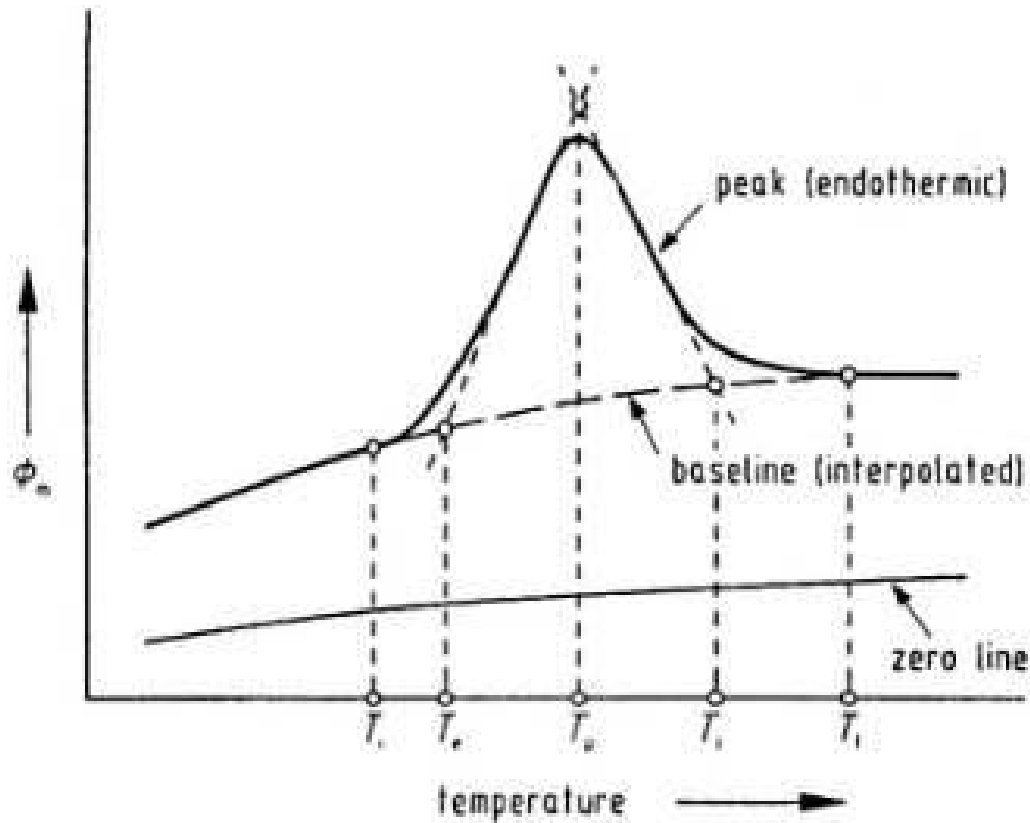
Modulated Temperature DSC



$$T = T_0 + \beta T + \gamma \sin \omega t$$

The heat flow $\frac{dq}{dt}$ is monitored for both the amplitude and phase difference with the temperature change!

Characteristics of DSC Curves



Y-axis – heat flux

Zero-line – is the curve obtained when the sample crucible is empty

Once the measurement is made,

A interpolated baseline is obtained when a peak is observed.

The line is drawn to match regions before and after a peak such that the peak was non-existent.

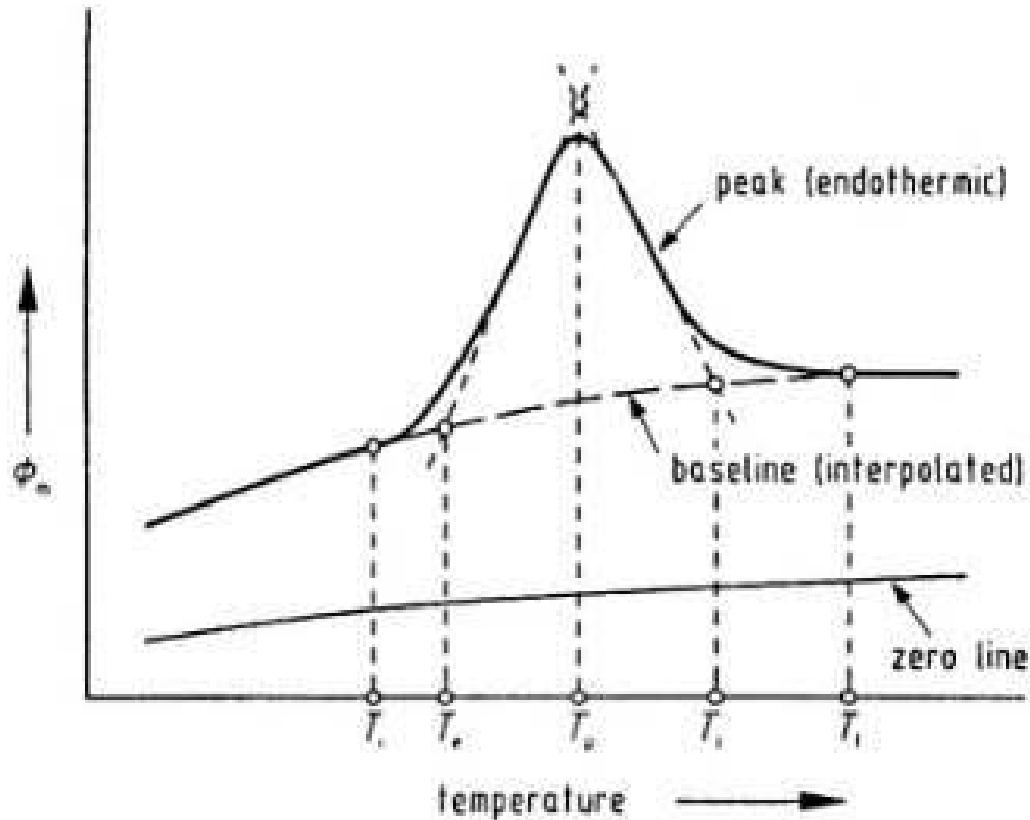
T_i – Initial temperature where the measured line deviates from baseline.

T_e - Actual onset is observed at intersection of baseline with the tangent to half-peak

T_c – extrapolated completion temperature

T_f – final peak temperature

Characteristics of DSC Curves



Zero-line – is the curve obtained when the sample crucible is empty

Once the measurement is made,

A interpolated baseline is obtained when a peak is observed.

Remember certain transitions such as glass transitions, just gives a change in the shape of the baseline

T_i – Initial temperature where the measured line deviates from baseline.

T_e - Actual onset is observed at intersection of baseline with the tangent to half-peak

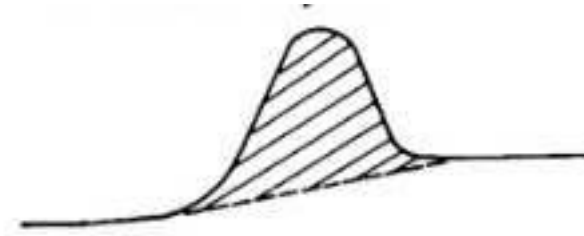
T_c – extrapolated completion temperature

T_f – final peak temperature

Measured Enthalpy: If A is the area of the peak

$$\Delta H = A \times \frac{k}{M}$$

M – mass of the sample, k – calibration factor



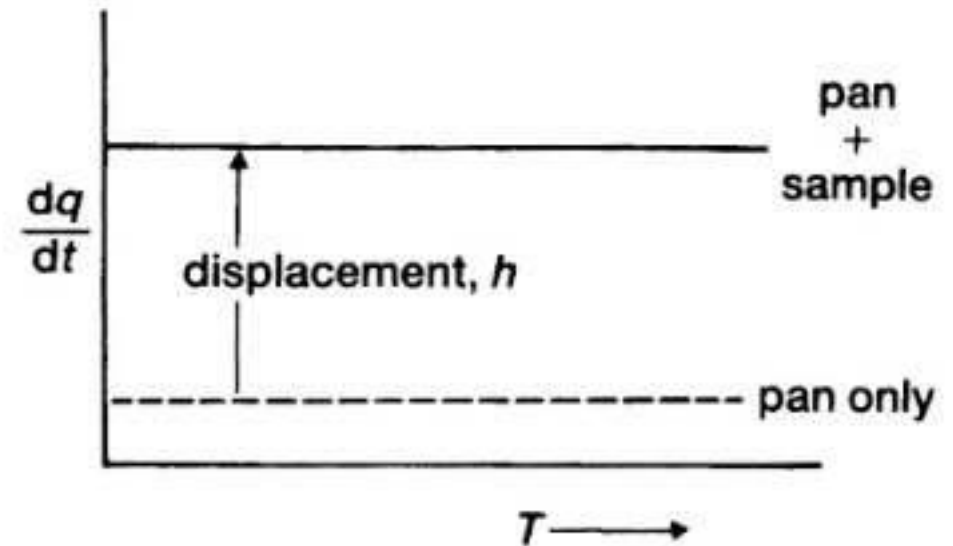
Measurement of heat Capacity

If h is the displacement after placing the sample,

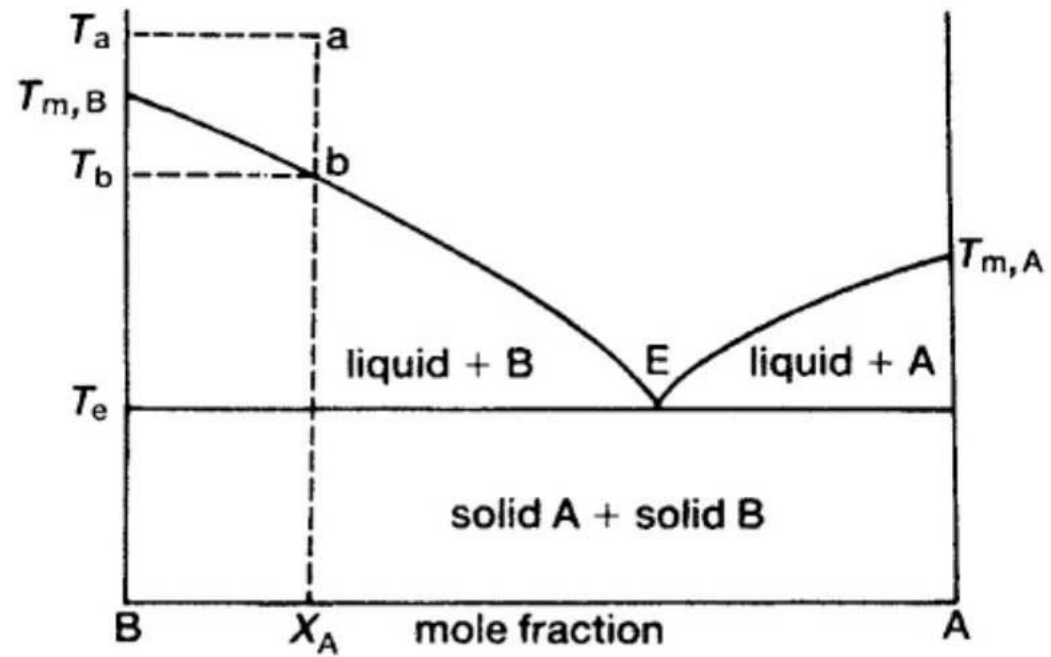
$$h \propto C_p$$

$$h = B\beta C_p$$

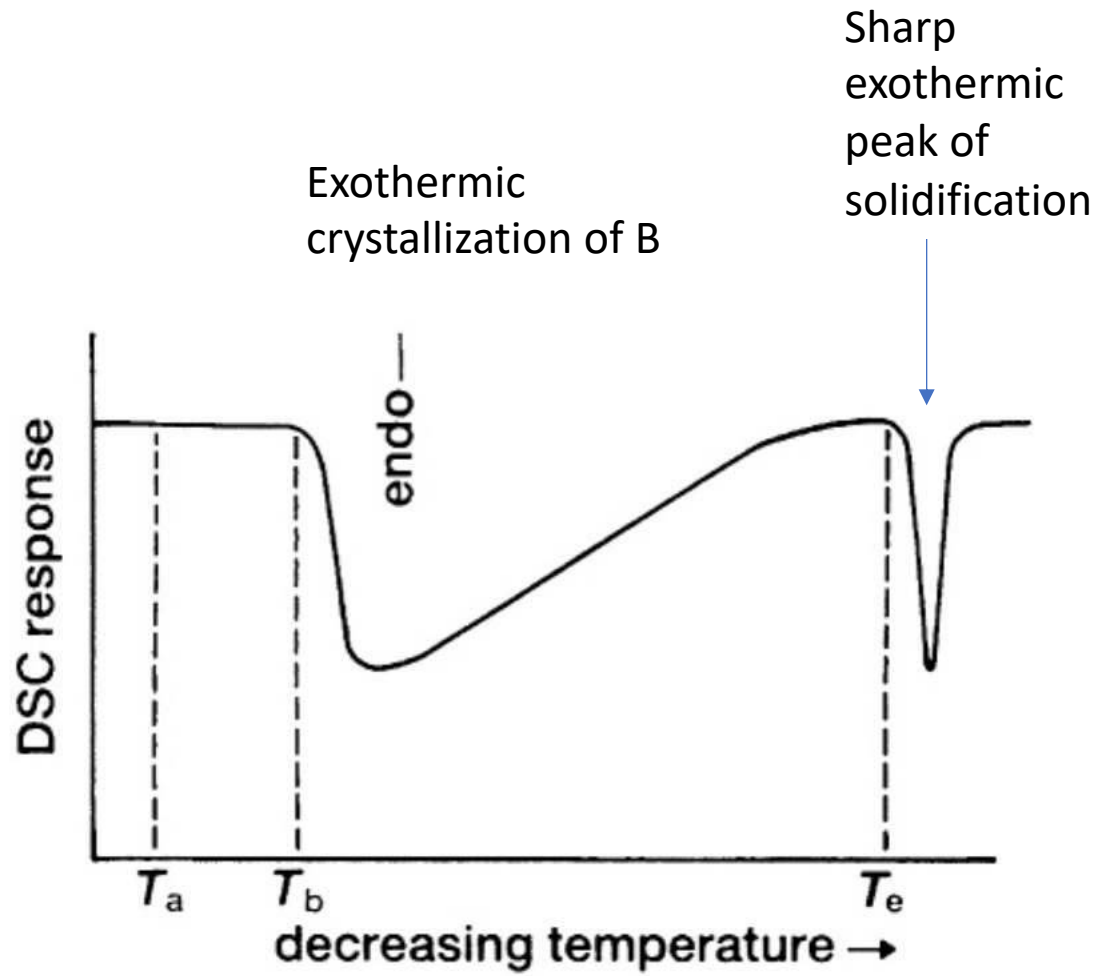
Where β is the heating rate, and B is the calibration factor.



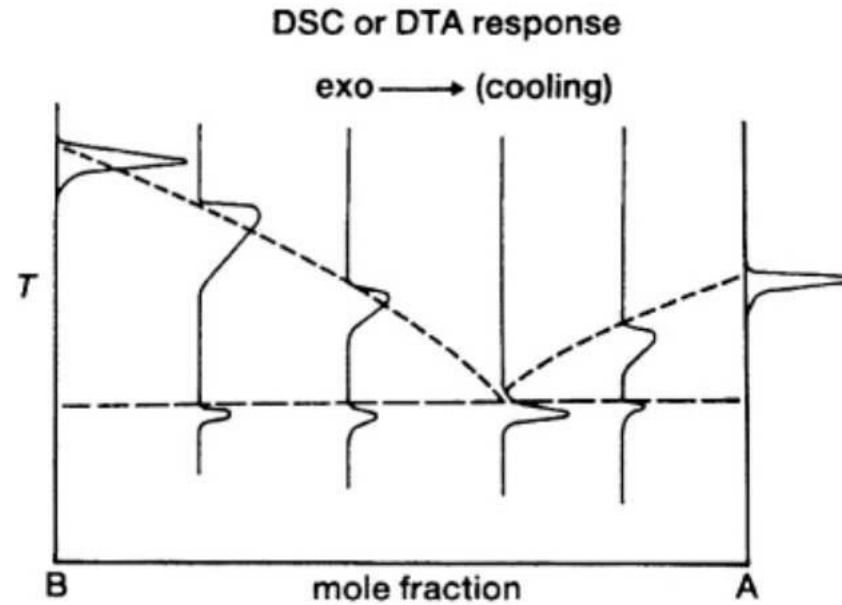
A case study:



If a composition X_A is taken, and cooled from $T_a \rightarrow T < T_e$, how will the DSc measurement look like ?

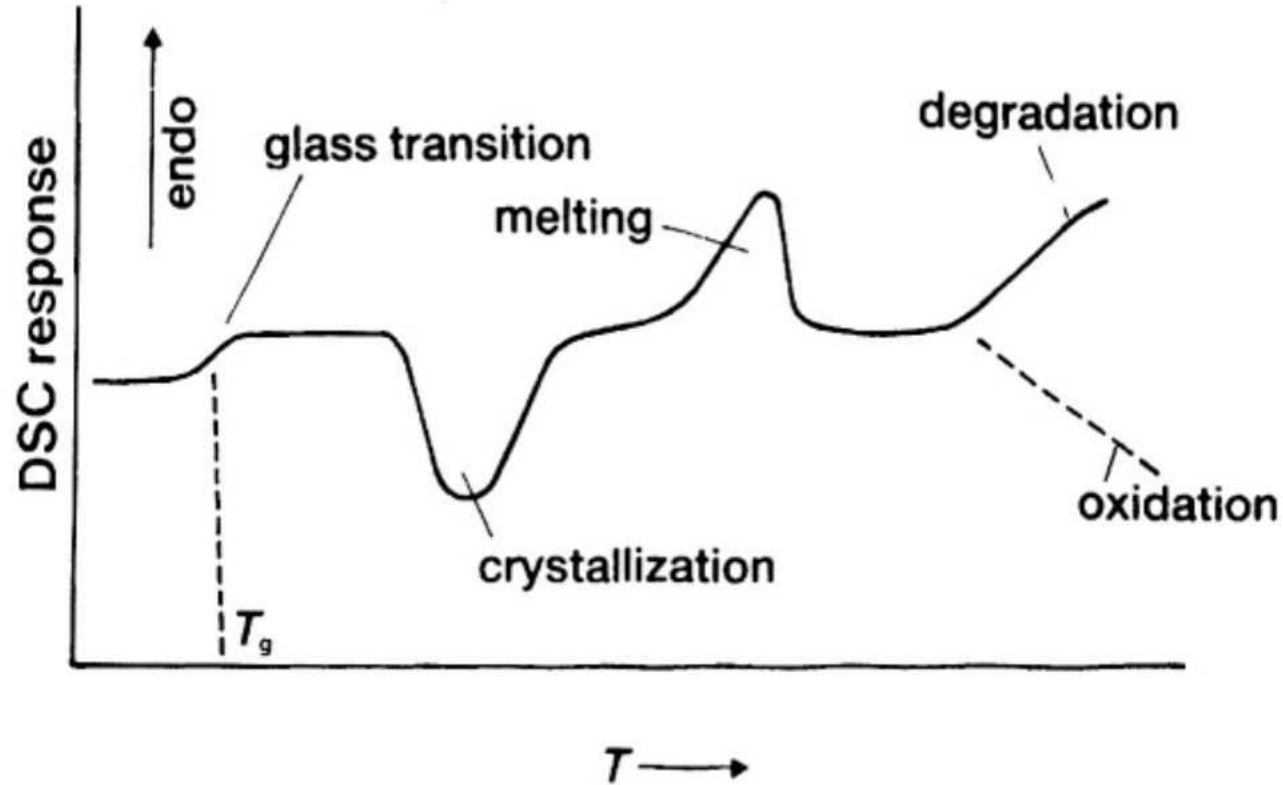


DSc – phase diagram relation



The ΔH enthalpy change – area under the peak gives a lot of information about the mixture

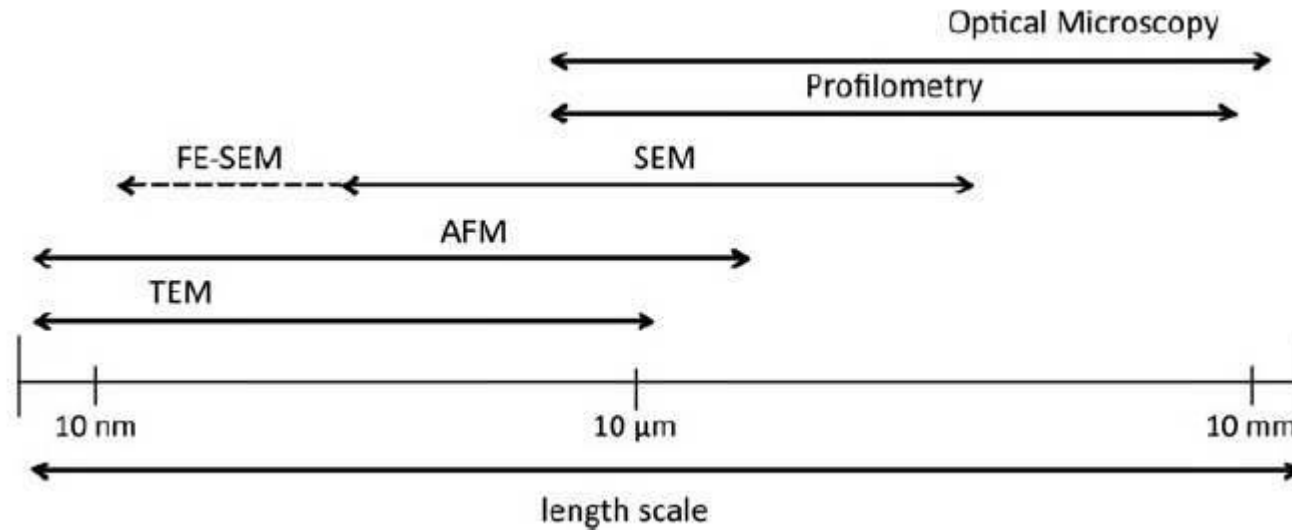
DSC of polymeric samples



Typical polymer is supercooled, hence at room temperature remain in a glassy state.
When heated, may undergo a glass transition to a rubbery state – associated with heat capacity change, but no enthalpy change

Atomic Force Microscope

Microscope at different length scales



Atomic Force Microscope has among the widest range of spatial resolution.

Its unique in its ability to magnify in the 3rd dimension (Z) as well.

Downside being, any scanning area greater than 100 μ m is largely not practical

How AFM compares with other Techniques ?

	AFM	SEM	TEM
Sample preparation	little or none	from little to a lot	from little to a lot
Resolution	0.1 nm	5 nm	0.1 nm
Relative cost	low	medium	high
Sample environment	any	vacuum(SEM) or gas (environmental SEM)	vacuum
Depth of field	poor	good	poor
Sample type	Conductive or insulating	conductive	conductive
Time for image	2–5 minutes	0.1–1 minute	0.1–1 minute
Maximum field of view	100 μm	1 mm	100 nm
Maximum sample size	unlimited	30 mm	2 mm
Measurements	3 dimensional	2 dimensional	2 dimensional

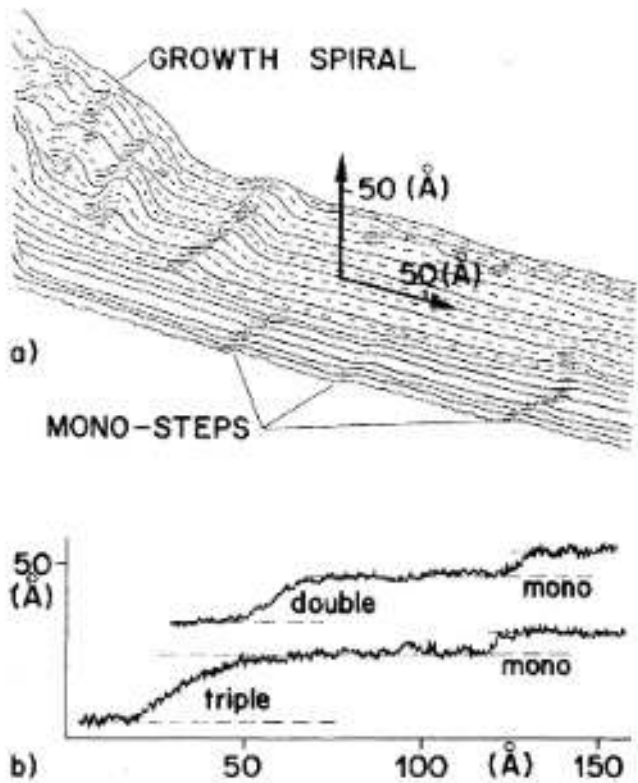
AFM with decent overlap with optical microscope spatial range is almost a logical next step

A study of Surfaces

- Surfaces are areas of abnormality in relation to the bulk material
- Atoms/ ions are subjected to asymmetric forces
- Surfaces are the only part of the material accessible to chemical change
- The ratio of the surface to the bulk atom density is a function of the material size and shape

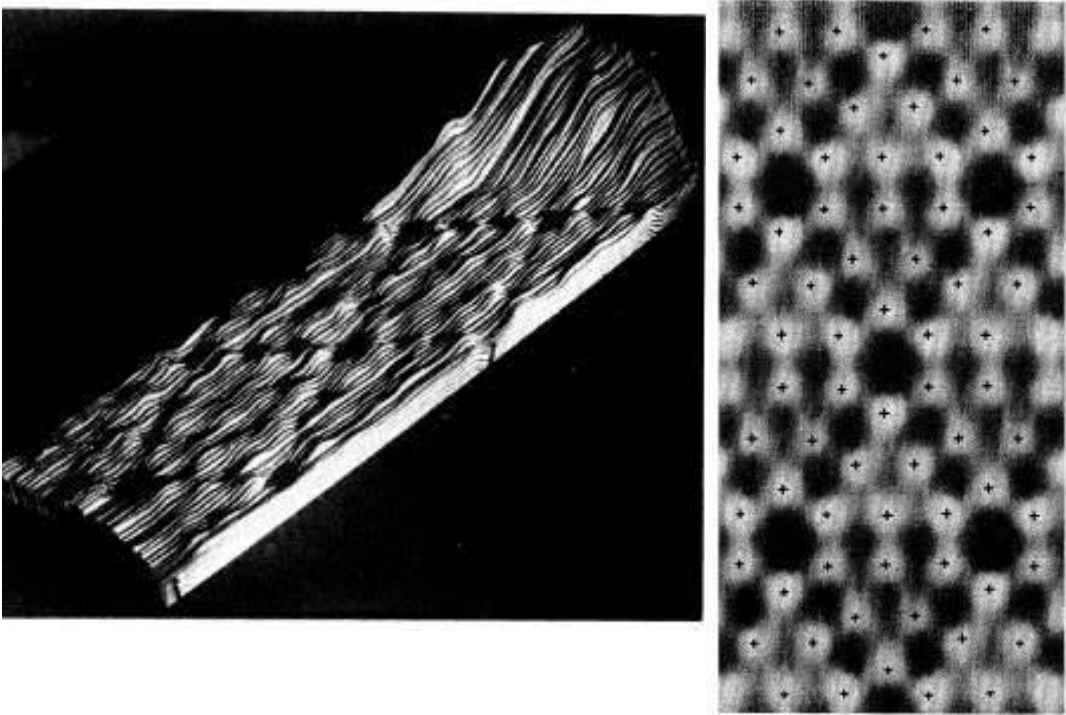
First atomic step scanned using tunneling microscope

Silicon 111 surface reconstruction

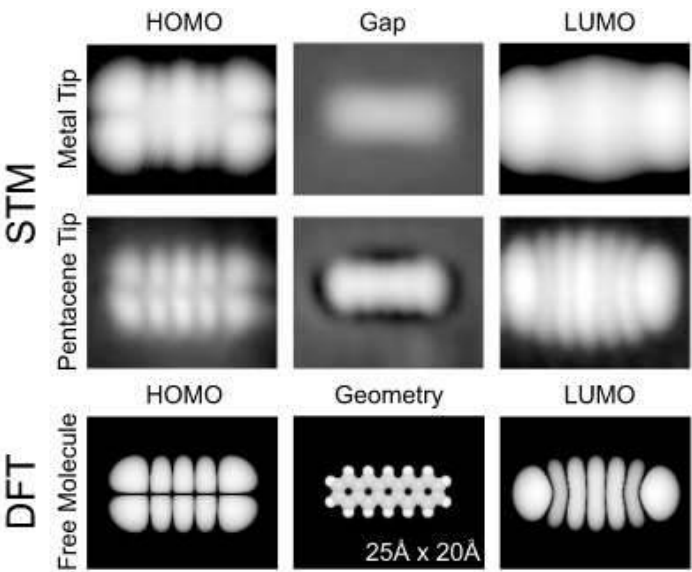


First scanning tunneling image taken by Binnig et. al in 1983

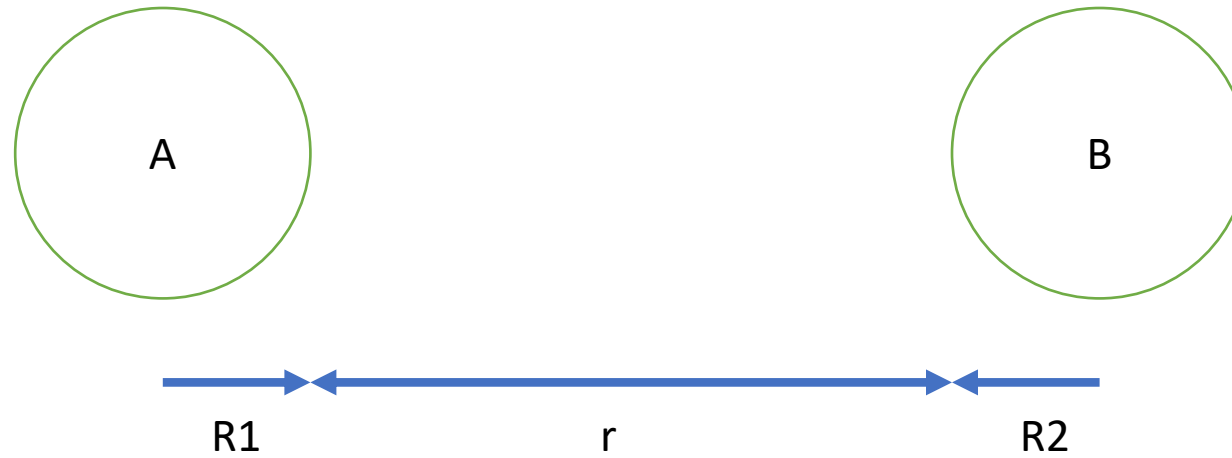
Binnig et.al., 1982



Electron orbital imaging on pentacene



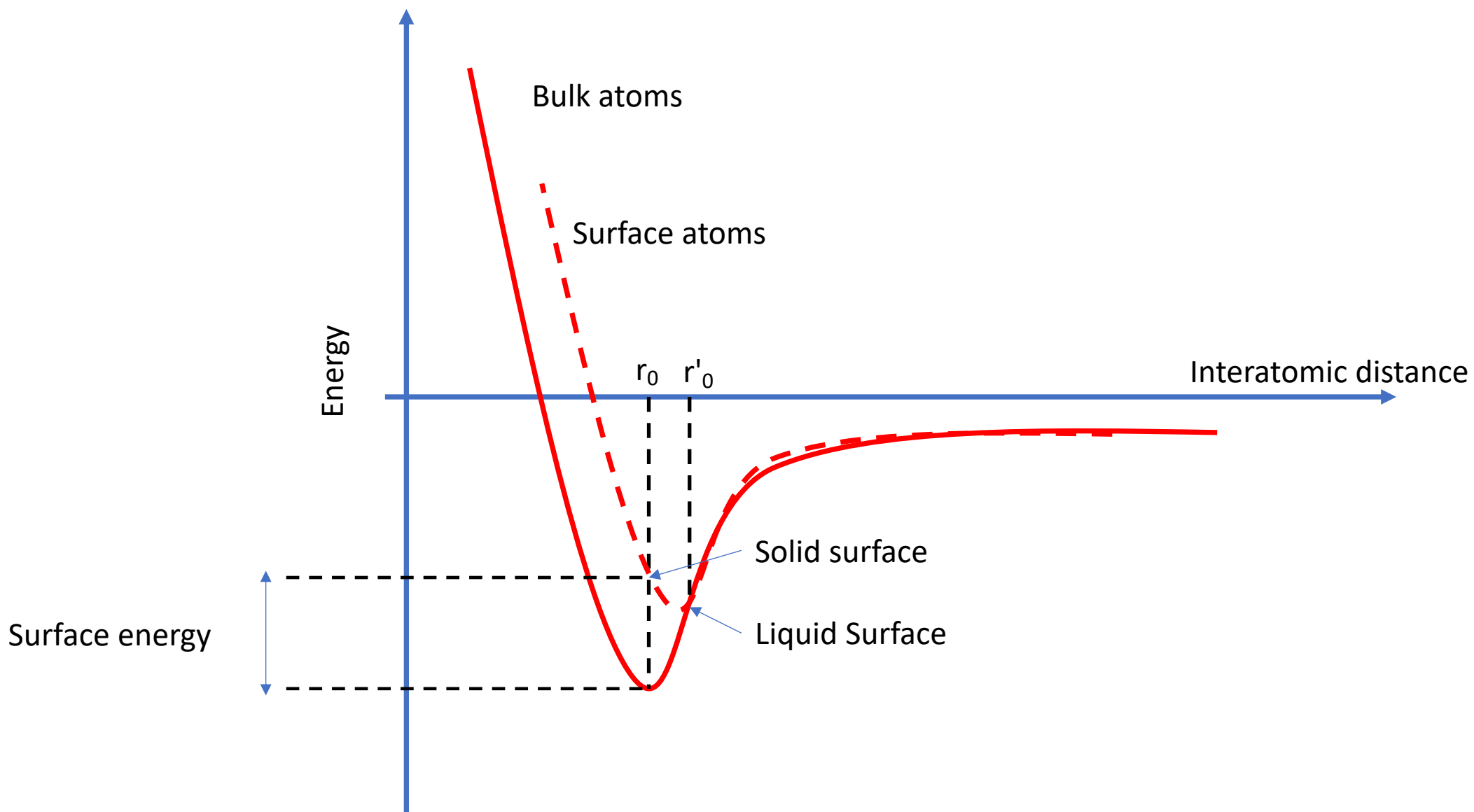
Jascha Reep, Gerhard Meyer, PRL, 2005



For the simplest case of two balls of radius R_1 and R_2 separated by r

The van der Waals' force $F_{VW} = -\frac{AR_1R_2}{(R_1+R_2)6r^2}$

A – Hamaker coefficient



Surface reconstruction

Si – 100 surface reconstructs to form dimers.

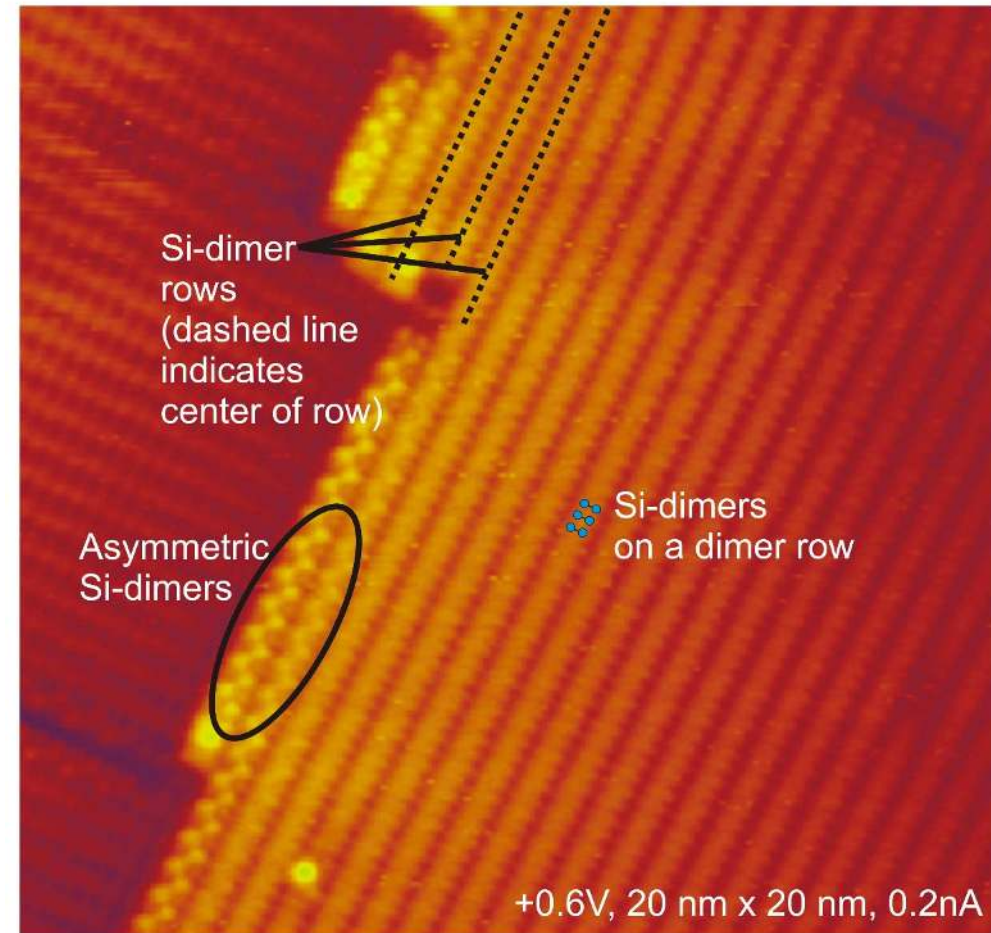
If 'a' is the side of the origin FCC cube,

The surface reconstructs such that the periodicity is 2x1

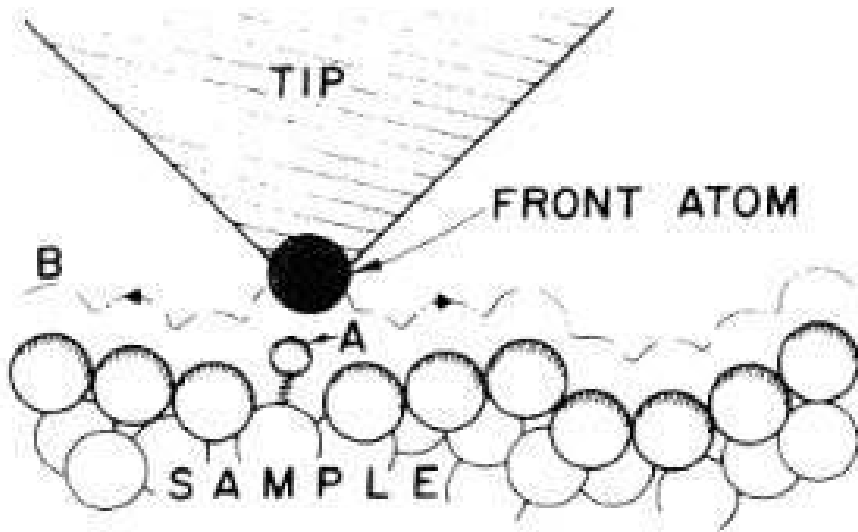
Si – 111 surface reconstructs to give multiple surfaces

With periodicity $\sqrt{3} \times \sqrt{3}$ and $\sqrt{31} \times \sqrt{31}$

Such notation of the surface is called as Wood's notation.



A tip : - An intentionally sharpened material to reduce the cross section of interaction



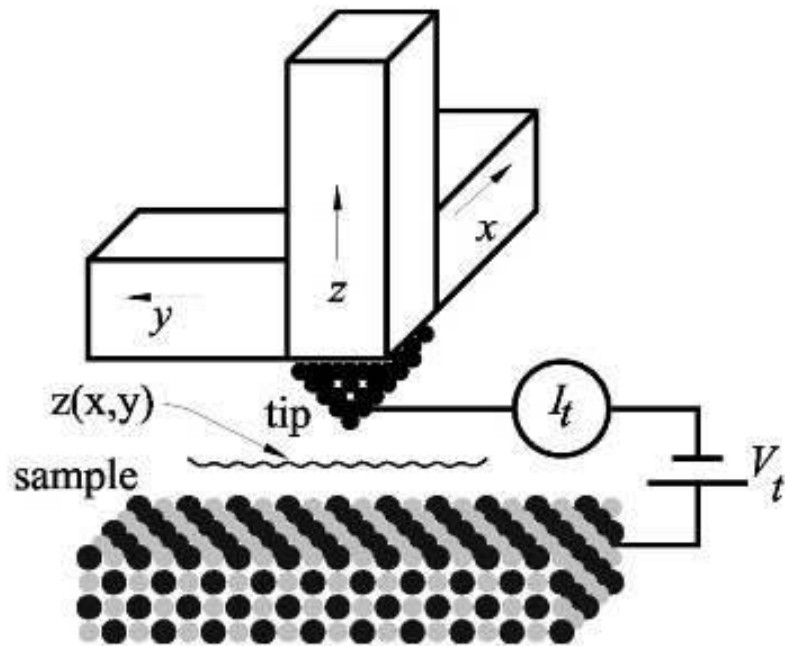
The tip follows the 'B' contour on the sample surface.

This is ensured by having the same tunneling current in STM

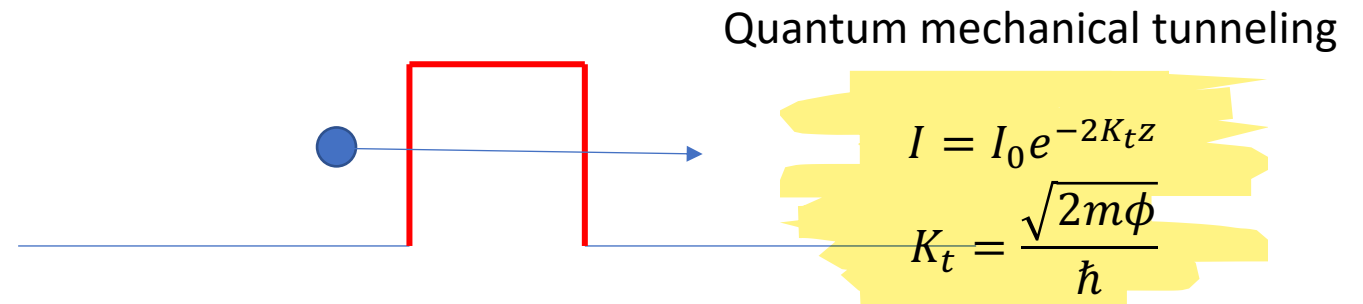
And by having the same force on the top in AFM.

But interatomic forces $\sim 10^{-18} \text{ N}$.

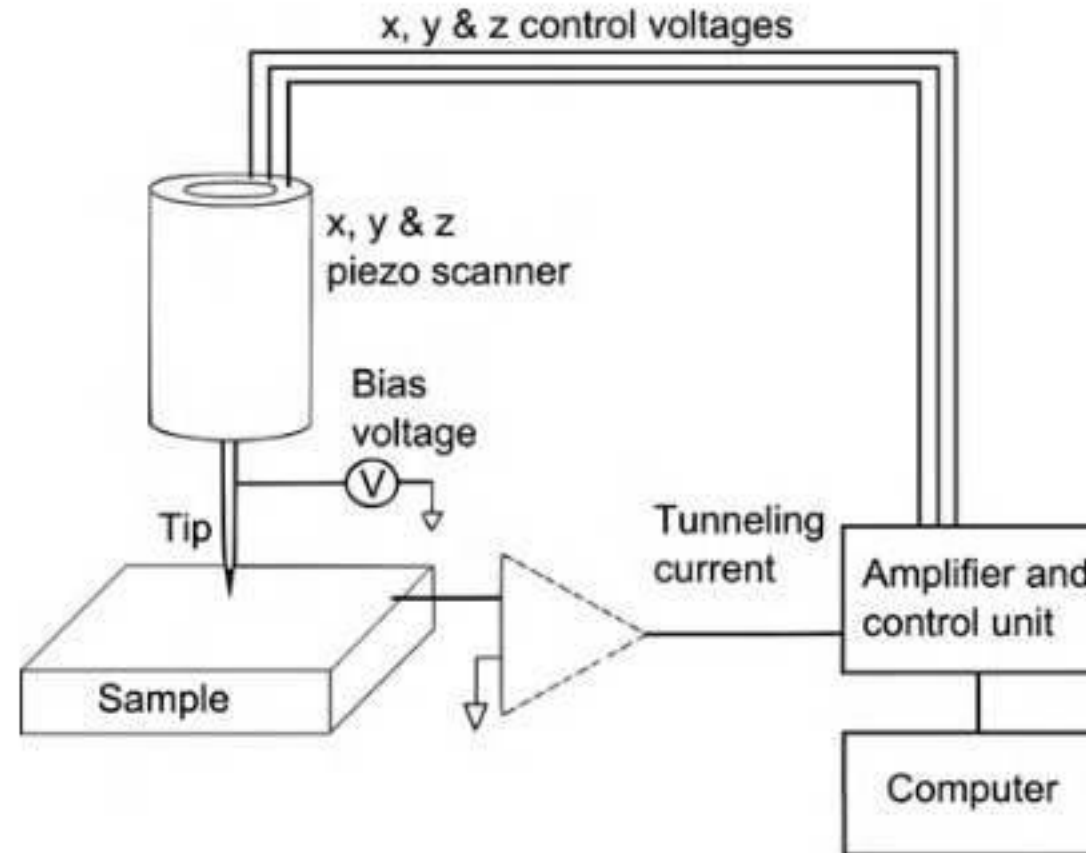
How can you make a spring so sensitive to measure that forces ?



The current in the tunneling microscope is proportional to the gap between the tip and the surface



A Typical STM set-up



One should ask: Why is it a microscope ?

As the tip moves across the sample, the tip height is changed to maintain the same current. This height of the tip maps the topography of the sample from the tunneling current expression we saw before

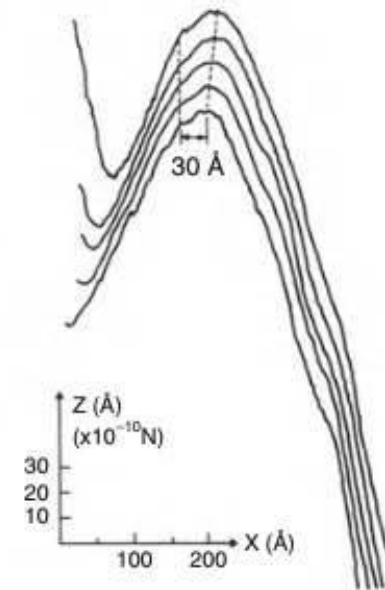
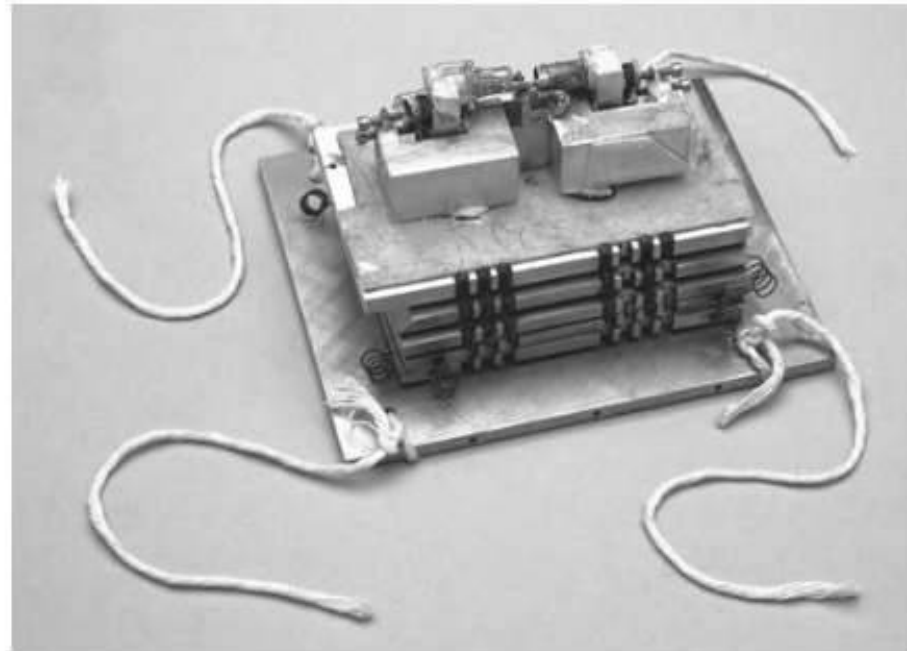
Atomic Force Microscope

While STM could provide atomic resolution of the surface, it requires the sample to be conducting!

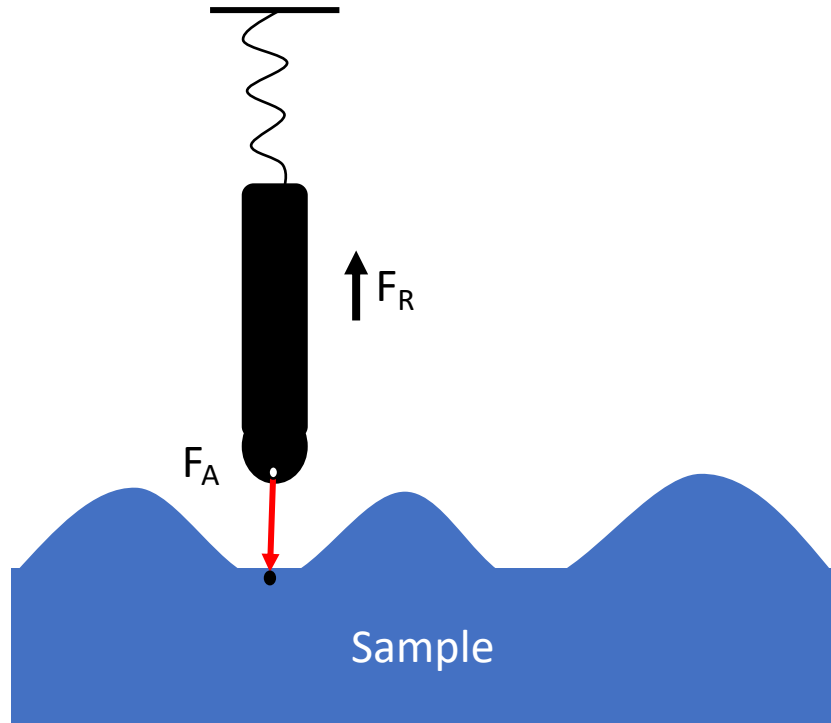
In addition, the sample, tip and the apparatus needs to be in vacuum and electronics cooled to reduce noise.

Force microscopy aims to obtain the same spatial resolution with the need for low temperatures, vacuum and the conducting samples

First AFM built by Binnig et. al., in 1986



Forces between Tip and the Sample



Meniscus in hydrophilic substrates

Attractive force:

1. van der Waals : $(F_{vdW} \propto \frac{1}{z^2})$

Van der Waals interaction is typically from neutral particles.

However, the underlying phenomena is Coulombic interaction between charge fluctuations in neutral particles

2. Electrostatic force: $(F_{ES} \propto U^2 \frac{1}{z})$

If there exists a potential difference U , then there is an attractive force between the tip and the sample

3. Capillary Force:

In hydrophilic substrates, moisture from room temperature can form a meniscus with the tip. This causes a capillary force driving the liquid up the tip due to surface tension effects

Repulsive Force



Pauli Exclusion / ionic repulsion:

Valence electron wavefunctions can extend few 10's of nm.

Pauli's idea excludes two electrons with same energy and spin in close proximity.

This leads to a repulsive potential $V_R \propto \left(\frac{\sigma}{z}\right)^{12}$

where σ is the distance at which the potential is zero.

For the simplest case where there is only van der Waal's attractive and Pauli's repulsion, the total potential

$$V_T = -K \left[\left(\frac{\sigma}{z}\right)^6 - \left(\frac{\sigma}{z}\right)^{12} \right]$$

This potential is called as Lennard-Jones potential.

The force due to the potential is then $F_{LJ} = -\frac{\partial V_T}{\partial z}$

Averaged van der Waal's force on the tip

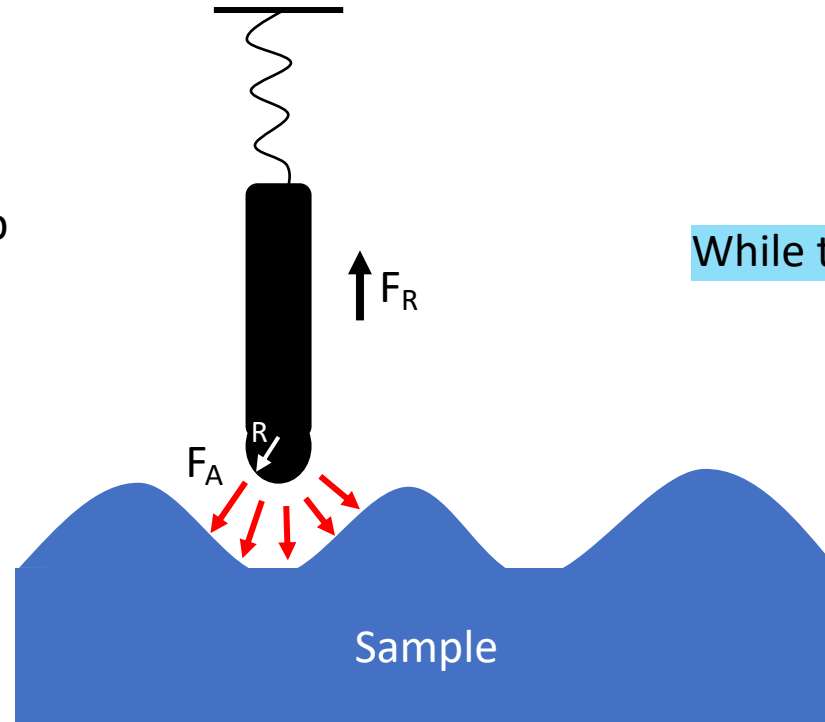
Hamaker's potential

$$V = -\frac{A_H R}{6z}$$

Thus the force

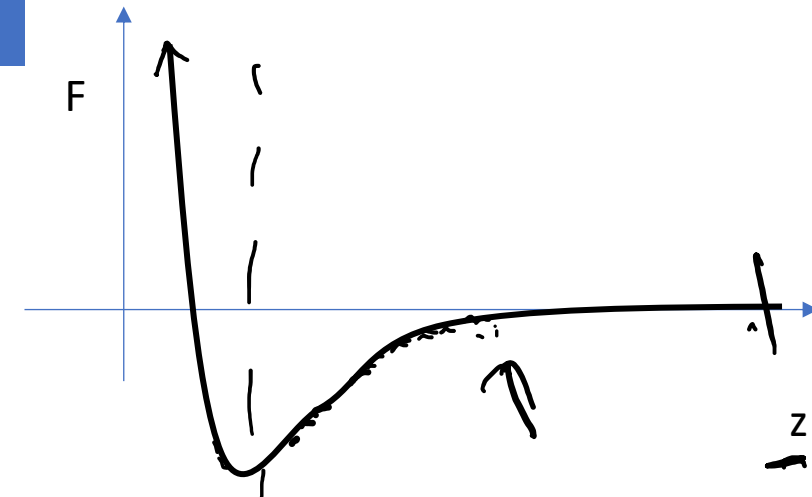
$$F_{vdW} \propto \frac{1}{z^2}$$

Its important to note that super short-range potential z^{-6} leads to a long-range force when averaged over several interacting atoms



While the repulsive forces is sharply short-range.

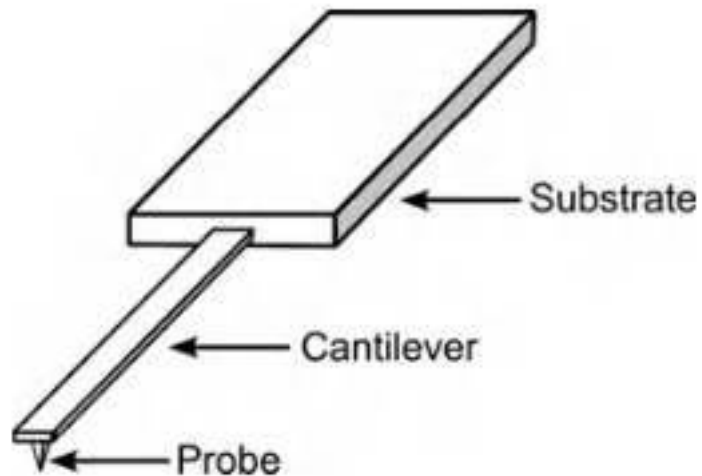
$$V_R \propto z^{-12}$$



Design of a tip

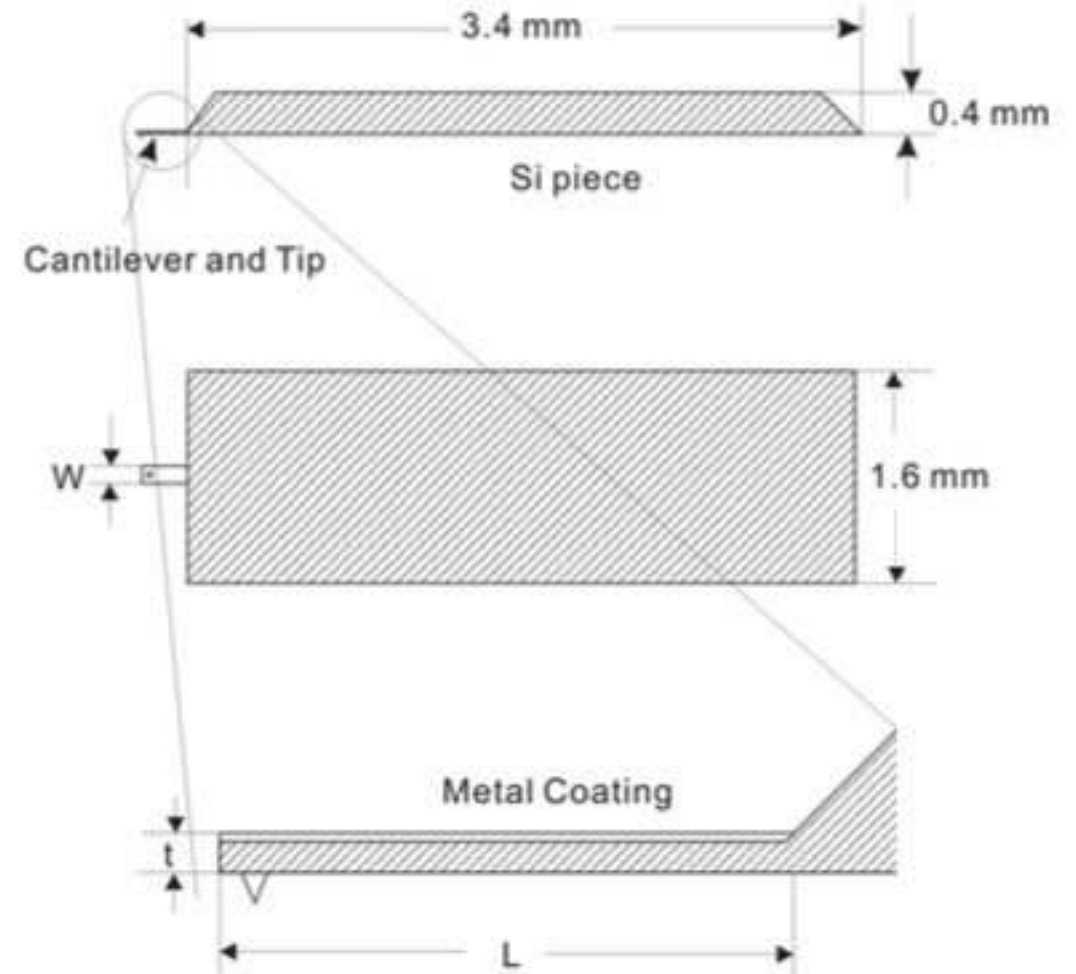
AFM tip is used to sense tip – sample interaction (Force)

Hence a mechanically pliant structure is required and is operated in elastic regime.



Spring constant $k = Ewt^3/4L^3$

Design of commercial AFM tips



Examples of Tips realized in experiments

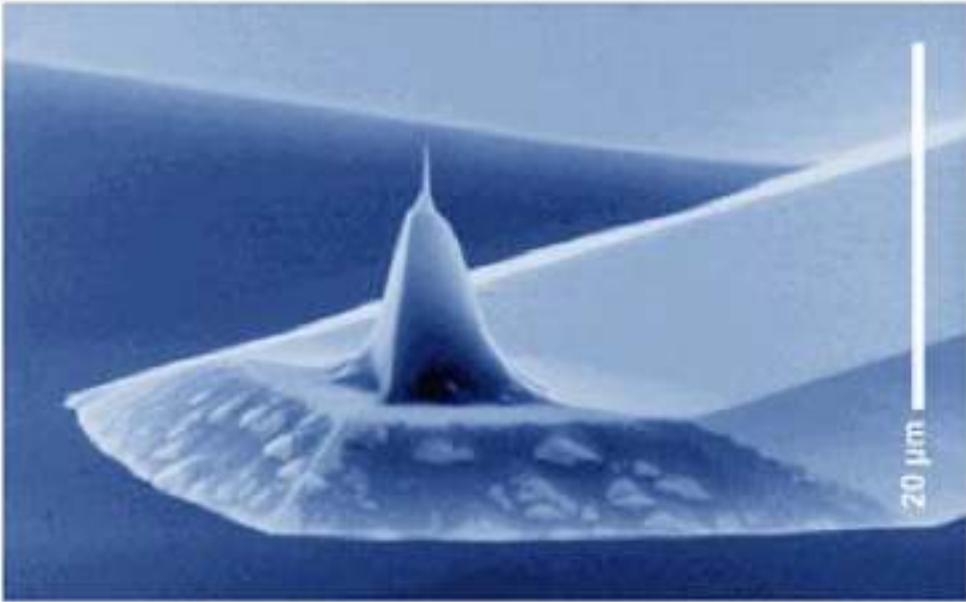


FIG. 8. (Color in online edition) Scanning electron micrograph of a micromachined silicon cantilever with an integrated tip pointing in the $[001]$ crystal direction (Wolter *et al.*, 1991). This is a Pointprobe sensor made by Nanosensors GmbH und Co. KG, Norderfriedrichskoog, Germany D-25870. Photo courtesy of Nanosensors GmbH & Co. KG.

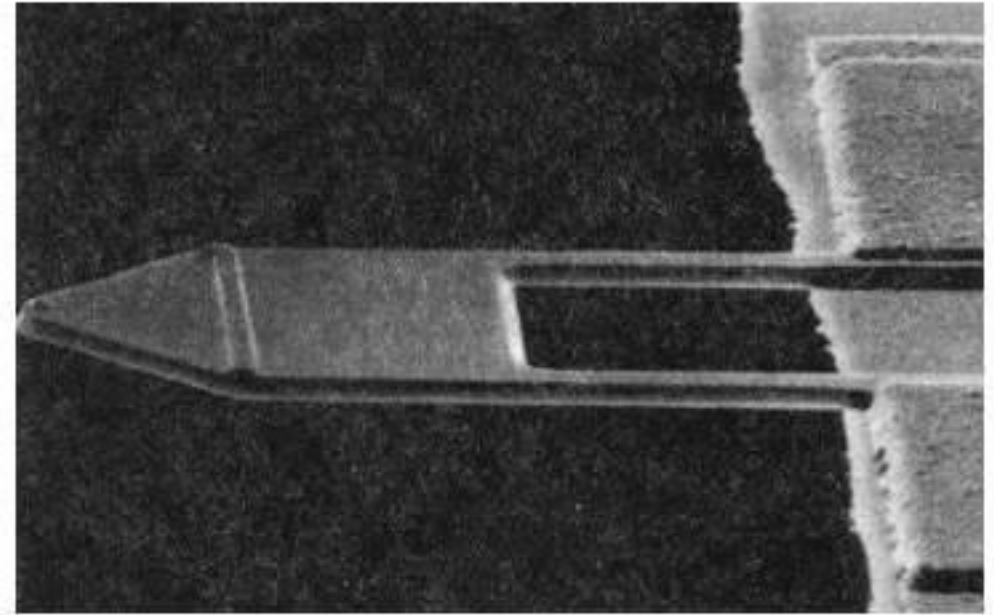


FIG. 10. Scanning electron micrograph of a piezoresistive cantilever built from silicon. Length, 250 μm; full width, 80 μm; thickness, 2 μm. From Tortonese *et al.*, 1993.

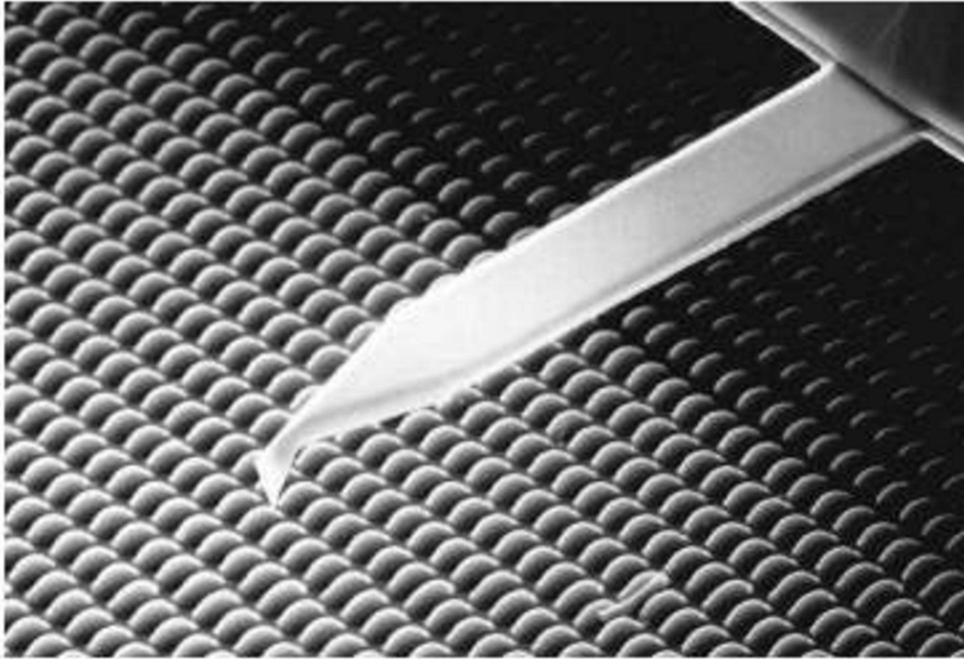


FIG. 9. Scanning electron micrograph of a micromachine — con cantilever with an integrated tip pointing in the $[001]$ crystal direction. In this type, the tip is etching free so that the sample area adjacent to the tip is visible in an optical microscope. Length, $120\ \mu\text{m}$; width, $30\ \mu\text{m}$; thickness, $2.8\ \mu\text{m}$; $k = 15\ \text{N/m}$; $f_0 = 300\ \text{kHz}$. Photo courtesy of Olympus Optical Co. Ltd, Hachioji, Tokyo 192-8507, Japan.

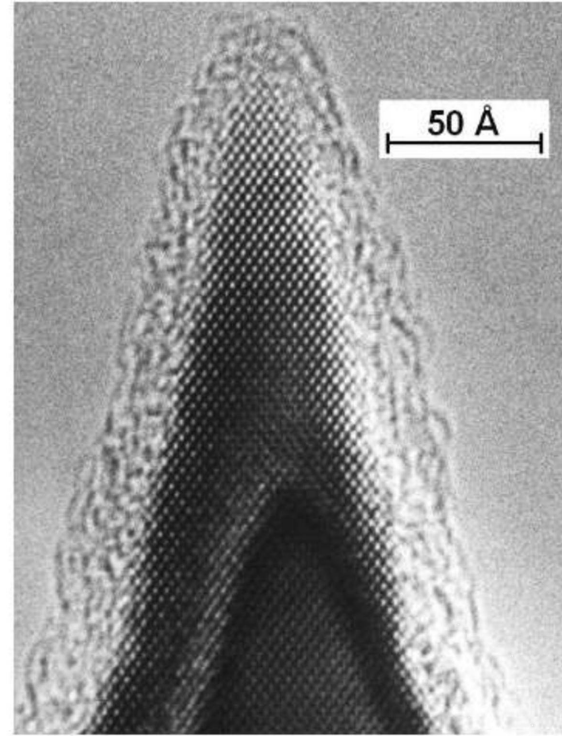


FIG. 13. Transmission electron micrograph of an extremely sharp silicon tip. The native oxide has been etched away with hydrofluoric acid before imaging. The $15\text{--}20\text{-\AA}$ -thick coating of the tip is mostly due to hydrocarbons which have been polymerized by the electron beam. Interestingly, the crystal structure appears to remain bulklike up to the apex of the tip. From Marcus *et al.*, 1990.

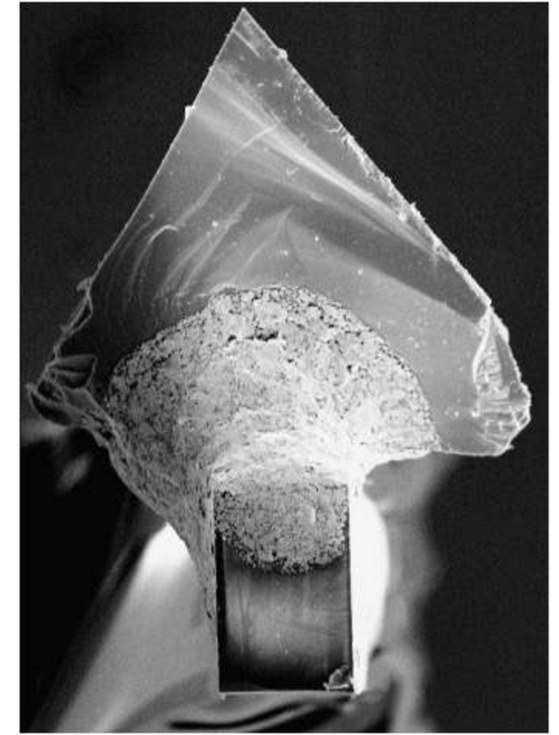


FIG. 15. Scanning electron micrograph of a cleaved single-crystal silicon tip attached to the free prong of a qPlus sensor. The rectangular section is the end of the free prong with a width of $130\ \mu\text{m}$ and a thickness of $214\ \mu\text{m}$. The tip is pointed in the $[111]$ direction and bounded by $(\bar{1}\bar{1}\bar{1})$, $(1\bar{1}\bar{1})$, and $(\bar{1}1\bar{1})$ planes according to the method of Giessibl *et al.* (2001b). Figure courtesy of Christian Schiller taken from Schiller, 2003.

AFM Instrumentation

Basic components:

1. Microscopic Stage
2. Control Electronics
3. Computer

Microscopic Stage:

1. Scanner head
2. Sample holder
3. Force Sensor
4. Integrated optical microscope

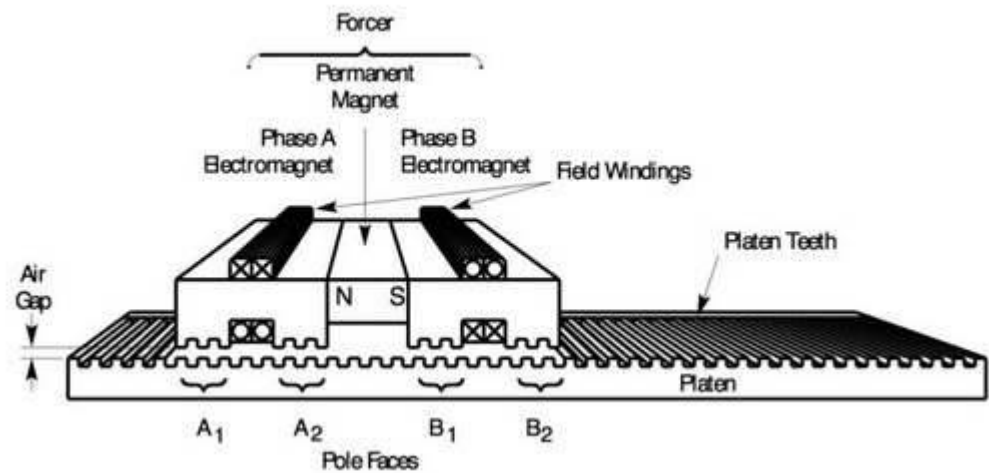


Linear Translations

However, for an AFM, there is a need for atomic resolution!

Linear translations in Å resolution is required.

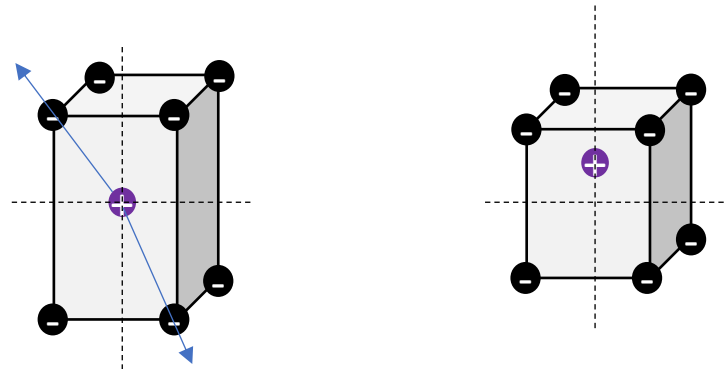
From inches - $\sim 1\mu m$



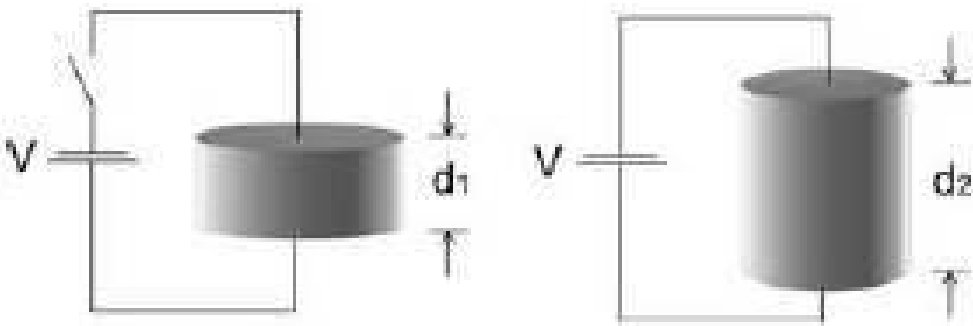
Piezoelectricity:

In non-centrosymmetric crystals, strain causes generation of electric dipoles

Uncompensated dipoles

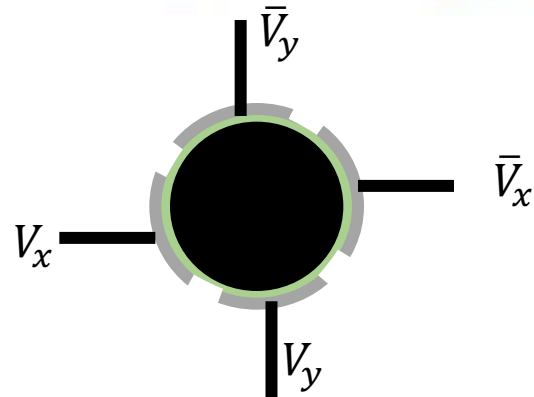
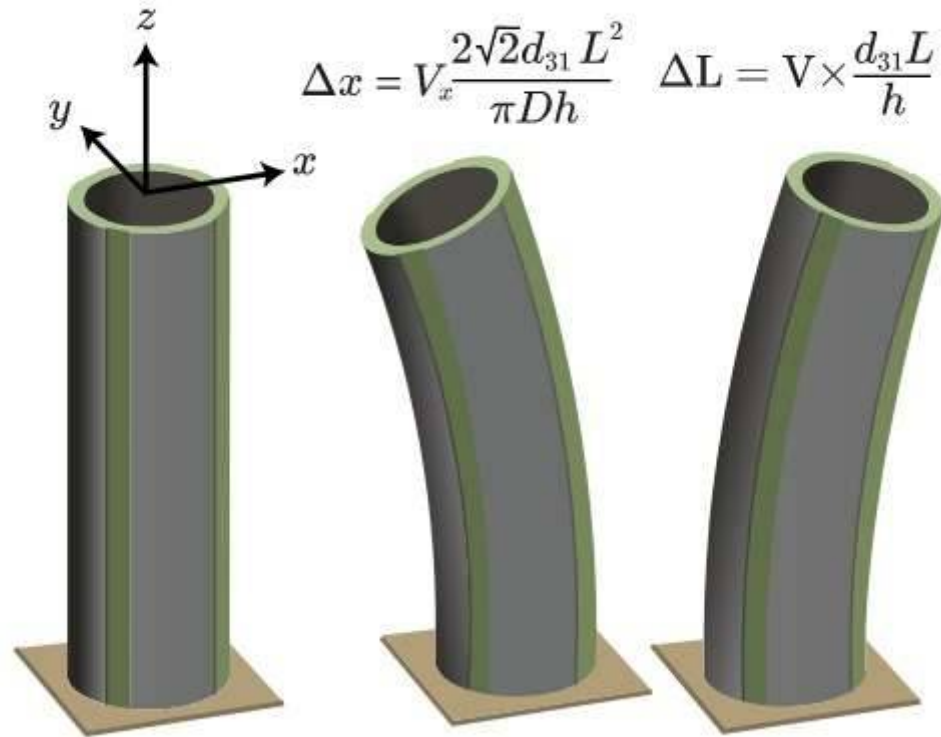


Equal and opposite dipoles



You either apply strain and get dipoles;
Or force a potential and observe mechanical strain

3-D motion using Piezoelectric transducers



When the base of the tube is fixed,
 V_x voltage along x axis, V_y voltage along Y

$$\Delta x = V_x \frac{2\sqrt{2}d_{31}L^2}{\pi D h}$$

$$\Delta y = V_y \frac{2\sqrt{2}d_{31}L^2}{\pi D h}$$

d_{31} is the piezo strain, L – length of tube, D – outside diameter, h – tube thickness.

If a potential V is applied to all four terminals,

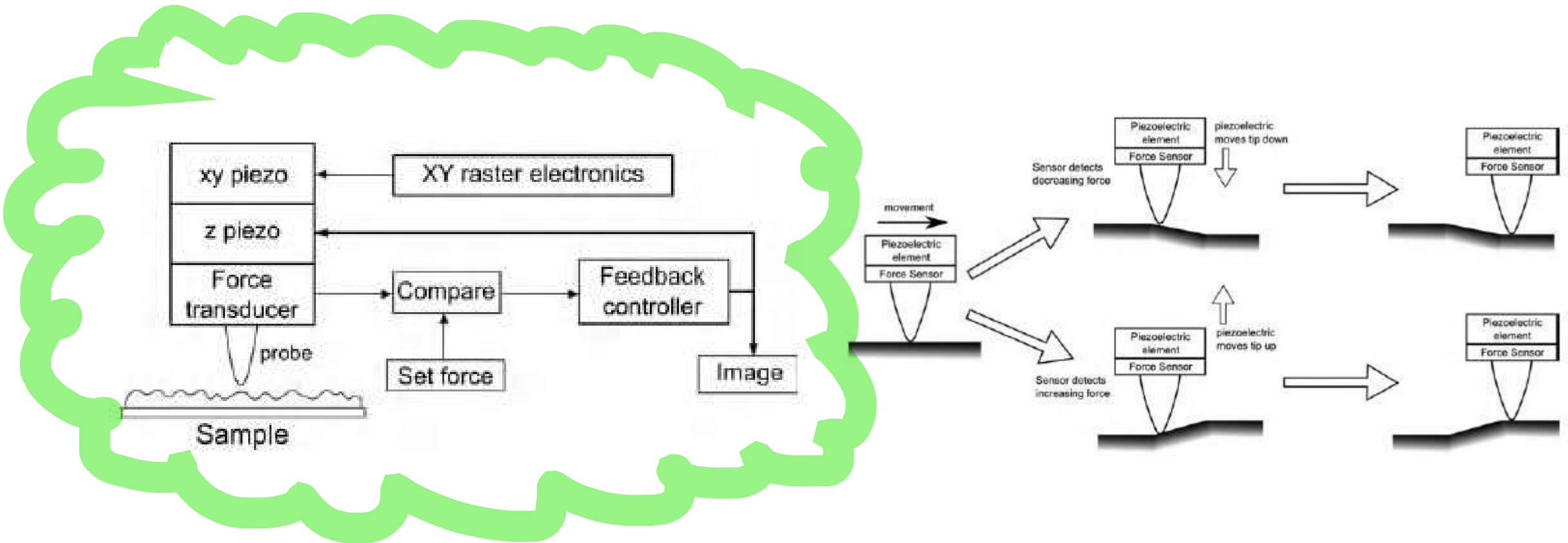
$$\Delta L = V \times \frac{d_{31}L}{h}$$

Since, + V_x and – V_x are applied to diametrically opposite sides, one side contracts and other side expands. The tube ‘bows’ in a fashion to accommodate the differential strain

Typical expansion coefficient: $\sim 0.1 \text{ nm/V}$

Thus, atomic scale movements are possible using Piezoelectric transducers

Typical Block diagram of AFM head

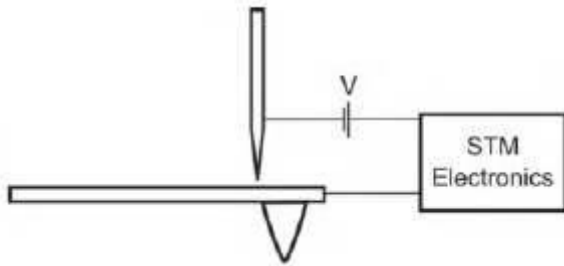


The XY Piezo in addition to linear transducers are used to raster the sample surface. At each point on the surface, the force transducer translates the force sample interaction

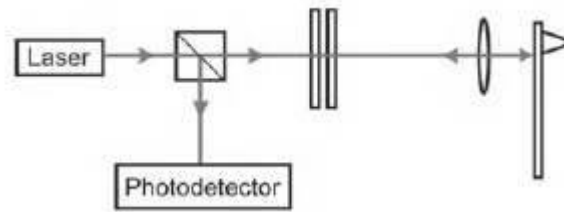
A feedback loop is used to ensure that the force between the tip and the probe remains the same. 'z' transduction need to bring the force to its set value is noted as the sample height

Force Sensors

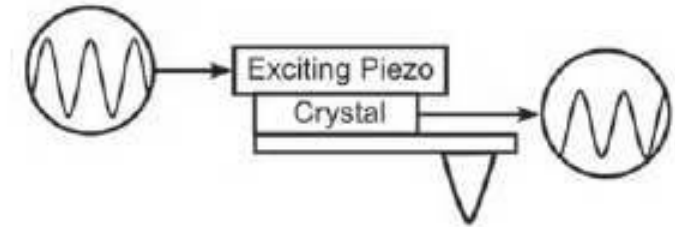
A force central to AFM design. Some common designs



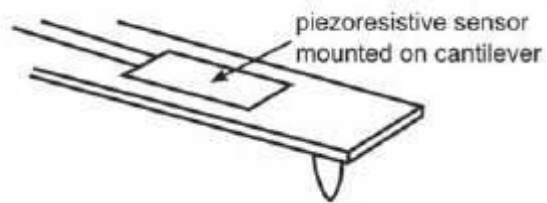
A STM based design



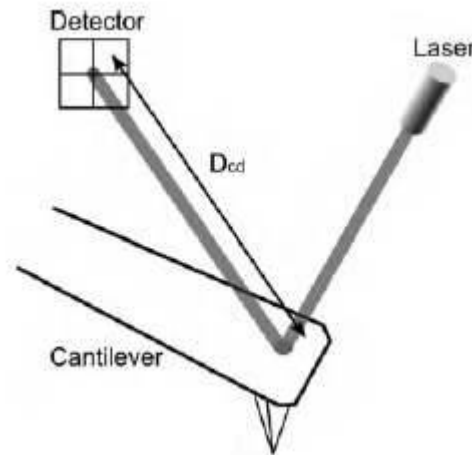
Traditional interferometer



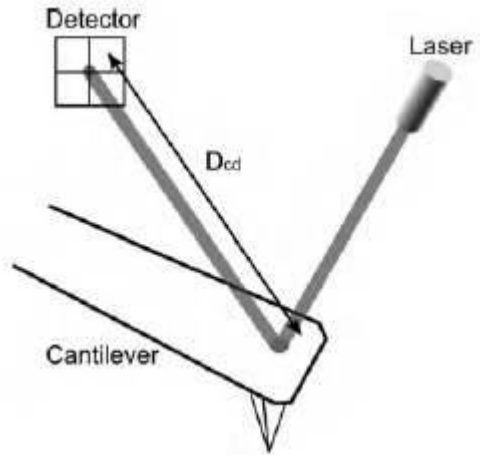
Dynamic measurement (Phase and amplitude)



Standard Piezoelectric sensor



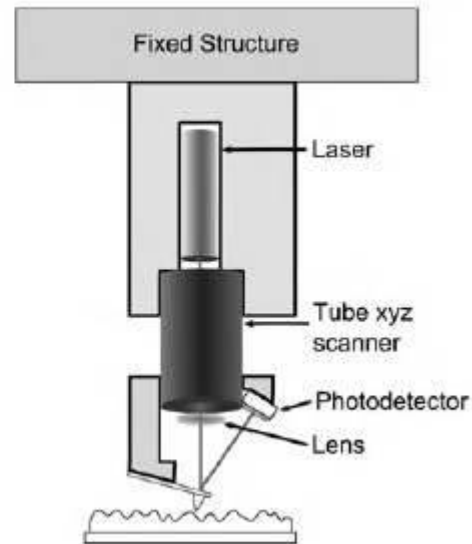
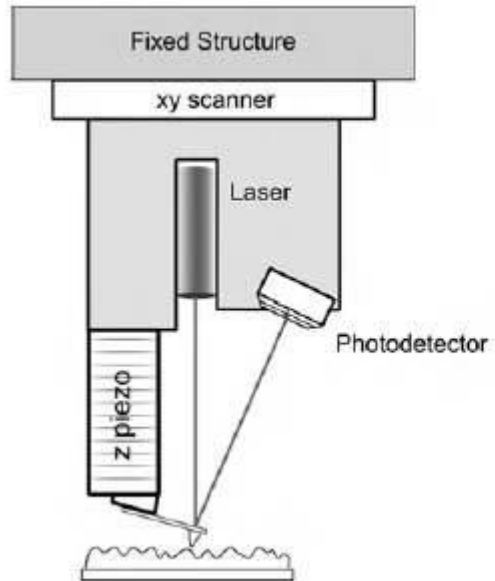
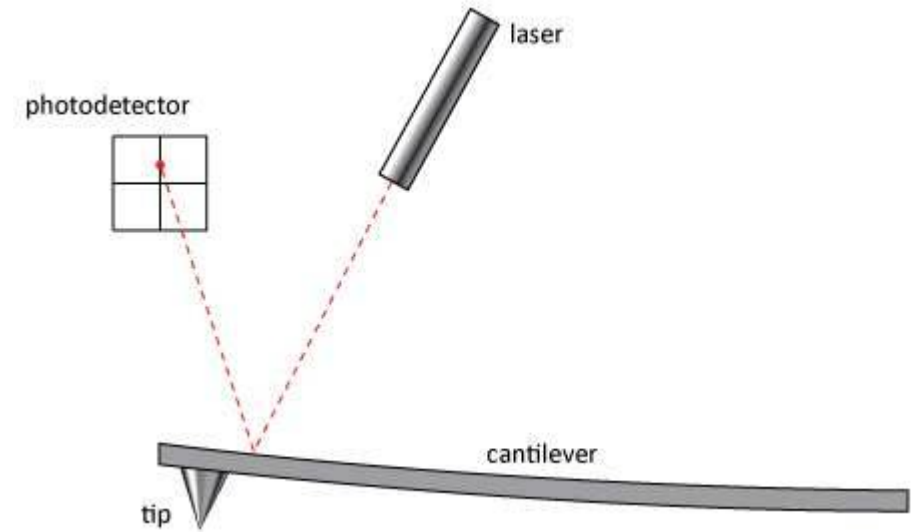
Most-commonly used technique



Cantilevers are typically 30 - 600 μm long

20 - 60 μm wide and 0.2 - 1 μm thick.

Position sensitive photo detection

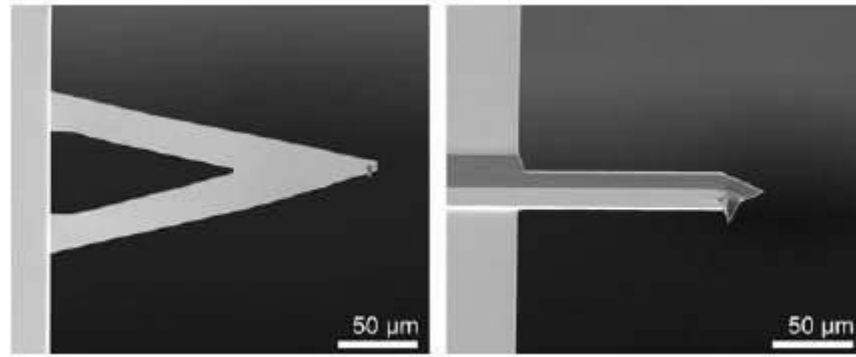


Either the laser and detector assembly is moved by X-Y stage.

Or the cantilever and the detector is moved. Lens focuses the light.

Typically the lens focuses the light at an angle to the cantilever to allow for imaging the sample and the cantilever together

Operational Modes of AFM



Contact

Typically a Triangular shaped less stiff cantilever is used ($K < 1 \text{ N/m}$)

Made of Si_3N_4 like material

The tip is brought to contact in a static mode

Bending of the cantilever is measured

Non-contact mode

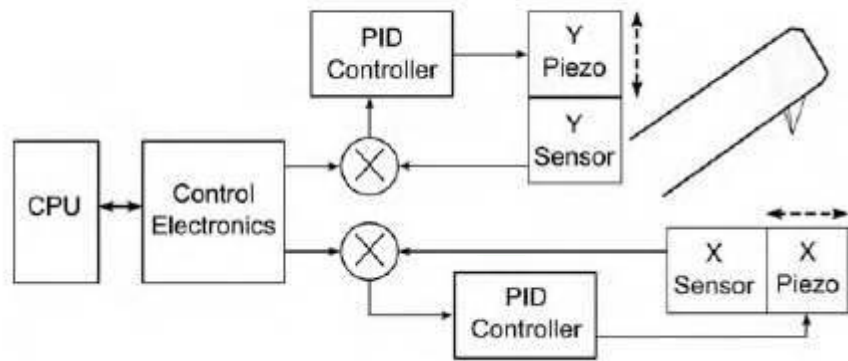
Typically a rectangular stiff material is used.

Like Si

The cantilever is made to oscillate close to the surface

The sample-tip interaction forces the oscillation amplitude and phase to change

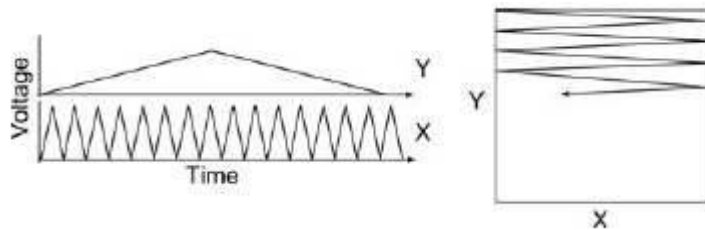
Generating an AFM Image



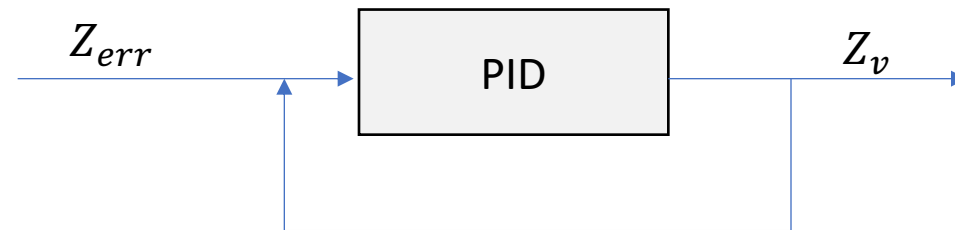
Steps:

1. Generate scanning signals for X-Y stage movement
2. Take input signal from force sensor and feed to z-piezo
3. Get output control to X-Y-Z motor
4. Generate signal for oscillating probe. Measure phase and amplitude
5. Collect signals and display on computer

Generating the X-Y raster



The Z potential is set using a feedback setup



$$Z = p * V_{err} + i * \int Z_{err} dt + D \times \frac{dZ_{err}}{dt}$$

Static Interaction of Tip with Sample

A cantilever with the sharp tip is moved close to the surface.

At each point, there is sample height determined force $F(z)$

The probe tip is assumed to be much stiffer than the cantilever and the sample

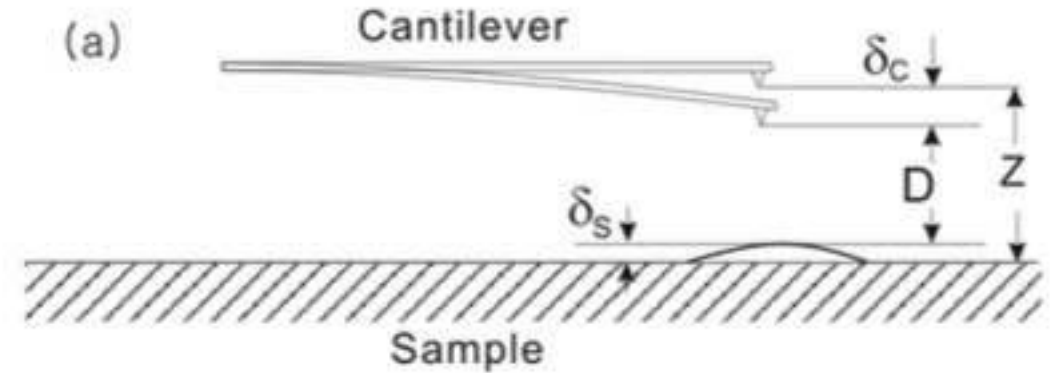
Typical inter-atomic force constants in solids – 10 – 100 N/m

In biological samples - ~ 0.1 N/m.

Hence the cantilever stiffness is typically set at 0.01 – 5 N/m

The cantilever deflection $\delta_c = -F_c(z)/k$ – Hooke's law

Where F_c is the elastic force on the cantilever.



Say, the cantilever is near a hill with D separation. There will be tip bending and a restoring force from the cantilever.

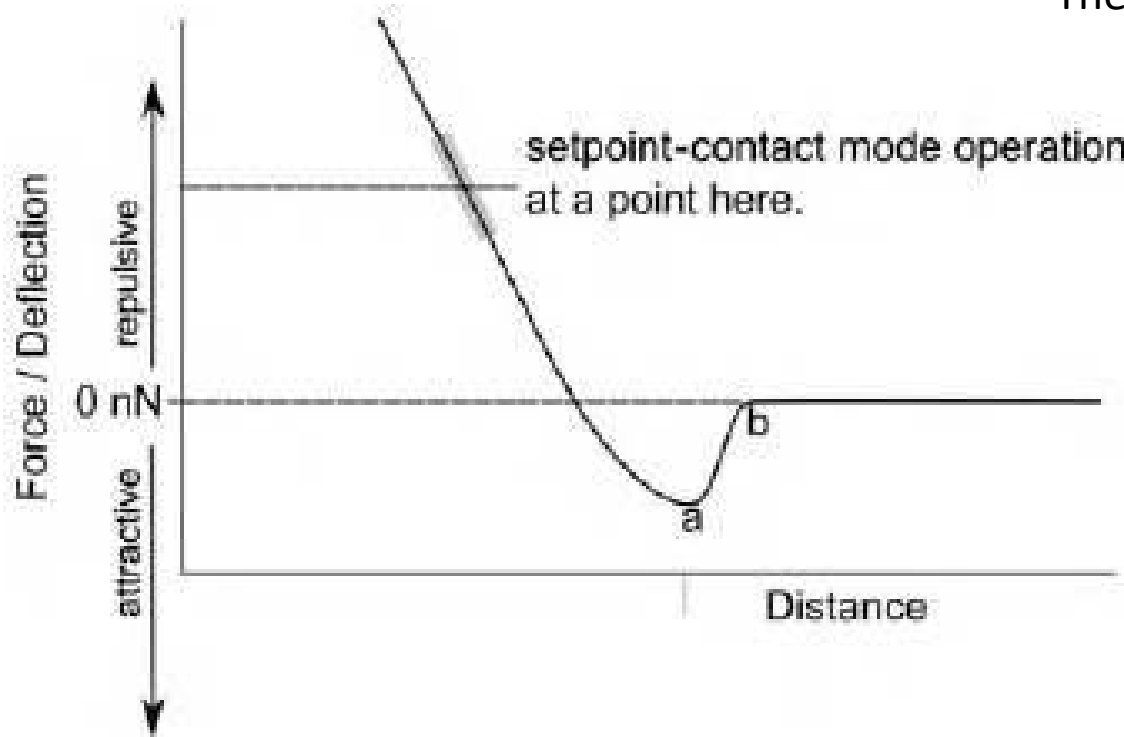
At equilibrium the cantilever restoring force is same as the sample-tip interaction force. So, the net force on the cantilever is 0. However, the cantilever at 0 force is at a different height (Z)

Force-distance curve

In contact mode, the tip Z is moved such that the force remains constant. This point is typically in the repulsive region as shown.

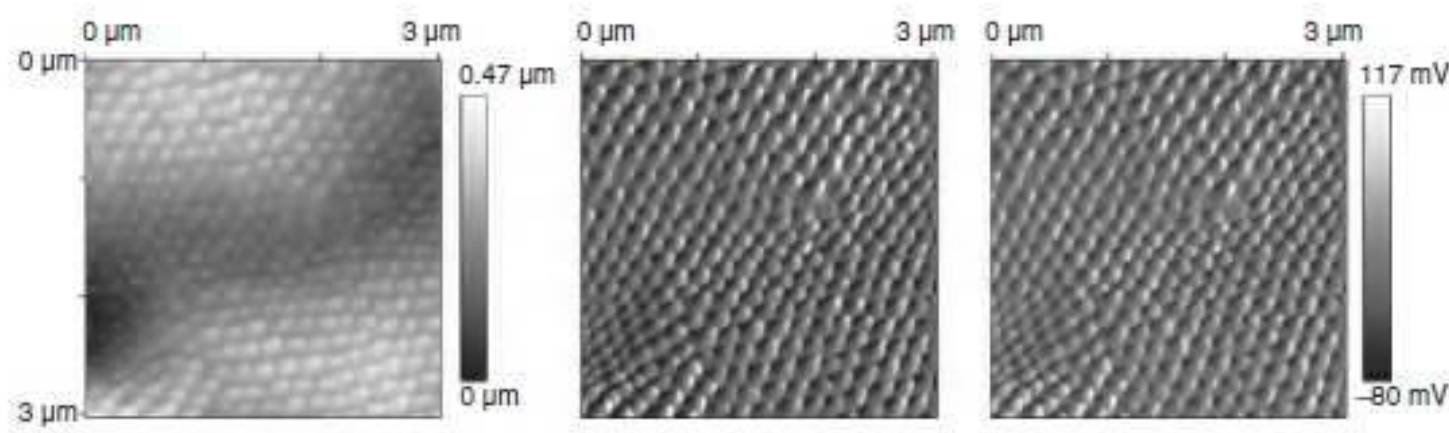
The tip is always touching.

At point b, an instability happens and the comes into contact



Force on the cantilever and its deflection are just scaled version of the same thing (related by Hooke's law)

Typical Contact Mode Image



Two modes of AFM operation in contact:

1. Constant Force mode: This is the most common mode of operation. The Z-height is changed using a feedback loop to get to the same force as seen in the previous slide. In most measurements, this is called as the height image.
2. Constant height mode: This is the deflection in the cantilever before the feedback kicks in. This is the raw deflection in the cantilever as a X-Y stage is moved. This is typically displayed as deflection image.

Many times, this image is used. However, if the feedback settings (PID) is correct, the height image and the deflection image should match.

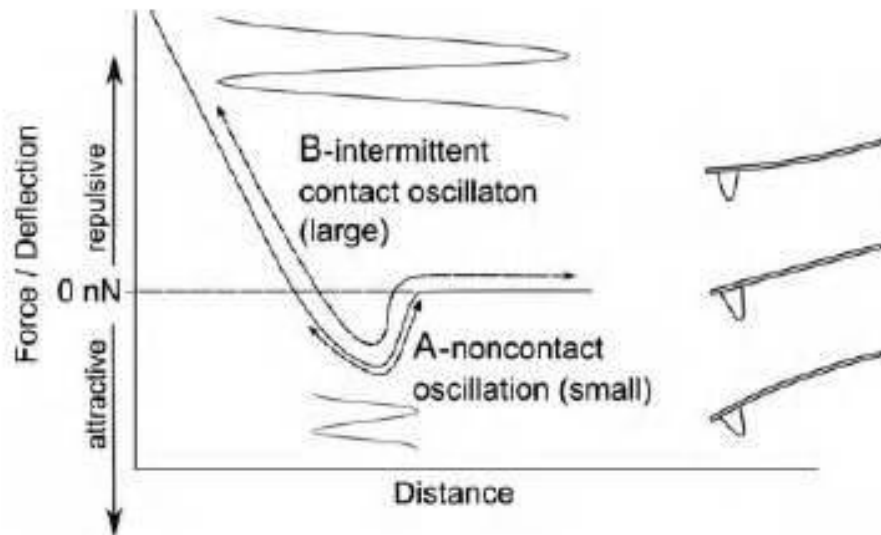
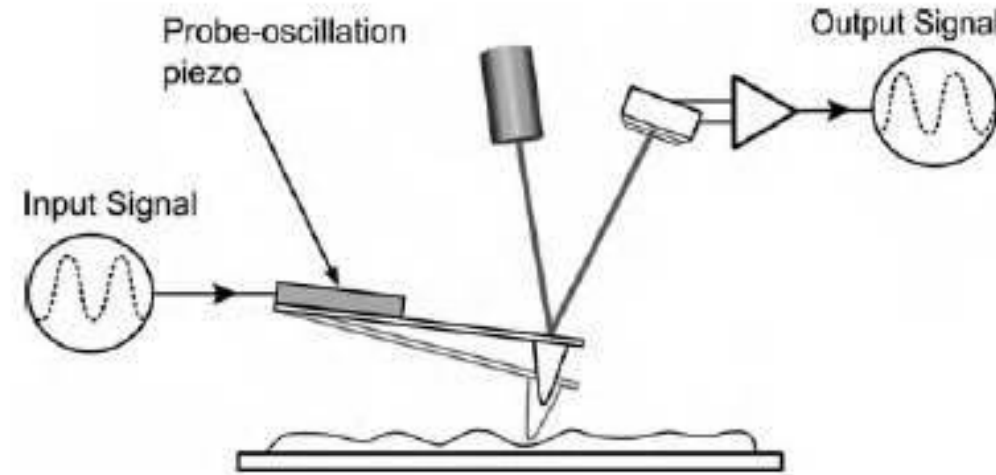
Dynamics Mode

An additional piezo patch is placed on the cantilever.

The potential of vibration is set close to mechanical resonance.

The tip oscillation is collected by optical means.

The sample interaction changes the amplitude, frequency and phase of the oscillation

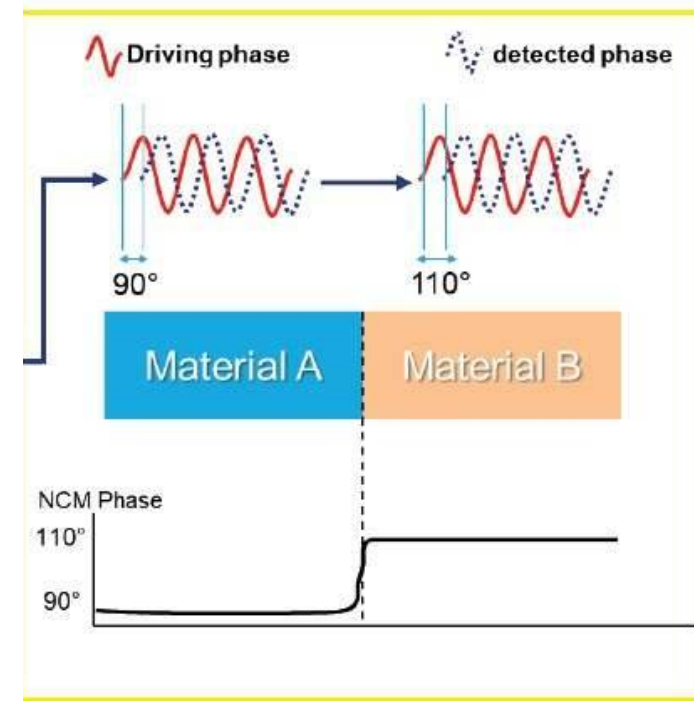
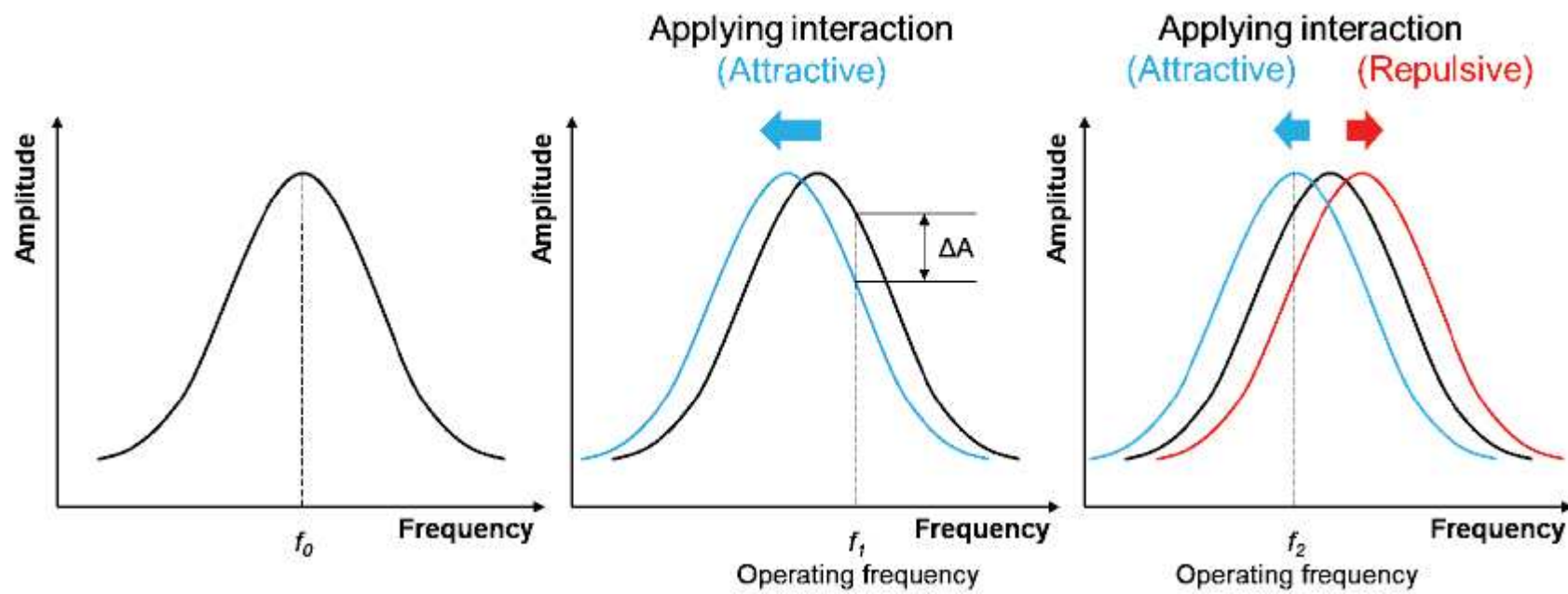


The amplitude of oscillation, determines the interaction method.

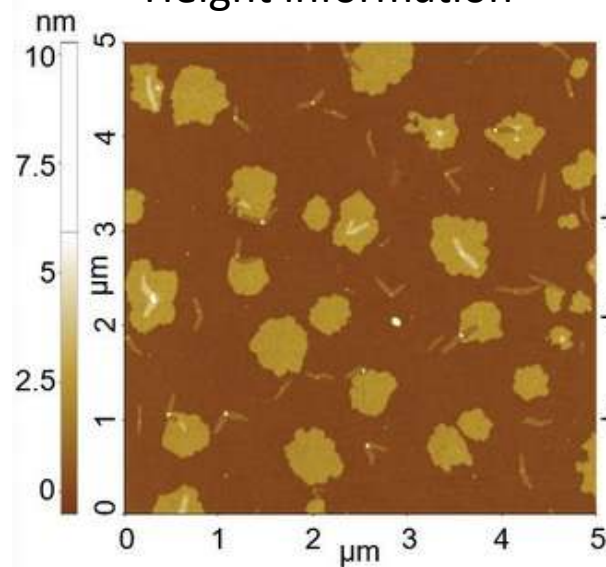
Small oscillations are mostly non contact

Large oscillators near the surface gives intermittent contact.

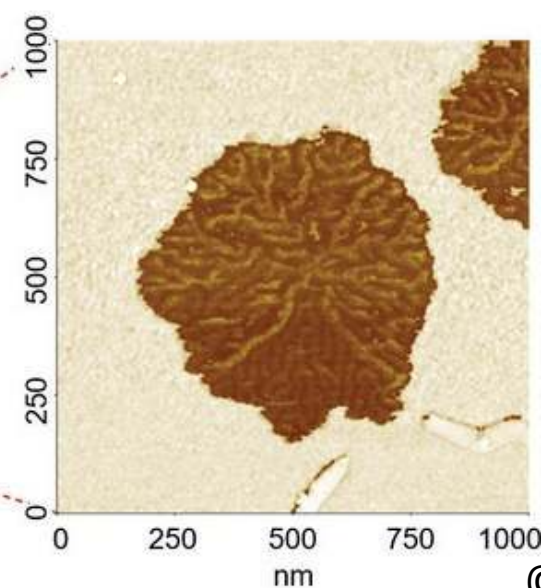
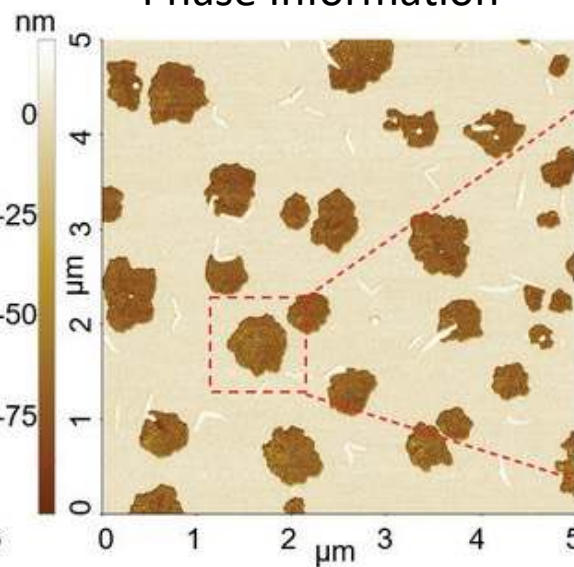
$$\text{Resonant frequency } f_0 = \sqrt{\frac{k}{m}}$$



Height information



Phase Information



Types and modes of Tip based microscopy

Contact modes
Non-contact and intermittent contact

Chemical Force microscopy

Force spectroscopy

Nanoindentation

Magnetic force microscopy

Kelvin-Probe microscopy

Nano-Lithography

Scanning Near Field optical microscopy (SNOM)

Aperture SNOM, non-aperture SNOM, transmission SNOM,
Collection SNOM

STM Based methods

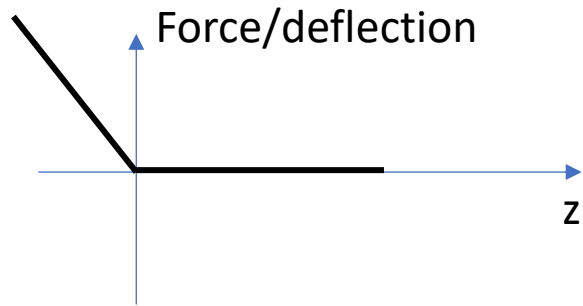
Scanning Tunnelling Spectroscopy

Scanning Tunnelling Electron microscopy (STOM)

Ballistic electron emission spectroscopy

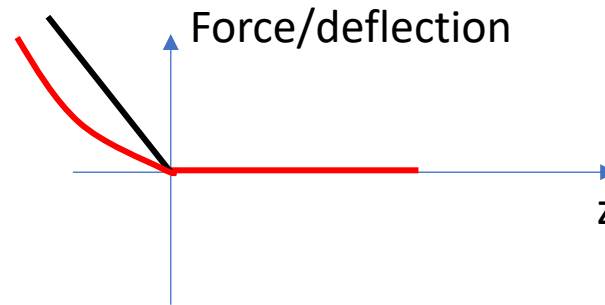
Force Spectroscopy:

Assume an ideal case: Infinitely hard sample and no surface forces

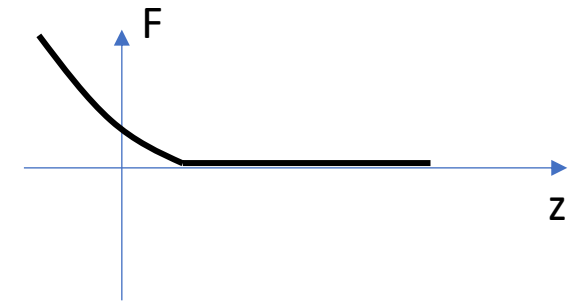


No force till contact!
After contact, there is linear deflection

Soft material no surface forces

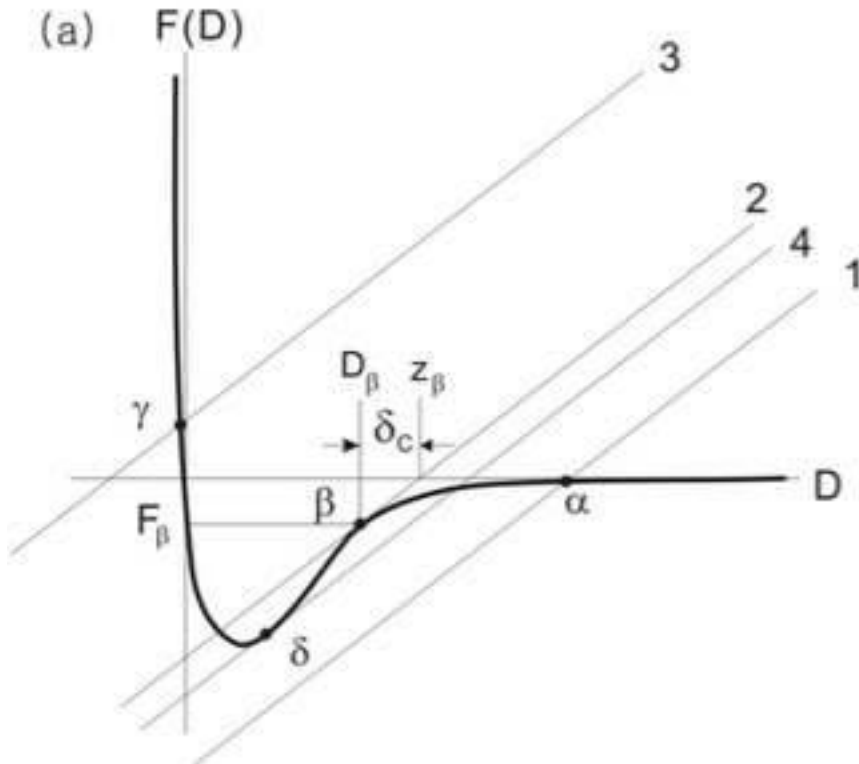


Infinitely hard with repulsive forces



As the tip comes a bit close, the repulsive force takes charge and bends the cantilever up. This happens even before contact

Force-Displacement Curve

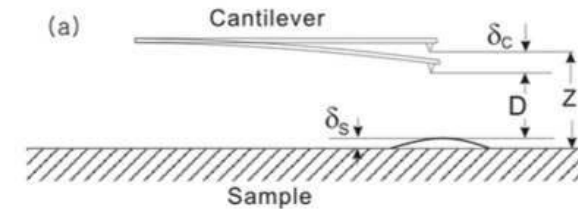
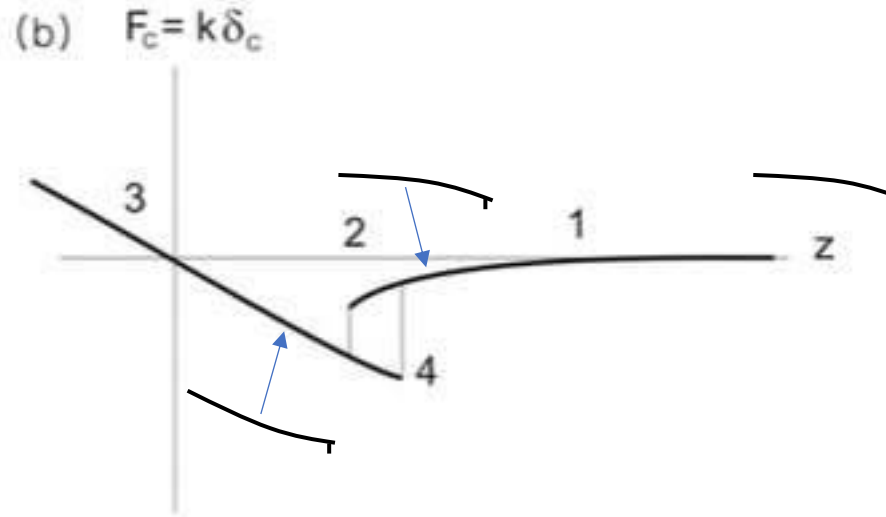


Till point 1, cantilever bends towards the sample
Right after 2, there is jump to contact. The cantilever bends away
due to contact. **There is onset of repulsive interaction.**

Further reduction in z , increases repulsive interaction quickly. Slightly before 3, repulsive force overtakes attractive force. You have displacement opposite direction to force!

Force-Displacement Curve

Force on the cantilever vs the final distance from surface



Many a times, when the tip is retracted,
There is hysteresis.

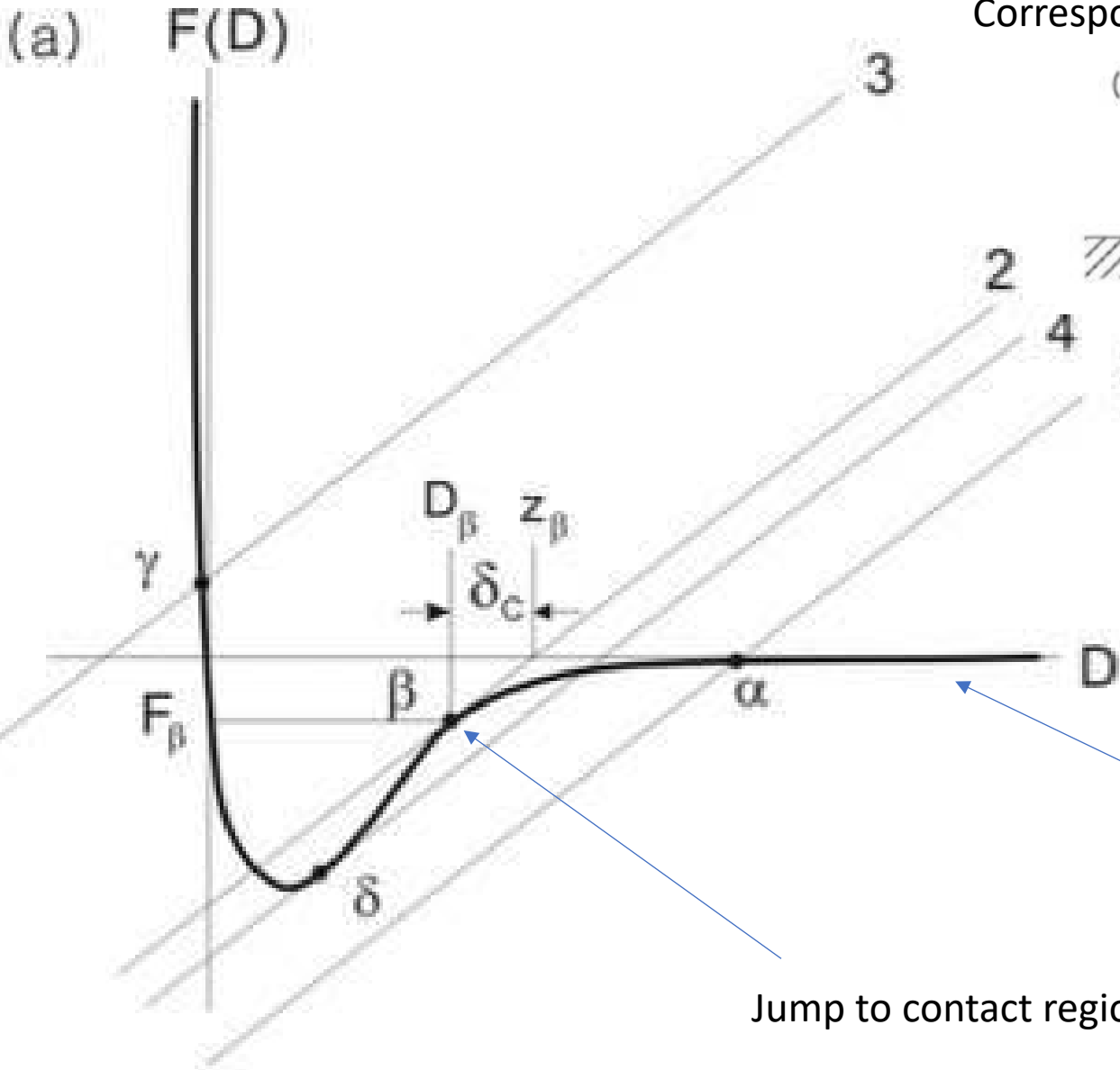
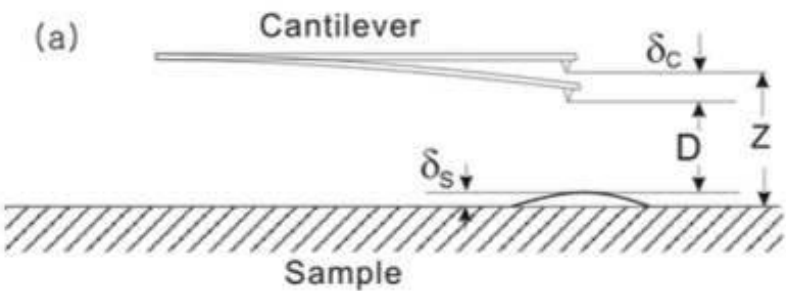
This is due to adhesion, and capillary forces.

The tip takes larger z than the jump to contact

Cantilever Force-Distance Curve

Lines 1- 4 are Hooke's law lines

Corresponding to $F_c = -k(D - z)$



No force, No displacement
 $F_c = \delta_c = 0$

Jump to contact region

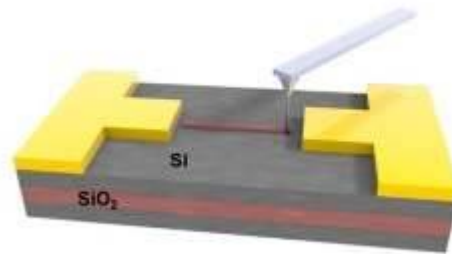
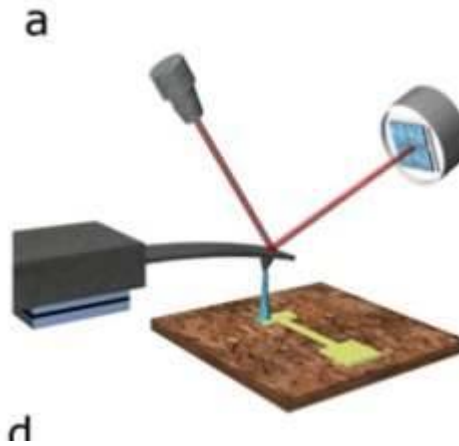
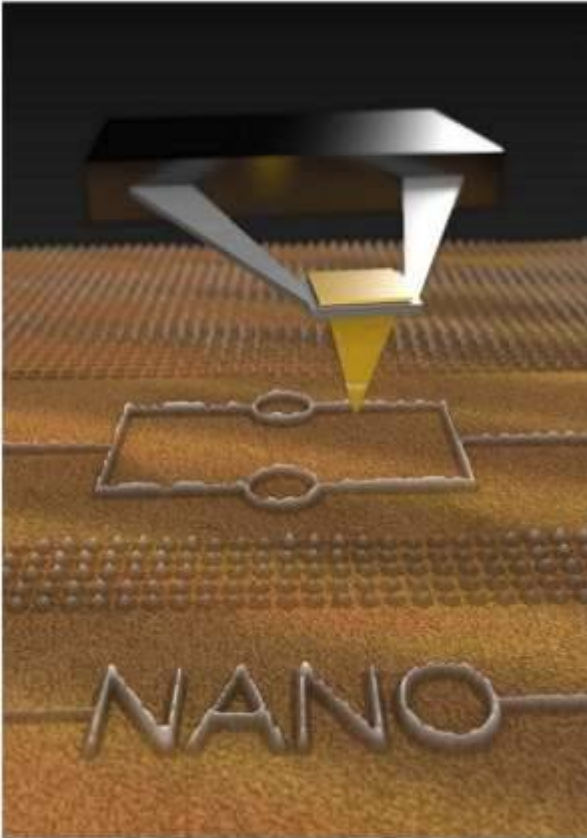
If you have moved the tip such that $D = \beta$, giving a interaction force F_β ,

The cantilever will have a spring restoring force.

The cantilever bends up, moving back to a position corresponding to z_β .

The deflection now is $\delta_c = D_\beta - z_\beta$

AFM Nanolithography



Lithography is a method of selectively removing material.

Nanoscale lithography is used in many places such as electronic devices, chemical synthesis and such

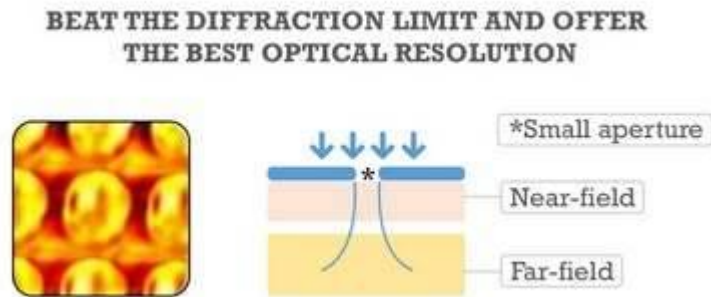
A probe can be used which can supply electric field modifying the properties of certain polymers in nanoscale.

The modified polymers are dissolved in suitable solvents.

Near Field Microscope

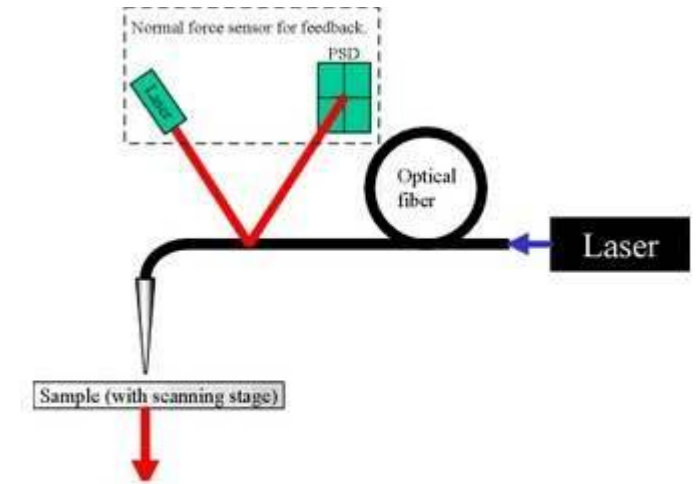
Abbe's limit: You can only resolve two features if $d > \frac{\lambda}{2 \sin \alpha}$ where d is the distance between points, α half angle of collection

So, use a AFM tip to go near field!
Go near the particle and collect the light at smaller wavelengths



A nanoparticle when irradiated with light forms a dipole.

The dipole oscillates with very small wavelength near the nanoparticle (near-field),
Which then diverges (diffracts) to larger wavelengths as you go far (far-field)



End of Atomic Force Microscopic Techniques

Introduction to Electronic Materials and Characterization



Electric Trains (1837)



Steam Turbines, 1884



Delhi-Faridabad
Electric Sub-station
1885



Transmission-Line cables



Electric Fans,
appliances
(1900)



Telephone lines
- The need for switching
voltages – fast (1876)



Birth of Transistors. 1947



1973 – First handheld
Motorola



Moto1973



Nokia 1987



Moto First Flip 1989



Nokia Communicator 1993



Seimens, Color Screen 1998



Nokia water proof 1999, 3G, GPS



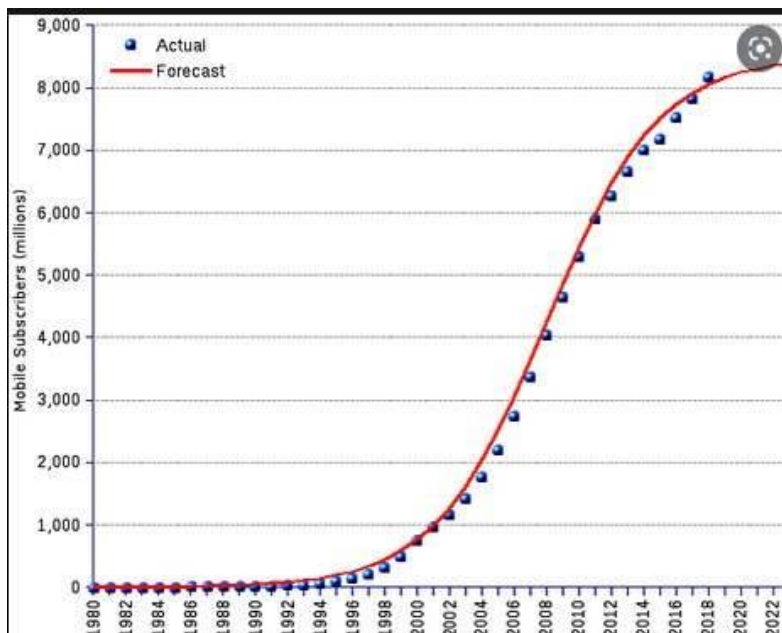
First camera phone was a Selfie one!



Year 2K



First iPhone 2007



The fold again! 2020

Electrically ?



Soot



Graphite

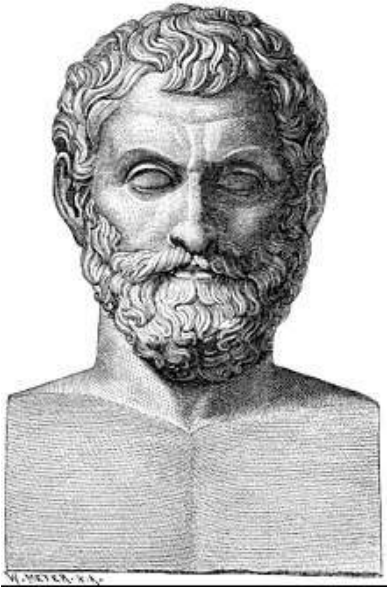


Diamond

Structurally different!

What is electric charge ?

Charge (q): It is an indivisible physical property which bears a force upon interaction with an electric field. Electrons are taken to be negatively charged and protons positively charged (convention).



Thales Miletus 400 – 600 BC

Amber and Fur are inanimate objects having soul



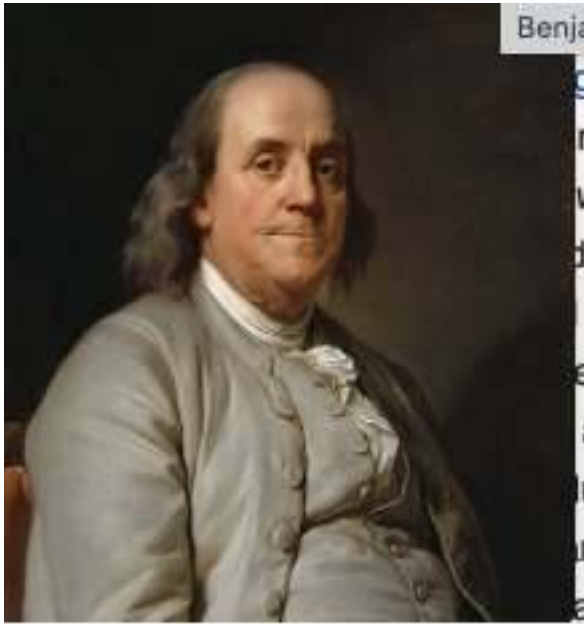
In 1600 **William Gilbert** coins the term *electrica* from Greek word *electron* for Amber



1729 The 'electron' in the Amber-fur flown across distances [Stephen Gray]

1733 Almost all materials can collect electrons by rubbing, other than metals! – C. Fay

Benjamin Franklin



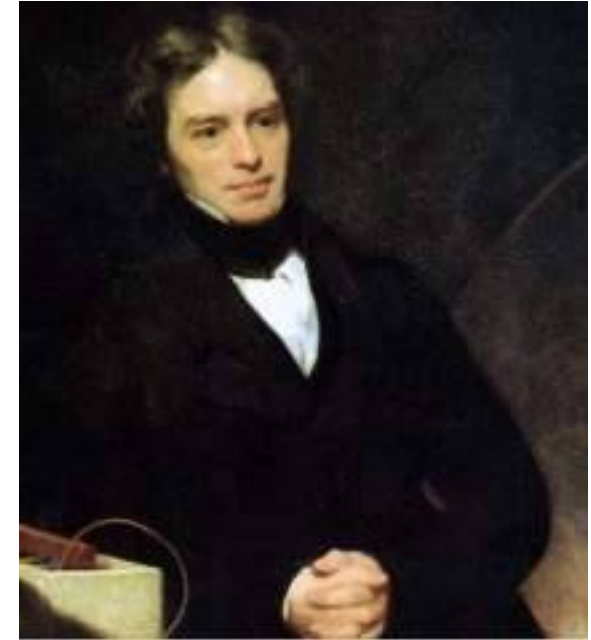
1750 – Rubbing glass with rubber develops equal 'electrons' on both and there exists attractive force. So two kinds of charges – B. Franklin

Alexandra Volta



1800 – Charges can be made to flow in a continuous loop. Alexander Volta

Michael Faraday

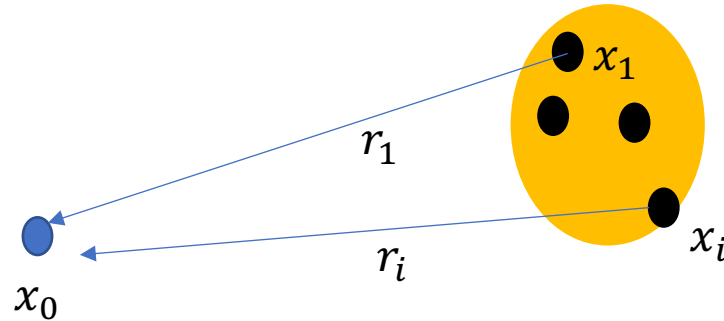


1833- 1838 There are only two kinds of charges. They are independent on its prepared. There is a force of attraction between the two kinds and they want to intermix in normal materials. When you rub one against other, the two types separate to different surfaces - M. Faraday

Charles Augustine Di Coulomb

In 1784:

- There exists a force between two charges
- The force was found to act on the line joining the charges, only either towards/away
 - Thereby giving only two kinds of charges
- The force is proportional to the number of charges
- The force is inversely proportional to the square of the distance between the two



Electric Field E : Force experienced by a unit positive charge

$$E = \frac{F}{q_0} \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{(x_i - x_0)^2} \hat{r}_i$$



Charge Dynamics

- Force on a charge $F = qE$
- Forces causes charges to move,
- moving charges are called current from hydrodynamics.
- One defines current density $J = \frac{d\rho}{dt}$

1820 – Orsted - flowing charge affects compass. There is an induced magnetic field.

- Magnetic field lines encircle the wire carrying current
- lie perpendicular to the plane of the wire.

Ampere also introduced the equations for the magnetic field and current density called as the circuital law:



Forms of the original circuital law written in SI units

	Integral form	Differential form
Using B -field and total current	$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{S} = \mu_0 I_{\text{enc}}$	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
Using H -field and free current	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \mathbf{J}_f \cdot d\mathbf{S} = I_{f,\text{enc}}$	$\nabla \times \mathbf{H} = \mathbf{J}_f$

James Clarke Maxwell



In dynamical theory of electromagnetic fields (1865)

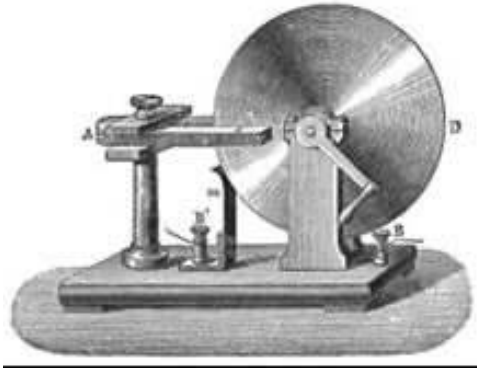
Name	Integral equations	Differential equations
Gauss's law	$\oiint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\varepsilon_0} \iiint_{\Omega} \rho dV$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$
Gauss's law for magnetism	$\oiint_{\partial\Omega} \mathbf{B} \cdot d\mathbf{S} = 0$	$\nabla \cdot \mathbf{B} = 0$
Maxwell–Faraday equation (Faraday's law of induction)	$\oint_{\partial\Sigma} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
Ampère's circuital law (with Maxwell's addition)	$\oint_{\partial\Sigma} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \left(\iint_{\Sigma} \mathbf{J} \cdot d\mathbf{S} + \varepsilon_0 \frac{d}{dt} \iint_{\Sigma} \mathbf{E} \cdot d\mathbf{S} \right)$	$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$

Electrical Components



Leiden Jar, 1745

First notion of storage of charges,
First unit of capacitance was Jar.



Faraday's Motor, 1820

Faraday thought a wave will flow between two coils wound on an iron core when one of them is carrying current. He also saw that current can flow in the coil as the magnet/iron rod can be moved fast enough. Hence he name inductance, where the wave is induced.



Morse code, 1844
Transmission of electric signals on long wires to communicate information.



Marconi,
Bose Radio waves
1886-1888

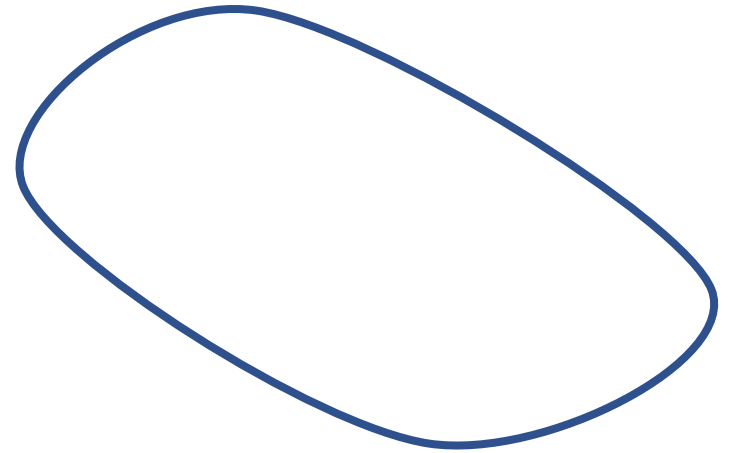
What is Potential/ Voltage ?

Consider a system with a large collection of chargeable particles.

The entire collection is entirely neutral – always!

However, some work can be done on the system to locally separate the charges.

Since work is done on the system, its total energy (potential) energy increases!



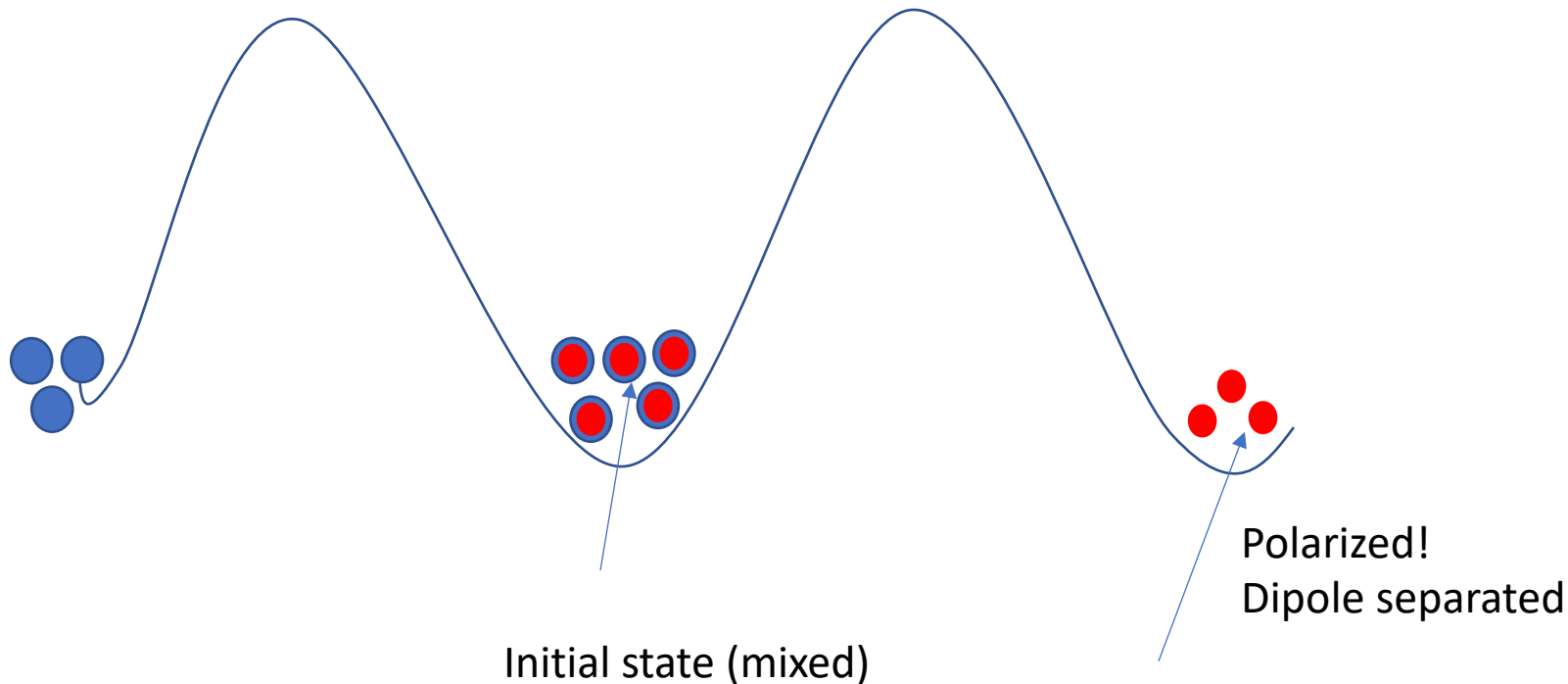
A battery is a special system in which the work is done to separate the charges.

Remember, just separation is not enough!

Remember a low energy state with increasing entropy exists in which the charges are mixed! .

So, if there exists a pathway for the charges to recombine, the system will recombine and reduce to lowest energy.

Excess energy will be delivered as heat!



A battery, potential source has the capacity to separate charges and retain the separation for enough time

Useful work can be done if the charges are allowed to recombine in a controllable fashion

Capacitor/ Charge accumulator



A charge density ρ separated by a distance d [m] develops a potential difference

$$V = \frac{d\rho}{\epsilon}$$

Defining a capacitance density $C = \epsilon/d$ [F/m]

Current – defined as the rate of flow of charges

Current through the plates: $i = C \frac{dV}{dt} = C \int \frac{dE}{dt} dl$

Resistor

If there exists a current density (j) due to a potential difference V

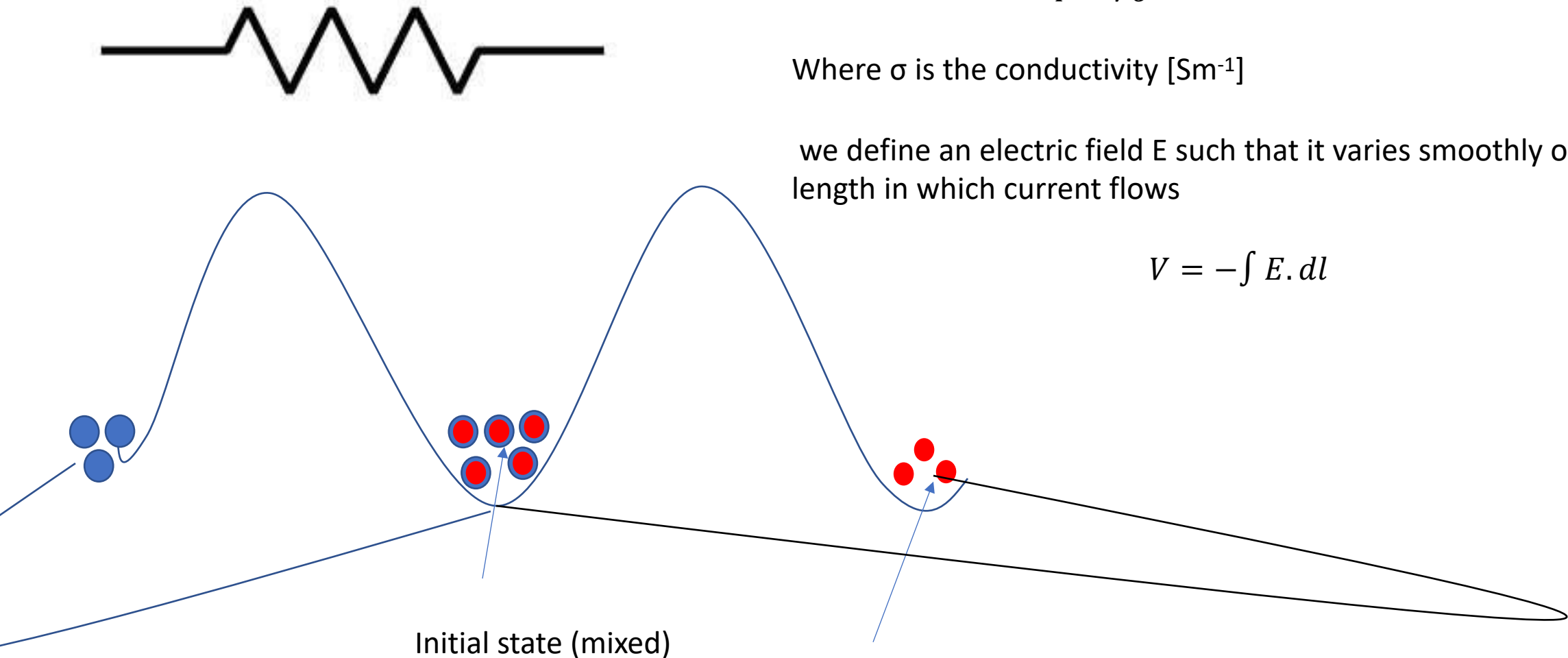
It was experimentally observed that

$$I = V\sigma$$

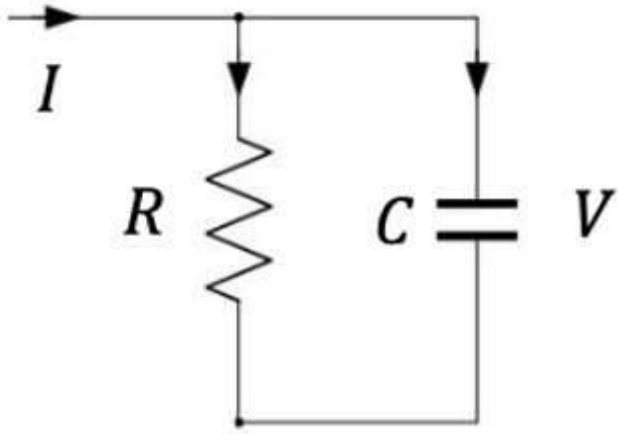
Where σ is the conductivity [Sm^{-1}]

we define an electric field E such that it varies smoothly over the length in which current flows

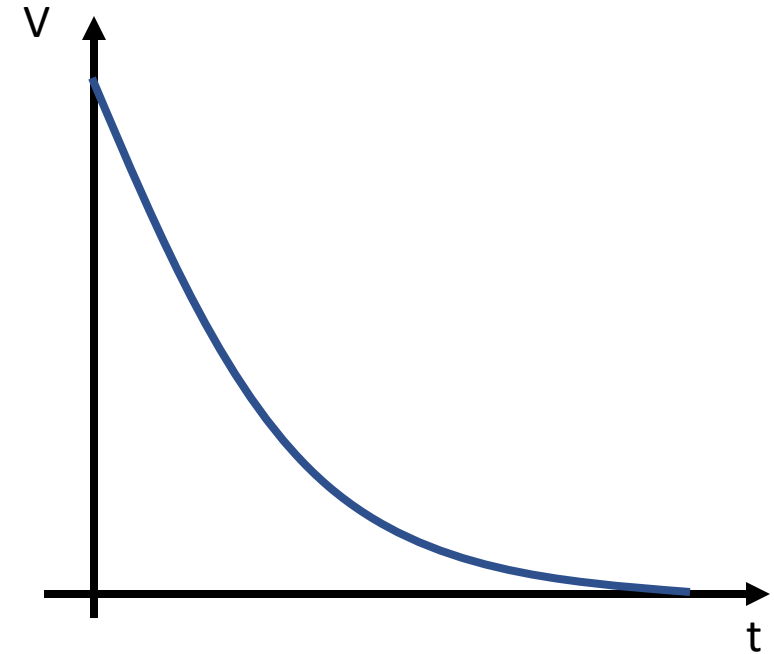
$$V = -\int E \cdot dl$$



At what rate will the charges now be recombined ?



$$I = \frac{V}{R} + C \frac{dV}{dt}$$
$$\frac{I}{C} = \frac{V}{RC} + \frac{dV}{dt}$$



Solution to the D.E $V(t) = V_0 e^{-\frac{t}{RC}}$

The potential will drop to $1/e$ times the initial value in a time determined by its time constant RC

Electrical Characterization of Materials

The following properties are usually measured.

1. Electrical conductivity
2. Dielectric constant/Strength
3. Ferro-electricity
4. Piezoelectricity
5. Pyroelectricity
6. Photosensitivity

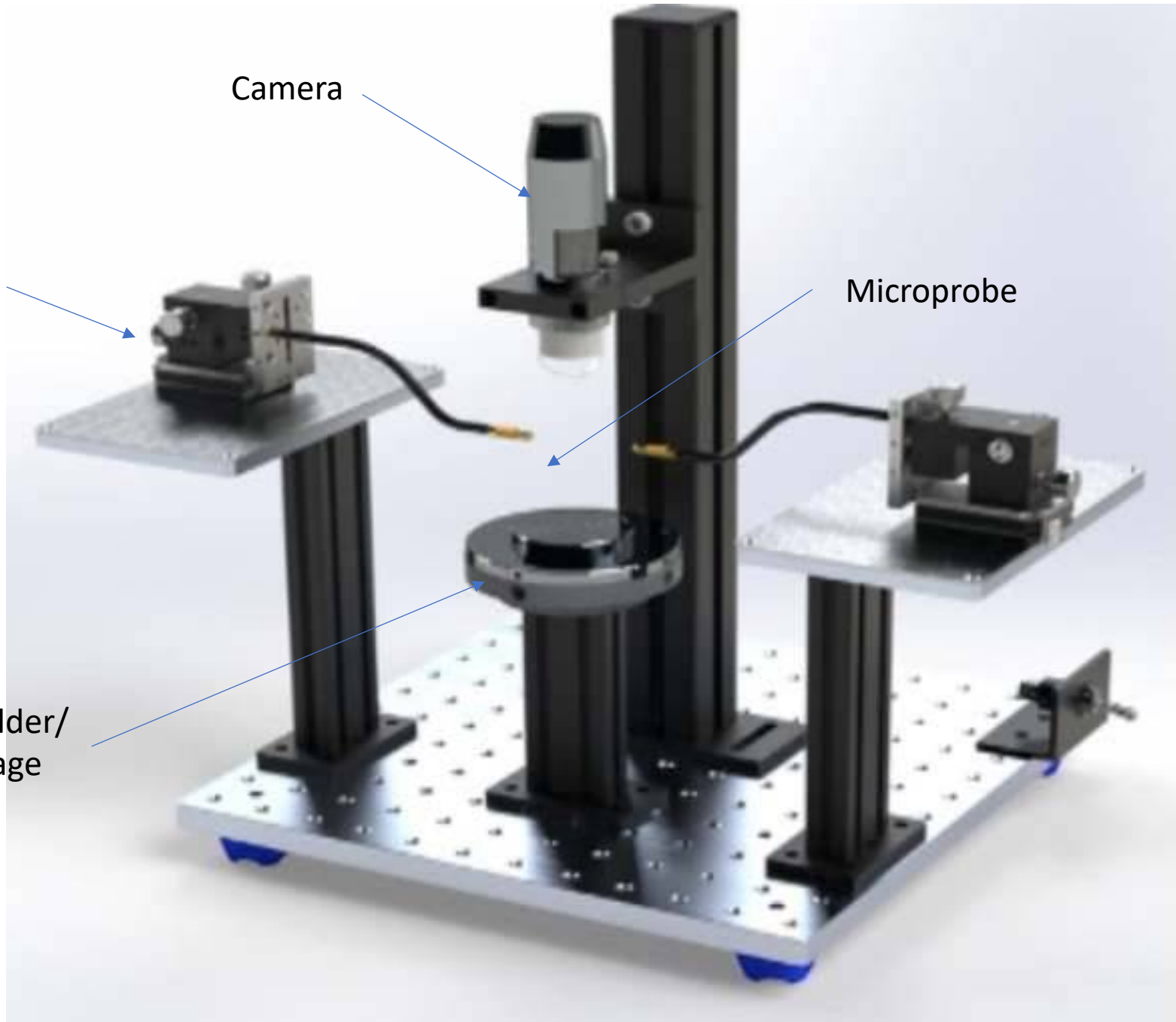
Probestation

Camera

Micro manipulator

Microprobe

Sample holder/
Heating stage



Resistivity Measurement

Two wires are connected to a material of a well-defined geometry. Preferably with a constant cross section.



Two methods:

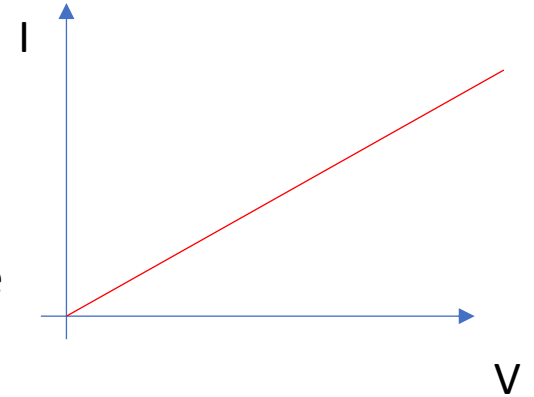
1. Constant potential:-

In this method, a constant voltage source is connected to the terminals and the current is measured.

Under such circumstances, we obtain the material resistance $R = \frac{V}{I}$

Typically, a series of measurements are taken such that the resistance is extracted as the slope

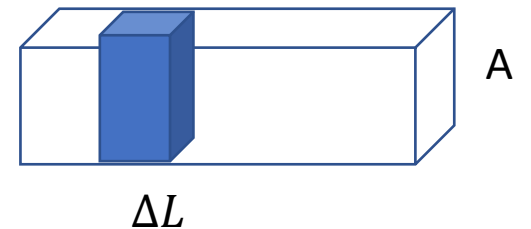
$$\text{Resistance } R = \frac{dV}{dI}$$



However, also remember, that the resistance is proportional to the geometry

$$\rho = R * \frac{A}{L}$$

Is an intrinsic material property



Resistance is measured in Ohms (Ω)

Length, and area in m and m^2 respectively.

The resistivity has the units of $\Omega - m$

At room temperature the measured resistances are given here.

Sl. No.	Element	Resistivity at 20°C in $\Omega - m$
1	Silver	1.59×10^{-8}
2	Copper	1.7×10^{-8}
3	Gold	2.44×10^{-8}
4	Aluminum	2.82×10^{-8}
5	Tungsten	5.6×10^{-8}
6	Iron	1.0×10^{-7}
7	Platinum	1.1×10^{-7}
8	Lead	2.2×10^{-7}
9	Manganin	4.82×10^{-7}
10	Constantan	4.9×10^{-7}
11	Mercury	9.8×10^{-7}
12	Carbon (Graphite)	3.5×10^{-5}
13	Germanium	4.6×10^{-1}
14	Silicon	6.4×10^2
15	Glass	10^{10} to 10^{14}
16	Quartz (fused)	7.5×10^{17}

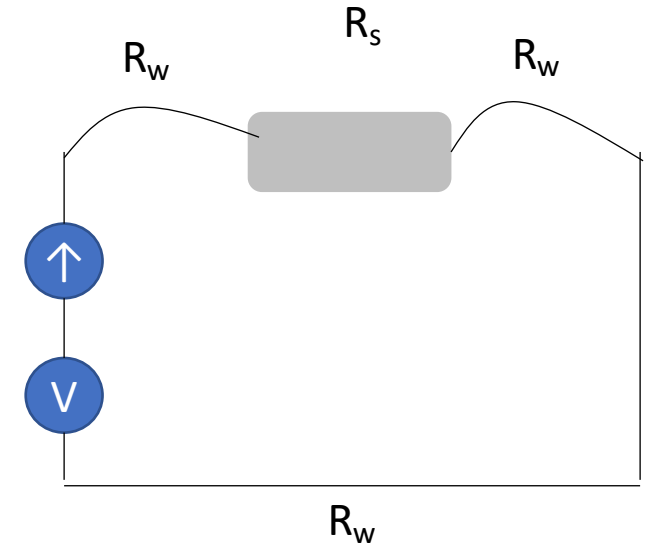


There is a problem here!

In an actual experiment, the system is like this

Therefore, the total resistance of the circuit is

$$R_M = R_S + n * R_W$$



In situations where the sample resistance is very small (particularly in metals), This method over-estimates the resistance

The solution is Via Kirchhoff's law.

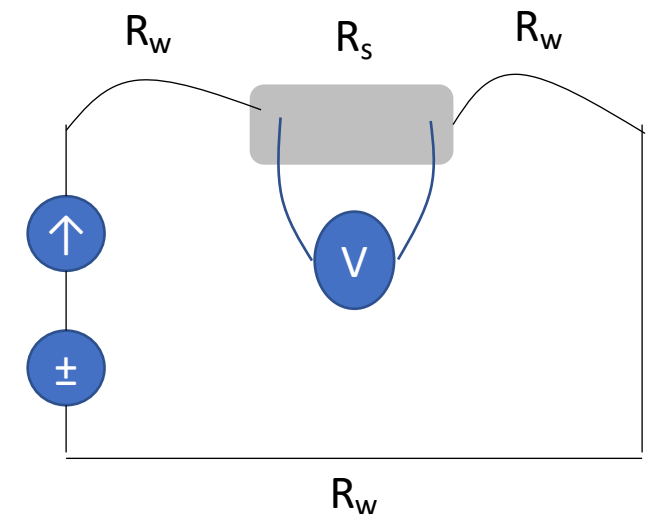
By Kirchhoff, all the resistive components give independent potential drops, which should sum up to the supply potential

$$V_T = V_S + n * V_W$$

Thus the solution to the problem is near! .

Since the current that passes through the wires and the samples is the same. All you need is to measure the potential difference on the sample

This method of measurement is called as four-probe Resistance measurement



Measure of Capacitance

For a class of materials that do not conduct current so well, resistance tends to very high values

No proper current sensor can measure such tiny currents

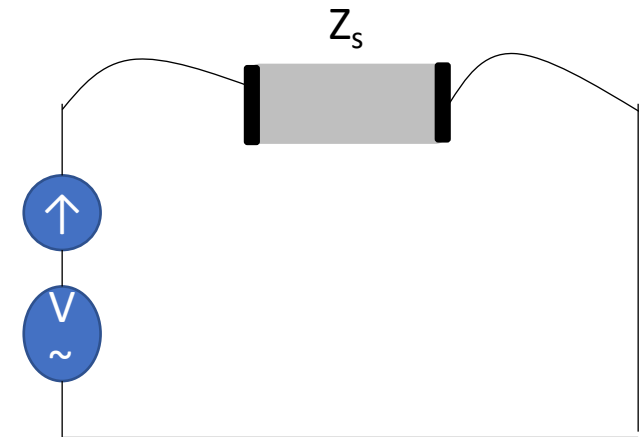
$R \rightarrow \infty$ as $I \rightarrow 0$ for some known small potential.

How do we characterize these materials ?

Since the resistance goes to ∞ for a steady DC potential,

We can measure the capacitance of the structure.

This is done by imposing an alternating potential of freq ω



Once again, the actual capacitance is measured by sweeping the frequency

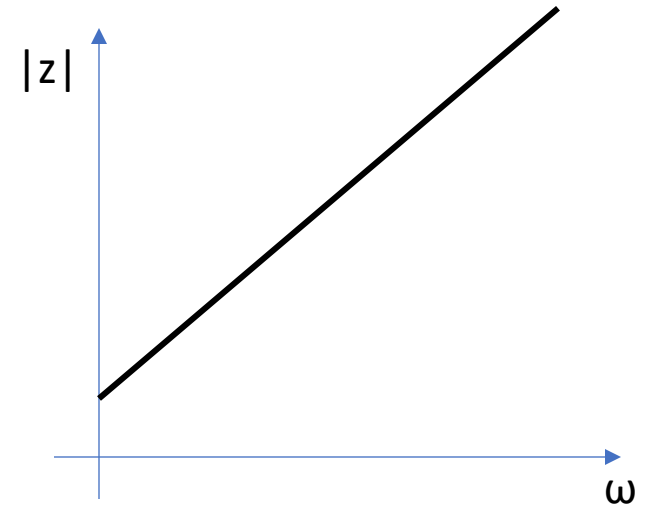
The slope of the impedance vs frequency plot gives the capacitance.

The Y-intercept gives the resistance!

For a typical device

$$Z = R + j\omega C$$

The impedance is complex as there is phase difference between the potential and the current.



The capacitance is once again a geometric parameter.

A fundamental property of the material, dielectric constant ϵ is then obtained by

$$\epsilon = \frac{CA}{d}$$

Where C is measured slope, A – cross section of the capacitor and d is the separation between the plates

A general table of dielectric constants

Material	Dielectric constant
Air (dry)	1.0
Bakelite	4.9
Mylar	3.2
Nylon	3.4
Paper	3.7
Paraffin-impregnated paper	3.5
Polypropylene	2.2
Polystyrene	2.6
Polyvinyl chloride	3.4
Porcelain	6.0
Pyrex glass	5.6
Strontium titanate	233.0
Water	80.0

For a general unknown material:

Suppose, I have an excitation signal such as

$$v = v_0 \sin \omega t$$

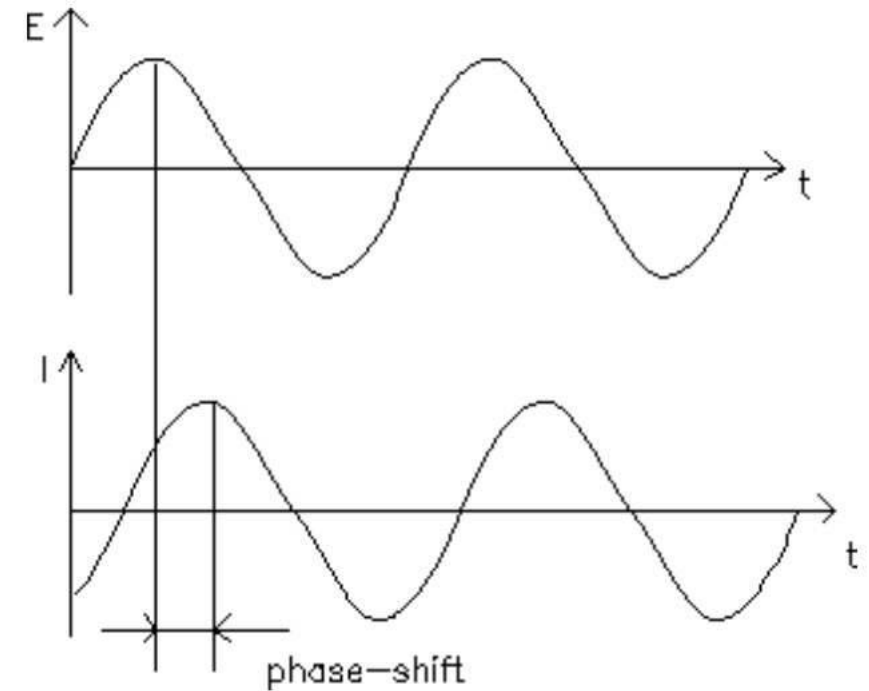
$$\omega \left(\frac{\text{rad}}{\text{sec}} \right) = 2\pi f - f \text{ being the frequency (hertz)}$$

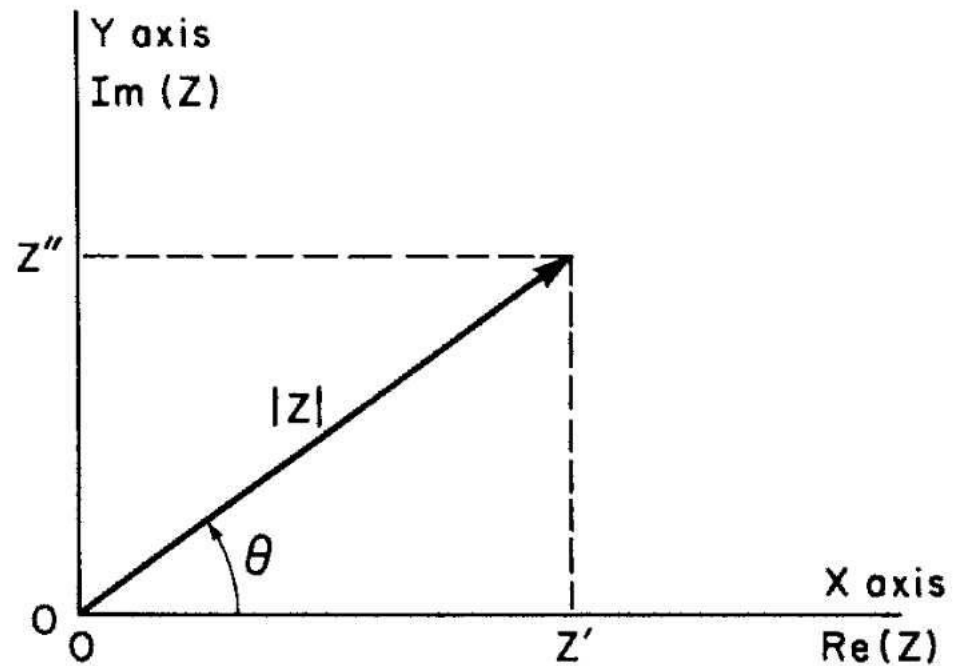
In a linear system, the output current is of the form

$$I = I_0 \sin \omega t + \phi$$

$$\text{An impedance } Z = \frac{v}{I} = \frac{v_0}{I_0} \frac{\sin \omega t}{\sin \omega t + \phi}$$

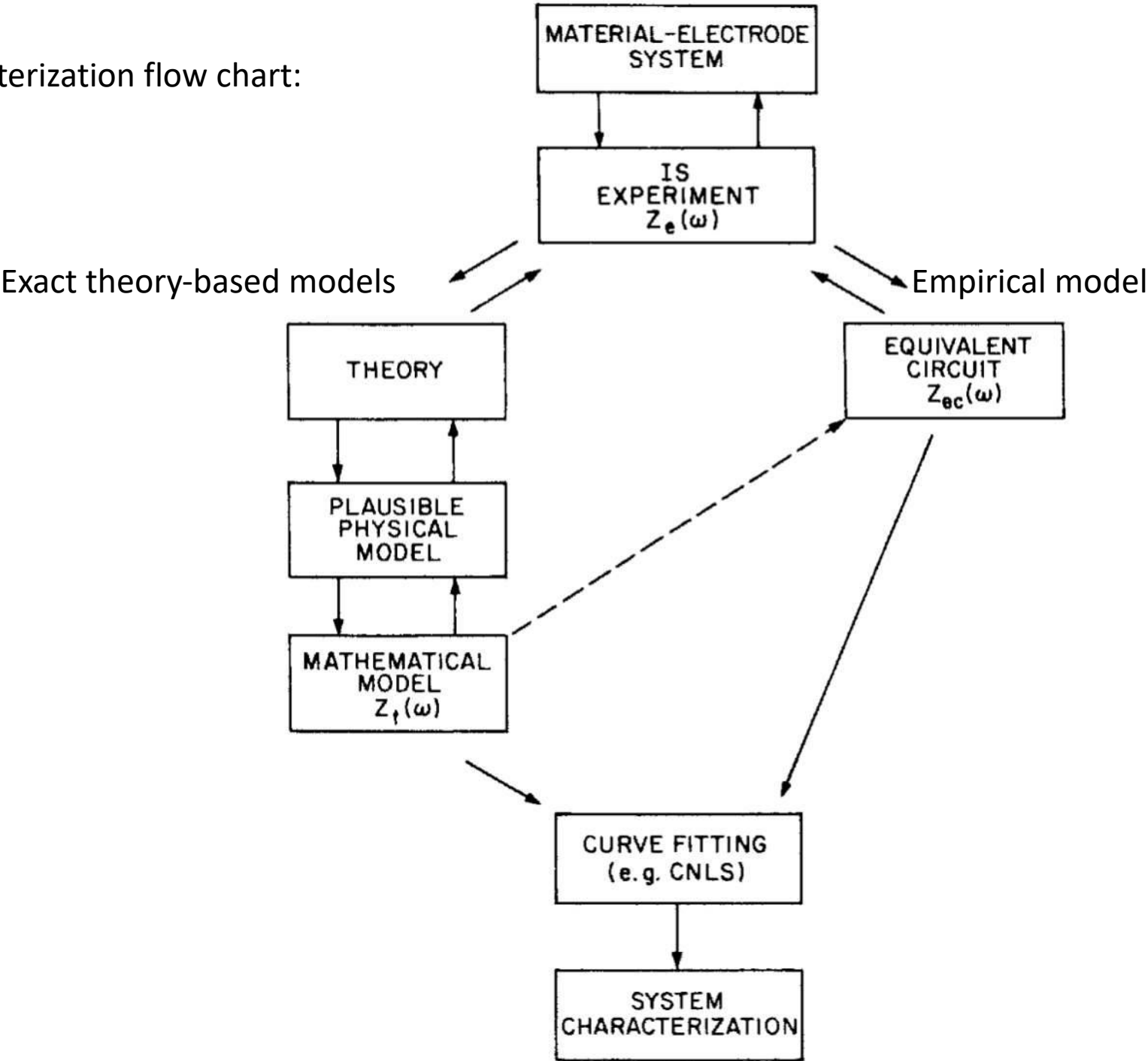
Thus, impedance is typically specified as Z_0, ϕ at ω



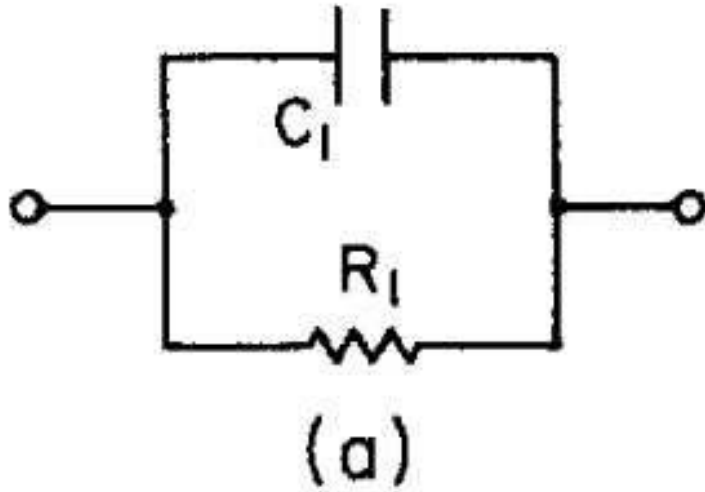


The impedance is a complex value $R(Z) = |Z| \cos \theta$ and $Im(Z) = |Z| \sin \theta$

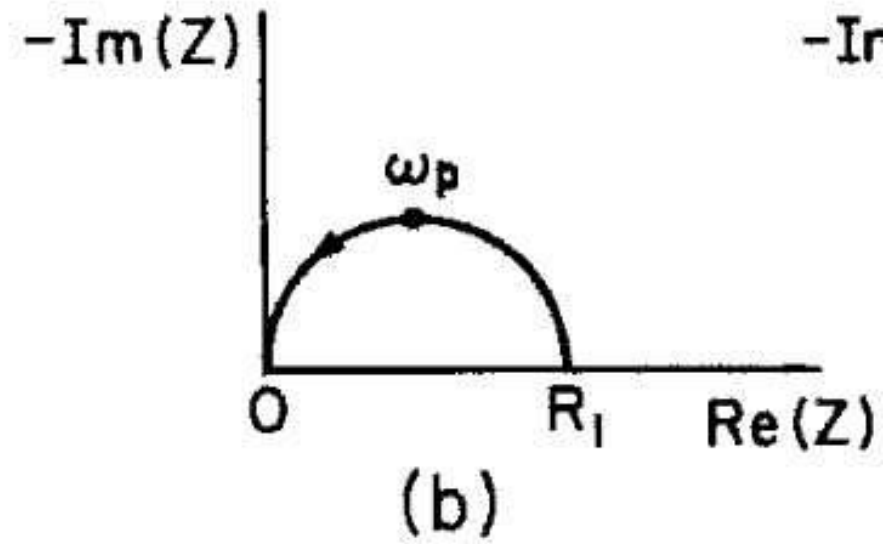
General characterization flow chart:



Simple equivalent circuit

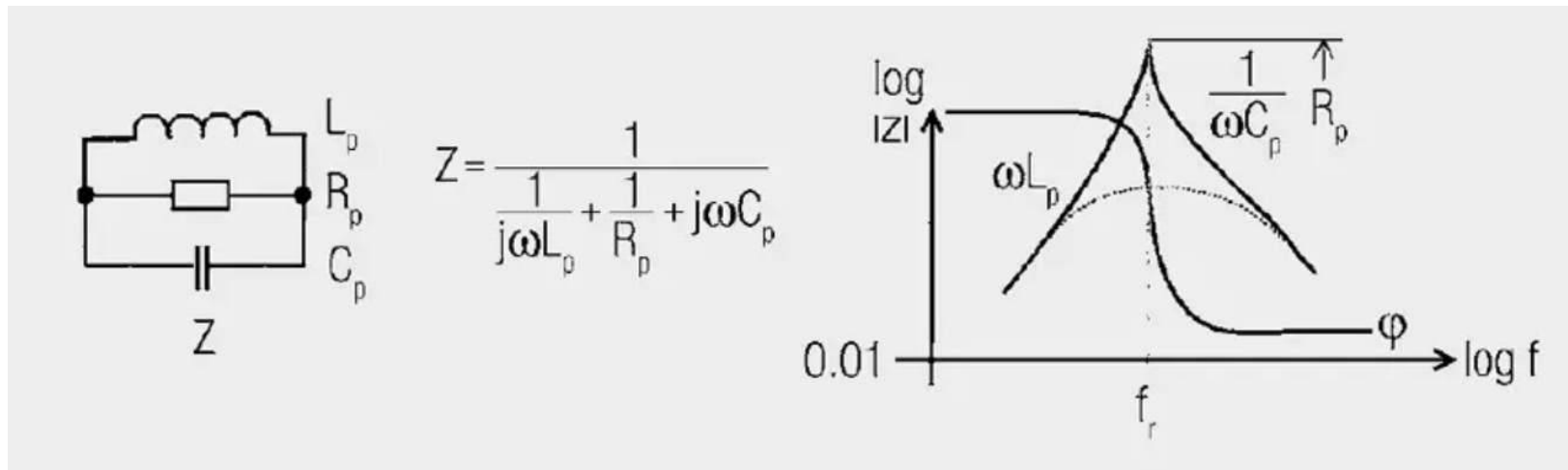
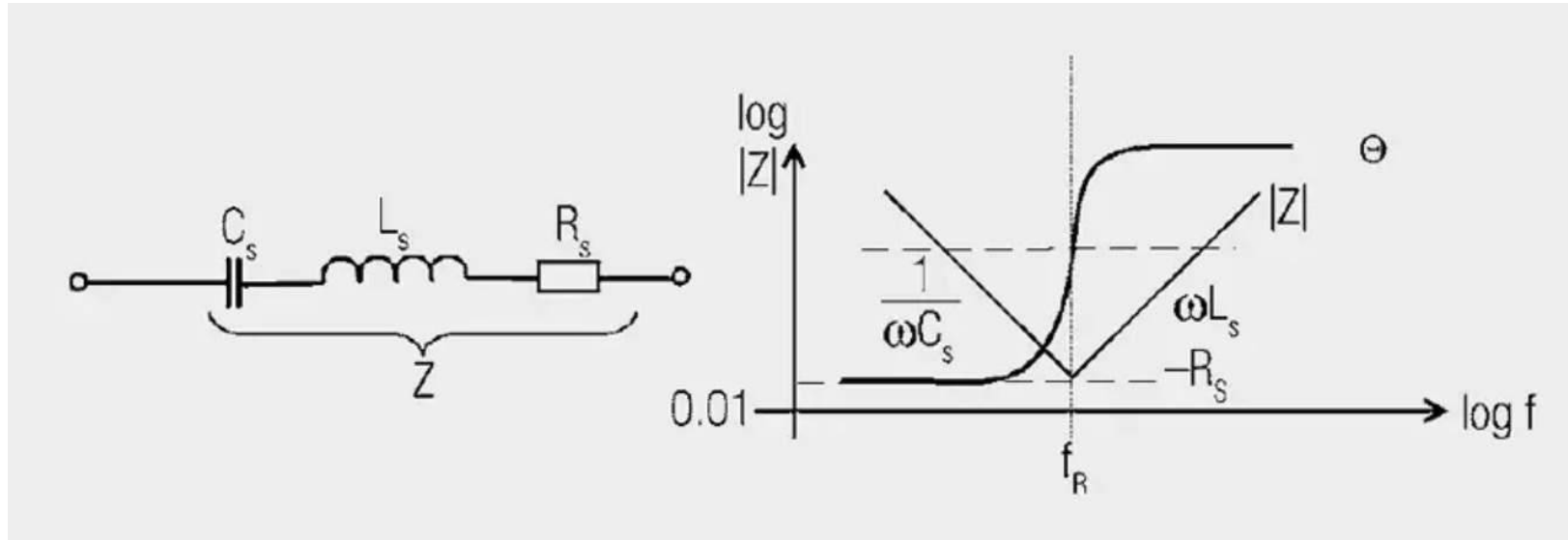


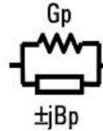
Nyquist plot



Maximal impedance frequency ω_p - associated with minimum time constant $\tau_D = 1/\omega_p$

Alternate equivalent circuits



Series mode		Parallel mode	
$R_s \pm jX_s$ 	$ Z = \sqrt{R_s^2 + X_s^2}$ $\theta = \tan^{-1} (X_s/R_s)$	$G_p \pm jB_p$ 	$ Y = \sqrt{G_p^2 + B_p^2}$ $\theta = \tan^{-1} (B_p/G_p)$
R_s: Series resistance X_s: Series reactance ($X_L = \omega L_s$, $X_C = -1/(\omega C_s)$) L_s: Series inductance ($= X_L/\omega$) C_s: Series capacitance ($= -1/(\omega X_C)$) D: Dissipation factor ($= R_s/X_s = R_s/(\omega L_s)$ or $\omega C_s R_s$) Q: Quality factor ($= X_s/R_s = \omega L_s/R_s$ or $1/(\omega C_s R_s)$)		G_p: Parallel conductance ($= 1/R_p$) B_p: Parallel susceptance ($B_C = \omega C_p$, $B_L = -1/(\omega L_p)$) L_p: Parallel inductance ($= -1/(\omega B_L)$) C_p: Parallel capacitance ($= B_C/\omega$) D: Dissipation factor ($= G_p/B_p = G_p/(\omega C_p)$ $= 1/(\omega C_p R_p)$ or $\omega L_p G_p = \omega L_p/R_p$) Q: Quality factor ($= B_p/G_p = \omega C_p/G_p$ $= \omega C_p R_p$ or $1/(\omega L_p G_p) = R_p/(\omega L_p)$)	

Dielectric Breakdown

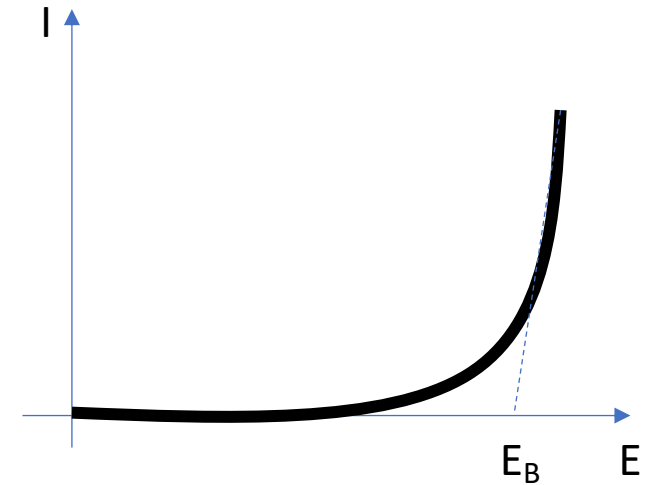
All the measurements before are true before a certain limit is reached.

A true dielectric blocks DC current and allows AC current to flow with a known impedance.

However, this blocking capacity is not infinite.

A typical current density vs DC potential for dielectrics look like

The field at which the current density increases drastically is called as the breakdown potential and the dielectric no longer blocks the DC potential



This property is important for insulators used in HV transmission, microelectronic devices and control systems

Some example dielectric strength of polymers

Dielectric strength in kilovolts per inch (kV/in):

Material*	Dielectric strength
Vacuum	20
Air	20 to 75
Porcelain	40 to 200
Paraffin Wax	200 to 300
Transformer Oil	400
Bakelite	300 to 550
Rubber	450 to 700
Shellac	900
Paper	1250
Teflon	1500
Glass	2000 to 3000
Mica	5000

Effect of Temperature

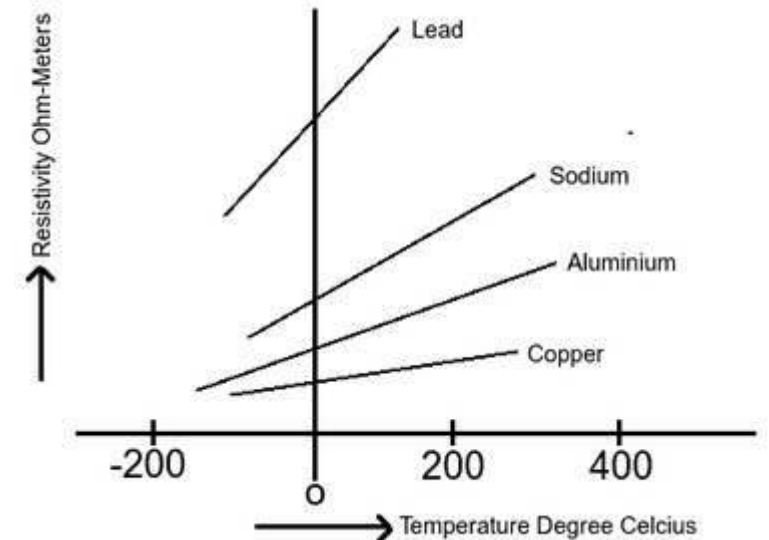
Temperature has an interesting effect on electrical properties

Experimentally, for many materials, if the resistance (R_{ref}) is measured at a reference temperature T_{ref}

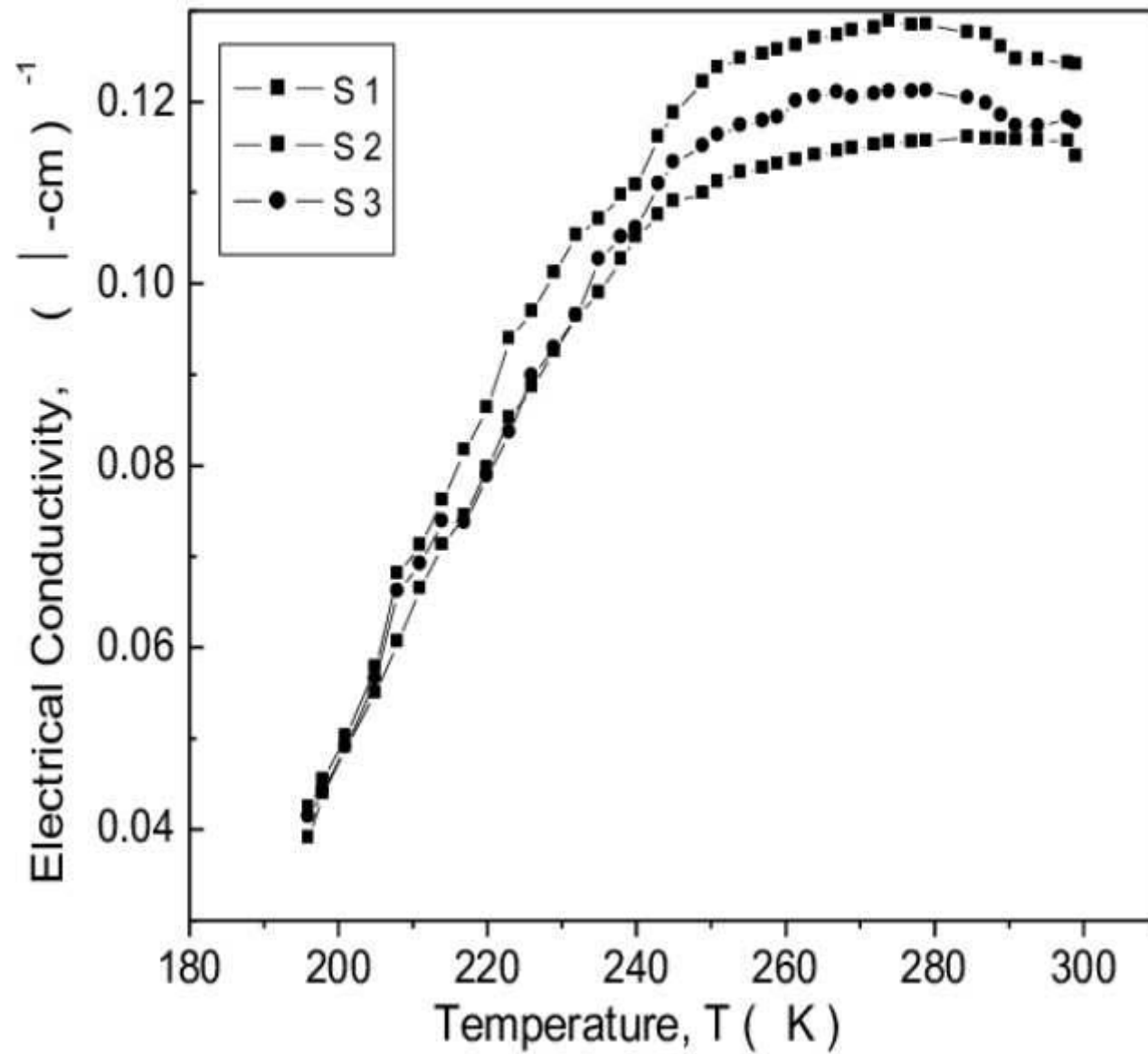
At any other temperature T , the resistance can be obtained from the expression

$$R = R_{ref}[1 + \alpha(T - T_{ref})]$$

Metals seems to have a positive temperature coefficient α



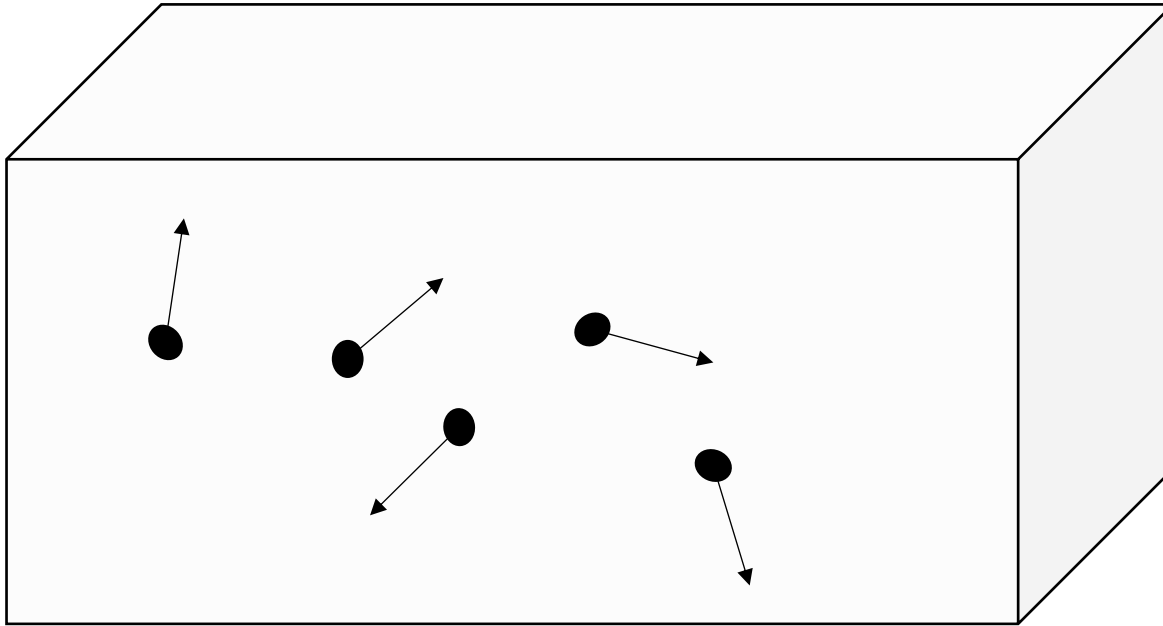
However, for another class of materials - Semiconductors



Experimental Temperature Co-efficient of Resistance

Material	Resistivity, ρ ($\Omega \cdot \text{m}$)	Temperature Coefficient, α ($^{\circ}\text{C}^{-1}$)
<i>Conductors</i>		
Silver	1.59×10^{-8}	0.0061
Copper	1.68×10^{-8}	0.0068
Gold	2.44×10^{-8}	0.0034
Aluminum	2.65×10^{-8}	0.00429
Tungsten	5.6×10^{-8}	0.0045
Iron	9.71×10^{-8}	0.00651
Platinum	10.6×10^{-8}	0.003927
Mercury	98×10^{-8}	0.0009
Nichrome (Ni, Fe, Cr alloy)	100×10^{-8}	0.0004
<i>Semiconductors[†]</i>		
Carbon (graphite)	$(3-60) \times 10^{-5}$	-0.0005
Germanium	$(1-500) \times 10^{-3}$	-0.05
Silicon	0.1-60	-0.07
<i>Insulators</i>		
Glass	10^9-10^{12}	
Hard rubber	$10^{13}-10^{15}$	

What is Resistance ?



Assume a material having n free electrons.

Electrons move in material like ideal gas in containers.

Only difference: Electrons scatter with ions in solids.

If mean scattering time is τ

Under these assumptions, one can extract resistivity $\frac{1}{\rho} = \frac{ne^2\tau}{m}$

This is the Drude's model for conductivity in metals

$$\frac{1}{\rho} = \frac{ne^2\tau}{m}$$

If the electron masses is assumed to be constant in all materials, the only variable then is

$$n * \tau$$

Thus, materials with higher number of electrons or those with longer scattering times have lower resistivity

Along the same lines: If temperature is increased, if the number of electrons is not changing, a increased resistance is observed because of reduction in the mean scattering time.

This is can be due to stronger ion oscillations at higher temperatures!

However, this is just the beginning! The story is not complete!